

Algebraic Geometry

A Functor-of-Points Viewpoint

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Developed from lectures by Marc Hoyois

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Preface

These notes grew from, and substantially extend, the *Algebraic Geometry* course taught by Professor Marc Hoyois at the University of Regensburg. The course supplies the original architecture, many guiding examples, and the functor-of-points viewpoint. The present text develops the foundational arguments in detail, adds material beyond the course, and identifies genuinely deep inputs as accepted or cited theorems with proof roadmaps.

What is Algebraic Geometry?

Algebraic geometry, from the perspective developed in these notes, is the study of functors

$$X : \mathbf{CAlg} \rightarrow \mathbf{Set}$$

from commutative rings to sets — or, in the derived variant, to the ∞ -category $\mathbf{An}$ of anima. Each such *algebraic functor* X assigns to every ring R a set $X(R)$ of “ R -points of X ”. Geometry then becomes the study of these functors: their subobjects, morphisms, gluing behaviour, and invariants.

To make this functorial picture concrete, Grothendieck’s *unstraightening* (equivalently, the category of elements construction) turns any X into a discrete opfibration

$$p : \int X \longrightarrow \mathbf{CAlg}.$$

Explicitly, $\int X$ has as objects the pairs (R, x) with $x \in X(R)$, and the fiber over each ring R is exactly the set of R -points:

$$p^{-1}(R) = X(R).$$

When X is the solution functor to a polynomial system $f_1, \dots, f_m \in \mathbb{Z}[x_1, \dots, x_n]$,

$$X(R) = \{a \in R^n \mid f_1(a) = \dots = f_m(a) = 0\},$$

the opfibration $\int X \rightarrow \mathbf{CAlg}$ assembles *all* solution sets over *all* rings into a single coherent geometric object, and ring homomorphisms $R \rightarrow S$ induce maps from R -points to S -points. Doing geometry on X is doing geometry on the whole family at once.

Toward Derived Algebraic Geometry. In derived algebraic geometry one replaces the source category $\mathbf{CAlg}$ by a category of “derived rings” (animated rings, or $\mathbb{E}_\infty$ -algebras), and the target $\mathbf{Set}$ by the ∞ -category $\mathbf{An}$. The unstraightening becomes a coCartesian fibration of ∞ -categories, and the examples below continue to make sense — producing richer invariants (e.g. homotopical tangent complexes, derived loop spaces) that are invisible in the classical setting.

Motivating Examples

The remainder of this preface — adapted from Marc Hoyois’s script — illustrates what algebraic geometry looks like **before** any formal machinery: solution functors of concrete polynomial systems, examined across rings. These examples guide our intuition for the entire development that follows.

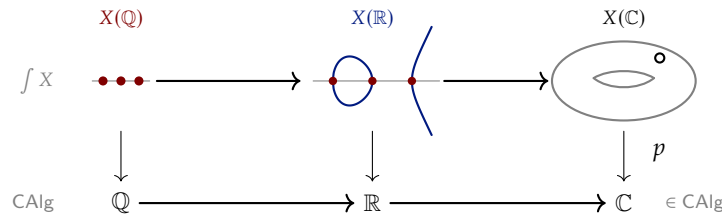
Punctured Elliptic Curves

Consider the equation $y^2 = x^3 - x$ and its solution functor

$$X(R) = \{(a, b) \in R^2 \mid b^2 = a^3 - a\}.$$

Different rings reveal different aspects of X .

Over $\mathbb{R}$. $X(\mathbb{R}) \subset \mathbb{R}^2$ is a 1-dimensional real submanifold with two connected components, diffeomorphic to $S^1 \sqcup \mathbb{R}$:



Over $\mathbb{Q}$ and $\mathbb{Z}$. The rational points are in fact integral:

$$X(\mathbb{Q}) = X(\mathbb{Z}) = \{(0, 0), (\pm 1, 0)\}.$$

Over $\mathbb{C}$. $X(\mathbb{C}) \subset \mathbb{C}^2$ is a 1-dimensional complex submanifold, diffeomorphic to a **punctured torus**. Holomorphically, it is the noncompact Riemann surface $(\mathbb{C}/\Lambda) \setminus \{0\}$, where $\Lambda = \mathbb{Z} \oplus \mathbb{Z}i$.

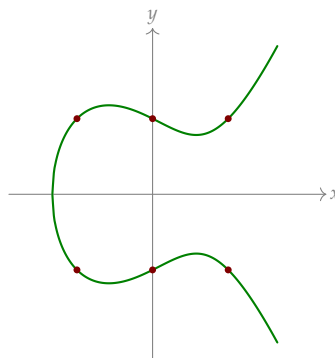
Over the dual numbers $R[\varepsilon]$, $\varepsilon^2 = 0$. A pair $(a + u\varepsilon, b + v\varepsilon)$ is a solution if and only if $(a, b) \in X(R)$ and (u, v) is a tangent vector at (a, b) . Hence $X(R[\varepsilon])$ is the **tangent bundle** of $X(R)$. Over a field k , the tangent space at every point of $X(k)$ is 1-dimensional, except at $(1, 0)$ in characteristic 2, where it becomes 2-dimensional. Accordingly, X has **bad reduction** at the prime 2 and **good reduction** at every other prime.

Dual Numbers and Derivatives. For $f \in R[x_1, \dots, x_n]$ and $a + u\varepsilon \in R[\varepsilon]^n$,

$$f(a + u\varepsilon) = f(a) + \left(\frac{\partial f}{\partial x_1}(a) \cdot u_1 + \dots + \frac{\partial f}{\partial x_n}(a) \cdot u_n \right) \varepsilon.$$

First-order differentiation emerges purely from the algebraic relation $\varepsilon^2 = 0$, which automatically annihilates every higher-order term. The tangent space at $a \in V(f)(R)$ is therefore the kernel of the gradient $\nabla f(a): R^n \rightarrow R$.

Compare this with the closely related equation $y^2 = x^3 - x + 1$, whose solution functor Y looks similar at first glance but is qualitatively different:



$Y(\mathbb{R})$ has a single connected component, diffeomorphic to $\mathbb{R}$; $Y(\mathbb{Z})$ already comprises 12 points,

$$Y(\mathbb{Z}) = \{(-1, \pm 1), (0, \pm 1), (1, \pm 1), (3, \pm 5), (5, \pm 11), (56, \pm 419)\},$$

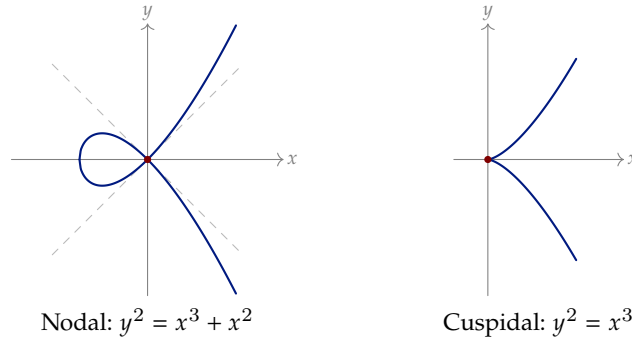
and $Y(\mathbb{Q})$ is infinite. Although $X(\mathbb{C})$ and $Y(\mathbb{C})$ are diffeomorphic — both are punctured tori — they are **not** biholomorphic: they correspond to non-isomorphic lattices in $\mathbb{C}$, a distinction invisible to real differential geometry.

Singular Cubic Curves

Consider the functors

$$N(R) = \{(a, b) \mid b^2 = a^3 + a^2\}, \quad C(R) = \{(a, b) \mid b^2 = a^3\}.$$

N is the **nodal cubic** and C the **cuspidal cubic**:



Both have a singularity at the origin, visible in the Jacobian: the gradient vanishes at $(0, 0)$, so the tangent space there is 2-dimensional rather than 1-dimensional.

There are two algebraic ways to address the nodal singularity:

- (i) **Deformation.** View N as the $\lambda = 0$ specialization of the family $N_\lambda = V(y^2 - x(x + \lambda)(x + 1))$. For $\lambda \neq 0, 1$, $N_\lambda(\mathbb{C})$ is a nonsingular punctured torus; singular fibers occur only at $\lambda = 0$ and $\lambda = 1$ (where two of the three roots $0, -\lambda, -1$ collide).
- (ii) **Blowup.** Introduce the slope coordinate $s = y/x$, in terms of which the equation rewrites as $s^2 = x + 1$ — now nonsingular. The functor $\tilde{N}(R) = \{(a, c) \mid c^2 = a + 1\}$, equipped with the map $\tilde{N} \rightarrow N$, $(a, c) \mapsto (a, ac)$, is the **blowup of N at the origin**.

In characteristic zero, any variety can be resolved by iterated blowups (Hironaka, 1964); whether the analogous statement holds in positive characteristic is a major open problem.

Compactifying Affine Curves

The affine curves of the previous subsections can be compactified by passing from affine n -space $\mathbb{A}^n$ to projective n -space $\mathbb{P}^n$. Over an arbitrary ring the coordinate formula must remember line bundles: an R -point is an isomorphism class of locally split rank-one quotients

$$R^{n+1} \twoheadrightarrow L, \quad L \text{ an invertible } R\text{-module.}$$

After choosing a trivialization $L \simeq R$ this quotient is a unimodular tuple $(a_0, \dots, a_n)$, and changing the trivialization multiplies all coordinates by an element of $R^\times$. If $U_n(R)$ denotes the set of unimodular tuples, the precise comparison is

$$\begin{array}{ccc} U_n(R) & \xrightarrow{\sim} & \{\text{surjections } R^{n+1} \twoheadrightarrow R\} \\ \downarrow /R^\times & & \downarrow \text{forget the target trivialization} \\ U_n(R)/R^\times & \xhookrightarrow{\text{trivial-line locus}} & \mathbb{P}^n(R). \end{array}$$

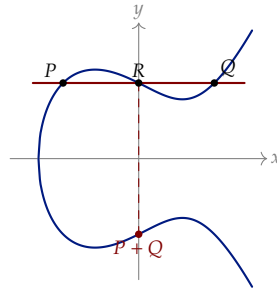
Thus the familiar formula by homogeneous coordinates is valid over a local ring (and, in particular, over a field), while for a general ring it describes precisely the points whose quotient line is trivial. A homogeneous polynomial f of degree d evaluates on the quotient to a section of $L^{\otimes d}$, so the condition $f = 0$ is independent of every trivialization. We retain the notation $[a_0 : \dots : a_n]$ for this quotient-line datum.

To compactify $y^2 = x^3 - x$, we **homogenize** with a new variable w : multiply each term by an appropriate power of w to equalise the total degree, yielding $wy^2 = x^3 - w^2x$. The resulting projective solution functor is

$$\bar{X}(R) = \{[w : x : y] \in \mathbb{P}^2(R) \mid wy^2 = x^3 - w^2x\}.$$

Setting $w = 1$ recovers the affine X . For field-valued, hence geometric, points, the equation with $w = 0$ forces $x = 0$ and the quotient condition forces $y \neq 0$; there is therefore a single point at infinity, $[0 : 0 : 1]$, which fills the puncture. This statement is deliberately about geometric points: over a nonreduced test ring, nonzero nilpotent values with $x^3 = 0$ may give additional infinitesimal R -points above the same point. Over $\mathbb{C}$, $\bar{X}(\mathbb{C})$ becomes a genuine torus, biholomorphic to $\mathbb{C}/(\mathbb{Z} \oplus \mathbb{Z}i)$.

Away from primes of bad reduction, the smooth projective closures of both X and the neighboring curve Y are **elliptic curves**, with the point at infinity $O = [0 : 0 : 1]$ as identity. A line meeting either cubic in P, Q, R , counted with intersection multiplicity, satisfies $P + Q + R = O$. To display this rule using three visible integral points, the following picture is drawn for the closure $\bar{Y}$ of $y^2 = x^3 - x + 1$, not for $\bar{X}$.



In the displayed affine chart of $\bar{Y}$, the line $y = 1$ meets the curve at $P = (-1, 1)$, $R = (0, 1)$, and $Q = (1, 1)$, and $P + Q$ is the reflection of R across the x -axis. The same chord–tangent construction defines the group law on $\bar{X}$. In the complex picture $\bar{X}(\mathbb{C}) \simeq \mathbb{C}/\Lambda$, that law is inherited from addition on $\mathbb{C}$.

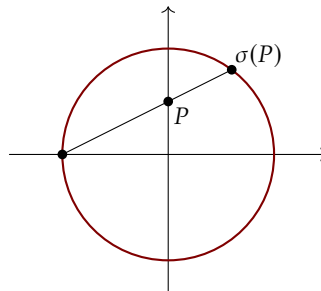
Fermat Curves and Pythagorean Triples

The Fermat functors $X_n(R) = \{[a : b : c] \in \mathbb{P}^2(R) \mid a^n = b^n + c^n\}$ interpolate between projective lines and profound number theory.

- $n = 0$: $X_0(R) = \emptyset$ for every nonzero ring R , so X_0 is the **empty scheme**.
- $n = 1$: $X_1 \simeq \mathbb{P}^1$ is a projective line.
- $n = 2$: The integral solutions $X_2(\mathbb{Z})$ are the (primitive) **Pythagorean triples**. Over any $\mathbb{Z}[\frac{1}{2}]$ -algebra R there is an explicit parameterization

$$\sigma : \mathbb{P}^1(R) \xrightarrow{\sim} X_2(R), \quad [s : t] \longmapsto [s^2 + t^2 : s^2 - t^2 : 2st],$$

specializing to Euclid's formula over $\mathbb{Z}$. Geometrically, σ is the **inverse stereographic projection** from the point $(-1, 0)$ on the unit circle $b^2 + c^2 = 1$:



- $n \geq 3$: Fermat's Last Theorem (Wiles, 1995) asserts

$$X_n(\mathbb{Q}) = X_n(\mathbb{Z}) = \begin{cases} \{[0 : 1 : -1], [1 : 0 : 1], [1 : 1 : 0]\} & n \text{ odd,} \\ \{[1 : 0 : \pm 1], [1 : \pm 1 : 0]\} & n \text{ even.} \end{cases}$$

Away from 3, $X_3 : \text{CAlg}_{\mathbb{Z}[1/3]} \rightarrow \text{Set}$ is another elliptic curve, with a unique abelian group law having $[0 : 1 : -1]$ as identity.

Affine, Projective, and General Schemes

The examples above touch on several overlapping classes of geometric objects, whose strict inclusions

$$\begin{array}{ccccccc} \{\text{affine schemes}\} & \subset & \{\text{quasi-affine schemes}\} & & & & \\ & & \cap & & & & \\ \{\text{projective schemes}\} & \subset & \{\text{quasi-projective schemes}\} & \subset & \{\text{schemes}\} & \subset & \text{Fun}(\text{CAlg}, \text{Set}) \end{array}$$

form a central organizing principle.

- Affine schemes are solution functors of polynomial systems in $\mathbb{A}^n$ (Chapter 1).
- Projective schemes are solution functors of **homogeneous** systems in $\mathbb{P}^n$ (Chapter 2).
- Quasi-affine and quasi-projective schemes are open subfunctors of the above.
- Schemes form the most general category — closed under gluing — and are defined in Chapter 6 as Zariski sheaves admitting an affine open cover.

A few boundary examples clarify these inclusions.

- (i) **Punctured affine n -space** $\mathbb{A}^n - 0$ for $n \geq 2$ — the functor $R \mapsto \{a \in R^n \mid (a) = R\}$ — is quasi-affine but not affine.
- (ii) **Projective n -space minus a point**. For $n \geq 2$ and the rational point $p = [1 : 0 : \cdots : 0]$, the open subscheme $\mathbb{P}^n \setminus \{p\}$ is quasi-projective but not projective. It is not quasi-affine either: it contains the closed positive-dimensional projective hyperplane $V(x_0) \simeq \mathbb{P}^{n-1}$.
- (iii) Non-quasi-projective schemes are harder to construct; the prototype is the **affine line with two origins**, obtained by gluing two copies of $\mathbb{A}^1$ along $\mathbb{A}^1 \setminus \{0\}$. This scheme is quasi-compact but not separated (Example 6.2.4).

All of these are schemes. In what follows we develop the full theory of affine schemes (Chapter 1), projective schemes (Chapter 2), and general schemes (Chapter 6), with detours through quasi-coherent modules (Chapter 3), topological realizations (Chapter 4), and the Grothendieck-topology language of sheaves (Chapter 5); their basic geometric properties are then taken up in Chapter 7.

Course Structure

The defining feature of this course is its systematic development of algebraic geometry through the *functor of points* perspective. These notes are organized into two course parts followed by appendices.

Part I: Algebraic Geometry I (Foundations of Scheme Theory)

The first part establishes the language of schemes using the functorial approach. It covers the construction of schemes from algebraic functors and the necessary sheaf theory.

- **Chapter 1 (Affine Geometry):** Define *affine schemes* as representable functors via the Yoneda embedding $\mathrm{Spec}: \mathrm{CAlg}^{\mathrm{op}} \rightarrow \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$; prove the functorial Nullstellensatz, classify open and closed subfunctors, establish Zariski descent.
- **Chapter 2 (Projective Geometry):** $\mathbb{N}$ -graded rings, projective space $\mathbb{P}^n$ and the Proj functor; functorial projective Nullstellensatz, open and closed subfunctors of $\mathrm{Proj}(A)$, projective closure, Grassmannians, Veronese and Segre embeddings.
- **Chapter 3 (Quasi-coherent Modules):** Linear algebra over algebraic functors via Mod_X ; classification of open/closed subfunctors by quasi-coherent ideals, relative Spec and Proj , Zariski descent for modules, Serre twisting sheaves $\mathcal{O}(d)$.
- **Chapter 4 (Locale):** *Locales* as point-free topology; $\mathrm{Open}(X)$ as a spatial locale for any algebraic functor X ; geometric realization $|X|$, characterization of $\mathrm{Prim}(R)$ as a spectral space.
- **Chapter 5 (Sheaves):** Grothendieck topologies, sieves, descent; sheafification, Comparison Lemma, Zariski / fppf / fpqc topologies on Aff .
- **Chapter 6 (Schemes):** Formal definition of *schemes* as Zariski sheaves with affine open covers; immersions, gluing via locally trivial equivalence relations, separated / qcqs morphisms, identification with classical locally ringed spaces.

Part II: Algebraic Geometry II (Geometry of Schemes)

The second part begins the systematic study of the geometric content of schemes and their morphisms.

- **Chapter 7 (Properties of Schemes):** Local and global properties; reduced, noetherian, irreducible, integral schemes; dimension theory, regular / Cohen–Macaulay / normal schemes, the Zariski (co)tangent space.
- **Chapter 8 (Smoothness):** Kähler differentials and the cotangent complex; Jacobian, infinitesimal, and fiberwise criteria for smooth, unramified, and étale morphisms; local étale structure and Bertini’s theorem.
- **Chapter 9 (Divisors and Blowups):** Effective, Cartier, and Weil divisors; Rees algebras, blowups, deformation to the normal cone, regular immersions, local complete intersections, and the divisor–Picard correspondence.
- **Chapter 10 (Properness):** Universally closed and proper morphisms; Prüfer schemes and valuation rings; valuative criteria, projectivity, Chow’s lemma, and finiteness theorems.

The second part follows the syllabus of the Summer Semester 2026 lectures. Because the present volume is intended to cover that syllabus rather than use the lecture script as mathematical evidence, the script serves only as an internal coverage checklist. Elementary arguments are expanded here, while deeper inputs such as Zariski’s Main Theorem, Bertini, Chow’s lemma, and Nagata compactification are explicitly identified as cited theorems with proof roadmaps. Independent references are supplied when available; when the precise formulation used here has not been independently sourced, it is instead marked explicitly as an accepted input, and its roadmap delineates what is and is not proved in this volume.

The compiled appendices contain two supplements. Appendix A develops descent, including the fpqc formalism used to complement the Zariski arguments in the main text. Appendix B develops the topos-theoretic extension of the theory of schemes. Its source baseline is Marc Hoyois’s *Lectures on Higher Topos Theory*, in Bastiaan Cnossen’s notes from the Regensburg 2025–2026 course [Hoy26]; separate working

notes on Gestalten are supplementary, not a replacement for that baseline. Results in either appendix that rely on a higher-categorical framework or another substantial external input are subject to the same source policy just stated.

Notation and Conventions

General Philosophy. We adopt the **functor-of-points** perspective: geometric objects are functors $\mathcal{C}\text{Alg} \rightarrow \text{Set}$. The locally ringed space perspective is reserved for Section 6.3. All rings are commutative with unit.

Typographical Conventions

Sans serif (A) Categories and categorical constructions.

Ex: $\text{Set}, \mathcal{C}\text{Alg}, \text{Aff}, \text{Sch}, \text{QCoh}(X), \text{Open}(X)$.

Roman (A) Functors, operators, and standard constructions.

Ex: $\text{Spec}(R), \text{Proj}(A), \text{Hom}(X, Y), \text{colim}, \text{Prim}(R)$.

Calligraphic ($\mathcal{A}$) Sheaves of rings, sieves, and specific categories.

Ex: $\mathcal{O}_X, \mathcal{R}, \mathcal{C}$.

Blackboard Bold ($\mathbb{A}$) Standard schemes and number systems.

Ex: $\mathbb{A}^n, \mathbb{P}^n, \mathbb{G}_m, \mathbb{Z}, \mathbb{Q}, \mathbb{C}$.

Fraktur ($\mathfrak{a}$) Ideals.

Ex: $\mathfrak{p}, \mathfrak{m}, \mathfrak{a}$.

Critical Distinctions

- **Linear algebra over a ring:** an R -“vector space” means a dualizable R -module, equivalently a finitely generated projective module; a line is such a module of constant rank one. It need not be free.
- **Finiteness in quasi-affine/projective language:** in this text, quasi-affine and quasi-projective objects are required to be cut out by finite open data. Thus they are quasi-compact; an arbitrary open subscheme of an affine or projective scheme is not automatically included in these classes.
- **Open Subobjects:** $\text{Open}(X)$ is the locale of open subfunctors of an algebraic functor X (categorical), while $\text{Open}(|X|)$ is the locale of open sets of its underlying space. They agree: $\text{Open}(X) \simeq \text{Open}(|X|)$.
- **Affine and projective notation:** $\text{Spec}(A)$ and $\text{Proj}(A)$ first denote functors of points and later the corresponding schemes or locally ringed spaces under the proved comparison equivalences. Likewise $D(f)$ and $V(I)$ denote open and closed subfunctors; their underlying subsets are written $|D(f)|$ and $|V(I)|$ when the distinction matters.
- **Structure Sheaf $\mathcal{O}_X$:** Depending on context, this is either the unit object in Mod_X or the sheaf of rings on $|X|$; these are identified in Section 6.3.
- **Equality:** We write $\simeq$ for isomorphism and $=$ for strict equality.

Abbreviations and Standard Notation

Abbr.	Meaning	Symbol	Meaning
qcqs	quasi-compact quasi-separated	$\text{Spec}(R)$	Affine scheme of R
ft	finite type	$\text{Proj}(A)$	Projective scheme of A
fp	finite presentation	$ X $	Topological space of X
fppf	faithfully flat finite presentation	$y_{\mathcal{C}}$	Yoneda embedding
Zar	Zariski topology	$X \times_S Y$	Fiber product
		$\text{El}(F)$	Category of elements

- **Loci:** $D(f)$ is the non-vanishing locus (open), $V(I)$ the vanishing locus (closed).
- **Localization:** R_f is the localization at f ; $A_{(f)}$ is the degree-zero part $(A_f)_0$.

Diagrammatic Conventions

- **Squares:** Pullbacks are marked with $\lrcorner$, pushouts with $\rceil$.
- **Adjunctions:** $F \dashv G$ indicates F is left adjoint to G .
- **Maps:** $f : X \rightarrow Y$ for morphisms, $x \mapsto f(x)$ for elements.

Main References

These notes are primarily based on the following standard textbooks.

- **U. Görtz, T. Wedhorn**, *Algebraic Geometry I: Schemes* [GW20]. The primary guideline for the structure of the course.
- **R. Vakil**, *The Rising Sea: Foundations of Algebraic Geometry* [Vak24]. A comprehensive resource with a geometric perspective.
- **D. Eisenbud, J. Harris**, *The Geometry of Schemes* [EH00]. Highly recommended for geometric intuition through examples.
- **R. Hartshorne**, *Algebraic Geometry* [Har77]. The classic reference.
- **Yu. I. Manin**, *Introduction to the Theory of Schemes* [Man18]. A profound conceptual introduction.
- **The Stacks Project Authors**, *The Stacks Project* [The25]. The ultimate reference for technical details.

Lecture Notes and Personal References

- **P. Scholze**, *Algebraic Geometry I* (typed by Jack Davies) [Sch17].
- **D. Clausen, L. Hesselholt**, *Scheme Theory* [CH22].
- **Y. Zhao**, *Scheme Theory I* [Zha24].
- **J. E. Rodríguez Camargo**, *Notes on D-Modules via Derived Algebraic Stacks* [Cam25].

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Part I.

Algebraic Geometry I (Foundations of Scheme Theory)

Chapter 1.

Affine Geometry

Affine geometry is where algebraic geometry begins. Classically one studies the zero sets of polynomials in k^n for k a field; modernly, the functor-of-points approach encourages us to study all solution sets at once, packaging $R \mapsto \{\text{solutions in } R^n\}$ into a single functor $\text{CAlg}_k \rightarrow \text{Set}$ on k -algebras.

This chapter develops the framework. We start by defining *algebraic functors* and identifying the representable ones as *affine schemes* (Yoneda embedding). We then classify their subfunctors: closed ones correspond to *ideals*, open ones to *radical ideals* — the functorial Nullstellensatz. Finally we prove *Zariski descent*, the sheaf condition that lets us glue local algebraic data, and establish finiteness properties (finite type, finite presentation) that govern the size of our objects.

The organising principle throughout is that everything is a functor: Spec turns rings into functors, V and D turn ideals into subfunctors, and all naturality conditions are automatic from the functor-category framework.

1.1. Algebraic Functors and Affine Schemes

Over a base ring k , our basic objects are functors $\text{CAlg}_k \rightarrow \text{Set}$, which we think of as “solution sets of polynomial equations parameterised by a k -algebra R ”. We take $k = \mathbb{Z}$ by default. The most concrete examples are the affine spaces $\mathbb{A}_k^I$; the representable ones form the category of *affine schemes over k* .

Definition 1.1.1 (Algebraic Functor). Let k be a ring.

- An *algebraic k -functor* is a functor $\text{CAlg}_k \rightarrow \text{Set}$. When $k = \mathbb{Z}$, we simply call it an *algebraic functor*.
- Given an algebraic k -functor X and a k -algebra R , we call elements of $X(R)$ the *R -points* of X .

The most fundamental algebraic functors are *affine spaces*.

Definition 1.1.2 (Affine Space). Let I be a set and k a ring. The *affine I -space over k* is the algebraic k -functor

$$\mathbb{A}_k^I : \text{CAlg}_k \rightarrow \text{Set}, \quad R \mapsto R^I.$$

When $k = \mathbb{Z}$, we simply write $\mathbb{A}^I$. For $n \geq 0$, we also write $\mathbb{A}_k^n = \mathbb{A}_k^{\{1, \dots, n\}}$ for the affine n -space over k . When $n = 1$, we call it the *affine line*; when $n = 2$, the *affine plane*.

Remark 1.1.3. Unwinding the definition:

- $\mathbb{A}_k^0$ is the terminal object of $\text{Fun}(\text{CAlg}_k, \text{Set})$.
- $\mathbb{A}_k^1$ is the forgetful functor $\text{CAlg}_k \rightarrow \text{Set}$.
- $\mathbb{A}_k^{(-)} : \text{Set}^{\text{op}} \rightarrow \text{Fun}(\text{CAlg}_k, \text{Set})$ is functorial.
- By the universal property of polynomial rings, $\mathbb{A}_k^I$ is represented by $k[x_i \mid i \in I]$, i.e., there is an isomorphism

$$\mathbb{A}_k^I \simeq \text{Hom}_{\text{CAlg}_k}(k[x_i \mid i \in I], -).$$

1.1.1. Presheaves and Yoneda Embedding

We begin by recalling the concept of presheaves.

Definition 1.1.4 (Presheaf). Let C be a category. A *presheaf* on C is a functor $C^{\text{op}} \rightarrow \text{Set}$. We denote

$$\text{PShv}(C) := \text{Fun}(C^{\text{op}}, \text{Set})$$

for the category of presheaves on C .

Note that algebraic k -functors defined earlier are precisely presheaves on $\text{CAlg}_k^{\text{op}}$.

Remark 1.1.5. In functor categories, limits and colimits are computed pointwise. Thus $\text{PShv}(C)$ inherits many properties of the category of sets, such as: filtered colimits commute with finite limits, epimorphisms are automatically effective, etc.

We now study the canonical functor from C to $\text{PShv}(C)$, called the *Yoneda embedding*.

Definition 1.1.6 (Yoneda Embedding). Let C be a category. The *Yoneda embedding* of C is the functor

$$y: C \rightarrow \text{PShv}(C), \quad y(X) := \text{Hom}_C(-, X).$$

A presheaf on C is called *representable* if it lies in the essential image of the Yoneda embedding.

For $C = \text{CAlg}^{\text{op}}$, the Yoneda embedding is

$$\text{CAlg}^{\text{op}} \rightarrow \text{PShv}(\text{CAlg}^{\text{op}}) = \text{Fun}(\text{CAlg}, \text{Set}),$$

sending a ring R to $\text{Hom}_{\text{CAlg}^{\text{op}}}(-, R) = \text{Hom}_{\text{CAlg}}(R, -)$. For this case, we use the special notation

$$\text{Spec}: \text{CAlg}^{\text{op}} \rightarrow \text{Fun}(\text{CAlg}, \text{Set})$$

to denote this embedding. Thus Remark 1.1.3 gives $\mathbb{A}^I \simeq \text{Spec}(\mathbb{Z}[x_i \mid i \in I])$ in the absolute case. When k is not understood as $\mathbb{Z}$, the same Yoneda construction is taken in $\text{CAlg}_k^{\text{op}}$; in that relative functor category, $\mathbb{A}_k^I$ is represented by the k -algebra $k[x_i \mid i \in I]$.

1.1.2. Category of Elements

For a functor $F: C \rightarrow \text{Set}$, we can encode its information as a category $\int_C F$. Let Set_* denote the category of pointed sets:

- Objects are pairs (X, x) where X is a set and $x \in X$ is a chosen point.
- Morphisms $(X, x) \rightarrow (Y, y)$ are maps $f: X \rightarrow Y$ satisfying $f(x) = y$.

Definition 1.1.7 (Category of Elements). Let C be a category and $F: C \rightarrow \text{Set}$ a functor. The *category of elements* $\int_C F$ is defined as the pullback:

$$\begin{array}{ccc} \int_C F & \longrightarrow & \text{Set}_* \\ \downarrow & \lrcorner & \downarrow \\ C & \longrightarrow & \text{Set} \end{array}$$

Concretely, $\int_C F$ is the category where:

- Objects are pairs (C, x) with $C \in C$ and $x \in F(C)$.

- A morphism $(C, x) \rightarrow (C', x')$ is a morphism $f: C \rightarrow C'$ in $\mathcal{C}$ such that $F(f): F(C) \rightarrow F(C')$ satisfies $F(f)(x) = x'$.

For presheaves $F: \mathcal{C}^{\text{op}} \rightarrow \text{Set}$, we define the category of elements similarly:

Definition 1.1.8 (Category of Elements for Presheaves). Let $\mathcal{C}$ be a category and $F: \mathcal{C}^{\text{op}} \rightarrow \text{Set}$ a functor. The *category of elements* $\int^{\mathcal{C}} F$ is defined as the pullback:

$$\begin{array}{ccc} \int^{\mathcal{C}} F & \longrightarrow & (\text{Set}_*)^{\text{op}} \\ \downarrow & \lrcorner & \downarrow \\ \mathcal{C} & \longrightarrow & \text{Set}^{\text{op}} \end{array}$$

Concretely, $\int^{\mathcal{C}} F$ is the category where:

- Objects are pairs (C, x) with $C \in \mathcal{C}$ and $x \in F(C)$.
- A morphism $(C, x) \rightarrow (C', x')$ is a morphism $f: C \rightarrow C'$ in $\mathcal{C}$ such that $F(f): F(C') \rightarrow F(C)$ satisfies $F(f)(x') = x$.

The two constructions are related by a canonical equivalence of categories

$$\left(\int_{\mathcal{C}} F \right)^{\text{op}} \simeq \int^{\mathcal{C}^{\text{op}}} F.$$

In this book we abbreviate the contravariant version by $\text{El}(F) := \int^{\mathcal{C}} F$.

For an algebraic functor $X: \text{CAlg} \rightarrow \text{Set}$, we regard X as a presheaf on CAlg^{op} and write $\text{El}(X)$ for its contravariant category of elements. With this convention density takes the form

$$X \simeq \text{colim}_{(R, x) \in \text{El}(X)} \text{Spec}(R).$$

Constructions obtained by contravariantly assigning algebraic data to affine charts—for example the module category $R \mapsto \text{Mod}_R$ in Definition 3.1.15—are consequently indexed by $\text{El}(X)^{\text{op}}$ when their values are assembled by a limit.

Remark 1.1.9 (Grothendieck Construction). This construction generalizes to category-valued functors $F: \mathcal{C} \rightarrow \text{Cat}$, yielding the *Grothendieck construction*. In the setting of ∞ -categories, this extends to $F: \mathcal{C} \rightarrow \text{An}$ and $F: \mathcal{C} \rightarrow \text{Cat}_{\infty}$, giving left fibrations and coCartesian fibrations respectively.

1.1.3. Yoneda Lemma

Theorem 1.1.10 (Yoneda Lemma). Let $\mathcal{C}$ be a category.

- (i) For $X \in \mathcal{C}$ and $F \in \text{PShv}(\mathcal{C})$, there is a bijection

$$\text{Hom}_{\text{PShv}(\mathcal{C})}(y(X), F) \simeq F(X), \quad f \mapsto f_X(\text{id}_X).$$

The inverse is given by $x \mapsto ((f: Y \rightarrow X) \mapsto f^*(x))$.

- (ii) The Yoneda embedding $y: \mathcal{C} \rightarrow \text{PShv}(\mathcal{C})$ is fully faithful.

- (iii) The Yoneda embedding preserves all limits that exist in $\mathcal{C}$.

(iv) (Density of Yoneda embedding) Every presheaf $F \in \mathbf{PShv}(C)$ can be written as a colimit of representable functors:

$$\operatorname{colim} \left(\operatorname{El}(F) \xrightarrow{\text{forget}} C \xrightarrow{y} \mathbf{PShv}(C) \right) \simeq F.$$

(v) (Universal property of presheaves) For any cocomplete category $\mathcal{E}$, there is a category equivalence

$$y^*: \operatorname{Fun}^{\operatorname{colim}}(\mathbf{PShv}(C), \mathcal{E}) \xrightarrow{\sim} \operatorname{Fun}(C, \mathcal{E}),$$

where $\operatorname{Fun}^{\operatorname{colim}}$ denotes colimit-preserving functors. The inverse is constructed via left Kan extension: for $F: C \rightarrow \mathcal{E}$, consider

$$\begin{array}{ccc} C & & \\ \downarrow y & \searrow F & \\ \mathbf{PShv}(C) & \xrightarrow{\operatorname{Lan}_y F} & \mathcal{E} \end{array}$$

Since $\mathcal{E}$ has all colimits, this extension exists automatically.

(vi) For a category $\mathcal{E}$ and a colimit-preserving functor $K: \mathbf{PShv}(C) \rightarrow \mathcal{E}$, then K has a right adjoint $N: \mathcal{E} \rightarrow \mathbf{PShv}(C)$ given by $e \mapsto \operatorname{Hom}_{\mathcal{E}}(K(y(-)), e)$. This is the realization-nerve adjunction.

Parts (1)–(4) of this theorem are the workhorses of the entire text: (1) turns natural transformations $y(X) \rightarrow F$ into points $F(X)$ and vice versa; (2) says that probing C by its Hom-functors loses no information; (4) is ubiquitous in extending a statement from representables to all presheaves; (5)–(6) provide the adjoint machinery underlying sheafification and colimit-preserving constructions.

Proof. (1) Yoneda bijection. The fundamental natural-transformation/element correspondence. Define $\Phi: \operatorname{Hom}_{\mathbf{PShv}(C)}(y(X), F) \rightarrow F(X)$ by $\Phi(\alpha) = \alpha_X(\operatorname{id}_X)$.

For the inverse, given $x \in F(X)$, define $\Psi(x): y(X) \rightarrow F$ as follows: for each $Y \in C$,

$$\Psi(x)_Y: y(X)(Y) = \operatorname{Hom}_C(Y, X) \rightarrow F(Y), \quad (f: Y \rightarrow X) \mapsto F(f)(x).$$

To verify $\Psi(x)$ is natural: for $g: Z \rightarrow Y$, we need

$$\begin{array}{ccc} \operatorname{Hom}(Y, X) & \xrightarrow{\Psi(x)_Y} & F(Y) \\ g^* \downarrow & & \downarrow F(g) \\ \operatorname{Hom}(Z, X) & \xrightarrow{\Psi(x)_Z} & F(Z) \end{array}$$

to commute. For $f \in \operatorname{Hom}(Y, X)$: $(F(g) \circ \Psi(x)_Y)(f) = F(g)(F(f)(x)) = F(f \circ g)(x) = \Psi(x)_Z(f \circ g) = (\Psi(x)_Z \circ g^*)(f)$.

Check $\Phi \circ \Psi = \operatorname{id}$: $\Phi(\Psi(x)) = \Psi(x)_X(\operatorname{id}_X) = F(\operatorname{id}_X)(x) = x$.

Check $\Psi \circ \Phi = \operatorname{id}$: For $\alpha: y(X) \rightarrow F$, we have $\Psi(\Phi(\alpha)) = \Psi(\alpha_X(\operatorname{id}_X))$. For any Y and $f: Y \rightarrow X$:

$$\Psi(\alpha_X(\operatorname{id}_X))_Y(f) = F(f)(\alpha_X(\operatorname{id}_X)) = \alpha_Y(y(X)(f)(\operatorname{id}_X)) = \alpha_Y(f),$$

where the second equality uses naturality of α .

(2) Full faithfulness. Yoneda is an embedding. Apply (1) with $F = y(X')$ to get $\operatorname{Hom}(y(X), y(X')) \simeq y(X')(X) = \operatorname{Hom}(X, X')$.

(3) Preservation of limits. Yoneda commutes with limits in C . Limits in $\mathbf{PShv}(C)$ are computed pointwise. If $\lim_i X_i$ exists in C , then for any Y :

$$y(\lim_i X_i)(Y) = \operatorname{Hom}(Y, \lim_i X_i) \simeq \lim_i \operatorname{Hom}(Y, X_i) = \lim_i y(X_i)(Y).$$

Thus $y(\lim_i X_i) \simeq \lim_i y(X_i)$ in $\text{PShv}(C)$.

(4) *Density. Every presheaf is a canonical colimit of representables — the slogan “ F is determined by its values on C ” made formal.* Recall that $\text{El}(F) = \int^C F$ is contravariant: a morphism $(C, x) \rightarrow (C', x')$ is $g: C \rightarrow C'$ with $F(g)(x') = x$. The colimit in question is

$$\text{colim}_{(C,x) \in \text{El}(F)} y(C),$$

which we evaluate pointwise. For any $Y \in C$,

$$(\text{colim}_{(C,x) \in \text{El}(F)} y(C))(Y) = \text{colim}_{(C,x) \in \text{El}(F)} \text{Hom}(Y, C).$$

A point in this colimit is represented by a triple (C, x, f) with $x \in F(C)$ and $f: Y \rightarrow C$. Two triples (C, x, f) and (C', x', f') are identified when there exists $g: C \rightarrow C'$ in $\text{El}(F)$ — that is, $g: C \rightarrow C'$ with $F(g)(x') = x$ — such that $g \circ f = f'$.

We construct a bijection with $F(Y)$. Send $(C, x, f) \mapsto F(f)(x) \in F(Y)$. This is well-defined on equivalence classes: if $(C, x, f) \sim (C', x', g \circ f)$ via g as above, then

$$F(g \circ f)(x') = F(f)(F(g)(x')) = F(f)(x).$$

Surjectivity is immediate, since $s \in F(Y)$ is the image of (Y, s, id_Y) . For injectivity, observe that any triple (C, x, f) is equivalent to $(Y, F(f)(x), \text{id}_Y)$: the map $f: Y \rightarrow C$ provides a morphism $(Y, F(f)(x)) \rightarrow (C, x)$ in $\text{El}(F)$ (since $F(f)(x) = F(f)(x)$), and $f \circ \text{id}_Y = f$.

(5) and (6) *Universal property and adjoints.* Colimit-preserving extensions of $F: C \rightarrow \mathcal{E}$ to $\text{PShv}(C)$ are determined by the values on C , computed via left Kan extension; the right adjoint is then given by Hom-sets. These follow from general Kan-extension and adjoint-functor theorem arguments. $\square$

1.2. Affine Schemes

The Yoneda embedding $y: \text{CAlg}^{\text{op}} \rightarrow \text{Fun}(\text{CAlg}, \text{Set})$ has a special name: the *Spec functor*. Its image consists of the *affine schemes*, which play the role in algebraic geometry that rings play in commutative algebra.

1.2.1. Polynomial Equations

Definition 1.2.1 (System of Polynomial Equations). Let k be a ring and let I and J be sets. A *system of J polynomial equations in I variables over k* is a J -tuple $\Sigma = (f_j)_{j \in J}$ in the polynomial ring $k[x_i \mid i \in I]$. We denote by (Σ) the ideal in $k[x_i \mid i \in I]$ generated by $(f_j)_{j \in J}$ and by $k[\Sigma]$ the k -algebra $k[x_i \mid i \in I]/(\Sigma)$.

Remark 1.2.2 (Systems of Equations and Presentations). Every k -algebra R admits an isomorphism $R \simeq k[\Sigma]$ for some system of polynomial equations Σ : choose a generating set $(x_i)_{i \in I}$ of R as a k -algebra, and let Σ be any generating set for the kernel $k[x_i] \twoheadrightarrow R$. Thus a choice of isomorphism $R \simeq k[\Sigma]$ is precisely a *presentation* of R by generators and relations. When I is finite, R is of finite type over k ; when Σ is also finite, R is of finite presentation (Definition 1.4.1).

Given a system of polynomial equations over k , we want to consider its solutions in an arbitrary k -algebra R . To that end, recall the *evaluation map*

$$k[x_i \mid i \in I] \times R^I \rightarrow R, \quad (f, a) \mapsto f(a),$$

defined as follows: for each tuple $a \in R^I$, the map $f \mapsto f(a)$ is the unique k -algebra homomorphism $k[x_i \mid i \in I] \rightarrow R$ sending x_i to a_i .

Definition 1.2.3 (Vanishing Locus). Let $F \subset k[x_i \mid i \in I]$ be a subset. The *vanishing locus* of F in $\mathbb{A}_k^I$ is the subfunctor $V(F) \subset \mathbb{A}_k^I$ given by

$$V(F)(R) = \{a \in R^I \mid f(a) = 0 \text{ for all } f \in F\} \subset R^I.$$

This is indeed a subfunctor: for any k -algebra map $R \rightarrow S$, the induced map $R^I \rightarrow S^I$ sends $V(F)(R)$ to $V(F)(S)$.

Remark 1.2.4. The vanishing locus of F manifestly depends only on the ideal it generates: if $(F) = (F')$ then $V(F) = V(F')$. Remarkably, the converse also holds — a functorial form of Hilbert’s Nullstellensatz that we prove below (Corollary 1.2.11).

Definition 1.2.5 (Solution Functor). Let $\Sigma = (f_j)_{j \in J}$ be a system of J polynomial equations in I variables over k . Its *solution functor* $\text{Sol}_\Sigma : \text{CAlg}_k \rightarrow \text{Set}$ is the vanishing locus of $\{f_j \mid j \in J\}$ in $\mathbb{A}_k^I$:

$$\text{Sol}_\Sigma = V(\{f_j \mid j \in J\}) \subset \mathbb{A}_k^I.$$

By the universal property of polynomial rings, there is a one-to-one correspondence between systems of J polynomial equations in I variables and k -algebra maps

$$k[x_j \mid j \in J] \rightarrow k[x_i \mid i \in I].$$

By the Yoneda lemma, these are in turn equivalent to natural transformations

$$\mathbb{A}_k^I \rightarrow \mathbb{A}_k^J.$$

Unraveling these equivalences, the map $\mathbb{A}_k^I \rightarrow \mathbb{A}_k^J$ corresponding to a system $\Sigma = (f_j)_{j \in J}$ is given on a k -algebra R by

$$R^I \rightarrow R^J, \quad a \mapsto (f_j(a))_{j \in J}.$$

By definition, the solution functor Sol_Σ is the kernel of this map, i.e., there is a pullback square

$$\begin{array}{ccc} \text{Sol}_\Sigma & \longrightarrow & \mathbb{A}_k^I \\ \downarrow & \lrcorner & \downarrow \\ 0 & \longrightarrow & \mathbb{A}_k^J \end{array} \quad (1.1)$$

where 0 is the subfunctor of $\mathbb{A}_k^J$ given by $0(R) = \{0\} \subset R^J$.

Definition 1.2.6 (Affine Scheme). A functor $\text{CAlg}_k \rightarrow \text{Set}$ is called an *affine k -scheme* if it is isomorphic to Sol_Σ for some system of polynomial equations Σ over k . We denote by $\text{Aff}_k \subset \text{Fun}(\text{CAlg}_k, \text{Set})$ the full subcategory spanned by the affine k -schemes. An *affine scheme* is an affine $\mathbb{Z}$ -scheme.

Example 1.2.7. The affine I -space $\mathbb{A}_k^I$ is an affine k -scheme, as it is the solution functor of the empty system of equations in I variables.

Lemma 1.2.8. Let Σ be a system of polynomial equations over k . Then the solution functor Sol_Σ is represented by the k -algebra $k[\Sigma]$, i.e., there is an isomorphism

$$\text{Sol}_\Sigma \simeq \text{Spec}(k[\Sigma]) : \text{CAlg}_k \rightarrow \text{Set}.$$

Proof. By definition, $k[\Sigma] = k[x_i \mid i \in I]/(\Sigma)$. For any k -algebra R , a k -algebra map $\varphi : k[\Sigma] \rightarrow R$ corresponds to a k -algebra map $\tilde{\varphi} : k[x_i \mid i \in I] \rightarrow R$ that sends each $f_j \in \Sigma$ to zero.

By the universal property of the polynomial ring, $\tilde{\varphi}$ is determined by the tuple $a = (\tilde{\varphi}(x_i))_{i \in I} \in R^I$. The condition that $\tilde{\varphi}(f_j) = 0$ for all $j \in J$ is exactly the condition that $f_j(a) = 0$ for all $j \in J$, i.e., $a \in \text{Sol}_\Sigma(R)$.

Thus we have a bijection

$$\mathrm{Hom}_{\mathrm{CAlg}_k}(k[\Sigma], R) \xrightarrow{\sim} \mathrm{Sol}_\Sigma(R), \quad \varphi \mapsto (\varphi(x_i))_{i \in I},$$

which is natural in R . Therefore $\mathrm{Sol}_\Sigma \simeq \mathrm{Spec}(k[\Sigma])$. $\square$

Theorem 1.2.9 (Characterization of Affine Schemes). *Let k be a ring. The following conditions are equivalent for an algebraic k -functor $X: \mathrm{CAlg}_k \rightarrow \mathrm{Set}$:*

- (i) X is an affine k -scheme.
- (ii) X is representable, i.e., isomorphic to $\mathrm{Spec}(A)$ for some k -algebra A .
- (iii) X preserves limits and is accessible.

Proof. (i) $\Rightarrow$ (ii): If X is an affine k -scheme, then $X \simeq \mathrm{Sol}_\Sigma$ for some system of polynomial equations Σ . By Lemma 1.2.8, $X \simeq \mathrm{Spec}(k[\Sigma])$, so X is representable.

(ii) $\Rightarrow$ (iii): If $X \simeq \mathrm{Spec}(A)$ for some k -algebra A , then $X(R) = \mathrm{Hom}_{\mathrm{CAlg}_k}(A, R)$. Since $\mathrm{Hom}_{\mathrm{CAlg}_k}(A, -)$ is a representable functor, it preserves all limits (by the Yoneda lemma).

For accessibility: A functor $F: \mathrm{CAlg}_k \rightarrow \mathrm{Set}$ is accessible if there exists a regular cardinal κ such that F preserves κ -filtered colimits. For the representable functor $\mathrm{Hom}(A, -)$ this holds because A is a κ -presentable object of CAlg_k whenever κ is a regular cardinal exceeding the cardinality of a presentation of A (a set of generators together with relations): for such κ , writing A as a κ -small colimit of free algebras shows that $\mathrm{Hom}(A, -)$ commutes with κ -filtered colimits, which in CAlg_k are computed pointwise on underlying sets.

(iii) $\Rightarrow$ (i): Suppose $X: \mathrm{CAlg}_k \rightarrow \mathrm{Set}$ preserves limits and is accessible. We invoke the *representability theorem* for Set -valued functors (a special case of the adjoint functor theorem): a functor from a presentable category to Set is representable if and only if it preserves small limits and is accessible. Since CAlg_k is presentable, the hypotheses give a k -algebra A together with a natural isomorphism $X \simeq \mathrm{Hom}_{\mathrm{CAlg}_k}(A, -) = \mathrm{Spec}(A)$. By Remark 1.2.2 we may write $A \simeq k[\Sigma]$ for some system of polynomial equations Σ , so $X \simeq \mathrm{Sol}_\Sigma$ by Lemma 1.2.8.

Concretely, the algebra A is constructed as follows. Accessibility lets us write $X = \mathrm{colim}_\alpha \mathrm{Spec}(A_\alpha)$ as a small colimit of representables in $\mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set})$; this colimit is computed pointwise and is generally not itself representable. However, X being limit-preserving forces it to admit a left adjoint $L: \mathrm{Set} \rightarrow \mathrm{CAlg}_k$, and we may take $A = L(\{*\})$. The Yoneda lemma then gives

$$X(R) = \mathrm{Hom}_{\mathrm{Set}}(\{*\}, X(R)) = \mathrm{Hom}_{\mathrm{CAlg}_k}(L(\{*\}), R) = \mathrm{Hom}_{\mathrm{CAlg}_k}(A, R). \quad \square$$

Corollary 1.2.10. *The Yoneda embedding of $\mathrm{CAlg}_k^{\mathrm{op}}$ induces an equivalence of categories*

$$\mathrm{Spec}: \mathrm{CAlg}_k^{\mathrm{op}} \xrightarrow{\sim} \mathrm{Aff}_k \subset \mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set}).$$

Under this equivalence, the affine k -scheme Sol_Σ corresponds to the k -algebra $k[\Sigma]$.

Proof. This follows immediately from Theorem 1.2.9. The Yoneda embedding is fully faithful and essentially surjective onto the representable functors, which by (i) $\Leftrightarrow$ (ii) are exactly the affine k -schemes. $\square$

Under the equivalence of Corollary 1.2.10, the embedding $\mathrm{Sol}_\Sigma \hookrightarrow \mathbb{A}_k^I$ of affine k -schemes corresponds to the quotient map $k[x_i \mid i \in I] \twoheadrightarrow k[\Sigma]$. This implies the following result:

Corollary 1.2.11 (Functorial Nullstellensatz). *Sending a subset $F \subset k[x_i \mid i \in I]$ to its vanishing locus $V(F) \subset \mathbb{A}_k^I$ induces an order-reversing bijection*

$$V: \{\text{ideals in } k[x_i \mid i \in I]\} \xrightarrow{\sim} \{\text{vanishing loci in } \mathbb{A}_k^I\}.$$

Proof. By Remark 1.2.4, $V(F)$ depends only on the ideal (F) , so we can view V as a map from ideals.

Injectivity: Suppose $V(F) = V(F')$ for ideals (F) and (F') . Under the equivalence $\text{Spec}: \text{CAlg}^{\text{op}} \xrightarrow{\sim} \text{Aff}$, the closed embedding $V(F) \hookrightarrow \mathbb{A}_k^I$ corresponds to the quotient map $k[x_i \mid i \in I] \twoheadrightarrow k[x_i \mid i \in I]/(F)$, and similarly for $V(F')$. Since $V(F) = V(F')$ as subfunctors, the two quotient maps are isomorphic over $k[x_i \mid i \in I]$, which forces $(F) = (F')$.

Surjectivity: Let $Z \subset \mathbb{A}_k^I$ be a vanishing locus. By definition, there exists a system of polynomial equations Σ such that $Z = \text{Sol}_{\Sigma} = V(\Sigma)$. Thus $Z = V((\Sigma))$ for the ideal (Σ) .

The order-reversing property is clear: if $(F) \subset (F')$, then $V(F') \subset V(F)$. $\square$

Example 1.2.12. Consider the following systems of polynomial equations over $\mathbb{R}$ in one variable:

$$\Sigma_1 = (x^2 + 1), \quad \Sigma_2 = ((x^2 + 1)^2), \quad \Sigma_3 = (x^2 + x + 1), \quad \Sigma_4 = (x^4 + 1).$$

Then:

$$\begin{aligned} \text{Sol}_{\Sigma_1}(\mathbb{R}) &= \emptyset, & \text{Sol}_{\Sigma_2}(\mathbb{R}) &= \emptyset, & \text{Sol}_{\Sigma_3}(\mathbb{R}) &= \emptyset, & \text{Sol}_{\Sigma_4}(\mathbb{R}) &= \emptyset, \\ \text{Sol}_{\Sigma_1}(\mathbb{C}) &= \{\pm i\}, & \text{Sol}_{\Sigma_2}(\mathbb{C}) &= \{\pm i\}, & \text{Sol}_{\Sigma_3}(\mathbb{C}) &= \{\zeta_3, \bar{\zeta}_3\}, & \text{Sol}_{\Sigma_4}(\mathbb{C}) &= \{\pm \zeta_8, \pm \bar{\zeta}_8\}, \end{aligned}$$

where $\zeta_n = \exp(2\pi i/n) \in \mathbb{C}$. All four equations have the same solutions in $\mathbb{R}$. However, as the four ideals (Σ_i) in $\mathbb{R}[x]$ are pairwise distinct, they define four different subfunctors of $\mathbb{A}_{\mathbb{R}}^1$ by Corollary 1.2.11. The solutions in $\mathbb{C}$ distinguish them, except for Sol_{Σ_1} and Sol_{Σ_2} . To see that $\text{Sol}_{\Sigma_1} \neq \text{Sol}_{\Sigma_2}$ as subfunctors of $\mathbb{A}_{\mathbb{R}}^1$, we can compute the solutions in the $\mathbb{R}$ -algebra $\mathbb{C}[\varepsilon]$ of dual complex numbers (where $\varepsilon^2 = 0$):

$$\text{Sol}_{\Sigma_1}(\mathbb{C}[\varepsilon]) = \{\pm i\}, \quad \text{Sol}_{\Sigma_2}(\mathbb{C}[\varepsilon]) = \{\pm i + a\varepsilon \mid a \in \mathbb{C}\}.$$

On the other hand, the associated $\mathbb{R}$ -algebras are

$$\mathbb{R}[\Sigma_1] \simeq \mathbb{C}, \quad \mathbb{R}[\Sigma_2] \simeq \mathbb{C}[\varepsilon], \quad \mathbb{R}[\Sigma_3] \simeq \mathbb{C}, \quad \mathbb{R}[\Sigma_4] \simeq \mathbb{C} \times \mathbb{C}.$$

By Lemma 1.2.8, Sol_{Σ_1} and Sol_{Σ_3} are both isomorphic to $\text{Spec}(\mathbb{C})$. The different ideals (Σ_1) and (Σ_3) correspond to two different embeddings of the affine $\mathbb{R}$ -scheme $\text{Spec}(\mathbb{C})$ into $\mathbb{A}_{\mathbb{R}}^1$, and the systems Σ_1 and Σ_3 themselves are two different presentations of the $\mathbb{R}$ -algebra $\mathbb{C}$.

Remark 1.2.13. In summary, given a system of polynomial equations Σ over k , we have the following relations between Σ and Sol_{Σ} :

- (i) The data of the pullback square (1.1) is equivalent to the data of Σ itself.
- (ii) The data of the embedding $\text{Sol}_{\Sigma} \hookrightarrow \mathbb{A}_k^I$ is equivalent to the data of the ideal (Σ) in the polynomial ring $k[x_i \mid i \in I]$.
- (iii) The data of the affine k -scheme Sol_{Σ} alone is equivalent to the data of the k -algebra $k[\Sigma]$.

This can be compared with the following types of data in differential geometry:

- (i) A smooth manifold M given as the vanishing locus of a smooth function $\mathbb{R}^n \rightarrow \mathbb{R}^m$.
- (ii) A smooth manifold M given as a closed submanifold of $\mathbb{R}^n$.
- (iii) A smooth manifold M .

Smooth manifolds are the basic objects of interest in differential geometry. Embedding a manifold M into a Euclidean space or realizing it as the vanishing locus of a function are often useful ways to understand M , but we do not consider this additional data to be part of the manifold M itself. The situation in algebraic geometry is entirely similar: the basic objects of interest are affine schemes. Any affine scheme X can be embedded into an affine space ($X \hookrightarrow \mathbb{A}^I$) or realized as the solution functor of a system of polynomial equations ($X \simeq \text{Sol}_{\Sigma}$), but this data is not part of the affine scheme X itself.

1.2.2. Examples

Example 1.2.14 (Terminal Scheme $*$). The algebraic functor $\mathrm{CAlg} \rightarrow \mathrm{Set}, R \mapsto \{*\}$ is isomorphic to $\mathrm{Spec}(\mathbb{Z})$, hence is an affine scheme. In fact, it is the terminal object of $\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$.

Example 1.2.15 (Initial Scheme $\emptyset$). This example shows that the inclusion $\mathrm{Aff} \hookrightarrow \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$ — equivalently, the Yoneda embedding $\mathrm{Spec}: \mathrm{CAlg}^{\mathrm{op}} \rightarrow \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$ after identifying its essential image with Aff — does not preserve colimits.

- In Aff , the initial object is the functor

$$\mathrm{CAlg} \rightarrow \mathrm{Set}, \quad R \mapsto \begin{cases} \emptyset, & R \neq 0 \\ \{*\}, & R = 0, \end{cases}$$

which is isomorphic to $\mathrm{Spec}(0)$.

- However, in $\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$, the initial object is the constant functor

$$\mathrm{CAlg} \rightarrow \mathrm{Set}, \quad R \mapsto \emptyset,$$

which is not representable, hence not an affine scheme.

Example 1.2.16 (Idempotent Classifier Idem). Let $\mathrm{Idem}: \mathrm{CAlg} \rightarrow \mathrm{Set}$ be the functor sending R to its set of idempotent elements $\mathrm{Idem}(R) = \{e \in R \mid e^2 = e\}$. There is a bijection

$$\begin{aligned} \mathrm{Hom}_{\mathrm{CAlg}}(\mathbb{Z} \times \mathbb{Z}, R) &\xrightarrow{\sim} \mathrm{Idem}(R) \\ \varphi &\mapsto \varphi(1, 0). \end{aligned}$$

Since this is natural in R , we have $\mathrm{Idem} \simeq \mathrm{Spec}(\mathbb{Z} \times \mathbb{Z})$, hence affine.

Example 1.2.17 (Multiplicative Group $\mathbb{G}_m$). Let $\mathbb{G}_m: \mathrm{CAlg} \rightarrow \mathrm{Ab}$ be the functor sending a ring R to its group of units $R^\times$. We claim this functor is represented by the *monoid ring* $\mathbb{Z}[\mathbb{Z}]$.

First, recall that the monoid ring functor $\mathbb{Z}[-]: \mathrm{CMon} \rightarrow \mathrm{CAlg}$ is left adjoint to the underlying multiplicative monoid functor $\mathrm{CAlg} \rightarrow \mathrm{CMon}, R \mapsto (R, \cdot)$. For any ring R and commutative monoid M :

$$\mathrm{Hom}_{\mathrm{CAlg}}(\mathbb{Z}[M], R) \simeq \mathrm{Hom}_{\mathrm{CMon}}(M, (R, \cdot)),$$

where $(R, \cdot)$ denotes the multiplicative monoid structure on R .

Taking $M = \mathbb{Z}$ (the additive group, viewed as a monoid): since $\mathbb{Z}$ is a group, any monoid homomorphism $\mathbb{Z} \rightarrow (R, \cdot)$ factors through $R^\times$. Moreover, $\mathbb{Z}$ is the free group on one generator, so group homomorphisms $\phi: \mathbb{Z} \rightarrow R^\times$ are determined by $\phi(1) \in R^\times$.

Therefore:

$$\mathbb{G}_m(R) = R^\times \simeq \mathrm{Hom}_{\mathrm{Grp}}(\mathbb{Z}, R^\times) \simeq \mathrm{Hom}_{\mathrm{CAlg}}(\mathbb{Z}[\mathbb{Z}], R).$$

This proves $\mathbb{G}_m \simeq \mathrm{Spec}(\mathbb{Z}[\mathbb{Z}])$. Concretely, $\mathbb{Z}[\mathbb{Z}] \simeq \mathbb{Z}[t, t^{-1}]$ (Laurent polynomials).

Remark 1.2.18 ($\mathbb{G}_m$ as Tate Circle). In higher category theory, the classifying space of $\mathbb{Z}$ is homotopy equivalent to the circle: $B\mathbb{Z} \simeq S^1$. For a presentable monoidal category $\mathcal{C}$, we have

$$\mathrm{Fun}(B\mathbb{Z}^{\mathrm{op}}, \mathcal{C}) \simeq \mathrm{Mod}_{\mathbb{Z}}(\mathcal{C}) \simeq \mathrm{Mod}_{\mathbb{1}[\mathbb{Z}]}(\mathcal{C}),$$

where $\mathbb{1}[\mathbb{Z}]$ is the monoid ring object in $\mathcal{C}$.

Since $\mathbb{Z}[\mathbb{Z}]$ is the coordinate ring $\mathcal{O}(\mathbb{G}_m)$, this suggests a deep duality between the topological circle S^1 (corresponding to $\mathbb{Z}$) and the algebraic $\mathbb{G}_m$ in algebraic geometry. Therefore, $\mathbb{G}_m$ is often called the *Tate circle*.

In motivic homotopy theory $\mathrm{H}(S)$, we have two types of “circles”:

- *Simplicial circle*: $S^{1,0} := S^1$ (constant sheaf).
- *Tate circle*: $S^{1,1} := \mathbb{G}_m$.

The motivic (i, j) -sphere is defined as:

$$S^{i,j} := (S^1)^{\wedge(i-j)} \wedge \mathbb{G}_m^{\wedge j}.$$

The projective line satisfies $\mathbb{P}^1 \simeq S^1 \wedge \mathbb{G}_m \simeq S^{2,1}$.

The stable motivic homotopy category $\mathrm{SH}(S)$ is constructed by formally inverting $(\mathbb{P}^1, \infty) \simeq S^{2,1}$ in the pointed motivic homotopy category $\mathrm{H}(S)_*$.

Example 1.2.19 (Additive Group $\mathbb{G}_a$). The functor $\mathbb{G}_a: \mathrm{CAlg} \rightarrow \mathrm{Ab}$ sending R to $(R, +)$ is called the *additive group*. The composition

$$\mathrm{CAlg} \xrightarrow{\mathbb{G}_a} \mathrm{Ab} \xrightarrow{\text{forget}} \mathrm{Set}$$

is simply the forgetful functor, a.k.a. the affine line $\mathbb{A}^1$. Hence $\mathbb{G}_a$ is an affine group scheme.

Example 1.2.20 (The Matrix Ring Mat_n). Let $n \geq 0$ and let $\mathrm{Mat}_n: \mathrm{CAlg} \rightarrow \mathrm{Alg}$ be the functor sending a ring R to the associative ring of $n \times n$ matrices over R . This is an associative ring object in the category of affine schemes. Indeed, since an $n \times n$ matrix over R is simply a family of n^2 elements of R , the composition of Mat_n with the forgetful functor $\mathrm{Alg} \rightarrow \mathrm{Set}$ is isomorphic to the affine space $\mathbb{A}^{n^2}$. Thus, Mat_n is representable by the polynomial ring $\mathbb{Z}[x_{ij} \mid 1 \leq i, j \leq n]$.

Example 1.2.21 (The General Linear Group GL_n). Let $n \geq 0$ and let $\mathrm{GL}_n: \mathrm{CAlg} \rightarrow \mathrm{Grp}$ be the functor sending R to the group $\mathrm{GL}_n(R)$ of invertible $n \times n$ matrices. Then GL_n is an affine group scheme. Let $A = \mathbb{Z}[x_{ij} \mid 1 \leq i, j \leq n]$ be the ring representing Mat_n , and let $X = (x_{ij})_{i,j}$ be the universal $n \times n$ matrix. A matrix $M \in \mathrm{Mat}_n(R)$ is invertible if and only if its determinant $\det(M) \in R$ is a unit. Hence, for any ring R , there is a natural isomorphism:

$$\mathrm{Hom}_{\mathrm{CAlg}}(A_{\det(X)}, R) \xrightarrow{\sim} \mathrm{GL}_n(R), \quad \varphi \mapsto (\varphi(x_{ij}))_{i,j}$$

where $A_{\det(X)}$ is the localization of A at the element $\det(X)$. Therefore, $\mathrm{GL}_n \simeq \mathrm{Spec}(A_{\det(X)})$.

Example 1.2.22 (The Special Linear Group SL_n). Let $n \geq 0$ and let $\mathrm{SL}_n: \mathrm{CAlg} \rightarrow \mathrm{Grp}$ be the functor sending R to the group $\mathrm{SL}_n(R)$ of $n \times n$ matrices with determinant 1. Using the universal matrix X as defined in Example 1.2.21, we have:

$$\mathrm{Hom}_{\mathrm{CAlg}}(A/(\det(X) - 1), R) \xrightarrow{\sim} \mathrm{SL}_n(R), \quad \varphi \mapsto (\varphi(x_{ij}))_{i,j}$$

Consequently, $\mathrm{SL}_n \simeq \mathrm{Spec}(A/(\det(X) - 1))$. The sequence of subfunctor inclusions $\mathrm{SL}_n \subset \mathrm{GL}_n \subset \mathrm{Mat}_n$ corresponds to the following sequence of ring maps:

$$A \longrightarrow A_{\det(X)} \longrightarrow A/(\det(X) - 1).$$

Remark 1.2.23 (Categorical Perspective of GL_n and SL_n). The group schemes GL_n and SL_n can be elegantly characterized as fiber products (pullbacks) in the category of algebraic functors. Specifically, GL_n is defined by the pullback square:

$$\begin{array}{ccc} \mathrm{GL}_n & \hookrightarrow & \mathrm{Mat}_n \\ \downarrow & \lrcorner & \downarrow \det \\ \mathbb{G}_m & \hookrightarrow & \mathbb{A}^1 \end{array}$$

Since the Spec functor preserves limits (and hence fiber products), and all other objects in the diagram are affine, it follows that GL_n is affine.

Similarly, SL_n is characterized by the following pullback:

$$\begin{array}{ccc} \mathrm{SL}_n & \longrightarrow & \mathrm{Mat}_n \\ \downarrow & \lrcorner & \downarrow \det \\ * & \xrightarrow{1} & \mathbb{A}^1 \end{array}$$

where $* \simeq \mathrm{Spec}(\mathbb{Z})$ is the final object, representing the condition that the determinant is fixed to 1.

Example 1.2.24 (The Affine Space of a Module $\mathbb{A}(M)$). Let k be a ring and M be a k -module. Consider the functor $\mathbb{A}(M): \mathrm{CAlg}_k \rightarrow \mathrm{Mod}_k$ defined by

$$\mathbb{A}(M)(R) = \{k\text{-linear maps } M \rightarrow R\} = (M \otimes_k R)^\vee.$$

Then $\mathbb{A}(M)$ is a k -module object in affine k -schemes. Indeed, if $\mathrm{Sym}_k(M)$ is the free k -algebra on M , there is a bijection

$$\mathrm{Hom}_{\mathrm{CAlg}_k}(\mathrm{Sym}_k(M), R) \xrightarrow{\sim} \mathbb{A}(M)(R), \quad \varphi \mapsto \varphi|_M,$$

so that $\mathbb{A}(M) \simeq \mathrm{Spec}(\mathrm{Sym}_k(M))$. The affine I -space is a special case of this construction $\mathbb{A}_k^I \simeq \mathbb{A}(k^{(I)})$.

Remark 1.2.25. Let k be a ring and let M be a k -module. We can consider the “predual” of $\mathbb{A}(M)$: define $\mathbb{A}^\vee(M): \mathrm{CAlg}_k \rightarrow \mathrm{Mod}_k$ by

$$\mathbb{A}^\vee(M)(R) = M \otimes_k R.$$

There is a canonical map $\mathbb{A}^\vee(M^\vee) \rightarrow \mathbb{A}(M)$, which is an isomorphism if and only if M is a vector space (Definition 2.1.5). Otherwise, $\mathbb{A}^\vee(M)$ does not preserve limits and hence is not an affine k -scheme (in fact, it is not even a scheme). For that reason, the functor $\mathbb{A}^\vee(M)$ is rarely used.

1.2.3. Base Change

Given a ring map $\varphi: k \rightarrow k'$, we can transform any system of polynomial equations Σ over k into a system $\varphi^*(\Sigma)$ over k' by applying φ to all the coefficients. More generally, many types of data over k can be transformed into data over k' using φ , a process known as *base change*, *change of coefficients*, or *extension of scalars*.

Other examples include:

- The functor $\varphi^*: \mathrm{Mod}_k \rightarrow \mathrm{Mod}_{k'}$ sending a k -module M to the k' -module $M \otimes_k k'$
- The functor $\varphi^*: \mathrm{CAlg}_k \rightarrow \mathrm{CAlg}_{k'}$ sending a k -algebra A to the k' -algebra $A \otimes_k k'$

Unraveling these constructions, we see that there is a canonical isomorphism of k' -algebras:

$$k'[\varphi^*(\Sigma)] \simeq \varphi^*(k[\Sigma]).$$

In this section, we investigate the related process of transforming an algebraic k -functor into an algebraic k' -functor.

Theorem 1.2.26 (Functoriality of Presheaves). Let $u: \mathcal{C} \rightarrow \mathcal{D}$ be a functor, and let

$$u^*: \mathrm{PShv}(\mathcal{D}) \rightarrow \mathrm{PShv}(\mathcal{C}), \quad F \mapsto F \circ u,$$

be the “restriction along u ” functor.

(i) u^* admits a left adjoint $u_{\#}$ and a right adjoint u_* given by:

$$u_{\#}(F)(d) = \operatorname{colim} \left((C \times_{\mathcal{D}} \mathcal{D}_{d/})^{\operatorname{op}} \rightarrow C^{\operatorname{op}} \xrightarrow{F} \operatorname{Set} \right) = \operatorname{colim}_{d \rightarrow u(c)} F(c),$$

$$u_*(F)(d) = \lim \left((C \times_{\mathcal{D}} \mathcal{D}_{/d})^{\operatorname{op}} \rightarrow C^{\operatorname{op}} \xrightarrow{F} \operatorname{Set} \right) = \lim_{u(c) \rightarrow d} F(c),$$

provided these colimits and limits exist (e.g., if C is small).

(ii) The functor $u_{\#}$ extends u : there is a canonical isomorphism

$$u_{\#} \circ y_C \simeq y_{\mathcal{D}} \circ u.$$

(iii) If u is fully faithful, then $u_{\#}$ and u_* are fully faithful.

(iv) If u is a localization, then u^* is fully faithful.

(v) If the functor u has a left adjoint u_L (resp. a right adjoint u_R), then there is a canonical isomorphism $u_{\#} \simeq u_L^*$ (resp. $u_* \simeq u_R^*$).

Proof. (i) The adjunctions $u_{\#} \dashv u^*$ and $u^* \dashv u_*$ follow from the general theory of Kan extensions. For any $F \in \operatorname{PShv}(\mathcal{D})$ and $G \in \operatorname{PShv}(C)$, we have:

$$\operatorname{Hom}_{\operatorname{PShv}(\mathcal{D})}(u_{\#}(F), G) \simeq \operatorname{Hom}_{\operatorname{PShv}(C)}(F, u^*(G))$$

$$\operatorname{Hom}_{\operatorname{PShv}(C)}(u^*(G), F) \simeq \operatorname{Hom}_{\operatorname{PShv}(\mathcal{D})}(G, u_*(F))$$

The explicit formulas follow from the pointwise computation of left Kan extensions. For $d \in \mathcal{D}$:

$$u_{\#}(F)(d) = \operatorname{colim}_{(c, \alpha: d \rightarrow u(c))} F(c)$$

where the colimit is taken over $(C \times_{\mathcal{D}} \mathcal{D}_{d/})^{\operatorname{op}}$, as in the formula in the theorem. Similarly for u_* .

(ii) By the universal property of the Yoneda embedding, for any $c \in C$:

$$u_{\#}(y_C(c))(d) = \operatorname{colim}_{d \rightarrow u(c')} \operatorname{Hom}_C(c', c) \simeq \operatorname{Hom}_{\mathcal{D}}(d, u(c)) = y_{\mathcal{D}}(u(c))(d).$$

(iii) Suppose that u is fully faithful. To prove that $u_{\#}$ is fully faithful it is enough to evaluate the unit $F \rightarrow u^*u_{\#}F$ at an object $c \in C$. In the pointwise formula for $(u_{\#}F)(u(c))$, full faithfulness identifies every arrow $u(c) \rightarrow u(c')$ uniquely with an arrow $c \rightarrow c'$; the identity of $u(c)$ therefore gives the final object in the correctly oriented comma category, and the unit is an isomorphism. Dually, the identity gives the initial object in the comma category computing $(u_*F)(u(c))$, so the counit $u^*u_*F \rightarrow F$ is an isomorphism. Hence both $u_{\#}$ and u_* are fully faithful. No assertion about comma categories over objects $d \notin u(C)$ is needed.

(iv) If u is a localization, then so is u^{op} , and restriction along a localization is fully faithful: u^* identifies $\operatorname{PShv}(\mathcal{D})$ with the full subcategory of presheaves on C that invert every morphism inverted by u .

(v) If $u_L \dashv u$, then by uniqueness of adjoints, we have $u_{\#} \simeq u_L^*$. Similarly for right adjoints.

□

Remark 1.2.27. (i) Given $F: C^{\operatorname{op}} \rightarrow \operatorname{Set}$, the presheaves $u_{\#}(F)$ and $u_*(F)$ are instances of *Kan extensions*: $u_{\#}(F)$ is the left Kan extension of F along u^{op} , and $u_*(F)$ is the right Kan extension of F along u^{op} .

(ii) By the universal property of y_C , the functor $u_{\#}$ is the unique colimit-preserving extension of u (up to unique isomorphism). There is no analogous characterization of u_* .

Corollary 1.2.28 (Base Change for Slice Categories). Let C be a category, let $Y \rightarrow X$ be a morphism in C , and let $u: C_{/Y} \rightarrow C_{/X}$ be the forgetful functor.

(i) The restriction functor $u^* : \text{PShv}(C_{/X}) \rightarrow \text{PShv}(C_{/Y})$ has a left adjoint $u_\#$ given by

$$u_\#(F)(U \rightarrow X) = \coprod_{\substack{\text{maps } U \rightarrow Y \\ \text{over } X}} F(U \rightarrow Y).$$

(ii) If pullbacks along $Y \rightarrow X$ exist in C , then u^* also has a right adjoint u_* , given by

$$u_*(F)(U \rightarrow X) = F(U \times_X Y \rightarrow Y).$$

Proof. This is Theorem 1.2.26 with the comma categories computed explicitly for slice categories. $\square$

For a ring k we have $\text{CAlg}_k \simeq \text{CAlg}_{k/}$, hence $\text{CAlg}_k^{\text{op}} \simeq (\text{CAlg}^{\text{op}})_{/k}$. Specializing Corollary 1.2.28 to $C = \text{CAlg}^{\text{op}}$ yields:

Corollary 1.2.29 (Base Change for Algebras). Let $\varphi : k \rightarrow k'$ be a ring map. Then there is a triple of adjoint functors

$$\begin{array}{ccc} & \varphi_\# & \\ \text{Fun}(\text{CAlg}_{k'}, \text{Set}) & \xleftarrow{\varphi^*} & \text{Fun}(\text{CAlg}_k, \text{Set}) \\ & \varphi_* & \end{array}$$

described as follows:

- φ^* is precomposition with the forgetful functor $\text{CAlg}_{k'} \rightarrow \text{CAlg}_k$; it is the unique colimit-preserving extension of $\varphi^* : \text{CAlg}_k^{\text{op}} \rightarrow \text{CAlg}_{k'}^{\text{op}}$.
- φ_* is precomposition with $\varphi^* : \text{CAlg}_k \rightarrow \text{CAlg}_{k'}$.
- $\varphi_\#$ is given by

$$\varphi_\#(X)(A) = \coprod_{\substack{k'\text{-algebra} \\ \text{structures on } A}} X(A);$$

it is the unique colimit-preserving extension of the forgetful functor $\text{CAlg}_{k'}^{\text{op}} \rightarrow \text{CAlg}_k^{\text{op}}$.

Proof. Apply Corollary 1.2.28 to the ring map $\varphi : k \rightarrow k'$ viewed as an object $k' \in \text{CAlg}_{k/}$. For the formula for $\varphi_\#$: a k' -algebra structure on A is the same data as a map $U \rightarrow k'$ in CAlg_k , where U is the underlying k -algebra of A . The coproduct ranges over all such structures. $\square$

Definition 1.2.30 (Base Change and Weil Restriction). Let $\varphi : k \rightarrow k'$ be a ring map, giving rise to the adjoint triple of Corollary 1.2.29.

- (i) The functor φ^* is called *base change* or *extension of scalars* along φ and is also denoted by $X \mapsto X_{k'}$.
- (ii) The functor φ_* is called *Weil restriction* or *restriction of scalars* along φ and is also denoted by R_φ or $R_{k'/k}$.

Remark 1.2.31 (Preservation of Affine Schemes). Corollary 1.2.29 implies in particular that φ^* and $\varphi_\#$ both preserve affine schemes:

- (i) For a k -algebra A , $\varphi^*(\text{Spec}(A)) \simeq \text{Spec}(A \otimes_k k')$ in $\text{Fun}(\text{CAlg}_{k'}, \text{Set})$. Consequently, for any system of polynomial equations Σ over k , we have $\varphi^*(\text{Sol}_\Sigma) \simeq \text{Sol}_{\varphi^*(\Sigma)}$.
- (ii) For a k' -algebra B , $\varphi_\#(\text{Spec}(B)) \simeq \text{Spec}(B)$ in $\text{Fun}(\text{CAlg}_k, \text{Set})$.

The third functor φ_* , however, does not always preserve affine schemes; the next example illustrates a case where it does.

Example 1.2.32 (Weil Restriction of $\mathbb{G}_m$). Consider $\mathbb{G}_m: \mathbf{CAlg} \rightarrow \mathbf{Set}$, $R \mapsto R^\times$. Its base change $\mathbb{G}_{m,\mathbb{C}}: \mathbf{CAlg}_{\mathbb{C}} \rightarrow \mathbf{Set}$ is the restriction of $\mathbb{G}_m$ along the forgetful functor $\mathbf{CAlg}_{\mathbb{C}} \rightarrow \mathbf{CAlg}$. The Weil restriction $R_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m,\mathbb{C}}): \mathbf{CAlg}_{\mathbb{R}} \rightarrow \mathbf{Set}$ is then

$$R_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m,\mathbb{C}})(A) = \mathbb{G}_{m,\mathbb{C}}(A \otimes_{\mathbb{R}} \mathbb{C}) = (A \otimes_{\mathbb{R}} \mathbb{C})^\times.$$

This functor is an affine $\mathbb{R}$ -scheme: for any $\mathbb{R}$ -algebra A , there is a natural bijection

$$\mathrm{Hom}_{\mathbf{CAlg}_{\mathbb{R}}}(\mathbb{R}[x, y, z, w]/(xz - yw - 1, yz + xw), A) \xrightarrow{\sim} (A \otimes_{\mathbb{R}} \mathbb{C})^\times, \quad \varphi \mapsto \varphi(x) + i\varphi(y).$$

The two relations encode the requirement that $\varphi(x) + i\varphi(y)$ have an inverse $\varphi(z) + i\varphi(w)$. More generally, for any finite field extension $k \subset k'$, the Weil restriction $R_{k'/k}$ sends affine k' -schemes to affine k -schemes.

Verification of the bijection. Given $\varphi: \mathbb{R}[x, y, z, w]/(xz - yw - 1, yz + xw) \rightarrow A$, set $\alpha = \varphi(x) + i\varphi(y)$ and $\beta = \varphi(z) + i\varphi(w)$ in $A \otimes_{\mathbb{R}} \mathbb{C}$. Direct computation gives

$$\begin{aligned} \alpha\beta &= \varphi(x)\varphi(z) - \varphi(y)\varphi(w) + i(\varphi(y)\varphi(z) + \varphi(x)\varphi(w)) \\ &= \varphi(xz - yw) + i\varphi(yz + xw) = 1 + i \cdot 0 = 1, \end{aligned}$$

so $\alpha \in (A \otimes_{\mathbb{R}} \mathbb{C})^\times$ with inverse β .

Conversely, given $\alpha = a + ib \in (A \otimes_{\mathbb{R}} \mathbb{C})^\times$ with inverse $\beta = c + id$, the relation $\alpha\beta = 1$ amounts to

$$ac - bd = 1, \quad ad + bc = 0.$$

Setting $\varphi(x) = a$, $\varphi(y) = b$, $\varphi(z) = c$, $\varphi(w) = d$ produces the inverse map. $\square$

A basic fact about sets is that maps and families are interchangeable: there is an equivalence of categories

$$\mathbf{Ar}(\mathbf{Set}) \simeq \mathbf{Fam}(\mathbf{Set}),$$

where $\mathbf{Ar}(\mathbf{Set}) = \mathbf{Fun}(\{0 \rightarrow 1\}, \mathbf{Set})$ is the arrow category, and $\mathbf{Fam}(\mathbf{Set})$ has as objects the set-indexed families $(X_i)_{i \in I}$ with morphisms $(X_i)_{i \in I} \rightarrow (Y_j)_{j \in J}$ consisting of a map $u: I \rightarrow J$ together with maps $X_i \rightarrow Y_{u(i)}$. A map $f: X \rightarrow I$ corresponds to the family of its fibers $(f^{-1}\{i\})_{i \in I}$, and a family $(X_i)_{i \in I}$ corresponds to the projection $\coprod_{i \in I} X_i \rightarrow I$. Fixing the target I specializes this to

$$\mathbf{Set}_{/I} \simeq \mathbf{Set}^I.$$

The following proposition generalizes this to presheaves of sets; we recover the equivalence above by taking $\mathbf{C} = *$.

Proposition 1.2.33 (Slices of Presheaf Categories). *Let $\mathbf{C}$ be a category and $F \in \mathbf{PShv}(\mathbf{C})$. Then there is an equivalence of categories*

$$\mathbf{PShv}(\mathbf{C})_{/F} \xrightleftharpoons[\mathrm{fib}_F]{\mathrm{II}_F} \mathbf{PShv}(\mathrm{El}(F))$$

described as follows. Let $u: \mathrm{El}(F) \rightarrow \mathbf{C}$ be the forgetful functor; the tautological map $ \rightarrow u^*(F)$ has adjoint $u_\#(*) \rightarrow F$, which is an isomorphism.*

- For $H \in \mathbf{PShv}(\mathrm{El}(F))$, the presheaf $\coprod_F H$ over F is $u_\#(H) \rightarrow u_\#(*) \simeq F$. Explicitly,

$$\left(\coprod_F H\right)(X) = \coprod_{x \in F(X)} (H(X, x) \rightarrow *).$$

- For $G \in \text{PShv}(C)_{/F}$, the presheaf $\text{fib}_F(G)$ on $\text{El}(F)$ is the pullback $u^*(G) \times_{u^*(F)} *$. Explicitly,

$$\text{fib}_F(G)(X, x) = G(X) \times_{F(X)} \{x\} \simeq \left\{ \begin{array}{ccc} & & G \\ & \nearrow & \downarrow \\ y(X) & \xrightarrow{x} & F \end{array} \right\}.$$

Proof. We verify that $\coprod_F$ and fib_F are mutually inverse.

Step 1: $\text{fib}_F \circ \coprod_F \simeq \text{id}$. For $H \in \text{PShv}(\text{El}(F))$,

$$\begin{aligned} \text{fib}_F\left(\coprod_F H\right)(X, x) &= \left(\coprod_F H\right)(X) \times_{F(X)} \{x\} \\ &= \left(\coprod_{y \in F(X)} H(X, y)\right) \times_{F(X)} \{x\} = H(X, x). \end{aligned}$$

Step 2: $\coprod_F \circ \text{fib}_F \simeq \text{id}$. For $G \in \text{PShv}(C)_{/F}$,

$$\begin{aligned} \left(\coprod_F \text{fib}_F(G)\right)(X) &= \coprod_{x \in F(X)} \text{fib}_F(G)(X, x) \\ &= \coprod_{x \in F(X)} (G(X) \times_{F(X)} \{x\}) = G(X), \end{aligned}$$

with the map to $F(X)$ given by the natural projection. $\square$

Specializing to the case of a representable presheaf, we get:

Corollary 1.2.34 (Slice Categories of Presheaves). *Let C be a category, $X \in C$, and $u : C_{/X} \rightarrow C$ the forgetful functor. Then $u_\#$ induces an equivalence*

$$\text{PShv}(C_{/X}) \xrightarrow{\sim} \text{PShv}(C)_{/y(X)}.$$

Proof. Apply Proposition 1.2.33 to $F = y(X)$ and use $\text{El}(y(X)) \simeq C_{/X}$. $\square$

Specializing to $C = \text{CAlg}^{\text{op}}$ gives the following key result:

Corollary 1.2.35 (Algebraic Geometry over a Base Ring). *Let k be a ring and $\varphi : \mathbb{Z} \rightarrow k$ the unique ring map. Then $\varphi_\#$ induces an equivalence*

$$\text{Fun}(\text{CAlg}_k, \text{Set}) \xrightarrow{\sim} \text{Fun}(\text{CAlg}, \text{Set})_{/\text{Spec}(k)}.$$

Proof. The special case of Corollary 1.2.34 with $C = \text{CAlg}^{\text{op}}$ and $X = \text{Spec}(k)$. $\square$

Remark 1.2.36 (Base-Independent Algebraic Geometry). By Corollary 1.2.35, algebraic geometry over a base ring k is subsumed by algebraic geometry over $\mathbb{Z}$: working in $\text{Fun}(\text{CAlg}, \text{Set})$ rather than $\text{Fun}(\text{CAlg}_k, \text{Set})$ costs no generality, and we will often do so from now on. Furthermore, the Spec functor is independent of the base, in the sense that the following square commutes:

$$\begin{array}{ccc} \text{CAlg}_k^{\text{op}} & \xrightarrow{\sim} & (\text{CAlg}^{\text{op}})_{/k} \\ \text{Spec} \downarrow & & \downarrow \text{Spec} \\ \text{Fun}(\text{CAlg}_k, \text{Set}) & \xrightarrow{\sim} & \text{Fun}(\text{CAlg}, \text{Set})_{/\text{Spec}(k)} \end{array}$$

1.2.4. Functions on Algebraic Functors

Recall that $\mathbb{A}^1: \mathbf{CAlg} \rightarrow \mathbf{Set}$ is the forgetful functor $R \mapsto R$. Thus $\mathbb{A}^1$ is a ring object in $\mathbf{Fun}(\mathbf{CAlg}, \mathbf{Set})$:

$$\begin{array}{ccc} & & \mathbf{CAlg} \\ & \nearrow \text{id} & \downarrow \text{forget} \\ \mathbf{CAlg} & \xrightarrow{\mathbb{A}^1} & \mathbf{Set} \end{array}$$

For any algebraic functor X , we have $\text{Hom}(X, \mathbb{A}^1) \in \mathbf{CAlg}$. Pointwise, for a ring R :

$$\text{Hom}(X(R), \mathbb{A}^1(R)) = \text{Hom}_{\mathbf{Set}}(X(R), R).$$

Each $f \in \text{Hom}(X, \mathbb{A}^1)$ functorially maps an R -point x to an element $f(x) \in R$. Thus we can view $\text{Hom}(X, \mathbb{A}^1)$ as “functions” on X .

Similarly, $\mathbb{G}_m: \mathbf{CAlg} \rightarrow \mathbf{Set}$ factors as

$$\mathbf{CAlg} \xrightarrow{R \mapsto R^\times} \mathbf{Ab} \xrightarrow{\text{forget}} \mathbf{Set}.$$

For $f \in \text{Hom}(X, \mathbb{G}_m)$, we have $f(x) \in R^\times$ for any $x: \text{Spec}(R) \rightarrow X$ (with $R \neq 0$), so $f(x) \neq 0$. Thus f represents a “nowhere-vanishing function” on X .

Definition 1.2.37 (Functions). Let X be an algebraic functor.

- (i) A *function* on X is a morphism $X \rightarrow \mathbb{A}^1$. We denote

$$\mathcal{O}(X) := \text{Hom}(X, \mathbb{A}^1)$$

for the ring of all functions on X , called the *coordinate ring* of X .

- (ii) A *nowhere-vanishing function* on X is a morphism $X \rightarrow \mathbb{G}_m$. We denote

$$\mathcal{O}^\times(X) := \text{Hom}(X, \mathbb{G}_m)$$

for the group of all nowhere-vanishing functions on X .

Remark 1.2.38. We note the following:

- Since $\mathbb{G}_m \subset \mathbb{A}^1$ is a monomorphism, $\mathcal{O}^\times(X) \subset \mathcal{O}(X)$.
- $\mathcal{O}(\text{Spec}(R)) = \text{Hom}(\text{Spec}(R), \mathbb{A}^1) \simeq \mathbb{A}^1(R) \simeq R$, by the Yoneda lemma.
- Writing any presheaf as a colimit of representables,

$$\mathcal{O}(X) = \text{Hom}\left(\text{colim}_{x: \text{Spec}(R) \rightarrow X} \text{Spec}(R), \mathbb{A}^1\right) \simeq \lim_{x: \text{Spec}(R) \rightarrow X} R,$$

so a function $f \in \mathcal{O}(X)$ is encoded by the family $(f \circ x)_x$. Similarly,

$$\mathcal{O}^\times(X) \simeq \lim_{x: \text{Spec}(R) \rightarrow X} R^\times \simeq \mathcal{O}(X)^\times,$$

where the last isomorphism uses that taking units commutes with limits (since $R^\times \simeq \text{Hom}(\mathbb{Z}[u^{\pm 1}], R)$ is itself a limit-preserving functor in R).

- The functor

$$\mathcal{O}: \mathbf{Fun}(\mathbf{CAlg}, \mathbf{Set})^{\text{op}} \rightarrow \mathbf{CAlg}, \quad X \mapsto \mathcal{O}(X)$$

preserves limits, and is the unique such extension of $\text{id}_{\mathbf{CAlg}}$ along Spec : under the universal property

$$\mathbf{Fun}^{\text{lim}}(\mathbf{Fun}(\mathbf{CAlg}, \mathbf{Set})^{\text{op}}, \mathbf{CAlg}) \simeq \mathbf{Fun}(\mathbf{CAlg}, \mathbf{CAlg}),$$

$\mathcal{O}$ corresponds to $\text{id}_{\mathbf{CAlg}}$.

Warning 1.2.39. In Remark 1.2.38, since $\text{Fun}(\text{CAlg}, \text{Set})$ is a large category, the indexing category for $\mathcal{O}(X)$ may become large, so $\mathcal{O}(X)$ might not be an object of CAlg but rather a large ring (an object of $\widehat{\text{CAlg}}$).

There are two approaches to resolve this issue:

- (i) Introduce $\widehat{\text{Set}}$ and $\widehat{\text{CAlg}}$, allowing us to describe $\mathcal{O}$ as a functor

$$\mathcal{O} : \text{Fun}(\text{CAlg}, \widehat{\text{Set}})^{\text{op}} \rightarrow \widehat{\text{CAlg}}.$$

This would be the unique large-limit-preserving extension of $\text{CAlg} \hookrightarrow \widehat{\text{CAlg}}$.

- (ii) Consider the full subcategory $\text{Fun}^{\text{acc}}(\text{CAlg}, \text{Set})$ of accessible functors. Recall that for a presentable category $\mathcal{C}$, a functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is *accessible* if there exists κ such that F preserves κ -filtered colimits. For $\mathcal{D} = \text{Set}$ (which is a topos), κ -filtered colimits commute with κ -small limits, so F can be written as a κ -small colimit of representables. For $X \in \text{Fun}^{\text{acc}}(\text{CAlg}, \text{Set})$, it can be written as a small colimit of $\text{Spec}(R_i)$, giving a functor

$$\mathcal{O} : \text{Fun}^{\text{acc}}(\text{CAlg}, \text{Set})^{\text{op}} \rightarrow \text{CAlg}.$$

This approach is more common in modern derived algebraic geometry.

For a ring A , we have:

$$\begin{aligned} \text{Hom}(X, \text{Spec}(A)) &\simeq \lim_{x : \text{Spec}(R) \rightarrow X} \text{Hom}(\text{Spec}(R), \text{Spec}(A)) \\ &\simeq \lim_{x : \text{Spec}(R) \rightarrow X} \text{Hom}(A, R) \\ &\simeq \text{Hom}(A, \lim_x R) \\ &\simeq \text{Hom}(A, \mathcal{O}(X)). \end{aligned}$$

Equivalently, regarding the same object assignment as a covariant functor $\mathcal{O} : \text{Fun}(\text{CAlg}, \text{Set}) \rightarrow \text{CAlg}^{\text{op}}$, the preceding calculation says that $\mathcal{O}$ is left adjoint to Spec :

$$\text{Fun}(\text{CAlg}, \text{Set}) \begin{array}{c} \xrightarrow{\mathcal{O}} \\ \dashv \\ \xleftarrow{\text{Spec}} \end{array} \text{CAlg}^{\text{op}}$$

1.2.5. Loci

The two distinguished subfunctors of $\mathbb{A}^1$,

$$0 \hookrightarrow \mathbb{A}^1 \hookleftarrow \mathbb{G}_m,$$

let us cut out two natural subfunctors from any algebraic functor X via a function $f : X \rightarrow \mathbb{A}^1$: the preimage $f^{-1}(0)$ — where f vanishes — and $f^{-1}(\mathbb{G}_m)$ — where f is invertible (*nowhere-vanishing* in our terminology). We package this for arbitrary subsets of functions.

Definition 1.2.40 (Loci). Let X be an algebraic functor and $F \subset \mathcal{O}(X)$ a subset of functions on X .

- (i) The *vanishing locus* of F is the subfunctor $V(F) \subset X$ defined by

$$V(F)(R) := \{x \in X(R) \mid f(x) = 0 \text{ for all } f \in F\}.$$

- (ii) The *non-vanishing locus* of F is the subfunctor $D(F) \subset X$ defined by

$$D(F)(R) := \{x \in X(R) \mid (f(x))_{f \in F} \text{ generates the unit ideal in } R\}.$$

Remark 1.2.41. Note that $V(F)$ depends only on the ideal $(F) \subset \mathcal{O}(X)$. If $\sqrt{(G)} = \sqrt{(F)}$, then $D(F) = D(G)$, so $D(F)$ depends only on $\sqrt{(F)}$.

Example 1.2.42 (Punctured Affine Space). We have $\mathbb{A}^I = \text{Spec}(\mathbb{Z}[x_i \mid i \in I])$. Define $\mathbb{A}^I - 0$ as the algebraic functor

$$(\mathbb{A}^I - 0)(R) = \{a \in R^I \mid (a_i)_{i \in I} \text{ generates } R\}.$$

Here (a) denotes the ideal generated by all components of $a \in R^I$.

The coordinate functions $x_i: \mathbb{A}^I \rightarrow \mathbb{A}^1$ send $a \in \mathbb{A}^I(R)$ to its i -th component a_i . Therefore:

$$\mathbb{A}^I - 0 = D(\{x_i \mid i \in I\}) = \{a \in R^I \mid (x_i(a))_{i \in I} \text{ generates } R\}.$$

Note that when $I = \{1\}$, we have $\mathbb{A}^1 - 0 \simeq \mathbb{G}_m$.

A subset $F \subset \mathcal{O}(X)$ assembles into a single morphism

$$\text{ev}_F: X \rightarrow \mathbb{A}^F, \quad (x: \text{Spec}(R) \rightarrow X) \mapsto (g(x))_{g \in F},$$

in terms of which

$$\begin{aligned} V(F) &= \text{ev}_F^{-1}(0) = \{x \in X(R) \mid g(x) = 0 \text{ for all } g \in F\}, \\ D(F) &= \text{ev}_F^{-1}(\mathbb{A}^F - 0) = \{x \in X(R) \mid (g(x))_{g \in F} \text{ generates } R\}. \end{aligned}$$

Warning 1.2.43. It is tempting to interpret $D(F)$ as the pointwise set-theoretic complement of $V(F)$ in X , but this is false. While it holds on points valued in fields, it fails over general rings. For example, for the zero ring one has $D(F)(0) = V(F)(0) = X(0)$, so the two subfunctors are not even pointwise disjoint.

When X is a scheme, the precise complementary statement is topological: $|D(F)| = |X| \setminus |V(F)|$ on the underlying space. It is not a statement that the two functors of points are set-theoretic complements. Indeed, $D(F) = D(F')$ can hold while the scheme-theoretic closed subfunctors $V(F)$ and $V(F')$ differ: D only remembers the radical, whereas V remembers the ideal and hence its nilpotent thickening.

Proposition 1.2.44 (Properties of Loci). *Let X be an algebraic functor.*

(i) *For any family $(F_i)_{i \in I}$ of subsets of $\mathcal{O}(X)$:*

$$\bigcap_{i \in I} V(F_i) = V\left(\bigcup_{i \in I} F_i\right)$$

and

$$\bigcup_{i \in I} D(F_i) \subset D\left(\bigcup_{i \in I} F_i\right).$$

The inclusion becomes equality when evaluated on local rings.

(ii) *For finitely many subsets $F_1, \dots, F_n \subset \mathcal{O}(X)$:*

$$D(F_1) \cap \dots \cap D(F_n) = D(F_1 \cdots F_n),$$

and

$$V(F_1) \cup \dots \cup V(F_n) \subset V(F_1 \cdots F_n).$$

The inclusion becomes equality when evaluated on integral domains.

Proof. We verify these pointwise on all rings R .

(1) For the vanishing loci:

$$\begin{aligned} \bigcap_{i \in I} V(F_i)(R) &= \bigcap_{i \in I} \{x \in X(R) \mid f_i(x) = 0 \text{ for all } f_i \in F_i\} \\ &= \{x \in X(R) \mid f(x) = 0 \text{ for all } f \in \bigcup_{i \in I} F_i\} \\ &= V\left(\bigcup_{i \in I} F_i\right)(R). \end{aligned}$$

For the non-vanishing loci:

$$\begin{aligned} \bigcup_{i \in I} D(F_i)(R) &= \bigcup_{i \in I} \{x \in X(R) \mid (f_i(x))_{f_i \in F_i} = R\} \\ &\subset \{x \in X(R) \mid (f(x))_{f \in \bigcup_{i \in I} F_i} = R\} \\ &= D\left(\bigcup_{i \in I} F_i\right)(R). \end{aligned}$$

The inclusion holds because if $(f_i(x))_{f_i \in F_i} = R$ for some i , then certainly $\bigcup_i F_i$ generates R .

When R is a local ring with unique maximal ideal $\mathfrak{m}$: if $(f(x))_{f \in \bigcup F_i} = R$, then at least one $f(x)$ is a unit. This f belongs to some F_i , so $(f_i(x))_{f_i \in F_i} = R$, hence $x \in D(F_i)(R)$. Thus equality holds.

(2) The equation $D(F_1) \cap \cdots \cap D(F_n) = D(F_1 \cdots F_n)$ is straightforward: x satisfies $(f_i(x))_{f_i \in F_i} = R$ for all i if and only if $(f_1(x) \cdots f_n(x))_{f_i \in F_i} = R$.

For the vanishing loci, the inclusion $V(F_1) \cup \cdots \cup V(F_n) \subset V(F_1 \cdots F_n)$ is clear.

When R is an integral domain, we prove the reverse inclusion $V(F_1 \cdots F_n) \subset V(F_1) \cup \cdots \cup V(F_n)$ by contraposition. Suppose $x \notin V(F_i)(R)$ for every i ; then for each i there is some $g_i \in F_i$ with $g_i(x) \neq 0$. The product $g_1 \cdots g_n$ lies in $F_1 \cdots F_n$, and since R is a domain, $(g_1 \cdots g_n)(x) = g_1(x) \cdots g_n(x) \neq 0$, so $x \notin V(F_1 \cdots F_n)(R)$. Equivalently, if $x \in V(F_1 \cdots F_n)(R)$ then $x \in V(F_i)(R)$ for some i . $\square$

Example 1.2.45. We have $\mathbb{Z} = (2, 3)$, but $V(2) \cap V(3) = V(1)$, while $D(2) \cup D(3) \neq D(1)$. This confirms that $D(F)$ is not the complement of $V(F)$.

Now we characterize open and closed subfunctors of affine schemes.

Proposition 1.2.46 (Subfunctors of $\text{Spec}(A)$). *Let A be a ring.*

(i) *For any subset $F \subset A$, the quotient map $A \rightarrow A/(F)$ induces an isomorphism*

$$\text{Spec}(A/(F)) \xrightarrow{\sim} V(F) \subset \text{Spec}(A).$$

Therefore $V(F)$ is an affine scheme.

(ii) *For any $f \in A$, the localization map $A \rightarrow A_f$ induces an isomorphism*

$$\text{Spec}(A_f) \xrightarrow{\sim} D(f) \subset \text{Spec}(A).$$

Therefore $D(f)$ is an affine scheme.

Proof. (1) For $F \subset A$:

$$V(F)(R) = \{\varphi \in \text{Hom}(A, R) \mid \varphi(f) = 0 \text{ for all } f \in F\}.$$

This condition is equivalent to φ factoring through $A/(F)$, i.e., $\varphi \in \text{Hom}(A/(F), R)$.

(2) For a ring R , a map $\psi: A_f \rightarrow R$ corresponds to a map $\tilde{\psi}: A \rightarrow R$ sending f to a unit in R . Thus $\text{Hom}(A_f, R)$ is in bijection with $D(f)(R)$. $\square$

Remark 1.2.47. For a general subset $F \subset A$, $D(F)$ is typically not affine. For instance, Example 1.2.42 shows $\mathbb{A}^I - 0 \simeq D(\{x_i \mid i \in I\})$ is not affine when $|I| > 1$.

1.2.6. Open and Closed Subfunctors

Definition 1.2.48 (Open and Closed Subfunctors). Let X be an algebraic functor and $Z \subset X$ a subfunctor.

- (i) Z is a *closed subfunctor* of X if for every R -point $x: \text{Spec}(R) \rightarrow X$, there exists $F \subset R$ such that $x^{-1}(Z) \simeq V(F)$.
- (ii) Z is an *open subfunctor* of X if for every R -point $x: \text{Spec}(R) \rightarrow X$, there exists $F \subset R$ such that $x^{-1}(Z) \simeq D(F)$.

Proposition 1.2.49 (Classification of Subfunctors of $\text{Spec}(A)$). Let A be a ring.

- (i) $F \mapsto V(F)$ induces an order-reversing bijection

$$\{\text{ideals in } A\} \xrightarrow{\sim} \{\text{closed subfunctors of } \text{Spec}(A)\}.$$

- (ii) $F \mapsto D(F)$ induces an order-preserving bijection

$$\{\text{radical ideals in } A\} \xrightarrow{\sim} \{\text{open subfunctors of } \text{Spec}(A)\}.$$

Proof. (1) First, $V(F) \subset \text{Spec}(A)$ is indeed a closed subfunctor: for any $x: \text{Spec}(R) \rightarrow \text{Spec}(A)$ (corresponding to $\varphi: A \rightarrow R$):

$$x^{-1}(V(F)) \simeq V(\varphi(F)) \subset \text{Spec}(R).$$

Injectivity: We claim that $f \in (F)$ if and only if $V(F) \subset V(f)$, for any $f \in A$ and ideal $(F) \subset A$. The forward direction is clear: if $f = \sum r_i f_i$ with $f_i \in F$, then f vanishes whenever all f_i do. For the converse, take the canonical point $\pi: A \rightarrow A/(F)$, which lies in $V(F)$ tautologically; if $V(F) \subset V(f)$, then $\pi(f) = 0$, i.e., $f \in (F)$.

Now suppose $V(F) = V(F')$. For any $f \in F$, the inclusion $V(F') = V(F) \subset V(f)$ gives $f \in (F')$, so $(F) \subset (F')$. By symmetry $(F') \subset (F)$, hence $(F) = (F')$.

Surjectivity: For any closed subfunctor $Z \subset \text{Spec}(A)$, consider $x = \text{id}_{\text{Spec}(A)}$. By definition, $\text{id}^{-1}(Z) \simeq V(F)$ for some $F \subset A$. Thus $Z \simeq V(F)$.

(2) For a radical ideal $F \subset A$, the well-definedness and surjectivity are clear from the definition.

Injectivity: We must show $f \in F$ if and only if $D(f) \subset D(F)$.

($\Leftarrow$) Assume $D(f) \subset D(F)$. Evaluating at $R = A_f$: the canonical localisation map $\varphi: A \rightarrow A_f$ corresponds, under $D(f) \simeq \text{Spec}(A_f)$, to the *universal* A_f -point $\text{id}_{A_f} \in D(f)(A_f)$ (it is the image of id_{A_f} along the equivalence; other A_f -points of $D(f)$ correspond to other endomorphisms of A_f , but φ is the distinguished one we need).

The inclusion $D(f) \subset D(F)$ means $\varphi \in D(F)(A_f)$, i.e. $(\varphi(F)) = A_f$. Therefore $1 = \sum_{i=1}^m \frac{r_i}{f^{n_i}} \varphi(a_i)$ in A_f for finitely many $a_i \in F$ and $r_i \in A$, $n_i \in \mathbb{N}$. Clearing denominators by multiplying through by a sufficiently large f^N ,

$$f^N = \sum_{i=1}^m r_i f^{N-n_i} a_i \in (F),$$

so $f^N \in (F) \subset F$. Since F is a radical ideal, this forces $f \in F$.

($\Rightarrow$) is immediate from the definition of D . □

Definition 1.2.50 (Immersion). Let $Y \rightarrow X$ be a morphism of algebraic functors.

- (i) $Y \rightarrow X$ is a *closed immersion* if it is a monomorphism with closed image.
- (ii) $Y \rightarrow X$ is an *open immersion* if it is a monomorphism with open image.

Definition 1.2.51 (Locally Closed Subfunctors). Let X be an algebraic functor.

- (i) A subfunctor $Y \subset X$ is *locally closed* if there exists an open subfunctor $U \subset X$ such that $Y \subset U$ is a closed subfunctor of U .
- (ii) A morphism $Y \rightarrow X$ is an *immersion* if it is a monomorphism with locally closed image.

Proposition 1.2.52 (Stability of Immersions). (i) (Base change) Consider a pullback square:

$$\begin{array}{ccc} Y' & \xrightarrow{f'} & X' \\ \downarrow & \lrcorner & \downarrow \\ Y & \xrightarrow{f} & X \end{array}$$

If f is a closed immersion, then so is f' . The same holds for open immersions and immersions.

(ii) (Composition) Consider a commutative diagram:

$$\begin{array}{ccc} Z & \xrightarrow{g} & Y \\ & \searrow h & \swarrow f \\ & X & \end{array}$$

If f and g are closed immersions, then so is h . If h is a closed immersion and f is a monomorphism, then g is a closed immersion. The same holds for open immersions and immersions.

Proof. (1) We prove for closed immersions; the other cases are similar.

If f is a closed immersion, then f is a monomorphism and has closed image. Monomorphisms are stable under pullback, so f' is a monomorphism.

For the image, let $x' : \text{Spec}(R) \rightarrow X'$ be an R -point, and let $x : \text{Spec}(R) \rightarrow X$ be the composition. By the pasting lemma:

$$\begin{array}{ccc} x'^{-1}(f'(Y')) & \longrightarrow & \text{Spec}(R) \\ \downarrow & \lrcorner & \downarrow x' \\ Y' & \xrightarrow{f'} & X' \\ \downarrow & \lrcorner & \downarrow \\ Y & \xrightarrow{f} & X \end{array}$$

the outer rectangle is a pullback, so $x^{-1}(f(Y)) \simeq x'^{-1}(f'(Y'))$. Since $f(Y)$ is closed in X , we have $x^{-1}(f(Y)) \simeq V(I)$ for some ideal $I \subset R$. Thus $f'(Y')$ is closed in X' .

(2 - Composition)

Closed case: If f and g are closed immersions, they are both monomorphisms, so $h = f \circ g$ is a monomorphism (monomorphisms are closed under composition).

For the image: for $x: \text{Spec}(R) \rightarrow X$, we have

$$x^{-1}(h(Z)) = x^{-1}(f(g(Z))).$$

Since f has closed image, $x^{-1}(f(Y)) \simeq \text{Spec}(R/I_1)$ for some ideal $I_1 \subset R$. This gives $y: \text{Spec}(R/I_1) \rightarrow Y$.

Since g has closed image in Y :

$$y^{-1}(g(Z)) \simeq \text{Spec}(R/(I_1 + I_2))$$

for some ideal I_2 . Thus $x^{-1}(f(g(Z))) \simeq \text{Spec}(R/(I_1 + I_2))$, which is closed.

Open case: Assume $X \simeq \text{Spec}(A)$. We must show: for any subset $F \subset A$ and open subfunctor $U \subset D(F)$, U is an open subfunctor of $\text{Spec}(A)$.

For each $f \in F$, we have $D(f) \subset \text{Spec}(A)$ with $D(f) \simeq \text{Spec}(A_f)$. Consider $U \cap D(f)$. Since $U \subset D(F)$ is open, $U \cap D(f)$ is an open subfunctor of $\text{Spec}(A_f)$, so we can write

$$U \cap D(f) = D(\tilde{H}_f) \subset \text{Spec}(A_f)$$

for some $\tilde{H}_f \subset A_f$. Choose $H_f \subset A$ whose image in A_f equals $\tilde{H}_f$ (essentially clearing denominators). Then

$$D(fH_f) = D(H_f) \cap D(f) = D(\tilde{H}_f) = U \cap D(f).$$

Let $H = \bigcup_{f \in F} fH_f \subset A$. We claim $U = D(H)$.

For $\varphi: A \rightarrow R$ (i.e., $\text{Spec}(\varphi): \text{Spec}(R) \rightarrow \text{Spec}(A)$):

($U \subset D(H)$): If $\text{Spec}(\varphi) \in U$, we need $(\varphi(H)) = R$. Since $U \subset D(F)$, we have $(\varphi(F)) = R$. For any $f \in F$, the map $\text{Spec}(R_{\varphi(f)}) \rightarrow \text{Spec}(A_f)$ lies in $U \cap D(f) = D(fH_f)$. Thus $(\varphi(fH_f)) = R_{\varphi(f)}$, so $\varphi(fH_f)$ contains some power of $\varphi(f)$, i.e., $\varphi(f) \in \sqrt{(\varphi(fH_f))}$.

Taking the union over all $f \in F$:

$$R = \sqrt{(\varphi(F))} \subset \sqrt{(\varphi(H))}.$$

($D(H) \subset U$): If $(\varphi(H)) = R$, then since $H \subset (F)$, we have $D(H) \subset D(F)$. Since $U \subset D(F)$ is open, $\text{Spec}(\varphi)^{-1}(U) = D(I)$ for some radical ideal $I \subset R$.

We need $I = R$. Consider the pullback with $D(f)$:

$$D(\varphi(f)I) = \text{Spec}(\varphi)^{-1}(U \cap D(f)) = D(\varphi(fH_f)).$$

Thus $\sqrt{(\varphi(f)I)} = \sqrt{(\varphi(fH_f))}$. Taking the union over $f \in F$:

$$I \supset \sqrt{(\varphi(F)I)} = \sqrt{(\varphi(H))} = R.$$

Therefore $I = R$.

immersion case: Let $X = \text{Spec}(A)$, $Z \subset X$ closed, and $U \subset Z$ open. We show $U \subset X$ is locally closed.

Write $Z = \text{Spec}(A/I)$ and $U = D(F)$ where $F \subset A/I$. Let $\hat{F}$ be the preimage of F in A and $\hat{U} = D(\hat{F})$. Then $\hat{U} \cap Z = U$:

$$\begin{array}{ccc} U & \longrightarrow & Z \\ \downarrow & \lrcorner & \downarrow \\ \hat{U} & \longrightarrow & X \end{array}$$

Since $Z \rightarrow X$ is a closed immersion, so is $U \rightarrow \hat{U}$. Thus $U \rightarrow X$ is locally closed.

(2 - Decomposition) Follows from Theorem 1.2.53 below. □

Many classes of morphisms in algebraic geometry (such as immersions) satisfy certain “cancellation laws”. To avoid repeating similar diagram chases for each property, we state a general result below. This theorem gives a criterion for when a property satisfies a “2-out-of-3” type condition, leveraging the behavior of the diagonal morphism.

Theorem 1.2.53 (The Diagonal Argument). *Let $\mathcal{C}$ be a category with fiber products. Let $\mathcal{P}$ be a property of morphisms in $\mathcal{C}$ that is:*

- (i) *Stable under composition.*
- (ii) *Stable under base change.*

Let $g: Z \rightarrow Y$ and $f: Y \rightarrow X$ be morphisms. If the composite $f \circ g$ has property $\mathcal{P}$ and the diagonal $\Delta_f: Y \rightarrow Y \times_X Y$ has property $\mathcal{P}$, then g has property $\mathcal{P}$.

Proof. Factor g through its graph Γ_g :

$$g: Z \xrightarrow{\Gamma_g} Z \times_X Y \xrightarrow{\text{pr}_Y} Y.$$

- (i) The projection $\text{pr}_Y: Z \times_X Y \rightarrow Y$ is the base change of $f \circ g: Z \rightarrow X$ along f :

$$\begin{array}{ccc} Z \times_X Y & \longrightarrow & Z \\ \downarrow \text{pr}_Y & \lrcorner & \downarrow f \circ g \\ Y & \xrightarrow{f} & X \end{array}$$

Since $f \circ g$ has property $\mathcal{P}$ and $\mathcal{P}$ is stable under base change, pr_Y has property $\mathcal{P}$.

- (ii) The graph morphism Γ_g is the base change of the diagonal Δ_f along $(f \circ g) \times \text{id}_Y$:

$$\begin{array}{ccc} Z & \xrightarrow{\Gamma_g} & Z \times_X Y \\ \downarrow g & \lrcorner & \downarrow \\ Y & \xrightarrow{\Delta_f} & Y \times_X Y \end{array}$$

(More precisely, it is the base change of Δ_f along the map $Z \times Y \rightarrow Y \times Y$ induced by $f \circ g$ and 1_Y). Since Δ_f has property $\mathcal{P}$ and $\mathcal{P}$ is stable under base change, Γ_g has property $\mathcal{P}$.

Since $\mathcal{P}$ is stable under composition, $g = \text{pr}_Y \circ \Gamma_g$ has property $\mathcal{P}$. □

1.2.7. Quasi-Affine Schemes

Definition 1.2.54 (Quasi-Affine Scheme). Let k be a ring. An algebraic k -functor X is a *quasi-affine k -scheme* if there exist a k -algebra A and a finite subset $F \subset A$ such that $X \simeq D(F) \subset \text{Spec}(A)$.

Denote $\text{QAff}_k \subset \text{Fun}(\text{CAlg}_k, \text{Set})$ for the full subcategory spanned by quasi-affine k -schemes.

Proposition 1.2.55 (Closed Subfunctors of Quasi-Affine Schemes). *Let X be a quasi-affine k -scheme and $Z \hookrightarrow X$ a closed immersion. Then Z is also a quasi-affine k -scheme.*

Proof. We prove for $k = \mathbb{Z}$; the general case is identical. Since X is quasi-affine, there exist a ring A and finite $F \subset A$ with $X \simeq D(F)$.

Let Y be the image of $Z \hookrightarrow X$, so Y is a closed subfunctor of $D(F)$. For any $f \in F$, $Y \cap D(f)$ is a closed subfunctor of $D(f)$. Let $J(f) \subset A_f$ be the corresponding ideal, and let $I(f) \subset A$ be its preimage under $A \rightarrow A_f$. Then $I(f)_f = J(f)$, so

$$V(I(f)) \cap D(f) = Y \cap D(f).$$

Define $I = \bigcap_{f \in F} I(f)$.

Claim 1.2.56. $Y \simeq D(\bar{F}) \subset \text{Spec}(A/I) = V(I)$, where $\bar{F}$ is the image of F in A/I .

Proof of Claim. Let $\bar{Y}$ be the closure of Y in $\text{Spec}(A)$ (the smallest closed subfunctor containing Y).

Step 1: $\bar{Y} = V(I)$.

First, $Y \subset V(I)$: Let $\varphi: A \rightarrow R$ be a point in $Y(R)$. Since $Y \subset D(F)$, we have $(\varphi(F)) = R$. We need $(\varphi(I)) = 0$.

Let $a \in I$. For every $f \in F$, the composite $A \xrightarrow{\varphi} R \rightarrow R_{\varphi(f)}$ is a point of $Y \cap D(f) = V(I(f)) \cap D(f)$, and $a \in I(f)$; hence the image of $\varphi(a)$ in $R_{\varphi(f)}$ is zero. The finite family $(\varphi(f))_{f \in F}$ generates the unit ideal, so $R \rightarrow \prod_{f \in F} R_{\varphi(f)}$ is injective. (Indeed, powers of the $\varphi(f)$ annihilating a given element still generate the unit ideal.) Thus $\varphi(a) = 0$. Since this holds for every $a \in I$, we obtain $\varphi(I) = 0$.

Conversely, $V(I) \subset \bar{Y}$: By Proposition 1.2.49, closed subfunctors of $\text{Spec}(A)$ correspond to ideals. If $Y \subset V(K)$, we need $K \subset I$. By definition of I , this reduces to showing $K_f \subset J(f)$ in A_f .

From $V(J(f)) = Y \cap D(f) \subset V(K_f)$, we get $K_f \subset J(f)$.

Step 2: $Y = \bar{Y} \cap D(F)$.

First, we show $I_g = J(g)$ for all $g \in F$. The inclusion $I_g \subset I(g)_g = J(g)$ is immediate. For the reverse inclusion, note that for any $f \in F$:

$$J(g)_f = J(f)_g \quad \text{in } A_{fg}$$

(both equal the ideal cutting out $Y \cap D(fg)$).

Since localization is exact, $I(f)_g$ is the preimage of $J(f)_g$ under $A_g \rightarrow A_{fg}$. Thus $J(g) \subset I(f)_g$ for all f . Because F is finite and localization commutes with finite intersections of submodules,

$$J(g) \subset \bigcap_{f \in F} I(f)_g = \left(\bigcap_{f \in F} I(f) \right)_g = I_g.$$

A point $x: \text{Spec}(R) \rightarrow \bar{Y} \cap D(F)$ corresponds to $\varphi: A \rightarrow R$ with $\varphi(I) = 0$ and $(\varphi(F)) = R$.

Since $Y \subset D(F)$ is closed in $D(F)$, the inclusion $Y \subset \bar{Y} \cap D(F)$ is closed. To show equality, consider the pullback $x^{-1}(Y) \simeq V(G)$ for an ideal $G \subset R$. For each $f \in F$, the equality $I_f = J(f)$ shows that the restriction of x to $D(\varphi(f))$ factors through $Y \cap D(f)$. Consequently $V(G_{\varphi(f)}) = \text{Spec}(R_{\varphi(f)})$, whence $G_{\varphi(f)} = 0$ by Proposition 1.2.49. The localizations at the finite unit-generating family $(\varphi(f))_{f \in F}$ are jointly faithful on R -modules, so $G = 0$. Thus x factors through Y , as required. $\square$

Since F is finite, so is $\bar{F}$, and hence $Y \simeq D(\bar{F}) \subset \text{Spec}(A/I)$ is quasi-affine. $\square$

1.3. Zariski Descent

Let $(f_i)_{i \in I}$ be a family of elements in a ring R generating the unit ideal. In this case,

$$\bigcap_{i \in I} V(f_i) = V\left(\bigcup_{i \in I} \{f_i\}\right) = \emptyset.$$

However, as mentioned in Warning 1.2.43, $D(f_i)$ is not the complement of $V(f_i)$. Thus, in general, the equality $\bigcup_{i \in I} D(f_i) = \text{Spec}(R)$ does not hold in $\text{Fun}(\text{CAlg}, \text{Set})$.

In this section, we explain that if we restrict to the category of affine schemes $\text{Aff} \subset \text{Fun}(\text{CAlg}, \text{Set})$, then $\text{Spec}(R)$ is indeed the union (i.e., colimit) of the $D(f_i)$. More precisely, a morphism $\varphi: \text{Spec}(R) \rightarrow \text{Spec}(S)$

is uniquely determined by a family $\{\varphi_i: D(f_i) \rightarrow \text{Spec}(S)\}_{i \in I}$ satisfying the gluing condition: for all $i, j \in I$,

$$\varphi_i|_{D(f_i) \cap D(f_j)} = \varphi_j|_{D(f_i) \cap D(f_j)}.$$

For two elements $f, g \in R$, the statement “ $D(f) \cup D(g) = \text{Spec}(R)$ in Aff” is equivalent to saying that

$$\begin{array}{ccc} D(f) \cap D(g) & \hookrightarrow & D(f) \\ \downarrow & \lrcorner & \downarrow \\ D(g) & \longrightarrow & \text{Spec}(R) \end{array}$$

is a pushout diagram in Aff. Equivalently, for any affine scheme X ,

$$\begin{array}{ccc} X(R) & \longrightarrow & X(R_f) \\ \downarrow & \lrcorner & \downarrow \\ X(R_g) & \longrightarrow & X(R_{fg}) \end{array}$$

is a pullback (i.e., limit) diagram in Set.

More generally, the condition “ $\bigcup_{i \in I} D(f_i) = \text{Spec}(R)$ in Aff” is equivalent to the statement that for any affine scheme X ,

$$X(R) \longrightarrow \prod_{i \in I} X(R_{f_i}) \rightrightarrows \prod_{i, j \in I} X(R_{f_i f_j})$$

is an equalizer diagram. We begin by establishing this for modules.

Theorem 1.3.1 (Zariski Descent for Modules). *Let R be a ring and let $(f_i)_{i \in I}$ be a family of elements in R generating the unit ideal.*

(i) (Descent for morphisms) *For R -modules M and N , the diagram*

$$\text{Hom}_R(N, M) \longrightarrow \prod_{i \in I} \text{Hom}_{R_{f_i}}(N_{f_i}, M_{f_i}) \rightrightarrows \prod_{i, j \in I} \text{Hom}_{R_{f_i f_j}}(N_{f_i f_j}, M_{f_i f_j})$$

is an equalizer in Set.

(ii) (Descent for objects) *Given the following data:*

- *For each $i \in I$, an R_{f_i} -module M_i ;*
- *For each pair $(i, j) \in I \times I$, an $R_{f_i f_j}$ -linear isomorphism $\alpha_{ij}: (M_i)_{f_j} \xrightarrow{\sim} (M_j)_{f_i}$;*
- *(Cocycle condition) For each triple $(i, j, k) \in I \times I \times I$, we have*

$$\alpha_{jk} \circ \alpha_{ij} = \alpha_{ik} \quad \text{in } \text{Hom}_{R_{f_i f_j f_k}}((M_i)_{f_j f_k}, (M_k)_{f_i f_j}).$$

Then there exists an R -module M and, for each $i \in I$, an R_{f_i} -linear isomorphism $\beta_i: M_{f_i} \xrightarrow{\sim} M_i$ such that $\alpha_{ij} \circ \beta_i = \beta_j$ for all i, j . Moreover, M is unique up to unique isomorphism.

Remark 1.3.2. For the higher categorical version, one uses faithfully flat descent, since the flat topology is finer than the Zariski topology. This reduces Zariski descent to faithfully flat descent via a standard argument.

Proof. (i) *Descent for morphisms:* We write the equalizer explicitly as

$$E = \left\{ (\varphi_i: N_{f_i} \rightarrow M_{f_i})_{i \in I} \mid \forall i, j \in I, (\varphi_i)_{f_j} = (\varphi_j)_{f_i} \text{ in } \text{Hom}_{R_{f_i f_j}}(N_{f_i f_j}, M_{f_i f_j}) \right\}.$$

We must show that the natural map $\text{Hom}_R(N, M) \rightarrow E$ is a bijection.

Setting $N = R$, the problem reduces to proving descent for elements of M . We establish this first.

Descent for elements (Existence): Given $(x_i)_{i \in I} \in \prod_{i \in I} M_{f_i}$ satisfying the compatibility condition $(x_i)_{f_j} = (x_j)_{f_i}$ for all $i, j \in I$.

Since $(f_i)_{i \in I}$ generates the unit ideal, we can choose a finite subset $J \subset I$ such that $(f_j)_{j \in J}$ still generates the unit ideal. For each $j \in J$, write $x_j = a_j / f_j^{n_j}$ where $a_j \in M$ and $n_j \geq 0$.

Choose a sufficiently large uniform integer $N \geq \max_j n_j$ such that for all $i, j \in J$:

- Each x_i can be written as $x_i = a'_i / f_i^N$ for some $a'_i \in M$ (by multiplying numerator and denominator by $f_i^{N-n_i}$);
- The compatibility condition $(x_i)_{f_j} = (x_j)_{f_i}$ translates to: there exists $m \geq 0$ such that

$$(f_i f_j)^m (a'_i f_j^N - a'_j f_i^N) = 0 \quad \text{in } M.$$

After possibly increasing N , we may assume this holds with $m = 0$, i.e.,

$$a'_j f_i^N = a'_i f_j^N \quad \text{for all } i, j \in J.$$

- There exist $b_i \in R$ such that $\sum_{i \in J} b_i f_i^N = 1$.

Define $x = \sum_{i \in J} b_i a'_i \in M$. For any $k \in J$, we verify that the image of x in M_{f_k} equals x_k :

$$\begin{aligned} x &= \frac{\sum_{i \in J} b_i a'_i}{1} = \frac{\sum_{i \in J} b_i a'_i f_k^N}{f_k^N} \\ &= \frac{\sum_{i \in J} b_i (a'_i f_k^N)}{f_k^N} \quad (\text{using } a'_i f_k^N = a'_k f_i^N) \\ &= \frac{a'_k \sum_{i \in J} b_i f_i^N}{f_k^N} = \frac{a'_k \cdot 1}{f_k^N} = \frac{a'_k}{f_k^N} = x_k. \end{aligned}$$

Thus $x|_{f_k} = x_k$ for all $k \in J$. For any $i \in I$, since $(f_j)_{j \in J}$ generates the unit ideal, by repeating the above argument on the covering $\{D(f_j) \cap D(f_i)\}_{j \in J}$ of $D(f_i)$, we conclude $x|_{f_i} = x_i$.

Descent for elements (Uniqueness):

Suppose $x \in M$ satisfies $x|_{f_i} = 0$ for all $i \in I$. Equivalently, for each i , there exists $n_i \geq 0$ such that $f_i^{n_i} x = 0$ in M .

Choose a finite subset $J \subset I$ such that $(f_j)_{j \in J}$ generates the unit ideal. Choose N large enough so that $f_j^N x = 0$ for all $j \in J$ and $\sum_{j \in J} c_j f_j^N = 1$ for some $c_j \in R$. Then:

$$x = 1 \cdot x = \left(\sum_{j \in J} c_j f_j^N \right) x = \sum_{j \in J} c_j (f_j^N x) = \sum_{j \in J} c_j \cdot 0 = 0.$$

General case for morphisms: Now let N be arbitrary and suppose given compatible morphisms $\varphi_i: N_{f_i} \rightarrow M_{f_i}$. Define $\varphi: N \rightarrow M$ as follows: for $n \in N$, take $\varphi(n) \in M$ to be the unique element (provided by descent for elements) whose image in each M_{f_i} is $\varphi_i(n|_{f_i})$. Uniqueness then forces φ to be R -linear.

(ii) *Descent for objects*: Define M as the equalizer of the following diagram:

$$M := \text{eq} \left(\prod_{i \in I} M_i \xrightleftharpoons[b]{a} \prod_{(i,j) \in I \times I} (M_i)_{f_j} \right),$$

where the morphisms are defined componentwise:

- The (i, j) -component of a is the localization $M_i \rightarrow (M_i)_{f_j}$;
- The (i, j) -component of b is the composition $M_j \rightarrow (M_j)_{f_i} \xrightarrow{\alpha_{ij}^{-1}} (M_i)_{f_j}$.

By construction, there are canonical morphisms $\beta_i: M_{f_i} \rightarrow M_i$ induced by the projection $M \rightarrow M_i$ followed by localization.

Case 1: I is finite.

Fix $0 \in I$. Since I is finite and localization commutes with finite products, we have $(-)_f(\prod_{i \in I} M_i) \simeq \prod_{i \in I} (M_i)_f$.

Consider the following commutative diagram:

$$\begin{array}{ccccc} M_{f_0} & \longrightarrow & \prod_{i \in I} (M_i)_{f_0} & \xrightleftharpoons[b_{f_0}]{a_{f_0}} & \prod_{i,j \in I} (M_i)_{f_j f_0} \\ \downarrow \beta_0 & (1) & \simeq \downarrow \alpha_{i0} & (2) & \simeq \downarrow (\alpha_{i0})_{f_j} \\ M_0 & \longrightarrow & \prod_{i \in I} (M_0)_{f_i} & \xrightarrow{\quad} & \prod_{i,j \in I} (M_0)_{f_i f_j} \end{array}$$

Two preliminary observations on the cocycle data, both forced by $\alpha_{jk} \circ \alpha_{ij} = \alpha_{ik}$:

- Setting $(i, j, k) = (i, i, i)$ gives $\alpha_{ii} \circ \alpha_{ii} = \alpha_{ii}$, so $\alpha_{ii} = \text{id}$.
- Setting $(i, j, k) = (i, j, i)$ then gives $\alpha_{ji} \circ \alpha_{ij} = \text{id}$, hence $\alpha_{ji} = \alpha_{ij}^{-1}$.

We verify that the diagram commutes square by square.

- (1) The left square commutes by the compatibility $\alpha_{ij} \circ \beta_i = \beta_j$ together with the definition of M as an equalizer.
- (2) For the right square, we compare both parallel maps componentwise. On the (i, j) -component, the a -path and the b -path of the top row are

$$\begin{aligned} a_{f_0}|_{(i,j)}: (M_i)_{f_0} &\xrightarrow{\text{loc}_{f_j}} (M_i)_{f_j f_0}, \\ b_{f_0}|_{(i,j)}: (M_j)_{f_0} &\xrightarrow{\text{loc}_{f_i}} (M_j)_{f_i f_0} \xrightarrow{(\alpha_{ij}^{-1})_{f_0}} (M_i)_{f_j f_0}, \end{aligned}$$

followed by $(\alpha_{i0})_{f_j}: (M_i)_{f_j f_0} \xrightarrow{\sim} (M_0)_{f_i f_j}$. Commutativity of the a -side is just naturality of localization. For the b -side, we must check that

$$(\alpha_{i0})_{f_j} \circ (\alpha_{ij}^{-1})_{f_0} = (\alpha_{j0})_{f_i}: (M_j)_{f_i f_0} \xrightarrow{\sim} (M_0)_{f_i f_j}.$$

Applying the cocycle condition with the triple $(j, i, 0)$ gives $\alpha_{i0} \circ \alpha_{ji} = \alpha_{j0}$ at the level of $R_{f_i f_j f_0}$ -modules. Substituting $\alpha_{ji} = \alpha_{ij}^{-1}$ and rearranging yields the desired identity.

The bottom row is the trivial descent diagram for the R_{f_0} -module M_0 — every module descends to itself — hence is an equalizer. Since the vertical maps are isomorphisms, $\beta_0: M_{f_0} \rightarrow M_0$ is an isomorphism.

Uniqueness: Given two solutions $(M, \{\beta_i\})$ and $(N, \{\gamma_i\})$, define for each $i \in I$:

$$\varphi_i: M_{f_i} \xrightarrow{\beta_i} M_i \xrightarrow{\gamma_i^{-1}} N_{f_i}.$$

The compatibility $\alpha_{ij} \circ \beta_i = \beta_j$ and $\alpha_{ij} \circ \gamma_i = \gamma_j$ ensures that $(\varphi_i)_{f_j} = (\varphi_j)_{f_i}$.

By descent for morphisms (part 1), there exists a unique morphism $\varphi: M \rightarrow N$ with $\varphi_{f_i} = \varphi_i$ for all i . Since $\ker(\varphi)_{f_i} = \ker(\varphi_i) = 0$ and $\operatorname{coker}(\varphi)_{f_i} = \operatorname{coker}(\varphi_i) = 0$ for all i , and both kernel and cokernel satisfy descent (as they are determined by exactness of sequences), we have $\ker(\varphi) = 0$ and $\operatorname{coker}(\varphi) = 0$. Thus φ is an isomorphism.

Case 2: I is arbitrary.

Choose a finite subset $J \subset I$ such that $(f_j)_{j \in J}$ generates the unit ideal. For any finite $J \subset J' \subset I$, define $M_{J'}$ as the equalizer constructed from the data restricted to J' . The inclusion $J \subset J'$ induces a map $M_{J'} \rightarrow M_J$ by forgetting the coordinates indexed by $J' \setminus J$. The finite case identifies the localizations of both modules over every $D(f_j)$, $j \in J$, with the prescribed M_j , compatibly on overlaps. By the uniqueness part of finite descent for the cover $(D(f_j))_{j \in J}$, this comparison map is therefore a canonical isomorphism $M_{J'} \simeq M_J$.

The poset of finite subsets $J' \subset I$ containing the chosen J is cofinal for this inverse-limit computation: every finite $K \subset I$ is contained in $K \cup J$. Hence, since limits commute with limits, we have

$$M = \operatorname{eq} \left(\prod_{i \in I} M_i \rightrightarrows \prod_{i, j \in I} (M_i)_{f_j} \right) \simeq \lim_{\substack{J \subset J' \subset I \\ J' \text{ finite}}} \operatorname{eq} \left(\prod_{i \in J'} M_i \rightrightarrows \prod_{i, j \in J'} (M_i)_{f_j} \right) \simeq M_J.$$

For any $i \in I$, the comparison map $\beta_i: M_{f_i} \rightarrow M_i$ is an isomorphism: applying the finite case to the enlarged finite cover $J' = J \cup \{i\}$ (which still generates the unit ideal) makes β_i an isomorphism in the construction for J' , and $M \simeq M_{J'} \simeq M_J$ canonically by the previous paragraph. As this holds for every $i \in I$, the pair $(M, \{\beta_i\}_{i \in I})$ is the required descent object. This completes the proof. $\square$

Corollary 1.3.3 (Descent for Module Elements). *Let R be a ring, and $(f_i)_{i \in I}$ a family of elements generating the unit ideal of R . For an R -module M , the diagram*

$$M \longrightarrow \prod_{i \in I} M_{f_i} \rightrightarrows \prod_{i, j \in I} M_{f_i f_j}$$

is an equalizer in Mod_R .

Proof. This is the special case of Theorem 1.3.1(1) with $N = R$. $\square$

Taking $M = R$ in Corollary 1.3.3, we obtain:

Corollary 1.3.4 (Zariski Descent for Rings). *Let R be a ring, and let $(f_i)_{i \in I}$ be a family of elements generating the unit ideal of R . Then the diagram*

$$R \longrightarrow \prod_{i \in I} R_{f_i} \rightrightarrows \prod_{i, j \in I} R_{f_i f_j}$$

is an equalizer in CAlg .

Corollary 1.3.5 (Zariski Descent for Affine Schemes). *Let R be a ring, and let $(f_i)_{i \in I}$ be a family of elements generating the unit ideal of R . Then for any affine scheme X , the diagram*

$$X(R) \longrightarrow \prod_{i \in I} X(R_{f_i}) \rightrightarrows \prod_{i, j \in I} X(R_{f_i f_j})$$

is an equalizer in Set .

Proof. Write $X \simeq \operatorname{Spec}(A)$ for some ring A . Since $\operatorname{Spec}(A)(R) = \operatorname{Hom}_{\operatorname{CAlg}}(A, R)$ and the functor $\operatorname{Hom}_{\operatorname{CAlg}}(A, -)$ preserves limits (being representable), the result follows from Corollary 1.3.4. $\square$

Corollary 1.3.6 (Descent for Non-vanishing Loci). *Let R be a ring and $F \subset R$ a subset. Then for any affine scheme X , the diagram*

$$\operatorname{Hom}(D(F), X) \longrightarrow \prod_{f \in F} X(R_f) \rightrightarrows \prod_{f, g \in F} X(R_{fg})$$

is an equalizer in Set.

Proof. Let $\eta: D(F) \rightarrow X$ be a natural transformation. Evaluating it on the canonical R_f -point $R \rightarrow R_f$ of $D(F)$ gives a point $x_f \in X(R_f)$ for every $f \in F$. Naturality shows that the two restrictions of x_f and x_g to $X(R_{fg})$ agree. This defines the displayed map into the equalizer.

Conversely, suppose that points $x_f \in X(R_f)$ are given and agree over every R_{fg} . For an R -algebra S and a point $y \in D(F)(S)$, write $y^*: R \rightarrow S$ for the corresponding ring map. By definition, the elements $y^*(f)$, $f \in F$, generate the unit ideal of S . Pulling x_f back along

$$R_f \longrightarrow S_{y^*(f)}$$

gives compatible points of $X(S_{y^*(f)})$; compatibility on double overlaps follows from the assumed equality over R_{fg} . Zariski descent for the affine scheme X (Corollary 1.3.5) therefore supplies a unique point $\eta_S(y) \in X(S)$.

If $S \rightarrow T$ is a morphism of R -algebras, the pullback of $\eta_S(y)$ and the point constructed from the induced element of $D(F)(T)$ have the same restrictions on all $T_{y^*(f)}$; uniqueness in descent makes them equal. Thus the assignments $y \mapsto \eta_S(y)$ are natural in S and define $\eta: D(F) \rightarrow X$. The two constructions are inverse, which proves the equalizer statement. $\square$

Example 1.3.7. In Corollary 1.3.6, let $X = \mathbb{A}^1$. For $|I| \geq 2$, we have an isomorphism induced by the inclusion $\mathbb{A}^1 - 0 \hookrightarrow \mathbb{A}^1$:

$$\mathcal{O}(\mathbb{A}^1) \xrightarrow{\sim} \mathcal{O}(\mathbb{A}^1 - 0).$$

By the equivalence of categories $\operatorname{Aff}^{\text{op}} \simeq \operatorname{CAlg}$ (Corollary 1.2.10), this shows that $\mathbb{A}^1 - 0$ is not an affine scheme when $|I| \geq 2$, even though it is a quasi-affine scheme. Thus we have a proper inclusion:

$$\operatorname{Aff} \subsetneq \operatorname{QAff}.$$

1.3.1. Zariski Local Properties

A key observation in the proof of Theorem 1.3.1 is that a morphism φ is an isomorphism if and only if all its localizations φ_{f_i} are isomorphisms. This property of being verifiable locally on R_{f_i} is ubiquitous in algebraic geometry. In this section, we systematically study such properties.

Definition 1.3.8 (Zariski Local Property). Let $\mathcal{P}$ be a property of modules (or linear maps, algebras, etc.). We say that $\mathcal{P}$ is *Zariski local* if for any ring R and any family $(f_i)_{i \in I}$ of elements generating the unit ideal, an R -module M (or R -linear map, R -algebra, etc.) satisfies $\mathcal{P}$ if and only if for each $i \in I$, the localized object M_{f_i} (or corresponding localized object) satisfies $\mathcal{P}$.

Proposition 1.3.9 (Examples of Zariski Local Properties). *The following properties are Zariski local:*

Properties of modules:

- (a) Being zero.

- (b) *Finite generation.*
- (c) *Finite presentation.*
- (d) *Projectivity.*
- (e) *Flatness.*

Properties of module homomorphisms:

- (f) *Being zero.*
- (g) *Injectivity.*
- (h) *Surjectivity.*
- (i) *Being an isomorphism.*

Properties of algebras:

- (j) *Finite generation.*
- (k) *Finite presentation.*

Properties of sequences:

- (l) *Exactness.*

Remark 1.3.10. Proving that projectivity satisfies Zariski descent is more subtle and beyond our current scope. For a classical reference (in French), see Raynaud-Gruson [RG71, p.81]. Modern approaches typically deduce Zariski descent from faithfully flat descent; see Stacks Project [The25, Tag 058B, Tag 05A5].

Proof. We prove several representative cases. Throughout, let $(f_i)_{i \in I}$ generate the unit ideal of R .

(a) *Being zero:* A module $M = 0$ if and only if $M_{f_i} = 0$ for all $i \in I$. This follows from the injectivity of $M \hookrightarrow \prod_{i \in I} M_{f_i}$ (Corollary 1.3.3).

(g), (h), (i) *Properties of maps:* For an R -linear map $\varphi : M \rightarrow N$:

- φ is injective $\Leftrightarrow \ker(\varphi) = 0 \Leftrightarrow \ker(\varphi)_{f_i} = 0$ for all i (by exactness of localization) $\Leftrightarrow \ker(\varphi_{f_i}) = 0$ for all $i \Leftrightarrow \varphi_{f_i}$ is injective for all i .
- Similarly for surjectivity using $\operatorname{coker}(\varphi)$.
- Being an isomorphism is equivalent to being both injective and surjective.

(e) *Flatness:* Assume M_{f_i} is a flat R_{f_i} -module for all $i \in I$. To show M is flat over R , consider any injection $\varphi : N \hookrightarrow N'$ of R -modules. We must show $\varphi \otimes \operatorname{id}_M : N \otimes_R M \rightarrow N' \otimes_R M$ is injective.

By Zariski locality of injectivity, it suffices to verify injectivity after localizing at each f_i . Since tensor product commutes with localization:

$$(\varphi \otimes \operatorname{id}_M)_{f_i} \simeq \varphi_{f_i} \otimes \operatorname{id}_{M_{f_i}} : N_{f_i} \otimes_{R_{f_i}} M_{f_i} \rightarrow N'_{f_i} \otimes_{R_{f_i}} M_{f_i}.$$

Since localization preserves injectivity and M_{f_i} is flat over R_{f_i} , this map is injective. Thus M is flat.

(b) *Finite generation:* Assume M_{f_i} is finitely generated over R_{f_i} for all i . Choose a finite subset $J \subset I$ such that $(f_j)_{j \in J}$ generates R .

For each $j \in J$, choose finitely many elements $m_{j,1}, \dots, m_{j,n_j} \in M$ whose images generate M_{f_j} (clearing denominators if necessary). Define

$$\phi_j : R^{n_j} \rightarrow M, \quad (r_1, \dots, r_{n_j}) \mapsto \sum_k r_k m_{j,k}.$$

Consider $\Phi = \sum_{j \in J} \phi_j: \bigoplus_{j \in J} R^{n_j} \rightarrow M$. To show Φ is surjective, it suffices to show Φ_{f_k} is surjective for each $k \in J$ (by Zariski locality of surjectivity). But $(\phi_k)_{f_k}$ is surjective by construction, so Φ_{f_k} is surjective. Thus M is finitely generated.

(c) *Finite presentation*: Assume M_{f_i} is finitely presented over R_{f_i} for all i . By (b), M is finitely generated. Choose a surjection $\pi: R^n \twoheadrightarrow M$ with kernel K , giving an exact sequence

$$0 \rightarrow K \rightarrow R^n \xrightarrow{\pi} M \rightarrow 0.$$

Localizing (exactness is preserved):

$$0 \rightarrow K_{f_i} \rightarrow R_{f_i}^n \rightarrow M_{f_i} \rightarrow 0.$$

Since M_{f_i} is finitely presented and $R_{f_i}^n$ is finitely generated free, the kernel K_{f_i} is finitely generated over R_{f_i} (this is the definition of finite presentation). By (b), K is finitely generated over R . Therefore M is finitely presented.

(l) *Exactness*: A sequence $L \xrightarrow{f} M \xrightarrow{g} N$ is exact at M if and only if $\ker(g) = \operatorname{im}(f)$, which holds if and only if $\ker(g)_{f_i} = \operatorname{im}(f)_{f_i}$ for all i (by part (a)), if and only if the localized sequence is exact at M_{f_i} for all i . $\square$

Proposition 1.3.11 (Zariski-Local Nature of Immersions). *Let A be a ring and $u: X \rightarrow \operatorname{Spec}(A)$ be a map of algebraic functors. Suppose that X satisfies Zariski descent. Let $(f_i)_{i \in I}$ generate the unit ideal in A , and let $u_i: X \times_{\operatorname{Spec}(A)} D(f_i) \rightarrow D(f_i)$ be the base change of u to $D(f_i)$. For each of the following classes of maps, if each u_i belongs to the class, so does u :*

- (i) *monomorphism*,
- (ii) *closed immersions*,
- (iii) *open immersions*,
- (iv) *immersions*

Proof. (i) *Monomorphisms*. Let $x, y \in X(R)$ be such that $u(x) = u(y) = \varphi \in \operatorname{Spec}(A)(R)$. We want to show that $x = y$. Since X is a Zariski sheaf, for any ring R and a family $(g_j)_{j \in J}$ generating the unit ideal, we have an equalizer diagram:

$$X(R) \longrightarrow \prod_{j \in J} X(R_{g_j}) \rightrightarrows \prod_{j, k \in J} X(R_{g_j g_k})$$

Thus, it suffices to show that for every $j \in J$, we have $x_{g_j} = y_{g_j}$.

Recall that we are given a family $(f_i)_{i \in I}$ generating the unit ideal in A . Since $\varphi \in \operatorname{Spec}(A)(R)$ corresponds to a ring map $\varphi^*: A \rightarrow R$, consider $g_i := \varphi^*(f_i)$. One can easily check that $(g_i)_{i \in I}$ generates the unit ideal of R . So, we can use the index set I for the equalizer diagram.

Fix $i \in I$ and consider the localization $R \rightarrow R_{g_i}$. The map $\varphi_{g_i}: \operatorname{Spec}(R_{g_i}) \rightarrow \operatorname{Spec}(A)$ factors through $\operatorname{Spec}(A_{f_i}) \simeq D(f_i)$ (by Proposition 1.2.46). Therefore, the pairs (x_{g_i}, φ_{g_i}) and (y_{g_i}, φ_{g_i}) define elements in the fiber product $(X \times_{\operatorname{Spec}(A)} D(f_i))(R_{g_i})$. By assumption, u_i is a monomorphism, which implies $x_{g_i} = y_{g_i}$. By the sheaf property, $x = y$.

- (ii) *Closed immersions*. Assume that each u_i is a closed immersion. Since $D(f_i) \simeq \operatorname{Spec}(A_{f_i})$, there exists an ideal $K_i \subset A_{f_i}$ such that $X \times_{\operatorname{Spec}(A)} D(f_i) \simeq \operatorname{Spec}(A_{f_i}/K_i)$.

Claim 1.3.12. *There is a unique ideal $K \subset A$ such that $K_{f_i} = K_i$ for all i .*

Proof of the claim. The functor $R \mapsto \{\text{Ideals in } R\}$ satisfies Zariski descent. Indeed, viewing ideals as submodules, by Theorem 1.3.1 (Zariski descent for modules), we can glue the modules K_i to a global module K . To see that K is an ideal (submodule of A), consider the inclusion maps $\iota_i: K_i \hookrightarrow A_{f_i}$. By full faithfulness of localization (descent for morphisms), these glue to a unique map $\iota: K \rightarrow A$. Since injectivity is a local property (kernel being zero is local), ι is injective. $\square$

Let $Y = \text{Spec}(A/K)$. There is a canonical map $Y \rightarrow X$ induced by the local isomorphisms $Y \times_{\text{Spec}(A)} D(f_i) \simeq \text{Spec}(A_{f_i}/K_i) \simeq X \times_{\text{Spec}(A)} D(f_i)$.

We verify $X \simeq Y$ by checking the functor of points. For every ring R and morphism $\varphi: \text{Spec}(R) \rightarrow \text{Spec}(A)$:

- *Factorization through Y :* By definition, φ factors through Y iff $\varphi^*(K) = 0$.
- *Factorization through X :* Since $X \subset \text{Spec}(A)$ is a subfunctor, φ factors through X iff it does so locally. Consider the cover $g_i = \varphi^*(f_i)$. The restriction $\varphi|_{D(g_i)}$ factors through X iff the map $A_{f_i} \rightarrow R_{g_i}$ kills K_i .

Since $K_{f_i} = K_i$, the condition $\varphi^*(K) = 0$ is equivalent to locally killing K_i . Thus φ factors through X iff it factors through Y . Hence $X \simeq Y$, so u is a closed immersion.

- (iii) *Open immersions.* By (1), u is a monomorphism. We must show its image corresponds to an open subscheme $D(J) \subset \text{Spec}(A)$.

Since u_i is an open immersion, its image in $D(f_i) \simeq \text{Spec}(A_{f_i})$ is of the form $D(J_i)$ for some radical ideal $J_i \subset A_{f_i}$. On underlying topological spaces, $D(J_i)$ is the complement of $V(J_i)$. The closed subsets $V(J_i)$ must glue to a global closed subset $V(J)$.

Claim 1.3.13. *There exists a unique radical ideal $J \subset A$ such that $\sqrt{J \cdot A_{f_i}} = J_i$ for all i .*

Proof of the claim. The functor $R \mapsto \{\text{Radical Ideals in } R\}$ satisfies Zariski descent. Specifically, the ideals J_i satisfy the compatibility condition on overlaps: $V(J_i) \cap D(f_j) = V(J_j) \cap D(f_i)$ inside $D(f_i f_j)$. Using the same logic as in the closed immersion case (descent for modules/ideals), we can glue the ideals J_i to a global ideal J' . Let $J = \sqrt{J'}$. Then J is the unique radical ideal with the required local property. $\square$

Let $U = D(J) \subset \text{Spec}(A)$. We verify $\text{im}(u) \simeq U$. Since u is a monomorphism, it suffices to check the image locally. The restriction of u to $D(f_i)$ is $D(J \cdot A_{f_i}) = D(J_i)$, which is exactly the image of u_i . Thus, the image of u is the open subfunctor $D(J)$, so u is an open immersion.

- (iv) *Immersion.* By (1), u is a monomorphism. For each i , choose a factorization

$$X_i := X \times_{\text{Spec}(A)} D(f_i) \xrightarrow{\text{closed}} W_i \xrightarrow{\text{open}} D(f_i).$$

Cover the open subscheme $W_i \subset \text{Spec}(A_{f_i})$ by principal opens $W_{i\alpha}$. After clearing denominators, every $W_{i\alpha}$ is also a principal open of $\text{Spec}(A)$. Put

$$W := \bigcup_{i,\alpha} W_{i\alpha} = \bigcup_i W_i \subset \text{Spec}(A).$$

This is an open subfunctor. The map u factors through W : after pullback to each $D(f_i)$ its image lies in W_i , and these local factorizations glue uniquely because X and open subfunctors satisfy Zariski descent.

It remains to prove that $X \rightarrow W$ is closed. Its restriction to each $W_{i\alpha}$ is the base change of the closed immersion $X_i \rightarrow W_i$, hence is closed. Given any point $\text{Spec}(R) \rightarrow W$, the inverse images of the $W_{i\alpha}$ form a principal-open cover of $\text{Spec}(R)$. The pullback of X to $\text{Spec}(R)$ still satisfies Zariski descent, so the closed-immersion case already proved in (2), applied to this principal-open cover, shows that

$X \times_W \text{Spec}(R) \rightarrow \text{Spec}(R)$ is closed. By the definition of a closed subfunctor, $X \rightarrow W$ is therefore a closed immersion. Thus $X \rightarrow W \rightarrow \text{Spec}(A)$ exhibits u as an immersion.

□

1.4. Finiteness Properties

Recall that a k -algebra is of *finite presentation* if it is isomorphic to $k[\Sigma]$ where Σ is a system of finitely many polynomial equations in finitely many variables, and it is of *finite type* if it is isomorphic to $k[\Sigma]$ where Σ has finitely many variables (but possibly infinitely many equations). We denote the corresponding full subcategories of CAlg_k by $\text{CAlg}_k^{\text{fp}}$ and $\text{CAlg}_k^{\text{ft}}$. It turns out that these finiteness conditions are encoded geometrically by properties of the functor $\text{Spec}(A): \text{CAlg}_k \rightarrow \text{Set}$.

Definition 1.4.1 (Locally of Finite Presentation/Type). Let $X: \text{CAlg}_k \rightarrow \text{Set}$ be an algebraic k -functor.

- (i) X is *locally of finite presentation* if it preserves filtered colimits.
- (ii) X is *locally of finite type* if it preserves colimits of filtered diagrams whose transition maps are all injective.

Proposition 1.4.2. Let k be a ring and A a k -algebra.

- (i) A is of finite presentation if and only if the functor $\text{Spec}(A): \text{CAlg}_k \rightarrow \text{Set}$ is locally of finite presentation (i.e., preserves filtered colimits).
- (ii) A is of finite type if and only if the functor $\text{Spec}(A): \text{CAlg}_k \rightarrow \text{Set}$ is locally of finite type (i.e., preserves filtered colimits of monomorphisms).

Proof. (i) (Finite Presentation)

Direction $\Rightarrow$: Assume that A is of finite presentation.

Step 1: Free algebras are compact. Let Σ be a finite set. The functor defined by the polynomial ring $k[\Sigma]$ is given by

$$\text{Hom}_{\text{CAlg}_k}(k[\Sigma], R) \simeq R^\Sigma \simeq \prod_{s \in \Sigma} R.$$

Since finite products commute with filtered colimits in Set , for any filtered diagram $F_\bullet: I \rightarrow \text{CAlg}_k$, we have

$$\begin{aligned} \text{Hom}(k[\Sigma], \text{colim}_i F_i) &\simeq (\text{colim}_i F_i)^\Sigma \\ &\simeq \text{colim}_i (F_i^\Sigma) \\ &\simeq \text{colim}_i \text{Hom}(k[\Sigma], F_i). \end{aligned}$$

Thus, $k[\Sigma]$ is a compact object.

Step 2: General case via Coequalizer. Since A is of finite presentation, we can choose a presentation

$$k[y_1, \dots, y_m] \rightrightarrows k[x_1, \dots, x_n] \rightarrow A,$$

which expresses A as a coequalizer of two maps between free algebras on finite sets. Let $P_1 = k[x_1, \dots, x_n]$ and $P_0 = k[y_1, \dots, y_m]$. We have an exact sequence of functors (equalizer diagram in Set):

$$\text{Hom}(A, -) \rightarrow \text{Hom}(P_1, -) \rightrightarrows \text{Hom}(P_0, -).$$

Now consider the filtered colimit. Since filtered colimits in $\mathbf{Set}$ commute with finite limits (specifically equalizers), we have the following commutative diagram where the rows are equalizers:

$$\begin{array}{ccccc} \mathrm{colim}_i \mathrm{Hom}(A, F_i) & \longrightarrow & \mathrm{colim}_i \mathrm{Hom}(P_1, F_i) & \rightrightarrows & \mathrm{colim}_i \mathrm{Hom}(P_0, F_i) \\ \downarrow \text{dashed} & & \downarrow \wr & & \downarrow \wr \\ \mathrm{Hom}(A, \mathrm{colim}_i F_i) & \longrightarrow & \mathrm{Hom}(P_1, \mathrm{colim}_i F_i) & \rightrightarrows & \mathrm{Hom}(P_0, \mathrm{colim}_i F_i) \end{array}$$

The vertical arrows for P_1 and P_0 are isomorphisms by Step 1. By the universal property of equalizers, the induced map for A (the dashed arrow) must also be an isomorphism.

Direction $\Leftarrow$: Conversely, assume that $\mathrm{Spec}(A)$ preserves filtered colimits. Recall that every k -algebra can be written as the canonical filtered colimit of its finitely presented objects over it:

$$A \simeq \mathrm{colim}_{(B, \varphi) \in (\mathrm{CAlg}_k^{\mathrm{fp}})_{/A}} B.$$

By assumption, the functor $\mathrm{Hom}(A, -)$ preserves this colimit:

$$\mathrm{Hom}_{\mathrm{CAlg}_k}(A, A) \simeq \mathrm{Hom}_{\mathrm{CAlg}_k}(A, \mathrm{colim}_B B) \simeq \mathrm{colim}_B \mathrm{Hom}_{\mathrm{CAlg}_k}(A, B).$$

Consider the identity morphism id_A on the left. It must map to some element in the colimit on the right. This implies there exists a finitely presented algebra A_0 (in the diagram) and a morphism $u: A \rightarrow A_0$ such that the composition

$$A \xrightarrow{u} A_0 \xrightarrow{\varphi} A$$

is equal to the identity id_A . Thus, A is a *retract* of the finitely presented algebra A_0 . Since a retract of a finitely presented object is finitely presented, A is of finite presentation.

(ii) (*Finite Type*)

Direction $\Rightarrow$: Assume A is of finite type. Then there exists a surjective homomorphism $\pi: k[\Sigma] \twoheadrightarrow A$ for some finite set Σ . Finite type corresponds to the property that $\mathrm{Spec}(A)$ is *locally of finite type*, which means the functor $\mathrm{Hom}(A, -)$ preserves filtered colimits of *monomorphisms*.

Let $F_\bullet: I \rightarrow \mathrm{CAlg}_k$ be a filtered diagram where all transition maps are injective. We want to show the map

$$\Phi: \mathrm{colim}_i \mathrm{Hom}(A, F_i) \rightarrow \mathrm{Hom}(A, \mathrm{colim}_i F_i)$$

is bijective.

Injectivity: Since $k[\Sigma] \rightarrow A$ is an epimorphism, the map $\mathrm{Hom}(A, R) \rightarrow \mathrm{Hom}(k[\Sigma], R)$ is injective. The injectivity of Φ follows from the injectivity of the corresponding map for $k[\Sigma]$ (which is an isomorphism) and the exactness properties of filtered colimits.

Surjectivity: Let $\varphi: A \rightarrow \mathrm{colim}_i F_i$ be a map. The composition $k[\Sigma] \rightarrow A \rightarrow \mathrm{colim}_i F_i$ lifts to some map $\psi_j: k[\Sigma] \rightarrow F_j$ (since $k[\Sigma]$ is compact). We check if ψ_j descends to A , i.e., if it vanishes on the kernel $I = \ker(k[\Sigma] \rightarrow A)$. Let $f \in I$ be any element. Its image maps to 0 in $\mathrm{colim}_i F_i$. Since the transition maps in $F_\bullet$ are *injective*, f must map to 0 in F_j itself (there is no “transient” kernel in the colimit). Since this holds for every element of I , ψ_j factors through A , providing the required lift.

Direction $\Leftarrow$: Conversely, assume that $\mathrm{Spec}(A)$ preserves filtered colimits of monomorphisms. Write A as the filtered colimit of its finitely generated sub- k -algebras,

$$A \simeq \mathrm{colim}_{A_\alpha \subseteq A} A_\alpha,$$

a diagram that is filtered (any two finitely generated subalgebras lie in a third, generated by their union) and whose transition maps are the inclusions, hence monomorphisms. By hypothesis $\mathrm{Hom}(A, -)$ preserves this colimit, so

$$\mathrm{Hom}_{\mathrm{CAlg}_k}(A, A) \simeq \mathrm{colim}_\alpha \mathrm{Hom}_{\mathrm{CAlg}_k}(A, A_\alpha).$$

The identity id_A is therefore represented by a morphism $u: A \rightarrow A_\alpha$ for some α whose composite with the inclusion is the identity:

$$A \xrightarrow{u} A_\alpha \hookrightarrow A \text{ equals } \text{id}_A.$$

Since the inclusion $A_\alpha \hookrightarrow A$ is injective while this composite is surjective, the inclusion is a bijection, i.e. $A = A_\alpha$. Hence A is finitely generated over k , that is, of finite type.

□

Example 1.4.3. (i) $\mathbb{A}_k^I$ is locally of finite type if and only if I is a finite set, in which case it is also locally of finite presentation. The same holds for the punctured affine spaces $\mathbb{A}_k^I - 0$.

(ii) The affine k -space $\mathbb{A}(M)$ (Example 1.2.24) is locally of finite presentation (resp. of finite type) if and only if the k -module M is of finite presentation (resp. of finite type).

(iii) The algebraic k -functor $\mathbb{A}^\vee(M)$ (Remark 1.2.25) is locally of finite presentation for any k -module M , since the tensor product functor is left adjoint.

(iv) The affine $\mathbb{Z}$ -schemes $*$, $\emptyset$, Idem , $\mathbb{G}_m$, $\mathbb{G}_a$, Mat_n , GL_n , and SL_n are all locally of finite presentation.

Remark 1.4.4. Filtered colimits commute with finite limits in Set , and finite limits in $\text{Fun}(\text{CAlg}, \text{Set})$ are computed pointwise. Consequently, being locally of finite presentation, or locally of finite type, is preserved under finite limits in $\text{Fun}(\text{CAlg}_k, \text{Set})$.

Remark 1.4.5 (Compatibility with Base Change). Let $\varphi: k \rightarrow k'$ be a ring map. Both the forgetful functor $\text{CAlg}_{k'} \rightarrow \text{CAlg}_k$ and its left adjoint $\varphi^*: \text{CAlg}_k \rightarrow \text{CAlg}_{k'}$ preserve filtered colimits, so by Corollary 1.2.29, base change along φ and Weil restriction along φ both preserve the property of being locally of finite presentation or locally of finite type. The third functor $\varphi_\#$ usually does not: for instance, $\text{Spec}(\mathbb{C})$ is locally of finite presentation as an affine $\mathbb{R}$ -scheme but not as an affine $\mathbb{Q}$ -scheme.

Recall that an algebraic k -functor can be thought of as a map of algebraic functors $X \rightarrow \text{Spec}(k)$ (Corollary 1.2.35). Remark 1.4.5 implies that, for any cartesian square

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow f' & & \downarrow f \\ \text{Spec}(k') & \longrightarrow & \text{Spec}(k), \end{array}$$

if f is locally of finite presentation or of finite type, so is f' . Consequently, we can extend Definition 1.4.1 to arbitrary maps of algebraic functors as follows:

Definition 1.4.6 (Morphism Locally of Finite Presentation/Type). Let $f: X \rightarrow S$ be a map of algebraic functors.

- (i) f is *locally of finite presentation* if, for every ring k and every k -point $\text{Spec}(k) \rightarrow S$, the algebraic k -functor $X \times_S \text{Spec}(k)$ is locally of finite presentation.
- (ii) f is *locally of finite type* if, for every ring k and every k -point $\text{Spec}(k) \rightarrow S$, the algebraic k -functor $X \times_S \text{Spec}(k)$ is locally of finite type.

Concretely, under the equivalence of Corollary 1.2.35, the base change of $f: X \rightarrow S$ along $s: \text{Spec}(k) \rightarrow S$ is the algebraic k -functor $\text{CAlg}_k \rightarrow \text{Set}$ given by

$$(\varphi: k \rightarrow R) \mapsto \{x \in X(R) \mid f(x) = s(\varphi) \text{ in } S(R)\} = \left\{ \begin{array}{ccc} \text{Spec}(R) & \xrightarrow{-x-} & X \\ \downarrow \text{Spec}(\varphi) & & \downarrow f \\ \text{Spec}(k) & \xrightarrow{s} & S \end{array} \right\}.$$

Example 1.4.7. Any open immersion is locally of finite presentation, and any closed immersion is locally of finite type. A closed immersion $i: Z \hookrightarrow X$ is locally of finite presentation if and only if, for every ring R and every $x: \text{Spec}(R) \rightarrow X$, there exists a finite subset $F \subset R$ such that $x^{-1}(i(Z)) = V(F)$.

Proposition 1.4.8 (Closure properties of Morphisms Locally of Finite Presentation/Type). (i) (**Base Change**) Consider a cartesian square of algebraic functors

$$\begin{array}{ccc} Y' & \xrightarrow{f'} & X' \\ \downarrow & \lrcorner & \downarrow \\ Y & \xrightarrow{f} & X. \end{array}$$

If f is locally of finite presentation, so is f' . The same holds for “locally of finite type”.

(ii) (**Cancellation / 2-out-of-3**) Consider a commutative triangle of algebraic functors

$$\begin{array}{ccc} Z & \xrightarrow{g} & Y \\ & \searrow h & \swarrow f \\ & X. & \end{array}$$

If f is locally of finite presentation, then g is locally of finite presentation if and only if h is. The same holds for “locally of finite type”.

Proof. (i) Let f be locally of finite presentation. We want to show that f' is also locally of finite presentation. For any ring k and any k -point $x: \text{Spec}(k) \rightarrow X'$, we must show that the fiber $X'_x := x^{-1}(Y') \simeq Y' \times_{X'} \text{Spec}(k)$ is locally of finite presentation (as a k -functor).

Consider the following diagram:

$$\begin{array}{ccccc} x^{-1}(Y') & \longrightarrow & Y' & \longrightarrow & Y \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow f \\ \text{Spec}(k) & \xrightarrow{x} & X' & \longrightarrow & X \end{array}$$

By the **Pasting Lemma**, the outer rectangle is a pullback square. Thus, the fiber $x^{-1}(Y')$ is isomorphic to the fiber of f over the composite point $\text{Spec}(k) \xrightarrow{x} X' \rightarrow X$. Since f is locally of finite presentation, its fibers are locally of finite presentation by definition. Thus f' satisfies the condition.

(ii) We prove the equivalence.

($\Rightarrow$) (**Composition**) Assume g is locally of finite presentation. We verify $h = f \circ g$ is locally of finite presentation. Let $z: \text{Spec}(k) \rightarrow X$ be a point. The fiber $z^{-1}(Z)$ can be computed as:

$$z^{-1}(Z) := Z \times_X \text{Spec}(k) \simeq Z \times_Y (Y \times_X \text{Spec}(k)).$$

Since f is LFP, the fiber $Y_z := z^{-1}(Y) = Y \times_X \text{Spec}(k)$ is LFP over k . Since g is LFP, the base change map $Z \times_Y Y_z \rightarrow Y_z$ is LFP (by Base Change, proved in part 1). LFP morphisms are closed under composition in this functorial formulation: over a filtered diagram, first descend the Y_z -coordinate to one stage, and then use preservation of the filtered colimit by the corresponding fiber of $Z \times_Y Y_z \rightarrow Y_z$; equality can likewise be checked after passing to a common stage. Hence the composite $Z \times_Y Y_z \rightarrow Y_z \rightarrow \text{Spec}(k)$ makes the fiber an LFP k -functor. Thus h is LFP.

($\Leftrightarrow$) **(Cancellation)** Assume h is locally of finite presentation. We must show that for every k -point $y: \operatorname{Spec}(k) \rightarrow Y$, the fiber $y^{-1}(Z) = Z \times_Y \operatorname{Spec}(k)$ is locally of finite presentation over k .

Set $x = f \circ y: \operatorname{Spec}(k) \rightarrow X$. Consider the following diagram of fibers:

$$\begin{array}{ccc} y^{-1}(Z) & \longrightarrow & x^{-1}(Z) \\ \downarrow & \lrcorner & \downarrow \\ \operatorname{Spec}(k) & \xrightarrow{\bar{y}} & x^{-1}(Y) \end{array}$$

where $x^{-1}(Y)$ is the fiber of f over x , $x^{-1}(Z)$ is the fiber of h over x , and $\bar{y}$ is the k -point of $x^{-1}(Y)$ induced by y . This square is cartesian because

$$y^{-1}(Z) = Z \times_Y \operatorname{Spec}(k) \simeq (Z \times_X \operatorname{Spec}(k)) \times_{(Y \times_X \operatorname{Spec}(k))} \operatorname{Spec}(k) = x^{-1}(Z) \times_{x^{-1}(Y)} \operatorname{Spec}(k).$$

We now verify the hypotheses needed to conclude:

Step 1: $\bar{y}$ is LFP. Put $Y_x = x^{-1}(Y)$. Since f is LFP, the k -functor Y_x preserves filtered colimits. Its diagonal $\Delta_{Y_x}: Y_x \rightarrow Y_x \times_{\operatorname{Spec}(k)} Y_x$ is LFP: after base change by a pair $a, b \in Y_x(R)$, its fiber on an R -algebra S is the equalizer of the two maps $\{*\} \rightrightarrows Y_x(S)$ determined by a and b . Filtered colimits commute with this finite limit in $\mathbf{Set}$. Write $p: Y_x \rightarrow \operatorname{Spec}(k)$ for the structure map. Any section of p is a base change of Δ_{Y_x} (along the map $Y_x \rightarrow Y_x \times_{\operatorname{Spec}(k)} Y_x$ given by $(\bar{y} \circ p, \operatorname{id}_{Y_x})$). Hence the section $\bar{y}: \operatorname{Spec}(k) \rightarrow Y_x$ is LFP.

Step 2: Conclude by base change and composition. By base change (part 1), since $\bar{y}$ is LFP, the map $y^{-1}(Z) \rightarrow x^{-1}(Z)$ is LFP. Since h is LFP, the fiber $x^{-1}(Z) \rightarrow \operatorname{Spec}(k)$ is LFP. By the composition result ($\Rightarrow$), the composite $y^{-1}(Z) \rightarrow x^{-1}(Z) \rightarrow \operatorname{Spec}(k)$ is LFP. Hence $y^{-1}(Z)$ is an LFP k -functor, so g is locally of finite presentation.

The same argument proves every assertion for “locally of finite type”: restrict throughout to filtered diagrams with injective transition maps; filtered colimits of sets still commute with finite limits. $\square$

1.5. Nullstellensatz

Consider a monic polynomial $f \in k[x]$ over a field k . The functorial Nullstellensatz (Corollary 1.2.11) tells us that f is determined by its zero sets in all k -algebras. On the other hand, elementary field theory tells us that f splits into linear factors over some finite field extension of k . Combining the two: knowing the zero sets of f over finite field extensions of k determines f provided it is separable (i.e., has no multiple roots), and in general determines the radical of f — the product of the distinct prime factors of f .

Hilbert’s Nullstellensatz generalizes this latter statement to systems of polynomial equations in several variables: given $f_1, \dots, f_m \in k[x_1, \dots, x_n]$, the sets of common zeros in all finite field extensions of k determine the radical ideal $\sqrt{(f_1, \dots, f_m)}$. This was the cornerstone of classical algebraic geometry, which restricted attention to solutions in fields. We record the result here, while pointing out some of its limitations.

For a ring k , define the maps:

$$\{\text{subsets of } k[x_1, \dots, x_n]\} \xrightleftharpoons[I]{V} \{\text{subsets of } k^n\}$$

as follows:

$$\begin{aligned} V(F) &= \{x \in k^n \mid f(x) = 0 \text{ for all } f \in F\}, \\ I(X) &= \{f \in k[x_1, \dots, x_n] \mid f(x) = 0 \text{ for all } x \in X\}. \end{aligned}$$

Note that both maps are order-reversing and that $F \subset I(V(F))$ and $X \subset V(I(X))$ (in other words, this is an adjunction between posets). Note also that $I(X)$ is always an ideal in $k[x_1, \dots, x_n]$, and it is even a radical ideal if k is reduced (if a power of f vanishes on X , so does f). Call a subset $X \subset k^n$ *algebraic* if it lies in the image of V , or equivalently if $X = V(I(X))$.

Theorem 1.5.1 (Hilbert's Nullstellensatz). *Let k be an algebraically closed field and let $n \in \mathbb{N}$. For any subset $F \subset k[x_1, \dots, x_n]$, we have:*

$$I(V(F)) = \sqrt{(F)}.$$

Consequently, the maps V and I define a one-to-one correspondence:

$$\{\text{radical ideals in } k[x_1, \dots, x_n]\} \xrightleftharpoons[I]{V} \{\text{algebraic subsets of } k^n\}$$

Proof. The inclusion $\sqrt{(F)} \subset I(V(F))$ is clear: if $f^m \in (F)$ for some $m \geq 1$, then f^m vanishes on $V(F)$, so f vanishes on $V(F)$ since k is a field (hence reduced).

For the reverse inclusion, we use the following key lemma:

Lemma 1.5.2 (Key Lemma for Nullstellensatz). *Let k be an algebraically closed field and let $I \subsetneq k[x_1, \dots, x_n]$ be a proper ideal. Then $V(I) \neq \emptyset$.*

Proof of Lemma. Let $\mathfrak{m}$ be a maximal ideal containing I . We claim that $\mathfrak{m} = (x_1 - a_1, \dots, x_n - a_n)$ for some $(a_1, \dots, a_n) \in k^n$.

The quotient $k[x_1, \dots, x_n]/\mathfrak{m}$ is a field extension of k . By Zariski's lemma, this is a finite extension. Since k is algebraically closed, we must have $k[x_1, \dots, x_n]/\mathfrak{m} = k$.

Let a_i be the image of x_i in k . Then $x_i - a_i \in \mathfrak{m}$ for all i , so $(x_1 - a_1, \dots, x_n - a_n) \subset \mathfrak{m}$. But the left ideal is maximal (its quotient is k), so equality holds.

Therefore, $(a_1, \dots, a_n) \in V(\mathfrak{m}) \subset V(I)$. □

Now suppose $f \in I(V(F))$ but $f \notin \sqrt{(F)}$. Consider the ideal $J = (F, 1 - yf) \subset k[x_1, \dots, x_n, y]$ where y is a new variable.

If J were the unit ideal, then we would have:

$$1 = \sum_i g_i f_i + h(1 - yf)$$

for some $g_i, h \in k[x_1, \dots, x_n, y]$ and $f_i \in F$. Setting $y = 1/f$ (where we work in the localization), we get:

$$1 = \sum_i g_i(x, 1/f) f_i$$

which means $f^N \in (F)$ for some N , contradicting $f \notin \sqrt{(F)}$.

Therefore, $J \neq k[x_1, \dots, x_n, y]$. By the lemma, there exists $(a_1, \dots, a_n, b) \in k^{n+1}$ with $(a_1, \dots, a_n, b) \in V(J)$. This means:

- $f_i(a_1, \dots, a_n) = 0$ for all $f_i \in F$
- $1 - bf(a_1, \dots, a_n) = 0$

The first condition says $(a_1, \dots, a_n) \in V(F)$, so $f(a_1, \dots, a_n) = 0$ by assumption. But then the second condition gives $1 = 0$, a contradiction. □

We can upgrade this result to an equivalence of categories as follows. Define the category AffSet_k of *affine algebraic sets* over k as follows:

- An object of AffSet_k is a pair (n, X) with $n \geq 0$ and $X \subset k^n$ an algebraic subset.
- A morphism $(n, X) \rightarrow (m, Y)$ is a map $f: X \rightarrow Y$ such that there exists a polynomial map $k^n \rightarrow k^m$ extending f .

Recall that a ring R is *reduced* if 0 is the only nilpotent element of R . We denote by $\text{CAlg}_k^{\text{red}}$ the category of reduced k -algebras.

Corollary 1.5.3 (Categorical Nullstellensatz). *Let k be an algebraically closed field. Then there is an equivalence of categories:*

$$\text{AffSet}_k \xrightarrow{\sim} (\text{CAlg}_k^{\text{ft,red}})^{\text{op}}, \quad (n, X) \mapsto k[x_1, \dots, x_n]/I(X).$$

Hence, AffSet_k is equivalent to the full subcategory of Aff_k spanned by the reduced affine k -schemes of finite type.

Proof. We must show this functor is fully faithful and essentially surjective.

Fully faithful: Let (n, X) and (m, Y) be affine algebraic sets. By Theorem 1.5.1, $I(X)$ and $I(Y)$ are radical, so the coordinate rings $k[X] := k[x_1, \dots, x_n]/I(X)$ and $k[Y] := k[y_1, \dots, y_m]/I(Y)$ are reduced; since k is algebraically closed, their elements are exactly the polynomial functions on X and Y , two of which are equal in the coordinate ring if and only if they agree at every point. We must show that

$$\{\text{polynomial-extendable } f: X \rightarrow Y\} \xrightarrow{f \mapsto f^*} \text{Hom}_{\text{CAlg}_k}(k[Y], k[X]), \quad f^*(\bar{h}) = \bar{h} \circ f,$$

is a bijection.

(*Well-defined and injective.*) If f is the restriction of a polynomial map $\tilde{f}: k^n \rightarrow k^m$ with $\tilde{f}(X) \subseteq Y$, then for every $g \in I(Y)$ the function $g \circ \tilde{f}$ vanishes on X , i.e. lies in $I(X)$; hence $f^*: \bar{y}_j \mapsto \overline{\tilde{f}_j}$ is a well-defined k -algebra map. If $f^* = f'^*$ for two such maps, then $\bar{y}_j \circ f = \bar{y}_j \circ f'$ for each j , so f and f' have the same coordinate functions on X and thus $f = f'$.

(*Surjective.*) Conversely, given $\varphi: k[Y] \rightarrow k[X]$, choose polynomials $\tilde{f}_j \in k[x_1, \dots, x_n]$ representing $\varphi(\bar{y}_j)$ and set $\tilde{f} = (\tilde{f}_1, \dots, \tilde{f}_m): k^n \rightarrow k^m$. For $x \in X$ and $g \in I(Y)$ we have $g(\tilde{f}(x)) = \varphi(\bar{g})(x) = 0$, since $\bar{g} = 0$ in $k[Y]$; thus $\tilde{f}(x) \in V(I(Y)) = Y$ by the Nullstellensatz. So $\tilde{f}$ restricts to a polynomial-extendable map $f: X \rightarrow Y$ with $f^* = \varphi$.

Essentially surjective: Any reduced finitely generated k -algebra has the form $k[x_1, \dots, x_n]/I$ where I is a radical ideal. By the Nullstellensatz, $I = I(V(I))$, so this algebra corresponds to the affine algebraic set $(n, V(I))$. $\square$

Remark 1.5.4 (Finite Type vs Finite Presentation). By Hilbert's Basissatz, “finite type” and “finite presentation” are equivalent for algebras over a field (and more generally over a Noetherian ring).

We can also formulate a Nullstellensatz for an arbitrary field k as follows. Denote by $\text{Field}_k^{\text{fin}} \subset \text{CAlg}_k$ the full subcategory of finite field extensions of k .

Corollary 1.5.5 (Nullstellensatz for Arbitrary Fields). *Let k be a field and let $n \in \mathbb{N}$. Then there is an order-reversing bijection:*

$$V: \{\text{radical ideals in } k[x_1, \dots, x_n]\} \xrightarrow{\sim} \{\text{vanishing loci in } \mathbb{A}^n: \text{Field}_k^{\text{fin}} \rightarrow \text{Set}\}.$$

Proof. A radical ideal $I \subset k[x_1, \dots, x_n]$ determines a vanishing locus:

$$V(I): \text{Field}_k^{\text{fin}} \rightarrow \text{Set}, \quad K \mapsto \{x \in K^n \mid f(x) = 0 \text{ for all } f \in I\}.$$

Conversely, given a vanishing locus $X: \text{Field}_k^{\text{fin}} \rightarrow \text{Set}$, we can recover the ideal:

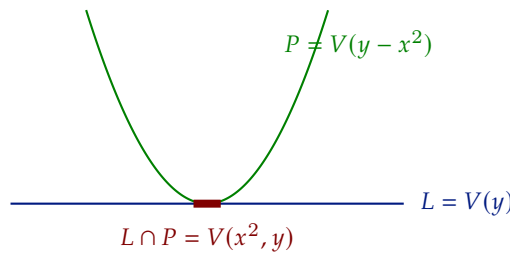
$$I(X) = \{f \in k[x_1, \dots, x_n] \mid f(x) = 0 \text{ for all } K \in \text{Field}_k^{\text{fin}}, x \in X(K)\}.$$

The key observation is that for any $f \in k[x_1, \dots, x_n]$, we have $f \in I(V(I))$ if and only if f vanishes on $V(I)(K)$ for all finite extensions K/k . This follows from the classical Nullstellensatz applied to the algebraic closure $\bar{k}$, since any zero of f in $\bar{k}$ lies in a finite extension of k . $\square$

Example 1.5.6 (Non-reduced Intersections). Even in the context of algebraic geometry over an algebraically closed field k , there are geometric phenomena that are not captured by only considering solutions in k . Consider for example the vanishing loci $L = V(y)$ and $P = V(y - x^2)$ in $\mathbb{A}_k^2 = \text{Spec}(k[x, y])$.

Since the k -algebras $k[x, y]/(y) \simeq k[t]$ and $k[x, y]/(y - x^2) \simeq k[t]$ are reduced, these affine k -schemes are determined by the algebraic sets $L(k)$ and $P(k)$ in k^2 (by the Nullstellensatz). The intersection $L(k) \cap P(k)$ is the algebraic set $\{(0, 0)\} \subset k^2$, which in turn corresponds to the subfunctor $V(x, y) \subset \mathbb{A}_k^2$.

However, the functorial intersection $L \cap P \subset \mathbb{A}_k^2$ is the subfunctor $V(x^2, y)$, which is isomorphic to $\text{Spec}(k[t]/(t^2))$. One can think of $V(x^2, y)$ as a first-order infinitesimal neighborhood of the origin $V(x, y)$ along the x -axis; this captures the fact that the line L is tangent (to first order) to the parabola P , so that they both contain the same infinitesimal horizontal segment at the origin. This residual tangency information in the intersection can only be seen by evaluating the functor $L \cap P$ on non-reduced k -algebras.



Assume here that $\text{char}(k) \neq 2$. This also resolves another issue in classical algebraic geometry, which is that intersections do not vary nicely in families. Consider for example the family of horizontal lines $L_a = V(y - a)$ for $a \in k$. The intersection $L_a(k) \cap P(k)$ has exactly two points for any $a \neq 0$ (since k is algebraically closed), but only a single point when $a = 0$. On the other hand, the scheme-theoretic intersection $L_a \cap P$ is given by a 2-dimensional k -algebra for all $a \in k$, namely $k \times k$ when $a \neq 0$ and $k[t]/(t^2)$ when $a = 0$.

Example 1.5.7 (Geometry in Mixed Characteristic). Another aspect not captured by algebraic geometry over a single field is geometry over arithmetic bases such as $\mathbb{Z}$. Local rings on such bases can have characteristic 0 while their residue fields have positive characteristic; this is the mixed-characteristic situation. It is especially relevant in number theory, which studies rings of integers in finite extensions of $\mathbb{Q}$. Such rings map to fields of different characteristics, which sometimes allows information to pass between characteristic 0 and characteristic p fibers.

As a very basic example, consider the following reformulation of the usual proof that $\sqrt{2}$ is irrational. If $a/b \in \mathbb{Q}$ in lowest terms satisfied $(a/b)^2 = 2$, then the coprime pair (a, b) would define a point $[a : b] \in X(\mathbb{Z})$, where $X \subset \mathbb{P}_{\mathbb{Z}}^1$ is cut out by the homogeneous equation $x^2 = 2y^2$. Conversely, every point of $X(\mathbb{Z})$ has $y \neq 0$ and therefore gives such a rational solution. Functoriality along $\mathbb{Z} \rightarrow \mathbb{Z}/4$ gives $X(\mathbb{Z}) \rightarrow X(\mathbb{Z}/4)$. Since the squares in $\mathbb{Z}/4$ are 0 and 1, no unimodular pair modulo 4 satisfies $x^2 = 2y^2$; hence $X(\mathbb{Z}/4) = \emptyset$. It follows that $X(\mathbb{Z}) = \emptyset$, and therefore that no rational number squares to 2.

Looking Ahead

This chapter has built affine geometry from the ground up: algebraic functors as the ambient arena, affine schemes as the representable ones, closed and open subfunctors classified by (radical) ideals, Zariski descent as the gluing mechanism, and the functorial Nullstellensatz as the dictionary translating between algebra and geometry.

Affine geometry is powerful but local. Many objects of central interest — projective curves, complete varieties, moduli spaces of compact objects — are not affine, and their natural home is the projective setting. The next chapter builds exactly the parallel theory: $\mathbb{N}$ -graded rings replace commutative rings, the Proj construction replaces Spec, saturated homogeneous ideals control scheme-theoretic closed loci while their saturated radicals control underlying closed sets and open complements, and the functorial projective Nullstellensatz replaces its affine predecessor. The entire web of correspondences will have to be reconstructed, but the guiding principle — “everything is a functor, everything is sheaf-theoretic” — carries across without change.

Chapter 2.

Projective Geometry

The goal of this chapter is to introduce projective geometry and transfer certain results from affine geometry to the projective setting. Specifically, we establish the following analogy:

<i>Affine Geometry</i>	<i>Projective Geometry</i>
Affine n -space $\mathbb{A}^n$	Projective n -space $\mathbb{P}^n$
Affine schemes	Projective schemes
Rings	Graded rings
$\text{Spec}: \text{CAlg}^{\text{op}} \rightarrow \text{Fun}(\text{CAlg}, \text{Set})$	$\text{Proj}: (\text{CAlg}^{\mathbb{N}, \text{es}})^{\text{op}} \rightarrow \text{Fun}(\text{CAlg}, \text{Set})$
Classify open/closed subfunctors of $\text{Spec}(A)$ via ideals in A	Classify open/closed subfunctors of $\text{Proj}(A)$ via homogeneous ideals in A

2.1. Projective Spaces

Remark 2.1.1 (Spec vs. Proj). Spec induces an anti-equivalence of categories between rings and affine schemes. By contrast, Proj is *not* fully faithful: many $\mathbb{N}$ -graded rings can yield the same Proj (e.g. a ring and any of its Veronese subrings). The classification of closed subschemes of $\text{Proj}(A)$ is correspondingly subtler: scheme-theoretic closed subschemes are controlled by *saturated homogeneous ideals*. Passing to saturated radical homogeneous ideals instead classifies the underlying closed subsets, and therefore their open complements; it forgets nilpotent thickenings.

Classically, projective space $\mathbb{P}^n(k)$ over a field is the set of lines through the origin in k^{n+1} . The functor-of-points version replaces “lines” by “line quotients”: an R -point of $\mathbb{P}^n$ is a quotient R -line of R^{n+1} . This presentation is manifestly functorial, requires no choice of homogeneous coordinates, and extends the naive picture from fields to arbitrary rings.

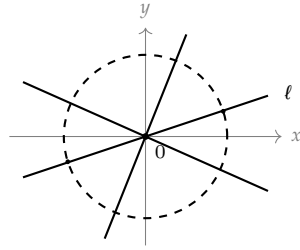
2.1.1. Projective Spaces over Fields

Projective space has two complementary descriptions: classically, as the set of lines through the origin; functorially, as the set of quotient lines. The two perspectives are linear-algebra duals of each other, and the move from “subspaces” to “quotients” is precisely what lets the definition extend from fields to arbitrary commutative rings, where “1-dimensional subspace” is an ill-defined notion but “quotient R -line” remains meaningful. We begin with the classical picture and transition to the functorial one at the end of this section.

Let k be a field and $n \geq -1$. The *projective n -space over k* is the set of all lines through the origin in k^{n+1} :

$$\mathbb{P}^n(k) = \{\text{one-dimensional subspaces of } k^{n+1}\}.$$

Concretely for $n = 1$ and $k = \mathbb{R}$, each point of $\mathbb{P}^1(\mathbb{R})$ corresponds to a line through the origin in the plane; each such line meets the unit circle in exactly two antipodal points, so $\mathbb{P}^1(\mathbb{R}) \simeq S^1/(\pm 1) \simeq S^1$.



Three lines through the origin, each meeting the unit circle in a pair of antipodal points; a single point of $\mathbb{P}^1(\mathbb{R})$ is a full line.

Given a nonzero $(n + 1)$ -tuple $(a_0, \dots, a_n) \in k^{n+1} - \{0\}$, we can consider the one-dimensional subspace it generates in k^{n+1} , denoted $[a_0 : \dots : a_n]$. That is,

$$[a_0 : \dots : a_n] = [b_0 : \dots : b_n] \Leftrightarrow \exists \lambda \in k^\times \text{ such that } \lambda(a_0, \dots, a_n) = (b_0, \dots, b_n).$$

In this way, we can write $\mathbb{P}^n(k)$ as the set of orbits of the free $k^\times$ -action on $k^{n+1} - \{0\}$:

$$(k^{n+1} - \{0\})/k^\times \xrightarrow{\sim} \mathbb{P}^n(k), \quad (a_0, \dots, a_n) \mapsto [a_0 : \dots : a_n].$$

In fact, we can view $\mathbb{P}^n(k)$ as the union of $n + 1$ subsets $U_0, \dots, U_n$, where

$$U_i = \{[a_0 : \dots : a_n] \in \mathbb{P}^n(k) \mid a_i \text{ is invertible}\}.$$

By the definition of $[a_0 : \dots : a_n]$, we can assume $a_i = 1$. Therefore, we can identify U_i via:

$$U_i \simeq \mathbb{A}^n(k), \\ [a_0 : \dots : a_n] \mapsto \left(\frac{a_0}{a_i}, \dots, \frac{\widehat{a_i}}{a_i}, \dots, \frac{a_n}{a_i} \right),$$

with inverse given by $(a_0, \dots, \widehat{a_i}, \dots, a_n) \mapsto [a_0 : \dots : 1 : \dots : a_n]$.

The complement $H_i = \mathbb{P}^n(k) - U_i$ is given by

$$H_i = \{[a_0 : \dots : a_n] \in \mathbb{P}^n(k) \mid a_i = 0\}.$$

According to the above characterization, we have

$$H_i \xrightarrow{\sim} \mathbb{P}^{n-1}(k), \quad [a_0 : \dots : a_n] \mapsto [a_0 : \dots : \widehat{a_i} : \dots : a_n].$$

Therefore, we can summarize:

- $\mathbb{P}^n(k) = \bigcup_{i=0}^n U_i$.
- For each $0 \leq i \leq n$, $\mathbb{P}^n(k) = U_i \sqcup H_i$ (disjoint as sets) with $U_i \simeq \mathbb{A}^n(k) = k^n$ and $H_i \simeq \mathbb{P}^{n-1}(k)$. Iterating, we get a stratification

$$\mathbb{P}^n(k) = k^n \sqcup k^{n-1} \sqcup \dots \sqcup k^1 \sqcup k^0.$$

- Taking $i = 0$, we can view $\mathbb{P}^n(k)$ as the space obtained by compactifying $U_0 = \mathbb{A}^n(k)$ by adding a “point at infinity” for each line through the origin:

$$[0 : a_1 : \cdots : a_n] = \lim_{\lambda \rightarrow \infty} (\lambda a_1, \dots, \lambda a_n),$$

with

$$(\lambda a_1, \dots, \lambda a_n) = [1 : \lambda a_1 : \cdots : \lambda a_n] = [\lambda^{-1} : a_1 : \cdots : a_n].$$

Some additional observations:

- Choosing a basis identifies k^n with its dual $(k^n)^\vee$. Two natural bijections then arise from a d -dimensional subspace $V \subset k^n$:
 - the orthogonal complement $V^\perp \subset k^n$ (an $(n - d)$ -dimensional subspace);
 - the quotient k^n / V (an $(n - d)$ -dimensional vector space).

Composing them gives the bijection

$$\{d\text{-dim subspaces of } k^n\} \xrightarrow{\sim} \{d\text{-dim quotients of } k^n\}, \quad V \mapsto k^n / V^\perp.$$

Specializing to lines: $\mathbb{P}^n(k)$ admits two equivalent descriptions, as either the set of 1-dimensional subspaces of k^{n+1} or the set of 1-dimensional quotients of k^{n+1} . Both send a representative $(a_0, \dots, a_n) \in k^{n+1} - \{0\}$ to the same point $[a_0 : \cdots : a_n]$.

The quotient-line description is the right one to generalize to arbitrary rings, since taking duals is well-behaved only for projective modules.

2.1.2. Vector Spaces and Lines over Rings

Proposition 2.1.2 (Characterization of Vector Spaces). *Let R be a ring and $M \in \text{Mod}_R$ a module. The following conditions are equivalent:*

- (i) M is a finitely generated projective module.
- (ii) M is a finitely presented flat module.
- (iii) There exists $n \in \mathbb{N}$ such that M is a direct summand of R^n .
- (iv) There exist $f_1, \dots, f_n \in R$ such that $(f_1, \dots, f_n) = R$ and each M_{f_i} is a finitely generated free R_{f_i} -module.
- (v) M is a dualizable module, i.e., there exists $M^\vee \in \text{Mod}_R$ and linear maps $\text{ev}: M^\vee \otimes M \rightarrow R$ and $\text{coev}: R \rightarrow M \otimes M^\vee$ such that the following diagrams commute:

$$\begin{array}{ccc} & M \otimes M^\vee \otimes M & \\ \text{coev} \otimes \text{id} \nearrow & & \searrow \text{id} \otimes \text{ev} \\ M & \xlongequal{\quad\quad\quad} & M \end{array}$$

and

$$\begin{array}{ccc} & M^\vee \otimes M \otimes M^\vee & \\ \text{id} \otimes \text{coev} \nearrow & & \searrow \text{ev} \otimes \text{id} \\ M^\vee & \xlongequal{\quad\quad\quad} & M^\vee \end{array}$$

Remark 2.1.3. In practice, condition (4) is most commonly used, as it is closely related to Zariski descent.

Proof of Proposition 2.1.2.

(1 $\Leftrightarrow$ 3 $\Rightarrow$ 2) Let M be a finitely generated projective module. We must show that there exists $n \in \mathbb{N}$ such that M is a retract of R^n .

Since M is finitely generated, there exists $n \in \mathbb{N}$ and a surjection $\pi: R^n \twoheadrightarrow M$. We now construct a section.

Since M is projective, $\text{Hom}_R(M, -)$ is an exact functor, hence preserves surjections. Therefore:

$$\text{Hom}_R(M, R^n) \twoheadrightarrow \text{Hom}_R(M, M).$$

Taking a preimage of id_M gives the desired section, so M is a retract of R^n .

Conversely, projective modules are closed under retracts, and R^n is obviously projective; this gives (3) $\Rightarrow$ (1). The implication (1) $\Rightarrow$ (2) is immediate as well: a retract of R^n is finitely presented and flat.

(2 $\Rightarrow$ 4) We first establish a local-freeness claim.

Claim 2.1.4. *If M is a finitely presented flat module, then for any maximal ideal $\mathfrak{m} \subset R$, $M_{\mathfrak{m}}$ is a free $R_{\mathfrak{m}}$ -module.*

Proof of claim. Take a maximal ideal $\mathfrak{m} \subset R$. We have $M/\mathfrak{m}M$ is an $R/\mathfrak{m}$ -module. Since $\mathfrak{m}$ is maximal, $R/\mathfrak{m}$ is a field. Therefore there exists $n \in \mathbb{N}$ such that $(R/\mathfrak{m})^n \simeq M/\mathfrak{m}M$.

Consider the commutative diagram:

$$\begin{array}{ccc} (R/\mathfrak{m})^n & \xrightarrow{\sim} & M/\mathfrak{m}M \\ \uparrow & & \uparrow \\ R_{\mathfrak{m}}^n & \xrightarrow{f} & M_{\mathfrak{m}} \end{array}$$

Surjectivity is immediate from the construction; we turn to injectivity.

By flatness of M , we have $\text{Tor}_1^R(R/\mathfrak{m}, M) = 0$. Therefore, the sequence

$$0 \rightarrow \ker(f) \otimes_R R/\mathfrak{m} \rightarrow (R/\mathfrak{m})^n \rightarrow M/\mathfrak{m}M \rightarrow 0$$

is exact.

Since M is finitely presented over R , $M_{\mathfrak{m}}$ is finitely presented over $R_{\mathfrak{m}}$. Consider the exact sequence:

$$R_{\mathfrak{m}}^m \rightarrow R_{\mathfrak{m}}^n \rightarrow M_{\mathfrak{m}} \rightarrow 0.$$

Localizing gives:

$$R_{\mathfrak{m}}^m \rightarrow R_{\mathfrak{m}}^n \rightarrow M_{\mathfrak{m}} \rightarrow 0,$$

which is also exact. Thus $\ker(f)$ is a finitely generated $R_{\mathfrak{m}}$ -module.

Therefore:

$$\ker(f) \otimes_{R_{\mathfrak{m}}} (R_{\mathfrak{m}}/\mathfrak{m}R_{\mathfrak{m}}) \simeq \ker(f) \otimes_R R/\mathfrak{m} = 0.$$

By Nakayama's lemma over $R_{\mathfrak{m}}$, we have $\ker(f) = 0$. □

Now we use the claim to prove (4). Recall that $M_{\mathfrak{m}} = M[(R - \mathfrak{m})^{-1}]$, so:

$$M_{\mathfrak{m}} = \text{colim}_{f \notin \mathfrak{m}} M_f.$$

Since this is a filtered colimit, a basis of $M_{\mathfrak{m}}$ lifts to M_f for some $f \notin \mathfrak{m}$. As $\mathfrak{m}$ ranges over all maximal ideals, the chosen elements f collectively avoid every maximal ideal, hence generate the unit ideal of R ; finitely many of them already suffice (since M is finitely presented). This produces the desired covering family.

- (4 $\Rightarrow$ 1) Use that finite generation and projectivity are Zariski local properties (see Proposition 1.3.9), together with the fact that free modules are automatically projective.
- (4 $\Rightarrow$ 5) Recall that M is dualizable iff the canonical map $N \otimes_R M^\vee \rightarrow \text{Hom}_R(M, N)$ is an isomorphism for all N . This is a Zariski-local condition on R , so by hypothesis we may pass to each R_{f_i} , where M_{f_i} is free of finite rank — and free modules are visibly dualizable.
- (5 $\Rightarrow$ 1) Use that $\text{Hom}_R(M, -)$ preserves colimits when M is dualizable.

□

The equivalent characterizations above justify treating dualizable R -modules as the algebro-geometric analogue of finite-dimensional vector spaces.

Definition 2.1.5 (Vector Spaces over Rings). Let R be a ring. We say that an R -module M is a *vector space* if M satisfies the equivalent conditions in Proposition 2.1.2. We denote by

$$\text{Vect}_R \subset \text{Mod}_R$$

the full subcategory spanned by all R -vector spaces.

Remark 2.1.6. Note that when $R = k$ is a field, the resulting vector spaces are precisely the *finite-dimensional* vector spaces over k .

Over a field, all bases of a vector space have the same cardinality, called its dimension or rank. To extend this notion over a general ring R , we test against R -fields — fields equipped with a ring homomorphism from R .

Definition 2.1.7 (Rank). Let R be a ring, M an R -module, and κ an R -field. The *rank* $\text{rk}_\kappa(M)$ of M is the rank of $M \otimes_R \kappa$.

We say that M has *constant rank* if there exists r such that for every R -field κ , we have $\text{rk}_\kappa(M) = r$.

Remark 2.1.8. We have the following relations:

- $\text{rk}_\kappa(M \oplus N) = \text{rk}_\kappa(M) + \text{rk}_\kappa(N)$.
- $\text{rk}_\kappa(M \otimes N) = \text{rk}_\kappa(M) \cdot \text{rk}_\kappa(N)$.
- $\text{rk}_\kappa(R^n) = n$.

Proposition 2.1.9 (Properties of Rank). Let R be a ring.

- (i) For an R -vector space V , we have $V = 0$ if and only if V has constant rank 0.
- (ii) For a short exact sequence in Vect_R :

$$0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0,$$

and any R -field κ , we have $\text{rk}_\kappa(V) = \text{rk}_\kappa(U) + \text{rk}_\kappa(W)$.

Proof. (i) The direction $V = 0 \Rightarrow \text{rk}_\kappa(V) = 0$ is obvious. Now take V with constant rank 0. By Proposition 2.1.2, choose $f_1, \dots, f_n$ such that $(f_1, \dots, f_n) = R$. Then for any κ :

$$\text{rk}_\kappa(V_{f_i}) = 0,$$

which means $V_{f_i} = 0$. By Zariski descent, $V = 0$.

(ii) By Proposition 2.1.2, vector spaces are projective modules. Therefore the short exact sequence

$$0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$$

splits, reducing to Remark 2.1.8. □

Remark 2.1.10. Note that having constant rank 0 implies $V = 0$ only for R -vector spaces. For general modules, there are counterexamples. For instance, take $R = \mathbb{Z}$; the $\mathbb{Z}$ -module $\mathbb{Q}/\mathbb{Z}$ has rank 0.

Recall that the *residue field* at a prime ideal $\mathfrak{p} \subset R$ is the R -field

$$\kappa(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}} \simeq \text{Frac}(R/\mathfrak{p}).$$

The residue fields are precisely the initial objects of the connected components of Field_R : for any R -field κ , the kernel $\mathfrak{p} := \ker(R \rightarrow \kappa)$ is prime, and the map $R \rightarrow \kappa$ factors uniquely through the residue field as

$$R \rightarrow \kappa(\mathfrak{p}) \hookrightarrow \kappa.$$

In other words, Field_R decomposes as

$$\text{Field}_R = \coprod_{\mathfrak{p} \subseteq R} \text{Field}_{\kappa(\mathfrak{p})}.$$

Using the fact that scalar extension between fields preserves dimension, we deduce that for κ lying in $\text{Field}_{\kappa(\mathfrak{p})}$, we have $\text{rk}_{\kappa}(M) = \text{rk}_{\kappa(\mathfrak{p})}(M)$.

Next, we discuss what “one-dimensional” vector spaces are. Rather than thinking of “one-dimensional” in terms of dimension, we prefer to think of it as something invertible.

Proposition 2.1.11 (Characterization of Lines). *Let R be a ring. For any R -module L , the following conditions are equivalent:*

- (i) L is a vector space with constant rank 1.
- (ii) L is an invertible module, i.e., there exists an R -module $L^{\vee}$ such that $L \otimes_R L^{\vee} \simeq R$.

Remark 2.1.12. An invertible module is automatically a vector space — as is its inverse.

Proof of Proposition 2.1.11(1 $\Rightarrow$ 2) Let L be an R -vector space. For $f \in R$, we show that $(L^{\vee})_f = (L_f)^{\vee}$.

By Proposition 2.1.2, L is finitely presented. Therefore we have a right exact sequence:

$$R^m \rightarrow R^n \rightarrow L \rightarrow 0.$$

Applying $\text{Hom}_R(-, R)$ gives a left exact sequence:

$$0 \rightarrow \text{Hom}_R(L, R) \rightarrow \text{Hom}_R(R^n, R) \rightarrow \text{Hom}_R(R^m, R).$$

Since R_f is flat over R , we obtain a left exact sequence:

$$0 \rightarrow \text{Hom}_R(L, R)_f \rightarrow \text{Hom}_R(R^n, R)_f \rightarrow \text{Hom}_R(R^m, R)_f.$$

Localizing the right exact sequence first and then applying $\text{Hom}_{R_f}(-, R_f)$ gives:

$$0 \rightarrow \text{Hom}_{R_f}(L_f, R_f) \rightarrow \text{Hom}_{R_f}(R_f^n, R_f) \rightarrow \text{Hom}_{R_f}(R_f^m, R_f).$$

Since $\text{Hom}_R(R^n, R) \simeq R^n$ for any ring R , using the five lemma gives the isomorphism:

$$\text{Hom}_R(L, R)_f \simeq \text{Hom}_{R_f}(L_f, R_f).$$

Since $L^\vee = \text{Hom}_R(L, R)$, the localization $(L^\vee)_f \simeq (L_f)^\vee$ commutes with $(-)^\vee$. By Proposition 2.1.2(4), choose $f_1, \dots, f_n$ with $(f_1, \dots, f_n) = R$ and each L_{f_i} free of rank 1 over R_{f_i} . Then $L_{f_i} \otimes_{R_{f_i}} L_{f_i}^\vee \simeq R_{f_i}$, and Zariski descent (Proposition 2.1.2) glues these isomorphisms to $L \otimes_R L^\vee \simeq R$.

(2 $\Rightarrow$ 1) Invertible modules are automatically dualizable, so it remains to check the rank. By multiplicativity of rank,

$$\text{rk}_K(L) \cdot \text{rk}_K(L^\vee) = \text{rk}_K(L \otimes_R L^\vee) = \text{rk}_K(R) = 1.$$

Since ranks are non-negative integers, both factors equal 1.

□

Therefore, we establish the principle: in R -vector spaces, “one-dimensional” = invertible.

Definition 2.1.13 (Lines). Let R be a ring. A *line over R* is an invertible R -module M . We denote by

$$\text{Line}_R \subset \text{Vect}_R$$

the full subcategory spanned by all R -lines.

Lines over R need not be trivial in general:

Example 2.1.14 (Non-trivial Line). Let $R = \mathbb{Z}[\sqrt{-5}] \subseteq \mathbb{Q}(\sqrt{-5}) = \mathbb{Q} \oplus \mathbb{Q}\sqrt{-5} \subseteq \mathbb{C}$. This is a classical example of a non-UFD, since:

$$(1 + \sqrt{-5})(1 - \sqrt{-5}) = 6 = 2 \cdot 3.$$

Consider the ideal:

$$I = (2, 1 + \sqrt{-5}) = (2, 1 - \sqrt{-5}) \subseteq R.$$

We can construct an R -linear map:

$$I \otimes_R I \rightarrow (2) \subseteq R, \quad (i, j) \mapsto i \cdot j.$$

Since (2) is principal, $(2) \simeq R$. We claim the above map is an isomorphism. Consider the map:

$$(2) \rightarrow I \otimes_R I, \quad 2 \mapsto (1 + \sqrt{-5}) \otimes (1 - \sqrt{-5}) - 2 \otimes 2.$$

First, the composition $(2) \rightarrow (2)$ is the identity. For the composition $I \otimes_R I \rightarrow I \otimes_R I$, write $I \otimes_R I = (2, 1 + \sqrt{-5}) \otimes (2, 1 - \sqrt{-5})$ and check on the four generators:

$$2 \otimes 2, \quad 2 \otimes (1 - \sqrt{-5}), \quad (1 + \sqrt{-5}) \otimes 2, \quad (1 + \sqrt{-5}) \otimes (1 - \sqrt{-5}).$$

To show I is non-trivial, we show it is not principal. If $a + b\sqrt{-5}$ divides 2, then so does its conjugate $a - b\sqrt{-5}$. Thus $a^2 + 5b^2$ divides 4. The only possibilities are $b = 0$ and $a = \pm 2$ or ± 1 . None of these combinations generate I .

2.1.3. Maps Between Vector Spaces

Next, we discuss morphisms between vector spaces. We want to understand what injectivity and surjectivity mean for maps between vector spaces.

In fact, for vector spaces, surjectivity remains surjectivity. Let M and N be R -vector spaces, and $f: M \rightarrow N$ a surjection. We have an exact sequence:

$$0 \rightarrow \ker(f) \rightarrow M \rightarrow N \rightarrow 0.$$

Since N is a vector space, hence projective, the sequence splits. Thus $\ker(f)$ is a retract of the projective module M , hence also projective, thus a vector space.

However, if f is injective, $\operatorname{coker}(f)$ need not be a vector space. For example, $\mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$, and $\mathbb{Z}/2\mathbb{Z}$ is not a vector space.

Concretely, we need a refined notion of injectivity dual to surjectivity — one that ensures the cokernel remains a vector space.

Definition 2.1.15 (Universal Injectivity). Let R be a ring, and M, N be R -modules. We say that an R -linear map $\varphi: M \rightarrow N$ is *universally injective* if for any ring homomorphism $R \rightarrow S$, the map $\varphi \otimes_R S$ is injective.

Remark 2.1.16. The prefix “universally” generally signals that a property is preserved under arbitrary base change.

Universal injectivity is the right notion of injectivity in the context of vector spaces, as the propositions below confirm.

Remark 2.1.17. • If $f: M \hookrightarrow N$ is universally injective, it is automatically injective (apply $\otimes_R R$). In fact, for any R -module P , $f \otimes_R P$ is injective. This follows by considering the symmetric algebra $\operatorname{Sym}_R(P)$ and the canonical map $R \rightarrow \operatorname{Sym}_R(P)$.

- Due to right exactness of tensor products, there is no need to consider “universal surjectivity.”

Example 2.1.18 (Universal Injectivity Examples). (i) Any map with a retraction is universally injective.

(ii) Maps without retractions can also be universally injective, e.g., $\bigoplus_{\mathbb{N}} \mathbb{Z} \hookrightarrow \prod_{\mathbb{N}} \mathbb{Z}$.

(iii) Let $f \in R$. The map $R \xrightarrow{f} R$ is injective if and only if f is not a zero divisor. Universal injectivity requires f to be a unit (invertible element), since you can always base change to $R/(f)$ to make f a zero divisor.

Finite presentation, finite generation, and projectivity are all Zariski-local conditions, so being an R -vector space (resp. an R -line) is itself Zariski-local. Since exactness of module sequences is also Zariski-local, the same holds for universal injectivity.

Definition 2.1.19 (Subspaces and Quotient Spaces). Let R be a ring and M an R -module.

- We say that a module N is a *subspace* of M if $N \subset M$ is a submodule, N is a vector space, and the embedding $N \hookrightarrow M$ is universally injective.
- We say that a module N is a *quotient space* of M if N is a quotient module of M and N is a vector space.

The following proposition characterizes subspaces and quotient spaces.

Proposition 2.1.20 (Characterization of Subspaces). *Let R be a ring, and U, V be R -vector spaces. For an R -linear map $f : U \rightarrow V$, the following are equivalent:*

- (i) f is universally injective.
- (ii) For any R -field κ , the base change $f \otimes_R \kappa$ is injective.
- (iii) f is injective and $\text{coker}(f)$ is a vector space.
- (iv) f has a retraction.
- (v) The dual map $f^\vee : V^\vee \rightarrow U^\vee$ is surjective.

Proof. • First, by Example 2.1.18, we immediately get $(4 \Rightarrow 1)$.

- By definition, $(1 \Rightarrow 2)$.
- Since $\text{coker}(f)$ is a vector space, the exact sequence

$$0 \rightarrow U \rightarrow V \rightarrow \text{coker}(f) \rightarrow 0$$

splits, so f has a retraction, giving $(3 \Rightarrow 4)$.

- If f has a retraction, this means the exact sequence

$$0 \rightarrow U \rightarrow V \rightarrow \text{coker}(f) \rightarrow 0$$

splits, so $\text{coker}(f)$ is also a retract of V , hence $\text{coker}(f)$ is a vector space, proving $(4 \Rightarrow 3)$.

- If the dual map $f^\vee : V^\vee \rightarrow U^\vee$ is surjective, then for any R -module M , $-\otimes_R M$ preserves surjections, so $\text{Hom}_R(V, N) \twoheadrightarrow \text{Hom}_R(U, N)$ is surjective. Taking $N = U$ and a preimage of $\text{id}_U \in \text{Hom}_R(U, U)$ gives the retraction, so $(5 \Rightarrow 4)$.
- For any R -field κ , $f \otimes_R \kappa$ is automatically a κ -linear map, so it is injective if and only if

$$(f \otimes_R \kappa)^\vee = f^\vee \otimes_R \kappa$$

is surjective. For a maximal ideal $\mathfrak{m}$, take $\kappa(\mathfrak{m}) = R_{\mathfrak{m}}/\mathfrak{m}R_{\mathfrak{m}}$. Then:

$$f^\vee \otimes_R \kappa(\mathfrak{m}) = f_{\mathfrak{m}}^\vee \otimes_{R_{\mathfrak{m}}} (R_{\mathfrak{m}}/\mathfrak{m}R_{\mathfrak{m}}) = V_{\mathfrak{m}}^\vee/\mathfrak{m}V_{\mathfrak{m}}^\vee \rightarrow U_{\mathfrak{m}}^\vee/\mathfrak{m}U_{\mathfrak{m}}^\vee$$

is surjective. Therefore $\text{coker}(f_{\mathfrak{m}}^\vee) \otimes_{R_{\mathfrak{m}}} R_{\mathfrak{m}}/\mathfrak{m}R_{\mathfrak{m}} = 0$. Since $U_{\mathfrak{m}}^\vee$ is finitely generated over $R_{\mathfrak{m}}$ (hence so is $\text{coker}(f_{\mathfrak{m}}^\vee)$), by Nakayama's lemma, $f_{\mathfrak{m}}^\vee$ is surjective.

Since localizing at $\mathfrak{m}$ is the filtered colimit of localizations at $g \notin \mathfrak{m}$, there exists $g \notin \mathfrak{m}$ such that $f^\vee \otimes_R R_g$ is surjective (more precisely, one should take stalks in the module sheaf). Using that surjectivity is a Zariski local property gives $(2 \Rightarrow 5)$.

□

Dually, we can give the quotient space version.

Proposition 2.1.21 (Characterization of Quotient Spaces). *Let R be a ring, and U, V be R -vector spaces. For an R -linear map $f : U \rightarrow V$, the following are equivalent:*

- (i) f is surjective.
- (ii) For any R -field κ , the base change $f \otimes_R \kappa$ is surjective.
- (iii) f is surjective and $\ker(f)$ is a vector space.
- (iv) f has a section.
- (v) The dual map $f^\vee : V^\vee \rightarrow U^\vee$ is universally injective.

We immediately obtain the following corollaries.

Corollary 2.1.22 (Isomorphisms Detected on Fields). *Let R be a ring, and V, W be R -vector spaces. Then $f : V \rightarrow W$ is an isomorphism if and only if for any R -field κ , $f \otimes_R \kappa$ is an isomorphism.*

Proof. Note that universal injectivity + surjectivity = isomorphism. $\square$

Corollary 2.1.23 (Pigeonhole for Vector Spaces). *Let R be a ring, and V, W be R -vector spaces with the same rank. For an R -linear map $f : V \rightarrow W$, the following are equivalent:*

- (i) f is an isomorphism.
- (ii) f is surjective.
- (iii) f is universally injective.

Proof. Since V and W have the same rank, for all R -fields κ , we have $\text{rk}_\kappa(V) = \text{rk}_\kappa(W)$. If $f \otimes_R \kappa$ is injective/surjective, by the pigeonhole principle, f is an isomorphism. $\square$

Corollary 2.1.24 (Duality Correspondence). *Let R be a ring and V an R -vector space. Then dualization induces bijections*

$$\begin{aligned} \{\text{subspaces of } V\} &\xrightarrow{\sim} \{\text{quotient spaces of } V^\vee\}, & (U \hookrightarrow V) &\mapsto (V^\vee \twoheadrightarrow U^\vee), \\ \{\text{quotient spaces of } V\} &\xrightarrow{\sim} \{\text{subspaces of } V^\vee\}, & (V \twoheadrightarrow W) &\mapsto (W^\vee \hookrightarrow V^\vee). \end{aligned}$$

Proof. By Definition 2.1.19, a subspace of V is a universally injective map $U \hookrightarrow V$ from an R -vector space, and a quotient space is a surjection $V \twoheadrightarrow W$ onto an R -vector space. Proposition 2.1.20(v) and Proposition 2.1.21(v) tell us that dualization swaps these two classes:

$$\text{universally injective } f \iff f^\vee \text{ surjective}, \quad f \text{ surjective} \iff f^\vee \text{ universally injective}.$$

Vector spaces are reflexive — by Proposition 2.1.2, the canonical map $V \rightarrow V^{\vee\vee}$ is an isomorphism — so applying $(-)^\vee$ twice recovers the original inclusion or surjection. The two assignments are therefore mutual inverses. $\square$

2.1.4. Projective Space

Now we can define projective space.

Definition 2.1.25 (Projective Space). *Let k be a ring and I a set. The *projective I -space over k* is the algebraic k -functor:*

$$\mathbb{P}_k^I : \text{CAlg}_k \rightarrow \text{Set}, \quad R \mapsto \{\text{quotient lines of } R^{(I)}\}.$$

When $k = \mathbb{Z}$, we simply write $\mathbb{P}^I$. For $n \geq -1$, the *projective n -space over k* is $\mathbb{P}_k^n = \mathbb{P}_k^{\{0, \dots, n\}}$. When $n = 1$, we also call it the *projective line*; when $n = 2$, the *projective plane*.

Remark 2.1.26. (i) $\mathbb{P}_k^{-1} = \mathbb{P}_k^\emptyset$ is the empty k -scheme $\emptyset$ (note that it is not the initial object of $\text{Fun}(\text{CAlg}_k, \text{Set})$).

(ii) $\mathbb{P}_k^0 = \text{Spec}(k)$ is the terminal k -scheme.

(iii) If I is finite, by Corollary 2.1.24, we have:

$$\mathbb{P}_k^I(R) \simeq \{\text{subspace lines of } R^I\}.$$

However, when I is infinite, $R^{(I)}$ is no longer a vector space, so the definition must use quotient lines.

Notation 2.1.27 (Projective Coordinates). Let R be a ring.

- Let L be a line over R .
- Let $(a_0, \dots, a_n) \in L^{n+1}$ be a family of elements in L that generate the module.

This is equivalent to saying that the R -linear map induced by the a_i :

$$R^{n+1} \rightarrow L, \quad e_i \mapsto a_i$$

is surjective. The *coimage* defines a *quotient line* of R^{n+1} , which we denote:

$$[a_0 : \dots : a_n] \in \mathbb{P}^n(R).$$

For another line M and generating family $(b_0, \dots, b_n) \in M^{n+1}$, we define:

$$[a_0 : \dots : a_n] = [b_0 : \dots : b_n]$$

if and only if there exists a unique isomorphism $\varphi: L \xrightarrow{\sim} M$ such that $\varphi(a_i) = b_i$ for all $0 \leq i \leq n$.

2.1.5. Relationship Between Affine and Projective Spaces

We now relate affine I -space and projective I -space. The bridge between them is the *punctured affine I -space* $\mathbb{A}_k^I - 0$.

Recall that the *punctured affine I -space* is the functor defined by:

$$(\mathbb{A}_k^I - 0): R \mapsto \{a \in R^I \mid (a) = R\}.$$

Here (a) denotes the ideal in R generated by all components of $a = (a_i)_i \in R^I$.

Clearly $\mathbb{A}_k^I - 0$ is a subfunctor of $\mathbb{A}_k^I$. By duality, an I -tuple in R generating the unit ideal is the same datum as a surjection $R^{(I)} \twoheadrightarrow R$, and remembering only the kernel produces a quotient line of $R^{(I)}$. This gives a canonical morphism

$$\mathbb{A}_k^I - 0 \longrightarrow \mathbb{P}_k^I.$$

In summary, we have the following relationship diagram:

$$\mathbb{A}_k^I \hookleftarrow (\mathbb{A}_k^I - 0) \rightarrow \mathbb{P}_k^I.$$

In particular, for $n \geq -1$, we have:

$$\mathbb{A}_k^{n+1} \hookleftarrow (\mathbb{A}_k^{n+1} - 0) \rightarrow \mathbb{P}_k^n.$$

Proposition 2.1.28 (Trivial Quotient Lines). Let I be a set and R a ring. Then the morphism $\mathbb{A}^I - 0 \rightarrow \mathbb{P}^I$ induces an injection:

$$(\mathbb{A}^I - 0)(R)/R^\times \hookrightarrow \mathbb{P}^I(R).$$

Here $R^\times$ acts on $(\mathbb{A}^I - 0)(R)$ by scalar multiplication. The image of this map is precisely the trivial quotient lines of $R^{(I)}$ (i.e., quotient lines isomorphic to R).

In particular, if all lines over R are trivial (e.g., R is a local ring or a principal ideal domain), then this map is a bijection.

Proof. Elements of $(\mathbb{A}^I - 0)(R)$ correspond to surjections $u: R^{(I)} \rightarrow R$, and the morphism $\mathbb{A}^I - 0 \rightarrow \mathbb{P}^I$ sends such a u to its quotient line $[u] \in \mathbb{P}^I(R)$ — manifestly the trivial line R .

Injectivity: We need to characterize when two surjections $a, b: R^{(I)} \rightarrow R$ define the same point in $\mathbb{P}^I(R)$. By definition, $[a] = [b]$ if and only if there exists an isomorphism $\varphi: R \xrightarrow{\sim} R$ making the following diagram commute:

$$\begin{array}{ccc} R^{(I)} & \xrightarrow{a} & R \\ & \searrow b & \swarrow \varphi \\ & R & \end{array}$$

Note that $\text{Aut}_R(R) \simeq R^\times$, so φ must be multiplication by some unit $\lambda \in R^\times$. Therefore, $[a] = [b]$ if and only if there exists $\lambda \in R^\times$ such that $b = \lambda a$ in $(\mathbb{A}^I - 0)(R)$. This is precisely the equivalence relation in $(\mathbb{A}^I - 0)(R)/R^\times$.

Image: Clearly, any element from $(\mathbb{A}^I - 0)(R)$ gives a quotient line with target R , i.e., a trivial quotient line. Conversely, if some quotient line L of $R^{(I)}$ is trivial, then there exists an isomorphism $L \simeq R$. Composing $R^{(I)} \rightarrow L \xrightarrow{\sim} R$ gives an element of $(\mathbb{A}^I - 0)(R)$. $\square$

Remark 2.1.29 ($\mathbb{G}_m$ -Action and Quotient Functors). The scalar action of $R^\times$ on $(\mathbb{A}^I - 0)(R)$ is functorial in R , so it upgrades to an action of the multiplicative group scheme $\mathbb{G}_m$ on $\mathbb{A}^I - 0$. The previous proposition then promotes the comparison map to an injection of algebraic functors

$$(\mathbb{A}^I - 0)/\mathbb{G}_m \hookrightarrow \mathbb{P}^I,$$

which becomes an isomorphism on $\text{Spec}(R)$ whenever every line over R is trivial.

By Proposition 2.1.2, every quotient line L over R becomes free after passing to a Zariski cover $\{R \rightarrow R_{f_i}\}_{i=1}^n$ with $(f_1, \dots, f_n) = R$. Consequently, $(\mathbb{A}^I - 0)/\mathbb{G}_m \hookrightarrow \mathbb{P}^I$ is surjective Zariski-locally, and after sheafification we obtain an isomorphism

$$(\mathbb{A}^I - 0)/\mathbb{G}_m \xrightarrow{\sim} \mathbb{P}^I$$

of Zariski sheaves.

Remark 2.1.30 (Functoriality and Topological Property Differences). (i) *Functoriality.* The assignment $I \mapsto \mathbb{A}^I$ upgrades to a contravariant functor $\mathbb{A}^-: \text{Set}^{\text{op}} \rightarrow \text{Fun}(\text{CAlg}, \text{Set})$, but $I \mapsto (\mathbb{A}^I - 0)$ and $I \mapsto \mathbb{P}^I$ do not extend that far: they are contravariant only on the subcategory Set^{surj} of sets and surjections. Indeed, pulling back a quotient line of $R^{(I)}$ to a quotient line of $R^{(J)}$ requires a surjection $J \rightarrow I$ to begin with.

(ii) *Quasi-compactness.* Topologically, $\mathbb{A}^I$ and $\mathbb{P}^I$ behave very differently: $\mathbb{A}^I$ is quasi-compact for every set I , even infinite, whereas $\mathbb{P}^I$ (and $\mathbb{A}^I - 0$) is quasi-compact if and only if I is finite. Quasi-compactness and quasi-separatedness are the central finiteness conditions in algebraic geometry — the six-functor formalism for étale cohomology, for instance, is typically developed for qcqs schemes — so most classical results in projective geometry are stated only in the finite case.

2.2. Graded Rings

Just as affine schemes are represented by rings, projective schemes will be represented by $\mathbb{N}$ -graded rings. This section gathers the basic vocabulary — homogeneous polynomials, Veronese subrings, localizations of graded rings, and eventually surjective maps — all tools we will need to define Proj properly.

Definition 2.2.1 (Homogeneous Polynomials). Let k be a ring, I a set, and $d \in \mathbb{N}$. We say that a polynomial $f \in k[x_i \mid i \in I]$ is a *homogeneous polynomial of degree d* if it is a k -linear combination of monomials of the form $\prod_{i \in I} x_i^{n_i}$ with $\sum_{i \in I} n_i = d$. We denote by $k[x_i \mid i \in I]_d \subset k[x_i \mid i \in I]$ the k -submodule spanned by all homogeneous polynomials of degree d .

Every polynomial decomposes uniquely as a sum of homogeneous components,

$$k[x_i \mid i \in I] = \bigoplus_{d \in \mathbb{N}} k[x_i \mid i \in I]_d,$$

and multiplication respects degrees: a degree- d polynomial times a degree- e polynomial is homogeneous of degree $d + e$.

This property motivates the concept of graded rings.

Definition 2.2.2 (Graded Rings). Let $(\Gamma, +, 0)$ be a commutative monoid (e.g., $\mathbb{N}$ or $\mathbb{Z}$). A Γ -graded ring is a ring A together with, for each $\gamma \in \Gamma$, a subgroup $A_\gamma \subset A$ such that:

- A decomposes as a direct sum: $\bigoplus_{\gamma \in \Gamma} A_\gamma \xrightarrow{\sim} A$.
- $1 \in A_0$ and for $\gamma, \delta \in \Gamma$, we have $A_\gamma A_\delta \subset A_{\gamma+\delta}$.

For $\gamma \in \Gamma$, we call elements of A_γ *homogeneous elements of degree γ* . For a ring k , a Γ -graded k -algebra is a Γ -graded ring A with a ring homomorphism $k \rightarrow A$ whose image lies in A_0 .

Definition 2.2.3 (Graded Modules). Let A be a Γ -graded ring. A Γ -graded A -module is an A -module M together with, for each $\gamma \in \Gamma$, a subgroup $M_\gamma \subset M$ such that:

- M decomposes as a direct sum: $\bigoplus_{\gamma \in \Gamma} M_\gamma \xrightarrow{\sim} M$.
- For all $\gamma, \delta \in \Gamma$, we have $A_\gamma M_\delta \subset M_{\gamma+\delta}$.

An ideal $I \subset A$ is *homogeneous* if $I = \bigoplus_{\gamma \in \Gamma} (I \cap A_\gamma)$ — equivalently, if I is a graded A -submodule of A .

Remark 2.2.4 (Properties of Graded Rings and Modules). Let A be a Γ -graded ring and M a Γ -graded A -module.

- A_0 is a subring of A , each A_γ is an A_0 -module, and each M_γ is also an A_0 -module.
- Let $\Gamma \rightarrow \Delta$ be a homomorphism of commutative monoids. Then A can be made into a Δ -graded ring by: for $\delta \in \Delta$, defining

$$A_\delta := \bigoplus_{\gamma \mapsto \delta} A_\gamma.$$

Similarly, M can be made into a Δ -graded A -module in the same way. For example, an $\mathbb{N}$ -graded ring A can be viewed as a $\mathbb{Z}$ -graded ring by taking $A_n = 0$ for $n < 0$.

- An ideal $I \subset A$ is homogeneous if and only if it is generated by homogeneous elements.
- For a homogeneous ideal $I \subset A$, the quotient ring A/I inherits a unique Γ -graded ring structure making $A \rightarrow A/I$ a graded map. Moreover, the A/I -module M/IM has a unique Γ -graded structure making $M \rightarrow M/IM$ a graded map.
- Let $S \subset A$ be the set of homogeneous elements whose degrees are invertible in Γ . Then the localization $A[S^{-1}]$ inherits a unique Γ -graded structure making $A \rightarrow A[S^{-1}]$ a graded map. Moreover, the $A[S^{-1}]$ -module $M[S^{-1}]$ inherits a unique Γ -graded structure making $M \rightarrow M[S^{-1}]$ a graded map.

Example 2.2.5 (Graded Rings). Let k be a ring.

- As mentioned earlier, for any set I , the polynomial ring $k[x_i \mid i \in I]$ is a $\mathbb{Z}$ -graded ring.

- The Laurent polynomial ring $k[x^{\pm 1}]$ is a $\mathbb{Z}$ -graded ring.
- Let M be a k -module. Then the symmetric algebra $\text{Sym}_k(M)$ has a canonical $\mathbb{N}$ -graded structure:

$$\text{Sym}_k(M) = \bigoplus_{d \in \mathbb{N}} \text{Sym}_k^d(M),$$

where $\text{Sym}_k^d(M)$ is the d -th symmetric power of M .

- Let L be a line over k . Then $\bigoplus_{d \in \mathbb{Z}} L^{\otimes d}$ is automatically a $\mathbb{Z}$ -graded ring. In fact, we have $L^{\otimes d} = \text{Sym}_k^d(L)$ and $L^{\otimes -1} = L^\vee$.

Notation 2.2.6 (Shifts, Veronese subrings, and homogeneous localization). Let A be a Γ -graded ring, M a Γ -graded A -module, and $\gamma \in \Gamma$.

- (i) We denote by $M(\gamma)$ the Γ -graded A -module defined as follows: its underlying A -module is M , and for $\delta \in \Gamma$, its graded component is:

$$(M(\gamma))_\delta = M_{\gamma+\delta}.$$

The assignment $M \mapsto M(\gamma)$ defines an endofunctor on graded modules. When $\gamma \in \Gamma$ is invertible, it is an equivalence of categories — the *shift by γ* .

- (ii) We denote by $A^{(\gamma)}$ the $\mathbb{N}$ -graded ring $\bigoplus_{n \in \mathbb{N}} A_{n\gamma}$, where $(A^{(\gamma)})_n = A_{n\gamma}$. Similarly, we define $M^{(\gamma)}$ to be the $\mathbb{N}$ -graded $A^{(\gamma)}$ -module.
- (iii) Suppose $\gamma \in \Gamma$ is invertible and $f \in A_\gamma$. By Remark 2.2.4(5), the localization A_f inherits a Γ -grading; write $A_{(f)} := (A_f)_0$ for its degree-zero part, called the *homogeneous localization*, and similarly $M_{(f)} := (M_f)_0$. If in addition the map $\mathbb{Z} \rightarrow \Gamma, n \mapsto n\gamma$, is injective, write

$$A_f^{[\gamma]} := \bigoplus_{n \in \mathbb{Z}} (A_f)_{n\gamma}, \quad (A_f^{[\gamma]})_n = (A_f)_{n\gamma},$$

for the $\mathbb{Z}$ -graded part supported on the subgroup generated by γ . There is a canonical isomorphism of $\mathbb{Z}$ -graded rings

$$A_{(f)}[u^{\pm 1}] \xrightarrow{\sim} A_f^{[\gamma]}, \quad u \mapsto f,$$

where $\deg(u) = 1$. In particular, if A is $\mathbb{N}$ -graded, $f \in A_d$, and A is viewed as $\mathbb{Z}$ -graded before localization, then

$$(A^{(d)})_f \simeq A_f^{[d]} \simeq A_{(f)}[u^{\pm 1}].$$

2.2.1. Projective Schemes

Now we can define projective schemes.

Construction 2.2.7 (Evaluation Map for Homogeneous Polynomials). Let R be a k -algebra, $L \in \text{Line}_R$ an R -line, I a set, and $d \in \mathbb{N}$. We define the *evaluation map*

$$k[x_i \mid i \in I]_d \times L^I \rightarrow L^{\otimes d}, \quad (f, a) \mapsto f(a).$$

Construction. An element $a \in L^I$ is the same datum as an R -linear map $a: R^{(I)} \rightarrow L$, uniquely determined by the images of the basis vectors. The map $(f, a) \mapsto f(a)$ is then defined as the composite

$$k[x_i \mid i \in I]_d = \text{Sym}_k^d(k^{(I)}) \rightarrow \text{Sym}_R^d(R^{(I)}) \xrightarrow{\text{Sym}_k^d(a)} \text{Sym}_R^d(L) \simeq L^{\otimes d}.$$

(For a rank-1 module L , the canonical map $\text{Sym}^d(L) \simeq L^{\otimes d}$ is an isomorphism.)

Concretely, this is a k -linear map uniquely determined by the images of monomials:

$$\prod_i x_i^{n_i} \mapsto \bigotimes_i a_i^{\otimes n_i} \in L^{\otimes d}.$$

In particular, when $L = R$ (i.e., L is a trivial R -line), we have $L^{\otimes d} \simeq R$, which recovers the usual polynomial evaluation.

Recall that we defined affine schemes as loci of polynomial systems. For projective schemes, we naturally need to use homogeneous polynomials.

Definition 2.2.8 (System of Homogeneous Polynomials). Let k be a ring, and I, J be sets. A *system of J homogeneous polynomials in I variables over k* is a J -tuple $\Sigma = (f_j)_{j \in J}$, where each f_j is a *homogeneous polynomial* in the polynomial ring $k[x_i \mid i \in I]$.

We denote by (Σ) the homogeneous ideal generated by Σ in $k[x_i \mid i \in I]$, and write

$$k[\Sigma] = k[x_i \mid i \in I]/(\Sigma)$$

for the corresponding $\mathbb{N}$ -graded k -algebra.

Definition 2.2.9 (Projective Vanishing Locus). Let $F \subset k[x_i \mid i \in I]$ be a family of homogeneous polynomials. The *vanishing locus of F in $\mathbb{P}_k^I$* is the subfunctor $V(F) \subset \mathbb{P}_k^I$ defined by:

$$V(F)(R) = \{a: R^{(I)} \twoheadrightarrow L \mid \text{for all } f \in F \cap k[x_i]_d, \text{ we have } f(a) = 0 \in L^{\otimes d}\}.$$

Verification as Subfunctor. We verify that for any k -algebra homomorphism $\varphi: R \rightarrow S$, the induced map $\mathbb{P}^I(\varphi)$ maps $V(F)(R)$ into $V(F)(S)$.

Let $a \in V(F)(R)$ be a quotient line $R^{(I)} \twoheadrightarrow L$ satisfying $f(a) = 0$. Let a_S be the base change of a along φ , i.e., the quotient line:

$$S^{(I)} \simeq R^{(I)} \otimes_R S \xrightarrow{a \otimes 1} L \otimes_R S =: L_S.$$

We must show $f(a_S) = 0 \in L_S^{\otimes d}$.

This follows from the *commutativity of evaluation with base change*. Consider the following commutative diagram (for $d = \deg f$):

$$\begin{array}{ccc} \text{Sym}_k^d(k^{(I)}) & \xrightarrow{f(a)} & L^{\otimes d} \\ & \searrow f(a_S) & \downarrow \varphi_{L^{\otimes d}} \\ & & L^{\otimes d} \otimes_R S \simeq (L \otimes_R S)^{\otimes d} \end{array}$$

where $\varphi_{L^{\otimes d}}$ is the canonical map sending x to $x \otimes 1$. Since $a \in V(F)(R)$, we have $f(a) = 0_R$ (the zero element in $L^{\otimes d}$). Since $\varphi_{L^{\otimes d}}$ is an additive group homomorphism, it maps 0_R to 0_S (the zero element in $L_S^{\otimes d}$). By commutativity of the diagram, $f(a_S) = 0_S$. Therefore $a_S \in V(F)(S)$. $\square$

The subfunctor $V(F) \subset \mathbb{P}_k^I$ manifestly depends only on the homogeneous ideal (F) . A subtlety distinguishes the projective case from the affine one: distinct homogeneous ideals can produce the same vanishing locus. We return to this point in Definition 2.3.20.

Now we can define the solution set functor.

Definition 2.2.10 (Homogeneous Solution Functor). Let $\Sigma = (f_j)_{j \in J}$ be a system of J homogeneous polynomials in I variables over k . Its *solution set functor* $\text{hSol}_\Sigma: \text{Alg}_k \rightarrow \text{Set}$ is the vanishing locus of $\{f_j \mid j \in J\}$ in $\mathbb{P}_k^I$, i.e.,

$$\text{hSol}_\Sigma = V(\{f_j \mid j \in J\}) \subset \mathbb{P}_k^I.$$

Remark 2.2.11. The “h” in hSol stands for “homogeneous”.

As in the affine case, we define projective schemes as algebraic functors isomorphic to solution set functors.

Definition 2.2.12 (Projective Schemes). Let k be a ring. We say that a k -algebraic functor $X: \text{CAlg}_k \rightarrow \text{Set}$ is a *projective k -scheme* if there exists a system of homogeneous polynomials in *finitely many variables* Σ such that $X \simeq \text{hSol}_\Sigma$.

We denote by $\text{Proj}_k \subset \text{Fun}(\text{CAlg}_k, \text{Set})$ the full subcategory spanned by all projective k -schemes.

Remark 2.2.13. The finiteness assumption is a convenience: it ensures that projective k -schemes are automatically of finite type and proper.

We now seek a projective analogue of the affine identity $\text{Spec}(k[\Sigma]) = \text{Sol}_\Sigma$. Not every $\mathbb{N}$ -graded k -algebra A has the form $k[\Sigma]$: this happens precisely when A is generated by A_1 , i.e., when $\text{Sym}_k(A_1) \twoheadrightarrow A$ is surjective. We restrict to this case for the remainder of this section, deferring the general construction to Section 2.3.

Construction 2.2.14 (The Proj Functor - Preliminary Version). Let k be a ring and A an $\mathbb{N}$ -graded k -algebra generated by A_1 . Define the algebraic k -functor $\text{Proj}(A): \text{CAlg}_k \rightarrow \text{Set}$ by:

$$\text{Proj}(A)(R) = \{ \text{quotient line } A_1 \otimes_k R \twoheadrightarrow L \mid \text{Sym}_k(A_1) \twoheadrightarrow \text{Sym}_R(L) \text{ factors through } A \}.$$

Proposition 2.2.15 (Comparison with Solution Functors). Let Σ be a system of homogeneous polynomials over k . Then there is a canonical isomorphism:

$$\text{hSol}_\Sigma \simeq \text{Proj}(k[\Sigma]): \text{CAlg}_k \rightarrow \text{Set}.$$

Proof. We construct the bijection for any k -algebra R .

$\Rightarrow$: Let $a \in \text{hSol}_\Sigma(R)$. By definition, a is a quotient line $R^{(I)} \twoheadrightarrow L$ satisfying $f(a) = 0$ for all $f \in \Sigma$. According to Construction 2.2.7 and the isomorphism $\text{Sym}_R(R^{(I)}) \simeq R[x_i \mid i \in I]$, giving a is equivalent to giving a graded surjection of $\mathbb{N}$ -graded R -algebras:

$$\varphi_a: \text{Sym}_R(R^{(I)}) \twoheadrightarrow \text{Sym}_R(L).$$

The condition “ $f(a) = 0$ for all $f \in \Sigma$ ” is equivalent to saying that the kernel of φ_a contains the ideal $(\Sigma) \otimes_k R$. This means φ_a factors through the quotient algebra $k[\Sigma] \otimes_k R$, i.e., there exists a unique factorization:

$$\begin{array}{ccccc} \text{Sym}_R(R^{(I)}) & \twoheadrightarrow & \text{Sym}_R(k[\Sigma]_1 \otimes_k R) & \twoheadrightarrow & k[\Sigma] \otimes_k R \\ & \searrow \scriptstyle \text{Sym}_R(a) & \downarrow & \swarrow \text{---} & \\ & & \text{Sym}_R(L) & & \end{array}$$

This corresponds precisely to an element of $\text{Proj}(k[\Sigma])(R)$.

$\Leftarrow$: Conversely, given an element of $\text{Proj}(k[\Sigma])(R)$, i.e., given $k[\Sigma]_1 \otimes_k R \twoheadrightarrow L$ such that the induced Sym map factors through $k[\Sigma] \otimes_k R$. Since $k[\Sigma]$ is generated by its degree 1 part (as a quotient of $k[x_i]$), we can pull this back to $R^{(I)} \twoheadrightarrow k[\Sigma]_1 \otimes_k R \twoheadrightarrow L$. Clearly this composition satisfies the equations in Σ . $\square$

Recall that for a k -module M , we can consider its symmetric algebra $\text{Sym}_k(M)$.

Example 2.2.16 (Projective Space as $\mathbb{P}(M)$). Let M be a k -module. Define $\mathbb{P}(M): \text{CAlg}_k \rightarrow \text{Set}$ by:

$$\mathbb{P}(M)(R) = \{\text{quotient line } M \otimes_k R \twoheadrightarrow L\}.$$

Comparing constructions yields the identification $\mathbb{P}(M) \simeq \text{Proj}(\text{Sym}_k(M))$; when M is finitely generated, $\mathbb{P}(M)$ is a projective k -scheme. For a free module $M = k^{(I)}$ we recover the earlier notation $\mathbb{P}_k^I \simeq \mathbb{P}(k^{(I)})$.

Similarly, we can consider the dual functor $\mathbb{P}^\vee(M): \text{CAlg}_k \rightarrow \text{Set}$ defined by:

$$\mathbb{P}^\vee(M)(R) = \{\text{subspace line } L \hookrightarrow M \otimes_k R\}.$$

Here “subspace” is meant in the sense of Definition 2.1.19: the embedding must be universally injective. This condition is essential, since an arbitrary injective map need not remain injective after base change.

If M is a vector space, we have $\mathbb{P}^\vee(M^\vee) \simeq \mathbb{P}(M)$. However, in general, $\mathbb{P}^\vee(M)$ is not a scheme.

Remark 2.2.17. If A is an $\mathbb{N}$ -graded k -algebra generated by A_1 , then $\text{Proj}(A)$ is a subfunctor of $\mathbb{P}(A_1)$.

2.2.2. Loci of Linear Maps

Consider the projective n -space $\mathbb{P}^n = \text{Proj}(\mathbb{Z}[x_0, \dots, x_n])$, whose R -points are quotient lines $a: R^{\{0, \dots, n\}} \twoheadrightarrow L$.

For a homogeneous polynomial $f \in \mathbb{Z}[x_0, \dots, x_n]_d = \text{Sym}^d(\mathbb{Z}^{\{0, \dots, n\}})$, the *non-vanishing locus* $D(f) \subset \mathbb{P}^n$ is the subfunctor whose R -points are quotient lines a for which $f(a)$ generates $L^{\otimes d}$ — equivalently, for which the composite

$$s_f: R \xrightarrow{f} \text{Sym}_R^d(R^{\{0, \dots, n\}}) \xrightarrow{\text{Sym}^d(a)} L^{\otimes d}$$

is an R -module isomorphism. This expresses the geometric condition “ $f(a)$ is non-degenerate” functorially: it is equivalent to $f(a) \neq 0$ in $L^{\otimes d} \otimes_R \kappa$ for every R -field κ .

Recall the affine analogue: for a k -algebra A and $f \in A$,

$$D(f)(R) = \{x \in \text{Hom}(A, R) \mid (x(f)) = R\}.$$

We can rephrase this in line with the projective case: at a point x , the “function” f gives an R -linear map

$$R \xrightarrow{\cdot x(f)} R, \quad 1 \mapsto x(f),$$

and $(x(f)) = R$ — i.e. $x(f)$ is a unit — exactly when this map is an isomorphism.

Summary (Unified Description of Non-vanishing Loci). In both the affine and projective settings, the non-vanishing locus $D(f)$ consists of the points x at which the evaluation

$$R \xrightarrow{f(x)} \mathcal{L}$$

is an isomorphism. The target line is $\mathcal{L} = R$ (trivial) in the affine case and $\mathcal{L} = L^{\otimes d}$ in the projective case.

There is a subtlety, however: $f \in \mathbb{Z}[x_0, \dots, x_n]_d$ is not a function on $\mathbb{P}^n$ but a section of the line bundle $\mathcal{O}(d)$ (Definition 3.4.6). Openness of $D(f) \subset \mathbb{P}^n$ is therefore not immediate. The goal of this section is to show that $D(f)$ is open and even *relatively affine* — every pullback along an affine morphism is affine (Proposition 3.2.15).

To do this, consider any A -point $x: \text{Spec}(A) \rightarrow \mathbb{P}^n$ (i.e., a quotient line $A^{n+1} \twoheadrightarrow L$ over A). Consider the pullback diagram:

$$\begin{array}{ccc} x^{-1}(D(f)) & \hookrightarrow & \text{Spec}(A) \\ \downarrow & \lrcorner & \downarrow x \\ D(f) & \hookrightarrow & \mathbb{P}^n \end{array}$$

We must show that $x^{-1}(D(f)) \subset \text{Spec}(A)$ is open.

First, note that we can actually characterize $x^{-1}(D(f))$ in the following way: for an A -algebra R , we have:

$$x^{-1}(D(f))(R) = \{\varphi: A \rightarrow R \mid \varphi^*(s): R \rightarrow L^{\otimes d} \otimes_A R \text{ is an isomorphism}\},$$

where s is the linear map over A :

$$s: A \xrightarrow{f} \text{Sym}_A^d(A^{\{0, \dots, n\}}) \xrightarrow{x} L^{\otimes d}.$$

Equivalently, $x^{-1}(D(f))$ is the locus in $\text{Spec}(A)$ where the linear map “ s is an isomorphism”.

We adopt a broader perspective. Let P be a property of modules stable under base change — e.g., being zero, finitely generated, or flat. For a ring A and an A -module M , the locus where $\varphi^*(M)$ has P defines a subfunctor of $\text{Spec}(A)$:

$$R \mapsto \{\varphi: A \rightarrow R \mid \text{the } R\text{-module } \varphi^*(M) \text{ has property } P\}.$$

Similarly, for an A -linear map $u: M \rightarrow N$, we can also define the locus where it has a specific property:

Definition 2.2.18 (Loci of Linear Maps). Let $u: M \rightarrow N$ be an A -module homomorphism. We define:

- *Vanishing locus* $V(u)$: the locus where u is zero.
- *Surjection locus* $\text{Epi}(u)$: the locus where u is surjective.
- *Injection locus* $\text{Mono}(u)$: the locus where u is universally injective.
- *Isomorphism locus* $\text{Iso}(u)$: the locus where u is an isomorphism.

Returning to the problem of $D(f)$, we find that $x^{-1}(D(f))$ is precisely the *isomorphism locus* $\text{Iso}(s)$ of the linear map $s: A \rightarrow L^{\otimes d}$.

Proposition 2.2.19 (Loci as Open/Closed Subfunctors). Let A be a ring and $f: M \rightarrow N$ an A -linear map. Then:

- (i) If N is a vector space, then $V(f) \subset \text{Spec}(A)$ is a closed subfunctor.
- (ii) If N is a vector space, then $\text{Epi}(f) \subset \text{Spec}(A)$ is an open subfunctor.
- (iii) If M and N are both vector spaces, then $\text{Mono}(f) \subset \text{Spec}(A)$ is an open subfunctor.
- (iv) If M and N are both vector spaces, then $\text{Iso}(f) \subset \text{Spec}(A)$ is an open subfunctor.

Remark 2.2.20. Part (4) immediately implies that $D(f) \subset \mathbb{P}^n$ is open, since $D(f)$ is precisely the isomorphism locus of $s: A \rightarrow L^{\otimes d}$.

Proof. (i) *Vanishing locus:* f is zero if and only if for any $x \in M$, the composite map $f(x): R \xrightarrow{x} M \xrightarrow{f} N$ is zero. That is, $V(f) = \bigcap_{x \in M} V(f(x))$. Since arbitrary intersections of closed subfunctors remain closed subfunctors, we may assume $M = R$. In this case, $f: R \rightarrow N$ corresponds to an element $n = f(1) \in N$.

Since N is a vector space, N is reflexive ($N \simeq N^{\vee\vee}$), so $n = 0$ if and only if for all $\lambda \in N^\vee$, we have $\lambda(n) = 0$. Therefore:

$$V(R \xrightarrow{f} N) = V(N^\vee \xrightarrow{f^\vee} R) = V(\text{im}(f^\vee)),$$

where $f^\vee: N^\vee \rightarrow R$ is the dual map induced by f . For any R -algebra homomorphism $\varphi: R \rightarrow S$, we have $\varphi \in V(f)$ if and only if $\varphi^*(\text{im}(f^\vee)) = 0$. This is precisely the closed subfunctor $V(\text{im}(f^\vee))$ defined by the ideal $\text{im}(f^\vee) \subset R$.

(ii) *Surjection locus.* Since N is a vector space, it is a direct summand of R^n : pick N' with $N \oplus N' \simeq R^n$. Surjectivity of f is equivalent to surjectivity of $f \oplus \text{id}_{N'}: M \oplus N' \rightarrow R^n$, so

$$\text{Epi}(f) = \text{Epi}(f \oplus \text{id}_{N'}: M \oplus N' \rightarrow R^n).$$

Replacing f with $f \oplus \text{id}_{N'}$, we may assume $N = R^n$, so $f: M \rightarrow R^n$.

Claim 2.2.21. *The surjection locus can be detected by the exterior power functor, i.e.,*

$$\text{Epi}(f) = \text{Epi}(\Lambda^n(f)).$$

Proof of Claim. We test f on n -tuples. For $m_1, \dots, m_n \in M$, the images $f(m_1), \dots, f(m_n)$ assemble into an $n \times n$ matrix in R , and $\det: R^{n \times n} \rightarrow R$ detects whether they form a basis. This is encoded by the commutative diagram

$$\begin{array}{ccccc} M^n & \xrightarrow{f^n} & (R^n)^n \simeq R^{n \times n} & \xrightarrow{\det} & R \\ \text{can} \downarrow & & \text{can} \downarrow & & \parallel \\ \Lambda^n M & \xrightarrow{\Lambda^n(f)} & \Lambda^n(R^n) & \xrightarrow{\simeq} & R \end{array}$$

- If f is surjective, then the standard basis $e_1, \dots, e_n$ of R^n all lie in the image of f . There exist m_i such that $f(m_i) = e_i$. Then in the bottom row of the diagram, $\Lambda^n(f)(m_1 \wedge \dots \wedge m_n) = 1$, so the image of $\Lambda^n(f)$ contains a unit, hence is surjective.
- If $\Lambda^n(f)$ is surjective, then its image is the ideal generated by all determinants $\det(f(m_1), \dots, f(m_n))$ with $m_1, \dots, m_n \in M$, and surjectivity means these determinants generate the unit ideal of R . (One cannot in general single out one n -tuple of unit determinant: $\Lambda^n(f)(\omega) = 1$ only writes 1 as an R -combination of such determinants.) At every prime $\mathfrak{p} \subset R$ some determinant is then invertible in $R_{\mathfrak{p}}$, so the columns $f(m_1), \dots, f(m_n)$ form a basis of $R_{\mathfrak{p}}^n$ and $f_{\mathfrak{p}}$ is surjective.

Since $\text{coker}(f)$ is finitely generated (a quotient of R^n), its vanishing at every prime forces $\text{coker}(f) = 0$; that is, f is surjective. □

Since $\Lambda^n(R^n) \simeq R$, the claim reduces $\text{Epi}(f)$ to a rank-one statement: $\Lambda^n(f): \Lambda^n M \rightarrow R$ is surjective iff its image generates the unit ideal. Hence

$$\text{Epi}(f) = \text{Epi}(\Lambda^n(f)) = D(\text{im}(\Lambda^n(f))),$$

an open subfunctor of $\text{Spec}(A)$.

(iii) *Injection locus:* When M, N are both vector spaces, $f: M \rightarrow N$ is universally injective if and only if $f^\vee: N^\vee \rightarrow M^\vee$ is surjective. Since $N^\vee, M^\vee$ are also vector spaces, by (2), $\text{Epi}(f^\vee)$ is an open subfunctor.

- (iv) *Isomorphism locus*: f is an isomorphism if and only if it is surjective and universally injective. That is, $\text{Iso}(f) = \text{Epi}(f) \cap \text{Mono}(f)$. The intersection of two open subfunctors is still an open subfunctor. $\square$

2.2.3. Localization at a Section

We now show that $D(s)$ is affine for any section $s: R \rightarrow L$.

More generally, given a map of R -lines $s: L' \rightarrow L$, tensoring with $(L')^\vee$ gives $D(s) = D(s \otimes_R (L')^\vee)$ — this reduces to the case $s: R \rightarrow L$, equivalently an element $s \in L$. When $L = R$ this is just $D(s) = \text{Spec}(R_s)$ (classical localization); the substance is in the case of a non-trivial line L .

Definition 2.2.22 (Periodic Modules). Let R be a ring, L an R -line, and $s \in L$. We say that an R -module M is *s-periodic* if the morphism:

$$\text{id}_M \otimes s: M \otimes_R R \rightarrow M \otimes_R L$$

is an isomorphism.

Proposition 2.2.23 (Localization at a Line Section). Let R be a ring, L an R -line (i.e., an invertible module over R), and $s \in L$ (i.e., a section $s: R \rightarrow L$). Then:

- (i) There exists an s -periodic R -algebra R_s that is initial among all s -periodic R -algebras.
- (ii) The forgetful functor $\text{Mod}_{R_s} \rightarrow \text{Mod}_R$ identifies Mod_{R_s} with the full subcategory spanned by s -periodic modules. Therefore, it has a left adjoint $\text{Mod}_R \rightarrow \text{Mod}_{R_s}$ sending M to $M_s := M \otimes_R R_s$.
- (iii) Concretely, M_s can be computed as the following colimit in Mod_R :

$$M \xrightarrow{\cdot s} M \otimes_R L \xrightarrow{\cdot s} M \otimes_R L^{\otimes 2} \xrightarrow{\cdot s} \dots$$

Here the map $M \otimes_R L^{\otimes n} \xrightarrow{\cdot s} M \otimes_R L^{\otimes n+1}$ is defined by $x \mapsto x \otimes s$ (using the isomorphism $M \otimes_R L^{\otimes n} \simeq M \otimes_R L^{\otimes n} \otimes R \xrightarrow{1 \otimes s} M \otimes_R L^{\otimes n} \otimes L$).

Proof. We construct R_s explicitly and verify the three claims simultaneously.

Construction. Define

$$R_s := \text{colim}(R \xrightarrow{\cdot s} L \xrightarrow{\cdot s} L^{\otimes 2} \xrightarrow{\cdot s} \dots)$$

as an R -module, where each transition map $L^{\otimes n} \rightarrow L^{\otimes n+1}$ sends $\xi \mapsto \xi \otimes s$. Because L is invertible, the natural isomorphisms $L^{\otimes m} \otimes_R L^{\otimes n} \xrightarrow{\sim} L^{\otimes(m+n)}$ are compatible with the transitions, and they assemble into an associative, unital multiplication

$$R_s \otimes_R R_s \rightarrow R_s,$$

making R_s a commutative R -algebra. The unit is the class of $1 \in R$.

R_s is *s-periodic*. The structure map $R \rightarrow R_s$ identifies $1 \in R$ with the class of 1 in the 0-th term of the colimit, while $s \otimes 1 \in L \otimes_R R_s$ identifies with the class of s in the 1-st term, shifted up by one through the structure isomorphism $L \otimes_R R_s \simeq R_s$ given by the transition maps. Concretely, multiplication by s on R_s corresponds to the shift $L^{\otimes n} \rightarrow L^{\otimes(n+1)}$, which is an isomorphism on the colimit. Hence $\text{id}_{R_s} \otimes s: R_s \rightarrow R_s \otimes_R L$ is an isomorphism, so R_s is s -periodic.

Universal property. Let A be any s -periodic R -algebra. We must show that $\text{Hom}_{\text{CAlg}_R}(R_s, A)$ has exactly one element. Given $\varphi: R \rightarrow A$ in CAlg_R , the s -periodicity of A means that $\cdot s: A \rightarrow A \otimes_R L$ is an isomorphism, with inverse $\sigma: A \otimes_R L \rightarrow A$. Iterating gives compatible maps $A \otimes_R L^{\otimes n} \rightarrow A$ that match the transitions in the colimit defining R_s , hence a unique R -algebra map $R_s \rightarrow A$ extending φ .

Module-theoretic version (parts 2 and 3). Repeat the construction with M in place of R : the colimit $M_s := \text{colim}(M \rightarrow M \otimes L \rightarrow M \otimes L^{\otimes 2} \rightarrow \dots)$ exhibits M_s as the universal s -periodic R -module receiving a map

from M . By the universal property, $M_s \simeq M \otimes_R R_s$, and the assignment $M \mapsto M_s$ defines a left adjoint to the forgetful functor $\text{Mod}_{R_s} \rightarrow \text{Mod}_R$. Conversely, any s -periodic R -module N admits a unique R_s -action extending its R -action — built by inverting the s -action — so the forgetful functor is fully faithful onto s -periodic modules. $\square$

Remark 2.2.24 (Classical Localization as Special Case). When $M = R$ and $L = R$ (trivial line), s corresponds to an element $f \in R$. In this case, the above colimit is $R \xrightarrow{f} R \xrightarrow{f} R \cdots$, whose colimit is precisely the classical localization $R_f = R[1/f]$.

For a general line bundle L one cannot literally write “ $1/s$ ” — s is not an element of a ring. The construction above instead universally inverts $\cdot s$, making it an isomorphism. This succeeds because L is invertible: tensoring with L is an autoequivalence of Mod_R , so it behaves formally like multiplication by an invertible scalar.

Therefore, we immediately obtain the following corollary:

Corollary 2.2.25 (Non-vanishing Locus is Affine). *Let R be a ring, L an R -line, and $s \in L$. Then $D(s) = \text{Epi}(R \xrightarrow{s} L) \simeq \text{Spec}(R_s)$.*

Proof. For an R -algebra $\varphi: R \rightarrow R'$, the pullback section $\varphi^*(s) \in L \otimes_R R'$ generates $L \otimes_R R'$ as an R' -module if and only if $\cdot \varphi^*(s): R' \rightarrow L \otimes_R R'$ is surjective; since $L \otimes_R R'$ is a line over R' and a surjection between line bundles is automatically an isomorphism (Corollary 2.1.23), this is equivalent to R' being s -periodic as an R -module. By Proposition 2.2.23(i), this in turn is equivalent to φ factoring uniquely through $R \rightarrow R_s$. Hence

$$D(s)(R') = \text{Hom}_{\text{CAlg}_R}(R_s, R') = \text{Spec}(R_s)(R'),$$

naturally in R' . $\square$

Summary (Relative Affineness of $D(f)$). Therefore, for $f \in \mathbb{Z}[x_0, \dots, x_n]_d$, we have that $D(f) \subset \mathbb{P}^n$ is an open subfunctor. Moreover, for any $x: \text{Spec}(R) \rightarrow \mathbb{P}^n$ corresponding to a quotient line $a: R^{n+1} \twoheadrightarrow L$, we have $x^{-1}D(f) = D(s)$, where $s = f(a) \in L^{\otimes d}$. Therefore, $D(f) \hookrightarrow \mathbb{P}^n$ is relatively affine.

2.2.4. Generalized Zariski Descent

The localization-at-a-section construction generalizes classical Zariski descent.

Proposition 2.2.26 (Generalized Zariski Descent). *Let R be a ring, L an R -line, and $(s_i)_{i \in I}$ a generating set for L . Then for any R -module M , the diagram:*

$$M \longrightarrow \prod_{i \in I} M_{s_i} \rightrightarrows \prod_{i, j \in I} M_{s_i s_j}$$

is an equalizer, where $s_i s_j = s_i \otimes s_j \in L^{\otimes 2}$.

Proof. The strategy is to reduce the descent over the line section s_i 's to standard Zariski descent over a suitable set of ring elements.

Step 1: From line sections to ring elements. For a single section $s: R \rightarrow L$, dualizing gives

$$s^\vee: L^\vee = R \otimes_R L^\vee \xrightarrow{s \otimes \text{id}} L \otimes_R L^\vee = R.$$

Fix a finite generating set $F \subset L^\vee$ and set $F_s := s^\vee(F) \subset R$. We claim that (F_s) generates the unit ideal in R_s .

Indeed, R_s is s -periodic, so $\text{id}_{R_s} \otimes s: R_s \rightarrow R_s \otimes_R L$ is an isomorphism. Tensoring its dual $s^\vee$ with R_s inherits this isomorphism property by base change, hence $s^\vee \otimes_R R_s: (L^\vee)_s \rightarrow R_s$ is surjective. Since F generates $L^\vee$, the image F_s generates R_s . The same argument applied to $st := s \otimes t \in L^{\otimes 2}$ shows that $F_s F_t$ generates R_{st} for any second section t .

Step 2: Translation to a Zariski cover of R . Apply Step 1 to the family $(s_i)_{i \in I}$ generating L . The two surjections

$$R^{(I)} \twoheadrightarrow L, \quad R^{(F)} \twoheadrightarrow L^\vee$$

tensor to a surjection onto $L \otimes_R L^\vee \simeq R$. Its images are precisely the evaluations $\phi(s_i)$ with $i \in I$ and $\phi \in F$. Consequently, the union

$$\mathcal{F} := \bigcup_{i \in I} F_{s_i} \subset R$$

generates the unit ideal of R , and similarly $F_{s_i} F_{s_j}$ generates $R_{s_i s_j}$ for each pair i, j . Standard Zariski descent for M along the cover $\mathcal{F}$ yields the equalizer

$$M \rightarrow \prod_{f \in \mathcal{F}} M_f \rightrightarrows \prod_{f, g \in \mathcal{F}} M_{fg}, \quad (2.1)$$

and within each $\text{Mod}_{R_{s_i}}$, descent along F_{s_i} exhibits M_{s_i} as an equalizer

$$M_{s_i} \rightarrow \prod_{f \in F_{s_i}} (M_{s_i})_f \rightrightarrows \prod_{f, g \in F_{s_i}} (M_{s_i})_{fg}. \quad (2.2)$$

Step 3: The two descent diagrams agree. Abbreviate $F_i := F_{s_i}$. Combining (2.2) for each i (and similarly for $M_{s_i s_j}$) gives the columns of

$$\begin{array}{ccccc} M & \longrightarrow & \prod_i M_{s_i} & \rightrightarrows & \prod_{i,j} M_{s_i s_j} \\ \parallel & & \downarrow & & \downarrow \\ M & \longrightarrow & \prod_{i, f \in F_i} M_f & \rightrightarrows & \prod_{i,j, f, g \in F_i F_j} M_{fg} \\ \parallel & & \Downarrow & & \Downarrow \\ M & \longrightarrow & \prod_{i, f, g \in F_i} M_{fg} & \rightrightarrows & \prod_{i,j, f, g, f' g' \in F_i F_j} M_{fg \cdot f' g'} \end{array}$$

The columns are equalizers (each is an instance of (2.2)), and limits commute, so the equalizer of the top row equals the equalizer of the equalizers of the lower rows. The latter is the descent diagram for M along $\mathcal{F} \cup (\mathcal{F} \cdot \mathcal{F})$, which by (2.1) computes M .

Finally, the lower descent only involves the products $F_i \cdot F_j$: every pair (f, g) with $f \in \mathcal{F}$ and $g \in \mathcal{F}$ has the form $f \in F_i, g \in F_j$ for some i, j , so the pairwise-compatibility conditions over $\mathcal{F}$ are exactly those recorded in $\prod_{i,j} \prod_{F_i F_j}$. Thus the top row is an equalizer, as claimed. $\square$

Example 2.2.27 (Projective completions of affine spaces). Let k be a ring and M a k -module. Recall the algebraic k -functors $\mathbb{A}(M)$ (Example 1.2.24) and $\mathbb{P}(M)$ (Example 2.2.16). There is a canonical embedding

$$\mathbb{A}(M) \hookrightarrow \mathbb{P}(M \oplus k), \quad (a: M \otimes_k R \rightarrow R) \mapsto ((a, \text{id}_R): (M \otimes_k R) \oplus R \twoheadrightarrow R).$$

Claim 2.2.28. *The canonical embedding above is an open immersion.*

Proof of Claim. Let $x: \text{Spec}(R) \rightarrow \mathbb{P}(M \oplus k)$ classify a quotient line $(M \otimes_k R) \oplus R \twoheadrightarrow L$ (Example 2.2.16). The pullback

$$\begin{array}{ccc} x^{-1}(\mathbb{A}(M)) & \longrightarrow & \text{Spec}(R) \\ \downarrow & & \downarrow \\ \mathbb{A}(M) & \hookrightarrow & \mathbb{P}(M \oplus k) \end{array}$$

is the locus in $\text{Spec}(R)$ where the composite $R \hookrightarrow (M \otimes_k R) \oplus R \twoheadrightarrow L$ is an isomorphism, namely

$$x^{-1}(\mathbb{A}(M)) = \text{Iso}(R \hookrightarrow (M \otimes_k R) \oplus R \twoheadrightarrow L),$$

which is open in $\text{Spec}(R)$ by Proposition 2.2.19(iv). $\square$

Taking $M = k^{(I)}$, we get a canonical open immersion

$$\mathbb{A}^I \hookrightarrow \mathbb{P}^{I \sqcup \{0\}},$$

identifying $\mathbb{A}^I$ with $D(x_0) \subset \mathbb{P}^{I \sqcup \{0\}}$. For $I = \{1, \dots, n\}$, this is an open immersion $\mathbb{A}^n \hookrightarrow \mathbb{P}^n$.

2.3. The Proj Construction

We now extend the Proj functor from rings generated in degree 1 to arbitrary $\mathbb{N}$ -graded rings. The construction is a colimit over the divisibility poset of Veronese subrings $A^{(d)}$; the structure theorem then says this colimit is glued from affine opens $D(f) \simeq \text{Spec}(A_{(f)})$ as f ranges over homogeneous elements.

Notation 2.3.1 (Graded Ring Notation). Let A be an $\mathbb{N}$ -graded ring and $d \in \mathbb{N}$ a natural number.

- $A^{(d)}$ denotes the $\mathbb{N}$ -graded ring $\bigoplus_{n \in \mathbb{N}} A_{nd}$ (Veronese subring), where $(A^{(d)})_n = A_{nd}$.
- $A_+ = \bigoplus_{d \geq 1} A_d$ is the irrelevant ideal.
- For a homogeneous element $f \in A$, A_f is a $\mathbb{Z}$ -graded ring, and $A_{(f)} := (A_f)_0$ is its degree 0 part.

We can now construct the general Proj functor — extending the preliminary version (Construction 2.2.14) to all $\mathbb{N}$ -graded rings, not only those generated in degree 1.

Definition 2.3.2 (Eventually Surjective). Let A, B be $\mathbb{N}$ -graded rings. We say that a graded map $A \rightarrow B$ is *eventually surjective* if for all $d \geq 1$ and every homogeneous element $b \in B_d$, there exists $n \geq 1$ such that b^n lies in the image of the map $A_{nd} \otimes_{A_0} B_0 \rightarrow B_{nd}$.

We denote by $\text{CAlg}^{\mathbb{N}, \text{es}}$ the category of $\mathbb{N}$ -graded rings and eventually surjective maps between them.

Construction 2.3.3 (The Proj Functor). We construct the functor:

$$\text{Proj}: (\text{CAlg}^{\mathbb{N}, \text{es}})^{\text{op}} \rightarrow \text{Fun}(\text{CAlg}, \text{Set}).$$

The construction also produces a natural transformation $\text{Proj}(-) \Rightarrow \text{Spec}((-)_0)$.

Step 1: Define $\text{Proj}_1(A)$. We define $\text{Proj}_1(A)$ as an object in $\text{Fun}(\text{CAlg}, \text{Set})_{/\text{Spec}(A_0)} \simeq \text{Fun}(\text{CAlg}_{A_0}, \text{Set})$. Given an A_0 -algebra R , the set $\text{Proj}_1(A)(R)$ consists of pairs (L, φ) where:

- L is a quotient line of $A_1 \otimes_{A_0} R$, i.e., a surjective R -module homomorphism $A_1 \otimes_{A_0} R \twoheadrightarrow L$;
- $\varphi: A \otimes_{A_0} R \rightarrow \text{Sym}_R(L)$ is a graded R -algebra morphism that extends the above quotient map (i.e., φ restricted to degree 1 is precisely $A_1 \otimes R \rightarrow L$).

Surjectivity of φ is automatic: its image contains L by surjectivity in degree 1, and the target $\text{Sym}_R(L)$ is generated as an R -algebra by L . Equivalently, $\text{Proj}_1(A)(R)$ is the set of graded quotients of $A \otimes_{A_0} R$ isomorphic to the symmetric algebra of a line.

Step 2: Define $\text{Proj}_d(A)$ and the colimit. For $d \in \mathbb{N}$, define $\text{Proj}_d(A) = \text{Proj}_1(A^{(d)})$. For $n \in \mathbb{N}$, there exists a canonical morphism:

$$\text{Proj}_d(A) \rightarrow \text{Proj}_{nd}(A), \quad (L, \varphi) \mapsto (L^{\otimes n}, \varphi^{(n)}),$$

where $\varphi^{(n)}$ is the restriction of φ to the Veronese subring (note that $\text{Sym}_R(L)^{(n)} \simeq \text{Sym}_R(L^{\otimes n})$).

This defines a functor:

$$\mathbb{N}^{\text{div}} \rightarrow \text{Fun}(\text{CAlg}, \text{Set}), \quad d \mapsto \text{Proj}_d(A),$$

where $\mathbb{N}^{\text{div}}$ is the poset ordered by divisibility. We define $\text{Proj}(A)$ as its colimit:

$$\text{Proj}(A) = \text{colim}_{d \in \mathbb{N}^{\text{div}}} \text{Proj}_d(A).$$

Since 1 divides every d , we also have $\text{Proj}(A) \simeq \text{colim}_{d \rightarrow \infty} \text{Proj}_1(A^{(d)})$.

Since 0 is the terminal object of $\mathbb{N}^{\text{div}}$, the colimit comes equipped with a canonical structure map

$$\text{Proj}(A) \rightarrow \text{Proj}_0(A) = \text{Proj}_1(A^{(0)}) = \text{Proj}_1(A_0[x]) \simeq \text{Spec}(A_0).$$

Step 3: Functoriality. Let $\alpha: A \rightarrow B$ be a graded map and (L, φ) an R -point of $\text{Proj}_d(B)$. Choose finitely many $b_1, \dots, b_r \in B_d$ whose images under φ_d generate L . Whenever some $n \geq 1$ makes every b_i^n lie in the image of $\alpha_{nd}: A_{nd} \otimes_{A_0} B_0 \rightarrow B_{nd}$, the line $L^{\otimes n}$ pulls back to a quotient line of $A^{(nd)}$, yielding the R -point $(L^{\otimes n}, \varphi^{(n)} \circ \alpha^{(nd)})$ of $\text{Proj}_{nd}(A)$.

Therefore, if α is *eventually surjective*, this process is well-defined for any R -point, inducing a morphism:

$$\text{Proj}(\alpha): \text{Proj}(B) \rightarrow \text{Proj}(A).$$

Remark 2.3.4 (Veronese Isomorphism). For $d \geq 1$, since $d\mathbb{N}^{\text{div}} \subset \mathbb{N}^{\text{div}}$ is cofinal, we have an isomorphism:

$$\text{Proj}(A) \simeq \text{Proj}(A^{(d)}).$$

Remark 2.3.5 (Degree d Maps). We can extend the domain of Proj in the following way.

For $\mathbb{N}$ -graded rings A and B and $d \in \mathbb{N}$, a *degree d map* from A to B is a graded ring map $A \rightarrow B^{(d)}$.

If it is *eventually surjective* and $d \geq 1$, it induces:

$$\text{Proj}(B) \simeq \text{Proj}(B^{(d)}) \rightarrow \text{Proj}(A).$$

This also holds for $d = 0$:

$$\text{Proj}(B) \rightarrow \text{Spec}(B_0) = \text{Proj}(B^{(0)}) \rightarrow \text{Proj}(A).$$

Composition of a degree- d map with a degree- e map gives a *degree- de map*, so we obtain an extended category — same objects, with $\text{Hom}(A, B) := \coprod_{d \in \mathbb{N}} \text{Hom}_{\text{CAlg}^{\mathbb{N}, \text{es}}}(A, B^{(d)})$ — to which Proj extends.

The central result of this section is the following structure theorem.

Theorem 2.3.6 (Proj Structure Theorem). *Let A be an $\mathbb{N}$ -graded ring and $d \geq 1$.*

- (i) *The canonical map $\text{Proj}_d(A) \rightarrow \text{Proj}(A)$ is an open immersion.*
- (ii) *If A is generated as an A_0 -algebra by homogeneous elements whose degrees divide d , then the canonical map $\text{Proj}_d(A) \rightarrow \text{Proj}(A)$ is an isomorphism.*

In particular, if A is generated by $A_{\leq 1}$ (i.e., A_0 and A_1), then $\text{Proj}_1(A) \simeq \text{Proj}(A)$.

Remark 2.3.7. Morally, the theorem says $\text{Proj}(A)$ is glued from the open subfunctors $\text{Proj}_d(A)$ as d varies.

As an immediate consequence, every weighted projective scheme embeds into an ordinary projective space.

Corollary 2.3.8 (Projective Schemes are Projective). *If A is finitely generated as an A_0 -algebra, then $\text{Proj}(A)$ is a projective A_0 -scheme: it is isomorphic to a closed subscheme of $\mathbb{P}_{A_0}^N$ for some N .*

Proof. Since A is finitely generated as an A_0 -algebra and A is an $\mathbb{N}$ -graded ring, choose finitely many homogeneous generators $x_1, \dots, x_n$ of positive degrees. For all sufficiently divisible d , the Veronese algebra $A^{(d)}$ is generated over A_0 by its degree-1 part A_d ; this is the finite monomial lemma recorded in the Stacks Project, [The25, Tag 0EGH]. The Veronese isomorphism of Remark 2.3.4 therefore gives $\text{Proj}(A) \simeq \text{Proj}(A^{(d)})$.

Since $A^{(d)}$ is generated by its degree-1 part, Theorem 2.3.6(2) gives

$$\text{Proj}(A) \simeq \text{Proj}(A^{(d)}) \simeq \text{Proj}_1(A^{(d)}).$$

There are only finitely many weighted monomials in the x_i of total degree d , and they span A_d over A_0 . Thus A_d is a finite A_0 -module; choose a surjection $A_0^{N+1} \twoheadrightarrow A_d$. A quotient line of A_d is a quotient line of A_0^{N+1} on which the kernel vanishes, so the induced map $\mathbb{P}(A_d) \hookrightarrow \mathbb{P}_{A_0}^N$ is a closed immersion, cut out by linear equations. Finally, $\text{Proj}_1(A^{(d)})$ is the closed subfunctor of $\mathbb{P}(A_d)$ cut out by the homogeneous relations in $A^{(d)}$. Hence $\text{Proj}(A)$ is a closed subscheme of $\mathbb{P}_{A_0}^N$. $\square$

Example 2.3.9 (Veronese Embedding). Let k be a ring, M a k -module, and $d \geq 1$. The d -fold Veronese embedding is a natural transformation of k -algebraic functors

$$\rho_d: \mathbb{P}(M) \rightarrow \mathbb{P}(\text{Sym}_k^d(M)),$$

which sends a quotient line $M \otimes_k R \twoheadrightarrow L$ to its d -th symmetric power:

$$\text{Sym}_k^d(M) \otimes_k R \twoheadrightarrow \text{Sym}_R^d(L) \simeq L^{\otimes d}.$$

(Since L is an R -line, $\text{Sym}_R^d(L) \simeq L^{\otimes d}$.)

Coordinate description: If $M = k^{(I)}$ is a free module, then the d -fold Veronese embedding is the map:

$$\rho_d: \mathbb{P}(k^{(I)}) \rightarrow \mathbb{P}(\text{Sym}^d(k^{(I)})).$$

In particular, when $M = k^{n+1}$ (i.e., $I = \{0, \dots, n\}$), $\text{Sym}^d(M)$ has rank $\binom{n+d}{n}$. The embedding map is:

$$\rho_d: \mathbb{P}_k^n \rightarrow \mathbb{P}_k^N, \quad \text{where } N = \binom{n+d}{n} - 1.$$

Its action on coordinates is given by:

$$[a_0 : \dots : a_n] \mapsto [\text{all degree } d \text{ monomials } M_J(a_0, \dots, a_n)].$$

Geometric property: We claim that for $d \geq 1$, ρ_d is a closed immersion.

We can prove this using the functoriality of Proj . Let $S = \text{Sym}_k(M)$. The map ρ_d decomposes as the composition of the following two maps:

$$\text{Proj}_1(S) \xrightarrow{\sim} \text{Proj}_d(S) \xrightarrow{i} \text{Proj}_1(\text{Sym}_k(S_d)).$$

Here $\text{Proj}_1(\text{Sym}_k(S_d))$ is precisely the target space $\mathbb{P}(\text{Sym}^d(M))$.

- (i) *First map:* $(L, \varphi) \mapsto (L^{\otimes d}, \varphi^{(d)})$ is an isomorphism. This is guaranteed by *Theorem 2.3.6* (Proj Structure Theorem), since S is generated by $S_1 = M$, so $S^{(d)}$ is generated by S_d .
- (ii) *Second map.* The morphism i is a closed immersion induced by a surjection of graded rings. Consider the canonical k -algebra map

$$\Phi: \text{Sym}_k(S_d) \twoheadrightarrow S^{(d)} = \bigoplus_{n \geq 0} S_{nd},$$

sending the formal generators of $\text{Sym}_k(S_d)$ to products of degree- d elements of S . Because S is generated by S_1 , the Veronese subring $S^{(d)}$ is generated by S_d , so Φ is surjective.

By the contravariant functoriality of Proj, the surjective ring homomorphism Φ induces a closed immersion:

$$\text{Proj}(\Phi): \text{Proj}(S^{(d)}) \hookrightarrow \text{Proj}(\text{Sym}_k(S_d)).$$

That is, $\text{Proj}_d(S) \hookrightarrow \mathbb{P}(\text{Sym}^d(M))$.

In summary, ρ_d is the composition of an isomorphism and a closed immersion, hence is a closed immersion.

We turn to Zariski descent for the Proj functor.

Proposition 2.3.10 (Zariski Descent for Proj). *Let A be an $\mathbb{N}$ -graded ring, $d \in \mathbb{N}$ a natural number, and X be $\text{Proj}_d(A)$ (or $\text{Proj}(A)$). Then for any ring R , R -line L , and generating set $(s_i)_{i \in I}$ of L , the diagram:*

$$X(R) \longrightarrow \prod_{i \in I} X(R_{s_i}) \rightrightarrows \prod_{i, j \in I} X(R_{s_i s_j})$$

is an equalizer.

Proof. We may assume $X = \text{Proj}_1(B)$ (taking $B = A^{(d)}$ in the $\text{Proj}_d(A)$ case). Set $M = B_1 \otimes_{B_0} R$.

Let $x_i = (L_i, \varphi_i) \in X(R_{s_i})$ be a family of local data — each L_i an R_{s_i} -line and $\varphi_i: B \otimes R_{s_i} \twoheadrightarrow \text{Sym}(L_i)$ — inducing in degree 1 a surjection $\pi_i: M_{s_i} \twoheadrightarrow L_i$.

Assume these data are compatible on overlaps: there exist isomorphisms $\theta_{ij}: (L_i)_{s_j} \xrightarrow{\sim} (L_j)_{s_i}$ identifying (L_i, φ_i) and (L_j, φ_j) after restriction to $R_{s_i s_j}$, that is, the triangle

$$\begin{array}{ccc} M_{s_i s_j} & \xrightarrow{\pi_j} & (L_j)_{s_i} \\ & \searrow \pi_i & \nearrow \theta_{ij} \\ & (L_i)_{s_j} & \end{array}$$

commutes. Since π_i is surjective, any such θ_{ij} is unique.

Step 1: Descent of the line L . We construct a global R -line L with quotient map $\pi: M \twoheadrightarrow L$. By generalized Zariski descent it suffices to show the cocycle condition $\theta_{jk} \circ \theta_{ij} = \theta_{ik}$ holds on $R_{s_i s_j s_k}$.

Consider the following diagram (all modules are localized at $R_{s_i s_j s_k}$):

$$\begin{array}{ccccc} & & (L_i) & \xrightarrow{\theta_{ij}} & (L_j) \\ & \nearrow \pi_i & & \searrow \theta_{jk} & \\ M & \xrightarrow{\pi_k} & (L_k) & & \end{array}$$

- The upper left triangle commutes (by definition of θ_{ij}).
- The right triangle commutes (by definition of θ_{jk}).

- The outer large triangle commutes (by definition of θ_{ik}).

Since $M \twoheadrightarrow (L_i)$ is surjective, we can verify by tracing the images of elements:

$$\theta_{jk}(\theta_{ij}(\pi_i(m))) = \theta_{jk}(\pi_j(m)) = \pi_k(m) = \theta_{ik}(\pi_i(m)).$$

Since π_i is surjective, $\theta_{jk} \circ \theta_{ij} = \theta_{ik}$.

By generalized Zariski descent for R -modules (Proposition 2.2.26), there exists a unique R -module L with isomorphisms $L_{s_i} \simeq L_i$. Moreover, since “being a line bundle (invertible module)” is a local property, and L is locally isomorphic to the line bundles L_i , L is an R -line.

Step 2: Descent of the morphism φ . Now we have the global line L and local isomorphisms. The local morphisms φ_i can be viewed as $\psi_i: B \otimes R_{s_i} \rightarrow \text{Sym}(L)_{s_i}$. The compatibility condition ensures that ψ_i and ψ_j agree on intersections.

By Zariski descent for morphisms, this family ψ_i uniquely glues to a global algebra morphism:

$$\varphi: B \otimes R \rightarrow \text{Sym}(L).$$

Surjectivity is local: each φ_i is surjective and φ agrees with φ_i after localizing at s_i , so φ is surjective (in the Proj_1 case this is just ordinary surjectivity; for Proj_d via $A^{(d)}$ it amounts to eventual surjectivity).

This exhibits the unique $(L, \varphi) \in X(R)$ restricting to the given local data. $\square$

The structure theorem rests on the following lemma about the affine open $D(f)$.

Lemma 2.3.11 (Properties of $D(f)$). *Let A be an $\mathbb{N}$ -graded ring, $d \geq 1$, and $f \in A_d$. Let $D(f) \subset \text{Proj}_d(A)$ be the subfunctor defined by: for a ring R , $D(f)(R) = \{(L, \varphi) \mid \varphi(f) \text{ generates } L\}$.*

- (i) *There is a canonical isomorphism $D(f) \simeq \text{Spec}(A_{(f)})$.*
- (ii) *For each $n \geq 1$, the map $\text{Proj}_d(A) \rightarrow \text{Proj}_{nd}(A)$ induces an isomorphism:*

$$D(f) \xrightarrow{\sim} D(f^n).$$

More precisely, the diagram

$$\begin{array}{ccc} D(f) & \xrightarrow{\sim} & D(f^n) \\ \downarrow & \lrcorner & \downarrow \\ \text{Proj}_d(A) & \longrightarrow & \text{Proj}_{nd}(A) \end{array}$$

is cartesian.

- (iii) *Let $f_1, \dots, f_r$ be homogeneous elements of positive degrees. If D is divisible by every $\deg(f_i)$ and by $\deg(f_1 \cdots f_r)$, use (2) to regard all the following loci as subfunctors of $\text{Proj}_D(A)$. Then*

$$D(f_1 \cdots f_r) = D(f_1) \cap \cdots \cap D(f_r).$$

Proof. (i) We construct mutually inverse maps in two steps.

- (a) *The map $D(f) \rightarrow \text{Spec}(A_{(f)})$.*

Let $(L, \varphi) \in D(f)(R)$. By definition, $\varphi(f)$ generates L , hence trivializes this line. After adjoining its inverse, the localization of $\text{Sym}_R(L)$ at the homogeneous element $\varphi(f)$ is the $\mathbb{Z}$ -graded algebra $\bigoplus_{n \in \mathbb{Z}} L^{\otimes n}$. This induces the following commutative diagram:

$$\begin{array}{ccc} A^{(d)} \otimes_{A_0} R & \xrightarrow{\varphi} & \text{Sym}_R(L) \\ f^{-1} \downarrow & & \downarrow \varphi(f)^{-1} \\ A_f^{[d]} \otimes_{A_0} R & \xrightarrow{\tilde{\varphi}} & \bigoplus_{n \in \mathbb{Z}} L^{\otimes n} \end{array}$$

Take the *degree 0* part of the bottom row. The left side is $A_{(f)} \otimes R$; since L is trivialized by $\varphi(f)$, the right side gives $L^{\otimes 0} \simeq R$. Therefore, $\tilde{\varphi}$ restricted to degree 0 gives a homomorphism $\psi: A_{(f)} \rightarrow R$.

(b) The map $\text{Spec}(A_{(f)}) \rightarrow D(f)$.

Conversely, given $\psi: A_{(f)} \rightarrow R$, take $L = R$ with generator T , so $\text{Sym}_R(L) = R[T]$. Using the graded isomorphism $(A^{(d)})_f \simeq A_f^{[d]} \simeq A_{(f)}[u, u^{-1}]$ (where u is the degree-1 generator corresponding to f), set $\tilde{\varphi}: A_{(f)}[u, u^{-1}] \rightarrow R[T, T^{-1}]$ to be ψ on coefficients and $u \mapsto T$ on the generator, then restrict to $A^{(d)} \rightarrow R[T]$:

$$\begin{array}{ccc} A^{(d)} & \xrightarrow{\tilde{\varphi}} & R[T] \\ \downarrow & & \downarrow \\ A_{(f)}[u, u^{-1}] & \xrightarrow{\tilde{\varphi}} & R[T, T^{-1}] \end{array}$$

Then $\varphi(f) = T$ generates L , so $(R, \varphi) \in D(f)(R)$. The two constructions are mutually inverse by direct verification.

- (ii) Localization of A at f and at f^n coincide as graded rings, $A_f = A_{f^n}$, hence so do their degree-0 parts $A_{(f)} = A_{(f^n)}$. Therefore, we have the following commutative diagram (where the vertical arrows are the isomorphisms established in part 1):

$$\begin{array}{ccc} D(f) & \xrightarrow{\quad} & D(f^n) \\ \downarrow \wr & & \downarrow \wr \\ \text{Spec}(A_{(f)}) & \xlongequal{\quad} & \text{Spec}(A_{(f^n)}) \end{array}$$

This proves that $D(f) \xrightarrow{\sim} D(f^n)$ is an isomorphism.

Finally, we show that the diagram:

$$\begin{array}{ccc} D(f) & \xrightarrow{\sim} & D(f^n) \\ \downarrow & & \downarrow \\ \text{Proj}_d(A) & \longrightarrow & \text{Proj}_{nd}(A) \end{array}$$

is a pullback. Since we are considering algebraic functors, we only need to check the fiber product property on sets for any ring R .

The map $\text{Proj}_d(A) \rightarrow \text{Proj}_{nd}(A)$ sends (L, φ) to $(L^{\otimes n}, \varphi^{(n)})$, so the cartesian-square claim reduces to the equivalence

$$\varphi(f) \text{ generates } L \iff \varphi(f)^{\otimes n} \text{ generates } L^{\otimes n}.$$

This is local on $\text{Spec}(R)$ and reduces to the elementary statement $u \in R^\times \iff u^n \in R^\times$ for $u \in R$.

(iii) Put $e_i = \deg(f_i)$ and $e = \sum_i e_i$. For a common multiple D as in the statement, set

$$a_i = f_i^{D/e_i} \in A_D, \quad b = (f_1 \cdots f_r)^{D/e} \in A_D.$$

In the Veronese ring $A^{(D)}$ these elements satisfy

$$b^e = \prod_i a_i^{e_i}.$$

At an R -point of $\text{Proj}_D(A)$, a section b generates the relevant line if and only if b^e generates its e -th tensor power. The displayed identity, checked after locally trivializing the line, says this happens if and only if every a_i generates the line: a product in a ring is a unit exactly when every factor is a unit. Via the power identifications of (2), these are precisely the asserted loci. $\square$

We are now ready to prove the structure theorem.

Proof of Theorem 2.3.6. We first analyze a transition map

$$j_{d,n} : \text{Proj}_d(A) \longrightarrow \text{Proj}_{nd}(A).$$

The loci $D(f)$, $f \in A_d$, form a Zariski cover of the source. If two points of $\text{Proj}_d(A)(R)$ have the same image under $j_{d,n}$, then on the open cover where the values of the f generate their quotient lines, Lemma 2.3.11(2) identifies both restrictions with the same point of $D(f^n)$. Zariski descent for $\text{Proj}_d(A)$ (Proposition 2.3.10) makes the two points equal. Thus $j_{d,n}$ is a monomorphism.

We claim that its image is the open subfunctor

$$U_{d,n} := \bigcup_{f \in A_d} D(f^n) \subset \text{Proj}_{nd}(A),$$

where the union is understood sheaf-theoretically. Indeed, if (M, ψ) is a point of $\text{Proj}_d(A)$, then the sections $\psi(f)$ generate M , and their n -th powers generate $M^{\otimes n}$. After trivializing M , this is the elementary implication $(a_f)_f = R \Rightarrow (a_f^n)_f = R$. Hence the image lies in $U_{d,n}$. Conversely, over each member $D(f^n)$ of the displayed cover, Lemma 2.3.11(2) gives a unique lift to $D(f)$. These lifts agree on overlaps because $j_{d,n}$ is a monomorphism, and Zariski descent glues them. Therefore $j_{d,n}$ identifies $\text{Proj}_d(A)$ with $U_{d,n}$ and is an open immersion.

All transition maps in the filtered diagram defining $\text{Proj}(A)$ are therefore open monomorphisms. We may regard the pointwise filtered colimit as their increasing union. More explicitly, a point of $\text{Proj}(A)$ is represented in some $\text{Proj}_e(A)$; after choosing a common multiple m of d and e , the pullback of $\text{Proj}_d(A) \rightarrow \text{Proj}(A)$ is computed by pulling back the open image of $\text{Proj}_d(A) \rightarrow \text{Proj}_m(A)$. It is consequently open, and injectivity follows from injectivity of the transitions. This proves (1).

For (2), let $(g_\alpha)_\alpha$ be homogeneous A_0 -algebra generators with $\deg(g_\alpha) \mid d$. Fix $e \geq 1$ and $f \in A_e$. Write f as a finite sum of homogeneous monomial terms

$$f = t_1 + \cdots + t_s, \quad t_j = c_j \prod_\alpha g_\alpha^{r_{j\alpha}}, \quad c_j \in A_0.$$

On $D(f)$ the values of the t_j generate the quotient line, since their sum does. Thus the loci $D(t_j)$ cover $D(f)$. Moreover, Lemma 2.3.11(3), together with the evident implication that a scalar multiple can generate only if the underlying section generates, gives

$$D(t_j) \subset \bigcap_{r_{j\alpha} > 0} D(g_\alpha).$$

For every generator occurring here, $D(g_\alpha) \simeq D(g_\alpha^{d/\deg(g_\alpha)})$ is a chart of $\text{Proj}_d(A)$ by Lemma 2.3.11(2). Hence every $D(t_j)$, and therefore every $D(f)$, lies in the image of the open immersion $\text{Proj}_d(A) \rightarrow \text{Proj}(A)$. The $D(f)$ with $f \in A_e$ cover $\text{Proj}_e(A)$, and this holds for every e . Since the stages $\text{Proj}_e(A)$ cover their colimit, the open immersion is surjective and therefore an isomorphism. $\square$

2.3.1. Comparison Between Spec and Proj

With the Proj construction in place, we can systematically compare it with Spec.

Remark 2.3.12 (Relationship Between Spec and Proj). For an $\mathbb{N}$ -graded ring A , we have the following canonical diagram over $\text{Spec}(A_0)$:

$$\begin{array}{ccccc} \text{Spec}(A) & \longleftrightarrow & D(A_+) & \longrightarrow & \text{Proj}(A) \\ \downarrow & & \downarrow & & \downarrow \\ \text{Spec}(A_0) & \xlongequal{\quad} & \text{Spec}(A_0) & \xlongequal{\quad} & \text{Spec}(A_0) \end{array}$$

Here:

- (i) $D(A_+) = \text{colim}_{d \geq 1} D(A_d)$ is the *non-vanishing locus of the irrelevant ideal* $A_+ = \bigoplus_{d \geq 1} A_d$ in $\text{Spec}(A)$.
- (ii) The map $D(A_+) \rightarrow \text{Proj}(A)$ is constructed as follows: For $f \in A_d$ with $d \geq 1$, we have $D(f) \subset \text{Spec}(A)$ corresponding to the localization $\text{Spec}(A_f)$. On the other hand, from Lemma 2.3.11, $D(f) \subset \text{Proj}_d(A)$ corresponds to $\text{Spec}(A_{(f)})$. The precise graded relationship is

$$(A^{(d)})_f \simeq A_f^{[d]} \simeq A_{(f)}[u, u^{-1}],$$

where u has regraded degree 1 and maps to f . In general $A_f^{[d]}$ is only the part of A_f in degrees divisible by d , not all of A_f . The inclusion of the degree-zero part $A_{(f)} \hookrightarrow A_f$ nevertheless gives the canonical map $\text{Spec}(A_f) \rightarrow \text{Spec}(A_{(f)})$, hence $D(f) \rightarrow D(f) \subset \text{Proj}_d(A)$.

- (iii) The map $D(f) \rightarrow \text{Proj}_d(A)$ sends an A -algebra homomorphism $\varphi: A \rightarrow R$ to the graded map:

$$\varphi^{(d)}: A^{(d)} \otimes_{A_0} R \rightarrow \bigoplus_{n \in \mathbb{N}} R \cdot \varphi(f)^n \simeq \text{Sym}_R(R),$$

where the right side is identified with the symmetric algebra of the trivial line R with generator $\varphi(f)$. This is precisely an element of $\text{Proj}_d(A)(R)$.

- (iv) $\mathbb{G}_m$ -equivariance: Note that $D(A_+) \subset \text{Spec}(A)$ carries a natural $\mathbb{G}_m$ -action (coming from the grading). The map $D(A_+) \rightarrow \text{Proj}(A)$ is $\mathbb{G}_m$ -equivariant. When $d = 1$ (i.e., A is generated by A_1), the map $D(A_1) \rightarrow \text{Proj}_1(A)$ has image precisely the *trivial quotient lines*, and induces an injection:

$$D(A_1)/\mathbb{G}_m \hookrightarrow \text{Proj}_1(A).$$

Remark 2.3.13 (Extension to Spec). We can extend the domain of the Proj functor to include Spec as follows: For any ring R (viewed as a trivially graded ring concentrated in degree 0), we can define $\text{Proj}(R[x])$ where x has degree 1. Then:

$$\text{Proj}(R[x]) \simeq \text{Proj}_1(R[x]) = \mathbb{P}_R^0 = \text{Spec}(R).$$

This shows that Proj can be viewed as a generalization of Spec.

Remark 2.3.14 (Comparison with Previous Definition). Earlier, we defined projective schemes using homogeneous polynomial systems. How does this relate to the current definition?

For an $\mathbb{N}$ -graded k -algebra A generated by A_1 , we have a canonical surjection:

$$\text{Sym}_{A_0}(A_1) \twoheadrightarrow A.$$

According to the previous definition, hSol_Σ for a homogeneous system Σ is defined as the set of quotient lines $A_1 \otimes_k R \twoheadrightarrow L$ through which the composition $\text{Sym}_k(A_1) \rightarrow A \rightarrow \text{Sym}_R(L)$ factors. By the universal property of the symmetric algebra, this is precisely the definition of $\text{Proj}_1(A)(R)$.

Therefore, the two definitions are completely consistent.

2.3.2. Vanishing Loci and Non-vanishing Loci in Proj

We now introduce vanishing and non-vanishing loci in $\text{Proj}(A)$, the analogues of the corresponding affine constructions.

Definition 2.3.15 (Vanishing Loci and Non-vanishing Loci). Let A be an $\mathbb{N}$ -graded ring and $F \subset A$ a set of homogeneous elements. We define:

- (i) The *vanishing locus* $V(F) \subset \text{Proj}(A)$ is defined as the colimit:

$$V(F) = \text{colim}_{d \geq 1} V_d(F),$$

where for each $d \geq 1$:

$$V_d(F)(R) = \{ \varphi : A^{(d)} \otimes_{A_0} R \twoheadrightarrow \text{Sym}_R(L) \mid \text{for all } f \in F^{(d)}, \varphi(f) = 0 \}.$$

- (ii) The *non-vanishing locus* $D(F) \subset \text{Proj}(A)$ is defined as the colimit:

$$D(F) = \text{colim}_{d \geq 1} D_d(F),$$

where for each $d \geq 1$:

$$D_d(F)(R) = \{ \varphi : A^{(d)} \otimes_{A_0} R \twoheadrightarrow \text{Sym}_R(L) \mid \exists n \geq 1 : \varphi((F)_{nd}) \text{ generates } L^{\otimes n} \}.$$

Here $(F)_{nd}$ denotes the degree nd part of the ideal generated by F in A .

Remark 2.3.16 (Properties of $V(F)$ and $D(F)$). These constructions enjoy properties parallel to the affine case:

- $\bigcap_i V(F_i) = V(\bigcup_i F_i)$.
- $\bigcup_i D(F_i) \subset D(\bigcup_i F_i)$ (not necessarily equality).
- $V(F)$ is determined by the ideal (F) generated by F .
- $D(F)$ is determined by the radical $\sqrt{(F)}$ of the ideal generated by F .
- When A is generated by $A_{\leq 1}$, we have simpler descriptions. For instance, for a single homogeneous element $f \in A_d$:

$$V(f)(R) = \{ \varphi \mid \varphi(f) = 0 \in L^{\otimes d} \},$$

$$D(f)(R) = \{ \varphi \mid \varphi(f) \text{ generates } L^{\otimes d} \}.$$

Remark 2.3.17 (Further Properties). (i) For a homogeneous ideal $I \subset A$, we have $\text{Proj}(A/I) \simeq V(I)$ canonically. This follows from the surjection $A \twoheadrightarrow A/I$ inducing a closed immersion $\text{Proj}(A/I) \hookrightarrow \text{Proj}(A)$.

(ii) For $f \in A_d$ with $d \geq 1$, we have $D(f^n) \simeq D(f) \simeq \text{Spec}(A_{(f)})$ for all $n \geq 1$. This was proved in Lemma 2.3.11.

(iii) For $f \in A_0$, we have $\text{Proj}(A_f) \simeq D(f)$, where $A_f = A[f^{-1}]$ is the ordinary localization (which inherits the $\mathbb{N}$ -grading since f has degree zero). Note that here we use A_f rather than $A_{(f)}$: the latter notation is reserved for the degree-zero part of A_f when $\deg f \geq 1$, where the two would clash.

Proposition 2.3.18 (Closedness and Openness). Let A be an $\mathbb{N}$ -graded ring and $F \subset A$ a set of homogeneous elements.

- (i) $V(F)$ is a closed subfunctor of $\text{Proj}(A)$.

(ii) $D(F)$ is an open subfunctor of $\text{Proj}(A)$.

Proof. We may assume A is generated by $A_{\leq 1}$ (the general case follows from passing to Veronese subrings and using cofinality).

(i) *Closedness of $V(F)$:* We can write $V(F) = \bigcap_{f \in F} V(f)$, so it suffices to show $V(f)$ is closed for a single homogeneous element $f \in A_d$.

For any R -point $x: \text{Spec}(R) \rightarrow \text{Proj}(A)$ corresponding to a quotient line $\varphi: A_1 \otimes_k R \rightarrow L$, we have:

$$x \in V(f) \iff \varphi(f) = 0 \in L^{\otimes d}.$$

Consider the pullback:

$$\begin{array}{ccc} x^{-1}(V(f)) & \longrightarrow & V(f) \\ \downarrow & \lrcorner & \downarrow \\ \text{Spec}(R) & \xrightarrow{x} & \text{Proj}(A) \end{array}$$

It suffices to show $x^{-1}(V(f)) \subset \text{Spec}(R)$ is closed. By definition,

$$x^{-1}(V(f))(S) = \{\psi: R \rightarrow S \mid \psi^*(\varphi(f)) = 0 \in L^{\otimes d} \otimes_R S\}.$$

Since L is an R -line, hence projective, $L^{\otimes d}$ is also projective, hence reflexive. Therefore:

$$\varphi(f) = 0 \iff \text{for all } \lambda \in (L^{\otimes d})^\vee, \lambda(\varphi(f)) = 0.$$

The condition $\lambda(\varphi(f)) = 0$ defines a closed subset $V(\lambda(\varphi(f))) \subset \text{Spec}(R)$ (since $\lambda(\varphi(f)) \in R$ generates an ideal). Therefore:

$$x^{-1}(V(f)) = \bigcap_{\lambda \in (L^{\otimes d})^\vee} V(\lambda(\varphi(f)))$$

is closed.

(ii) *Openness of $D(F)$:* For an R -point x corresponding to $\varphi: A \otimes R \rightarrow \text{Sym}_R(L)$, we have:

$$x \in D(F) \iff \exists n \geq 1 : \varphi((F)_n) \text{ generates } L^{\otimes n},$$

where $(F)_n \subset A_n$ denotes the degree- n part of the ideal (F) generated by F .

This is equivalent to:

$$x^{-1}(D(F)) = \bigcup_{n \geq 1} \text{Epi}(R^{((F)_n)} \rightarrow L^{\otimes n}),$$

where the map sends the basis vector e_h to $\varphi(h) \in L^{\otimes n}$ for each $h \in (F)_n$.

Each $\text{Epi}(R^{((F)_n)} \rightarrow L^{\otimes n})$ is an open subfunctor by Proposition 2.2.19. More compactly,

$$x^{-1}(D(F)) = D\left(\bigcup_{n \geq 1} I_n\right),$$

where $I_n \subset R$ is the radical ideal cut out by the image of $(F)_n$ under φ . Since φ is surjective, $\varphi(A_{n-m})$ generates $L^{\otimes(n-m)}$, so $(F)_m \cdot A_{n-m} \subset (F)_n$ forces $I_m \subset I_n$ for $m \leq n$; the chain $I_1 \subset I_2 \subset \dots$ has union $\bigcup_n I_n$ which is itself an ideal, so $x^{-1}(D(F))$ is the open subfunctor it cuts out.

□

Example 2.3.19 (Non-uniqueness of Vanishing Loci). Unlike the affine case, in the projective case, $V(F) = V(F')$ does *not* imply $(F) = (F')$ in general.

Example: Consider $A = \mathbb{Z}[x, y]$ with the standard grading. We have:

$$V(x) = V(x^2, xy) \subset \mathbb{P}^1.$$

Verification. The inclusion $(x^2, xy) \subset (x)$ gives $V(x) \subset V(x^2, xy)$.

For the reverse inclusion, suppose $\varphi: \mathbb{Z}[x, y] \rightarrow \text{Sym}_R(L)$ satisfies $\varphi(x^2) = 0$ and $\varphi(xy) = 0$ in $L^{\otimes 2}$. Write $\varphi(x) = a$ and $\varphi(y) = b$ where $a, b \in L$ generate L (i.e., $L = \langle a, b \rangle$).

From $\varphi(x^2) = a^{\otimes 2} = 0$ and $\varphi(xy) = a \otimes b = 0$ in $L^{\otimes 2}$, we deduce:

- Since $a^{\otimes 2} = 0$, we have $a \cdot a = 0$ (viewing multiplication in the symmetric algebra).
- Since $a \otimes b = 0$ and $L = \langle a, b \rangle$, we must have $a \cdot L = 0$.

Consequently the map $a \otimes (-): L \rightarrow L^{\otimes 2}$, $c \mapsto a \otimes c$, vanishes on both a and b ; since a, b generate L , this map is zero. Now $a \otimes (-)$ is the image of $a: R \rightarrow L$ under the functor $-\otimes_R L$, which is an equivalence because L is invertible; hence $a \otimes (-) = 0$ forces $a = 0$. (Note that b alone need not generate L — only a, b jointly do — so one cannot directly invert $-\otimes b$.) Hence $\varphi(x) = a = 0$, i.e. $\varphi \in V(x)$. $\square$

This shows that the ideals (x) and (x^2, xy) define the same vanishing locus in $\mathbb{P}^1$, even though they are different ideals.

This phenomenon leads us to introduce the concept of *saturation*.

2.3.3. Saturation of Homogeneous Ideals

Definition 2.3.20 (Saturation). Let A be an $\mathbb{N}$ -graded ring and $H \subset A$ a homogeneous ideal. The *saturation* of H is:

$$H^{\text{sat}} = \{x \in A \mid \text{for all } f \in A_+, \exists n \geq 1 : f^n x \in H\}.$$

We say H is *saturated* if $H = H^{\text{sat}}$.

Remark 2.3.21 (Properties of Saturation). (i) If A is generated by $A_{\leq 1}$, we can replace A_+ with A_1 in the definition of saturation.

(ii) For a radical ideal H (i.e., $\sqrt{H} = H$), the saturation H^{sat} is also radical.

(iii) A homogeneous prime ideal $\mathfrak{p} \subset A$ is saturated if and only if $A_+ \not\subset \mathfrak{p}$.

Example 2.3.22 (Examples of Saturation). (i) For $A = \mathbb{Z}[x, y]$, $(x) = (x^2, xy)^{\text{sat}}$: any $g \in A$ killed by some power of every element of $A_+ = (x, y)$ — equivalently, satisfying $y^n g, x^m g \in (x^2, xy)$ for some n, m — must lie in (x) , by inspecting the y -degree.

(ii) For a polynomial ring $A = k[x_i \mid i \in I]$ with $I \neq \emptyset$ and each variable in positive degree, the zero ideal (0) is saturated. Indeed, fix one variable x_j . If $g \in (0)^{\text{sat}}$, then $x_j^n g = 0$ for some $n \geq 1$; multiplication by the monomial x_j^n is injective on the free k -module of polynomials, even when k has zero divisors, so $g = 0$.

Our goal is to show that $V(H) = V(H')$ if and only if $H^{\text{sat}} = H'^{\text{sat}}$.

The key tool is the theory of *I-nilpotent* and *I-local* modules, which we now develop.

Definition 2.3.23 (*I*-nilpotent and *I*-local Modules). Let R be a ring, $I \subset R$ a subset, and M an R -module.

- (i) An element $x \in M$ is *I-nilpotent* if for all $f \in I$, there exists $n \geq 1$ such that $f^n x = 0$.
- (ii) We denote by $\Gamma_I(M) \subset M$ the *I-nilpotent submodule*, i.e., the submodule consisting of all *I-nilpotent* elements.
- (iii) M is *I-nilpotent* if $\Gamma_I(M) = M$.
- (iv) M is *I-local* if for any homomorphism $h: P \rightarrow Q$ with *I-nilpotent* kernel and cokernel, the induced map:

$$h^*: \operatorname{Hom}_R(Q, M) \xrightarrow{\sim} \operatorname{Hom}_R(P, M)$$

is an isomorphism.

Proposition 2.3.24 (Categories of *I-nilpotent* and *I-local* Modules). *Let R be a ring and $I \subset R$ a subset.*

- (i) $\operatorname{Mod}_R^{I\text{-nil}}$ and $\operatorname{Mod}_R^{I\text{-loc}}$ are Grothendieck abelian categories. For the second category, the abelian structure is the one transported from the Gabriel quotient $\operatorname{Mod}_R / \operatorname{Mod}_R^{I\text{-nil}}$; in particular, its cokernels need not be ambient module cokernels.
- (ii) The inclusion $\iota: \operatorname{Mod}_R^{I\text{-nil}} \hookrightarrow \operatorname{Mod}_R$ has a right adjoint Γ_I , and the inclusion $\iota': \operatorname{Mod}_R^{I\text{-loc}} \hookrightarrow \operatorname{Mod}_R$ has a left adjoint L_I . As a functor into the Gabriel quotient category, L_I is exact. The underlying-module endofunctor $\iota' L_I$ is left exact but need not be right exact. The functor Γ_I is left exact.
- (iii) M is *I-nilpotent* if and only if $L_I(M) = 0$.
- (iv) $\Gamma_I(M) = \ker(M \rightarrow L_I(M))$, and the cokernel of $M \rightarrow L_I(M)$ is *I-nilpotent*.
- (v) For a morphism $h: M \rightarrow N$ in Mod_R , $L_I(h)$ is an isomorphism if and only if $h_f: M_f \rightarrow N_f$ is an isomorphism for every $f \in I$.

Proof. (1) and (2). Let $\mathcal{T} = \operatorname{Mod}_R^{I\text{-nil}}$. The class $\mathcal{T}$ is closed under submodules, quotients, extensions, and arbitrary direct sums: each condition $f^n x = 0$ is preserved by these operations. Thus $\mathcal{T}$ is a localizing Serre subcategory. The Serre quotient construction gives an exact functor

$$Q_I: \operatorname{Mod}_R \longrightarrow \operatorname{Mod}_R / \mathcal{T}$$

with kernel $\mathcal{T}$; see [The25, Tag 02MS]. The Gabriel localization theorem for a localizing subcategory of a Grothendieck abelian category gives a fully faithful right adjoint S_I to Q_I . Its essential image consists exactly of the *I-local* modules of Definition 2.3.23. Consequently $\operatorname{Mod}_R^{I\text{-loc}} \simeq \operatorname{Mod}_R / \mathcal{T}$ is Grothendieck abelian, and

$$L_I = S_I Q_I: \operatorname{Mod}_R \longrightarrow \operatorname{Mod}_R^{I\text{-loc}}$$

is the left adjoint to the inclusion. It is exact when its target has this quotient exact structure. Since S_I is a right adjoint, the composite underlying-module functor is left exact; it is not generally right exact.

For example, let $R = k[x, y]$ and $I = \{x, y\}$. For $q: R^2 \rightarrow R$, $(a, b) \mapsto ax + by$, one has $L_I(R) = R$, $L_I(R^2) = R^2$, and $L_I(R/(x, y)) = 0$. Applying the underlying-module reflector to $R^2 \xrightarrow{q} R \rightarrow R/(x, y) \rightarrow 0$ therefore leaves the nonsurjective map $R^2 \xrightarrow{q} R$. This rules out ambient right exactness.

Finally, $\Gamma_I(M)$ is the maximal *I-nilpotent* submodule of M , so $\iota \dashv \Gamma_I$; as a right adjoint it is left exact. It need not be right exact: for $R = \mathbb{Z}$ and $I = \{p\}$, apply it to $\mathbb{Z} \xrightarrow{p} \mathbb{Z} \twoheadrightarrow \mathbb{Z}/p$.

(3). Since Q_I is exact with kernel $\mathcal{T}$, $L_I(M) = 0$ if and only if M is *I-nilpotent*. To see the orthogonality directly, if T is *I-nilpotent* and N is *I-local*, apply the definition to the map $0 \rightarrow T$: it gives $\operatorname{Hom}_R(T, N) \simeq \operatorname{Hom}_R(0, N) = 0$.

(4). The unit $\eta_M: M \rightarrow L_I(M)$ is an L_I -isomorphism, so its kernel and cokernel are *I-nilpotent* — this gives the second assertion. For the first, $\ker(\eta_M)$ is an *I-nilpotent* submodule of M , hence $\ker(\eta_M) \subset \Gamma_I(M)$. Conversely, the composite $\Gamma_I(M) \hookrightarrow M \xrightarrow{\eta_M} L_I(M)$ factors through $L_I(\Gamma_I(M)) = 0$ (by (3)), so $\Gamma_I(M) \subset \ker(\eta_M)$.

(5). Exactness of Q_I shows that $L_I(h)$ is an isomorphism exactly when $\ker(h)$ and $\operatorname{coker}(h)$ are I -nilpotent. For any module T ,

$$T \text{ is } I\text{-nilpotent} \iff T_f = 0 \text{ for every } f \in I :$$

the reverse implication says precisely that each element of T is killed by some power of each f . Since localization is exact, the kernel and cokernel condition is equivalent to every $h_f: M_f \rightarrow N_f$ being an isomorphism. $\square$

Remark 2.3.25 (Explicit Description for Finite I). When $I = \{f_1, \dots, f_n\}$ is finite, $\operatorname{Mod}_R^{I\text{-loc}}$ can be identified with the category of quasi-coherent modules on the open subset $D(I) \subset \operatorname{Spec}(R)$.

In this case, $L_I(M)$ can be computed as the equalizer:

$$L_I(M) \rightarrow \prod_{i=1}^n M_{f_i} \rightrightarrows \prod_{i,j} M_{f_i f_j}.$$

In particular, $\mathcal{O}(D(I)) = L_I(R)$.

Example 2.3.26 (Special Cases). (i) When $I = \{f\}$ consists of a single element, “ I -nilpotent” means “ f -torsion”, and $L_I(M) = M_f$ (the localization at f).

(ii) For an $\mathbb{N}$ -graded ring A and homogeneous ideal $H \subset A$, we have:

$$H^{\operatorname{sat}} = \ker(A \rightarrow A/H \rightarrow L_{A_+}(A/H)).$$

(iii) In particular, $(0)^{\operatorname{sat}} = \Gamma_{A_+}(A)$.

We can now state the key proposition relating Proj to I -local modules.

Proposition 2.3.27 (Characterization of Proj via I -local Modules). Let A, B be $\mathbb{N}$ -graded rings and $\alpha: A \rightarrow B$ an eventually surjective graded map. Then the following are equivalent:

(i) $\operatorname{Proj}(B) \rightarrow \operatorname{Proj}(A)$ is an isomorphism.

(ii) For all $d \geq 1$, the map:

$$L_{A_d}(A^{(d)}) \rightarrow L_{A_d}(B^{(d)})$$

is an isomorphism.

If α is surjective, these are also equivalent to:

(3) For all $d \geq 1$, $\ker(A^{(d)} \rightarrow B^{(d)})$ is A_d -nilpotent as an $A^{(d)}$ -module.

If A is generated by $A_{\leq 1}$, then it suffices to check condition (2) or (3) for $d = 1$ only.

Proof. For $d \geq 1$ and $f \in A_d$, the inverse image of the chart $D(f) \subset \operatorname{Proj}_d(A)$ is $D(\alpha(f))$, and Lemma 2.3.11 identifies the induced chart map with

$$\operatorname{Spec}(B_{(\alpha(f))}) \longrightarrow \operatorname{Spec}(A_{(f)}).$$

The charts for all d and $f \in A_d$ cover $\operatorname{Proj}(A)$. Hence, by Zariski descent, $\operatorname{Proj}(B) \rightarrow \operatorname{Proj}(A)$ is an isomorphism if and only if

$$A_{(f)} \longrightarrow B_{(\alpha(f))} \tag{*}$$

is an isomorphism for every such d and f .

The corrected Laurent description of Notation 2.2.6 gives canonical graded isomorphisms

$$(A^{(d)})_f \simeq A_f^{[d]} \simeq A_{(f)}[u^{\pm 1}], \quad (B^{(d)})_{\alpha(f)} \simeq B_{(\alpha(f))}[u^{\pm 1}],$$

compatible with α and with $u \mapsto f$, respectively $u \mapsto \alpha(f)$. Thus $(*)$ is an isomorphism if and only if

$$(A^{(d)})_f \longrightarrow (B^{(d)})_{\alpha(f)}$$

is an isomorphism. By Proposition 2.3.24(5), this holds for every $f \in A_d$ if and only if

$$L_{A_d}(A^{(d)}) \longrightarrow L_{A_d}(B^{(d)})$$

is an isomorphism. This proves $(1) \Leftrightarrow (2)$.

If α is surjective, so is $A^{(d)} \rightarrow B^{(d)}$. Exactness of the Gabriel quotient, together with Proposition 2.3.24(3), says that its image under L_{A_d} is an isomorphism exactly when its kernel is A_d -nilpotent. This proves the equivalence with (3).

Finally, if A is generated by $A_{\leq 1}$, then $\text{Proj}(A) = \text{Proj}_1(A)$ by Theorem 2.3.6, and the charts $D(f)$ for $f \in A_1$ already cover it. The same argument therefore requires only $d = 1$. $\square$

Corollary 2.3.28 (Functorial Projective Nullstellensatz). *For a ring k , the map $F \mapsto V(F)$ induces an order-reversing bijection:*

$$\{\text{saturated homogeneous ideals in } k[x_i \mid i \in I]\} \xrightarrow{\sim} \{\text{vanishing loci in } \mathbb{P}_k^I\}.$$

Proof. Set $A = k[x_i \mid i \in I]$ and let $H, H' \subset A$ be homogeneous ideals. We show that $V(H) = V(H')$ iff $H^{\text{sat}} = H'^{\text{sat}}$.

Suppose first that $V(H) = V(H')$, and put $K = H + H'$. Then $V(K) = V(H) \cap V(H')$ equals both $V(H)$ and $V(H')$. The two surjections

$$A/H \twoheadrightarrow A/K, \quad A/H' \twoheadrightarrow A/K$$

therefore induce isomorphisms on Proj . By Proposition 2.3.27, their kernels are A_+ -nilpotent. Equivalently, $K \subset H^{\text{sat}}$ and $K \subset (H')^{\text{sat}}$. Thus $H' \subset H^{\text{sat}}$ and $H \subset (H')^{\text{sat}}$; saturating gives $H^{\text{sat}} = (H')^{\text{sat}}$.

Conversely, the surjection $A/H \twoheadrightarrow A/H^{\text{sat}}$ has A_+ -nilpotent kernel, so Proposition 2.3.27 gives $V(H) = V(H^{\text{sat}})$. The same holds for H' , and equality of the two saturations therefore implies $V(H) = V(H')$. $\square$

We can now classify the open and closed subfunctors of $\text{Proj}(A)$.

Proposition 2.3.29 (Classification of Open and Closed Subfunctors). *Let A be an $\mathbb{N}$ -graded ring generated by $A_{\leq 1}$.*

(i) *The map $F \mapsto V(F)$ induces an order-reversing injection:*

$$\{\text{saturated homogeneous ideals in } A\} \hookrightarrow \{\text{closed subfunctors of } \text{Proj}(A)\}.$$

This is a bijection if A is finitely generated over A_0 .

(ii) *The map $F \mapsto D(F)$ induces an order-preserving bijection:*

$$\{\text{saturated homogeneous radical ideals in } A\} \xrightarrow{\sim} \{\text{open subfunctors of } \text{Proj}(A)\}.$$

Proof. (i) *Injectivity* follows from the same $H + H'$ argument used in the proof of Corollary 2.3.28, now applied directly to the two surjections from A .

Surjectivity (assuming A finitely generated over A_0). Pick a finite generating set $F \subset A_1$, so that $\{D(f)\}_{f \in F}$ is an affine open cover of $\text{Proj}(A)$. For each $f \in F$, pull Z back along $D(f) \simeq \text{Spec}(A_{(f)}) \hookrightarrow \text{Proj}(A)$ to obtain a closed subfunctor $Z_f \subset \text{Spec}(A_{(f)})$, which by the affine Nullstellensatz has the form $Z_f = V(I_f)$ for a unique ideal $I_f \subset A_{(f)}$.

Now lift each I_f back to A . Since here $f \in A_1$, all degrees belong to $\mathbb{Z} \cdot \deg(f)$ and the corrected localization formula gives $A_f = A_f^{[1]} \simeq A_{(f)}[t^{\pm 1}]$. Thus the homogeneous ideal $\tilde{I}_f := I_f[t^{\pm 1}] \subset A_f$ is the unique homogeneous ideal whose degree-zero part is I_f . Set

$$H(f) := \ker(A \longrightarrow A_f / \tilde{I}_f), \quad H := \bigcap_{f \in F} H(f).$$

Each $H(f)$ is homogeneous and f -saturated ($x \in H(f) \iff f^n x \in H(f)$ for some $n \geq 0$), since it is pulled back from a localization at f .

$Z = V(H)$. Fix $f \in F$. On each overlap $D(fg)$, $g \in F$, compatibility of the closed subfunctors Z_f and Z_g says that the localized homogeneous ideals $\tilde{I}_f$ and $\tilde{I}_g$ agree. The opens $D(fg)$ cover $D(f)$, so descent for ideals gives

$$H_f = \left(\bigcap_{g \in F} H(g) \right)_f = \bigcap_{g \in F} H(g)_f = \tilde{I}_f;$$

here the middle equality uses finiteness of F . Thus $V(H) \cap D(f) = V(I_f) = Z_f$ on every chart. Zariski descent for Proj (Proposition 2.3.10) now gives $V(H) = Z$.

H is saturated. It is enough to test homogeneous x . If $x \in H^{\text{sat}}$, then for every $f \in F$ some power $f^{n(f)}x$ lies in $H \subset H(f)$. By f -saturation of $H(f)$ this forces $x \in H(f)$ for every f , hence $x \in H$.

- (ii) *Injectivity.* The key observation is that for any homogeneous subset $F \subset A_+$ and homogeneous element $a \in A_+$,

$$D(a) \subset D(F) \iff a \in \sqrt{(F)}^{\text{sat}}.$$

The implication “ $\Leftarrow$ ” is immediate, since $D(\cdot)$ is preserved by sums, powers, and saturation. For “ $\Rightarrow$ ”: pulling back to $D(a) \simeq \text{Spec}(A_{(a)})$, the hypothesis says that the ideal generated by the degree-zero homogeneous fractions coming from (F) is the unit ideal of $A_{(a)}$. A finite unit relation, followed by clearing the powers of a in its denominators (and then clearing any a -torsion), gives $a^N \in (F)$ for some $N \geq 1$. In particular $a \in \sqrt{(F)} \subset \sqrt{(F)}^{\text{sat}}$.

Now $D(F) = D(F')$ forces $a \in \sqrt{(F')}^{\text{sat}}$ for every positive-degree $a \in F$, and symmetrically. Saturating gives equality of the saturated radicals generated by the positive-degree parts. For saturated ideals in a ring generated by $A_{\leq 1}$, the positive-degree part determines the degree-zero part as well: if $a \in A_0$ and aA_1 lies in the ideal, saturation forces a itself to lie in it. Thus $\sqrt{(F)}^{\text{sat}} = \sqrt{(F')}^{\text{sat}}$ without losing degree-zero elements.

Surjectivity. Let $U \subset \text{Proj}(A)$ be open. Define a graded subset degree by degree by

$$H_d := \{a \in A_d \mid D(a) \subset U\}, \quad H := \bigoplus_{d \geq 0} H_d.$$

This is a homogeneous ideal. Indeed, multiplication is handled by $D(ca) = D(c) \cap D(a)$ for homogeneous a, c ; and if $a, b \in H_d$, then on $D(a+b)$ the two sections a and b generate the same trivialized line jointly, so $D(a+b)$ is covered by $D(a)$ and $D(b)$ and is contained in U .

The standard opens $D(a)$ form a basis for $\text{Proj}(A)$. To see this in the present notation, work inside a chart $D(f) \simeq \text{Spec}(A_{(f)})$ with $f \in A_1$: a principal affine open chosen inside $U \cap D(f)$ and defined by a fraction $a/f^n \in A_{(f)}$ is $D(a) \cap D(f) = D(af)$, hence is itself a standard open contained in U . Consequently U is the sheaf-theoretic union of the $D(a)$ contained in it, and hence $U = D(H)$.

Finally, H is radical: if a homogeneous a has $a^n \in H$, then $D(a) = D(a^n) \subset U$. It is saturated as well. If a homogeneous a lies in H^{sat} , then for every $f \in A_1$ some power $f^{n(f)}a$ lies in H , so

$$D(a) \cap D(f) = D(f^{n(f)}a) \subset U.$$

The $D(f)$, $f \in A_1$, cover $\text{Proj}(A)$, whence $D(a) \subset U$ and $a \in H$. Thus H is a saturated homogeneous radical ideal, completing surjectivity. $\square$

Remark 2.3.30 (General $\mathbb{N}$ -graded Rings). For an $\mathbb{N}$ -graded ring A that is *not* generated by $A_{\leq 1}$, saturation must be controlled simultaneously at all the Veronese subrings $A^{(d)}$. Concretely, declare $H \subset A$ to be *saturated* when

$$H = \ker\left(A \rightarrow \prod_{d \geq 1} L_{A_d}(A/H)\right),$$

i.e. a homogeneous element lies in H iff its image in A/H is annihilated by some power of A_d for every $d \geq 1$. With this definition, Proposition 2.3.29 continues to hold for general $\mathbb{N}$ -graded A .

2.3.4. Quasi-projective Schemes

Just as quasi-affine schemes are open subfunctors of affine ones, quasi-projective schemes are open subfunctors of projective ones. By Proposition 2.3.29(2), every such open subfunctor of $\text{Proj}(A)$ has the form $D(F)$ for a homogeneous subset $F \subset A_+$. One may take F finite precisely when the open is quasi-compact; finiteness is therefore imposed explicitly in the following definition.

Definition 2.3.31 (Quasi-projective Schemes). Let k be a ring. An algebraic k -functor X is a *quasi-projective k -scheme* if there exist an $\mathbb{N}$ -graded k -algebra A finitely generated over A_0 and a finite homogeneous subset $F \subset A_+$ with $X \simeq D(F) \subset \text{Proj}(A)$. Write $\text{QProj}_k \subset \text{Fun}(\text{CAlg}_k, \text{Set})$ for the full subcategory of quasi-projective k -schemes.

Remark 2.3.32 (Relations Among the Four Categories). The valid inclusions are

$$\text{Aff}_k \subset \text{QAff}_k \subset \text{QProj}_k, \quad \text{Proj}_k \subset \text{QProj}_k.$$

Indeed, $\text{Spec}(R) \simeq \text{Proj}(R[t]) = D(t)$ for $\deg(t) = 1$, and a principal open $D(f) \subset \text{Spec}(R)$ corresponds to $D(ft) \subset \text{Proj}(R[t])$; a finite union gives the inclusion $\text{QAff}_k \subset \text{QProj}_k$ under our definition. This does *not* give $\text{Aff}_k \subset \text{Proj}_k$: the displayed Proj is relative to $\text{Spec}(R) = \text{Spec}((R[t])_0)$, not a closed embedding into a finite projective space over k . For example, $\mathbb{A}_k^1$ is not projective over k . The categories are also distinct in the other direction: $\mathbb{P}_k^1$ is projective but not quasi-affine.

2.4. Projective Closure

Recall from Example 2.2.27 that the affine space $\mathbb{A}^I$ embeds into $\mathbb{P}^{I \sqcup \{0\}}$ as the non-vanishing locus $D(x_0)$. This section compares vanishing and non-vanishing loci across this embedding.

Proposition 2.4.1. Let I be a set not containing 0, let $F \subset k[x_0, x_i \mid i \in I]$ be a homogeneous subset, and let $F_0 \subset k[x_i \mid i \in I]$ be obtained from F by setting $x_0 = 1$. Consider the vanishing loci $V(F) \subset \mathbb{P}_k^{I \sqcup \{0\}}$ and $V(F_0) \subset \mathbb{A}_k^I$ and the nonvanishing loci $D(F) \subset \mathbb{P}_k^{I \sqcup \{0\}}$ and $D(F_0) \subset \mathbb{A}_k^I$. Then

$$V(F) \cap \mathbb{A}_k^I = V(F_0) \quad \text{and} \quad D(F) \cap \mathbb{A}_k^I = D(F_0).$$

Proof. We calculate the R -points for an arbitrary ring R . Recall that an R -point of $\mathbb{P}_k^{I \sqcup \{0\}}$ is a tuple $(L, s_0, (s_i)_{i \in I})$ where L is an R -line and s_0 together with $(s_i)_{i \in I}$ are sections of L generating it.

The intersection with $\mathbb{A}_k^I \simeq D(x_0)$ corresponds to the subfunctor of points where the section s_0 generates L . If s_0 generates L , the map $R \rightarrow L$ defined by $1 \mapsto s_0$ is an isomorphism. Under this trivialization $L \simeq R$, the sections correspond to elements in R :

$$s_0 \mapsto 1, \quad s_i \mapsto a_i \in R.$$

Thus, R -points of $\mathbb{A}_k^I$ are identified with tuples $(1, (a_i)_{i \in I}) \in R \times R^I$.

Now consider a homogeneous polynomial $f \in F$ of degree d . Evaluated at such a point, $\varphi(f) \in L^{\otimes d}$ corresponds under the trivialization $L^{\otimes d} \simeq R$ (induced by $s_0^{\otimes d} \mapsto 1$) to the element:

$$f(1, (a_i)_{i \in I}) \in R.$$

By definition of F_0 , this value is exactly $f_0((a_i)_{i \in I})$.

We now check the conditions for $V(F)$ and $D(F)$ (via Remark 2.3.16):

- **For $V(F)$:** A point $\varphi \in \mathbb{A}_k^I(R)$ lies in $V(F)(R)$ if and only if $\varphi(f) = 0$ in $L^{\otimes \deg(f)}$ for all $f \in F$. Under the trivialization, this is equivalent to $f_0((a_i)) = 0$ in R for all $f_0 \in F_0$. This is precisely the definition of $V(F_0)(R)$.
- **For $D(F)$:** Use s_0 to identify $L^{\otimes \deg(f)}$ with R separately for every $f \in F$. A point $\varphi \in \mathbb{A}_k^I(R)$ lies in $D(F)(R)$ if and only if the resulting scalars $\{\varphi(f)\}_{f \in F}$ generate the unit ideal in R . Under these trivializations, this means the ideal generated by $\{f_0((a_i))\}_{f_0 \in F_0}$ is the unit ideal in R (i.e., $1 \in (f_0((a_i)))$). This is precisely the definition of $D(F_0)(R)$.

Since this holds for any ring R , the functors are isomorphic. $\square$

To describe the projective closure of an affine vanishing locus, we first record the two relevant notions: the closure of a subfunctor, and the homogenization of an ideal.

Definition 2.4.2 (Closure of subfunctor). Let X be an algebraic functor and $Y \subset X$ a subfunctor. The *closure* of Y in X is the smallest closed subfunctor of X containing Y .

Definition 2.4.3 (Homogenization of Polynomials). Let k be a ring and I a set (not containing 0).

- (i) Set $0^h = 0$. For a nonzero polynomial $f \in k[x_i \mid i \in I]$, let d be the maximal degree of a monomial with nonzero coefficient in f . The *homogenization* of f is the homogeneous polynomial

$$f^h = x_0^d f \left(\left(\frac{x_i}{x_0} \right)_{i \in I} \right) \in k[x_0, x_i \mid i \in I],$$

where equivalently d is the smallest integer for which the above expression is a polynomial.

- (ii) Let $F \subset k[x_i \mid i \in I]$ be an ideal. The *homogenization* of F is the homogeneous ideal

$$F^h = (f^h \mid f \in F) \subset k[x_0, x_i \mid i \in I].$$

Theorem 2.4.4 (Projective closure via homogenization). Let I be finite and let $F_0 \subset k[x_i \mid i \in I]$ be an ideal. Under the standard chart $\mathbb{A}_k^I = D(x_0) \subset \mathbb{P}_k^{I \sqcup \{0\}}$, the scheme-theoretic closure of $V(F_0)$ is

$$\overline{V(F_0)} = V(F_0^h).$$

Proof. Setting $x_0 = 1$ in every homogenized equation recovers F_0 , so Proposition 2.4.1 gives

$$V(F_0^h) \cap D(x_0) = V(F_0).$$

Thus $V(F_0^h)$ is one closed subfunctor containing $V(F_0)$, and it remains to prove minimality.

Let Z be any closed subfunctor of $\mathbb{P}_k^{I \sqcup \{0\}}$ containing $V(F_0)$. Since I is finite, Proposition 2.3.29 writes $Z = V(G)$ for a saturated homogeneous ideal $G \subset k[x_0, x_i \mid i \in I]$. On the chart $D(x_0)$, dehomogenize at $x_0 = 1$. The inclusion $V(F_0) \subset V(G) \cap D(x_0)$, together with the functorial affine Nullstellensatz Corollary 1.2.11, implies

$$(g(1, (x_i)) \mid g \in G \text{ homogeneous}) \subset F_0.$$

If $0 \neq g \in G$ is homogeneous of degree d , put $\bar{g} = g(1, (x_i))$ and $e = \deg(\bar{g})$. Homogeneity gives

$$g = x_0^{d-e} \bar{g}^h.$$

Since $\bar{g} \in F_0$, its homogenization belongs to F_0^h , and hence $g \in F_0^h$. Therefore $G \subset F_0^h$, so $V(F_0^h) \subset V(G) = Z$. This holds for every closed Z containing $V(F_0)$, proving the claimed minimality. $\square$

2.4.1. Examples of Projective Schemes

We present three classical constructions — the Segre embedding, Grassmannians, and flag schemes — illustrating how the functor-of-points language handles them.

Example 2.4.5 (Segre Embedding). Let k be a ring and let M and N be k -modules. The **Segre embedding** is the map of algebraic k -functors:

$$\varsigma : \mathbb{P}(M) \times \mathbb{P}(N) \rightarrow \mathbb{P}(M \otimes_k N)$$

that sends a pair of quotient lines $M \otimes_k R \twoheadrightarrow K$ and $N \otimes_k R \twoheadrightarrow L$ to their tensor product $(M \otimes_k N) \otimes_k R \twoheadrightarrow K \otimes_R L$. This is well-defined since the tensor product $\otimes$ preserves lines and epimorphisms.

If $M = k^{(I)}$ and $N = k^{(J)}$, the Segre embedding is a map:

$$\varsigma : \mathbb{P}_k^I \times \mathbb{P}_k^J \rightarrow \mathbb{P}_k^{I \times J}.$$

If $M = k^{m+1}$ and $N = k^{n+1}$, this becomes the map:

$$\varsigma : \mathbb{P}_k^m \times \mathbb{P}_k^n \rightarrow \mathbb{P}_k^{mn+m+n},$$

which is given in coordinates by the formula:

$$([a_0 : \dots : a_m], [b_0 : \dots : b_n]) \mapsto [\text{all products } a_i b_j].$$

More generally, we have a Segre embedding:

$$\varsigma : \prod_{i=1}^n \mathbb{P}(M_i) \rightarrow \mathbb{P}\left(\bigotimes_{i=1}^n M_i\right)$$

for any finite family of k -modules $M_1, \dots, M_n$.

Claim 2.4.6. *The Segre embedding is a closed immersion.*

Proof of Claim. Step 1: Reduction to free modules. Choose free modules M', N' and surjections $M' \twoheadrightarrow M$ and $N' \twoheadrightarrow N$. These induce closed immersions $\mathbb{P}(M) \hookrightarrow \mathbb{P}(M')$ and $\mathbb{P}(N) \hookrightarrow \mathbb{P}(N')$. Consider the following commutative diagram:

$$\begin{array}{ccc} \mathbb{P}(M) \times \mathbb{P}(N) & \xrightarrow{\zeta} & \mathbb{P}(M \otimes N) \\ \downarrow & & \downarrow \\ \mathbb{P}(M') \times \mathbb{P}(N') & \xrightarrow{\zeta'} & \mathbb{P}(M' \otimes N') \end{array}$$

Both vertical arrows are closed immersions, being products of closed immersions. If ζ' is a closed immersion (for free modules), then the composite $\mathbb{P}(M) \times \mathbb{P}(N) \rightarrow \mathbb{P}(M' \otimes N')$ is also a closed immersion, and since the right vertical arrow is separated, ζ itself is a closed immersion (cancellation for closed immersions through a separated map). It thus suffices to treat the free case.

Step 2: Local verification. Assume $M = k^{(I)}$ and $N = k^{(J)}$ with bases $(e_i)_{i \in I}$ and $(f_j)_{j \in J}$. Then $M \otimes N$ is free with basis $(e_i \otimes f_j)$. The property of being a closed immersion is local on the target (Proposition 1.3.11). The target $\mathbb{P}(M \otimes N)$ is covered by the standard affine open subfunctors $D(e_i \otimes f_j)$ for $(i, j) \in I \times J$. It suffices to show that for each pair (i, j) , the restriction

$$\zeta_{ij}: \zeta^{-1}(D(e_i \otimes f_j)) \rightarrow D(e_i \otimes f_j)$$

is a closed immersion.

Step 3: Coordinate description. Fix indices $i \in I$ and $j \in J$. The open subfunctor $D(e_i \otimes f_j)$ is isomorphic to the affine space $\mathbb{A}^{(I \times J) \setminus \{(i, j)\}}$. Its coordinate ring is the polynomial ring in variables z_{kl} where $(k, l) \neq (i, j)$, corresponding to the ratio $\frac{e_k \otimes f_l}{e_i \otimes f_j}$.

The preimage is given by:

$$\zeta^{-1}(D(e_i \otimes f_j)) = D(e_i) \times D(f_j) \simeq \mathbb{A}^{I \setminus \{i\}} \times \mathbb{A}^{J \setminus \{j\}}.$$

Its coordinate ring is $k[x_k \mid k \neq i] \otimes k[y_l \mid l \neq j]$, corresponding to ratios $x_k = \frac{e_k}{e_i}$ and $y_l = \frac{f_l}{f_j}$.

The morphism ζ_{ij} corresponds to the ring homomorphism:

$$\begin{aligned} \varphi: k[z_{kl} \mid (k, l) \neq (i, j)] &\rightarrow k[x_k, y_l \mid k \neq i, l \neq j] \\ z_{kl} &\mapsto x_k y_l. \end{aligned}$$

(Here we adopt the convention $x_i = 1$ and $y_j = 1$).

To check that ζ_{ij} is a closed immersion, we must show that φ is surjective. We check the generators of the target ring:

- For any $k \neq i$, we have $\varphi(z_{kj}) = x_k y_j = x_k \cdot 1 = x_k$. Thus $x_k \in \text{im}(\varphi)$.
- For any $l \neq j$, we have $\varphi(z_{il}) = x_i y_l = 1 \cdot y_l = y_l$. Thus $y_l \in \text{im}(\varphi)$.

Since all generators x_k and y_l are in the image, φ is surjective. Therefore, ζ is a closed immersion. $\square$

Corollary 2.4.7 (Products of Projective Schemes). *Let X and Y be projective k -schemes. Then $X \times Y$ is a projective k -scheme.*

Proof. Since X and Y are projective, there exist closed immersions $X \hookrightarrow \mathbb{P}(M)$ and $Y \hookrightarrow \mathbb{P}(N)$ for some k -modules M and N . The product map:

$$X \times Y \hookrightarrow \mathbb{P}(M) \times \mathbb{P}(N) \xrightarrow{\zeta} \mathbb{P}(M \otimes_k N)$$

is a closed immersion (as a composition of closed immersions), showing that $X \times Y$ is projective. $\square$

Remark 2.4.8 (Veronese via Segre). The Veronese embedding (Example 2.3.9) factors through the Segre embedding: for any k -module M and $d \geq 1$, there is a commutative square

$$\begin{array}{ccc} \mathbb{P}(M) & \xrightarrow{\rho_d} & \mathbb{P}(\mathrm{Sym}_k^d(M)) \\ \downarrow \Delta & & \downarrow \\ \mathbb{P}(M)^{\times d} & \xrightarrow{\varsigma} & \mathbb{P}(M^{\otimes d}) \end{array}$$

where the left vertical map is the diagonal and the right vertical map is the closed immersion induced by the quotient $M^{\otimes d} \twoheadrightarrow \mathrm{Sym}_k^d(M)$. This gives another proof that ρ_d is a closed immersion.

Replacing quotient lines by quotient spaces of arbitrary rank yields the Grassmannian.

Definition 2.4.9 (Grassmannian). Let k be a ring, M a k -module, and $n \in \mathbb{N}$. The **rank n Grassmannian** of M is the algebraic k -functor $\mathrm{Gr}_n(M)$ given by:

$$\mathrm{Gr}_n(M)(R) = \{\text{quotient spaces of } M \otimes_k R \text{ of constant rank } n\}.$$

Remark 2.4.10 (Properties of Grassmannians). (i) By definition, $\mathrm{Gr}_1(M) = \mathbb{P}(M)$.

(ii) If M is a vector space of constant rank r , then duality induces isomorphisms:

$$\mathrm{Gr}_n(M) \simeq \mathrm{Gr}_{r-n}(M^\vee)$$

Example 2.4.11 (Plücker Embedding). Let M be a k -module and let $n \in \mathbb{N}$. The **Plücker embedding** is the map of algebraic k -functors:

$$\omega: \mathrm{Gr}_n(M) \rightarrow \mathbb{P}(\Lambda_k^n(M))$$

that sends a quotient space of rank n to its n -th exterior power, which is a quotient line.

Claim 2.4.12. *The Plücker embedding is a closed immersion.*

Proof of Claim. Step 1: Reduction to free modules. For any surjective map $M' \twoheadrightarrow M$, we have a commutative diagram:

$$\begin{array}{ccc} \mathrm{Gr}_n(M) & \xrightarrow{\omega} & \mathbb{P}(\Lambda_k^n(M)) \\ \downarrow & & \downarrow \\ \mathrm{Gr}_n(M') & \xrightarrow{\omega'} & \mathbb{P}(\Lambda_k^n(M')) \end{array}$$

The vertical maps are closed immersions. For projective spaces, the surjection $M' \twoheadrightarrow M$ induces a surjection $\Lambda_k^n(M') \twoheadrightarrow \Lambda_k^n(M)$, whence $\mathbb{P}(\Lambda_k^n(M)) \hookrightarrow \mathbb{P}(\Lambda_k^n(M'))$ is a closed immersion. For Grassmannians, set $K' = \ker(M' \twoheadrightarrow M)$; a rank- n quotient $q: M' \otimes_k R \twoheadrightarrow Q$ lies in the subfunctor $\mathrm{Gr}_n(M) \subset \mathrm{Gr}_n(M')$ precisely when q annihilates the image of $K' \otimes_k R$, i.e. when the composite $K' \otimes_k R \rightarrow M' \otimes_k R \xrightarrow{q} Q$ vanishes. This is a closed condition (the vanishing of the induced section of $\mathcal{H}om(K', Q)$), so $\mathrm{Gr}_n(M) \hookrightarrow \mathrm{Gr}_n(M')$ is a closed immersion, established independently of the Plücker map. If ω' is a closed immersion, then the composite $\mathrm{Gr}_n(M) \rightarrow \mathbb{P}(\Lambda_k^n(M'))$ is a closed immersion. Since the right vertical map is a separated morphism, this implies ω is a closed immersion. Thus, we may assume M is a free module with basis $(e_i)_{i \in I}$.

Step 2: Local verification. Let $(e_J)_{J \in \binom{I}{n}}$ be the induced basis for $\Lambda_k^n(M)$. The target $\mathbb{P}(\Lambda_k^n(M))$ is covered by the affine open subfunctors $D(e_J)$. By Proposition 1.3.11 (Closed immersion is local on the target), it suffices to show that for each $J \in \binom{I}{n}$, the restriction

$$\omega_J: \omega^{-1}(D(e_J)) \rightarrow D(e_J)$$

is a closed immersion.

Step 3: Coordinate description. Fix a subset $J \subset I$ of size n . The open subfunctor $D(e_J) \subset \mathbb{P}(\Lambda_k^n(M))$ is the affine space with coordinate ring generated by variables X_K (representing x_K/x_J) for $K \in \binom{I}{n} \setminus \{J\}$.

An R -point of the preimage is a locally split quotient $q: R^{(I)} \twoheadrightarrow Q$ of constant rank n for which

$$q(e_{j_1}) \wedge \cdots \wedge q(e_{j_n})$$

trivializes $\det(Q)$. Equivalently, the restriction $q|_{R^J}: R^J \rightarrow Q$ is an isomorphism. Using this isomorphism to identify Q with R^J , the quotient q has a unique matrix whose columns indexed by J form the identity and whose remaining columns are arbitrary:

$$A = \begin{pmatrix} I_n & Z \end{pmatrix}.$$

Thus this chart of the quotient Grassmannian is the affine space $\text{Mat}_{J \times (I \setminus J)}$, with coordinates z_{ji} for $j \in J$ and $i \notin J$.

The Plücker embedding sends q to the quotient line $\Lambda^n R^{(I)} \twoheadrightarrow \det(Q)$; after the displayed trivialization, its coordinates are the $n \times n$ column minors of A . The restriction ω_J corresponds to the ring homomorphism:

$$\varphi: k[X_K \mid K \neq J] \longrightarrow k[z_{ji} \mid j \in J, i \notin J],$$

where $\varphi(X_K)$ is the minor of A using the columns indexed by K , normalized so that the J -minor is 1.

Step 4: Surjectivity of ring map. To prove ω_J is a closed immersion, we show φ is surjective by finding preimages for the generators z_{ji} . Fix $i_0 \in I \setminus J$ and $j_0 \in J$, and put $K = (J \setminus \{j_0\}) \cup \{i_0\}$. The corresponding column minor replaces the identity column indexed by j_0 with the i_0 -column of Z . Its determinant is $\pm z_{j_0 i_0}$. Consequently $\varphi(X_K) = \pm z_{j_0 i_0}$. Every coordinate generator occurs in this way, so φ is surjective.

Hence, ω_J is a closed immersion, and so is ω .

□

Definition 2.4.13 (Flag Schemes). Let M be a k -module and let $\mathbf{n} = (n_1, \dots, n_s)$ be an increasing sequence of natural numbers.

- (i) A **flag of type $\mathbf{n}$** on M is a sequence of quotient modules:

$$M \twoheadrightarrow F_s \twoheadrightarrow \cdots \twoheadrightarrow F_1$$

where each F_i is a vector space of constant rank n_i .

- (ii) The **type $\mathbf{n}$ flag scheme** of M is the algebraic k -functor $\text{Flag}_{\mathbf{n}}(M)$ given by:

$$\text{Flag}_{\mathbf{n}}(M)(R) = \{\text{flags of type } \mathbf{n} \text{ on } M \otimes_k R\}.$$

Remark 2.4.14 (Special Cases of Flag Schemes). (i) By definition, $\text{Flag}_{(n)}(M) = \text{Gr}_n(M)$.

- (ii) If M is a vector space of constant rank r , then duality induces isomorphisms:

$$\text{Flag}_{(n_1, \dots, n_s)}(M) \simeq \text{Flag}_{(r-n_s, \dots, r-n_1)}(M^\vee).$$

Example 2.4.15 (Projectivity of Flag Schemes). Sending a flag to its components defines a monomorphism:

$$\text{Flag}_{\mathbf{n}}(M) \hookrightarrow \prod_{i=1}^s \text{Gr}_{n_i}(M).$$

Claim 2.4.16. *It is a closed immersion.*

In particular, $\text{Flag}_n(M)$ is a projective k -scheme if M is of finite type.

Proof. For any R -point $x: \text{Spec}(R) \rightarrow \prod_{i=1}^s \text{Gr}_{n_i}(M)$ classifying quotient spaces $F_1, \dots, F_s$ of $M \otimes_k R$, the preimage of $\text{Flag}_n(M)$ by x is exactly the joint vanishing locus of the R -linear maps:

$$\ker(M \otimes_k R \twoheadrightarrow F_{i+1}) \hookrightarrow M \otimes_k R \twoheadrightarrow F_i$$

for $1 \leq i \leq s-1$, which is a closed subfunctor of $\text{Spec}(R)$ by Proposition 2.2.19(i). $\square$

2.5. The Projective Nullstellensatz

Hilbert's Nullstellensatz (Theorem 1.5.1) has a homogeneous counterpart for the projective space. We record the classical statement, which complements the functorial Corollary 2.3.28.

For a ring k , define the maps:

$$\{\text{homogeneous subsets of } k[x_0, \dots, x_n]\} \xrightleftharpoons[I]{V} \{\text{subsets of } \mathbb{P}^n(k)\}$$

as follows:

$$\begin{aligned} V(F) &= \{x \in \mathbb{P}^n(k) \mid f(x) = 0 \text{ for all } f \in F\}, \\ I(X) &= \{f \in k[x_0, \dots, x_n] \text{ homogeneous} \mid f(x) = 0 \text{ for all } x \in X\}. \end{aligned}$$

Note that both maps are order-reversing and that $F \subset I(V(F))$ and $X \subset V(I(X))$ (in other words, this is an adjunction between posets). Note also that $I(X)$ is always a saturated homogeneous ideal in $k[x_0, \dots, x_n]$, which is radical if k is reduced. Call a subset $X \subset \mathbb{P}^n(k)$ **algebraic** if it lies in the image of V , or equivalently if $X = V(I(X))$.

Proposition 2.5.1 (Projective Nullstellensatz). *Let k be an algebraically closed field and let $n \in \mathbb{N}$. For any homogeneous subset $F \subset k[x_0, \dots, x_n]$, we have:*

$$I(V(F)) = \sqrt{(F)}^{\text{sat}}.$$

Consequently, the maps V and I define a one-to-one correspondence:

$$\{\text{saturated radical homogeneous ideals in } k[x_0, \dots, x_n]\} \xrightleftharpoons[I]{V} \{\text{algebraic subsets of } \mathbb{P}^n(k)\}$$

Proof. The inclusion $\sqrt{(F)}^{\text{sat}} \subset I(V(F))$ is clear: if $f \in \sqrt{(F)}^{\text{sat}}$, then there exist $m, N \geq 1$ such that $x_i^N f^m \in (F)$ for every $i = 0, \dots, n$. For any $x = [a_0 : \dots : a_n] \in V(F)$, every element of (F) vanishes at x , so $a_i^N f(x)^m = 0$ for all i . Since x is a projective point, some $a_i \neq 0$, and as k is a field this forces $f(x)^m = 0$, hence $f(x) = 0$.

For the reverse inclusion, we use the affine Nullstellensatz. Consider the standard affine cover of $\mathbb{P}_k^n$ given by $D(x_i) \simeq \mathbb{A}_k^n$ for $i = 0, \dots, n$. For each i , let $F_i \subset k[y_1, \dots, y_n]$ be the dehomogenization of F with respect to x_i (setting $x_i = 1$).

By the affine Nullstellensatz (Theorem 1.5.1), we have:

$$I(V(F_i)) = \sqrt{(F_i)}$$

in $k[y_1, \dots, y_n]$, where $V(F_i)$ denotes the vanishing locus in k^n .

Now suppose $f \in I(V(F))$ is homogeneous of degree d . For each i , the dehomogenization $f_i = f|_{x_i=1} \in k[y_1, \dots, y_n]$ vanishes on $V(F_i)$, so $f_i \in \sqrt{(F_i)}$ by the affine Nullstellensatz: $f_i^{m_i} \in (F_i)$ for some $m_i \geq 1$. Setting $m = \max_i m_i$, we have $f_i^m \in (F_i)$ for every i .

Re-homogenizing lifts $f_i^m \in (F_i)$ back to $k[x_0, \dots, x_n]$: there exist integers $N_i \geq 0$ such that

$$x_i^{N_i} \cdot f_i^m \in (F) \quad \text{for every } i = 0, \dots, n.$$

Indeed, f_i^m is a polynomial in $y_1, \dots, y_n$ expressible as a combination of dehomogenized generators of F at x_i ; homogenizing both sides by reintroducing x_i yields a relation in (F) multiplied by a suitable power $x_i^{N_i}$ that accounts for any degree mismatch.

Hence $f^m \in (F : (x_0, \dots, x_n)^\infty) = (F)^{\text{sat}}$. Since $\sqrt{(F)}$ is also closed under taking radicals, $f \in \sqrt{(F)^{\text{sat}}} = \sqrt{(F)}^{\text{sat}}$, completing the reverse inclusion. $\square$

Looking Ahead

The projective setting is now in place: $\text{Proj}(A)$ of an $\mathbb{N}$ -graded ring, its open cover by $D(f) \simeq \text{Spec}(A_{(f)})$, the Proj structure theorem, saturation of homogeneous ideals, the projective-closure theorem, and examples ranging from Grassmannians to Segre and Veronese embeddings. We also proved the functorial projective Nullstellensatz, completing the picture of how algebra translates to projective geometry.

Two natural directions open up. First, we need to do *linear algebra* over our schemes: modules, ideals, quasi-coherent sheaves, relative constructions. The next chapter develops this machinery via the category Mod_X of quasi-coherent modules — defined for any algebraic functor X as a categorical limit of the Mod_R . Second, we eventually need to assemble affine and projective schemes into a single category that is closed under gluing: this is the category of schemes, treated in Chapter 6.

Chapter 3.

Quasi-coherent Modules

For a ring R , the category Mod_R of R -modules is the natural home for linear algebra over R . We now ask: given an algebraic functor $X \in \text{Fun}(\text{CAlg}, \text{Set})$, what is the “linear algebra” attached to X ?

The answer is the category Mod_X of *quasi-coherent modules on X* . The idea is compelling in its simplicity: since X assembles, via its R -points, out of affine pieces $\text{Spec}(R) \rightarrow X$, the category Mod_X should assemble out of the individual Mod_R along the same diagram. This is a right Kan extension of the functor $R \mapsto \text{Mod}_R$ along Spec , making Mod_X a categorical *limit*:

$$\text{Mod}_X \simeq \lim_{(R, x) \in \text{El}(X)^{\text{op}}} \text{Mod}_R.$$

The universal property of this limit automatically gives us pullback functors, base change, and compatibility with all the constructions we already know for rings.

What follows is a careful unpacking of this picture. After reviewing limits in Cat , we define quasi-coherent modules and ideals, classify open and closed subfunctors via quasi-coherent ideals, construct the *relative* Spec and Proj over a base, prove Zariski descent for modules, and introduce Serre’s twisting sheaves $\mathcal{O}(d)$ on projective schemes — the building blocks of projective geometry’s cohomology theory.

3.1. Quasi-coherence

Before extending $R \mapsto \text{Mod}_R$ along the Yoneda embedding, we need a mild amount of categorical machinery: how to take limits of category-valued functors. The resulting category Mod_X is best understood as a limit of ordinary module categories along the affine points of X .

Throughout this chapter, a “limit in Cat ” means a *pseudolimit*: diagrams are pseudofunctors, cones commute up to specified natural isomorphisms, and morphisms of cones respect those isomorphisms. Equivalently, we work in the $(2, 1)$ -category obtained from the 2-category of categories by retaining functors as 1-morphisms and natural *isomorphisms* as 2-morphisms. This convention is essential for descent: transition maps are canonical only up to coherent isomorphism. A strict 1-categorical limit will always be called *strict* explicitly.

3.1.1. Category-valued Functors

We begin with limits of category-valued functors, the categorical machinery underlying the definition of Mod_X .

Definition 3.1.1 (Category-valued Pseudofunctors). Let $\mathcal{J}$ be a category. A *category-valued pseudofunctor* $F: \mathcal{J} \rightarrow \text{Cat}$ consists of the following data:

- (i) For each object $I \in \mathcal{J}$, a category $F(I)$.
- (ii) For each morphism $f: I \rightarrow J$ in $\mathcal{J}$, a functor $F(f): F(I) \rightarrow F(J)$.
- (iii) For each object $I \in \mathcal{J}$, a natural isomorphism $\eta_I: \text{id}_{F(I)} \simeq F(\text{id}_I)$.

(iv) For morphisms $f: I \rightarrow J$ and $g: J \rightarrow K$ in $\mathcal{J}$, a natural isomorphism $\mu_{f,g}: F(g) \circ F(f) \xrightarrow{\sim} F(g \circ f)$.

These data must satisfy the following compatibility conditions:

5. For any morphism $f: I \rightarrow J$ in $\mathcal{J}$, both of the following triangles commute:

$$\begin{array}{ccc} F(f) & \xrightarrow{\eta_I} & F(f) \circ F(\text{id}_I) \\ & \searrow \text{id} & \downarrow \mu_{\text{id}_I, f} \\ & & F(f), \end{array} \quad \text{and} \quad \begin{array}{ccc} F(f) & \xrightarrow{\eta_J} & F(\text{id}_J) \circ F(f) \\ & \searrow \text{id} & \downarrow \mu_{f, \text{id}_J} \\ & & F(f). \end{array}$$

6. For morphisms $f: I \rightarrow J$, $g: J \rightarrow K$, and $h: K \rightarrow L$ in $\mathcal{J}$, the following square commutes:

$$\begin{array}{ccc} F(h) \circ F(g) \circ F(f) & \xrightarrow{\mu_{f,g}} & F(h) \circ F(g \circ f) \\ \downarrow \mu_{g,h} & & \downarrow \mu_{g \circ f, h} \\ F(h \circ g) \circ F(f) & \xrightarrow{\mu_{f, h \circ g}} & F(h \circ g \circ f). \end{array}$$

An ordinary strict functor is the special case in which the displayed coherence isomorphisms are identities. Below, “functor to Cat ” is shorthand for a pseudofunctor in this sense.

Example 3.1.2 (Slice Categories). Let $\mathcal{C}$ be a category with pullbacks. We can consider the functor:

$$C_{/-}: \mathcal{C}^{\text{op}} \rightarrow \text{Cat}, \quad X \mapsto C_{/X},$$

called the *self-indexing* of $\mathcal{C}$. For a morphism $f: X \rightarrow Y$, it sends f to the pullback functor:

$$f^*: C_{/Y} \rightarrow C_{/X}, \quad (A \rightarrow Y) \mapsto (X \times_Y A \rightarrow X).$$

Remark 3.1.3 (Topos). If $\mathcal{C}$ is a presentable category and the above functor satisfies the property that for any small category I and functor $F: I \rightarrow \mathcal{C}$:

$$C_{/\text{colim}_{i \in I} F(i)} \simeq \lim_{i \in I} C_{/F(i)},$$

then $\mathcal{C}$ is called a *topos*.

Example 3.1.4 (Module Categories). We can construct the functor $R \mapsto \text{Mod}_R$ in three different ways:

(i) *Base change*: There exists a functor:

$$\text{Mod}^*: \text{CAlg} \rightarrow \text{Cat}, \quad R \mapsto \text{Mod}_R,$$

which sends a ring homomorphism $f: R \rightarrow S$ to the base change functor:

$$f^*: \text{Mod}_R \rightarrow \text{Mod}_S, \quad M \mapsto M \otimes_R S.$$

(ii) *Restriction of scalars*: There exists a functor:

$$\text{Mod}_*: \text{CAlg}^{\text{op}} \rightarrow \text{Cat}, \quad R \mapsto \text{Mod}_R,$$

which sends a ring homomorphism $f: R \rightarrow S$ to the forgetful functor:

$$f_*: \text{Mod}_S \rightarrow \text{Mod}_R, \quad M \mapsto M,$$

where we view the S -module M as an R -module via composition with f .

(iii) *Internal Hom*: There exists a functor:

$$\mathrm{Mod}^!: \mathrm{CAlg} \rightarrow \mathrm{Cat}, \quad R \mapsto \mathrm{Mod}_R,$$

which sends a ring homomorphism $f: R \rightarrow S$ to the functor:

$$f^!: \mathrm{Mod}_R \rightarrow \mathrm{Mod}_S, \quad M \mapsto \mathrm{Hom}_R(S, M).$$

Remark 3.1.5. In the above example, we can replace Mod_R with CAlg_R . Then constructions (i) and (ii) work for CAlg_R , but (iii) does not.

Example 3.1.6 (Properties Preserved by Base Change). Any property P of modules that is preserved by base change (such as being finitely generated, finitely presented, flat, projective, free, a vector space, or a line) defines a subfunctor of Mod^* .

Example 3.1.7 (Ideal Functors). For a poset P , we can automatically view it as a category. Therefore, for a category C , we can consider functors landing in Pos (the category of posets):

$$C \rightarrow \mathrm{Pos}.$$

For example, taking $C = \mathrm{CAlg}$:

- For a ring R , consider the poset Id_R of all ideals of R ordered by inclusion. For $R \rightarrow S$, we have $\mathrm{Id}_R \rightarrow \mathrm{Id}_S$ sending $I \mapsto IS$.
- For a ring R , consider the poset Rad_R of all radical ideals of R ordered by inclusion. For $R \rightarrow S$, we have $\mathrm{Rad}_R \rightarrow \mathrm{Rad}_S$ sending $I \mapsto \sqrt{IS}$.

Notation 3.1.8 (Left and Right Cones). Let $\mathcal{J}$ be a category. We denote by $\mathcal{J}^\triangleleft$ the category $[0] \star \mathcal{J}$. In other words, it is $\mathcal{J}$ with an initial object $-\infty$ added, such that for any $I \in \mathcal{J}$, we have $\mathrm{Hom}_{\mathcal{J}^\triangleleft}(-\infty, I) = *$. We call this the *left cone* over $\mathcal{J}$.

Similarly, we can define the *right cone* $\mathcal{J}^\triangleright$.

We can now formalize the limit construction.

Definition 3.1.9 (Pseudolimits of Categories). Let $F: \mathcal{J} \rightarrow \mathrm{Cat}$ be a pseudofunctor. Its *pseudolimit* $\lim F$ (or $\lim_{I \in \mathcal{J}} F(I)$) is the category defined as follows:

- *Objects*: An object of $\lim F$ is a pair (x, α) where:
 - For each $I \in \mathcal{J}$, we specify an object $x_I \in F(I)$.
 - For each morphism $f: I \rightarrow J$ in $\mathcal{J}$, we specify an isomorphism $\alpha_f: F(f)(x_I) \xrightarrow{\sim} x_J$.

These data must satisfy the following *cocycle condition*:

- For each $I \in \mathcal{J}$, the following triangle commutes:

$$\begin{array}{ccc} x_I & \xrightarrow{\eta_I(x_I)} & F(\mathrm{id}_I)(x_I) \\ & \searrow \mathrm{id} & \downarrow \alpha_{\mathrm{id}_I} \\ & & x_I. \end{array}$$

(ii) For morphisms $f: I \rightarrow J$ and $g: J \rightarrow K$ in $\mathcal{J}$, the following diagram commutes:

$$\begin{array}{ccc} F(g)(F(f)(x_I)) & \xrightarrow{F(g)(\alpha_f)} & F(g)(x_J) \\ \downarrow \mu_{f,g}(x_I) & & \downarrow \alpha_g \\ F(g \circ f)(x_I) & \xrightarrow{\alpha_{g \circ f}} & x_K. \end{array}$$

• *Morphisms*: A morphism φ from (x, α) to (y, β) in $\lim F$ consists of:

- (i) For each $I \in \mathcal{J}$, a morphism $\varphi_I: x_I \rightarrow y_I$ in $F(I)$.
- (ii) For each morphism $f: I \rightarrow J$ in $\mathcal{J}$, the following diagram commutes:

$$\begin{array}{ccc} F(f)(x_I) & \xrightarrow{F(f)(\varphi_I)} & F(f)(y_I) \\ \downarrow \alpha_f & & \downarrow \beta_f \\ x_J & \xrightarrow{\varphi_J} & y_J. \end{array}$$

• *Composition*: Identity morphisms and composition of morphisms are defined pointwise.

Definition 3.1.10 (Limit Diagrams). Let $F: \mathcal{J} \rightarrow \text{Cat}$ be a pseudofunctor. Consider a pseudofunctorial extension $\bar{F}: \mathcal{J}^\triangleleft \rightarrow \text{Cat}$. We say $\bar{F}$ is a *limit diagram* if the canonical map $\bar{F}(-\infty) \rightarrow \lim F$ is a categorical equivalence.

Remark 3.1.11 (Colimits). Similarly, we define pseudocolimits of category-valued pseudofunctors. Given $F: \mathcal{J} \rightarrow \text{Cat}$ and an extension $\bar{F}: \mathcal{J}^\triangleright \rightarrow \text{Cat}$, the extension is a colimit diagram if and only if for every category $\mathcal{E}$ the diagram

$$\text{Fun}(\bar{F}(-), \mathcal{E}): (\mathcal{J}^{\text{op}})^\triangleleft \rightarrow \text{Cat}$$

is a limit diagram. (Note: in synthetic category theory, we define pushouts in a similar way.)

Example 3.1.12 (Pullbacks of Categories). Given functors $\mathcal{C} \xrightarrow{f} \mathcal{E} \xleftarrow{g} \mathcal{D}$, their pullback under the standing convention is the *iso-comma category*. Its objects are triples (x, y, α) with $x \in \mathcal{C}$, $y \in \mathcal{D}$, and $\alpha: f(x) \xrightarrow{\sim} g(y)$ in $\mathcal{E}$; its morphisms are pairs compatible with α . This need not agree with the strict pullback of the underlying 1-category Cat .

Example 3.1.13 (The Loop Category as a Limit). Let $\mathcal{C}$ be a category. Consider the diagram in Cat :

$$\mathcal{C} \begin{array}{c} \xrightarrow{\text{id}_{\mathcal{C}}} \\ \xrightarrow{\text{id}_{\mathcal{C}}} \end{array} \mathcal{C}.$$

The *pseudolimit* of this diagram is equivalent to the functor category $\text{Fun}(S^1, \mathcal{C})$, where S^1 (often denoted $B\mathbb{Z}$) is the groupoid with one object and $\mathbb{Z}$ as the automorphism group. By contrast, the strict equalizer of the two identity functors in the underlying 1-category is just $\mathcal{C}$; the automorphism below is precisely the extra coherence retained by the pseudolimit.

Derivation from Definition 3.1.9: Let's check what an object in this limit looks like according to the definition.

- The index category $\mathcal{J}$ has two objects (say $0, 1$) and two non-trivial morphisms $s, t: 0 \rightarrow 1$.
- The functor F maps $0, 1 \mapsto \mathcal{C}$ and $s, t \mapsto \text{id}_{\mathcal{C}}$.
- An object in $\lim F$ is a pair (x, α) , which consists of:
 - Objects $x_0, x_1 \in \mathcal{C}$.

- Isomorphisms $\alpha_s: \text{id}(x_0) \xrightarrow{\sim} x_1$ and $\alpha_t: \text{id}(x_0) \xrightarrow{\sim} x_1$ in C .

Since α_s is an isomorphism, we can identify x_1 with x_0 . The data then reduces to a single object $x := x_0$ and a single automorphism formed by the loop $\gamma = \alpha_t^{-1} \circ \alpha_s: x \xrightarrow{\sim} x$.

Thus, the category $\lim F$ is equivalent to the category where:

- **Objects:** Pairs (x, γ) where $x \in C$ and $\gamma \in \text{Aut}_C(x)$.
- **Morphisms:** A morphism $f: (x, \gamma) \rightarrow (y, \delta)$ is a map $f: x \rightarrow y$ in C such that $\delta \circ f = f \circ \gamma$.

This is precisely the description of $\text{Fun}(S^1, C)$, where a functor from S^1 selects an object and an automorphism.

Example 3.1.14 (Zariski Descent for Modules). Limits of categories allow us to reformulate Theorem 1.3.1 more succinctly: it is exactly the statement that, if $(f_i)_{i \in I}$ generates the unit ideal in R , then the diagram of categories

$$\text{Mod}_R \longrightarrow \prod_{i \in I} \text{Mod}_{R_{f_i}} \rightrightarrows \prod_{i,j \in I} \text{Mod}_{R_{f_i f_j}} \rightrightarrows \prod_{i,j,k \in I} \text{Mod}_{R_{f_i f_j f_k}}$$

obtained by restricting the functor Mod^* is a limit diagram. In the case $I = \{1, 2\}$, this is equivalent to the simpler statement that the square of categories

$$\begin{array}{ccc} \text{Mod}_R & \longrightarrow & \text{Mod}_{R_{f_1}} \\ \downarrow & & \downarrow \\ \text{Mod}_{R_{f_2}} & \longrightarrow & \text{Mod}_{R_{f_1 f_2}} \end{array}$$

is cartesian. Moreover, these results hold for any subfunctor of Mod^* defined by a Zariski-local property, such as $R \mapsto \text{Vect}_R$ and $R \mapsto \text{Line}_R$.

Faithfully flat descent replaces the principal-localization diagram by the Amitsur diagram of a faithfully flat algebra. The distinction between descent for modules, affine morphisms, and arbitrary schemes is treated in Section 5.2.4.

3.1.2. Quasi-coherent Modules over Algebraic Functors

We are now ready to define modules over a general algebraic functor.

Idea (Modules via Kan Extension). Our core idea is to extend the notion of “module” contravariantly to algebraic functors. The variance is visible in the diagram

$$\begin{array}{ccc} \text{CAlg} & \xrightarrow{\text{Mod}^*} & \text{Cat} \\ R \mapsto \text{Spec}(R) \downarrow & \nearrow \text{Ran} & \\ \text{Fun}(\text{CAlg}, \text{Set})^{\text{op}} & & \end{array}$$

because a ring map $R \rightarrow S$ gives both base change $\text{Mod}_R \rightarrow \text{Mod}_S$ and, in the opposite geometric direction, $\text{Spec}(S) \rightarrow \text{Spec}(R)$. Thus $R \mapsto \text{Spec}(R)$ is covariant only after the target category of algebraic functors is replaced by its opposite.

The pointwise right-Kan-extension formula is concrete. Density gives

$$X \simeq \text{colim}_{(R,x) \in \text{El}(X)} \text{Spec } R,$$

and hence the value of the dashed functor at X is the pseudolimit

$$\mathrm{Mod}_X \simeq \lim_{(R,x) \in \mathrm{El}(X)^{\mathrm{op}}} \mathrm{Mod}_R.$$

Equivalently, $\mathrm{El}(X)$ is the category Aff/X of affine points of X , and the diagram being limited is

$$(\mathrm{Aff}/X)^{\mathrm{op}} \longrightarrow \mathrm{Cat}, \quad (\mathrm{Spec}(R) \xrightarrow{x} X) \longmapsto \mathrm{Mod}_R.$$

An arrow in $(\mathrm{Aff}/X)^{\mathrm{op}}$ represented by a triangle $\mathrm{Spec}(S) \rightarrow \mathrm{Spec}(R) \rightarrow X$ comes from a ring map $R \rightarrow S$ and is sent to $(-) \otimes_R S : \mathrm{Mod}_R \rightarrow \mathrm{Mod}_S$. This fixes both the variance and the coherence data in the limit formula.

Definition 3.1.15 (Quasi-coherent Modules). Let X be an algebraic functor. A *quasi-coherent module* over X (or *quasi-coherent $\mathcal{O}_X$ -module*) is an element of the limit category:

$$\mathrm{Mod}_X := \lim_{(R,x) \in \mathrm{El}(X)^{\mathrm{op}}} \mathrm{Mod}_R,$$

Explicitly, a quasi-coherent module M over X consists of:

- (i) For each ring R and each R -point $x : \mathrm{Spec}(R) \rightarrow X$, an R -module $M(x) \in \mathrm{Mod}_R$.
- (ii) For each ring homomorphism $\varphi : R \rightarrow S$ and commutative triangle:

$$\begin{array}{ccc} \mathrm{Spec}(S) & \xrightarrow{y} & X \\ & \searrow \mathrm{Spec}(\varphi) \quad \nearrow x & \\ & \mathrm{Spec}(R) & \end{array}$$

an isomorphism $\alpha_\varphi : M(x) \otimes_R S \xrightarrow{\sim} M(y)$ in Mod_S .

These data must satisfy:

- *Transitivity*: For composable ring homomorphisms $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ with compatible points x, y, z , we have:

$$\alpha_{\psi \circ \varphi} = \alpha_\psi \circ (\alpha_\varphi \otimes_S T).$$

- *Identity*: For $\varphi = \mathrm{id}_R$, we have $\alpha_{\mathrm{id}} = \mathrm{id}$.

Remark 3.1.16 (Functoriality). The assignment $X \mapsto \mathrm{Mod}_X$ is contravariant functorial. For a morphism $f : Y \rightarrow X$ of algebraic functors, we obtain a pullback functor:

$$f^* : \mathrm{Mod}_X \rightarrow \mathrm{Mod}_Y, \quad M \mapsto f^*M,$$

where $(f^*M)(y) = M(f \circ y)$ for any $y : \mathrm{Spec}(R) \rightarrow Y$.

Similarly, we can define quasi-coherent algebras, ideals, and radical ideals.

Definition 3.1.17 (Quasi-coherent Algebras and Ideals). Let X be an algebraic functor.

- (i) A *quasi-coherent $\mathcal{O}_X$ -algebra* is an element of:

$$\mathrm{CAlg}_X := \lim_{x : \mathrm{Spec}(R) \rightarrow X} \mathrm{CAlg}_R.$$

- (ii) A *quasi-coherent ideal* is an element of:

$$\mathrm{Id}_X := \lim_{x : \mathrm{Spec}(R) \rightarrow X} \mathrm{Id}_R.$$

(iii) A *quasi-coherent radical ideal* is an element of:

$$\text{Rad}_X := \lim_{x: \text{Spec}(R) \rightarrow X} \text{Rad}_R.$$

Warning 3.1.18 (Incompatibility of forgetful functors). It should be noted that the forgetful functors

$$\text{Rad}_R \hookrightarrow \text{Id}_R \rightarrow \text{Mod}_{\text{Spec}(R)}$$

are not compatible with base change. Therefore, they cannot automatically extend to general algebraic functors X (although they will when X is a scheme). Specifically:

- A quasi-coherent radical ideal on X does not automatically induce a quasi-coherent ideal structure.
- A quasi-coherent ideal on X does not automatically induce a quasi-coherent module structure.

This can be seen by observing base change. For a ring homomorphism $\varphi: R \rightarrow S$ and a radical ideal I in R , we have the following **non**-commutative diagram:

$$\begin{array}{ccc} \text{Rad}_R & \xrightarrow{\text{forget}} & \text{Id}_R \\ I \mapsto \sqrt{IS} \downarrow & & \downarrow I \mapsto IS \\ \text{Rad}_S & \xrightarrow{\text{forget}} & \text{Id}_S \end{array}$$

(In general $\sqrt{IS} \neq IS$, so the distinction matters.)

Notation 3.1.19 ($\mathcal{O}_X$). Let X be an algebraic functor. Denote by $\mathcal{O}_X \in \text{CAlg}_X$ the initial object of the category of quasi-coherent algebras on X , i.e., for $x: \text{Spec}(R) \rightarrow X$, we have $\mathcal{O}_X(x) = R$. It is not hard to see that its underlying quasi-coherent module is the unit of Mod_X under the tensor product. Furthermore, for any morphism $f: Y \rightarrow X$, we have $f^*(\mathcal{O}_X) = \mathcal{O}_Y$.

Definition 3.1.20 (Global Sections). Let X be an algebraic functor and M be a quasi-coherent module on X . Then a *global section* of M refers to a morphism $\mathcal{O}_X \rightarrow M$ in Mod_X . The set formed by the global sections of M can be characterized as follows:

$$M(X) = \Gamma(X, M) = \lim_{(R, x) \in \text{El}(X)^{\text{op}}} \text{Hom}_{\text{Mod}_R}(R, M(x)) = \lim_{(R, x) \in \text{El}(X)^{\text{op}}} M(x).$$

Notation 3.1.21 (Sections over an Open Subscheme). More generally, for any morphism $g: U \rightarrow X$ of algebraic functors — in particular for an open subscheme $U \subseteq X$, with g its inclusion — and any $M \in \text{Mod}_X$, we write

$$M(U) := \Gamma(U, g^*M) = \text{Hom}_{\text{Mod}_U}(\mathcal{O}_U, g^*M) = \lim_{(R, x) \in \text{El}(U)^{\text{op}}} M(g \circ x)$$

for the global sections of the restriction $g^*M \in \text{Mod}_U$ (Remark 3.1.16). Being a Hom out of the unit, this is always defined. It recovers the two cases above: for an affine point $g = x: \text{Spec}(R) \rightarrow X$ it is the value $M(x) \in \text{Mod}_R$ of Definition 3.1.15 (the object $\text{id}_{\text{Spec } R}$ is terminal in $\text{El}(\text{Spec } R)$, hence initial in the displayed limit's index $\text{El}(\text{Spec } R)^{\text{op}}$), and for $U = X$ it is $\Gamma(X, M)$. Thus $M(-)$ extends from affine points to all X -schemes, and restriction along $V \hookrightarrow U$ makes $U \mapsto M(U)$ a presheaf on the opens of X .

Remark 3.1.22 (Functions as Global Sections). One can find that a global section of $\mathcal{O}_X$ is exactly a function on X in the sense of Definition 1.2.37: $\Gamma(X, \mathcal{O}_X) \simeq \mathcal{O}(X)$. Hence for any $M \in \text{Mod}_X$, the abelian group $M(X)$ has a canonical structure of module over the ring $\mathcal{O}(X)$.

Remark 3.1.23 (Properties Compatible with Base Change). Since quasi-coherent objects are essentially collections of local data compatible with base change, any property or construction in the category of R -modules (or algebras, ideals) that is compatible with base change can be extended to the category of quasi-coherent objects.

The following lists some important properties and constructions:

- **Properties of modules:** Properties such as finite type, finite presentation, flat, projective, free, etc., can all be defined. In particular, vector bundles (Vect_X) and line bundles (Line_X) form full subcategories of Mod_X .
- **Properties of module morphisms:** Zero morphism, surjection, universal injection, etc.
- **Sum of morphisms:** The additive structure on $\text{Mod}_X(M, N)$ makes it an abelian group.
- **Colimits of modules:** Can be computed pointwise in Mod_X , i.e., $(\text{colim}_i M_i)(x) = \text{colim}_i (M_i(x))$.
- **Tensor product:** Mod_X has a symmetric monoidal structure, given by the pointwise tensor product $(M \otimes N)(x) = M(x) \otimes N(x)$.
- **Power operations on modules:** Symmetric powers Sym^d , exterior powers Λ^d , and divided powers Γ^d can all be defined.
- **Operations on ideals:** The sum $I + J$ and product IJ of quasi-coherent ideals are both defined by pointwise operations.
- **Radical of an ideal:** The radical $\sqrt{I}$ of a quasi-coherent ideal is defined as $(\sqrt{I})(x) = \sqrt{I(x)}$.
- **Properties of algebras:** Finite type, finite presentation, finite, flat, free, etc. For $\mathbb{N}$ -graded algebras, properties such as “generated by A_1 ” can also be defined.
- **Properties of algebra morphisms:** Surjection, localization, having nilpotent kernel, eventually surjective, etc.
- **Other constructions:** Including the forgetful functor from algebras to modules, the symmetric algebra construction $\text{Sym}(M)$, etc.

Note: Injectivity is usually not defined in Mod_X , because injectivity is not necessarily preserved under base change (unless the module is flat). Similarly, limits of modules usually exist (if X is accessible), but cannot necessarily be computed pointwise.

3.2. Classification of Open and Closed Subfunctors

We have seen for affine and projective schemes that open and closed subfunctors correspond bijectively to radical and arbitrary ideals respectively (with saturation in the projective case). We now extend this correspondence to general algebraic functors.

Proposition 3.2.1 (Classification Theorem). *Let X be an algebraic functor.*

(i) *There exists an isomorphism of posets:*

$$V: \text{Id}_X^{\text{op}} \xrightarrow{\sim} \{\text{closed subfunctors of } X\}.$$

(ii) *There exists an isomorphism of posets:*

$$D: \text{Rad}_X \xrightarrow{\sim} \{\text{open subfunctors of } X\}.$$

Proof. Strategy. The affine case is already in hand: closed subfunctors of $\text{Spec}(R)$ correspond to ideals via V , and open subfunctors to radical ideals via D (Proposition 1.2.49). We reduce the general case to the affine one using the canonical colimit presentation $X \simeq \text{colim}_{x: \text{Spec}(R) \rightarrow X} \text{Spec}(R)$ and the fact that subfunctors of X are uniquely determined by their pullbacks to each affine chart.

Let $\text{Sub}(X)$ denote the poset of all subfunctors of X . Since $X \simeq \text{colim}_{x: \text{Spec}(R) \rightarrow X} \text{Spec}(R)$, we have an isomorphism:

$$\begin{aligned} \text{Sub}(X) &\xrightarrow{\sim} \lim_{x: \text{Spec}(R) \rightarrow X} \text{Sub}(\text{Spec}(R)) \\ Y &\mapsto (x^{-1}(Y) \subset \text{Spec}(R))_x. \end{aligned}$$

The inverse is given by $(Y_x \subset \text{Spec}(R))_x \mapsto (Y(R) = \{x: \text{Spec}(R) \rightarrow X \mid x^{-1}(Y) = Y_x = \text{Spec}(R)\})$.

(i) Consider the diagram:

$$\begin{array}{ccc} \text{Sub}(X) & \xrightarrow{\sim} & \lim_x \text{Sub}(\text{Spec}(R)) \\ \uparrow & & \uparrow \\ \text{Closed}(X) & \xrightarrow{\sim} & \lim_x \text{Closed}(\text{Spec}(R)) \end{array}$$

By the definition of closed subfunctors and $X \simeq \text{colim}_x \text{Spec}(R)$, the bottom row is an isomorphism. Therefore, we reduce to showing:

$$\lim_x \text{Closed}(\text{Spec}(R)) \simeq \lim_x \text{Id}_R^{\text{op}}.$$

For this, we must show that the bijection $V: \text{Id}_R^{\text{op}} \xrightarrow{\sim} \text{Closed}(\text{Spec}(R))$ is functorial. For $\varphi: R \rightarrow S$, consider the diagram:

$$\begin{array}{ccc} \text{Id}_R & \xrightarrow{V_R} & \text{Closed}(\text{Spec}(R)) \\ \downarrow \varphi^* & & \downarrow \text{Spec}(\varphi)^{-1} \\ \text{Id}_S & \xrightarrow{V_S} & \text{Closed}(\text{Spec}(S)). \end{array}$$

We verify commutativity by showing $V(IS) = \text{Spec}(\varphi)^{-1}(V(I))$. The right side is:

$$V(I) \times_{\text{Spec}(R)} \text{Spec}(S) \simeq V(S \otimes_R I) \simeq \text{Spec}(S \otimes_R R/I).$$

Using $S/IS \simeq S \otimes_R R/I$, we get:

$$\text{Spec}(\varphi)^{-1}(V(I)) \simeq \text{Spec}(S/IS) = V(IS) = V(\varphi^*(I)).$$

This shows V is a natural transformation from $\text{Id}_{(-)}^{\text{op}}$ to $\text{Closed}(\text{Spec}(-))$.

(ii) Similarly, using $X \simeq \text{colim} \text{Spec}(R)$ and the definition of open subfunctors, we reduce to proving $D: \text{Rad}_R \xrightarrow{\sim} \text{Open}(\text{Spec}(R))$ is functorial.

For $\varphi: R \rightarrow S$ (let $f = \text{Spec}(\varphi)$), we must show the following diagram commutes:

$$\begin{array}{ccc} \text{Rad}_R & \xrightarrow{D_R} & \text{Open}(\text{Spec}(R)) \\ \downarrow \varphi^* & & \downarrow f^{-1} \\ \text{Rad}_S & \xrightarrow{D_S} & \text{Open}(\text{Spec}(S)) \end{array}$$

Note: In the category Rad , for $J \in \text{Rad}_R$, the pullback is defined as $\varphi^*(J) := \sqrt{J}S$.

We verify $f^{-1}(D(J)) = D(\sqrt{J}S)$:

- *Left side:* By the definition of open subfunctor pullback:

$$f^{-1}(D(J)) = D(J) \times_{\text{Spec}(R)} \text{Spec}(S).$$

For affine schemes, the pullback of $D(J)$ is the open subfunctor defined by the extended ideal JS , so $f^{-1}(D(J)) = D(JS)$.

- *Right side:* By definition, this is $D(\sqrt{JS})$.

Since $D(K)$ depends only on the radical of K (i.e., $D(K) = D(\sqrt{K})$), we have $D(JS) = D(\sqrt{JS})$ identically. This means the diagram commutes, inducing an isomorphism at the limit level:

$$\text{Rad}_X = \lim_x \text{Rad}_R \xrightarrow{\sim} \lim_x \text{Open}(\text{Spec}(R)) \simeq \text{Open}(X).$$

□

Definition 3.2.2 (Open Complement). Let $Z \subset X$ be a closed subfunctor determined by a quasi-coherent ideal I . The *open complement* $X - Z$ is the open subfunctor corresponding to the quasi-coherent radical ideal $\sqrt{I}$.

- Example 3.2.3** (Open Complements). • Let A be a ring and $F \subset A$ a subset. Then in $\text{Spec}(A)$, $D(F)$ is the open complement of $V(F)$. For example, for a set I , take $A = \mathbb{Z}[x_i \mid i \in I]$ and $F = \{x_i\}_{i \in I}$. Then $D(F) = \mathbb{A}^I - 0$ is the open complement of $V(F) = 0$.
- Let A be an $\mathbb{N}$ -graded ring and $F \subset A$ a subset of homogeneous elements. Then in $\text{Proj}(A)$, $D(F)$ is the open complement of $V(F)$. For example, for a set I , in $\mathbb{P}^{I \sqcup \{0\}}$, the affine space $D(x_0) \simeq \mathbb{A}^I$ is the open complement of $V(x_0) \simeq \mathbb{P}^I$.

Warning 3.2.4 (No Closed Complement). Different closed subfunctors can have the same open complement, because different ideals can have the same radical. Therefore, there is no notion of “closed complement” of an open subfunctor.

Example 3.2.5 (Scheme-theoretic Image). For any ring homomorphism $\varphi: A \rightarrow B$, we can factor it as a surjection followed by an injection:

$$A \twoheadrightarrow A/\ker(\varphi) \hookrightarrow B.$$

Therefore, $V(\ker(\varphi)) \subset \text{Spec}(A)$ is the smallest closed subfunctor through which $\text{Spec}(\varphi): \text{Spec}(B) \rightarrow \text{Spec}(A)$ factors.

This construction generalizes without introducing a pushforward. Let $f: Y \rightarrow X$ be a morphism of algebraic functors and set

$$I_f := \bigvee \{J \in \text{Id}_X \mid f \text{ factors through } V(J)\}.$$

Joins of quasi-coherent ideals are computed pointwise, and extension of ideals along a ring map preserves joins. Hence I_f is again quasi-coherent. Moreover, if f factors through every $V(J)$ occurring above, it factors pointwise through their intersection $V(I_f)$. Thus

$$\overline{f(Y)}^{\text{sch}} := V(I_f)$$

is the smallest closed subfunctor of X through which f factors. This universal property is the definition of the *scheme-theoretic image* in the present functor-of-points framework. It avoids the notation $f_*\mathcal{O}_Y$, which has not been defined for a general algebraic functor; after passing to schemes and their structure sheaves, the familiar kernel description is a separate theorem with its own hypotheses.

Warning 3.2.6. Do not replace I_f here by a “kernel of $\mathcal{O}_X \rightarrow f_*\mathcal{O}_Y$ ”: neither the displayed pushforward nor its quasi-coherence has been established at this level of generality. The universal largest-ideal description above is the statement available for arbitrary algebraic functors.

Remark 3.2.7 (Loci of Maps Between Quasi-coherent Modules). Recall the loci of linear maps $u : M \rightarrow N$ from Section 2.3: we set

$$V(u)(R) := \{\varphi : A \rightarrow R \mid \varphi^*(u) = 0\}.$$

We can now generalize this to general algebraic functors X . For a morphism $f : M \rightarrow N$ of quasi-coherent modules over X , we define $V(f)$ as:

$$V(f)(R) := \{x : \text{Spec}(R) \rightarrow X \mid f(x) = 0\}.$$

Note: we have $f(x) : M(x) \rightarrow N(x)$, so $f(x) = 0$ means the zero homomorphism. We can similarly define $\text{Epi}(f)$, $\text{Mono}(f)$, and $\text{Iso}(f)$.

We can establish the following analogous proposition: If $N \in \text{Vect}_X$ is a vector bundle, then:

- $V(f)$ is a closed subfunctor.
- $\text{Epi}(f)$ is an open subfunctor.

If M is also a vector bundle, then:

- $\text{Mono}(f)$ is an open subfunctor.
- $\text{Iso}(f)$ is an open subfunctor.

3.2.1. Relative Spec and Relative Proj

We now globalize Spec and Proj, working relative to an arbitrary algebraic functor as base.

Observation 3.2.8 (Base Change Compatibility). Note that the functor:

$$\text{Spec} : \text{CAlg}_R^{\text{op}} \rightarrow \text{Fun}(\text{CAlg}, \text{Set})_{/\text{Spec}(R)}$$

is compatible with base change in the following sense: For an R -algebra A and a ring homomorphism $f : R \rightarrow S$, we have a pullback diagram:

$$\begin{array}{ccc} \text{Spec}(A \otimes_R S) & \longrightarrow & \text{Spec}(A) \\ \downarrow & \lrcorner & \downarrow \\ \text{Spec}(S) & \longrightarrow & \text{Spec}(R) \end{array}$$

Therefore, we can extend the Spec functor to quasi-coherent algebras.

Definition 3.2.9 (Relative Spec). Let X be an algebraic functor and $A \in \text{CAlg}_X$ a quasi-coherent algebra over X . We define $\text{Spec}(A) \in \text{Fun}(\text{CAlg}, \text{Set})_X$ by:

$$\text{Spec}(A)(R) := \{(x, a) \mid x \in X(R) \text{ and } a \text{ is a section of } \text{Spec}(A(x)) \rightarrow \text{Spec}(R)\}.$$

This construction is functorial, giving a functor:

$$\text{Spec} : \text{CAlg}_X^{\text{op}} \rightarrow \text{Fun}(\text{CAlg}, \text{Set})_X.$$

Similarly, for an $\mathbb{N}$ -graded R -algebra A and a ring homomorphism $f : R \rightarrow S$, we have a pullback diagram:

$$\begin{array}{ccc} \text{Proj}(A \otimes_R S) & \longrightarrow & \text{Proj}(A) \\ \downarrow & \lrcorner & \downarrow \\ \text{Spec}(S) & \longrightarrow & \text{Spec}(R) \end{array}$$

Therefore, we can also extend the Proj functor to quasi-coherent algebras.

Definition 3.2.10 (Relative Proj). Let X be an algebraic functor and $A \in \text{CAlg}_X$ a quasi-coherent $\mathbb{N}$ -graded algebra over X . We define $\text{Proj}(A) \in \text{Fun}(\text{CAlg}, \text{Set})/X$ by:

$$\text{Proj}(A)(R) := \{(x, a) \mid x \in X(R) \text{ and } a \text{ is a section of } \text{Proj}(A(x)) \rightarrow \text{Spec}(R)\}.$$

This construction is functorial, giving a functor:

$$\text{Proj}: \text{CAlg}_X^{\mathbb{N}, \text{es}, \text{op}} \rightarrow \text{Fun}(\text{CAlg}, \text{Set})/X.$$

We can also define relative affine and projective spaces.

Definition 3.2.11 (Relative Affine and Projective Spaces). Let X be an algebraic functor and M a quasi-coherent module over X .

(i) The *affine space over X* associated to M is $\mathbb{A}(M) := \text{Spec}(\text{Sym}(M))$. Explicitly:

$$\mathbb{A}(M)(R) = \{(x, a) \mid x \in X(R) \text{ and } a: M(x) \rightarrow R \text{ is an } R\text{-linear map}\}.$$

(ii) The *punctured affine space over X* is $\mathbb{A}(M) - 0 := D(M) \subset \text{Spec}(\text{Sym}(M))$. Explicitly:

$$(\mathbb{A}(M) - 0)(R) = \{(x, a) \mid x \in X(R) \text{ and } a: M(x) \rightarrow R \text{ is surjective}\}.$$

(iii) The *projective space over X* associated to M is $\mathbb{P}(M) := \text{Proj}(\text{Sym}(M))$. Explicitly:

$$\mathbb{P}(M)(R) = \{(x, a) \mid x \in X(R) \text{ and } a: M(x) \twoheadrightarrow L \text{ is a quotient } R\text{-line}\}.$$

Remark 3.2.12 (Functorial Characterization). Let X and Y be algebraic functors.

(i) *Relative Spec*: For any quasi-coherent algebra $A \in \text{CAlg}_X$, a morphism $Y \rightarrow \text{Spec}(A)$ consists of:

- A morphism of algebraic functors $f: Y \rightarrow X$;
- An algebra homomorphism in CAlg_Y : $\varphi: f^*(A) \rightarrow \mathcal{O}_Y$.

(ii) *Relative Proj*: Let A be a quasi-coherent $\mathbb{N}$ -graded algebra over X generated by A_1 . A morphism $Y \rightarrow \text{Proj}(A)$ consists of:

- A morphism of algebraic functors $f: Y \rightarrow X$;
- A quotient line bundle on Y , i.e., a surjection $q: f^*(A_1) \twoheadrightarrow L$;
- A compatibility condition: the induced map $\text{Sym}(f^*(A_1)) \rightarrow \text{Sym}(L)$ must factor through $f^*(A)$. That is, there exists a unique dashed arrow making the following diagram commute:

$$\begin{array}{ccc} \text{Sym}(f^*(A_1)) & \xrightarrow{\text{Sym}(q)} & \text{Sym}(L) \\ \downarrow & \nearrow \exists! & \\ f^*(A) & & \end{array}$$

(This is equivalent to saying that homogeneous relations in A must vanish when mapped to $L^{\otimes n}$.)

In fact, relative Spec and relative Proj allow us to extend the notions of affine schemes, quasi-affine schemes, projective schemes, and quasi-projective schemes to the relative setting. For an algebraic functor V , let Rad_V^{ft} denote the category of finitely generated radical ideals over V .

Definition 3.2.13 (Classification of Relative Morphisms). Let $f: Y \rightarrow X$ be a morphism of algebraic functors.

- (i) *Affine morphism*: If there exists $A \in \text{CAlg}_X$ such that $Y \simeq \text{Spec}(A)$, then f is called an *affine morphism*, or we say Y is *affine over* X . We denote by $\text{Aff}_X \subset \text{Fun}(\text{CAlg}, \text{Set})_{/X}$ the full subcategory spanned by affine schemes over X .
- (ii) *Quasi-affine morphism*: If there exist $V \in \text{Aff}_X$ and $I \in \text{Rad}_V^{\text{ft}}$ such that $Y \simeq D(I)$, then f is called a *quasi-affine morphism*, or we say Y is *quasi-affine over* X . We denote by $\text{QAff}_X \subset \text{Fun}(\text{CAlg}, \text{Set})_{/X}$ the full subcategory of quasi-affine schemes over X .
(Note: The condition $I \in \text{Rad}_V^{\text{ft}}$ means I is the radical of an ideal generated by a finite subset $F \subset \mathcal{O}_V$.)
- (iii) *Projective morphism*: If there exists $A \in \text{CAlg}_X^{\mathbb{N}, \text{es}}$ with a specified identification $A_0 \simeq \mathcal{O}_X$, such that A_1 is a finitely generated $\mathcal{O}_X$ -module, A is generated as a graded $\mathcal{O}_X$ -algebra by A_1 , and $Y \simeq \text{Proj}(A)$, then f is called a *projective morphism*, or we say Y is *projective over* X . We denote by $\text{Proj}_X \subset \text{Fun}(\text{CAlg}, \text{Set})_{/X}$ the full subcategory of projective schemes over X .
- (iv) *Quasi-projective morphism*: If there exist $V \in \text{Proj}_X$ and $I \in \text{Rad}_V^{\text{ft}}$ such that $Y \simeq D(I)$, then f is called a *quasi-projective morphism*, or we say Y is *quasi-projective over* X . We denote by $\text{QProj}_X \subset \text{Fun}(\text{CAlg}, \text{Set})_{/X}$ the full subcategory of quasi-projective schemes over X .

We characterize affine morphisms via their fibers.

Observation 3.2.14 (Fibers of Affine Morphisms). Let $f: Y \rightarrow X$ be an affine morphism. Then there exists $A \in \text{CAlg}_X$ such that $\text{Spec}(A) \simeq Y$. For any R -point $x: \text{Spec}(R) \rightarrow X$, consider the pullback:

$$\begin{array}{ccc} Y_x & \longrightarrow & Y \\ \downarrow & \lrcorner & \downarrow \\ \text{Spec}(R) & \xrightarrow{x} & X. \end{array}$$

For any ring S , by the universal property of fiber products, we have $Y_x(S) = Y(S) \times_{X(S)} \text{Hom}_{\text{CAlg}}(R, S)$. Concretely, elements of $Y_x(S)$ have the form $((r, a), r')$ where:

- $(r, a) \in Y(S) = \text{Spec}(A)(S)$;
- $r' \in \text{Hom}_{\text{CAlg}}(R, S)$,

satisfying $x \circ r' = r$ and a is a section of $\text{Spec}(A(r)) \rightarrow \text{Spec}(S)$.

By the characterization of quasi-coherent modules (more precisely, quasi-coherent algebras), we have $(r')^*(A(x)) \simeq A(r)$. Therefore, a can be interpreted as an S -algebra homomorphism:

$$\varphi: (r')^*(A(x)) \simeq A(x) \otimes_R S \rightarrow S.$$

Using the fact that tensor product is the pushout in the category of rings, the above map corresponds to an R -algebra homomorphism:

$$\psi: A(x) \rightarrow S.$$

Therefore, we obtain an isomorphism of sets:

$$Y_x(S) \simeq \text{Hom}_{\text{CAlg}_R}(A(x), S) \simeq \text{Spec}(A(x))(S),$$

yielding an isomorphism:

$$Y_x \simeq \text{Spec}(A(x)).$$

Conversely, if for any ring R and any morphism $x: \text{Spec}(R) \rightarrow X$, the pullback Y_x is an affine scheme, we define $A(x) := \mathcal{O}(Y_x)$. Since Y_x satisfies pullback properties with respect to x and the $\mathcal{O}$ functor preserves

base change, the assignment $x \mapsto A(x)$ forms a quasi-coherent X -algebra A . Recalling the definition of relative Spec, we have:

$$\mathrm{Spec}(A)(R) = \{(x, \sigma) \mid x \in X(R), \sigma \in \mathrm{Hom}_{\mathrm{CAlg}}(A(x), R)\}.$$

Since Y_x is affine, we have $\mathrm{Hom}_{\mathrm{CAlg}}(A(x), R) \simeq Y_x(R)$ (the fiber of $Y(R)$ projecting to x). Thus we establish $\mathrm{Spec}(A) \simeq Y$.

Proposition 3.2.15 (Affine Morphisms). *Let X be an algebraic functor. We have a categorical equivalence:*

$$\mathrm{Spec}: \mathrm{CAlg}_X^{\mathrm{op}} \xrightarrow{\sim} \mathrm{Aff}_X.$$

The inverse sends an affine morphism $f: Y \rightarrow X$ to the quasi-coherent X -algebra A_f defined at a point $x: \mathrm{Spec}(R) \rightarrow X$ by

$$A_f(x) := \mathcal{O}(Y \times_X \mathrm{Spec}(R)).$$

For a ring map $R \rightarrow S$, affineness identifies the pullback over $\mathrm{Spec}(S)$ with the base change of the pullback over $\mathrm{Spec}(R)$, and hence $A_f(x) \otimes_R S \simeq A_f(x_S)$, where x_S is the base-changed point. Thus A_f is quasi-coherent, as in the preceding observation.

Proof. Everything reduces to showing:

$$\mathrm{Aff}_X \xrightarrow{\sim} \lim_x \mathrm{Aff}_R.$$

Note that we have a pullback:

$$\begin{array}{ccc} \mathrm{Aff}_X & \xrightarrow{\quad} & \lim_x \mathrm{Aff}_R \\ \downarrow & \lrcorner & \downarrow \\ \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})/X & \longrightarrow & \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})/\mathrm{colim}_x \mathrm{Spec}(R). \end{array}$$

Since $\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$ is a topos, all colimits in it are van Kampen. This means the bottom row is an isomorphism, hence the top row is also an isomorphism. $\square$

3.3. Zariski Descent for Quasi-coherent Modules

Recall the generalized Zariski descent for an R -line L with generating sections $(s_i)_{i \in I}$:

$$\mathrm{Mod}_R \longrightarrow \prod_{i \in I} \mathrm{Mod}_{R_{s_i}} \rightrightarrows \prod_{i, j \in I} \mathrm{Mod}_{R_{s_i s_j}} \rightrightarrows \prod_{i, j, k \in I} \mathrm{Mod}_{R_{s_i s_j s_k}}$$

is a limit diagram.

For a general algebraic functor, the corresponding local lifting square has the following form: after pulling a family $(Y_i \rightarrow X)$ back along an affine point $x: \mathrm{Spec}(R) \rightarrow X$, a finite principal cover admits lifts to members of that family,

$$\begin{array}{ccc} \mathrm{Spec}(R_f) & \dashrightarrow & Y_i \\ \downarrow & & \downarrow \\ \mathrm{Spec}(R) & \xrightarrow{x} & X. \end{array}$$

This motivates the following definition:

Definition 3.3.1 (Zariski-Local Epimorphism). A morphism $Y \rightarrow X$ of algebraic functors is called a *Zariski-local epimorphism* if for any affine scheme $\text{Spec}(R)$ and morphism $\text{Spec}(R) \rightarrow X$, there exist elements $f_1, \dots, f_n \in R$ satisfying $(f_1, \dots, f_n) = R$ (i.e., they generate the unit ideal) such that for each i , the following dashed lift exists:

$$\begin{array}{ccc} & & Y \\ & \nearrow \text{dashed} & \downarrow \\ \text{Spec}(R_{f_i}) & \longrightarrow & \text{Spec}(R) \longrightarrow X. \end{array}$$

Example 3.3.2 (Examples of Zariski-Local Epimorphism). (i) Every epimorphism is a Zariski-local epimorphism. In particular, morphisms with sections, such as $\mathbb{A}^n \times X \rightarrow X$, are also Zariski-local epimorphism.

(ii) Let $(F_i)_{i \in I}$ be a family of subsets of $\mathcal{O}(X)$ such that $(\bigcup_{i \in I} F_i)$ generates $\mathcal{O}(X)$. Then $\coprod_{i \in I} D(F_i) \rightarrow X$ is a Zariski-local epimorphism.

(iii) Let L be a line bundle over X . We say L is *generated by a family of global sections* $(s_i)_{i \in I}$ if for any $x \in \text{El}(X)$, the elements $(s_i(x))_{i \in I}$ generate $L(x)$. In this case, $\coprod_{i \in I} D(s_i) \rightarrow X$ is a Zariski-local epimorphism.

(iv) The morphism $\mathbb{A}^I - 0 \rightarrow \mathbb{P}^I$ is a Zariski-local epimorphism. In fact, for a general $\mathbb{N}$ -graded algebra A , we have $D(A_1) \rightarrow \text{Proj}_1(A)$ is a Zariski-local epimorphism.

Proposition 3.3.3 (Zariski Descent for Quasi-coherent Modules). Let $(Y_i \rightarrow X)_{i \in I}$ be a family of morphisms such that $\coprod_{i \in I} Y_i \rightarrow X$ is a Zariski-local epimorphism. Then the following diagram is a limit diagram:

$$\text{Mod}_X \longrightarrow \prod_{i \in I} \text{Mod}_{Y_i} \rightrightarrows \prod_{i,j \in I} \text{Mod}_{Y_i \times_X Y_j} \rightrightarrows \prod_{i,j,k \in I} \text{Mod}_{Y_i \times_X Y_j \times_X Y_k}$$

In particular, $\text{Mod}_X \rightarrow \prod_{i \in I} \text{Mod}_{Y_i}$ is a conservative functor.

Proof. Set $Y = \coprod_i Y_i$ and let $p: Y \rightarrow X$ be the induced map. We prove directly, using only the affine module-descent theorem Theorem 1.3.1, that restriction identifies Mod_X with the category of descent data for the Čech nerve of p .

Full faithfulness. Let $M, N \in \text{Mod}_X$, and suppose that a morphism $p^*M \rightarrow p^*N$ is given together with its compatibility over $Y \times_X Y$. Fix an affine point $x: \text{Spec}(R) \rightarrow X$. By Zariski-local surjectivity, there are $f_1, \dots, f_n$ generating R and lifts

$$\tilde{x}_a: \text{Spec}(R_{f_a}) \longrightarrow Y$$

of $x|_{D(f_a)}$. The given morphism restricts to compatible maps

$$M(x)_{f_a} \longrightarrow N(x)_{f_a}.$$

Compatibility follows on $D(f_a f_b)$ from the map determined by the pair $(\tilde{x}_a, \tilde{x}_b)$ to $Y \times_X Y$. Descent for morphisms in Theorem 1.3.1(i) therefore produces a unique map $M(x) \rightarrow N(x)$. A second choice of principal cover and lifts gives the same map after passage to the common refinement, hence gives the same map globally. For a ring map $R \rightarrow S$ and a compatible point over S , the required base-change square commutes after localizing at the images of the f_a ; affine descent then makes it commute over S . Thus the maps just constructed form a morphism in Mod_X . This construction is inverse to restriction, proving full faithfulness.

Essential surjectivity. A descent datum consists of $M_Y \in \text{Mod}_Y$, an isomorphism between its two pullbacks to $Y \times_X Y$, and the cocycle identity over $Y \times_X Y \times_X Y$. For the affine point x and lifts $\tilde{x}_a$ above, put

$$M_a := M_Y(\tilde{x}_a) \in \text{Mod}_{R_{f_a}}.$$

On $D(f_a f_b)$, the pair of lifts defines a point of $Y \times_X Y$, so the descent isomorphism gives $(M_a)_{f_b} \simeq (M_b)_{f_a}$; the triple cocycle is exactly the cocycle required in Theorem 1.3.1(ii). That theorem glues the M_a to an R -module $M(x)$, unique up to unique isomorphism. Common refinements show that $M(x)$ is independent of all choices. Pulling a chosen principal cover along $R \rightarrow S$ and using uniqueness in affine descent gives the base-change isomorphism

$$M(x) \otimes_R S \xrightarrow{\sim} M(y).$$

Its identity and transitivity constraints may be checked after the same principal localization, where they are those of M_Y ; uniqueness in affine descent then gives them globally. Hence $x \mapsto M(x)$ is an object of Mod_X , and its restriction to Y recovers the original datum. This proves essential surjectivity and the asserted limit description.

Finally, an equivalence with a descent category detects isomorphisms on its degree-zero term. Therefore $\text{Mod}_X \rightarrow \prod_i \text{Mod}_{Y_i}$ is conservative. $\square$

Corollary 3.3.4 (Zariski Descent for Functions). *Let $(Y_i \rightarrow X)_{i \in I}$ be a family of morphisms such that $\coprod_{i \in I} Y_i \rightarrow X$ is a Zariski-local epimorphism. Then the following diagram of rings is an equalizer*

$$\mathcal{O}(X) \longrightarrow \prod_{i \in I} \mathcal{O}(Y_i) \rightrightarrows \prod_{i, j \in I} \mathcal{O}(Y_i \times_X Y_j).$$

Corollary 3.3.5 (Zariski Descent for $D(I)$). *Let A be a ring and $I \subset A$ a subset. Then $\coprod_{f \in I} D(f) \rightarrow D(I)$ is automatically a Zariski-local epimorphism, and:*

$$\text{Mod}_{D(I)} \longrightarrow \prod_{f \in I} \text{Mod}_{D(f)} \rightrightarrows \prod_{f, g \in I} \text{Mod}_{D(fg)} \rightrightarrows \prod_{f, g, h \in I} \text{Mod}_{D(fgh)}$$

is a limit diagram.

Proof. Set $X = D(I)$ and $Y_f = D(f)$. The claim reduces to the identification $D(fg) \simeq D(f) \times_{D(I)} D(g)$, which follows from

$$D(f) \times_{D(I)} D(g) \simeq D(f) \times_{\text{Spec}(A)} D(g) \simeq D(fg).$$

$\square$

We can now describe quasi-coherent modules on $\text{Proj}(A)$.

Example 3.3.6 (Modules over Proj). Let A be an $\mathbb{N}$ -graded ring.

- (i) Let $I \subset A_+$ be a subset of homogeneous elements satisfying $A_+ \subset \sqrt{(I)}$. Then $\coprod_{f \in I} \text{Spec}(A_{(f)}) \rightarrow \text{Proj}(A)$ is a Zariski-local epimorphism. Therefore, Proposition 3.3.3 shows that we have a limit diagram:

$$\text{Mod}_{\text{Proj}(A)} \longrightarrow \prod_{f \in I} \text{Mod}_{A_{(f)}} \rightrightarrows \prod_{f, g \in I} \text{Mod}_{A_{(fg)}} \rightrightarrows \prod_{f, g, h \in I} \text{Mod}_{A_{(fgh)}}.$$

- (ii) Consider $D(A_1) \subset \text{Spec}(A)$. The map $D(A_1) \rightarrow \text{Proj}_1(A)$ is a Zariski-local epimorphism inducing an injection $D(A_1)/\mathbb{G}_m \hookrightarrow \text{Proj}_1(A)$. The injectivity means that the iterated fiber product

$$D(A_1) \times_{\text{Proj}_1(A)} \cdots \times_{\text{Proj}_1(A)} D(A_1)$$

(n factors) coincides with the iterated fiber product over $D(A_1)/\mathbb{G}_m$. Since the $\mathbb{G}_m$ -action on $D(A_1)$ is free, the latter is just $D(A_1) \times_{\mathbb{G}_m}^{n-1}$. We thus obtain the limit diagram:

$$\text{Mod}_{\text{Proj}_1(A)} \longrightarrow \text{Mod}_{D(A_1)} \rightrightarrows \text{Mod}_{D(A_1) \times \mathbb{G}_m} \rightrightarrows \text{Mod}_{D(A_1) \times \mathbb{G}_m \times \mathbb{G}_m}.$$

Example 3.3.7 (Modules over $\mathbb{P}^1$). Let k be a ring. Consider the projective line $\mathbb{P}_k^1 = \text{Proj}(k[x, y])$. Set $u = y/x$ and $v = x/y$, so $uv = 1$. The degree-zero parts of the relevant localizations are

$$k[x, y]_{(x)} = k[u], \quad k[x, y]_{(y)} = k[v], \quad k[x, y]_{(xy)} = k[u, v]/(uv - 1) \simeq k[t^{\pm 1}],$$

where the last identification sets $t = u$ (equivalently $v = t^{-1}$). Therefore, the limit diagram from Example 3.3.6 can be rewritten as a pullback diagram:

$$\begin{array}{ccc} \text{Mod}_{\mathbb{P}_k^1} & \longrightarrow & \text{Mod}_{k[u]} \\ \downarrow & \lrcorner & \downarrow \\ \text{Mod}_{k[v]} & \longrightarrow & \text{Mod}_{k[t^{\pm 1}]} \end{array}$$

Thus we can write $\text{Mod}_{\mathbb{P}_k^1}$ as:

$$\text{Mod}_{\mathbb{P}_k^1} = \left\{ (M, N, \alpha) \left| \begin{array}{c} M \in \text{Mod}_{k[u]}, N \in \text{Mod}_{k[v]}, \\ \alpha: M[u^{\pm 1}] \xrightarrow{\sim} N[v^{\pm 1}] \text{ is a } k[t^{\pm 1}]\text{-linear isomorphism} \end{array} \right. \right\}.$$

Example 3.3.8 (Modules over the Affine Line with Two Origins). Let X be the affine line with two origins. From previous exercises, we know the explicit characterization of X is:

$$X(R) := \{(f, e) \mid f \in R, e \in R/(f), e^2 = e\}.$$

Let $X_0 \subset X$ and $X_1 \subset X$ be the loci where $e = 0$ and $e = 1$, respectively. Then we can verify:

- X_0 and X_1 are both open subfunctors of X , and $X_0 \sqcup X_1 \rightarrow X$ is a Zariski-local epimorphism.
- The projection $X \rightarrow \mathbb{A}_k^1$ given by $(f, e) \mapsto f$ induces isomorphisms:

$$X_0 \xrightarrow{\sim} \mathbb{A}_k^1, \quad X_1 \xrightarrow{\sim} \mathbb{A}_k^1, \quad X_0 \cap X_1 \xrightarrow{\sim} \mathbb{G}_{m,k}.$$

Therefore, combining with Proposition 3.3.3, we obtain a pullback diagram:

$$\begin{array}{ccc} \text{Mod}_X & \longrightarrow & \text{Mod}_{k[x]} \\ \downarrow & \lrcorner & \downarrow x \mapsto x \\ \text{Mod}_{k[x]} & \xrightarrow{x \mapsto x} & \text{Mod}_{k[x^{\pm 1}]} \end{array}$$

That is:

$$\text{Mod}_X = \left\{ (M, N, \alpha) \left| \begin{array}{c} M \in \text{Mod}_{k[x]}, N \in \text{Mod}_{k[x]}, \\ \alpha: M[x^{-1}] \xrightarrow{\sim} N[x^{-1}] \text{ is a } k[x^{\pm 1}]\text{-linear isomorphism} \end{array} \right. \right\}.$$

3.4. Modules over Quasi-projective Schemes

We now specialize the descent formalism to quasi-affine and quasi-projective schemes. The constructions below use the finite affine charts of $D(I)$ and $\text{Proj}(A)$, and culminate in Serre's twisting modules.

Recall the notion of I -local modules from Section 2.3: for a subset $I \subset A$ there is a localization functor

$$L_I: \text{Mod}_A \rightarrow \text{Mod}_A^{I\text{-loc}}.$$

Moreover, for a homomorphism $h: M \rightarrow N$, we have $L_I(h)$ is an isomorphism if and only if for all $f \in I$, the map $h_f: M_f \rightarrow N_f$ is an isomorphism.

Only this universal localization property is used below. In particular, the proof does not require the multi-generator reflector L_I to be exact; the exact functors used in the local calculation are the ordinary single-element localizations $(-)_f$.

Theorem 3.4.1 (Modules over $D(I)$). *Let A be a ring and $I \subset A$ a finite subset. Then the base change functor $\text{Mod}_A \rightarrow \text{Mod}_{D(I)}$ induced by $D(I) \hookrightarrow \text{Spec}(A)$ gives a categorical equivalence:*

$$\text{Mod}_A^{I\text{-loc}} \xrightarrow{\sim} \text{Mod}_{D(I)}.$$

Moreover, we have:

$$L_I(M) = \lim \left(\prod_{f \in I} M_f \rightrightarrows \prod_{f, g \in I} M_{fg} \right).$$

Remark 3.4.2 (Infinite Sets). When I is infinite, full faithfulness may fail: the argument for finite I uses that finite products commute with localization, while infinite products need not.

Proof. Let $i: D(I) \hookrightarrow \text{Spec}(A)$. We first construct the factorization in

$$\begin{array}{ccc} \text{Mod}_A & \xrightarrow{i^*} & \text{Mod}_{D(I)} \\ L_I \downarrow & \nearrow F & \\ \text{Mod}_A^{I\text{-loc}} & & \end{array}$$

Suppose $a: M \rightarrow N$ is inverted by L_I . By the characterization recalled above, every a_f is an isomorphism. The restriction of i^*a to the covering open $D(f)$ is precisely a_f , so Corollary 3.3.5 implies that i^*a is an isomorphism. Thus i^* inverts every L_I -equivalence, and the universal property of the reflective localization L_I supplies the dashed functor F , uniquely up to unique isomorphism. The same argument shows that F is conservative: if $F(a)$ is an isomorphism, then every a_f is an isomorphism, hence a is an isomorphism between I -local objects.

Write an object of $\text{Mod}_{D(I)}$ as descent data $(M_f, \alpha_{fg})_{f, g \in I}$ on the finite cover $(D(f))_{f \in I}$. Define

$$G(M_\bullet) := \text{eq} \left(\prod_{f \in I} M_f \rightrightarrows \prod_{f, g \in I} (M_f)_g \right)$$

Each M_f , regarded as an A -module, is f -local. It is therefore I -local: a map with I -nilpotent kernel and cokernel becomes an isomorphism after localizing at f , and maps into an f -local module may be computed after that localization. The full subcategory of I -local modules is closed under products and equalizers, so $G(M_\bullet)$ is I -local. Descent for morphisms on the cover $(D(f))$ gives natural bijections

$$\text{Hom}_{\text{Mod}_{D(I)}}(F(N), M_\bullet) \simeq \text{Hom}_A(N, G(M_\bullet)) \simeq \text{Hom}_{\text{Mod}_A^{I\text{-loc}}}(N, G(M_\bullet)),$$

so $F \dashv G$.

We now compute the counit locally. For $h \in I$, exactness of the ordinary localization $(-)_h$ and finiteness of I allow localization to pass through the displayed equalizer and its finite products. The affine module-descent calculation of Theorem 1.3.1, applied after localizing at h , then gives

$$G(M_\bullet)_h \xrightarrow{\sim} M_h.$$

These isomorphisms respect the gluing maps α_{fg} , so the counit $FG(M_\bullet) \rightarrow M_\bullet$ is an isomorphism. Hence G is fully faithful. Since F is conservative, the unit is also an isomorphism: applying F to it gives an isomorphism by the triangle identities and the already invertible counit. Thus F is an equivalence.

Applying the inverse equivalence G to i^*M gives exactly the displayed equalizer formula for $L_I(M)$. Notice again that this last calculation uses exactness only of each single-element localization $(-)_h$, not exactness of L_I . $\square$

Corollary 3.4.3 (Affine Completion of Quasi-affine Schemes). *Let X be a quasi-affine scheme. Then the canonical morphism $X \rightarrow \text{Spec}(\mathcal{O}(X))$ is an open immersion cut out by some finitely generated radical ideal.*

Proof. Since X is a quasi-affine scheme, there exist a ring A and a finite subset $I \subset A$ such that $X \simeq D(I)$. By Theorem 3.4.1, we have a categorical equivalence:

$$\mathrm{Mod}_A^{I\text{-loc}} \simeq \mathrm{Mod}_{D(I)} \simeq \mathrm{Mod}_X.$$

This actually says that $\mathcal{O}_X \in \mathrm{Mod}_X$ corresponds to $L_I(A)$. In other words, the global sections ring $\mathcal{O}(X) \in \mathrm{CAlg}$ corresponds to $\mathrm{End}_A(L_I(A))$. Moreover, we can see that the ring homomorphism $\lambda: A \rightarrow L_I A$ induces the canonical morphism:

$$X \rightarrow \mathrm{Spec}(\mathcal{O}(X)).$$

Next, consider the pullback:

$$\begin{array}{ccc} D(\lambda(I)) & \hookrightarrow & \mathrm{Spec}(\mathcal{O}(X)) \\ \downarrow & \lrcorner & \downarrow \mathrm{Spec}(\lambda) \\ D(I) & \hookrightarrow & \mathrm{Spec}(A). \end{array}$$

Everything reduces to showing $D(\lambda(I)) \rightarrow D(I)$ is an isomorphism. For this, we only need to show that for any $\varphi: A \rightarrow R$, if φ lies in $D(I)(R)$, then R can be viewed as an I -local A -module via φ . This follows from $D(I)(R) = \{\varphi: A \rightarrow R \mid (\varphi(I)) = R\}$ and Zariski descent. $\square$

Construction 3.4.4 (Quasi-coherent Module from $\mathbb{Z}$ -graded Module). Let A be an $\mathbb{N}$ -graded ring generated by $A_{\leq 1}$. We define a functor:

$$\{\mathbb{Z}\text{-graded } A\text{-modules}\} \rightarrow \mathrm{Mod}_{\mathrm{Proj}(A)}, \quad M \mapsto \tilde{M}.$$

By the structure theorem, $\mathrm{Proj}(A) \simeq \mathrm{Proj}_1(A)$. Thus an R -point $x: \mathrm{Spec}(R) \rightarrow \mathrm{Proj}(A)$ is represented by a quotient of $\mathbb{N}$ -graded algebras

$$\varphi: A \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L).$$

Let $\tilde{\varphi}$ be the induced map of $\mathbb{Z}$ -graded rings $A \rightarrow \bigoplus_{n \in \mathbb{Z}} L^{\otimes n}$. For a $\mathbb{Z}$ -graded A -module M , define

$$\tilde{M}(x) = \tilde{\varphi}^*(M)_0 \in \mathrm{Mod}_R.$$

For $f \in A_1$, restriction along $D(f) \simeq \mathrm{Spec}(A_{(f)})$ sends $\tilde{M}$ to $M_{(f)}$. Example 3.3.6 then shows that $M \mapsto \tilde{M}$ factors through L_{A_+} .

Theorem 3.4.5 (Quasi-coherent Modules on Quasi-projective Schemes). Let A be an $\mathbb{N}$ -graded ring generated by $A_{\leq 1}$, and let $I \subset A_1$ be a finite subset such that $D(I) \subset \mathrm{Proj}(A)$. Then the functor $M \mapsto \tilde{M}$ induces an equivalence of categories

$$\{I\text{-local } \mathbb{Z}\text{-graded } A\text{-modules}\} \xrightarrow{\sim} \mathrm{Mod}_{D(I)}.$$

In particular, if A is finitely generated as an A_0 -algebra, then

$$\{A_1\text{-local } \mathbb{Z}\text{-graded } A\text{-modules}\} \xrightarrow{\sim} \mathrm{Mod}_{\mathrm{Proj}(A)}.$$

Proof. Since A is generated by $A_{\leq 1}$, the open subschemes $\mathrm{Spec}(A_{(f)}) \simeq D(f)$ for $f \in I$ form an affine open cover of $D(I) \subset \mathrm{Proj}(A)$. By Example 3.3.6(i) (an instance of Proposition 3.3.3), we have a limit description

$$\mathrm{Mod}_{D(I)} \xrightarrow{\sim} \lim \left(\prod_{f \in I} \mathrm{Mod}_{A_{(f)}} \rightrightarrows \prod_{f, g \in I} \mathrm{Mod}_{A_{(fg)}} \rightrightarrows \prod_{f, g, h \in I} \mathrm{Mod}_{A_{(fgh)}} \right). \quad (3.1)$$

On the other side, write $\mathrm{Mod}_A^{\mathbb{Z}\text{-gr}}$ for $\mathbb{Z}$ -graded A -modules. For each $f \in A_1$, the degree-zero functor $N \mapsto N_0$ on graded A_f -modules gives an equivalence $\mathrm{Mod}_{A_f}^{\mathbb{Z}\text{-gr}} \xrightarrow{\sim} \mathrm{Mod}_{A_{(f)}}$, with inverse $P \mapsto \bigoplus_{n \in \mathbb{Z}} P \otimes_{A_{(f)}} A_{(f)}^{\otimes n}$.

$(A_f)_n$ (using the trivialization of the line A_f -module $(A_f)_1 \simeq A_{(f)} \cdot f$). Composing with the localization $\text{Mod}_A^{\mathbb{Z}\text{-gr}} \rightarrow \text{Mod}_{A_f}^{\mathbb{Z}\text{-gr}}, M \mapsto M_f$, yields a functor

$$\Phi_f: \text{Mod}_A^{\mathbb{Z}\text{-gr}} \rightarrow \text{Mod}_{A_{(f)}}, \quad M \mapsto (M_f)_0 = M_{(f)}.$$

Bringing these together, Construction 3.4.4 sends a graded A -module M to the descent datum $((M_{(f)})_f, \text{evident gluings})$ on the cover $\{D(f)\}_{f \in I}$. Under the explicit equivalence (3.1), this datum corresponds exactly to $\tilde{M}|_{D(I)}$. Thus the construction is compatible at every stage of the displayed Čech limit, without introducing any abbreviated localization notation.

It remains to identify the kernel and essential image of $(-)$. By Theorem 3.4.1, the right-hand limit is equivalent to I -local A -modules where I is regarded as a subset of A ; localizing degree-by-degree on the graded level gives the same equivalence at the graded level. Concretely, $\tilde{M} \simeq \tilde{N}$ iff $M_f \simeq N_f$ as graded A_f -modules for every $f \in I$, which by Proposition 2.3.24(v) is exactly $L_I M \simeq L_I N$. Restricting to I -local graded modules makes $(-)$ fully faithful.

For essential surjectivity: any descent datum $(N_f, \alpha_{fg})_{f \in I}$ with $N_f \in \text{Mod}_{A_{(f)}}$ assembles into a graded A -module via

$$M = \bigoplus_{n \in \mathbb{Z}} \lim \left(\prod_f N_f \otimes_{A_{(f)}} (A_f)_n \rightrightarrows \prod_{f,g} N_f \otimes_{A_{(f)}} (A_{fg})_n \right),$$

whose localizations at each $f \in I$ recover $\bigoplus_n N_f \otimes_{A_{(f)}} (A_f)_n$ — i.e., the graded module corresponding to N_f . By construction M is I -local, and $\tilde{M}$ recovers the original descent datum.

The “in particular” statement: when A is finitely generated over A_0 and generated by $A_{\leq 1}$, the degree-1 part A_1 is itself finitely generated. Choosing I to be a finite generating set of A_1 , we have $D(I) = D(A_1) = \text{Proj}(A)$ inside $\text{Proj}(A)$, and “ I -local” coincides with “ A_1 -local”. $\square$

3.4.1. Serre Twisting

Let A be a $\mathbb{Z}$ -graded ring, M a $\mathbb{Z}$ -graded A -module, and $d \in \mathbb{Z}$. Recall the shift notation $M(d) := \bigoplus_{n \in \mathbb{Z}} M_{n+d}$ (so $(M(d))_n = M_{n+d}$).

Definition 3.4.6 (Serre Twisting). Let A be an $\mathbb{N}$ -graded ring generated by $A_{\leq 1}$ and let $d \in \mathbb{Z}$. We denote by $\mathcal{O}(d)$ the quasi-coherent module corresponding to the $\mathbb{Z}$ -graded A -module $A(d)$. For a given $M \in \text{Mod}_{\text{Proj}(A)}$, the d -th Serre twist of M is:

$$M(d) := M \otimes \mathcal{O}(d).$$

Remark 3.4.7 (Explicit Description). If A is generated by $A_{\leq 1}$, an R -point $x: \text{Spec}(R) \rightarrow \text{Proj}(A)$ corresponds to a quotient line $A_1 \otimes_{A_0} R \twoheadrightarrow L$ such that $A \otimes_{A_0} R \rightarrow \text{Sym}_R(L)$ factors through A . Under this correspondence $\mathcal{O}(d)(x) = L^{\otimes d}$.

Remark 3.4.8 (Notation Variation). In some references, $\mathcal{O}(-d)$ may be used to denote what we call $\mathcal{O}(d)$.

Definition 3.4.9 (Tautological Line Bundle). Let A be an $\mathbb{N}$ -graded ring generated by $A_{\leq 1}$. We call $\mathcal{O}(1)$ the *tautological line bundle* on $\text{Proj}(A)$.

We collect the basic properties of Serre twisting.

Proposition 3.4.10 (Properties of Serre Twisting). Let A be an $\mathbb{N}$ -graded ring generated by $A_{\leq 1}$.

- (i) For any $d \in \mathbb{Z}$, $\mathcal{O}(d)$ is a line bundle on $\text{Proj}(A)$.

- (ii) For a $\mathbb{Z}$ -graded A -module M and $d \in \mathbb{Z}$, there exists an isomorphism $\widetilde{M}(d) \xrightarrow{\sim} \widetilde{M(d)}$. In particular, for all $d, e \in \mathbb{Z}$, there exist isomorphisms:

$$\mathcal{O}(d) \otimes \mathcal{O}(e) \simeq \mathcal{O}(d+e) \quad \text{and} \quad \mathcal{O}(1)^{\otimes d} \xrightarrow{\sim} \mathcal{O}(d).$$

Therefore, $\bigoplus_{d \in \mathbb{Z}} \mathcal{O}(d)$ can be viewed as a $\mathbb{Z}$ -graded quasi-coherent algebra on $\text{Proj}(A)$.

- (iii) There exist canonical maps of $\mathbb{Z}$ -graded rings:

$$L_{A_1} A \rightarrow \bigoplus_{d \in \mathbb{Z}} \Gamma(\text{Proj}(A), \mathcal{O}(d)),$$

and for any $\mathbb{Z}$ -graded A -module M , there exist canonical maps of $\mathbb{Z}$ -graded A -modules:

$$L_{A_1} M \rightarrow \bigoplus_{d \in \mathbb{Z}} \Gamma(\text{Proj}(A), \widetilde{M}(d)).$$

When A is finitely generated as an A_0 -algebra, these are isomorphisms.

Proof. (i) For $x: \text{Spec}(R) \rightarrow \text{Proj}(A)$, we know it corresponds to $\varphi: A \otimes_{A_0} R \twoheadrightarrow \text{Sym}_R(L)$. Therefore, we can consider the commutative diagram:

$$\begin{array}{ccc} A & \xrightarrow{\quad \widetilde{\varphi} \quad} & \bigoplus_{n \in \mathbb{Z}} L^{\otimes n} \\ \downarrow & & \uparrow \\ A \otimes_{A_0} R & \xrightarrow{\quad \varphi \quad} & \text{Sym}_R(L). \end{array}$$

This gives the dashed $\widetilde{\varphi}$. By definition:

$$\mathcal{O}(d)(x) = \widetilde{A(d)}(x) = \widetilde{\varphi}^*(A(d))_0 = \left(\bigoplus_{n \in \mathbb{Z}} L^{\otimes n}(d) \right)_0 = L^{\otimes d}.$$

- (ii) We have:

$$(\widetilde{M} \otimes_{\mathcal{O}_X} \mathcal{O}(d))(x) = \widetilde{M}(x) \otimes_R \mathcal{O}(d)(x) = \widetilde{\varphi}^*(M) \otimes_R L^{\otimes d} = \widetilde{\varphi}^*(M(d))_0.$$

- (iii) *Construction of the canonical maps.* For each R -point $x: \text{Spec}(R) \rightarrow \text{Proj}(A)$, parts (1)–(2) furnish the extension $\widetilde{\varphi}: A \rightarrow \bigoplus_n L^{\otimes n}$ and the identification $\widetilde{M}(d)(x) = \widetilde{\varphi}^*(M(d))_0$; the resulting evaluation $m \mapsto \widetilde{\varphi}^*(m)$ is natural in x . Letting x range over all points and passing to the limit that defines global sections, we obtain $\mathbb{Z}$ -graded maps

$$L_{A_1} A \rightarrow \bigoplus_{d \in \mathbb{Z}} \Gamma(\text{Proj}(A), \mathcal{O}(d)), \quad L_{A_1} M \rightarrow \bigoplus_{d \in \mathbb{Z}} \Gamma(\text{Proj}(A), \widetilde{M}(d)),$$

the first a ring map and the second A -linear. This construction uses only that A is generated by $A_{\leq 1}$, so the maps exist in the stated generality.

The isomorphism case. When A is finitely generated as an A_0 -algebra, $\text{Proj}(A)$ is a quasi-projective scheme, so by Theorem 3.4.5 there is an isomorphism:

$$\text{Hom}_A^{A_1\text{-loc}, \mathbb{Z}\text{-graded}}(L_{A_1} A, L_{A_1} A(d)) \xrightarrow[\substack{\sim \\ M \mapsto \widetilde{M}}]{} \text{Hom}_{\text{Mod}_{\text{Proj}(A)}}(\mathcal{O}, \mathcal{O}(d)) = \Gamma(\text{Proj}(A), \mathcal{O}(d)).$$

The left side is nothing but $\text{Hom}_{\text{Mod}_A^{\mathbb{Z}}}(A, L_{A_1} A(d)) \simeq (L_{A_1} A)_d$, and under these identifications the isomorphism is the canonical map constructed above; the module version is identical with A replaced by M .

□

Looking Ahead

We have now constructed the ambient linear algebra of algebraic functors: Mod_X , together with the separately defined posets Id_X of quasi-coherent ideals and Rad_X of quasi-coherent radical ideals, the relative Spec and Proj over a base, Zariski descent for modules, and Serre's twisting sheaves $\mathcal{O}(d)$ on projective schemes. In general there is no inclusion $\text{Rad}_X \subseteq \text{Id}_X$: as explained in Warning 3.1.18, forgetting the radical structure is not compatible with base change. Likewise, the pointwise radical in the list of Remark 3.1.16 naturally defines an object of Rad_X , not automatically one of Id_X . Together with the content of Chapters 1 and 2, the sheaf-theoretic / module-theoretic vocabulary is complete.

What is missing is *topology*. To talk about open covers, generic points, connected components, or dimension, we need to attach to each algebraic functor X a genuine topological space $|X|$ compatible with its subfunctor structure. This is the subject of the next chapter, where we develop point-free topology (locales) and prove that the poset $\text{Open}(X)$ of open subfunctors is always a spatial locale, realized by the space $|X|$. From Chapter 5 onward the topological and categorical pictures will be available in parallel.

Chapter 4.

Locale

In this chapter we introduce *pointless topology*, otherwise known as the theory of *locales*.

When studying a topological space, we often care more about the lattice of open sets than the underlying points. Locales formalize this — open-set data, abstracted away from any concrete point set. The goal is to construct the diagram

$$\begin{array}{ccc} & & \text{Loc} \\ & & \downarrow \\ \text{Top} & \xrightarrow{X \mapsto \text{Open}(X)} & \text{Pos}^{\text{op}}. \end{array}$$

Not every space is recoverable from its open-set lattice; those that are constitute the class of *sober spaces*, and we have

$$\text{Top}^{\text{Haus}} \subset \text{Top}^{\text{sober}} \hookrightarrow \text{Loc}.$$

4.1. Pointless Topology

In fact, M. Hochster proved the following theorem in 1969:

Theorem 4.1.1 (Hochster [Hoc69]). *Let X be a topological space. The following are equivalent:*

- (i) *There exists a ring A such that $X \simeq \text{Prim}(A) = |\text{Spec}(A)|$.*
- (ii) *X is quasi-compact and has a basis $\mathcal{B}$ of quasi-compact open sets that is closed under finite intersections, and X is sober.*
- (iii) *X can be written as an inverse limit of finite T_0 spaces.*

The proof — that every space satisfying (ii) arises as a prime spectrum — is delicate; we refer to the original paper.

Definition 4.1.2 (Spectral Space). *A spectral space is a topological space satisfying any of the equivalent conditions in Theorem 4.1.1.*

For algebraic geometry, sober spaces will be sufficient throughout.

The main goal of this chapter is to show that for any algebraic functor X , the poset $\text{Open}(X)$ of open subfunctors is a locale, and it lies in the essential image of:

$$\text{Top} \xrightarrow{\text{Open}} \text{Loc}.$$

That is, there exists $|X|$ such that $\text{Open}(X) \simeq \text{Open}(|X|)$. We call $|X|$ the *geometric realization* or *underlying topological space* of X .

The starting point of “pointless topology” is the observation that most topological spaces (including all Hausdorff spaces) are determined by their posets of open subsets. Since posets are ubiquitous in

mathematics, this observation lets topology appear in unexpected places. Our goal here is to develop the formalism that will later express $\text{Open}(X)$ of an algebraic functor X as the open-set poset of a topological space $|X|$.

Notation 4.1.3. Let P be a poset. Given a family $(x_i)_{i \in I}$ in P , we write $\bigvee_{i \in I} x_i$ for its supremum (least upper bound) and $\bigwedge_{i \in I} x_i$ for its infimum (greatest lower bound), when they exist.

Remark 4.1.4 (Posets as Categories). (i) Viewing a poset as a category, suprema are colimits and infima are limits.

(ii) The supremum of the empty family is the smallest element (initial object); dually, the infimum of the empty family is the largest element.

(iii) A poset admits all suprema iff it admits all infima: the infimum of a family equals the supremum of all elements below it.

Definition 4.1.5 (Locale). A *locale* is a poset O satisfying:

(i) (*Completeness*) O admits all suprema (hence all infima).

(ii) (*Distributivity*) For any $u \in O$ and family $(v_i)_{i \in I}$,

$$u \wedge \left(\bigvee_{i \in I} v_i \right) = \bigvee_{i \in I} (u \wedge v_i).$$

A *morphism of locales* $f: O' \rightarrow O$ is a morphism of posets $f^*: O \rightarrow O'$ (in the opposite direction) that preserves colimits (arbitrary suprema) and finite limits (finite infima). We denote the category of locales by Loc .

Example 4.1.6 (Totally Ordered Sets). Any totally ordered set with suprema is a locale. For instance, $\mathbb{N} \cup \{+\infty\}$ and $\mathbb{R} \cup \{\pm\infty\}$ are locales.

Example 4.1.7 (The Locale of a Topological Space). Let T be a topological space. The poset $\text{Open}(T)$ of open subsets of T is a locale:

- Suprema: $\bigvee_i U_i = \bigcup_i U_i$; infima: $\bigwedge_i U_i = \text{interior}(\bigcap_i U_i)$.
- Distributivity follows from the set-theoretic identity $U \cap (\bigcup_i V_i) = \bigcup_i (U \cap V_i)$ combined with the fact that finite intersections of opens are open.

A continuous map $f: T \rightarrow S$ induces a locale morphism $\text{Open}(T) \rightarrow \text{Open}(S)$ via $f^{-1}: \text{Open}(S) \rightarrow \text{Open}(T)$ (which preserves colimits and finite limits). We thus obtain a functor

$$\text{Open}: \text{Top} \rightarrow \text{Loc}.$$

Remark 4.1.8 (Frames). A *frame* is the same lattice-theoretic object as a locale, but a frame morphism is the map f^* itself, preserving arbitrary joins and finite meets. Thus, with these morphism conventions,

$$\text{Loc} = \text{Frm}^{\text{op}}.$$

A frame is equivalently a complete Heyting algebra. Keeping the words “locale” and “frame” separate will prevent variance errors below.

Proposition 4.1.9 (Limits and Colimits of Locales). The category Loc admits limits and colimits, and the inclusion $\text{Loc} \hookrightarrow \text{Pos}^{\text{op}}$ preserves colimits. The initial object is $0 = \text{Open}(\emptyset) = \{\emptyset\}$ and the terminal object is $1 = \text{Open}(\{*\}) = \{\emptyset \rightarrow *\}$.

Proof. The variance can be summarized as follows:

$$\begin{aligned} \operatorname{colim}_{i \in I}^{\operatorname{Loc}} P_i &\xrightarrow{\operatorname{Loc}=\operatorname{Frm}^{\operatorname{op}}} \operatorname{Frm} \lim_{i \in I^{\operatorname{op}}} P_i, \\ \operatorname{Loc} \lim_{i \in I} P_i &\xrightarrow{\operatorname{Loc}=\operatorname{Frm}^{\operatorname{op}}} \operatorname{colim}_{i \in I^{\operatorname{op}}}^{\operatorname{Frm}} P_i. \end{aligned}$$

Colimits in Loc. View $(P_i)_{i \in I}$ as an I^{op} -diagram of frames and form its limit P in Pos . Limits of frames are created by the forgetful functor to posets: joins and finite meets are computed pointwise,

$$\left(\bigvee_{\alpha} p_{\alpha} \right)_i = \bigvee_{\alpha} (p_{\alpha})_i, \quad (p \wedge q)_i = p_i \wedge q_i,$$

and the frame distributivity law is therefore checked pointwise. This constructs $\operatorname{colim}_I^{\operatorname{Loc}} P_i$ and shows that $\operatorname{Loc} \hookrightarrow \operatorname{Pos}^{\operatorname{op}}$ preserves colimits.

Limits in Loc. These correspond to colimits of frames and are *not* generally computed pointwise on the underlying posets. To construct such a frame colimit, take one generator $[i, u]$ for every $u \in P_i$, form the free frame on this set, and quotient by the relations expressing all joins and finite meets already present in each P_i together with the relations imposed by the transition maps. A map out of the resulting quotient is exactly a compatible family of frame maps out of the P_i , which is the required colimit universal property. Hence frame colimits, and therefore locale limits, exist.

Initial and terminal objects. For any locale L , $\operatorname{Hom}_{\operatorname{Loc}}(\{\emptyset\}, L)$ and $\operatorname{Hom}_{\operatorname{Loc}}(L, \{\emptyset \rightarrow *\})$ are singletons by direct verification. $\square$

Definition 4.1.10 (Points of a Locale). Let O be a locale. The *point space* $\operatorname{Pt}(O)$ is the topological space defined by:

- Points: morphisms of locales $p: 1 \rightarrow O$.
- Open subsets: for each $u \in O$, the subset $\operatorname{Pt}(u) := \{p \mid p^*(u) = *\}$.

A locale morphism $f: O' \rightarrow O$ induces a continuous map $\operatorname{Pt}(O') \rightarrow \operatorname{Pt}(O)$ (since $\operatorname{Pt}(u)$ pulls back to $\operatorname{Pt}(f^*(u))$), giving a functor

$$\operatorname{Pt}: \operatorname{Loc} \rightarrow \operatorname{Top}.$$

Remark 4.1.11 (Points as Completely Prime Filters). Unwinding the definition: a poset map $O \rightarrow \{\emptyset \rightarrow *\}$ is specified by the preimage $F \subset O$ of $*$. Such an F is a *filter* when (i) upward-closed and (ii) nonempty + closed under binary meets; a filter is *completely prime* when (iii) $\bigvee_i x_i \in F$ implies $x_i \in F$ for some i . The points of a locale O correspond bijectively to completely prime filters on O .

Proposition 4.1.12 ($\operatorname{Open} \dashv \operatorname{Pt}$). The functor $\operatorname{Open}: \operatorname{Top} \rightarrow \operatorname{Loc}$ is left adjoint to $\operatorname{Pt}: \operatorname{Loc} \rightarrow \operatorname{Top}$. The adjunction is idempotent: it restricts to an equivalence between the essential images of both functors.

Proof. Adjunction. We exhibit mutually inverse natural bijections

$$\operatorname{Hom}_{\operatorname{Top}}(T, \operatorname{Pt}(L)) \rightleftarrows \operatorname{Hom}_{\operatorname{Loc}}(\operatorname{Open}(T), L).$$

A continuous map $f: T \rightarrow \operatorname{Pt}(L)$ goes to the locale morphism whose underlying poset map $L \rightarrow \operatorname{Open}(T)$ sends $u \mapsto f^{-1}(\operatorname{Pt}(u))$. Conversely, a locale morphism $g: \operatorname{Open}(T) \rightarrow L$ goes to the continuous map $T \rightarrow \operatorname{Pt}(L)$, $t \mapsto g \circ \operatorname{Open}(\iota_t)$, where $\iota_t: \{*\} \hookrightarrow T$ is the inclusion of the point t .

Idempotency. The unit at T is

$$\eta_T: T \rightarrow \operatorname{Pt}(\operatorname{Open}(T)), \quad t \mapsto (\iota_t^{-1}: \operatorname{Open}(T) \rightarrow \operatorname{Open}(\{*\})).$$

By construction $\eta_T^{-1}(\text{Pt}(u)) = u$ for every $u \in \text{Open}(T)$, so $\text{Open}(\eta_T): \text{Open}(T) \rightarrow \text{Open}(\text{Pt}(\text{Open}(T)))$ is inverse to the canonical map $u \mapsto \text{Pt}(u)$, hence an isomorphism. The dual statement that $\text{Pt}(\varepsilon_L)$ is an isomorphism for every locale L follows analogously, completing idempotency. $\square$

Definition 4.1.13 (Sober Spaces and Spatial Locales). (i) A topological space T is *sober* if it lies in the essential image of Pt , equivalently if the unit $T \rightarrow \text{Pt}(\text{Open}(T))$ is a homeomorphism.
(ii) A locale O is *spatial* if it lies in the essential image of Open , equivalently if the counit $\text{Open}(\text{Pt}(O)) \rightarrow O$ is an isomorphism.

We write $\text{Top}^{\text{sob}} \subset \text{Top}$ and $\text{Loc}^{\text{spa}} \subset \text{Loc}$ for the respective full subcategories.

Remark 4.1.14 (Soberification and Spatialization). By Proposition 4.1.12, the adjunction $\text{Open} \dashv \text{Pt}$ restricts to an equivalence $\text{Top}^{\text{sob}} \simeq \text{Loc}^{\text{spa}}$. The inclusion $\text{Top}^{\text{sob}} \hookrightarrow \text{Top}$ has a left adjoint $\text{Pt} \circ \text{Open}$ called *soberification*; the inclusion $\text{Loc}^{\text{spa}} \hookrightarrow \text{Loc}$ has a right adjoint $\text{Open} \circ \text{Pt}$ called *spatialization*. Consequently, limits of sober spaces are sober and colimits of spatial locales are spatial.

Proposition 4.1.15 (Characterization of Sober Spaces). *Let T be a topological space. The points of the locale $\text{Open}(T)$ are in bijection with the irreducible closed subsets of T , and the unit $T \rightarrow \text{Pt}(\text{Open}(T))$ sends $t \mapsto \overline{\{t\}}$. Hence T is sober iff the map*

$$\{\text{points of } T\} \rightarrow \{\text{irreducible closed subsets of } T\}, \quad t \mapsto \overline{\{t\}},$$

is a bijection — i.e., every irreducible closed subset has a unique generic point.

Remark 4.1.16 (Sober Examples and Locality). (i) Every Hausdorff space is sober, since its irreducible subsets are singletons.

(ii) Sobriety is a local property: if T admits a cover by sober open subsets, then T is sober.

4.2. Locales of Radical Ideals

Let R be a ring. Recall (Proposition 3.2.1) that open subfunctors of $\text{Spec}(R)$ correspond to radical ideals in R via $D: \text{Rad}_R \xrightarrow{\sim} \text{Open}(\text{Spec}(R))$. We verify that Rad_R is a locale and compute its points.

Proposition 4.2.1 (Rad_R is a Locale). *For any ring R , the poset Rad_R of radical ideals is a locale. Explicitly:*

- The supremum of a family $(K_i)_{i \in I}$ is $\sqrt{\sum_i K_i}$.
- The infimum (distributing over sums) satisfies

$$K \cap \sqrt{\sum_i L_i} = \sqrt{\sum_i (K \cap L_i)}.$$

The distributivity rests on the identities $K \cap L = \sqrt{KL}$ for radical ideals and the distributivity of the product of ideals over sums. (The full lattice Id_R of arbitrary ideals is not a locale, since intersection does not distribute over sums.)

Remark 4.2.2 (Base Change). For a ring map $R \rightarrow S$, the base-change map $\text{Id}_R \rightarrow \text{Id}_S, I \mapsto IS$, preserves products and sums of ideals. Consequently the induced $\text{Rad}_R \rightarrow \text{Rad}_S, I \mapsto \sqrt{IS}$, preserves finite infima and suprema, hence is a morphism of locales $\text{Rad}_S \rightarrow \text{Rad}_R$ (note the direction).

Definition 4.2.3 (Prime Spectrum). Let R be a ring. The *prime spectrum* $\text{Prim}(R)$ is the topological space with

- underlying set: the set of prime ideals of R ;

- open subsets: $\text{Prim}(I) := \{\mathfrak{p} \mid I \not\subset \mathfrak{p}\}$ for $I \subset R$.

Since prime ideals are radical, $\text{Prim}(I) = \text{Prim}(\sqrt{I})$ depends only on the radical.

Proposition 4.2.4 (The Locale of a Ring). *Let R be a ring.*

(i) *There is a homeomorphism*

$$\text{Prim}(R) \xrightarrow{\sim} \text{Pt}(\text{Rad}_R), \quad \mathfrak{p} \mapsto \{I \in \text{Rad}_R \mid I \not\subset \mathfrak{p}\},$$

under which $\text{Pt}(I)$ corresponds to $\text{Prim}(I)$.

(ii) *The locale Rad_R is spatial: the map in (i) induces $\text{Open}(\text{Prim}(R)) \xrightarrow{\sim} \text{Rad}_R$.*

(iii) *$\text{Prim}(R)$ is a spectral space: it is sober, and its quasi-compact open subsets form a basis closed under finite intersections.*

Proof. (i) *Homeomorphism.* Fix a prime $\mathfrak{p} \subset R$ and set

$$F_{\mathfrak{p}} = \{I \in \text{Rad}_R \mid I \not\subset \mathfrak{p}\}.$$

We verify $F_{\mathfrak{p}}$ is a completely prime filter:

- *Upward-closed:* if $I \in F_{\mathfrak{p}}$ and $I \subset J$, then any element of $I \setminus \mathfrak{p}$ also lies in J , so $J \notin F_{\mathfrak{p}}$.
- *Closed under finite meets:* if $I, J \in F_{\mathfrak{p}}$, pick $i \in I \setminus \mathfrak{p}$ and $j \in J \setminus \mathfrak{p}$; primality of $\mathfrak{p}$ gives $ij \notin \mathfrak{p}$, while $ij \in IJ \subset I \cap J = \sqrt{IJ}$.
- *Completely prime:* if $\bigvee_i K_i = \sqrt{\sum_i K_i} \in F_{\mathfrak{p}}$, then $\sum_i K_i \not\subset \mathfrak{p}$ (since $\mathfrak{p}$ is radical), so some $K_i \not\subset \mathfrak{p}$.

For the inverse, given a completely prime filter F , set

$$q_F := \bigvee_{\text{Rad}_R} \{J \in \text{Rad}_R \mid J \notin F\} = \sqrt{\sum_{J \notin F} J}.$$

The complement $F^c \subset \text{Rad}_R$ is downward-closed (since F is upward-closed) and closed under arbitrary suprema (the contrapositive of complete primality), so the displayed supremum is itself in F^c , i.e. q_F is a radical ideal of R not in F . To see it is prime: if $a, b \in R$ with $ab \in q_F$, then $\sqrt{(ab)} \subset q_F$, so $\sqrt{(ab)} \notin F$ — otherwise upward-closure would force $q_F \in F$, contradicting that $q_F = \sup_{\text{Rad}_R} F^c \notin F$. Now $\sqrt{(ab)} = \sqrt{(a)} \cap \sqrt{(b)}$, and F is closed under finite meets, so $\sqrt{(a)} \notin F$ or $\sqrt{(b)} \notin F$, i.e. $a \in q_F$ or $b \in q_F$.

The two constructions are mutually inverse: $F_{q_F} = F$ unfolds to a tautology, and $q_{F_{\mathfrak{p}}} = \mathfrak{p}$ uses that a radical ideal equals the intersection of the primes containing it.

(ii) *Spatiality.* It suffices to show $I \mapsto \text{Prim}(I)$ is a bijection $\text{Rad}_R \rightarrow \text{Open}(\text{Prim}(R))$. Surjectivity is the definition of the topology; injectivity follows from $I = \bigcap_{I \subset \mathfrak{p}} \mathfrak{p}$.

(iii) *Spectrality.* Sobriety is automatic: by (i), $\text{Prim}(R) \cong \text{Pt}(\text{Rad}_R)$, and point spaces of locales are sober. The basic opens $\text{Prim}(f) \cong \text{Prim}(R_f)$ for $f \in R$ form a basis: any open is $\text{Prim}(I) = \bigcup_{f \in I} \text{Prim}(f)$. Each $\text{Prim}(f)$ is quasi-compact, and the basis is closed under finite intersections via $\text{Prim}(f) \cap \text{Prim}(g) = \text{Prim}(fg)$. Quasi-compactness of $\text{Prim}(R)$ itself: if $(\text{Prim}(f_i))_{i \in I}$ is any cover, then $\bigcup_i \text{Prim}(f_i) = \text{Prim}(R)$, equivalently $\sqrt{(f_i \mid i \in I)} = R$, equivalently $(f_i \mid i \in I) = R$. Writing $1 = \sum_{j=1}^n c_j f_{i_j}$ as a finite sum exhibits a finite subcover $(\text{Prim}(f_{i_j}))_{j=1}^n$. $\square$

Remark 4.2.5 (Converse to Hochster). Conversely, every spectral space is homeomorphic to $\text{Prim}(R)$ for some ring R (the R is far from unique; see Theorem 4.1.1).

Now let A be an $\mathbb{N}$ -graded ring, and write hRad_A for the poset of *saturated radical homogeneous* ideals of A . By the projective Nullstellensatz, $\text{hRad}_A \simeq \text{Open}(\text{Proj}(A))$. This poset is again a locale: distributivity inherits from Rad_A combined with the fact that saturation preserves finite intersections.

Definition 4.2.6 (Homogeneous Prime Spectrum). Let A be an $\mathbb{N}$ -graded ring. The *homogeneous prime spectrum* $\text{hPrim}(A)$ is the topological space with

- underlying set: the saturated homogeneous prime ideals of A (equivalently, the homogeneous primes not containing A_+);
- open subsets: $\text{hPrim}(I) := \{\mathfrak{p} \mid I \not\subset \mathfrak{p}\}$ for $I \subset A$ homogeneous.

$\text{hPrim}(I)$ depends only on $\sqrt{(I)}^{\text{sat}}$.

Proposition 4.2.7 (The Locale of an $\mathbb{N}$ -Graded Ring). Let A be an $\mathbb{N}$ -graded ring.

- (i) There is a homeomorphism $\text{hPrim}(A) \xrightarrow{\sim} \text{Pt}(\text{hRad}_A)$, under which $\text{Pt}(I)$ corresponds to $\text{hPrim}(I)$.
- (ii) hRad_A is spatial: $\text{Open}(\text{hPrim}(A)) \xrightarrow{\sim} \text{hRad}_A$.
- (iii) $\text{hPrim}(A)$ is sober with a basis of quasi-compact open subsets closed under binary intersections. If A is finitely generated over A_0 , $\text{hPrim}(A)$ is spectral.

Proof. Items (i) and (ii) follow by an exact parallel of Proposition 4.2.4, replacing Rad_R by hRad_A throughout. We focus on (iii).

Sobriety follows from (i): $\text{hPrim}(A) \cong \text{Pt}(\text{hRad}_A)$, and point spaces of locales are sober. For the basis, fix a homogeneous $f \in A_+$ of degree d and set

$$\text{Prim}(f) = \{\mathfrak{p} \in \text{hPrim}(A) \mid f \notin \mathfrak{p}\}.$$

We claim a homeomorphism

$$\text{Prim}(f) \cong \text{Prim}((A_f)_0),$$

where $(A_f)_0$ is the degree-zero part of A_f . The map $\mathfrak{p} \mapsto (\mathfrak{p}A_f)_0$ has inverse Ψ defined by graded components

$$\Psi(\mathfrak{q})_n := \left\{ a \in A_n \mid \frac{a^d}{f^n} \in \mathfrak{q} \right\}.$$

$\Psi(\mathfrak{q})$ is a homogeneous ideal. For *additive closure*, take $a, b \in \Psi(\mathfrak{q})_n$ and consider $(a+b)^{2d}$: every binomial term contains $a^k b^{2d-k}$ with $k \geq d$ or $2d-k \geq d$, hence is divisible by a^d or b^d . So $(a+b)^{2d}/f^{2n} \in \mathfrak{q}$, and primality of $\mathfrak{q}$ gives $(a+b)^d/f^n \in \mathfrak{q}$. For *absorption*, take $a \in \Psi(\mathfrak{q})_n$ and $u \in A_m$:

$$\frac{(ua)^d}{f^{n+m}} = \frac{u^d}{f^m} \cdot \frac{a^d}{f^n} \in \mathfrak{q},$$

so $ua \in \Psi(\mathfrak{q})_{n+m}$.

$\Psi(\mathfrak{q})$ is prime. For $x \in A_n, y \in A_m$ with $xy \in \Psi(\mathfrak{q})$,

$$\frac{(xy)^d}{f^{n+m}} = \frac{x^d}{f^n} \cdot \frac{y^d}{f^m} \in \mathfrak{q},$$

and primality of $\mathfrak{q}$ forces $x \in \Psi(\mathfrak{q})$ or $y \in \Psi(\mathfrak{q})$.

$f \notin \Psi(\mathfrak{q})$: since $f^d/f^d = 1 \notin \mathfrak{q}$.

By the standard correspondence for graded localizations, Ψ is inverse to the contraction map. Continuity in both directions reduces to checking that basic opens correspond: for $g = a/f^k \in (A_f)_0$ with $a \in A_{kd}$, the basic open $\text{Prim}(g) \subset \text{Prim}((A_f)_0)$ pulls back under Ψ to the basic open $\text{Prim}(a) \subset \text{Prim}(f)$.

Conclusion. The basic opens $\text{Prim}(f)$ for $f \in A_+$ homogeneous form a basis of $\text{hPrim}(A)$, are quasi-compact (each homeomorphic to a prime spectrum), and are closed under binary intersection: $\text{Prim}(f) \cap \text{Prim}(g) = \text{Prim}(fg)$. If A is finitely generated over A_0 , then A_+ is generated by finitely many homogeneous elements $f_1, \dots, f_n$, so $\text{hPrim}(A) = \bigcup_{i=1}^n \text{Prim}(f_i)$ is a finite union of quasi-compact opens, hence quasi-compact; this gives spectrality. $\square$

4.3. Geometric Realization of Algebraic Functors

We now turn to a general algebraic functor X and the poset $\text{Open}(X)$ of its open subfunctors. The classification theorem (Proposition 3.2.1) gives equivalences

$$\begin{array}{ccc} \text{Rad}_X & \xrightarrow{\sim} & \text{Open}(X) \\ \downarrow \simeq & & \downarrow \simeq \\ \lim_{\text{El}(X)^{\text{op}}} \text{Rad}_R & \longrightarrow & \lim_{\text{El}(X)^{\text{op}}} \text{Open}(\text{Spec}(R)). \end{array}$$

The limit is taken in $\text{Pos} \subset \text{Cat}$. Each Rad_R is a locale, and the embedding $\text{Loc} \hookrightarrow \text{Pos}^{\text{op}}$ preserves colimits — so this Pos-limit is a colimit in Loc . Concretely, the left Kan extension of Open along $\text{Aff} \hookrightarrow \text{Fun}(\text{CAlg}, \text{Set})$ stays inside Loc :

$$\begin{array}{ccc} \text{Aff} & & \\ \downarrow & \searrow \text{Open} & \\ \text{Fun}(\text{CAlg}, \text{Set}) & \xrightarrow{\text{Open}} & \text{Loc} \end{array}$$

Since left Kan extension preserves colimits, the resulting functor $\text{Open}: \text{Fun}(\text{CAlg}, \text{Set}) \rightarrow \text{Loc}$ preserves colimits.

Thus, we immediately obtain:

Corollary 4.3.1 (Open Subfunctors Form Spatial Locales). *For any algebraic functor X , $\text{Open}(X)$ is a spatial locale.*

Proof. Each $\text{Rad}_R \simeq \text{Open}(\text{Prim}(R))$ is spatial (Proposition 4.2.4(ii)), and the colimit-preserving extension above sends X to a colimit of spatial locales. The inclusion $\text{Loc}^{\text{spa}} \hookrightarrow \text{Loc}$ is a left adjoint (its right adjoint is the spatialization $\text{Open} \circ \text{Pt}$), hence preserves colimits — so this colimit lies in Loc^{spa} , i.e. $\text{Open}(X)$ is spatial. $\square$

Remark 4.3.2 (Properties of the Open Set Locale). For an algebraic functor X , $\text{Open}(X)$ as a locale has the following properties:

- *Terminal object:* $D(1) = X$ (the entire functor).
- *Initial object:* $D(0) = \emptyset_X$, where $\emptyset_X(0) = X(0)$ (a singleton over the zero ring) and $\emptyset_X(R) = \emptyset$ for $R \neq 0$.
- *Meet operation:* For open subfunctors $U, V \in \text{Open}(X)$, the locale meet is the functorial intersection: $U \wedge V = U \cap V$.

Our goal is to find a topological space $|X|$ with $\text{Open}(|X|) \simeq \text{Open}(X)$ — i.e., to fill in the dashed functor $|-|$ below, compatibly with the locale-theoretic data:

$$\begin{array}{ccc}
 \text{Aff} & & \\
 \text{Spec} \downarrow & \searrow \text{Prim} & \\
 \text{Fun}(\text{CAlg}, \text{Set}) & \dashrightarrow & \text{Top} \\
 & \searrow \text{Open} & \downarrow \text{Open} \\
 & & \text{Loc}^{\text{spa}}
 \end{array}$$

Since Top is cocomplete and $\text{Fun}(\text{CAlg}, \text{Set})$ is generated by representables, the universal property identifies $|-|: \text{Fun}(\text{CAlg}, \text{Set}) \rightarrow \text{Top}$ as the *left Kan extension* of Prim along Spec . Explicitly,

$$|X| \simeq \text{colim}_{(R,x) \in \text{El}(X)} \text{Prim}(R).$$

Because $\text{Open}: \text{Top} \rightarrow \text{Loc}$ is a left adjoint (Proposition 4.1.12), it preserves this colimit. The classification of open subfunctors in Proposition 3.2.1 identifies $\text{Open}(X)$ with the same locale colimit. Therefore there is a natural equivalence

$$\text{Open}(X) \simeq \text{Open}(|X|),$$

not merely a bijection of underlying sets.

Concretely, an open subfunctor $U \subset X$ pulls back along every point $x: \text{Spec}(R) \rightarrow X$ to a compatible open subset $U_x \subset \text{Prim}(R)$. The colimit topology makes these chartwise opens descend to a unique open subset of $|X|$. Conversely, an open subset of $|X|$ pulls back to compatible chartwise opens, which the classification theorem glues to an open subfunctor. These constructions are inverse because both can be checked after pullback to every affine chart; they are functorial because a map $X \rightarrow Y$ acts by inverse image on both sides. Finally, the soberization of $|X|$ recovers the point space of this locale:

$$|X|^{\text{sob}} \simeq \text{Pt}(\text{Open}(X)).$$

We now describe $|X|$ explicitly. Recall first that a category C is *connected* if any two objects $X, Y \in C$ can be joined by a zigzag

$$X = i_0 \leftarrow i_1 \rightarrow i_2 \leftarrow \cdots \rightarrow i_{2n} = Y.$$

Proposition 4.3.3 (Explicit Description of Geometric Realization). *Let $\text{Field}_X \subset \text{El}(X)$ be the full subcategory spanned by pairs (k, x) , where k is a field and $x: \text{Spec}(k) \rightarrow X$ is a k -point of X . Then the map $(k, x) \mapsto (|x|: |\text{Spec}(k)| \rightarrow |X|)$ defines a bijection:*

$$\{\text{connected components of Field}_X\} \xrightarrow{\sim} |X|.$$

Proof. First, consider the case where X is an affine scheme. Let $X \simeq \text{Spec}(R)$. We know that:

$$\text{Field}_R \simeq \coprod_{\mathfrak{p} \in \text{Prim}(R)} \text{Field}_{\kappa(\mathfrak{p})}.$$

Hence we get a bijection

$$\begin{aligned}
 \{\text{connected components of Field}_R\} &\xrightarrow{\sim} \text{Prim}(R), \\
 (R \rightarrow k) &\mapsto \ker(R \rightarrow k),
 \end{aligned}$$

with inverse $\mathfrak{p} \mapsto (R \rightarrow \kappa(\mathfrak{p}))$.

For a general algebraic functor $X \simeq \text{colim}_{\text{El}(X)} \text{Spec}(R)$, we have $|X| \simeq \text{colim}_{\text{El}(X)} \text{Prim}(R)$, so it remains to exhibit a bijection

$$\text{colim}_{(R,x) \in \text{El}(X)} \{\text{connected components of Field}_R\} \xrightarrow{\sim} \{\text{connected components of Field}_X\}.$$

Surjectivity: For any object (k, x) in Field_X , i.e., $x: \text{Spec}(k) \rightarrow X$, since X is a colimit of affine schemes, this map must factor as $\text{Spec}(k) \rightarrow \text{Spec}(R) \xrightarrow{x'} X$. This means the connected component comes from some Field_R , so the map is surjective.

Injectivity: Suppose two elements of the left-hand colimit map to the same connected component of Field_X ; we show they already coincide in the colimit.

First, every element of the colimit is represented by a field point: an element coming from a chart (R, x) and a prime $\mathfrak{p} \in \text{Prim}(R) \simeq \pi_0(\text{Field}_R)$ agrees with the one coming from the residue-field point $(\kappa(\mathfrak{p}), x_{\mathfrak{p}})$, because the quotient $R \rightarrow \kappa(\mathfrak{p})$ is a morphism of $\text{El}(X)$ taking $\mathfrak{p}$ to the unique prime of $\kappa(\mathfrak{p})$, and the colimit cocone identifies the two classes.

So we may assume the two elements are field points (k_1, x_1) and (k_2, x_2) joined by a zigzag of morphisms of Field_X . Each edge of the zigzag is a morphism of $\text{El}(X)$ between two field objects, hence a morphism of the very index category of the colimit; the colimit cocone therefore forces the classes of consecutive field points to coincide (in either orientation, as these are equalities of elements). By transitivity along the zigzag, (k_1, x_1) and (k_2, x_2) represent the same element of the colimit. This proves injectivity.

The geometric mechanism behind a single zigzag step is worth recording, as it also underlies Lemma 4.3.5 below: a span $k_1 \leftarrow k_0 \rightarrow k_2$ in CAlg (corresponding to $x_1 \rightarrow x_0 \leftarrow x_2$ in Field_X) gives rise to the pushout

$$\begin{array}{ccc} k_0 & \longrightarrow & k_1 \\ \downarrow & \lrcorner & \downarrow \\ k_2 & \longrightarrow & k_1 \otimes_{k_0} k_2 \end{array}$$

Since the k_i are fields, $A = k_1 \otimes_{k_0} k_2$ is non-zero, so it has a prime $\mathfrak{p}$ with residue field $K = \kappa(\mathfrak{p})$ and a composite $A \rightarrow K$, yielding the commutative square

$$\begin{array}{ccc} k_0 & \longrightarrow & k_1 \\ \downarrow & & \downarrow \\ k_2 & \longrightarrow & K \end{array}$$

In Field_X this provides an object (K, x_K) with morphisms $(K, x_K) \rightarrow (k_1, x_1)$ and $(K, x_K) \rightarrow (k_2, x_2)$, exhibiting x_1 and x_2 in a common connected component through x_K . $\square$

Remark 4.3.4 (Two Meanings of “Point”). In algebraic geometry, “point” has two different meanings:

- For any ring R , an “ R -point” is an element of $X(R)$. We know it gives the element category $\text{El}(X)$.
- A point of X generally refers to a point of $|X|$, i.e., an equivalence class of field-valued points of X , which gives the topological space $|X|$.

Therefore, we obtain the following relationship:

$$\text{El}(X) \hookleftarrow \text{Field}_X \twoheadrightarrow |X|.$$

The functor $| - |$ does not preserve limits in general — for example, the pullback

$$\begin{array}{ccc} \text{Spec}(\mathbb{C} \times \mathbb{C}) & \longrightarrow & \text{Spec}(\mathbb{C}) \\ \downarrow & \lrcorner & \downarrow \\ \text{Spec}(\mathbb{C}) & \longrightarrow & \text{Spec}(\mathbb{R}) \end{array}$$

But after applying $|-|$, we get the diagram:

$$\begin{array}{ccc} S^0 & \longrightarrow & * \\ \downarrow & & \downarrow \\ * & \longrightarrow & * \end{array}$$

is plainly not a pullback. We now characterize when realization preserves pullbacks.

Lemma 4.3.5 (Pullback Preservation). *Consider a diagram in $\text{Fun}(\text{CAlg}, \text{Set})$:*

$$Y \rightarrow X \leftarrow Z.$$

Then there is a canonical map:

$$|Y \times_X Z| \rightarrow |Y| \times_{|X|} |Z|.$$

This map is surjective, and is a bijection when $Y \rightarrow X$ is a monomorphism.

Proof. Surjectivity: For $y: \text{Spec}(k) \rightarrow Y$ and $z: \text{Spec}(k') \rightarrow Z$ such that they map to the same point in $|X|$, i.e., given $(y, z) \in |Y| \times_{|X|} |Z|$, we need to find $\text{Spec}(k'') \rightarrow Y \times_X Z$ corresponding to it. We know that y and z mapping to the same point in $|X|$ is equivalent to there existing a field k'' such that the following diagram commutes:

$$\begin{array}{ccccc} & & \text{Spec}(k) & \longrightarrow & Y \\ & \nearrow & & & \searrow \\ \text{Spec}(k'') & & & & X \\ & \searrow & & & \nearrow \\ & & \text{Spec}(k') & \longrightarrow & Z \end{array}$$

This is equivalent to saying there is a canonical morphism:

$$\text{Spec}(k'') \rightarrow Y \times_X Z.$$

Therefore, the map is surjective.

Bijection when $Y \hookrightarrow X$ is a monomorphism. Pullbacks of monomorphisms are monomorphisms, so $Y \times_X Z \hookrightarrow Z$ is a monomorphism. The commutative diagram

$$\begin{array}{ccc} |Y \times_X Z| & \hookrightarrow & |Z| \\ & \searrow & \nearrow \\ & |Y| \times_{|X|} |Z| & \end{array}$$

reduces the claim to showing $|Y| \times_{|X|} |Z| \rightarrow |Z|$ is a monomorphism, equivalently that $|Y| \rightarrow |X|$ is injective. We prove this directly from the field-point description (Proposition 4.3.3). Let (k_1, y_1) and (k_2, y_2) be field points of Y whose images in $|X|$ coincide. As in the surjectivity argument above, this provides a field k'' with maps $\text{Spec}(k'') \rightarrow \text{Spec}(k_1)$ and $\text{Spec}(k'') \rightarrow \text{Spec}(k_2)$ whose composites to X agree at a single point $x'': \text{Spec}(k'') \rightarrow X$. The two induced elements $\text{Spec}(k'') \rightarrow \text{Spec}(k_i) \xrightarrow{y_i} Y$ of $Y(k'')$ then have the same image x'' in $X(k'')$; since $Y \hookrightarrow X$ is a monomorphism, $Y(k'') \rightarrow X(k'')$ is injective, so these elements coincide in a single $y'': \text{Spec}(k'') \rightarrow Y$. Thus $(k_1, y_1) \leftarrow (k'', y'') \rightarrow (k_2, y_2)$ is a span in Field_Y , so (k_1, y_1) and (k_2, y_2) lie in the same connected component, i.e. define the same point of $|Y|$. Hence $|Y| \rightarrow |X|$ is injective. $\square$

Proposition 4.3.6 (Geometric Realization of Immersions). *Let X be an algebraic functor, and $Y \hookrightarrow X$ an open immersion/closed immersion/immersion. Then it induces an open immersion/closed immersion/immersion $|Y| \hookrightarrow |X|$.*

Proof. We treat the closed case (open and locally-closed are analogous). For a closed immersion $Y \rightarrow X$ and any R -point $x: \text{Spec}(R) \rightarrow X$, the pullback

$$\begin{array}{ccc} Y_R & \longrightarrow & \text{Spec}(R) \\ \downarrow & \lrcorner & \downarrow \\ Y & \longrightarrow & X \end{array}$$

satisfies $Y_R \simeq \text{Spec}(R/I)$ for some $I \subset R$, hence $|Y_R| \simeq \text{Prim}(R/I) \hookrightarrow \text{Prim}(R)$. By Lemma 4.3.5 (applied to the monomorphism $Y \hookrightarrow X$), the geometric realization gives a pullback square in Set:

$$\begin{array}{ccc} |Y_R| & \hookrightarrow & \text{Prim}(R) \\ \downarrow & \lrcorner & \downarrow \\ |Y| & \hookrightarrow & |X| \end{array}$$

It remains to verify that $|Y| \hookrightarrow |X|$ is closed. A closed $Z \subset |Y|$ pulls back to a closed $Z_R \subset |Y_R| \subset \text{Prim}(R)$ for each R . Since $|X|$ has the colimit topology, $Z \subset |X|$ is closed iff each $Z_R \subset \text{Prim}(R)$ is closed, which we have just verified. $\square$

Remark 4.3.7 (Closed Subfunctors vs Closed Subspaces). Proposition 4.3.6 yields a map of posets

$$\text{Closed}(X) \rightarrow \text{Closed}(|X|), \quad Z \mapsto |Z|.$$

This map is rarely a bijection. It is, however, surjective when $X = \text{Spec}(A)$ or $X = \text{Proj}(A)$: any closed subset of $|X|$ is the realization of the vanishing locus of the radical ideal cutting out its open complement.

In general, if $Z \subset X$ is a closed subfunctor with open complement U , then $|Z| = |X| \setminus |U|$.

We can transport topological properties from spaces to algebraic functors via geometric realization.

Definition 4.3.8 (Topological Properties of Algebraic Functors). Let P be a property of topological spaces (such as Hausdorff, connected, locally connected, irreducible, discrete). An algebraic functor X is said to have property P if its underlying space $|X|$ does.

Example 4.3.9 (Properties of Affine Schemes). For an affine scheme $X \simeq \text{Spec}(R)$, with $|X| \simeq \text{Prim}(R)$:

- (i) X is connected iff the only idempotents of R are 0 and 1 (which may coincide when R is the zero ring).
- (ii) X is irreducible iff $\sqrt{(0)} \subset R$ is prime, equivalently $R/\sqrt{(0)}$ is an integral domain.
- (iii) X is discrete iff $R/\sqrt{(0)}$ is a finite product of fields — equivalently, R has finitely many primes and all of them are maximal. (Since $|X| = \text{Prim}(R)$ is always quasi-compact, a discrete $|X|$ is necessarily finite; for Noetherian R the condition reduces to R being artinian.)

Definition 4.3.10 (Topological Properties of Morphisms). Let P be a property of continuous maps of topological spaces (e.g. surjective, injective, bijective, dominant, submersive, open, closed, homeomorphism). For a morphism $f: Y \rightarrow X$ of algebraic functors:

- f is P if $|f|: |Y| \rightarrow |X|$ is P .
- f is *universally* P if every base change $Y \times_X X' \rightarrow X'$ along an arbitrary morphism $X' \rightarrow X$ is P .

Proposition 4.3.11 (Universal Topological Properties). *Let $f : Y \rightarrow X$ be a morphism of algebraic functors.*

- (i) *For $P \in \{\text{surjective, injective, bijective, dominant, submersive, open, closed, homeomorphism}\}$, f is universally P iff for every ring R and every R -point $x : \text{Spec}(R) \rightarrow X$, the base change $Y \times_X \text{Spec}(R) \rightarrow \text{Spec}(R)$ is P .*
- (ii) *f is universally surjective iff it is surjective.*
- (iii) *If f is a monomorphism, it is universally injective.*
- (iv) *If f is a Zariski-local epimorphism, it is universally submersive; in particular, universally surjective.*

Proof. (i) We treat $P = \text{submersive}$; the other cases are analogous. Recall that a continuous map $\varphi : T \rightarrow S$ is *submersive* if it is surjective and S carries the quotient topology.

For each ring R and R -point $x : \text{Spec}(R) \rightarrow X$, consider the pullback

$$\begin{array}{ccc} Y_R & \xrightarrow{f_R} & \text{Spec}(R) \\ \downarrow & \lrcorner & \downarrow \\ Y & \xrightarrow{f} & X. \end{array}$$

Surjectivity of $|f|$. Suppose each f_R is surjective. Since $X \simeq \text{colim}_{(R,x) \in \text{El}(X)} \text{Spec}(R)$, taking colimits over $\text{El}(X)$ in the pullback diagram gives

$$\begin{array}{ccc} \text{colim}_{(R,x)} Y_R & \longrightarrow & \text{colim}_{(R,x)} \text{Spec}(R) \\ \downarrow & \lrcorner & \downarrow \\ Y & \longrightarrow & X. \end{array}$$

The right vertical map is an isomorphism, so $Y \simeq \text{colim}_{(R,x)} Y_R$. Since $|-|$ preserves colimits, $|Y| \simeq \text{colim}_{(R,x)} |Y_R|$. Each point $\xi \in |X|$ comes from some $|\text{Spec}(R)|$, and surjectivity of $|f_R|$ produces a preimage in $|Y_R|$, hence in $|Y|$. So $|f|$ is surjective.

Quotient topology. Now suppose each f_R is submersive; we show $|X|$ carries the quotient topology under $|f|$. Let $U \subset |X|$ with $|f|^{-1}(U)$ open in $|Y|$. Since $|X|$ carries the colimit topology of the $|\text{Spec}(R)|$, the set U is open precisely when its preimage $c_R^{-1}(U)$ under each cone map $c_R : |\text{Spec}(R)| \rightarrow |X|$ is open. Now f_R submersive means $c_R^{-1}(U)$ is open as soon as $|f_R|^{-1}(c_R^{-1}(U))$ is open in $|Y_R|$; by the commuting square $c_R \circ |f_R| = |f| \circ \iota_{Y_R}$ (with $\iota_{Y_R} : |Y_R| \rightarrow |Y|$ the cone map), this preimage equals $\iota_{Y_R}^{-1}(|f|^{-1}(U))$, which is open since $|f|^{-1}(U)$ is. Hence $c_R^{-1}(U)$ is open for every R -point, so U is open in $|X|$.

(ii) The forward implication is immediate. Conversely, suppose $|f|$ is surjective. For any $X' \rightarrow X$, Lemma 4.3.5 gives a surjection

$$|Y \times_X X'| \twoheadrightarrow |Y| \times_{|X|} |X'|,$$

and $|Y| \times_{|X|} |X'| \rightarrow |X'|$ is surjective because $|f|$ is. Composing, $|Y \times_X X'| \rightarrow |X'|$ is surjective.

(iii) Suppose f is a monomorphism. Monomorphisms are stable under base change, so it suffices to show $|f|$ itself is injective. The fact that f is mono means $Y \times_X Y \xrightarrow{\sim} Y$, and Lemma 4.3.5 then gives $|Y| \xrightarrow{\sim} |Y| \times_{|X|} |Y|$, i.e., $|f|$ is injective.

(iv) By (i) we may assume $X = \text{Spec}(R)$. The Zariski-local epimorphism hypothesis yields $(f_1, \dots, f_n) = R$ with lifts

$$\begin{array}{ccc} & & Y \\ & \nearrow & \downarrow f \\ \text{Spec}(R_{f_i}) & \hookrightarrow & \text{Spec}(R). \end{array}$$

Then $\text{Prim}(R) = \bigcup_{i=1}^n \text{Prim}(R_{f_i})$, exhibiting surjectivity of $|f|$.

For submersivity, take $U \subset \text{Prim}(R)$ with $|f|^{-1}(U) \subset |Y|$ open. The lifts make $U \cap \text{Prim}(R_{f_i})$ open in $\text{Prim}(R_{f_i})$ for each i , so U is open in $\text{Prim}(R)$. $\square$

Example 4.3.12 (Failures of Universality). Among the properties listed in Proposition 4.3.11, surjectivity is essentially the only one that automatically is universal. Three illustrative failures:

Homeomorphism is not universal. Consider $\text{Spec}(\mathbb{C}) \rightarrow \text{Spec}(\mathbb{R})$. Topologically this is a map of one-point spaces, hence trivially a homeomorphism. But it is not universally injective: base-changing along itself uses $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$, giving

$$\text{Spec}(\mathbb{C} \times \mathbb{C}) \rightarrow \text{Spec}(\mathbb{C}),$$

topologically $\{*\} \sqcup \{*\} \rightarrow \{*\}$, which is not a homeomorphism.

Closedness is not universal. For a field k , $|\text{Spec}(k)| = \{*\}$ is a one-point space, so any morphism $X \rightarrow \text{Spec}(k)$ is automatically closed. The morphism $\mathbb{A}_k^1 \rightarrow \text{Spec}(k)$ is closed, but pulling back along itself gives the projection $p: \mathbb{A}_k^2 \rightarrow \mathbb{A}_k^1$, under which the closed subset $V(xy - 1) \subset \mathbb{A}_k^2$ has image $\mathbb{A}_k^1 \setminus \{0\}$ — open, not closed.

Openness is not universal. For k of characteristic $\neq 2$, consider the normalization of the nodal cubic

$$f: \text{Spec}(k[t]) \rightarrow \text{Spec}(k[x, y]/(y^2 - x^2 - x^3)), \quad x \mapsto t^2 - 1, \quad y \mapsto t(t^2 - 1)$$

It is finite and surjective, is an isomorphism away from the node, and identifies the two closed points $t = \pm 1$. Since every closed subset of $\text{Spec}(k[t])$ other than the whole space is finite, this quotient map sends opens to opens; hence $|f|$ is open.

The failure after base change is visible without a flatness criterion. Base-change f along itself. With coordinates s, t on the two copies of the normalization, the underlying closed set of the fiber product is

$$\begin{aligned} V(s^2 - t^2, s(s^2 - 1) - t(t^2 - 1)) &= V(s - t) \\ &\cup V(s + t, s^2 + st + t^2 - 1) \\ &= V(s - t) \cup \{(-1, 1), (1, -1)\}. \end{aligned}$$

The two off-diagonal points are open-and-closed components of this fiber product. Under the projection to the t -line, the open singleton $\{(-1, 1)\}$ maps to the closed point $t = 1$, which is not open in $\text{Spec}(k[t])$. Thus this base change of f is not open. This also shows why the tempting claim “a finite morphism is universally open exactly when it is flat” cannot be used here: for example, a nontrivial nilpotent closed immersion is finite and universally a homeomorphism, hence universally open, but need not be flat.

Corollary 4.3.13 (Zariski Codescent). *Let $(Y_i \rightarrow X)_{i \in I}$ be a family of morphisms such that $\coprod_{i \in I} Y_i \rightarrow X$ is a Zariski-local epimorphism. Then*

$$\coprod_{i, j \in I} |Y_i \times_X Y_j| \rightrightarrows \coprod_{i \in I} |Y_i| \xrightarrow{q} |X|$$

is a coequalizer diagram in Top.

Proof. Set $Y = \coprod_{i \in I} Y_i$. By Proposition 4.3.11(iv), the induced map $q: |Y| \rightarrow |X|$ is submersive, so $|X|$ carries the quotient topology under q , i.e.

$$|Y| \times_{|X|} |Y| \rightrightarrows |Y| \xrightarrow{q} |X|$$

is a coequalizer in Top, where $|Y| \times_{|X|} |Y|$ is computed in Top.

On the algebraic side, $Y \times_X Y \cong \coprod_{i, j} Y_i \times_X Y_j$, and the canonical map

$$\phi: |Y \times_X Y| \rightarrow |Y| \times_{|X|} |Y|$$

is surjective by Lemma 4.3.5.

A general point-set fact: if Q is the quotient of X by an equivalence relation R (so $R \rightrightarrows X \rightarrow Q$ is a coequalizer) and $\phi: R' \rightarrow R$ is surjective, then $R' \rightrightarrows X \rightarrow Q$ is also a coequalizer. Applying this with $R = |Y| \times_{|X|} |Y|$ and $R' = |Y \times_X Y|$ yields the asserted coequalizer

$$\coprod_{i,j} |Y_i \times_X Y_j| \rightrightarrows \coprod_i |Y_i| \rightarrow |X|.$$

□

4.3.1. Open Coverings

For a topological space T , the notion of an open cover depends only on the locale $\text{Open}(T)$: a family $(U_i \subset T)_{i \in I}$ is an open cover iff its supremum is T . We extend this to algebraic functors.

Proposition 4.3.14 (Characterization of Open Covers). *Let X be an algebraic functor, $(U_i \subset X)_{i \in I}$ a family of open subfunctors, and $K_i \in \text{Rad}_X$ the corresponding radical ideals. The following are equivalent:*

- (i) $\bigvee_{i \in I} U_i = X$ in $\text{Open}(X)$.
- (ii) $\bigvee_{i \in I} K_i = \mathcal{O}_X$ in Rad_X .
- (iii) For every ring R and every $x: \text{Spec}(R) \rightarrow X$, $\bigcup_{i \in I} K_i(x)$ generates R .
- (iv) For every local ring R , $X(R) = \bigcup_{i \in I} U_i(R)$.
- (v) For every field k , $X(k) = \bigcup_{i \in I} U_i(k)$.
- (vi) $\coprod_{i \in I} U_i \rightarrow X$ is a Zariski-local epimorphism.
- (vii) $\coprod_{i \in I} U_i \rightarrow X$ is surjective.

Proof. We follow the implication scheme

$$(iv) \Rightarrow (v) \Rightarrow (vii), \quad (i) \Leftrightarrow (ii) \Leftrightarrow (iii), \quad (i) \Leftrightarrow (vii), \quad (iii) \Rightarrow (iv), \quad (vi) \Rightarrow (vii), \quad (iii) \Rightarrow (vi).$$

(i) $\Leftrightarrow$ (ii): immediate from the equivalence $\text{Open}(X) \simeq \text{Rad}_X$.

(ii) $\Leftrightarrow$ (iii): suprema in $\text{Rad}_X = \lim_{(R,x) \in \text{El}(X)^{\text{op}}} \text{Rad}_R$ are computed pointwise, so $\bigvee_i K_i = \mathcal{O}_X$ iff $\bigvee_i K_i(x) = R$ in Rad_R for every R -point x , iff $\bigcup_i K_i(x)$ generates R .

(i) $\Leftrightarrow$ (vii): via $\text{Open}(X) \simeq \text{Open}(|X|)$, $\bigvee_i U_i = X$ corresponds to $\bigcup_i |U_i| = |X|$, equivalently to surjectivity of $\coprod_i U_i \rightarrow X$.

(iii) $\Rightarrow$ (iv): for R local, if $\sum_i K_i(x) = R$ then some $K_i(x)$ contains a unit, hence equals R , so $x \in U_i(R)$.

(iv) $\Rightarrow$ (v): fields are local rings.

(v) $\Rightarrow$ (vii): surjectivity is detected on field-valued points (Proposition 4.3.11(ii) combined with the description of $|X|$ as field-valued points modulo equivalence).

(vi) $\Rightarrow$ (vii): Proposition 4.3.11(iv) shows Zariski-local epimorphisms are surjective.

(iii) $\Rightarrow$ (vi): given $x: \text{Spec}(R) \rightarrow X$, condition (iii) yields $\sum_i K_i(x) = R$, hence $1 = \sum_{j=1}^n f_j$ for finitely many $f_j \in K_{i_j}(x) \subset R$. The principal opens $D(f_j) \cong \text{Spec}(R_{f_j})$ then cover $\text{Spec}(R)$. For each j , $f_j \in K_{i_j}(x)$ becomes a unit in R_{f_j} , so $K_{i_j}(x) \cdot R_{f_j} = R_{f_j}$; by the correspondence between U_{i_j} and K_{i_j} (the open locus where K_{i_j} generates the unit ideal), this means $x|_{D(f_j)}$ factors as

$$\text{Spec}(R_{f_j}) \rightarrow U_{i_j} \hookrightarrow X.$$

Hence $\coprod_i U_i \rightarrow X$ is a Zariski-local epimorphism.

□

Definition 4.3.15 (Open Cover). A family of open subfunctors $(U_i \hookrightarrow X)_{i \in I}$ satisfying the equivalent conditions of Proposition 4.3.14 is called an *open cover* of X .

Looking Ahead

We have constructed the underlying space functor $|-|: \text{Fun}(\text{CAlg}, \text{Set}) \rightarrow \text{Top}$, shown that it is a left Kan extension of Prim , and proved that $\text{Open}(X) \simeq \text{Open}(|X|)$ as locales — so every algebraic functor has a genuine topological space attached to it. Morphisms, immersions, open covers, and topological properties (connectedness, irreducibility) all transfer cleanly between the functorial and the topological pictures.

Locales and topological spaces remain, however, *set-valued* gadgets. To talk about covering data at the level of sheaves — where one really does local-to-global analysis with sections and descent — we need the full generality of Grothendieck topologies. The next chapter introduces sieves as the categorical analogue of open covers, defines Grothendieck topologies and sheaves on arbitrary sites, constructs the sheafification functor, and sets up the language in which “scheme” will eventually mean “Zariski sheaf on Aff with an affine open cover”.

Chapter 5.

Sheaves

This chapter aims to establish the theory of *Grothendieck topologies* and *sheaves* on them.

For any category C , the presheaf category $\mathbf{PShv}(C)$ is too coarse to capture the “local-to-global” properties of geometric objects. The right notion is to single out those presheaves with a *descent* property — these are τ -*sheaves*, where τ is a Grothendieck topology on C encoding the local data.

In algebraic geometry, we are primarily concerned with the following scenarios:

- (i) *The case of topological spaces*: Let $C = \mathbf{Open}(T)$ be the category of open sets (poset) of a topological space T . Endow it with the *canonical Grothendieck topology*. The resulting sheaves are precisely the classical sheaves on T . This concept was originally introduced by Leray.
- (ii) *The case of affine schemes*: Let $C = \mathbf{Aff}_k \simeq \mathbf{CAlg}_k^{\text{op}}$. The main topologies we use include:
 - *Zariski topology* (i.e., satisfying Zariski descent);
 - *Étale topology*;
 - *Nisnevich topology*;
 - *fppf topology* (fidèlement plat de présentation finie — faithfully flat of finite presentation);
 - *fpqc topology* (fidèlement plat quasi-compact — faithfully flat quasi-compact).
- (iii) *Extension and generalization*: We can naturally extend the above topologies from $\mathbf{Aff}_k$ to the category of k -schemes $\mathbf{Sch}_k$, and even further generalize them to the functor category $\mathbf{Fun}(\mathbf{CAlg}_k, \mathbf{Set})$ (i.e., the presheaf category).

5.1. Introduction: Descent and Sheaves

We sketch the main ideas before proceeding to the formal definitions.

We know that colimits in categories generally represent “gluing”. For example, in the category of topological spaces $\mathbf{Top}$, we use colimits to glue spaces together.

Consider a category C , its presheaf category $\mathbf{PShv}(C) = \mathbf{Fun}(C^{\text{op}}, \mathbf{Set})$, and a presheaf $F: C^{\text{op}} \rightarrow \mathbf{Set}$. We ask whether F preserves the gluing information of C : given a diagram $X: \mathcal{I} \rightarrow C$, how does $F(\text{colim}_{i \in \mathcal{I}} X_i)$ compare with $\lim_{i \in \mathcal{I}} F(X_i)$? (Note that colimits in C become limits when viewed in C^{op} .)

More precisely, we want the canonical map

$$\text{Hom}_{\mathbf{PShv}(C)}(y(\text{colim}_{i \in \mathcal{I}} X_i), F) \longrightarrow \text{Hom}_{\mathbf{PShv}(C)}(\text{colim}_{i \in \mathcal{I}} y(X_i), F),$$

equivalently $F(\text{colim}_i X_i) \rightarrow \lim_i F(X_i)$, to be an isomorphism. The direction is forced by applying the contravariant functor $\text{Hom}(-, F)$ to $\text{colim}_i y(X_i) \rightarrow y(\text{colim}_i X_i)$. From F ’s perspective, this asks whether the family $(X_i)_i$ glues into $\text{colim}_i X_i$. This isn’t true in general, but it is the defining condition for those F we are interested in: we say such an F *satisfies descent with respect to X* .

In the category of topological spaces, this local-to-global property is entirely controlled by the concept of open covers. Specifically, for a topological space X , we do not need to concern ourselves with the complex gluing data inside it; we only need to focus on a family of open embeddings $\{U_i \rightarrow X\}_{i \in I}$ whose union equals X . This family of maps is called an *open cover* of X . A section s of a sheaf on X is then determined by its restrictions $s|_{U_i}$, provided these agree on the overlaps $U_i \cap U_j$.

To do local-to-global analysis on a general category C , we generalize the geometric notion of open cover. In C open subsets get replaced by arbitrary morphisms $U \rightarrow X$, and a cover is then a family of morphisms with target X .

To treat such families of morphisms rigorously — and in particular to allow refinements and base changes — we introduce *sieves*, which play the role in category theory that open sets play in topology. This is the starting point of Grothendieck topologies.

5.2. Grothendieck Topology

A Grothendieck topology on a category C is a coherent rule for saying “a family of morphisms $(U_i \rightarrow X)$ covers X ”. Using sieves — subfunctors of representable presheaves — we make this rule functorial, descriptive, and compatible with base change.

5.2.1. Sieves and Descent

Definition 5.2.1 (Sieves and Pullbacks). Let C be a category and $X \in C$ an object.

- (i) *Sieve*: A *sieve* on X is a full subcategory $\mathcal{R} \subset C_{/X}$ of the slice category satisfying: for any morphism $h: V' \rightarrow V$ in $C_{/X}$ (i.e., a morphism making the diagram commute), if $V \in \mathcal{R}$, then $V' \in \mathcal{R}$. (In other words, sieves are families of morphisms closed under precomposition.)
- (ii) *Pullback*: Let $f: X' \rightarrow X$ be a morphism in C , and $\mathcal{R}$ a sieve on X . The *pullback* $f^*\mathcal{R}$ of $\mathcal{R}$ along f is a sieve on X' defined by:

$$f^*\mathcal{R} := \{(g: Y \rightarrow X') \in C_{/X'} \mid (f \circ g) \in \mathcal{R}\}.$$

Remark 5.2.2 (Right Fibration Composition and Subfunctor Viewpoint). Note that *right fibrations are closed under composition*.

We can view a sieve $\mathcal{R}$ on X as the composite of the following maps:

$$\mathcal{R} \hookrightarrow C_{/X} \rightarrow C.$$

Where:

- (i) The inclusion $\mathcal{R} \hookrightarrow C_{/X}$ is a right fibration. (This follows directly from the definition of a sieve: a sieve is a full subcategory closed under precomposition, which is precisely the right fibration property for full subcategory inclusions.)
- (ii) The forgetful functor (domain projection) $C_{/X} \rightarrow C$ is also a right fibration (this is a standard property of slice categories).

Therefore, the composite functor $\mathcal{R} \rightarrow C$ is still a *right fibration*.

By straightening, there is an equivalence between right fibrations over C and its presheaf category:

$$\mathrm{St}^R: \mathrm{RFib}(C) \xrightarrow{\sim} \mathrm{Fun}(C^{\mathrm{op}}, \mathbf{An}).$$

Under this correspondence:

- The right fibration $C_{/X} \rightarrow C$ corresponds to the representable functor $y(X)$.
- The right fibration $\mathcal{R} \rightarrow C$ corresponds to a *subfunctor* of $y(X)$.

Furthermore, if we consider the straightening of the inclusion $\mathcal{R} \hookrightarrow C_{/X}$ itself, we get a functor taking values in (-1) -truncated objects:

$$\chi_{\mathcal{R}}: (C_{/X})^{\text{op}} \rightarrow \mathbf{An}_{\leq -1} \simeq [1].$$

Here we view $[1]$ as the full subcategory consisting of $\{\emptyset, [0]\}$ (i.e., Boolean values in the homotopy sense: false and true). In this viewpoint:

- Objects belonging to the sieve $\mathcal{R}$ are mapped to $[0]$ (true/one-point space);
- Objects not belonging to the sieve $\mathcal{R}$ are mapped to $\emptyset$ (false/empty space).

Functoriality ensures that: if $V \rightarrow X$ is mapped to $[0]$ (i.e., in the sieve), then any factorization $V' \rightarrow V \rightarrow X$ must also be mapped to $[0]$ (i.e., also in the sieve).

This is the higher categorical interpretation of why we usually define sieves as subfunctors of $y(X)$ (i.e., $\mathcal{R}(V) \subset \text{Hom}(V, X)$).

It is not hard to see that pullbacks of sieves along morphisms are precisely pullbacks in the presheaf category $\text{PShv}(C)$. The viewpoint of $\mathcal{R} \subset y(X)$ as a subfunctor is sometimes more useful.

We now describe sieves generated by morphisms.

Definition 5.2.3 (Generated Sieve). Let C be a category and $X \in C$ an object. The sieve *generated* by a family of morphisms $\{f_i: Y_i \rightarrow X\}_{i \in I}$ is defined as the image of the natural transformation:

$$\coprod_{i \in I} y(Y_i) \rightarrow y(X).$$

Concretely, this sieve $\mathcal{R} \subset C_{/X}$ contains precisely those morphisms that can be factored through some f_i . That is:

$$\mathcal{R} = \{(g: V \rightarrow X) \in C_{/X} \mid \exists i \in I, \exists h: V \rightarrow Y_i \text{ such that } g = f_i \circ h\}.$$

We now introduce the descent condition for a sieve.

Definition 5.2.4 (Descent Along a Sieve). Let C be a category, $X \in C$ an object, and $\mathcal{R} \subset C_{/X}$ a sieve on X . Let $F: C^{\text{op}} \rightarrow \text{Set}$ be a presheaf.

- (i) *Descent data*: The *descent data* of F with respect to the sieve $\mathcal{R}$ is defined as the following limit:

$$\text{Desc}(\mathcal{R}, F) := \lim_{(V \rightarrow X) \in \mathcal{R}^{\text{op}}} F(V) \simeq \text{Nat}(\mathcal{R}, F).$$

(Note: If we view the sieve $\mathcal{R}$ as a subfunctor of $y(X)$, we can understand this limit as the set of natural transformations from $\mathcal{R}$ to F .)

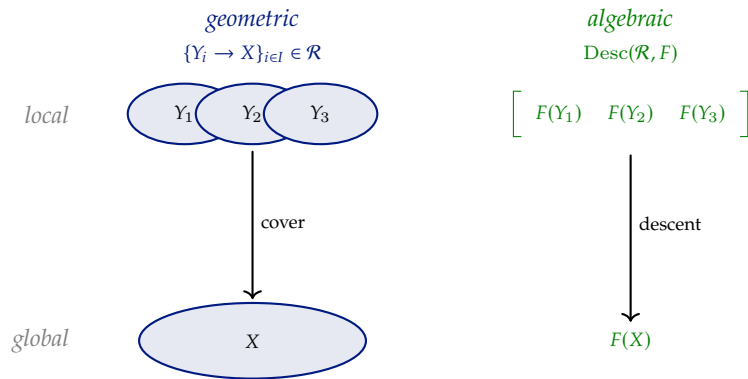
- (ii) *Descent condition*: We say F *satisfies descent along $\mathcal{R}$* if the canonical map:

$$F(X) \xrightarrow{\sim} \text{Desc}(\mathcal{R}, F)$$

is an isomorphism.

- (iii) *Universal descent*: We say F *universally descends* along $\mathcal{R}$ if for any morphism $f: Y \rightarrow X$, F satisfies descent along the pullback sieve $f^*\mathcal{R}$.

Remark 5.2.5 (Intuitive Picture). Intuitively, we can explain “descent” through the following correspondence diagram:



- On the left is the *geometric* picture: a family of “pieces” $\{Y_i\}$ (belonging to the sieve $\mathcal{R}$) covers X .
- On the right is the *algebraic* picture: $\text{Desc}(\mathcal{R}, F)$ contains all local sections on the pieces (satisfying compatibility).

Descent asks: given a compatible family of local data (top row), can it be glued uniquely into a global section (bottom row)? When the canonical map is an isomorphism, the local data fully determines the global.

We list several examples of sieves.

Example 5.2.6 (Empty Sieve). For any $X \in \mathcal{C}$, the empty subfunctor $\emptyset \subset y(X)$ is a sieve. A presheaf F satisfies descent along $\emptyset$ iff $F(X)$ is a singleton.

Example 5.2.7 (Sieve Generated by Two Subobjects). Let $X \in \mathcal{C}$ and $U, V \subset X$ be subobjects. Let $\mathcal{R}$ be the sieve on X generated by U and V . If $U \cap V$ exists, then $\mathcal{R}$ is actually the pushout in $\text{PShv}(\mathcal{C})$:

$$\begin{array}{ccc} y(U \cap V) & \longrightarrow & y(U) \\ \downarrow & & \downarrow \\ y(V) & \longrightarrow & \mathcal{R} \end{array}$$

Therefore, a presheaf F on $\mathcal{C}$ satisfying descent along $\mathcal{R}$ is equivalent to the following diagram being a pullback:

$$\begin{array}{ccc} F(X) & \longrightarrow & F(U) \\ \downarrow & \lrcorner & \downarrow \\ F(V) & \longrightarrow & F(U \cap V) \end{array}$$

In a 1-category, the descent condition can be reinterpreted as an equalizer diagram.

Lemma 5.2.8 (Sieve as Coequalizer). Let $\mathcal{C}$ be a category and $X \in \mathcal{C}$ an object. Let $\mathcal{R}$ be the sieve generated by a family $(Y_i \rightarrow X)_{i \in I}$. Then the canonical maps $y(Y_i) \rightarrow \mathcal{R}$ induce an isomorphism in the presheaf category:

$$\text{colim} \left(\coprod_{i,j \in I} y(Y_i) \times_{y(X)} y(Y_j) \rightrightarrows \coprod_{i \in I} y(Y_i) \right) \xrightarrow{\sim} \mathcal{R}.$$

Proof. By Definition 5.2.3, the sieve $\mathcal{R}$ is the image of:

$$\coprod_{i \in I} y(Y_i) \rightarrow y(X).$$

The *coimage* of a morphism $f: A \rightarrow B$ is the coequalizer

$$\text{coim}(f) := \text{colim}(A \times_B A \rightrightarrows A).$$

In $\text{PShv}(\mathcal{C})$ (which inherits image factorizations pointwise from Set), every morphism has $\text{im}(f) \simeq \text{coim}(f)$. Substituting $A = \coprod_i y(Y_i)$ and $B = y(X)$ yields the claim. $\square$

Remark 5.2.9 (Higher Categorical Perspective: Image and Čech Nerve). In higher categories (∞ -categories), we directly use the definition of image from Kerodon Tag 04X1.

Let $p: \coprod_{i \in I} y(Y_i) \rightarrow y(X)$ be the canonical morphism. The sieve $\mathcal{R}$ generated by $\{Y_i \rightarrow X\}$ is the image $\text{im}(p)$. In this case, $\text{im}(p)$ is identified with the *geometric realization* of the Čech nerve $\check{C}(p)_\bullet$ of p :

$$\mathcal{R} \simeq \text{im}(p) \simeq |\check{C}(p)_\bullet| := \text{colim}_{\Delta^{\text{op}}} \left(\cdots \rightrightarrows U \times_{y(X)} U \times_{y(X)} U \rightrightarrows U \times_{y(X)} U \rightarrow U \right).$$

Where $U = \coprod_{i \in I} y(Y_i)$. Since $\text{Fun}(\mathcal{C}^{\text{op}}, \mathbf{An})$ is an ∞ -topos, colimits in it are universal, i.e., coproducts commute with fiber products. Therefore, we recover the form of Lemma 5.2.8.

Alternatively, one can verify cofinality using Quillen's Theorem A.

Applying Lemma 5.2.8 to descent yields the following.

Proposition 5.2.10 (Equivalent Forms of Descent). *Let $\mathcal{C}$ be a category, $\mathcal{R}$ a sieve on $X \in \mathcal{C}$, and $(Y_i \rightarrow X)_{i \in I}$ a family of morphisms generating $\mathcal{R}$. For any presheaf $F \in \text{PShv}(\mathcal{C})$, the following conditions are equivalent:*

(i) *F satisfies descent along $\mathcal{R}$.*

(ii) *The canonical morphism:*

$$F(X) \rightarrow \lim_{Y \in \text{El}(\mathcal{R})^{\text{op}}} F(Y)$$

is an isomorphism.

(iii) *There is an equalizer diagram in which the intersections are taken in the presheaf category:*

$$F(X) \longrightarrow \prod_{i \in I} F(Y_i) \rightrightarrows \prod_{i, j \in I} \text{Hom}_{\text{PShv}(\mathcal{C})}(y(Y_i) \times_{y(X)} y(Y_j), F)$$

Proof. (i) $\Leftrightarrow$ (ii): By definition, F satisfies descent along $\mathcal{R}$ if $F(X) = \text{Hom}(y(X), F) \xrightarrow{\sim} \text{Hom}(\mathcal{R}, F)$. Since $\mathcal{R} = \text{colim}_{Y \in \text{El}(\mathcal{R})} y(Y)$ (the canonical Yoneda density colimit), we have

$$\text{Hom}(\mathcal{R}, F) = \lim_{Y \in \text{El}(\mathcal{R})^{\text{op}}} \text{Hom}(y(Y), F) = \lim_{Y \in \text{El}(\mathcal{R})^{\text{op}}} F(Y),$$

the second equality by Yoneda. This is exactly the limit in (ii).

(ii) $\Leftrightarrow$ (iii): By Lemma 5.2.8, $\mathcal{R}$ is the coequalizer

$$\text{colim} \left(\prod_{i, j \in I} y(Y_i) \times_{y(X)} y(Y_j) \rightrightarrows \prod_{i \in I} y(Y_i) \right).$$

Applying $\text{Hom}(-, F)$ converts this to the equalizer

$$F(X) \rightarrow \prod_{i \in I} F(Y_i) \rightrightarrows \prod_{i, j \in I} \text{Hom}(y(Y_i) \times_{y(X)} y(Y_j), F),$$

which is the equalizer in (iii). If every fiber product $Y_i \times_X Y_j$ exists in C , Yoneda identifies its last term with $\prod_{i, j} F(Y_i \times_X Y_j)$, recovering the usual overlap formula. No fiber products in C are needed for the presheaf-level statement. $\square$

Example 5.2.11 (Monogenic Sieve). Let C be a category and $\mathcal{R}$ a sieve on $X \in C$. We say $\mathcal{R}$ is *monogenic* if it is generated by a single morphism $Y \rightarrow X$. Assuming the pullback $Y \times_X Y$ exists in C , a presheaf F satisfies descent along $\mathcal{R}$ if and only if

$$F(X) \longrightarrow F(Y) \rightrightarrows F(Y \times_X Y)$$

is an equalizer diagram.

Remark 5.2.12 (Descent for Categories). Descent extends to category-valued pseudofunctors $F: C^{\text{op}} \rightarrow \text{Cat}$. Here we use the convention of Definition 3.1.9: limits are pseudolimits and the 2-cells in descent data are natural isomorphisms. We are therefore using the $(2, 1)$ -categorical core of the 2-category of categories, not asserting that arbitrary natural transformations are invertible. When the displayed fiber products exist in C , the first three stages of the resulting Čech diagram are

$$F(X) \longrightarrow \prod_{i \in I} F(Y_i) \rightrightarrows \prod_{i, j \in I} F(Y_i \times_X Y_j) \rightrightarrows \prod_{i, j, k \in I} F(Y_i \times_X Y_j \times_X Y_k)$$

If they do not exist, the same pseudolimit is formed using the corresponding pullbacks of representable presheaves, as in Proposition 5.2.10.

5.2.2. Grothendieck Topologies

To capture the structure of open covers axiomatically, we now distinguish a class of sieves with three closure properties.

Definition 5.2.13 (Grothendieck Topology). Let C be a category. A *Grothendieck topology* on C (or just a *topology*, for short) is a choice, for each $X \in C$, of a family of sieves $\tau(X)$ on X — the τ -covering sieves — satisfying:

- (i) (*Pullback stability*) If $\mathcal{R}$ is a τ -covering sieve on X and $f: Y \rightarrow X$ is any morphism, then the pullback sieve $f^*\mathcal{R}$ is also a τ -covering sieve on Y .
- (ii) (*Local character*) Let $\mathcal{S}$ be a sieve on X and $\mathcal{R}$ a τ -covering sieve on X . If for every $f \in \mathcal{R}$, we have $f^*\mathcal{S}$ is a τ -covering sieve, then $\mathcal{S}$ is a τ -covering sieve.
- (iii) (*Maximality*) For any $X \in C$, $y(X)$ is a τ -covering sieve.

We call a category C equipped with a Grothendieck topology a *site*.

Remark 5.2.14 (Additional Properties). Two further closure properties follow from the three axioms above:

- (iv) (*Refinement*) If $\mathcal{R}$ is a τ -covering sieve on X and $\mathcal{S}$ is a sieve containing $\mathcal{R}$, then $\mathcal{S}$ is also a τ -covering sieve.
- (v) (*Finite intersections*) Any finite intersection of τ -covering sieves on X is still a τ -covering sieve.

We can now define sheaves on a site.

Definition 5.2.15 (Sheaves). Let (C, τ) be a site. A presheaf F is called a τ -*sheaf* if for any τ -covering sieve $\mathcal{R}$, F satisfies descent along $\mathcal{R}$. We denote by:

$$\mathrm{Shv}_\tau(C) \subset \mathrm{PShv}(C)$$

the full subcategory of τ -sheaves.

Remark 5.2.16 (Poset of Topologies). Given two topologies τ, ρ on C , we say τ is *coarser* than ρ (or ρ *finer* than τ), written $\tau \leq \rho$, if every τ -covering sieve is also ρ -covering. The topologies on C form a poset under $\leq$.

In particular, C admits a coarsest and a finest topology.

Example 5.2.17 (Extreme Topologies). For any category C , consider the following topologies:

- (i) The finest topology on C is the *discrete topology*: all sieves are covering sieves. The terminal object $*$ in $\mathrm{PShv}(C)$ is the *only* sheaf in the discrete topology (the empty sieve is then covering, which forces $F(X)$ to be a singleton for every X).
- (ii) The coarsest topology on C is the *indiscrete topology* (or *trivial topology*): only maximal sieves are covering sieves. In this case, all presheaves are sheaves.

Example 5.2.18 (Canonical Topology). Let O be a locale, viewed as a category. Define a topology by declaring a sieve on $u \in O$ to be a covering sieve iff its supremum is u . The pullback-stability and local-character axioms follow from the distributive law (note that pullback in O is meet, $\wedge$); maximality is automatic. This is the *canonical topology* on O .

For a general topological space T , we denote by $\mathrm{Shv}(T)$ the sheaf category $\mathrm{Shv}_{\mathrm{can}}(\mathrm{Open}(T))$, i.e., the category of sheaves on $\mathrm{Open}(T)$ with respect to the canonical topology. We also simply call this the *sheaf category* on T . The resulting sheaves are precisely the classical sheaves on topological spaces.

Example 5.2.19 (κ -Weiss Topology). For a topological space $T \in \mathbf{Top}$ and a cardinal κ , we can consider the κ -Weiss topology on the category of open sets $\mathrm{Open}(T)$.

In this topology, a sieve $\mathcal{R}$ on an open set U is a covering sieve if and only if every subset $S \subset U$ of cardinality less than κ (i.e., $|S| < \kappa$) is *contained* in some member of $\mathcal{R}$. Formally:

$$\forall S \subset U, \text{ if } |S| < \kappa, \text{ then } \exists V \in \mathcal{R} \text{ such that } S \subseteq V.$$

(Note: When $\kappa = \aleph_0$, this is the standard *Weiss topology*, which requires that covers can capture all finite point configurations.) Additionally:

- When $\kappa = 0$, we get the discrete topology.
- When $\kappa = 1$, we get the topology consisting of non-empty sieves. The corresponding sheaf category is the category of constant sheaves.
- When $\kappa = 2$, the condition agrees with the canonical topology on nonempty opens. On the empty open it requires a nonempty sieve; to obtain the canonical topology one must additionally declare the empty sieve to cover $\emptyset$. This standard empty-cover convention forces every sheaf to satisfy $F(\emptyset) = *$.
- When $\kappa > |T|$, we get the trivial topology.

Remark 5.2.20 (Motivation for Grothendieck Topologies). Grothendieck topologies are not a direct generalization of point-set topology — Example 5.2.18 shows the relation is more subtle. Historically, Grothendieck introduced the notion entirely from the sheaf-theoretic side: classical sheaves on a topological space depend only on the lattice of open sets and their covers, not on the underlying points. Grothendieck’s insight was: since sheaves only care about “covers”, we can abandon the underlying space and directly axiomatize the concept of “covering” on any category.

The next proposition is the cornerstone of the section.

Proposition 5.2.21 (Adjunction Between Topologies and Sheaf Categories). *Let $\mathcal{C}$ be a category. Then the map between posets:*

$$\begin{aligned} \{\text{topologies on } \mathcal{C}\}^{\text{op}} &\rightarrow \{\text{subcategories of } \mathbf{PShv}(\mathcal{C})\} \\ \tau &\mapsto \mathbf{Shv}_\tau(\mathcal{C}) \end{aligned}$$

has a left adjoint $\mathcal{E} \mapsto \tau_{\mathcal{E}}$. That is, for any subcategory $\mathcal{E}$ of the presheaf category, there exists a finest topology $\tau_{\mathcal{E}}$ making $\mathcal{E}$ into sheaves.

Concretely, a sieve $\mathcal{R}$ on X is $\tau_{\mathcal{E}}$ -covering iff every $F \in \mathcal{E}$ universally descends along $\mathcal{R}$; explicitly, for every $F \in \mathcal{E}$ and every $g: X' \rightarrow X$ the map

$$F(X') \longrightarrow \text{Hom}(g^*\mathcal{R}, F)$$

must be an isomorphism. Taking $g = \text{id}_X$ gives the necessary special case $F(X) \simeq \text{Hom}(\mathcal{R}, F)$, but that single case alone is not equivalent to universal descent.

Proof. It suffices to check that $\tau_{\mathcal{E}}$ is a Grothendieck topology. Conditions (i) and (iii) of Definition 5.2.13 are immediate; the content lies in condition (ii).

Let $\mathcal{S}, \mathcal{R} \subset y(X)$ be two sieves on X . Suppose $\mathcal{R}$ is a $\tau_{\mathcal{E}}$ -covering sieve, and for every $(f: Y \rightarrow X) \in \mathcal{R}$, the pullback $f^*\mathcal{S}$ is a $\tau_{\mathcal{E}}$ -covering sieve. We prove that $\mathcal{S}$ is also a $\tau_{\mathcal{E}}$ -covering sieve.

This means for any $F \in \mathcal{E}$, we prove the isomorphism:

$$F(X) \xrightarrow{\sim} \text{Hom}_{\mathbf{PShv}(\mathcal{C})}(\mathcal{S}, F).$$

We establish this isomorphism through the intermediate term $\text{Hom}(\mathcal{R} \cap \mathcal{S}, F)$ in three steps:

(i) *Using $\mathcal{R}$ as covering sieve:* Since $\mathcal{R}$ is a covering sieve and $F \in \mathcal{E}$, we have an isomorphism:

$$F(X) \xrightarrow{\sim} \text{Hom}_{\mathbf{PShv}(\mathcal{C})}(\mathcal{R}, F).$$

(ii) *Using condition (ii):* Since $\mathcal{R}$ is a presheaf, we can write $\mathcal{R} = \text{colim}_{Y \in \text{El}(\mathcal{R})} y(Y)$. Since $\text{Hom}(-, F)$ takes colimits to limits, the restriction map from $\mathcal{R}$ to $\mathcal{R} \cap \mathcal{S}$

$$\text{Hom}(\mathcal{R}, F) \rightarrow \text{Hom}(\mathcal{R} \cap \mathcal{S}, F).$$

Note that $\mathcal{R} \cap \mathcal{S} = \mathcal{R} \times_{y(X)} \mathcal{S} \simeq \text{colim}_{Y \in \mathcal{R}} (y(Y) \times_{y(X)} \mathcal{S})$. The above map can be expanded as a limit:

$$\lim_{Y \in \text{El}(\mathcal{R})^{\text{op}}} \left(F(Y) \rightarrow \text{Hom}_{\mathbf{PShv}(\mathcal{C})}(y(Y) \times_{y(X)} \mathcal{S}, F) \right).$$

For each $Y \in \mathcal{R}$ (i.e., morphism $f: Y \rightarrow X$ in $\mathcal{R}$), the term $y(Y) \times_{y(X)} \mathcal{S}$ is precisely the pullback sieve $f^*\mathcal{S}$. By assumption, $f^*\mathcal{S}$ is a covering sieve, so $F(Y) \xrightarrow{\sim} \text{Hom}(f^*\mathcal{S}, F)$ is an isomorphism. Since limits preserve isomorphisms, we get $\text{Hom}(\mathcal{R}, F) \xrightarrow{\sim} \text{Hom}(\mathcal{R} \cap \mathcal{S}, F)$.

(iii) *Establishing* $\text{Hom}(\mathcal{S}, F) \simeq \text{Hom}(\mathcal{R} \cap \mathcal{S}, F)$: Finally, we use the colimit decomposition of $\mathcal{S}$: $\mathcal{S} \simeq \text{colim}_{Z \in \text{El}(\mathcal{S})} y(Z)$. Similarly, the map $\text{Hom}(\mathcal{S}, F) \rightarrow \text{Hom}(\mathcal{R} \cap \mathcal{S}, F)$ can be written as:

$$\lim_{Z \in \text{El}(\mathcal{S})^{\text{op}}} \left(F(Z) \rightarrow \text{Hom}_{\text{PShv}(\mathcal{C})}(\mathcal{R} \times_{y(X)} y(Z), F) \right).$$

For $Z \in \mathcal{S}$ (i.e., $g: Z \rightarrow X$ in $\mathcal{S}$), the term $\mathcal{R} \times_{y(X)} y(Z)$ is precisely the pullback sieve $g^*\mathcal{R}$. Since $\mathcal{R}$ is a covering sieve, $g^*\mathcal{R}$ is also a covering sieve on Z . This means $F(Z) \rightarrow \text{Hom}(g^*\mathcal{R}, F)$ is an isomorphism. Again, since limits preserve isomorphisms, we get $\text{Hom}(\mathcal{S}, F) \simeq \text{Hom}(\mathcal{R} \cap \mathcal{S}, F)$.

Combining the three steps gives $F(X) \simeq \text{Hom}(\mathcal{S}, F)$. To obtain universal descent, apply the same argument after an arbitrary base change $g: X' \rightarrow X$. The sieve $g^*\mathcal{R}$ universally descends, and for every arrow $Z \rightarrow X'$ in it the composite $Z \rightarrow X$ lies in $\mathcal{R}$, so the required pullback of $g^*\mathcal{S}$ universally descends by hypothesis. Hence $F(X') \simeq \text{Hom}(g^*\mathcal{S}, F)$ for every g . $\square$

Corollary 5.2.22 (Supremum of Topologies). *Let $\mathcal{C}$ be a category and $(\tau_i)_{i \in I}$ a family of topologies on $\mathcal{C}$. Let $\tau = \bigvee_{i \in I} \tau_i$ be their supremum. Then:*

$$\text{Shv}_\tau(\mathcal{C}) = \bigcap_{i \in I} \text{Shv}_{\tau_i}(\mathcal{C}).$$

Proof. Simply take $\mathcal{E} = \{F \in \text{PShv}(\mathcal{C}) \mid F \text{ descends along } \tau_i \text{ for each } i \in I\}$. $\square$

There is a simpler way to specify topologies, with two caveats:

- It requires the existence of corresponding pullbacks in $\mathcal{C}$.
- The relationship with topologies is not injective, meaning there may be multiple pretopologies determining the same topology.

Definition 5.2.23 (Pretopology). *Let $\mathcal{C}$ be a category. A pretopology π on $\mathcal{C}$ consists of the following data: for each object X in $\mathcal{C}$, we specify a family of objects in $\mathcal{C}/_X$, called π -coverings, satisfying:*

- (Pullback stability) For a π -covering $(U_i \rightarrow X)_{i \in I}$ and any morphism $f: Y \rightarrow X$, the pullbacks $U_i \times_X Y$ exist and $(U_i \times_X Y \rightarrow Y)_{i \in I}$ is a π -covering.
- (Transitivity) For a π -covering $(U_i \rightarrow X)_{i \in I}$ and π -coverings $(V_{ij} \rightarrow U_i)_{j \in J_i}$, the family $(V_{ij} \rightarrow X)_{i \in I, j \in J_i}$ is a π -covering.
- (Identity) $(\text{id}_X: X \rightarrow X)$ is a π -covering.

The topology generated by the pretopology π is the coarsest topology containing all sieves generated by π -coverings.

Remark 5.2.24 (From Topology to Pretopology). *Let τ be a topology on $\mathcal{C}$. The τ -coverings produce a pretopology on $\mathcal{C}$ only when $\mathcal{C}$ has pullbacks.*

Proposition 5.2.25 (Pretopology and Descent). *Let $\mathcal{C}$ be a category, π a pretopology on $\mathcal{C}$, and τ the topology generated by π . Then:*

- A sieve is a τ -covering sieve if and only if it contains at least one π -covering.
- A presheaf F is a τ -sheaf if and only if for each π -covering $(U_i \rightarrow X)_{i \in I}$, there is an equalizer diagram:

$$F(X) \longrightarrow \prod_{i \in I} F(U_i) \rightrightarrows \prod_{i, j \in I} F(U_i \times_X U_j)$$

Proof. This follows immediately from the fact that τ is the coarsest topology containing all covering sieves generated by π -coverings. $\square$

5.2.3. Examples

We now turn to concrete examples.

Example 5.2.26 (Standard Topology on Top). On the category of topological spaces $\mathbf{Top}$, open covers form a pretopology. Let τ be the corresponding topology. Then:

- (i) A sieve on X is a τ -covering sieve if and only if it contains an open cover of X . Therefore, $(Y_i \rightarrow X)_{i \in I}$ is a τ -covering if and only if it locally admits sections.
- (ii) A presheaf $F: \mathbf{Top}^{\text{op}} \rightarrow \mathbf{Set}$ is a τ -sheaf if and only if for each topological space T , the restriction $F|_{\text{Open}(T)}$ is a sheaf on T .

We now describe the most commonly used topologies in algebraic geometry. (We omit the Nisnevich, étale, and smooth topologies, which require étale and smooth morphisms not yet introduced.)

Example 5.2.27 (Zariski Topology). On $\mathbf{Aff} \simeq \mathbf{CAlg}^{\text{op}}$, declare a family $\{\text{Spec}(R_{f_i}) \rightarrow \text{Spec}(R)\}_{i \in I}$ to be a covering whenever $(f_i)_{i \in I}$ generates the unit ideal of R . These families form a pretopology, and the topology it generates is the *Zariski topology*.

We have the following characterizations:

- (i) *Covering sieve criterion*: A sieve $\mathcal{S}$ on $\text{Spec}(R)$ is a Zariski-covering sieve iff the set

$$\{f \in R \mid (\text{Spec}(R_f) \rightarrow \text{Spec}(R)) \in \mathcal{S}\}$$

generates the unit ideal of R .

- (ii) *Sheaf condition*: An algebraic functor $X: \mathbf{CAlg} \rightarrow \mathbf{Set}$ is a Zariski sheaf if and only if for any ring R and elements $(f_i)_{i \in I}$ generating the unit ideal, there is an equalizer diagram:

$$X(R) \rightarrow \prod_{i \in I} X(R_{f_i}) \rightrightarrows \prod_{i, j \in I} X(R_{f_i f_j}).$$

Example 5.2.28 (fpqc and fppf Topologies). The Zariski topology is a special case of a more general construction. Fix a class E of ring maps containing all isomorphisms and closed under composition and pushouts. On affine schemes, the topology is generated by the standard finite covering families $(R \rightarrow R_i)_{i \in I}$ satisfying:

- The index set I is finite;
- The induced map $\coprod_{i \in I} \text{Spec}(R_i) \rightarrow \text{Spec}(R)$ is surjective;
- Each map $R \rightarrow R_i$ belongs to the class E .

Depending on the choice of E , we obtain different topologies:

- (i) If E is the class of localizations $R \rightarrow R_f$, this recovers the Zariski topology.
- (ii) If E is the class of flat morphisms, the corresponding topology is called the *fpqc topology*.
- (iii) If E is the class of finitely presented flat morphisms, the corresponding topology is called the *fppf topology*.

Since localizations $R \rightarrow R_f$ are flat and finitely presented, the topologies are ordered

$$\text{Zariski} \leq \text{fppf} \leq \text{fpqc}.$$

In fact, we have

$$\text{Zariski} \leq \text{Nisnevich} \leq \text{Étale} \leq \text{Smooth} \leq \text{fppf} \leq \text{fpqc}$$

5.2.4. Faithfully Flat Descent: One Map, a Family, and Effectivity

An fpqc covering family of general schemes need not be globally finite. The precise convention is local on the base: for every affine open $U \subset X$, there must be finitely many affine opens $V_j \subset U_{i_j} \times_X U$ whose images cover U ; the maps $V_j \rightarrow U$ are flat. This is the scheme-level fpqc covering condition of [The25, Tag 022B]. Thus, after restricting to an affine open of the base and refining the sources by affine opens, one reaches a finite affine family. Only this finite affine refinement is replaced by a single product algebra below.

The fpqc topology is useful because flat base change preserves linear relations, while faithful flatness detects whether such a relation already held over the base. Recall that a ring map $R \rightarrow S$ is *flat* if $- \otimes_R S$ is exact, and *faithfully flat* if it is flat and $M \otimes_R S = 0$ implies $M = 0$. Equivalently, it is flat and $\text{Spec}(S) \rightarrow \text{Spec}(R)$ is surjective.

There is no essential difference between a finite affine fpqc family and one faithfully flat algebra. If $(R \rightarrow R_i)_{i=1}^m$ is a standard fpqc cover, put $S = \prod_i R_i$. A finite product of modules is a finite direct sum, so S is flat over R , and

$$\text{Spec}(S) = \coprod_i \text{Spec}(R_i) \longrightarrow \text{Spec}(R)$$

is surjective; hence $R \rightarrow S$ is faithfully flat. Conversely, a faithfully flat map $R \rightarrow S$ is a one-member affine fpqc cover. Moreover,

$$S^{\otimes_R(n+1)} \simeq \prod_{i_0, \dots, i_n} R_{i_0} \otimes_R \cdots \otimes_R R_{i_n}, \quad (*)$$

so the family Čech diagram is exactly the Amitsur diagram of $R \rightarrow S$. If the R_i are finitely presented over R , then so is S , giving the corresponding fppf statement. This reduction is the standard affine fpqc covering convention [The25, Tag 022F].

Definition 5.2.29 (Descent datum and effectivity). Let $R \rightarrow S$ be a ring map and let $p_1, p_2: S \rightrightarrows S \otimes_R S$ be the two structure maps. An S -module descent datum consists of an S -module N and an isomorphism

$$\theta: p_1^* N \xrightarrow{\sim} p_2^* N$$

whose three pullbacks to $S \otimes_R S \otimes_R S$ satisfy

$$p_{23}^* \theta \circ p_{12}^* \theta = p_{13}^* \theta.$$

It is *effective* if it is isomorphic to the canonical datum on $S \otimes_R M$ for some R -module M .

Set-valued descent glues compatible sections. Category-valued descent also asks whether an object carrying a cocycle actually comes from the base: this is essential surjectivity, or *effectivity*. The pseudolimit convention is the one fixed in Remark 5.2.12.

Theorem 5.2.30 (Classical faithfully flat descent). Let $R \rightarrow S$ be faithfully flat.

(i) Base change induces an equivalence

$$\text{Mod}_R \xrightarrow{\sim} \lim_{[n] \in \Delta} \text{Mod}_{S^{\otimes_R(n+1)}}.$$

Equivalently, every S -module descent datum is effective, uniquely up to a unique compatible isomorphism.

(ii) The analogous functor for commutative algebras is an equivalence:

$$\text{CAlg}_R \xrightarrow{\sim} \lim_{[n] \in \Delta} \text{CAlg}_{S^{\otimes_R(n+1)}}.$$

(iii) After taking opposites and applying Spec , affine schemes—and, more generally, affine morphisms—have effective fpqc descent.

Proof. For a datum (N, θ) , form the equalizer

$$M = \{n \in N \mid \theta(n \otimes 1) = 1 \otimes n\}. \quad (**)$$

There is a canonical S -linear map $S \otimes_R M \rightarrow N$. Faithful flatness allows us to test whether it is an isomorphism after tensoring once more with S . After this base change the covering map $S \rightarrow S \otimes_R S$ is split by multiplication. For a split covering, the cocycle identity supplies an explicit inverse to the base-changed map; hence $S \otimes_R M \simeq N$. Applying the same equalizer to morphisms proves full faithfulness. This is the usual Amitsur proof of faithfully flat module descent [The25, Tag 023N].

The resulting equivalence is symmetric monoidal. Units, multiplications, and their associativity and commutativity diagrams therefore descend along with the underlying modules, proving (ii). Commutative R -algebras are opposite to affine schemes over $\mathrm{Spec}(R)$; applying this observation locally on the base proves (iii), whose global formulation is [The25, Tag 0245]. $\square$

Corollary 5.2.31 (fpqc descent for quasi-coherent modules). *For an fpqc covering $(U_i \rightarrow X)$, quasi-coherent modules on X are equivalent to quasi-coherent modules on the U_i equipped with compatible cocycle data on all overlaps.*

Proof. On an affine base and for a standard affine fpqc cover this is Theorem 5.2.30, using (*). Refine over affine opens of X , apply the affine result, and glue the descended modules Zariski locally. The complete global argument is recorded in [The25, Tag 023T]. After Proposition 6.4.5, this is equally a statement about Mod_X . $\square$

Warning 5.2.32 (Morphisms descend more generally than objects). Two assertions must not be conflated. Subcanonicity says that compatible maps from an fpqc cover into a fixed scheme glue uniquely. Equivalently, morphisms between schemes already descended to the base satisfy fpqc descent; this holds for every scheme [The25, Tag 023Q]. Effective descent for objects asks whether an arbitrary scheme over the cover, equipped with a cocycle, itself comes from a scheme over the base. That stronger assertion is false in general.

Object over the cover	Morphisms descend	Objects are effective
modules and quasi-coherent modules	yes	yes
commutative algebras	yes	yes
affine morphisms	yes	yes
quasi-affine morphisms	yes	yes
arbitrary schemes	yes	not in general
projective morphisms	yes	not in general

Affine and quasi-affine effectivity are [The25, Tags 0245 and 0247]. Schemes over the base form a full subcategory of the category of fpqc descent data [The25, Tag 02W0], but that inclusion need not be essentially surjective. In fact, projective descent data can fail to be effective even for an étale cover; the standard example descends naturally to an algebraic space rather than a scheme [The25, Tag 08KE].

5.3. Sheafification

Every presheaf has a universal sheaf approximation, its *sheafification*. We construct this functor $(-)^{\sharp}: \mathrm{PShv}(C) \rightarrow \mathrm{Shv}_{\tau}(C)$ via a small-object-argument style plus construction and verify its universal property — the left adjoint to the inclusion of sheaves.

Theorem 5.3.1 (Sheafification). *Let C be a small category and τ be a topology on C . Then the inclusion functor $\mathrm{Shv}_\tau(C) \hookrightarrow \mathrm{PShv}(C)$ admits a left adjoint:*

$$a_\tau : \mathrm{PShv}(C) \rightarrow \mathrm{Shv}_\tau(C),$$

called τ -sheafification. Moreover:

- (i) *the functor a_τ is left exact (i.e., it preserves finite limits);*
- (ii) *a sieve $\mathcal{R} \subset y(X)$ is a τ -covering sieve if and only if $a_\tau(\mathcal{R}) \rightarrow a_\tau(y(X))$ is an isomorphism.*

Proof. Rather than reproduce the full construction, we sketch the idea and refer to [The25, Tag 00ZG] for details.

The *plus construction* $F \mapsto F^+$ sends a presheaf F to the presheaf

$$F^+(X) := \operatorname{colim}_{\mathcal{R} \in \tau(X)} \operatorname{Hom}_{\mathrm{PShv}}(\mathcal{R}, F),$$

where the colimit is over all τ -covering sieves on X . Informally: $F^+(X)$ consists of “sections of F compatible along some cover of X ”. The plus construction is *not* a sheaf in general; applying it twice is required for F^{++} to satisfy descent (this iteration is the “ a_τ ” — it is a small-object argument in disguise). Left exactness follows because filtered colimits commute with finite limits, and (ii) is essentially the definition of a covering sieve. $\square$

An immediate corollary:

Corollary 5.3.2. *Let (C, τ) be a small site. Then $\mathrm{Shv}_\tau(C)$ admits limits and colimits:*

- *Limits are computed as in $\mathrm{PShv}(C)$.*
- *Colimits are computed as in $\mathrm{PShv}(C)$ and then sheafified.*

Combining Theorem 5.3.1 with Proposition 5.2.21 yields an adjunction between topologies and categories of sheaves.

Corollary 5.3.3. *Let C be a small category. Then the map between posets*

$$\begin{aligned} \{\text{topologies on } C\}^{\mathrm{op}} &\rightarrow \{\text{full subcategories of } \mathrm{PShv}(C) \text{ admitting left exact left adjoint}\} \\ \tau &\mapsto \mathrm{Shv}_\tau(C) \end{aligned}$$

is an isomorphism.

Proof. Injectivity follows from Theorem 5.3.1(ii). For surjectivity, let $i : \mathcal{E} \hookrightarrow \mathrm{PShv}(C)$ be a full reflective subcategory with left exact reflector $L \dashv i$, and declare

$$\mathcal{R} \in \tau(X) \iff L(\mathcal{R}) \xrightarrow{\sim} L(y(X)).$$

The topology axioms. Maximality is immediate. If $f : Y \rightarrow X$, left exactness gives

$$L(f^*\mathcal{R}) \simeq L(y(Y)) \times_{L(y(X))} L(\mathcal{R}),$$

so covering sieves are stable under pullback. For local character, suppose $\mathcal{R}$ covers X and $f^*\mathcal{S}$ covers Y for every $(f : Y \rightarrow X) \in \mathcal{R}$. Every $E \in \mathcal{E}$ is orthogonal to every morphism inverted by L , because

$$\operatorname{Hom}_{\mathrm{PShv}}(B, iE) \simeq \operatorname{Hom}_{\mathcal{E}}(L(B), E).$$

The intersection argument in the proof of Proposition 5.2.21, applied to $\mathcal{R}$ and $\mathcal{S}$, therefore gives

$$\operatorname{Hom}(y(X), iE) \xrightarrow{\sim} \operatorname{Hom}(\mathcal{S}, iE)$$

for every $E \in \mathcal{E}$. By adjunction and Yoneda inside $\mathcal{E}$, this says $L(\mathcal{S}) \rightarrow L(y(X))$ is an isomorphism. Thus τ is a Grothendieck topology.

Identification of local objects. If $E \in \mathcal{E}$ and $\mathcal{R} \in \tau(X)$, the same adjunction shows

$$iE(X) = \text{Hom}(y(X), iE) \xrightarrow{\sim} \text{Hom}(\mathcal{R}, iE),$$

so iE is a τ -sheaf. Hence $\mathcal{E} \subseteq \text{Shv}_\tau(C)$.

Conversely, let F be a τ -sheaf and consider the unit $\eta_F: F \rightarrow iL(F)$. Applying L makes η_F an isomorphism. For a section $t \in iL(F)(X)$, form the presheaf of its lifts $P_t = y(X) \times_{iL(F)} F$ and let $\mathcal{R}_t \subset y(X)$ be the image sieve. Left exactness identifies $L(P_t)$ with $L(y(X))$, while L preserves the epimorphism $P_t \twoheadrightarrow \mathcal{R}_t$ (presheaf epimorphisms are regular) and the monomorphism $\mathcal{R}_t \hookrightarrow y(X)$. It follows that $L(\mathcal{R}_t) \rightarrow L(y(X))$ is an isomorphism, so t lifts locally along a covering sieve. Likewise, if two sections of $F(X)$ have the same image under η_F , their equality sieve is the pullback of the diagonal of η_F ; left exactness again shows that this sieve is covering. Thus η_F is locally surjective and locally injective. Both its source and target are τ -sheaves, so local lifts glue uniquely and η_F is an isomorphism. Therefore $F \in \mathcal{E}$, proving $\mathcal{E} = \text{Shv}_\tau(C)$. $\square$

Remark 5.3.4 (Non-topological localizations). The isomorphism established in Corollary 5.3.3 relies essentially on the fact that objects in a 1-category are 0-truncated, which implies that for any morphism f , the diagonal Δ_f is a monomorphism.

In the context of ∞ -topoi, the situation is different. As pointed out by Marc (see [MO 273085](#)), there are many natural examples of ∞ -topoi that arise as left exact localizations but have *a priori* nothing to do with Grothendieck topologies. Examples include:

- The ∞ -category of n -excisive functors (in the sense of Goodwillie calculus) with values in an ∞ -topos.
- The ∞ -category of coalgebras for a left exact comonad in an ∞ -topos.

While these ∞ -topoi might happen to be equivalent to sheaf ∞ -categories “by accident”, their defining localizations involve inverting morphisms that are not monomorphisms (e.g., hypercovers), a feature that is invisible in the 1-categorical setting.

We can now describe sheaf-theoretic properties of morphisms in terms of local properties at the presheaf level.

Definition 5.3.5 (Local epimorphism, monomorphism, isomorphism). Let (C, τ) be a site and let $f: F \rightarrow G$ be a morphism in $\text{PShv}(C)$. Let $\Delta_f: F \rightarrow F \times_G F$ denote the diagonal morphism.

- f is called a τ -local epimorphism if for every $X \in C$ and every section $s \in G(X)$, the sieve on X consisting of all morphisms $u: Y \rightarrow X$ such that $u^*(s)$ lies in the image of the map $f_Y: F(Y) \rightarrow G(Y)$ is a τ -covering sieve.
- f is called a τ -local monomorphism if the diagonal Δ_f is a τ -local epimorphism.
- f is called a τ -local isomorphism if it is both a τ -local epimorphism and a τ -local monomorphism.

Proposition 5.3.6 (Stability of τ -local Morphisms under Base Change). Let $f: F \rightarrow G$ be a morphism of presheaves, and let $h: H \rightarrow G$ be any morphism. Consider the base change $f': F \times_G H \rightarrow H$ fitting into the cartesian diagram:

$$\begin{array}{ccc} F \times_G H & \longrightarrow & F \\ f' \downarrow & \lrcorner & \downarrow f \\ H & \xrightarrow{h} & G \end{array}$$

If f is a τ -local epimorphism (resp. τ -local monomorphism, τ -local isomorphism), then f' is a τ -local epimorphism (resp. τ -local monomorphism, τ -local isomorphism).

Proof. (i) **Stability of τ -local epimorphisms.** Let $f : F \rightarrow G$ be a τ -local epimorphism. We must show that $f' : F \times_G H \rightarrow H$ is a τ -local epimorphism. Let $X \in \mathcal{C}$ and let $t \in H(X)$. We consider the sieve S' on X generated by f' over t :

$$S' = \{u : Y \rightarrow X \mid u^*(t) \in \text{im}(f'_Y)\}.$$

We must show S' is a τ -covering sieve. Let $s = h_X(t) \in G(X)$. Since f is a τ -local epimorphism, the sieve

$$S = \{u : Y \rightarrow X \mid u^*(s) \in \text{im}(f_Y)\}$$

is a τ -covering sieve on X . We claim that $S \subseteq S'$.

Indeed, let $u : Y \rightarrow X$ be in S . Then there exists $a \in F(Y)$ such that $f_Y(a) = u^*(s)$. Since $s = h_X(t)$, we have:

$$f_Y(a) = u^*(h_X(t)) = h_Y(u^*(t)).$$

By the universal property of the fiber product, the pair $(a, u^*(t))$ determines a unique section $b \in (F \times_G H)(Y)$ such that $f'_Y(b) = u^*(t)$. Consequently, $u^*(t) \in \text{im}(f'_Y)$, so $u \in S'$.

Since S is a τ -covering sieve and $S \subseteq S'$, S' is also a τ -covering sieve. Thus f' is a τ -local epimorphism.

- (ii) **Stability of τ -local monomorphisms.** Let $f : F \rightarrow G$ be a τ -local monomorphism. By definition, this means the diagonal $\Delta_f : F \rightarrow F \times_G F$ is a τ -local epimorphism. We must show that $\Delta_{f'} : F \times_G H \rightarrow (F \times_G H) \times_H (F \times_G H)$ is a τ -local epimorphism.

There is a canonical isomorphism of presheaves:

$$(F \times_G H) \times_H (F \times_G H) \cong (F \times_G F) \times_G H.$$

Under this identification, $\Delta_{f'}$ corresponds to the morphism $\Delta_f \times_G \text{id}_H$. This fits into the following diagram, which is a pullback square in the category of presheaves:

$$\begin{array}{ccc} F \times_G H & \xrightarrow{\quad} & F \\ \Delta_{f'} \downarrow & \lrcorner & \downarrow \Delta_f \\ (F \times_G F) \times_G H & \xrightarrow{\quad} & F \times_G F \end{array}$$

Since Δ_f is a τ -local epimorphism, and we established in part (i) that τ -local epimorphisms are stable under base change, it follows that $\Delta_{f'}$ is a τ -local epimorphism. Therefore, f' is a τ -local monomorphism.

- (iii) **Stability of τ -local isomorphisms.** If f is a τ -local isomorphism, it is both a τ -local epimorphism and a τ -local monomorphism. By parts (i) and (ii), the base change f' retains both properties. Hence, f' is a τ -local isomorphism.

□

Proposition 5.3.7. *Let $(\mathcal{C}, \tau)$ be a small site and let f be a morphism in $\text{PShv}(\mathcal{C})$. Then $a_\tau(f)$ is an epimorphism (resp. monomorphism, isomorphism) in $\text{Shv}_\tau(\mathcal{C})$ if and only if f is a τ -local epimorphism (resp. monomorphism, isomorphism).*

Proof. We treat the three cases in turn; throughout, write $i : \text{Shv}_\tau(\mathcal{C}) \hookrightarrow \text{PShv}(\mathcal{C})$ for the inclusion, with adjunction $a_\tau \dashv i$.

Epimorphisms. Let $I = \text{im}(f) \subseteq F'$ be the pointwise image presheaf. Since a_τ preserves finite limits and all colimits, it preserves image factorizations: an image is the coequalizer of the kernel pair followed by a monomorphism. Hence

$$\text{im}(a_\tau f) \simeq a_\tau I \subseteq a_\tau F',$$

and $a_\tau(f)$ is an epimorphism exactly when $a_\tau I \rightarrow a_\tau F'$ is an isomorphism.

For a subpresheaf $I \subseteq F'$, the plus-construction description of sheafification says precisely that $a_\tau I \rightarrow a_\tau F'$ is an isomorphism iff every section $s \in F'(X)$ belongs to I after restriction along a covering sieve of X : elements of the sheafification are locally represented by elements of the original presheaf, and equality after sheafification is likewise local. Applied to the pointwise image I , this condition says that the sieve of maps $u: Y \rightarrow X$ for which $u^*s \in \text{im}(f_Y)$ is covering. By Definition 5.3.5(i), that is exactly the assertion that f is a τ -local epimorphism.

Monomorphisms. A morphism h is a monomorphism iff its diagonal Δ_h is an isomorphism. Sheafification preserves finite limits (Theorem 5.3.1), so $\Delta_{a_\tau(f)} \simeq a_\tau(\Delta_f)$. Hence $a_\tau(f)$ is a monomorphism iff $a_\tau(\Delta_f)$ is an isomorphism, equivalently a sheaf-theoretic epimorphism (since Δ_f is automatically a monomorphism). By the epimorphism case, this is equivalent to Δ_f being a τ -local epimorphism, i.e., to f being a τ -local monomorphism (Definition 5.3.5(ii)).

Isomorphisms. An iso is both a mono and an epi; combining the two cases above yields the claim. $\square$

5.4. Pushforward and Pullback of Sheaves

Let $f: X \rightarrow Y$ be a continuous map of topological spaces. The induced map of locales corresponds to the order-preserving map $f^{-1}: \text{Open}(Y) \rightarrow \text{Open}(X)$ sending U to $f^{-1}(U)$, which preserves open coverings. It follows that the pushforward functor

$$f_*^{\text{pre}}: \text{PShv}(\text{Open}(X)) \rightarrow \text{PShv}(\text{Open}(Y)), \quad f_*^{\text{pre}}(F)(V) = F(f^{-1}(V)),$$

preserves sheaves, so it lifts to a functor between categories of sheaves:

$$\begin{array}{ccc} \text{PShv}(\text{Open}(X)) & \xrightarrow{f_*^{\text{pre}}} & \text{PShv}(\text{Open}(Y)) \\ \uparrow & & \downarrow a \\ \text{Shv}(X) & \xrightarrow{f_*} & \text{Shv}(Y). \end{array}$$

Proposition 5.4.1 (Pullbacks of Sheaves). *Let $f: X \rightarrow Y$ be a continuous map between topological spaces. Then the pushforward functor $f_*: \text{Shv}(X) \rightarrow \text{Shv}(Y)$ admits a left exact left adjoint $f^*: \text{Shv}(Y) \rightarrow \text{Shv}(X)$.*

Proof. First, we define the pullback functor in the context of presheaves. Let G be a presheaf on Y . We define its presheaf pullback $f_{\text{pre}}^* G$ via the left Kan extension along the functor $f^{-1}: \text{Open}(Y)^{\text{op}} \rightarrow \text{Open}(X)^{\text{op}}$ induced by f :

$$\begin{array}{ccc} \text{Open}(Y)^{\text{op}} & \xrightarrow{G} & \text{Set} \\ f^{-1} \downarrow & \nearrow f_{\text{pre}}^* G = \text{Lan}_{f^{-1}} G & \\ \text{Open}(X)^{\text{op}} & & \end{array}$$

By the pointwise formula for left Kan extensions, we obtain an explicit description:

$$(f_{\text{pre}}^* G)(U) = \text{colim}_{U \subset f^{-1}(V)} G(V).$$

The indexing category $\{V \in \text{Open}(Y) \mid U \subset f^{-1}(V)\}$ is filtered (since the intersection of two open sets containing $f(U)$ is again an open set containing $f(U)$). Since filtered colimits in Set commute with finite limits, the functor $f_{\text{pre}}^*: \text{PShv}(Y) \rightarrow \text{PShv}(X)$ is left exact. By the universal property of left Kan extensions, f_{pre}^* is the left adjoint to the presheaf pushforward.

Now, consider the diagram of adjoint pairs relating presheaves and sheaves:

$$\begin{array}{ccc}
 \mathbf{PShv}(X) & \xleftarrow{f_{\text{pre}}^*} & \mathbf{PShv}(Y) \\
 \downarrow a \quad \uparrow i & & \downarrow a \quad \uparrow i \\
 \mathbf{Shv}(X) & \xleftarrow{f^*} \xrightarrow{f_*} & \mathbf{Shv}(Y)
 \end{array}$$

We define the functor f^* as the composition:

$$f^* = a \circ f_{\text{pre}}^* \circ i.$$

Since f^* is a composition of left adjoints (a and f_{pre}^*), it is left adjoint to f_* on sheaves. Explicitly:

$$\text{Hom}_{\mathbf{Shv}(X)}(f^*G, \mathcal{F}) \simeq \text{Hom}_{\mathbf{PShv}(X)}(f_{\text{pre}}^*(iG), i\mathcal{F}) \simeq \text{Hom}_{\mathbf{PShv}(Y)}(iG, f_*(i\mathcal{F})) \simeq \text{Hom}_{\mathbf{Shv}(Y)}(G, f_*\mathcal{F}).$$

Regarding exactness: By Theorem 5.3.1, the sheafification functor a is left exact (preserves finite limits). The inclusion i preserves limits. Since f_{pre}^* is left exact (as shown above via filtered colimits), the composition f^* is left exact. $\square$

Remark 5.4.2. In fact, the construction f_{pre}^* and f_*^{pre} above can be extended into the general definition of a *morphism of sites*.

A morphism of sites $f: (C, \tau) \rightarrow (D, \sigma)$ is defined not by a functor $C \rightarrow D$, but by a functor in the *inverse* direction:

$$f^{-1}: D \rightarrow C,$$

which is *continuous* in the sense that:

- (i) f^{-1} is left exact (preserves finite limits);
- (ii) f^{-1} preserves covering families (it sends sieves in σ to sieves in τ).

Given such a continuous functor, we obtain an adjunction on the categories of presheaves and sheaves exactly as we did for topological spaces:

- The *direct image* $f_*: \mathbf{PShv}(C) \rightarrow \mathbf{PShv}(D)$ is given by pre-composition: $f_*(F) = F \circ f^{-1}$.
- The *inverse image* $f^*: \mathbf{PShv}(D) \rightarrow \mathbf{PShv}(C)$ is the left Kan extension along f^{-1} (followed by sheafification).

In the context of topological spaces, a continuous map $f: X \rightarrow Y$ corresponds to the continuous functor $f^{-1}: \text{Open}(Y) \rightarrow \text{Open}(X)$, thus defining a morphism of sites.

Stalks are the special case of sheaf pullback along the inclusion of a single point.

Definition 5.4.3 (Stalk). Let $x \in X$ be a point of a topological space. Consider the one-point space $*$ and the inclusion map $i_x: * \hookrightarrow X$ mapping the unique point to x . For any sheaf F on X , the *stalk* of F at x , denoted by F_x , is defined as the pullback of F along i_x :

$$F_x := i_x^* F.$$

Since $\mathbf{Shv}(*) \simeq \mathbf{Set}$, we view the stalk naturally as a set (or an algebraic structure, depending on the target category).

Using the explicit formula for f^* from Proposition 5.4.1, we can compute this pullback explicitly. Since every open neighborhood U of the point $*$ is just $*$ itself, and $i_x(U) = \{x\}$, the colimit is taken over open sets $V \subset X$ such that $V \supset \{x\}$. Thus we recover the classical definition:

$$F_x = \operatorname{colim}_{V \ni x} F(V).$$

This perspective explains why stalks commute with pullbacks: the identity $(f^*G)_x \simeq G_{f(x)}$ is simply the functoriality of pullbacks for the composition $* \xrightarrow{x} X \xrightarrow{f} Y$.

We can rephrase the criterion of Proposition 5.3.7 in terms of stalks:

Proposition 5.4.4 (Stalkwise characterization). *Let T be a topological space, let $f: F \rightarrow G$ be a map in $\operatorname{PShv}(\operatorname{Open}(T))$, and let $a: \operatorname{PShv}(\operatorname{Open}(T)) \rightarrow \operatorname{Shv}(T)$ be the sheafification functor. Then $a(f)$ is an epimorphism/monomorphism/isomorphism in $\operatorname{Shv}(T)$ if and only if, for every $x \in T$, the induced map of stalks $f_x: F_x \rightarrow G_x$ is surjective/injective/bijective.*

Proof. By Proposition 5.3.7, $a(f)$ is an epimorphism (resp. monomorphism) in $\operatorname{Shv}(T)$ if and only if f is a τ -local epimorphism (resp. monomorphism). We analyze each case using the properties of the standard topology τ on Top .

1. Epimorphisms ($\Rightarrow$) Suppose f is a τ -local epimorphism. Let $x \in T$ and let $y \in G_x$ be an element of the stalk. By description of the stalk, y is represented by a section $s \in G(U)$ for some open neighborhood U of x , such that $s_x = y$. Since f is a τ -local epimorphism, there exists a τ -covering sieve S on U such that for any $V \rightarrow U$ in S , the restriction $s|_V$ is in the image of $f_V: F(V) \rightarrow G(V)$. In the standard topology, a sieve is a covering sieve if and only if it contains an open cover. Thus, there exists an open cover $\{U_i\}_{i \in I}$ of U such that for each i , there exists $t_i \in F(U_i)$ with $f(t_i) = s|_{U_i}$. Since $\{U_i\}$ covers U , there exists an index j such that $x \in U_j$. Passing to the stalk at x , we have:

$$f_x((t_j)_x) = (f(t_j))_x = (s|_{U_j})_x = s_x = y.$$

Thus, f_x is surjective.

($\Leftarrow$) Suppose f_x is surjective for all $x \in T$. Let U be an open set and $s \in G(U)$. We must show that the sieve of morphisms $V \rightarrow U$ such that $s|_V$ lifts to F is a τ -covering sieve. For every $x \in U$, consider the germ $s_x \in G_x$. Since f_x is surjective, there exists $t_x \in F_x$ such that $f_x(t_x) = s_x$. The germ t_x is represented by a section $\tilde{t} \in F(V_x)$ for some open neighborhood $V_x \subseteq U$ of x . The equality $f_x(t_x) = s_x$ implies that $f(\tilde{t})$ and $s|_{V_x}$ have the same germ at x . Therefore, there exists a possibly smaller open neighborhood $W_x \subseteq V_x$ of x such that $f(\tilde{t}|_{W_x}) = s|_{W_x}$. The collection $\{W_x \hookrightarrow U\}_{x \in U}$ generates a sieve on U . Since $\bigcup_{x \in U} W_x = U$, this is a τ -covering sieve. Thus, f is a τ -local epimorphism.

2. Monomorphisms Recall that f is a τ -local monomorphism if and only if the diagonal $\Delta_f: F \rightarrow F \times_G F$ is a τ -local epimorphism. The functor taking a presheaf to its stalk at x commutes with finite limits (it is a filtered colimit). Therefore, there is a natural isomorphism $(F \times_G F)_x \simeq F_x \times_{G_x} F_x$. Under this isomorphism, the stalk of the diagonal, $(\Delta_f)_x$, corresponds to the diagonal of the stalks, $\Delta_{f_x}: F_x \rightarrow F_x \times_{G_x} F_x$. Applying the result from part 1 to the morphism Δ_f , we see that f is a τ -local monomorphism if and only if Δ_{f_x} is surjective for all $x \in T$. In the category of sets, the diagonal map $A \rightarrow A \times_B A$ is surjective if and only if the map $A \rightarrow B$ is injective. Therefore, f is a τ -local monomorphism if and only if f_x is injective for all $x \in T$.

3. Isomorphisms $a(f)$ is an isomorphism if and only if it is both a monomorphism and an epimorphism. Combining the results above, this holds if and only if f_x is both injective and surjective (bijective) for all $x \in T$. $\square$

5.5. Locally Ringed Space

The classical definition of a scheme involves a pair $(X, \mathcal{O}_X)$ where X is a topological space and $\mathcal{O}_X$ is a sheaf of rings whose stalks are local rings. We collect the vocabulary — ringed spaces, locally ringed spaces, their morphisms — that bridges our functorial approach to this classical picture in the next chapter.

Definition 5.5.1 (Ringed Space). Let k be a ring.

- A k -ringed space $(X, \mathcal{O}_X)$ is a topological space X together with a sheaf of k -algebras $\mathcal{O}_X$ on X .
- A morphism of k -ringed spaces $(f, \varphi): (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a continuous map $f: X \rightarrow Y$ together with a map of sheaves of k -algebras $\varphi: \mathcal{O}_Y \rightarrow f_*(\mathcal{O}_X)$.

We denote by RingSpace_k the category of k -ringed spaces. When $k = \mathbb{Z}$, we write $\text{RingSpace} = \text{RingSpace}_{\mathbb{Z}}$ for the category of ringed spaces.

By the $f^* \dashv f_*$ adjunction, the datum $\varphi: \mathcal{O}_Y \rightarrow f_*(\mathcal{O}_X)$ is equivalent to a map $f^*(\mathcal{O}_Y) \rightarrow \mathcal{O}_X$ of sheaves of k -algebras.

Definition 5.5.2 (Local Map). A ring map $\varphi: A \rightarrow B$ is *local* if it detects units, i.e., if $A^\times = \varphi^{-1}(B^\times)$.

Remark 5.5.3. Let A and B be local rings with maximal ideals $\mathfrak{m}_A$ and $\mathfrak{m}_B$. For a ring map $\varphi: A \rightarrow B$, the following are equivalent:

- (i) φ is local;
- (ii) $\mathfrak{m}_A = \varphi^{-1}(\mathfrak{m}_B)$.
- (iii) $\varphi(\mathfrak{m}_A) \subset \mathfrak{m}_B$.

We now introduce locally ringed spaces.

Definition 5.5.4 (Locally Ringed Space). Let k be a ring.

- A *locally k -ringed space* is a k -ringed space $(X, \mathcal{O}_X)$ such that, for each $x \in X$, the stalk $\mathcal{O}_{X,x}$ is a local ring.
- If $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ are locally ringed k -spaces, a *morphism of locally k -ringed spaces* $(f, \varphi): (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a morphism of k -ringed spaces such that, for each $x \in X$, the map $\varphi_x: \mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$ is a local map of local rings (i.e., $\varphi_x(\mathfrak{m}_{f(x)}) \subset \mathfrak{m}_x$).

We denote by LocRingSpace_k the category of locally k -ringed spaces, which is a (non-full!) subcategory of RingSpace_k . When $k = \mathbb{Z}$, we write $\text{LocRingSpace} = \text{LocRingSpace}_{\mathbb{Z}}$ for the category of *locally ringed spaces*.

Remark 5.5.5 (Another description of Locally Ringed Spaces). Let $(X, \mathcal{O}_X)$ be a ringed space. For any open subset $U \subset X$ and any section $s \in \mathcal{O}_X(U)$, we denote by U_s the locus where the germ of s is invertible:

$$U_s := \{x \in U \mid s_x \in \mathcal{O}_{X,x}^\times\}.$$

Since the property of being a unit is open in the stalk topology, U_s is an open subset of U .

(i) **Objects:** A ringed space $(X, \mathcal{O}_X)$ is a locally ringed space if and only if:

- $X_0 = \emptyset$.

- For all open subsets $U \subset X$ and sections $s, t \in \mathcal{O}_X(U)$:

$$U_{s+t} \subseteq U_s \cup U_t.$$

(This is the geometric equivalent of the local ring axiom: if a sum is a unit, at least one summand must be a unit).

- (ii) **Morphisms:** A morphism (f, φ) in RingSpace is a morphism in LocRingSpace if and only if for every open set $V \subset Y$ and every section $t \in \mathcal{O}_Y(V)$:

$$f^{-1}(V_t) = (f^{-1}(V))_{\varphi(t)}.$$

(Geometrically: the pullback of the unit locus is the unit locus of the pullback section).

Locally ringed spaces unify many flavors of geometric object. We isolate the relevant pattern by naming spaces built from local models.

Notation 5.5.6. Let k be a ring and let $\mathcal{M}$ be a collection (set) of locally k -ringed spaces, which we call *local models*. We denote by $\text{LocRingSpace}(\mathcal{M})$ the full subcategory of LocRingSpace_k consisting of objects $(X, \mathcal{O}_X)$ with the following property: every point $x \in X$ admits an open neighborhood U such that the restricted locally ringed space $(U, \mathcal{O}_X|_U)$ is isomorphic to an object in $\mathcal{M}$.

Example 5.5.7 (Manifolds as Locally Ringed Spaces). (i) **Real Manifolds (C^r , Smooth, Analytic).** Let $r \in \mathbb{N} \cup \{\infty, \omega\}$. For any open subset $U \subset \mathbb{R}^n$, denote by C_U^r the sheaf of $\mathbb{R}$ -valued functions of class C^r on U (continuous for $r = 0$, smooth for $r = \infty$, real-analytic for $r = \omega$).

Let $\mathcal{D}isk_{C^r}$ be the collection of locally $\mathbb{R}$ -ringed spaces consisting of open unit disks in Euclidean spaces:

$$\mathcal{D}isk_{C^r} = \{(D^n, C_{D^n}^r) \mid D^n \subset \mathbb{R}^n \text{ is the open unit disk, } n \in \mathbb{N}\}.$$

The category of C^r -manifolds can be identified with the subcategory of locally ringed spaces locally isomorphic to these models. Specifically, we have an equivalence:

$$\{C^r\text{-manifolds}\} \simeq \{X \in \text{LocRingSpace}(\mathcal{D}isk_{C^r}) \mid X \text{ is Hausdorff and second-countable}\}.$$

In particular:

- $r = 0$ yields *topological manifolds*;
- $r = \infty$ yields *smooth manifolds*;
- $r = \omega$ yields *analytic manifolds*.

See [Wed16] for a sheaf-theoretic introduction to the theory of manifolds.

- (ii) **Complex Manifolds.** For an open subset $U \subset \mathbb{C}^n$, denote by $\mathcal{H}_U$ (or simply $\mathcal{O}_U$) the sheaf of holomorphic functions on U . Let $\mathcal{D}isk_{\mathbb{C}}$ be the collection of locally $\mathbb{C}$ -ringed spaces $(D^n, \mathcal{H}_{D^n})$ where D^n is the open unit disk in $\mathbb{C}^n$ for some $n \in \mathbb{N}$. Then there is an equivalence of categories:

$$\{\text{complex manifolds}\} \simeq \{X \in \text{LocRingSpace}(\mathcal{D}isk_{\mathbb{C}}) \mid X \text{ is Hausdorff and second-countable}\}.$$

- (iii) **Generalizations.** More generally, many categories of manifolds—such as manifolds with boundary, manifolds with corners—are equivalent to categories of the form $\text{LocRingSpace}_{\mathbb{R}/\mathbb{C}}(\mathcal{M})$ (usually with additional Hausdorff/second-countable constraints).

Remark 5.5.8 (Colimits and Limits of Ringed Spaces). The category RingSpace_k admits both arbitrary colimits and limits. Let $(X_i, \mathcal{O}_{X_i})_{i \in I}$ be a diagram of k -ringed spaces. Let $X = \text{colim}_{i \in I} X_i$ and $Y = \lim_{i \in I} X_i$ be the colimit and limit in Top , with canonical maps $\iota_j: X_j \rightarrow X$ and $\pi_j: Y \rightarrow X_j$. Then the colimit and limit in RingSpace_k are given by:

$$\begin{aligned} \text{colim}_{i \in I} (X_i, \mathcal{O}_{X_i}) &= \left(X, \lim_{i \in I} (\iota_i)_* \mathcal{O}_{X_i} \right), \\ \lim_{i \in I} (X_i, \mathcal{O}_{X_i}) &= \left(Y, \text{colim}_{i \in I} (\pi_i)^* \mathcal{O}_{X_i} \right). \end{aligned}$$

Note the reversal of limits and colimits for the structure sheaf: the structure sheaf of the colimit space is the limit of the pushforward sheaves, while the structure sheaf of the limit space is the colimit of the pullback sheaves (in the category of sheaves on Y).

We now discuss colimits in LocRingSpace_k .

Proposition 5.5.9. *The category LocRingSpace_k admits colimits, and the inclusion functor $\text{LocRingSpace}_k \hookrightarrow \text{RingSpace}_k$ preserves colimits.*

Proof. Since LocRingSpace_k is a category with all small colimits if it admits coproducts and coequalizers, it suffices to show that the coproducts and coequalizers formed in RingSpace_k actually lie in LocRingSpace_k and satisfy the universal property.

- (i) **Coproducts.** Let $\{(X_i, \mathcal{O}_{X_i})\}_{i \in I}$ be a family of locally k -ringed spaces. The coproduct in RingSpace_k is given by

$$(X, \mathcal{O}_X) = \coprod_{i \in I}^{\text{RS}} (X_i, \mathcal{O}_{X_i}) = \left(\coprod_{i \in I} X_i, \prod_{i \in I} \alpha_{i,*}(\mathcal{O}_{X_i}) \right),$$

where X is the disjoint union topological space, and $\alpha_i: X_i \hookrightarrow X$ are the canonical inclusions. Explicitly, for any open set $U \subset X$, let $U_i = U \cap X_i$. The structure sheaf is:

$$\mathcal{O}_X(U) = \prod_{i \in I} \mathcal{O}_{X_i}(U_i).$$

Stalks: Let $x \in X$. Then $x \in X_j$ for a unique index j . The stalk is:

$$\mathcal{O}_{X,x} \simeq \mathcal{O}_{X_j,x}.$$

Since $(X_j, \mathcal{O}_{X_j})$ is a locally ringed space, $\mathcal{O}_{X_j,x}$ is a local ring. Thus $(X, \mathcal{O}_X) \in \text{LocRingSpace}_k$.

Universal Property: Consider maps $(f_i, \varphi_i): (X_i, \mathcal{O}_{X_i}) \rightarrow (Y, \mathcal{O}_Y)$ in LocRingSpace_k . The unique induced map $(f, \varphi): (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is defined such that its restriction to each component is (f_i, φ_i) . Since the map on stalks φ_x is simply $(\varphi_i)_x$ (where $x \in X_i$), and $(\varphi_i)_x$ is a local map, φ_x is local.

- (ii) **Coequalizers.** Consider a pair of morphisms in LocRingSpace_k :

$$(Z, \mathcal{O}_Z) \xrightleftharpoons[(g, \psi)]{(f, \varphi)} (Y, \mathcal{O}_Y).$$

Let $h: Y \rightarrow X$ be the coequalizer of f, g in Top (so X has the quotient topology). The structure sheaf $\mathcal{O}_X$ is defined as the equalizer in the category of sheaves of k -algebras on X :

$$\mathcal{O}_X \xrightarrow{\chi} h_* \mathcal{O}_Y \xrightleftharpoons[h_*(\psi)]{h_*(\varphi)} h_* f_* \mathcal{O}_Z = h_* g_* \mathcal{O}_Z.$$

Explicitly, for an open set $U \subset X$ (letting $V = h^{-1}(U)$ and $W = f^{-1}(V) = g^{-1}(V)$):

$$\mathcal{O}_X(U) = \{s \in \mathcal{O}_Y(V) \mid \varphi_V(s) = \psi_V(s) \text{ in } \mathcal{O}_Z(W)\}.$$

Verification of Locally Ringed Space: We use Remark 5.5.5. Let $r \in \mathcal{O}_X(U)$. By the definition above, r corresponds to a section $s \in \mathcal{O}_Y(V)$ such that $\varphi(s) = \psi(s)$.

Claim 5.5.10. $h^{-1}(U_r) = V_s$.

Proof of Claim. Invertibility is constant along fibres. Suppose $s_y \in \mathcal{O}_{Y,y}^\times$ for some y with $h(y) = x$, and let y' be any other point of the fibre $h^{-1}(x)$. Since h is the coequalizer of f and g , the points y and y' are linked by a finite chain of relations $f(z) \sim g(z)$ in Y ; along each such relation the identity $\varphi_z(s) = \psi_z(s)$ together with the locality of φ_z, ψ_z (they reflect units) carries invertibility from one point to the next. Hence $s_{y'} \in \mathcal{O}_{Y,y'}^\times$ for every $y' \in h^{-1}(x)$. Equivalently, the open set $V_s = \{y' \in V : s_{y'} \in \mathcal{O}_{Y,y'}^\times\}$ is *saturated*: $V_s = h^{-1}(h(V_s))$.

$h^{-1}(U_r) \subseteq V_s$. If $h(y) = x \in U_r$, then $r_x \in \mathcal{O}_{X,x}^\times$, and its image s_y under the stalk map $\mathcal{O}_{X,x} \rightarrow \mathcal{O}_{Y,y}$ is a unit; so $y \in V_s$.

$V_s \subseteq h^{-1}(U_r)$. As V_s is open and saturated and X carries the quotient topology, $U' := h(V_s)$ is open with $h^{-1}(U') = V_s$. The restriction $s|_{V_s} \in \mathcal{O}_Y(V_s)^\times$ has an inverse $t \in \mathcal{O}_Y(V_s)$, and since $\varphi(s) = \psi(s)$ are units on $f^{-1}(V_s) = g^{-1}(V_s)$, their inverses agree: $\varphi(t) = \psi(t)$. Thus t lies in $\mathcal{O}_X(U') = \{s' \in \mathcal{O}_Y(V_s) \mid \varphi(s') = \psi(s')\}$ and inverts $r|_{U'}$, so every $x \in U'$ lies in U_r ; hence $V_s = h^{-1}(U') \subseteq h^{-1}(U_r)$.

Combining the two inclusions, $h^{-1}(U_r) = V_s$. In particular the stalk map $\mathcal{O}_{X,x} \rightarrow \mathcal{O}_{Y,y}$ reflects units. $\square$

Since Y is a locally ringed space, V_s is open. Because X has the quotient topology, $h^{-1}(U_r) = V_s$ being open implies U_r is open in X . Furthermore, for any $r, r' \in \mathcal{O}_X(U)$ corresponding to $s, s' \in \mathcal{O}_Y(V)$:

$$h^{-1}(U_{r+r'}) = V_{s+s'} \subseteq V_s \cup V_{s'} = h^{-1}(U_r) \cup h^{-1}(U_{r'}) = h^{-1}(U_r \cup U_{r'}).$$

Since h is surjective, taking images yields $U_{r+r'} \subseteq U_r \cup U_{r'}$. Also, $X_0 = \emptyset$ follows from $Y_0 = \emptyset$. Thus $(X, \mathcal{O}_X)$ is a locally ringed space.

Universal Property: Let $(k, \lambda): (X, \mathcal{O}_X) \rightarrow (T, \mathcal{O}_T)$ be the unique map in RingSpace_k induced by a compatible map from Y . For any $x \in X$, the map on stalks $\lambda_x: \mathcal{O}_{T,k(x)} \rightarrow \mathcal{O}_{X,x}$ fits into a commutative triangle with the local map from Y . Since the map $\mathcal{O}_{X,x} \rightarrow \mathcal{O}_{Y,y}$ reflects units (as shown in the Claim), λ_x must be local. $\square$

The category LocRingSpace_k also admits limits, but this is more subtle as the inclusion $\text{LocRingSpace}_k \hookrightarrow \text{RingSpace}_k$ does not preserve them.

5.6. The Comparison Lemma

In this section we explain how to compare the sheaf categories of two related sites.

Definition 5.6.1 (Dense Subcategory). Let (C, τ) be a site. A *dense subcategory* of (C, τ) is a full subcategory $\mathcal{D} \subset C$ such that every $X \in C$ admits a τ -cover $(U_i \rightarrow X)_{i \in I}$ with $U_i \in \mathcal{D}$.

Proposition 5.6.2 (The “Comparison Lemma”). Let (C, τ) be a site and $\mathcal{D} \subset C$ a dense subcategory. Let $\tau|_{\mathcal{D}}$ be the induced topology on $\mathcal{D}$. Then the restriction functor $u^*: \text{Shv}_\tau(C) \rightarrow \text{Shv}_{\tau|_{\mathcal{D}}}(\mathcal{D})$ is an equivalence of categories.

Proof. Let $u: \mathcal{D} \hookrightarrow \mathcal{C}$ be the inclusion. Restriction of presheaves has right adjoint

$$u_*^{\text{pre}} G := \text{Ran}_{u^{\text{op}}} G, \quad (u_*^{\text{pre}} G)(X) = \lim_{(D, u(D) \rightarrow X) \in (u \downarrow X)^{\text{op}}} G(D).$$

The opposite on the comma category is forced by the fact that G is contravariant. In particular, when $X = u(D)$ the full subcategory indexed by arrows in $\mathcal{D}$ is $(\mathcal{D}_{/D})^{\text{op}}$; its initial object is id_D . Consequently

$$(u^* u_*^{\text{pre}} G)(D) \xrightarrow{\sim} G(D),$$

so the presheaf counit is an isomorphism. This is the fully faithful right-Kan-extension calculation; using a limit over $\mathcal{D}_{/D}$ without the opposite would reverse the variance.

We next verify that this adjunction restricts to sheaves. Restriction preserves sheaves by definition of the induced topology. Let G be a sheaf on $\mathcal{D}$ and let $\mathcal{R} \subset y(X)$ be a τ -covering sieve. By adjunction, descent for $u_*^{\text{pre}} G$ along $\mathcal{R}$ is equivalent to the assertion that

$$\text{Hom}(u^* y(X), G) \longrightarrow \text{Hom}(u^* \mathcal{R}, G)$$

is an isomorphism. Pulling $u^* \mathcal{R} \rightarrow u^* y(X)$ back along a point $y(D) \rightarrow u^* y(X)$ gives the restriction to $\mathcal{D}$ of the covering sieve on D obtained by pulling $\mathcal{R}$ back along $u(D) \rightarrow X$. Density refines that covering sieve by objects of $\mathcal{D}$, so this representable pullback is covering for $\tau|_{\mathcal{D}}$. Writing $u^* y(X)$ as its Yoneda colimit and applying the sheaf condition for G to these pullbacks proves the displayed isomorphism. Hence $u_*^{\text{pre}} G$ is a τ -sheaf; denote the restricted functor by u_* . This is the standard dense-site calculation, also recorded in [The25, Tag 03A0].

The counit $u^* u_* G \rightarrow G$ is already an isomorphism by the pointwise calculation above, so u_* is fully faithful. Conversely, for a sheaf F on $\mathcal{C}$, the unit

$$F \longrightarrow u_* u^* F$$

is an isomorphism after restriction to every $D \in \mathcal{D}$. Every object of $\mathcal{C}$ has a covering by such D 's, and both sides are sheaves; the sheaf condition therefore makes the unit an isomorphism on every object. Thus the unit and counit are isomorphisms, and $u^* \dashv u_*$ is an equivalence. $\square$

Remark 5.6.3. For the higher categorical version (specifically in the context of ∞ -sites and ∞ -sheaves), we refer the reader to [Hoy14, Lemma C.3]. The proof presented there follows the same strategy as ours, relying on the right Kan extension along the embedding of the dense subcategory.

We collect a few applications of the Comparison Lemma.

Example 5.6.4 (Comparison Lemma for Presheaves). Let $\mathcal{D} \subset \mathcal{C}$ be a full subcategory. Then $\mathcal{D}$ is dense with respect to the indiscrete topology if and only if every object of $\mathcal{C}$ is a retract of an object of $\mathcal{D}$. In this case, the Comparison Lemma says that $\text{PShv}(\mathcal{C}) \xrightarrow{\sim} \text{PShv}(\mathcal{D})$.

Example 5.6.5 (Basis of a Topological Space). Let T be a topological space and let $\mathcal{B} \subset \text{Open}(T)$ be a basis of its topology. By definition, $\mathcal{B}$ is dense with respect to canonical topology on $\text{Open}(T)$, and the restriction of the canonical topology to $\mathcal{B}$ consists of sieves generated by open coverings in $\mathcal{B}$. Hence, the Comparison Lemma provides an equivalence $\text{Shv}(T) \xrightarrow{\sim} \text{Shv}(\mathcal{B})$.

Example 5.6.6 (Sheaves on Manifolds). Let Man_n be the category of n -dimensional topological manifolds, equipped with the standard topology (induced by the pretopology of open coverings). Every n -manifold admits an open covering by manifolds homeomorphic to $\mathbb{R}^n$, so the full subcategory $\{\mathbb{R}^n\} \subset \text{Man}_n$ is dense, and the Comparison Lemma yields an equivalence $\text{Shv}(\text{Man}_n) \simeq \text{Shv}(\{\mathbb{R}^n\})$.

Example 5.6.7 (Sheaves on $\text{Spec}(R)$). Let R be a ring. Consider the following two full subcategories of the locale $\text{Open}(\text{Spec}(R)) \simeq \text{Open}(\text{Prim}(R)) \simeq \text{Rad}_R$:

- (i) $\text{Open}^{\text{pr}}(\text{Spec}(R))$ consists of open subfunctors of the form $D(f)$ with $f \in R$ (called *principal* open subschemes of $\text{Spec}(R)$);
- (ii) $\text{Open}^{\text{aff}}(\text{Spec}(R))$ consists of the *affine* open subfunctors of $\text{Spec}(R)$.

We have $\text{Open}^{\text{pr}}(\text{Spec}(R)) \subset \text{Open}^{\text{aff}}(\text{Spec}(R))$, and both subcategories are dense in $\text{Open}(\text{Spec}(R))$, since $D(I) = \bigvee_{f \in I} D(f)$. By the Comparison Lemma, the three categories of sheaves are equivalent. In particular, every sheaf on $\text{Open}^{\text{pr}}(\text{Spec}(R))$ *extends uniquely* to a sheaf on $\text{Open}(\text{Spec}(R))$.

Example 5.6.8 (Sheaves on Presheaves). Let C be a small category, which we view as a full subcategory of $\text{PShv}(C)$ via the Yoneda embedding. Any topology τ on C extends canonically to a topology $\hat{\tau}$ on $\text{PShv}(C)$ as follows: A sieve $\mathcal{R}$ on presheaf F is a $\hat{\tau}$ -covering if, for any $y(X) \rightarrow F$, the sieve of all $Y \rightarrow X$ such that $y(Y) \rightarrow y(X) \rightarrow F$ is in $\mathcal{R}$ is τ -covering.

Since every object in $\text{PShv}(C)$ is a quotient of a coproduct of representable presheaves, the Comparison Lemma applies to the Yoneda embedding $C \hookrightarrow \text{PShv}(C)$, so that

$$\text{Shv}_{\tau}(C) \simeq \text{Shv}_{\hat{\tau}}(\text{PShv}(C)).$$

Applying this to the category of affine schemes $C = \text{Aff}$ equipped with the Zariski topology τ_{Zar} , the Comparison Lemma yields an equivalence of categories:

$$\text{Shv}_{\tau_{\text{Zar}}}^{\text{Cat}}(\text{Aff}) \simeq \text{Shv}_{\hat{\tau}_{\text{Zar}}}^{\text{Cat}}(\text{Fun}(\text{CAlg}, \text{Set})).$$

This equivalence says that, at the level of categorical sheaves, extending from affine schemes to arbitrary algebraic functors and enlarging the topology to $\hat{\tau}_{\text{Zar}}$ leaves the category of sheaves unchanged.

Remark 5.6.9 (Comparison Lemma for Stacks). The comparison lemma extends to category-valued sheaves: if $\mathcal{B} \subset C$ is a basis closed under finite limits with the induced topology $\tau|_{\mathcal{B}}$, then restriction induces an equivalence

$$\text{Shv}_{\tau}^{\text{Cat}}(C) \simeq \text{Shv}_{\tau|_{\mathcal{B}}}^{\text{Cat}}(\mathcal{B}).$$

Here Shv^{Cat} means pseudofunctors with descent computed by the pseudolimits of Definition 3.1.9; all comparison 2-cells are natural isomorphisms. With that convention, the right-Kan-extension proof above carries over verbatim, with ordinary limits replaced by pseudolimits.

Looking Ahead

With Grothendieck topologies, sheaves, sheafification, and the Comparison Lemma in hand, we now have the full language of descent. Along the way we set up the Zariski, fppf, and fpqc topologies on Aff , and verified that a Zariski sheaf on affine schemes extends canonically to one on all algebraic functors.

This is exactly the right tool to define schemes. In the next chapter, a scheme is simply a Zariski sheaf $X \in \text{Shv}_{\text{Zar}}(\text{Aff})$ admitting an open cover by affines — all the categorical scaffolding developed here (sheafification, the Comparison Lemma, covering sieves) feeds directly into the theory of Sch .

Chapter 6.

Schemes

Everything in the previous chapters — affine schemes, projective schemes, quasi-coherent modules, Grothendieck topologies — has been building towards this definition. A *scheme* is an algebraic functor that (a) satisfies Zariski descent and (b) is locally affine. In one formula:

$$\text{scheme} = \text{Zariski sheaf} + \text{affine open cover}.$$

Condition (a) says the scheme can be glued from local data; condition (b) says it admits local coordinates, i.e. is built up out of familiar commutative-algebra pieces $\text{Spec}(R)$.

Schemes generalize both affine schemes (which are always schemes) and projective ones (which we saw are coverable by the affines $D(f) \simeq \text{Spec}(A_{(f)})$). But the category of schemes is richer still: it is closed under gluing along open subfunctors, so it also contains non-affine global objects and constructions that do not come equipped with a single global Proj presentation.

This chapter develops the formal theory. We (i) verify that schemes form a full subcategory $\text{Sch} \subset \text{Shv}_{\text{Zar}}(\text{Aff})$ closed under immersions, finite limits, cofiltered affine limits, and coproducts; (ii) construct schemes by gluing along locally trivial equivalence relations; (iii) introduce *separatedness* and the quasi-compact / quasi-separated (qcqs) conditions that make finiteness work globally; and (iv) relate the functorial picture to the classical definition of a scheme as a locally ringed space.

6.1. The Category of Schemes

We first verify that schemes form a well-behaved subcategory: closed under immersions, with good properties inherited from the ambient Zariski sheaves. The principle that recurs throughout is that everything about schemes reduces, locally on an affine cover, to commutative algebra.

Definition 6.1.1 (Scheme). A *scheme* is a functor $X: \text{CAlg} \rightarrow \text{Set}$ satisfying:

- (i) X is a sheaf with respect to the Zariski topology;
- (ii) X admits an open cover by affine schemes.

We write $\text{Sch} \subset \text{Shv}_{\text{Zar}}(\text{Aff})$ for the full subcategory of schemes. For a scheme S , the objects of the slice $\text{Sch}_S := \text{Sch}_{/S}$ are called *S-schemes*; when $S = \text{Spec}(k)$ for a ring k , we abbreviate to *k-schemes*.

Since Sch is only a full subcategory of $\text{Shv}_{\text{Zar}}(\text{Aff})$, properties of the latter need not survive the restriction. We now investigate which constructions and limits do remain in Sch .

Proposition 6.1.2 (Immersions of Schemes). *Let X be a scheme. If $Y \hookrightarrow X$ is an open immersion, a closed immersion, or an immersion, then Y is also a scheme.*

Proof. Choose an affine open cover $(U_i \hookrightarrow X)_{i \in I}$ with each U_i an affine scheme. By stability of open covers under base change, $(U_i \cap Y \hookrightarrow Y)_{i \in I}$ is an open cover of Y . So it suffices to check, in each of the three cases, that (a) every $U_i \cap Y$ admits an affine open cover, and (b) Y is a Zariski sheaf.

The open case. For each i , the open subfunctor $U_i \cap Y \subset U_i$ corresponds to a quasi-coherent radical ideal J_i , so $U_i \cap Y = D(J_i)$. The family $(D(f) \hookrightarrow D(J_i))_{f \in J_i}$ is an affine open cover (each $D(f) \simeq \operatorname{Spec}((R_i)_f)$), and refining the original cover gives an affine open cover $(D(f) \hookrightarrow Y)_{i \in I, f \in J_i}$ of Y .

For Zariski descent of Y : fix a ring R and a Zariski cover $(R \rightarrow R_{f_k})_{k \in K}$. Consider the diagram of subfunctor inclusions

$$\begin{array}{ccccc} Y(R) & \longrightarrow & \prod_k Y(R_{f_k}) & \rightrightarrows & \prod_{k,l} Y(R_{f_k f_l}) \\ \downarrow & & \downarrow & & \downarrow \\ X(R) & \longrightarrow & \prod_k X(R_{f_k}) & \rightrightarrows & \prod_{k,l} X(R_{f_k f_l}). \end{array}$$

The bottom row is an equalizer because X is a Zariski sheaf, so the top row is at least a sub-equalizer; we must verify that any compatible family lying in X in fact lies in Y . Take $x \in X(R)$ in the equalizer of the bottom row, and assume $x_k := x|_{R_{f_k}} \in Y(R_{f_k})$ for every k . Pulling back $Y \hookrightarrow X$ along $x: \operatorname{Spec}(R) \rightarrow X$ yields an open subfunctor of $\operatorname{Spec}(R)$, i.e., an open subscheme $D(I) \subset \operatorname{Spec}(R)$ for some radical ideal $I \subset R$. The hypothesis $x_k \in Y(R_{f_k})$ says $D(I \cdot R_{f_k}) = \operatorname{Spec}(R_{f_k})$, equivalently $I \cdot R_{f_k} = R_{f_k}$ for each k . Since (f_k) generates the unit ideal in R , this forces $I = R$, so $D(I) = \operatorname{Spec}(R)$, which means $x \in Y(R)$.

The closed case. Each $U_i \cap Y \subset U_i$ is a closed subfunctor, hence of the form $V(J_i) = \operatorname{Spec}(R_i/J_i)$ for some ideal $J_i \subset R_i$. In particular $U_i \cap Y$ is itself affine, providing an affine open cover of Y .

For Zariski descent, the same diagram-chase as above applies with V in place of D : pulling back $Y \hookrightarrow X$ along an R -point x of X gives a closed subscheme $V(I) \subset \operatorname{Spec}(R)$. The condition $V(I \cdot R_{f_k}) = \operatorname{Spec}(R_{f_k})$ means $I \cdot R_{f_k} = 0$ for every k . Since vanishing of an ideal is detected by a Zariski cover (modules satisfy Zariski descent), $I = 0$, so $V(I) = \operatorname{Spec}(R)$ and x factors through Y .

The immersion case. By Definition 1.2.51, an immersion $Y \hookrightarrow X$ factors as $Y \hookrightarrow U \hookrightarrow X$ where the first map is closed and the second is open. The open case shows U is a scheme, and applying the closed case to $Y \hookrightarrow U$ shows Y is a scheme. $\square$

Example 6.1.3 (Examples of Schemes). (i) For any ring A , $\operatorname{Spec}(A)$ is a scheme. By Proposition 6.1.2, every quasi-affine scheme is also a scheme.

(ii) For an $\mathbb{N}$ -graded ring A , $\operatorname{Proj}(A)$ is a scheme, with affine cover $(D(f) \simeq \operatorname{Spec}(A_{(f)}))_{f \in A_+}$. In particular every (quasi-)projective k -scheme is a scheme.

(iii) The affine line with two origins is a scheme, covered by two copies of $\mathbb{A}^1$.

(iv) For a scheme X and an affine, quasi-affine, projective, or quasi-projective morphism $Y \rightarrow X$, the source Y is again a scheme.

(v) For a ring k , a k -module M , and $n \in \mathbb{N}$, the Grassmannian $\operatorname{Gr}_n(M)$ is a scheme.

We turn to gluing schemes — a categorical analogue of gluing topological spaces. We first generalize the notion of equivalence relation to an arbitrary category.

Definition 6.1.4 (Equivalence Relation). Let $\mathcal{C}$ be a category. An *equivalence relation* in $\mathcal{C}$ is a pair of morphisms $s, t: R \rightrightarrows X$ such that, for every $Y \in \mathcal{C}$, the induced map

$$(s_*, t_*): \operatorname{Hom}_{\mathcal{C}}(Y, R) \rightarrow \operatorname{Hom}_{\mathcal{C}}(Y, X) \times \operatorname{Hom}_{\mathcal{C}}(Y, X)$$

is injective with image an equivalence relation on $\operatorname{Hom}_{\mathcal{C}}(Y, X)$.

If the coequalizer of s, t exists, it is called the *quotient* of X by R , denoted X/R .

Remark 6.1.5 (Higher Categorical Analogue: Groupoid Objects). The higher-categorical analogue of an equivalence relation is a *groupoid object*, namely a simplicial object $X_\bullet$ in $\mathcal{C}$

$$\cdots \rightrightarrows X_2 \rightrightarrows X_1 \rightrightarrows X_0$$

satisfying the *groupoid (Kan) condition*: for any $[n] \in \Delta$ and any partition (not necessarily order-preserving) $[n] \simeq \{i_0, \dots, i_k\} \cup \{i_k, \dots, i_n\}$, the diagram

$$\begin{array}{ccc} X_n & \xrightarrow{(i_0, \dots, i_k)^*} & X_k \\ (i_k, \dots, i_n)^* \downarrow & \lrcorner & \downarrow i_k^* \\ X_{n-k} & \xrightarrow{i_k^*} & X_0 \end{array}$$

is a pullback square.

This condition encodes both core properties of groupoids. For order-preserving partitions (such as $[2] = \{0, 1\} \cup \{1, 2\}$) it gives the *Segal condition* $X_2 \simeq X_1 \times_{X_0} X_1$, ensuring composition. For non-order-preserving partitions it forces invertibility of morphisms.

Specialized to $\mathcal{C} = \mathbf{Set}$: a groupoid object is a classical groupoid, and the image of $(s, t): \mathcal{G}_1 \rightarrow \mathcal{G}_0 \times \mathcal{G}_0$ defines an equivalence relation on $\mathcal{G}_0$ (reflexivity from identities, transitivity from composition, symmetry from inversion). Conversely, an equivalence relation $R \subseteq X \times X$ corresponds bijectively to a groupoid $\mathcal{G}$ in $\mathbf{Set}$ with $\mathcal{G}_0 = X$ and $(s, t): \mathcal{G}_1 \hookrightarrow \mathcal{G}_0 \times \mathcal{G}_0$ injective; under this correspondence, the colimit $|\mathcal{G}|$ recovers the quotient X/R .

Remark 6.1.6 (Effective Epimorphisms). Any morphism $f: X \rightarrow Y$ in a category with pullbacks gives rise to an equivalence relation $X \times_Y X \rightrightarrows X$ (in the higher setting, a groupoid object via the Čech nerve). We call f an *effective epimorphism* if the diagram

$$X \times_Y X \rightrightarrows X \xrightarrow{f} Y$$

is a coequalizer.

In a 1-topos every epimorphism is effective; this fails in general, e.g. $\mathbb{Z} \hookrightarrow \mathbb{Q}$ in $\mathbf{CAlg}$ is an epimorphism but not an effective one.

We isolate the relevant class of equivalence relations for gluing schemes.

Definition 6.1.7 (Locally Trivial Relation). An equivalence relation $s, t: R \rightrightarrows X$ in $\mathbf{Shv}_{\mathbf{Zar}}(\mathbf{Aff})$ is *locally trivial* if there exists an open cover $(U_i \hookrightarrow X)_{i \in I}$ such that, for every i , the composition $s^{-1}(U_i) \hookrightarrow R \xrightarrow{t} X$ is an open immersion.

Concretely, gluing two schemes U_1 and U_2 along $U_{12} = U_{21}$ is captured by the following data.

$$\begin{array}{ccc} U_{12} = U_{21} & \xhookrightarrow{\text{open}} & U_1 \\ \text{open} \downarrow & & \downarrow \\ U_2 & \xrightarrow{\quad} & X/R \end{array} \rightsquigarrow \begin{array}{c} \text{Diagram of two overlapping ellipses } U_1 \text{ (blue) and } U_2 \text{ (green) with intersection } U_{12} \text{ (red).} \\ \text{The quotient } X/R \text{ is indicated below.} \end{array}$$

Here $X = U_1 \sqcup U_2$ and $R = \coprod_{i,j \in \{1,2\}} U_{ij} \rightrightarrows X$, with the convention $U_{ii} = U_i$. If both $U_{12} \hookrightarrow U_1$ and $U_{21} \hookrightarrow U_2$ are open immersions, then R is locally trivial. This motivates:

Definition 6.1.8 (Gluing Datum). A *gluing datum* in $\mathbf{Shv}_{\mathbf{Zar}}(\mathbf{Aff})$ consists of families $(U_i)_{i \in I}$ and $(U_{ij})_{i,j \in I}$ together with morphisms $U_{ij} \rightarrow U_i \times U_j$ satisfying:

- the projections $U_{ij} \hookrightarrow U_i$ and $U_{ij} \hookrightarrow U_j$ are open immersions;
- the diagram $\coprod_{i,j \in I} U_{ij} \rightrightarrows \coprod_{i \in I} U_i$ is an equivalence relation in $\text{Shv}_{\text{Zar}}(\text{Aff})$.

Remark 6.1.9 (Open Covers and Gluing Data). The notions of open cover and gluing datum are interconvertible:

- If X is the quotient of the equivalence relation given by gluing data $((U_i), (U_{ij}))$, then $(U_i \rightarrow X)_{i \in I}$ is an open cover with $U_{ij} \xrightarrow{\sim} U_i \times_X U_j$.
- Conversely, given a Zariski sheaf X and an open cover $(U_i \rightarrow X)_{i \in I}$, the families (U_i) and $(U_i \cap U_j)$ form a gluing datum.

We now identify which constructions are preserved by the inclusion $\text{Sch} \subset \text{Shv}_{\text{Zar}}(\text{Aff})$.

Theorem 6.1.10 (Limits and Colimits of Schemes). *The full subcategory $\text{Sch} \subset \text{Shv}_{\text{Zar}}(\text{Aff})$ is closed under:*

- (i) *finite limits;*
- (ii) *cofiltered diagrams of affine morphisms;*
- (iii) *arbitrary coproducts;*
- (iv) *quotients by locally trivial equivalence relations.*

We first establish a lemma.

Lemma 6.1.11 (Coproducts Are Clopen). *Let $(X_i)_{i \in I}$ be a family of Zariski sheaves with coproduct $X = \coprod_{i \in I} X_i$. Then each canonical map $X_i \hookrightarrow X$ is both an open and a closed immersion. In particular $(X_i \hookrightarrow X)_{i \in I}$ is an open cover of X .*

Proof. Decompose $X_i \hookrightarrow X$ as $X_i \hookrightarrow X_i \sqcup (\coprod_{j \neq i} X_j) = X$. Everything reduces to showing that for any Zariski sheaves X, Y , the inclusion $X \hookrightarrow X \sqcup Y$ is clopen.

Form the pullback in the presheaf category

$$\begin{array}{ccc} X \sqcup^{\text{pre}} \emptyset_Y & \longrightarrow & \mathbb{G}_m \\ \downarrow & \lrcorner & \downarrow \\ X \sqcup^{\text{pre}} Y & \xrightarrow{(1,0)} & \mathbb{A}^1 \end{array}$$

where $\sqcup^{\text{pre}}$ is the presheaf coproduct and $(1, 0)$ is the algebraic-functor map taking constant value 1 on X and 0 on Y . The upper-left corner has explicit description

$$\emptyset_Y(R) = \begin{cases} \emptyset, & R \neq 0, \\ Y(0), & R = 0. \end{cases}$$

There is a unique map $\emptyset_Y \rightarrow \emptyset = \text{Spec}(0)$, which is an isomorphism on every nonzero ring. Since the empty sieve covers $\text{Spec}(0)$ in the Zariski topology, $\emptyset_Y \rightarrow \emptyset$ is a Zariski-local isomorphism, so $a_{\text{Zar}}(\emptyset_Y) = \emptyset$.

Sheafification preserves finite limits, so applying a_{Zar} to the diagram yields a pullback square in $\text{Shv}_{\text{Zar}}(\text{Aff})$:

$$\begin{array}{ccc} X & \longrightarrow & \mathbb{G}_m \\ \downarrow & \lrcorner & \downarrow \\ X \sqcup Y & \xrightarrow{(1,0)} & \mathbb{A}^1. \end{array}$$

This realizes $X \hookrightarrow X \sqcup Y$ as the base change of the open immersion $\mathbb{G}_m \hookrightarrow \mathbb{A}^1$ along the section $(1, 0)$. Geometrically: the open subscheme on which $(1, 0)$ is invertible is exactly X , and its complement $V(1, 0) = Y$ is the closed locus where it vanishes. Hence $X \hookrightarrow X \sqcup Y$ is simultaneously open and closed. $\square$

By Lemma 6.1.11, condition (ii) of Definition 6.1.8 amounts to a locally trivial equivalence relation. Concretely, it unfolds into the following compatibility data, each asserting a factorization through some $U_{kl} \hookrightarrow U_k \times U_l$:

- *Reflexivity*: the diagonal $U_i \rightarrow U_i \times U_i$ factors through U_{ii} ;
- *Symmetry*: the composite $U_{ij} \rightarrow U_i \times U_j \xrightarrow{\sim} U_j \times U_i$ factors through U_{ji} ;
- *Transitivity*: the composite $U_{ij} \times_{U_j} U_{jk} \rightarrow U_i \times U_k$ factors through U_{ik} .

Proof of Theorem 6.1.10. (i) Finite limits. It suffices to handle the terminal object and pullbacks. The terminal object is $\text{Spec}(\mathbb{Z})$, a scheme. For pullbacks, let $\alpha: X \rightarrow Z$ and $\beta: Y \rightarrow Z$ be morphisms of schemes, and set $P = X \times_Z Y$. Given a pair $x \in |X|$, $y \in |Y|$ with common image $z \in |Z|$, choose an affine open $W \subset Z$ containing z , an affine open $U \subset X$ contained in $\alpha^{-1}(W)$ and containing x , and an affine open $V \subset Y$ contained in $\beta^{-1}(W)$ and containing y . Then $U \times_W V$ is affine (fiber product of affines is affine), and inclusion gives an open subfunctor

$$U \times_W V \hookrightarrow X \times_Z Y = P.$$

Varying (x, y, z) produces an affine open cover of P , so P is a scheme.

(ii) Cofiltered limits of affine morphisms. Recall that $f: Y \rightarrow X$ is affine iff for any $\text{Spec}(R) \rightarrow X$ the pullback $Y \times_X \text{Spec}(R)$ is affine, and there is an equivalence $\text{Spec}: \text{CAlg}_X^{\text{op}} \xrightarrow{\sim} \text{Aff}_X$.

Let I be cofiltered and $F: I \rightarrow \text{Shv}_{\text{Zar}}(\text{Aff})$ a diagram with affine transition maps. Pick any basepoint $x_0 \in I$; cofilteredness gives $\lim_I F \simeq \lim_{I/x_0} F$, so we may assume I has a terminal object $*$. Set $X = F(*)$. The diagram restricts to a diagram in Aff_X , and since $\text{Aff}_X \hookrightarrow \text{Fun}(\text{CAlg}, \text{Set})/X$ preserves limits while CAlg admits filtered colimits, the limit is itself an affine X -scheme, hence a scheme.

(iii) Arbitrary coproducts. For $X = \coprod_{i \in I} X_i$ with each X_i a scheme: by Lemma 6.1.11 each $X_i \hookrightarrow X$ is an open immersion and these jointly cover X . Refining each X_i by an affine open cover yields one for X , so X is a scheme.

(iv) Quotients by locally trivial equivalence relations. Let $R \rightrightarrows X$ be a locally trivial equivalence relation with X a scheme, and form the quotient X/R . Choose an open cover $(U_i \hookrightarrow X)_{i \in I}$ as in Definition 6.1.7, so that for each i the composite $s^{-1}(U_i) \hookrightarrow R \xrightarrow{t} X$ is an open immersion.

Pulling back the inclusion $U_i \hookrightarrow X/R$ along the Zariski-local epimorphism $X \rightarrow X/R$ gives the cartesian square

$$\begin{array}{ccc} U_i \times_{X/R} X & \longrightarrow & X \\ \downarrow & \lrcorner & \downarrow \\ U_i & \longrightarrow & X/R. \end{array}$$

Since $X \times_{X/R} X = R$ (definition of the quotient), $U_i \times_{X/R} X \simeq s^{-1}(U_i)$ via s , with second projection given by t . The locally trivial hypothesis says this base change $s^{-1}(U_i) \xrightarrow{t} X$ is an open immersion. The property “open immersion” descends along Zariski-local epimorphisms, so $U_i \rightarrow X/R$ is itself an open immersion. The family $(U_i \rightarrow X/R)_{i \in I}$ jointly covers X/R , since (U_i) covers X and $X \rightarrow X/R$ is a Zariski-local epimorphism. Refining each U_i by an affine open cover yields an affine open cover of X/R . $\square$

Theorem 6.1.10 thus implies: given a gluing datum in which every U_i and U_{ij} is a scheme, the resulting quotient X is again a scheme.

6.1.1. Examples

Example 6.1.12 (Obtaining Proj by Gluing). Let A be an $\mathbb{N}$ -graded ring and $I \subset A_+$ a subset of homogeneous elements with $A_+ \subset \sqrt{(I)}$. The family $(D(f) \subset \text{Proj}(A))_{f \in I}$ is an open cover, with pairwise intersections

$$D(f) \cap D(g) = \text{Spec}(A_{(fg)}) \times_{\text{Proj}(A)} \text{Spec}(A_{(g)}) \simeq D(fg).$$

Hence $\text{Proj}(A)$ is the quotient of the equivalence relation

$$\coprod_{f,g \in I} \text{Spec}(A_{(fg)}) \rightrightarrows \coprod_{f \in I} \text{Spec}(A_{(f)}).$$

For $\mathbb{P}^n = \text{Proj}(\mathbb{Z}[x_0, \dots, x_n])$, take $I = \{x_0, \dots, x_n\}$. Then

$$D(x_i) = \text{Spec}(\mathbb{Z}[x_0, \dots, \widehat{x_i}, \dots, x_n]) \simeq \mathbb{A}^n,$$

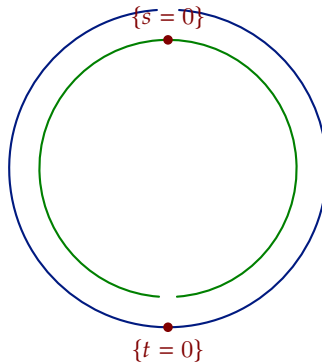
and for $i \neq j$, $D(x_i x_j) \simeq \mathbb{A}^{n-1} \times \mathbb{G}_m$. The presentation becomes

$$\coprod_{i < j} \mathbb{A}^{n-1} \times \mathbb{G}_m \rightrightarrows \coprod_{i=0}^n \mathbb{A}^n \longrightarrow \mathbb{P}^n.$$

For $n = 1$, this collapses to the pushout

$$\begin{array}{ccc} \mathbb{G}_m & \xrightarrow{t \mapsto t} & \mathbb{A}^1 \\ t \mapsto t^{-1} \downarrow & \lrcorner & \downarrow \\ \mathbb{A}^1 & \longrightarrow & \mathbb{P}^1, \end{array}$$

visualized as gluing two copies of $\mathbb{A}^1$ along $\mathbb{G}_m$ via the identification $t \leftrightarrow t^{-1}$:



Remark 6.1.13 (Group Quotients and Affine Decomposition).

- If a group object G acts on X , then

$$X/G = \text{colim} \left(X \times G \begin{array}{c} \xrightarrow{\text{act}} \\ \xrightarrow{\text{proj}} \end{array} X \right).$$

When $X = *$, the resulting object is called a *classifying stack*. In the higher-categorical setting, by Remark 6.1.5, the colimit must be taken as a geometric realization over the full groupoid object.

- Every scheme has the canonical affine-probe presentation

$$X \simeq \text{colim}_{(R,x) \in \text{El}(X)} \text{Spec}(R).$$

An affine open cover gives another presentation via its full Čech nerve. Its multiple intersections are schemes but need not be affine; refining those intersections by affine opens produces a diagram of affines. Pairwise intersections by themselves, without the higher cocycle maps, are not the complete gluing diagram.

Example 6.1.14 (Obtaining $\mathbb{P}^n$ via a Group Action). The canonical action $\mathbb{G}_m \curvearrowright \mathbb{A}^{n+1} - 0$ induces a monomorphism

$$(\mathbb{A}^{n+1} - 0)/\mathbb{G}_m \hookrightarrow \mathbb{P}^n,$$

which is a Zariski-local epimorphism (verified earlier), so in $\mathrm{Shv}_{\mathrm{Zar}}(\mathrm{Aff})$ it becomes an isomorphism:

$$(\mathbb{A}^{n+1} - 0)/\mathbb{G}_m \xrightarrow{\sim} \mathbb{P}^n.$$

More generally, for any $\mathbb{N}$ -graded ring generated in degree ≤ 1 ,

$$D(A_+)/\mathbb{G}_m \xrightarrow{\sim} \mathrm{Proj}(A).$$

Example 6.1.15 (Coproducts of Affine Schemes). (i) *Finite coproducts.* The initial scheme is $\emptyset = \mathrm{Spec}(0) = a_{\mathrm{Zar}}(\emptyset^{\mathrm{pre}})$.

For $R, S \in \mathrm{CAlg}$, the product $R \times S$ in CAlg fits into a commutative square (pullback in CAlg , pushout in Aff):

$$\begin{array}{ccc} R \times S & \xrightarrow{(0,1)} & S \\ (1,0) \downarrow & \lrcorner & \downarrow \\ R & \longrightarrow & 0, \end{array}$$

where $R \times S \rightarrow R$ and $R \times S \rightarrow S$ are localizations at the orthogonal idempotents $e_1 = (1, 0)$ and $e_2 = (0, 1)$ (which satisfy $e_1 + e_2 = 1$). Let $\mathcal{T}$ be the sieve on $R \times S$ generated by $\{R \times S \rightarrow R, R \times S \rightarrow S\}$. Since (e_1, e_2) generate the unit ideal, $\mathcal{T}$ is a Zariski-covering sieve, so for any $X \in \mathrm{Shv}_{\mathrm{Zar}}(\mathrm{Aff})$,

$$X(R \times S) \xrightarrow{\sim} \mathrm{Hom}(\mathcal{T}, X) \simeq X(R) \times X(S).$$

This shows $\mathrm{Spec}(R \times S)$ represents the coproduct of $\mathrm{Spec}(R)$ and $\mathrm{Spec}(S)$ in the sheaf category, so $\mathrm{Aff} \hookrightarrow \mathrm{Sch}$ preserves finite coproducts.

(ii) *Counterexample for arbitrary coproducts.* The inclusion $\mathrm{Aff} \hookrightarrow \mathrm{Sch}$ does not preserve infinite coproducts. For an infinite family $(\mathrm{Spec}(R_i))_{i \in I}$, the coproduct in Sch is the disjoint union $X = \coprod_{i \in I} \mathrm{Spec}(R_i)$ (a scheme by Theorem 6.1.10, but generally not affine).

If $X \simeq \mathrm{Spec}(A)$ were affine, we would have $A \simeq \mathcal{O}(X) = \prod_{i \in I} R_i$. The idempotents $e_i = (\delta_{ij})_{j \in I} \in A$ (1 in position i , 0 elsewhere) determine standard opens $D(e_i) \simeq \mathrm{Spec}(R_i)$. However, when infinitely many $R_i \neq 0$, the ideal generated by $(e_i)_{i \in I}$ is $\bigoplus_{i \in I} R_i \subsetneq \prod_{i \in I} R_i$ (it does not contain $1 = (1, 1, \dots)$). Hence

$$\bigcup_{i \in I} D(e_i) \subsetneq \mathrm{Spec}\left(\prod_{i \in I} R_i\right),$$

i.e., the affine product contains “extra” points (corresponding to non-principal ultrafilters on I) not present in the coproduct in Sch .

6.2. Separatedness and Quasi-compactness

This section introduces three finiteness/separation conditions on morphisms: *separated*, *quasi-compact*, and *quasi-separated*. They are the scheme-theoretic substitutes for Hausdorff and compactness in topology, but the analogy is indirect: schemes are almost never Hausdorff in the usual topological sense (the underlying space $|X|$ has lots of non-closed points), so the right substitute looks at the diagonal $\Delta_f: Y \rightarrow Y \times_X Y$.

The key idea: f is *separated* when Δ_f is a closed immersion — mimicking the topological theorem “ f is Hausdorff iff Δ_f is closed”. There are two natural strengths, depending on whether Δ_f is required to be a closed immersion (*separated*) or merely an immersion (*locally separated*). The latter turns out to be automatic for morphisms of schemes, so the genuinely restrictive notion is separatedness.

6.2.1. Separatedness

Definition 6.2.1 (Local Separatedness, Separatedness). For a morphism $f : Y \rightarrow X$ of algebraic functors with diagonal $\Delta_f : Y \rightarrow Y \times_X Y$:

- f is *locally separated* if Δ_f is an immersion;
- f is *separated* if Δ_f is a closed immersion.

Proposition 6.2.2 (All Scheme Morphisms Are Locally Separated). *Any morphism $f : Y \rightarrow X$ between schemes is locally separated.*

Proof. Strategy. We exhibit Δ_f as a composition $Y \rightarrow W \hookrightarrow Y \times_X Y$ in which the first map is a closed immersion and the second is open. Locally on affine opens $V_{ij} \subset Y$ over $U_i \subset X$, the diagonal of an affine morphism $\text{Spec}(B) \rightarrow \text{Spec}(A)$ is the closed immersion dual to the multiplication $B \otimes_A B \rightarrow B$; the open W is the union of the affine opens $V_{ij} \times_{U_i} V_{ij} \subset Y \times_X Y$.

Choose an affine open cover $(U_i)_{i \in I}$ of X . For each i , choose an affine open cover $(V_{ij})_{j \in J_i}$ of $f^{-1}(U_i)$, so that each $V_{ij} \rightarrow U_i$ is a morphism of affine schemes. For each pair (i, j) , the natural open immersion

$$W_{ij} := V_{ij} \times_{U_i} V_{ij} \hookrightarrow f^{-1}(U_i) \times_{U_i} f^{-1}(U_i) \hookrightarrow Y \times_X Y$$

exhibits W_{ij} as an affine open subscheme. Let $|W|$ be the union of all $|W_{ij}|$ inside $|Y \times_X Y|$. Since each $|W_{ij}|$ is open, so is $|W|$. Via the correspondence $\text{Open}(Y \times_X Y) \xrightarrow{\sim} \text{Open}(|Y \times_X Y|)$, there is a unique open subscheme $W \subset Y \times_X Y$ with underlying space $|W|$, and $(W_{ij})_{i,j}$ is an affine open cover of it.

Step 1: Δ_f **factors through** W . Since $(V_{ij})_{i,j}$ covers Y , it suffices to show $\Delta_f(V_{ij}) \subset W$ for each (i, j) . The relative diagonal of the affine morphism $V_{ij} \rightarrow U_i$ has image landing in W_{ij} :

$$\Delta_f|_{V_{ij}} : V_{ij} \xrightarrow{\Delta_{V_{ij}/U_i}} W_{ij} \hookrightarrow W.$$

Gluing yields a factorization $\Delta' : Y \rightarrow W$.

Step 2: $\Delta' : Y \rightarrow W$ **is a closed immersion**. Closed immersions are local on the target, so it suffices to check $\Delta'^{-1}(W_{ij}) \rightarrow W_{ij}$ is closed for each (i, j) . We claim $\Delta'^{-1}(W_{ij}) = V_{ij}$: for $y \in Y$, the condition $\Delta'(y) = (y, y) \in V_{ij} \times_{U_i} V_{ij}$ is equivalent to $y \in V_{ij}$ (the over- U_i constraint is automatic, both projections being $f(y)$). The restricted morphism is then exactly the affine relative diagonal

$$\Delta_{V_{ij}/U_i} : V_{ij} \rightarrow V_{ij} \times_{U_i} V_{ij}.$$

Writing $V_{ij} = \text{Spec}(B_{ij})$ and $U_i = \text{Spec}(A_i)$, this morphism is dual to the multiplication $B_{ij} \otimes_{A_i} B_{ij} \rightarrow B_{ij}$, $b \otimes b' \mapsto bb'$, which is surjective. Hence the corresponding scheme morphism is a closed immersion.

Combining the two steps, Δ_f factors as $Y \xrightarrow{\text{closed}} W \xrightarrow{\text{open}} Y \times_X Y$, so it is an immersion. $\square$

We therefore focus on the genuinely restrictive notion.

Remark 6.2.3 (Interpretation of Separatedness). The condition that Δ_f is a closed immersion can be understood as follows.

- In topology, a closed relative diagonal $\Delta_f : Y \rightarrow Y \times_X Y$ forces every fiber of f to be Hausdorff, but Hausdorff fibers alone do not force the relative diagonal to be closed. If the base X is Hausdorff, then Δ_f is closed if and only if Y is Hausdorff.
- For schemes, separatedness is checked on affine covers: $f : Y \rightarrow X$ is separated iff for any affine open $U = \text{Spec}(A) \subset X$ and any affine opens $V_1 = \text{Spec}(B_1), V_2 = \text{Spec}(B_2) \subset f^{-1}(U)$, the intersection $V_1 \cap V_2$ is affine and the natural map $B_1 \otimes_A B_2 \rightarrow \mathcal{O}(V_1 \cap V_2)$ is surjective.

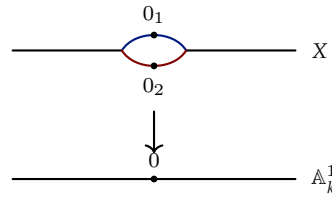
- The affine line with two origins makes the obstruction concrete. On the cross-chart $U_1 \times U_2$, the diagonal meets the chart in the graph of the identification $D(t) \rightarrow D(t)$. Its closure also contains $(0_1, 0_2)$, so the relative diagonal is not closed.

Example 6.2.4. (i) *Affine morphisms are separated.* If $f: Y \rightarrow X$ is affine, then $Y \times_X Y \rightarrow X$ is also affine (affine morphisms are stable under base change), and Δ_f corresponds to the multiplication $\mathcal{O}_Y \otimes_{\mathcal{O}_X} \mathcal{O}_Y \rightarrow \mathcal{O}_Y$, which is surjective; hence Δ_f is a closed immersion.

(ii) *Projective morphisms are separated.* More generally, any projective morphism $f: Y \rightarrow X$ is separated; this includes $\mathbb{P}_X^n \rightarrow X$ for any scheme X .

(iii) *Open immersions are separated.* If $U \hookrightarrow X$ is an open immersion, $\Delta: U \rightarrow U \times_X U$ is an isomorphism.

(iv) *Counterexample: affine line with two origins.* Let X be the affine line with two origins, glued from two copies of $\mathbb{A}^1$ along $\mathbb{A}^1 \setminus \{0\}$:



The morphism $X \rightarrow \operatorname{Spec}(k)$ is not separated: $\Delta: X \rightarrow X \times_k X$ is an immersion but not closed, because the two origins fail to be separated by disjoint affine neighborhoods. Equivalently, $U_1 \cap U_2 = \mathbb{A}^1 \setminus \{0\}$ is not closed in either U_i .

Proposition 6.2.5 (Properties of Separated Morphisms).

(i) Consider a commutative triangle of algebraic functors

$$\begin{array}{ccc} Z & \xrightarrow{g} & Y \\ & \searrow h & \swarrow f \\ & X & \end{array}$$

If f and g are separated, so is h . If h is separated, so is g . If h is a closed immersion and f is separated, then g is a closed immersion. The same statements hold with “separated” replaced by “locally separated” and “closed immersion” by “immersion”.

(ii) Consider a cartesian square of algebraic functors

$$\begin{array}{ccc} Y' & \xrightarrow{f'} & X' \\ \downarrow & \lrcorner & \downarrow g \\ Y & \xrightarrow{f} & X. \end{array}$$

If f is (locally) separated, so is f' . The converse holds if X and Y are Zariski sheaves and g is a Zariski-local epimorphism.

Proof. (i) *Cancellation of “closed immersion”.* Suppose h is a closed immersion and f is separated. Closed immersions are stable under composition and base change, so by Theorem 1.2.53 applied to the property “is a closed immersion” and the sequence $Z \xrightarrow{g} Y \xrightarrow{f} X$ (whose diagonal Δ_f is a closed immersion by hypothesis), we conclude g is a closed immersion.

Cancellation of “separated”. Suppose h is separated; we show g is separated. The diagonal $\Delta_g: Z \rightarrow Z \times_Y Z$ fits into the cartesian square

$$\begin{array}{ccc} Z & \xrightarrow{\Delta_g} & Z \times_Y Z \\ \parallel & \lrcorner & \downarrow j \\ Z & \xrightarrow{\Delta_h} & Z \times_X Z, \end{array}$$

where j is the canonical inclusion (any pair commuting over Y commutes over X). The cartesianness is direct: pulling back the inclusion j along Δ_h imposes $(z_1, z_2) = (z, z)$, i.e., the diagonal. Since Δ_h is a closed immersion by hypothesis, its base change Δ_g is too. Hence g is separated.

Composition. Suppose f and g are separated; we show $h = f \circ g$ is too. The diagonal Δ_h factors as Δ_g followed by the canonical inclusion $j: Z \times_Y Z \hookrightarrow Z \times_X Z$, where j fits into the cartesian square

$$\begin{array}{ccc} Z \times_Y Z & \xrightarrow{j} & Z \times_X Z \\ \downarrow & \lrcorner & \downarrow g \times_X g \\ Y & \xrightarrow{\Delta_f} & Y \times_X Y. \end{array}$$

Since f is separated, Δ_f is a closed immersion, hence so is its base change j . Composing with the closed immersion Δ_g (since g is separated) gives $\Delta_h = j \circ \Delta_g$ a closed immersion.

The locally separated case is identical, replacing “closed immersion” with “immersion” throughout.

(ii) *Forward direction.* The diagonals fit into the cartesian square

$$\begin{array}{ccc} Y' & \xrightarrow{\Delta_{f'}} & Y' \times_{X'} Y' \\ \downarrow & \lrcorner & \downarrow \\ Y & \xrightarrow{\Delta_f} & Y \times_X Y \end{array}$$

(the diagonal of a base change is the base change of the diagonal). If Δ_f is a closed immersion, so is $\Delta_{f'}$ by stability under base change. The locally separated case is identical.

Converse, g a Zariski-local epimorphism. Assume f' is separated; we show f is. From the cartesian square above, $\Delta_{f'}$ is the base change of Δ_f along $Y' \times_X Y' \rightarrow Y \times_X Y$. The latter map is itself the base change of $X' \rightarrow X$, since

$$Y' \times_{X'} Y' = (Y \times_X X') \times_{X'} (Y \times_X X') \simeq (Y \times_X Y) \times_X X'.$$

Zariski-local epimorphisms are stable under base change, so $Y' \times_{X'} Y' \rightarrow Y \times_X Y$ is a Zariski-local epimorphism. Closed immersion property descends along Zariski-local epimorphisms (Proposition 1.3.11), so $\Delta_{f'}$ being a closed immersion forces Δ_f to be one. The locally separated case is identical. $\square$

Corollary 6.2.6 (Sections of Morphisms of Schemes). *Let $f: X \rightarrow S$ be a morphism of schemes and $s: S \rightarrow X$ a section of f . Then s is an immersion. If f is separated, s is a closed immersion.*

Proof. The section s fits into a cartesian square

$$\begin{array}{ccc} S & \xrightarrow{s} & X \\ s \downarrow & \lrcorner & \downarrow \Delta_f \\ X & \xrightarrow{(\text{id}_X, s \circ f)} & X \times_S X \end{array}$$

identifying S with the locus $\{x \in X : x = s(f(x))\}$. By Proposition 6.2.2, Δ_f is an immersion (and a closed immersion if f is separated). Stability of (closed) immersions under base change (Proposition 1.2.52) gives the conclusion. $\square$

6.2.2. Quasi-compactness and Quasi-separatedness

Every scheme is a colimit of a diagram of affine schemes and open immersions (Remark 6.1.13); a scheme is *quasi-compact and quasi-separated (qcqs)* when such a diagram can be taken to be finite. We now study these finiteness conditions in detail, working at the level of general algebraic functors where they still make sense.

Definition 6.2.7 (Quasi-compact and Quasi-separated).

- (i) An algebraic functor X is *quasi-compact* if every Zariski-local epimorphism $\coprod_{i \in I} Y_i \rightarrow X$ admits a finite subset $J \subset I$ for which $\coprod_{i \in J} Y_i \rightarrow X$ is still a Zariski-local epimorphism.
- (ii) A morphism $f: Y \rightarrow X$ is *quasi-compact* if for every $Z \rightarrow X$ with Z quasi-compact, the pullback $Y \times_X Z$ is quasi-compact.
- (iii) An algebraic functor X is *quasi-separated* if for every diagram $Y \rightarrow X \leftarrow Z$ with Y, Z quasi-compact, the pullback $Y \times_X Z$ is quasi-compact.
- (iv) A morphism $f: Y \rightarrow X$ is *quasi-separated* if its diagonal $\Delta_f: Y \rightarrow Y \times_X Y$ is quasi-compact.

Lemma 6.2.8 (Affine schemes are quasi-compact). *For every ring A , the algebraic functor $\text{Spec}(A)$ is quasi-compact in the sense of Definition 6.2.7.*

Proof. Let $q: \coprod_{j \in J} Z_j \rightarrow \text{Spec}(A)$ be a Zariski-local epimorphism. Apply Definition 3.3.1 to the identity point of $\text{Spec}(A)$. We obtain elements $f_1, \dots, f_n$ generating the unit ideal and, for each a , a lift of $\text{Spec}(A_{f_a}) \rightarrow \text{Spec}(A)$ through some Z_{j_a} . The finite subfamily indexed by $J_0 = \{j_1, \dots, j_n\}$ is still Zariski-locally surjective: after any base change $A \rightarrow B$, the images of the f_a generate B and the same lifts base change to the corresponding principal cover of $\text{Spec}(B)$. Hence q admits a finite subcover. $\square$

Proposition 6.2.9 (Characterizations of Quasi-compactness). *For an algebraic functor X , the following are equivalent:*

- (i) X is quasi-compact;
- (ii) the unique map $X \rightarrow \text{Spec}(\mathbb{Z})$ is quasi-compact;
- (iii) there exists a Zariski-local epimorphism $\coprod_{i \in I} Y_i \rightarrow X$ with I finite and each Y_i affine.

Proof. (i) $\Rightarrow$ (iii). By the Yoneda Lemma, X is the colimit of representables over its covariant category of elements: $X \simeq \text{colim}_{(R,x) \in \text{El}(X)} \text{Spec}(R)$. The canonical map

$$p: \coprod_{(R,x) \in \text{El}(X)} \text{Spec}(R) \rightarrow X$$

is a Zariski-local epimorphism. By (i), it admits a finite subcover, yielding (iii).

(iii) $\Rightarrow$ (i). Suppose $p: U = \coprod_{i=1}^n U_i \rightarrow X$ is a Zariski-local epimorphism with each U_i affine. Let $q: \coprod_{j \in J} Z_j \rightarrow X$ be any Zariski-local epimorphism. For each $i \in \{1, \dots, n\}$, Lemma 6.2.8 shows that U_i is quasi-compact, so the pulled-back cover $\coprod_j (Z_j \times_X U_i) \rightarrow U_i$ admits a finite subcover indexed by $J_i \subset J$. The finite union $J' = \bigcup_{i=1}^n J_i \subset J$ then gives a finite Zariski-local epimorphism to X , since this can be checked after pullback along the Zariski-local epimorphism p .

(ii) $\Leftrightarrow$ (i). If $f: X \rightarrow \text{Spec}(\mathbb{Z})$ is quasi-compact, then Lemma 6.2.8 makes $\text{Spec}(\mathbb{Z})$ quasi-compact, so the pullback $X \times_{\text{Spec}(\mathbb{Z})} \text{Spec}(\mathbb{Z}) \simeq X$ is quasi-compact.

Conversely, suppose X is quasi-compact, and let $Z \rightarrow \operatorname{Spec}(\mathbb{Z})$ with Z quasi-compact. Choose finite affine covers $U = \coprod_{i \in I} \operatorname{Spec}(A_i) \rightarrow X$ and $V = \coprod_{j \in J} \operatorname{Spec}(B_j) \rightarrow Z$ via (iii). Stability of Zariski-local epimorphisms under products gives a Zariski-local epimorphism

$$U \times_{\operatorname{Spec}(\mathbb{Z})} V \simeq \coprod_{(i,j) \in I \times J} \operatorname{Spec}(A_i \otimes_{\mathbb{Z}} B_j) \rightarrow X \times_{\operatorname{Spec}(\mathbb{Z})} Z.$$

The left-hand side is a finite coproduct of affines, so by (iii) the pullback is quasi-compact. $\square$

Proposition 6.2.10 (Characterizations of Quasi-separatedness). *For an algebraic functor X , the following are equivalent:*

- (i) X is quasi-separated;
- (ii) the unique map $X \rightarrow \operatorname{Spec}(\mathbb{Z})$ is quasi-separated;
- (iii) for any $Y \rightarrow X$ and $Z \rightarrow X$ with Y, Z affine, the fiber product $Y \times_X Z$ is quasi-compact.

Proof. (i) $\Rightarrow$ (iii). Affine schemes are quasi-compact, so this is a special case of the definition.

(iii) $\Rightarrow$ (ii). We must show the diagonal $\Delta_X: X \rightarrow X \times X$ is quasi-compact. Take any $W \rightarrow X \times X$ with W quasi-compact, and form the pullback $P = W \times_{X \times X} X$. By Proposition 6.2.9(iii), W admits a finite affine cover $\coprod_{i=1}^n U_i \rightarrow W$; quasi-compactness is preserved by finite coproducts, so it suffices to handle each $P_i = U_i \times_{X \times X} X$ separately.

A map $U_i \rightarrow X \times X$ corresponds to a pair $u, v: U_i \rightarrow X$, and P_i fits into the cartesian square

$$\begin{array}{ccc} P_i & \longrightarrow & U_i \\ \downarrow & \lrcorner & \downarrow \Delta_{U_i} \\ U_i \times_X U_i & \longrightarrow & U_i \times U_i. \end{array}$$

Now (a) $U_i \times_X U_i$ is quasi-compact by hypothesis (iii); (b) Δ_{U_i} is a closed immersion since U_i is affine (the multiplication $A \otimes A \rightarrow A$ is surjective); (c) the base change $P_i \rightarrow U_i \times_X U_i$ is therefore a closed immersion. A closed subfunctor of a quasi-compact functor is quasi-compact (Proposition 6.2.9(iii)), so P_i is quasi-compact.

(ii) $\Rightarrow$ (i). For the structure map $X \rightarrow \operatorname{Spec}(\mathbb{Z})$, the relative diagonal coincides with the absolute $\Delta_X: X \rightarrow X \times X$ (since $\operatorname{Spec}(\mathbb{Z})$ is terminal), so (ii) says exactly that Δ_X is quasi-compact. Let $Y, Z \rightarrow X$ be quasi-compact; then $Y \times Z$ is quasi-compact (qc is preserved under finite products), and the cartesian square

$$\begin{array}{ccc} Y \times_X Z & \longrightarrow & Y \times Z \\ \downarrow & \lrcorner & \downarrow \\ X & \xrightarrow{\Delta_X} & X \times X \end{array}$$

exhibits $Y \times_X Z$ as the pullback of $Y \times Z$ along Δ_X . Since Δ_X is qc by (ii), this pullback is qc by definition of qc morphism, giving (i). $\square$

Proposition 6.2.11 (Composition of qc/qs Morphisms). *Consider a commutative triangle of algebraic functors*

$$\begin{array}{ccc} & Z & \\ g \swarrow & & \searrow h \\ Y & \xrightarrow{f} & X. \end{array}$$

- (i) If f and g are quasi-compact, then h is quasi-compact.
- (ii) If f and g are quasi-separated, then h is quasi-separated.

- (iii) If h is quasi-compact and f is quasi-separated, then g is quasi-compact.
- (iv) If h is quasi-separated, then g is quasi-separated.
- (v) If h is quasi-compact and g is a Zariski-local epimorphism, then f is quasi-compact.
- (vi) If h is quasi-separated and g is quasi-compact and a Zariski-local epimorphism, then f is quasi-separated.

Proof. We freely use Definition 6.2.7.

(i) Let $W \rightarrow X$ with W quasi-compact. Then $W \times_X Y$ is qc since f is qc, and $W \times_X Z \simeq (W \times_X Y) \times_Y Z$ is qc since g is qc.

(ii) The diagonal Δ_h factors as

$$Z \xrightarrow{\Delta_g} Z \times_Y Z \xrightarrow{\iota} Z \times_X Z,$$

where ι is the base change of Δ_f along $g \times_X g: Z \times_X Z \rightarrow Y \times_X Y$. Since Δ_f is qc (f is qs) and qc is stable under base change, ι is qc; combined with Δ_g qc (g is qs), (i) gives Δ_h qc.

(iii) Factor g through its graph: $g: Z \xrightarrow{\Gamma_g} Z \times_X Y \xrightarrow{\text{pr}_Y} Y$. The projection is the base change of h , hence qc. The graph Γ_g fits into the cartesian square

$$\begin{array}{ccc} Z & \xrightarrow{\Gamma_g} & Z \times_X Y \\ g \downarrow & \lrcorner & \downarrow g \times_X \text{id} \\ Y & \xrightarrow{\Delta_f} & Y \times_X Y, \end{array}$$

so Γ_g is the base change of Δ_f (qc since f is qs), hence qc. Composition gives g qc.

(iv) The diagonal $\Delta_g: Z \rightarrow Z \times_Y Z$ is the base change of $\Delta_h: Z \rightarrow Z \times_X Z$ along the canonical map $Z \times_Y Z \rightarrow Z \times_X Z$. By hypothesis Δ_h is qc, and qc is stable under base change, so Δ_g is qc; i.e., g is qs.

(v) Let $W \rightarrow X$ with W quasi-compact; we show $W \times_X Y$ is qc. Pulling back $g: Z \rightarrow Y$ along $W \times_X Y \rightarrow Y$ gives a Zariski-local epimorphism

$$W \times_X Z \rightarrow W \times_X Y$$

(Zariski-local epimorphisms are stable under base change). The source $W \times_X Z$ is qc since h is qc and W is qc. By Proposition 6.2.9(iii), it admits a finite affine cover; composing with the surjection yields a finite Zariski-local epimorphism from a finite coproduct of affines onto $W \times_X Y$, so $W \times_X Y$ is qc.

(vi) We reduce to (v) applied to the diagonal. First, $g \times_X g: Z \times_X Z \rightarrow Y \times_X Y$ is qc, being the composition of two base changes of g . Next, $\Delta_h: Z \rightarrow Z \times_X Z$ is qc by hypothesis. By (i),

$$(g \times_X g) \circ \Delta_h: Z \rightarrow Y \times_X Y \text{ is qc.}$$

But this composition equals $\Delta_f \circ g$, since both send z to $(g(z), g(z)) \in Y \times_X Y$. Hence $\Delta_f \circ g$ is qc and g is a Zariski-local epimorphism, so (v) applied to the triangle $Z \xrightarrow{g} Y \xrightarrow{\Delta_f} Y \times_X Y$ yields Δ_f qc, i.e., f is qs. $\square$

Proposition 6.2.12 (Base Change of qc/qs Morphisms). Consider a cartesian square of algebraic functors:

$$\begin{array}{ccc} Y' & \longrightarrow & X' \\ f' \downarrow & \lrcorner & \downarrow g \\ Y & \xrightarrow{f} & X. \end{array}$$

If f is quasi-compact (resp. quasi-separated), so is f' . The converse holds if g is a Zariski-local epimorphism.

Proof. Forward direction (qc). For any $W \rightarrow X'$ with W qc, the pasting lemma gives $Y' \times_{X'} W \simeq Y \times_X W$, qc since f is qc.

Forward direction (qs). The cartesian square

$$\begin{array}{ccc} Y' & \xrightarrow{\Delta_{f'}} & Y' \times_{X'} Y' \\ \downarrow & \lrcorner & \downarrow \\ Y & \xrightarrow{\Delta_f} & Y \times_X Y \end{array}$$

exhibits $\Delta_{f'}$ as the base change of Δ_f along $Y' \times_{X'} Y' \rightarrow Y \times_X Y$. Since Δ_f is qc and qc is preserved under base change (just proved), $\Delta_{f'}$ is qc, so f' is qs.

Converse, g a Zariski-local epimorphism (qc). Suppose f' is qc; we show f is qc. Let $W \rightarrow X$ with W qc. By Proposition 6.2.9(iii), W admits a finite affine cover $\coprod_k \text{Spec}(A_k) \twoheadrightarrow W$. Base change to Y gives a Zariski-local epimorphism $\coprod_k Y \times_X \text{Spec}(A_k) \twoheadrightarrow Y \times_X W$, so it suffices to show each $Y \times_X \text{Spec}(A_k)$ is qc. We may therefore reduce to the case $W = \text{Spec}(A)$ affine.

The Zariski-local epimorphism g pulls back to a Zariski-local epimorphism $\text{Spec}(A) \times_X X' \rightarrow \text{Spec}(A)$, so there exist $f_1, \dots, f_n \in A$ with $(f_1, \dots, f_n) = A$ such that each $\text{Spec}(A_{f_k}) \rightarrow \text{Spec}(A) \rightarrow X$ factors through X' . For each such k ,

$$Y \times_X \text{Spec}(A_{f_k}) \simeq Y' \times_{X'} \text{Spec}(A_{f_k}),$$

which is qc since f' is qc and $\text{Spec}(A_{f_k})$ is affine, hence qc. The finite coproduct $\coprod_k Y \times_X \text{Spec}(A_{f_k}) \twoheadrightarrow Y \times_X \text{Spec}(A)$ is a Zariski-local epimorphism with qc source, so $Y \times_X \text{Spec}(A)$ is qc by Proposition 6.2.9(iii).

Converse, g a Zariski-local epimorphism (qs). From the forward (qs) cartesian square above, $\Delta_{f'}$ is the base change of Δ_f along $Y' \times_{X'} Y' \rightarrow Y \times_X Y$, which is itself the base change of $X' \rightarrow X$ (using $Y' \times_{X'} Y' \simeq (Y \times_X Y) \times_{X'} X'$). Zariski-local epimorphisms are stable under base change, so $Y' \times_{X'} Y' \rightarrow Y \times_X Y$ is a Zariski-local epimorphism. Since $\Delta_{f'}$ is qc by hypothesis, the qc converse just proved (applied to the cartesian square defining $\Delta_{f'}$) gives Δ_f qc, so f is qs. $\square$

Example 6.2.13 (Quasi-compact Immersions). (i) By Proposition 6.2.9(iii), a closed subfunctor of a quasi-compact algebraic functor is itself quasi-compact. It follows that any closed immersion is quasi-compact. In particular, as the terminology suggests, every separated morphism is quasi-separated.

(ii) Let R be a ring and $I \subset R$. The cover $(D(f) \subset D(I))_{f \in I}$ shows that $D(I)$ is quasi-compact iff $\sqrt{(I)} = \sqrt{(J)}$ for some finite $J \subset I$. Hence an open immersion is quasi-compact iff its associated quasi-coherent radical ideal is finitely generated.

(iii) Any monomorphism is quasi-separated, since its diagonal is an isomorphism.

Example 6.2.14 (Quasi-affine and Quasi-projective Schemes). (i) By Example 6.2.13(ii), any quasi-affine scheme is quasi-compact; it is quasi-separated since it is separated (Example 6.2.4).

(ii) Let A be an $\mathbb{N}$ -graded ring and $I \subset A$ a homogeneous subset. As in Example 6.2.13(ii), $D(I) \subset \text{Proj}(A)$ is quasi-compact iff the saturated radical ideal generated by I is finitely generated. In particular, every quasi-projective k -scheme is quasi-compact, and is quasi-separated since it is separated.

(iii) By (i), (ii), and Proposition 6.2.12, any quasi-affine or quasi-projective morphism of algebraic functors is quasi-compact and quasi-separated.

For schemes, these finiteness conditions depend only on the underlying topological space and can be tested on any affine open cover.

Proposition 6.2.15 (Quasi-compactness for Schemes). *Let X be a scheme and $(U_i \subset X)_{i \in I}$ an open cover by affine schemes. The following are equivalent:*

- (i) X is quasi-compact;
- (ii) $|X|$ is quasi-compact as a topological space;
- (iii) there exists a finite $J \subset I$ such that $(U_i)_{i \in J}$ already covers X .

Proof. (i) $\Leftrightarrow$ (iii). The cover $(U_i \hookrightarrow X)_{i \in I}$ is a Zariski-local epimorphism with each U_i affine (hence qc). By Proposition 6.2.9(iii), X is qc iff the cover admits a finite refinement, i.e., a finite J as in (iii).

(ii) $\Leftrightarrow$ (iii). The space $|X|$ is the colimit of $(|U_i|)$ in topological spaces (Corollary 4.3.13), and each $|U_i|$ is qc (the prime spectrum of an affine scheme is qc). Such a colimit is qc iff a finite subfamily covers, which is (iii). $\square$

Proposition 6.2.16 (Quasi-separatedness for Schemes). *Let X be a scheme and $(U_i \subset X)_{i \in I}$ an open cover by affine schemes. The following are equivalent:*

- (i) X is quasi-separated;
- (ii) the intersection of any two quasi-compact open subsets of $|X|$ is quasi-compact;
- (iii) for every $i, j \in I$, the intersection $U_i \cap U_j$ is quasi-compact.

Proof. (i) $\Rightarrow$ (ii). Two qc opens $V, W \subset |X|$ correspond to qc subfunctors of X , and their intersection is the pullback $V \times_X W$, qc by qs of X .

(ii) $\Rightarrow$ (iii). Each affine U_i is qc, so $U_i \cap U_j$ is qc by hypothesis.

(iii) $\Rightarrow$ (i). We must show $\Delta_X: X \rightarrow X \times X$ is qc. The cover $(U_i \times U_j \hookrightarrow X \times X)_{i, j \in I}$ is a Zariski-local epimorphism, so by Proposition 6.2.12 (qc converse), it suffices to show each base change

$$\Delta_X \times_{X \times X} (U_i \times U_j): U_i \cap U_j \rightarrow U_i \times U_j$$

is qc. The target $U_i \times U_j$ is affine, hence qs, and the source $U_i \cap U_j$ is qc by hypothesis (iii). For any qc $V \rightarrow U_i \times U_j$, the fiber product $(U_i \cap U_j) \times_{U_i \times U_j} V$ is qc by qs of $U_i \times U_j$, so the morphism $U_i \cap U_j \rightarrow U_i \times U_j$ is qc. $\square$

Remark 6.2.17 (QCQS Schemes). Combining Propositions 6.2.15 and 6.2.16: a scheme is qcqs iff it admits a finite affine cover whose pairwise intersections are themselves quasi-compact. This finiteness propagates to many invariants — for instance, the underlying space of a qcqs scheme is spectral, and its category of quasi-coherent modules is compactly generated.

As Example 6.2.14 shows, most schemes of interest are qcqs. Examples failing quasi-compactness include infinite coproducts of nonempty schemes and projective spaces $\mathbb{P}_k^I$ for infinite I (over $k \neq 0$). An infinite-dimensional affine space with doubled origin is quasi-compact but not quasi-separated.

Remark 6.2.18 (Tannaka Duality and 1-Affineness). There is a deep conceptual reason why qcqs morphisms are central. An affine scheme $X = \operatorname{Spec}(A)$ is completely determined by its ring of global functions $A \simeq \Gamma(X, \mathcal{O}_X)$; one might say affine schemes are “0-affine”.

For a general scheme X , the global section ring loses too much information. However, if X is qcqs, it can be reconstructed from the symmetric monoidal category $\operatorname{QCoh}(X)$. Specifically, a form of Tannaka duality proved by Brandenburg [BC14] says that, for a qcqs scheme X and any scheme Y , pullback induces an equivalence

$$\operatorname{Hom}_{\operatorname{Sch}}(Y, X) \xrightarrow{\sim} \operatorname{Fun}^{\otimes, \operatorname{cocont}}(\operatorname{QCoh}(X), \operatorname{QCoh}(Y)), \quad f \mapsto f^*,$$

where the functors on the right are cocontinuous strong symmetric monoidal functors. Notice the variance: a morphism $Y \rightarrow X$ induces pullback from X to Y . Thus, at the cost of passing from rings to symmetric monoidal abelian categories, the geometry of qcqs schemes is fully captured.

This phenomenon is formalized by Gaitsgory [Gai15] via *1-affineness*. Here one must pass to derived categories. Write $D_{\text{qc}}(X)$ for the stable presentable symmetric monoidal ∞ -category of complexes with quasi-coherent cohomology on X . An algebraic functor X is 1-affine if the ∞ -category of quasi-coherent sheaves of categories on X , denoted $\text{ShvCat}(X)$, is equivalent to modules over $D_{\text{qc}}(X)$:

$$\text{ShvCat}(X) \xrightarrow{\sim} \text{Mod}_{D_{\text{qc}}(X)}(\text{Pr}_{\text{st}}^{\text{L}}).$$

The category $D_{\text{qc}}(X)$ is the stable derived enhancement relevant here; it is not the ordinary category Mod_X developed in Definition 3.1.15. Just as a 0-affine object is controlled by a commutative algebra A , a 1-affine object is controlled by a symmetric monoidal stable ∞ -category. The theorem for qcqs schemes says that they are 1-affine in this sense.

Scholze and Stefanich generalize 1-affineness to n -affineness in the context of Stefanich rings; see [Sch26].

The following standard definition combines the local “algebraic” finiteness properties of Definition 1.4.1 with the global “topological” finiteness properties of this section.

Definition 6.2.19 (Morphism of Finite Presentation/Type). Let $f: X \rightarrow S$ be a morphism of algebraic functors.

- (i) f is of *finite presentation* if it is locally of finite presentation, quasi-compact, and quasi-separated.
- (ii) f is of *finite type* if it is locally of finite type and quasi-compact.

For example, every quasi-projective morphism is of finite type.

6.3. Schemes as Locally Ringed Spaces

We now show that every scheme can be canonically regarded as a locally ringed space (Definition 5.5.4), and that this construction induces an equivalence onto a natural full subcategory of LocRingSpace .

Let X be an algebraic functor. Denote by $\mathcal{O}_X$ the presheaf of functions on $\text{Open}(X) \simeq \text{Open}(|X|)$, given by

$$\mathcal{O}_X(U) = \mathcal{O}(U) = \text{Hom}_{\text{Fun}(\text{CAlg}, \text{Set})}(U, \mathbb{A}^1).$$

By Corollary 3.3.4, $\mathcal{O}_X$ is a sheaf of rings on $|X|$. This yields a functor

$$(|-|, \mathcal{O}_{(-)}): \text{Fun}(\text{CAlg}, \text{Set}) \rightarrow \text{RingSpace}, \quad X \mapsto (|X|, \mathcal{O}_X). \quad (6.1)$$

Warning 6.3.1 (Ambiguity of Notation $\mathcal{O}_X$). The symbol $\mathcal{O}_X$ is overloaded:

- In Notation 3.1.19, $\mathcal{O}_X$ denotes the initial object of CAlg_X (the category of quasi-coherent algebras on X); it serves as the monoidal unit of Mod_X .
- Above, $\mathcal{O}_X$ denotes the sheaf of rings $\mathcal{O}|_{\text{Open}(X)}$ on $|X|$.

The two objects arise from the same functions construction but are indexed in different ways: the first is specified on all affine probes of X , whereas the second is evaluated on open subfunctors of X . They are therefore not literally two restrictions along the same indexing category. We will see below that, for a scheme, the equivalence between quasi-coherent modules and quasi-coherent sheaves identifies the first object with the second structure sheaf.

To justify that this functor produces the “correct” geometry, we first inspect the affine case. Recall from Example 5.6.7 the description of sheaves on $\text{Spec}(R)$.

Example 6.3.2 (Prime Spectra as Locally Ringed Spaces). Let $X = \text{Spec}(R)$. The ringed space $(|X|, \mathcal{O}_X)$ has the explicit description:

- $|X| = \text{Prim}(R)$, the prime spectrum;
- under the equivalence $\text{Shv}(\text{Prim}(R)) \simeq \text{Shv}(\text{Open}^{\text{pr}}(\text{Spec}(R)))$, we have $\mathcal{O}_X(D(f)) = R_f$.

For a prime $\mathfrak{p} \in \text{Prim}(R)$, the stalk is the local ring

$$\mathcal{O}_{X,\mathfrak{p}} = \text{colim}_{\mathfrak{p} \in |D(f)|} \mathcal{O}_X(D(f)) = \text{colim}_{f \notin \mathfrak{p}} R_f = R_{\mathfrak{p}}.$$

Since $R_{\mathfrak{p}}$ is local, $(\text{Prim}(R), \mathcal{O}_{\text{Spec}(R)})$ is in fact a locally ringed space.

For any ring map $\varphi: R \rightarrow S$, the induced map $(\text{Prim}(S), \mathcal{O}_{\text{Spec}(S)}) \rightarrow (\text{Prim}(R), \mathcal{O}_{\text{Spec}(R)})$ is a morphism of locally ringed spaces: at any $\mathfrak{q} \in \text{Prim}(S)$, the map on stalks is the local homomorphism $\varphi_{\mathfrak{q}}: R_{\varphi^{-1}(\mathfrak{q})} \rightarrow S_{\mathfrak{q}}$.

Proposition 6.3.3 (Factorization through Sheaves). *There is a unique factorization*

$$\begin{array}{ccc} \text{Fun}(\text{CAlg}, \text{Set}) & \xrightarrow{\Phi := (|-|, \mathcal{O}_{(-)})} & \text{RingSpace} \\ \downarrow a_{\text{Zar}} & & \uparrow \\ \text{Shv}_{\text{Zar}}(\text{CAlg}^{\text{op}}) & \xrightarrow{\Phi} & \text{LocRingSpace}. \end{array}$$

In particular, the locally ringed space associated to an algebraic functor depends only on its associated Zariski sheaf.

Proof. We construct Φ and verify the factorization in three steps.

Step 1: Construction of the locally ringed space. Since LocRingSpace is cocomplete (Proposition 5.5.9), the universal property of presheaves (Theorem 1.1.10(v)) extends $R \mapsto (\text{Prim}(R), \mathcal{O}_{\text{Spec}(R)})$ uniquely along the Yoneda embedding to a colimit-preserving functor

$$\Phi := (|-|, \mathcal{O}_{(-)}): \text{Fun}(\text{CAlg}, \text{Set}) \rightarrow \text{LocRingSpace}.$$

Explicitly, using the canonical presentation $X \simeq \text{colim}_{(R,x) \in \text{El}(X)} \text{Spec}(R)$,

$$(|X|, \mathcal{O}_{|X|}) := \text{colim}_{(R,x) \in \text{El}(X)}^{\text{RingSpace}} (\text{Prim}(R), \mathcal{O}_{\text{Spec}(R)}).$$

By Proposition 5.5.9, $\text{LocRingSpace} \hookrightarrow \text{RingSpace}$ preserves colimits, so $(|X|, \mathcal{O}_{|X|})$ is in fact a locally ringed space. Concretely, the structure sheaf is the limit of pushforwards

$$\mathcal{O}_{|X|} := \lim_{(R,x) \in \text{El}(X)^{\text{op}}} (\alpha_x)_* \mathcal{O}_{\text{Spec}(R)},$$

which is itself a sheaf on $|X|$ (a limit of sheaves).

Step 2: Identification with the algebraic structure sheaf. We show the topologically constructed sheaf $\mathcal{O}_{|X|}$ coincides with the algebraically defined presheaf $\mathcal{O}_X$ ($\mathcal{O}_X(U) = \text{Hom}_{\text{Fun}}(U, \mathbb{A}^1)$).

Let $U \subset |X|$ be open, defining an open subfunctor $U \subset X$. Since colimits are stable under base change,

$$U \simeq \text{colim}_{(R,x) \in \text{El}(X)} x^{-1}(U),$$

where $x^{-1}(U) = \alpha_x^{-1}(U) \subset \text{Spec}(R)$. Computing the algebraic side:

$$\begin{aligned} \mathcal{O}_X(U) &= \text{Hom}_{\text{Fun}(\text{CAlg}, \text{Set})}(U, \mathbb{A}^1) \\ &\simeq \text{Hom}_{\text{Fun}}\left(\text{colim}_{(R,x) \in \text{El}(X)} x^{-1}(U), \mathbb{A}^1\right) \\ &\simeq \lim_{(R,x) \in \text{El}(X)^{\text{op}}} \text{Hom}_{\text{Fun}}(x^{-1}(U), \mathbb{A}^1) \\ &\simeq \lim_{(R,x) \in \text{El}(X)^{\text{op}}} \mathcal{O}_{\text{Spec}(R)}(U_R), \end{aligned}$$

which is exactly the section $\mathcal{O}_{|X|}(U)$ from Step 1. Hence $\mathcal{O}_X \simeq \mathcal{O}_{|X|}$ as sheaves on $|X|$. This shows in particular that $\mathcal{O}_X$ is a sheaf, and that $(|X|, \mathcal{O}_X)$ is a locally ringed space.

Step 3: Factorization through Zariski sheafification. By construction, Φ is the left Kan extension along $\text{Aff} \hookrightarrow \text{Fun}(\text{CAlg}, \text{Set})$ of $R \mapsto (\text{Prim}(R), \mathcal{O}_{\text{Spec}(R)})$. Affine descent says this functor inverts every Zariski-covering family in the locally ringed space sense, so Φ inverts all Zariski-local isomorphisms. Since a_{Zar} is the universal such inversion, Φ factors uniquely through a_{Zar} . $\square$

Theorem 6.3.4 (Adjunction between Rings and Locally Ringed Spaces). *The global section functor*

$$\Gamma: \text{LocRingSpace} \rightarrow \text{CAlg}^{\text{op}}, \quad (X, \mathcal{O}_X) \mapsto \mathcal{O}_X(X)$$

has a right adjoint given by the prime spectrum functor

$$\text{Prim}: \text{CAlg}^{\text{op}} \rightarrow \text{LocRingSpace}, \quad R \mapsto (\text{Prim}(R), \mathcal{O}_{\text{Spec}(R)}).$$

Remark 6.3.5. Since $\mathcal{O}(\text{Spec}(R)) \simeq R$, the counit of this adjunction is an isomorphism, so Prim is fully faithful.

Proof. We must show that for every ring R and locally ringed space $X = (T, \mathcal{O}_T)$ there is a natural bijection

$$\text{Hom}_{\text{CAlg}}(R, \mathcal{O}_T(T)) \xrightarrow{\sim} \text{Hom}_{\text{LocRingSpace}}((T, \mathcal{O}_T), (\text{Prim}(R), \mathcal{O}_{\text{Spec}(R)})).$$

Construction $\psi \mapsto (f, \varphi)$. Given $\psi: R \rightarrow \mathcal{O}_T(T)$, we build a morphism (f, φ) of locally ringed spaces. For $x \in T$, consider the composite $R \xrightarrow{\psi} \mathcal{O}_T(T) \rightarrow \mathcal{O}_{T,x}$, and let $\mathfrak{m}_x$ be the maximal ideal of the local ring $\mathcal{O}_{T,x}$. Define $f: T \rightarrow \text{Prim}(R)$ by

$$f(x) := \{r \in R : \psi(r)_x \in \mathfrak{m}_x\},$$

clearly a prime ideal.

Step 1: f is continuous. For a basic open $D(g) \subset \text{Prim}(R)$,

$$f^{-1}(D(g)) = \{x \in T : g \notin f(x)\} = \{x \in T : \psi(g)_x \in \mathcal{O}_{T,x}^\times\},$$

which is the open locus $T_{\psi(g)}$ where the section $\psi(g) \in \mathcal{O}_T(T)$ is invertible.

Step 2: Construction of the sheaf morphism φ . The sheaf $\mathcal{O}_{\text{Spec}(R)}$ is determined by its values $\mathcal{O}_{\text{Spec}(R)}(D(g)) \simeq R_g$ on the basis of principal opens. The map $\psi(g)|_{T_{\psi(g)}}$ is invertible by definition, so the universal property of localization produces a unique factorization

$$\begin{array}{ccccc} R & \xrightarrow{\psi} & \mathcal{O}_T(T) & \xrightarrow{\text{res}} & \mathcal{O}_T(T_{\psi(g)}) \\ \downarrow & & & \nearrow & \\ R_g & & & \exists! \varphi_{D(g)} & \end{array}$$

The compatibility on intersections $D(g) \cap D(h) = D(gh)$ follows from uniqueness of the localization, giving a sheaf morphism $\varphi: \mathcal{O}_{\text{Spec}(R)} \rightarrow f_*\mathcal{O}_T$.

Step 3: (f, φ) is a morphism of locally ringed spaces. For $x \in T$ with $\mathfrak{p} = f(x)$, the induced map on stalks

$$\varphi_x: R_{\mathfrak{p}} \rightarrow \mathcal{O}_{T,x}, \quad \frac{a}{s} \mapsto \psi(a)_x \cdot \psi(s)_x^{-1}$$

is well-defined: $s \notin \mathfrak{p}$ means $\psi(s)_x \notin \mathfrak{m}_x$, hence $\psi(s)_x \in \mathcal{O}_{T,x}^\times$. To see locality, note that $\frac{a}{s} \in \mathfrak{p}R_{\mathfrak{p}}$ requires $a \in \mathfrak{p}$, hence $\psi(a)_x \in \mathfrak{m}_x$, so $\varphi_x(\frac{a}{s}) \in \mathfrak{m}_x$.

Verification of bijection. Let $\Theta: (h, \theta) \mapsto \theta_{\text{Spec}(R)}: R \simeq \mathcal{O}_{\text{Spec}(R)}(\text{Prim}(R)) \rightarrow \mathcal{O}_T(T)$ be the candidate inverse, and $\Lambda: \psi \mapsto (f, \varphi)$ as above.

$\Theta \circ \Lambda = \text{id}$. Setting $g = 1$ in Step 2, the unique factorization $R_1 \simeq R \rightarrow \mathcal{O}_T(T)$ recovers exactly ψ .

$\Lambda \circ \Theta = \text{id}$. Given (h, θ) , set $\psi = \Theta(h, \theta)$ and $(\tilde{h}, \tilde{\theta}) = \Lambda(\psi)$. For locality: by hypothesis, $\theta_x: R_{h(x)} \rightarrow \mathcal{O}_{T,x}$ is local, meaning $r \in h(x)$ iff $\theta_x(r) \in \mathfrak{m}_x$ iff $\psi(r)_x \in \mathfrak{m}_x$. Hence $h(x) = \tilde{h}(x)$, so $h = \tilde{h}$. Since h and $\tilde{h}$ agree, and the sheaf maps θ and $\tilde{\theta}$ are determined on the basis $\{D(g)\}$ by the global section map $\psi = \Theta(h, \theta)$ via Step 2 (a property of the localization construction), $\theta = \tilde{\theta}$. $\square$

Recall from Notation 5.5.6 that for a collection $\mathcal{M}$ of (geometric) objects, $\text{LocRingSpace}(\mathcal{M})$ denotes locally ringed spaces that locally look like objects in $\mathcal{M}$.

Theorem 6.3.6 (Schemes as Locally Ringed Spaces). *Let $\mathcal{P}rim$ be the collection of locally ringed spaces $(\text{Prim}(R), \mathcal{O}_{\text{Spec}(R)})$ for R any ring. Then the realization functor $(|-|, \mathcal{O}): \text{Fun}(\text{CAlg}, \text{Set}) \rightarrow \text{LocRingSpace}$ restricts to an equivalence of categories*

$$\text{Sch} \xrightarrow{\sim} \text{LocRingSpace}(\mathcal{P}rim).$$

Proof. By the realization–nerve adjunction (Theorem 1.1.10(vi)), the colimit-preserving functor $\Phi = (|-|, \mathcal{O})$ has a right adjoint

$$G: \text{LocRingSpace} \rightarrow \text{Fun}(\text{CAlg}, \text{Set}), \quad G(T, \mathcal{O}_T)(R) = \text{Hom}_{\text{LocRingSpace}}((\text{Prim}(R), \mathcal{O}_{\text{Spec}(R)}), (T, \mathcal{O}_T)).$$

Let η and ε denote the unit and counit. We show η_X is an isomorphism for every scheme X and ε_Y is an isomorphism for every $Y \in \text{LocRingSpace}(\mathcal{P}rim)$.

Step 1: Full faithfulness on schemes.

If $X = \text{Spec}(A)$, the unit η_X is an isomorphism by Theorem 6.3.4 and Remark 6.3.5. Let now X be a scheme with affine open cover $(U_i)_{i \in I}$. We show directly that, for every ring R , the map

$$\eta_X(R): X(R) \longrightarrow \text{Hom}_{\text{LocRingSpace}}(\Phi(\text{Spec}(R)), \Phi(X))$$

is a bijection.

Given a locally ringed space morphism $h: \Phi(\text{Spec}(R)) \rightarrow \Phi(X)$, the inverse images $h^{-1}(|U_i|)$ cover $\text{Prim}(R)$. Refine this cover by principal opens $D(f_a)$, choosing for each a an index $i(a)$ such that $D(f_a) \subset h^{-1}(|U_{i(a)}|)$. The restricted morphism

$$\Phi(D(f_a)) \longrightarrow \Phi(U_{i(a)})$$

comes from a unique morphism of affine schemes $D(f_a) \rightarrow U_{i(a)}$ by the affine case. On a double overlap the two realizations are both the restriction of h ; in particular, their images lie in $U_{i(a)} \cap U_{i(b)}$. Covering this target intersection and the source overlap by affine opens reduces equality of the two scheme maps to the affine case again. Hence the local maps agree and glue, because X is a Zariski sheaf, to a unique element of $X(R)$. Its realization is h , since morphisms of locally ringed spaces also glue over open covers. The same local argument shows uniqueness. Thus $\eta_X(R)$ is bijective for every R , naturally in R , and η_X is an isomorphism.

Step 2: Essential surjectivity. Let $Y = (T, \mathcal{O}_T) \in \text{LocRingSpace}(\mathcal{P}rim)$ and choose an open cover $(U_i)_{i \in I}$ with $U_i \simeq \Phi(\text{Spec}(R_i))$. Put $X = G(Y)$.

We first prove that $G(U_i) \rightarrow X$ is an open subfunctor, rather than inferring this merely from preservation of monomorphisms. Take an affine point $x: \text{Spec}(A) \rightarrow X$, corresponding to a locally ringed space morphism $h_x: \Phi(\text{Spec}(A)) \rightarrow Y$. The inverse image $h_x^{-1}(U_i)$ is an open subset of $\text{Prim}(A)$, hence corresponds to an open subfunctor $D(J_{x,i}) \subset \text{Spec}(A)$ for a radical ideal $J_{x,i} \subset A$. A morphism $\text{Spec}(B) \rightarrow \text{Spec}(A)$ factors through $D(J_{x,i})$ exactly when the associated locally ringed space morphism has image in U_i . Therefore the square

$$\begin{array}{ccc} D(J_{x,i}) & \longrightarrow & G(U_i) \\ \downarrow & \lrcorner & \downarrow \\ \text{Spec}(A) & \xrightarrow{x} & G(Y) \end{array}$$

is cartesian. This affine test proves that $G(U_i) \rightarrow X$ is open.

The family of these open subfunctors is Zariski-locally surjective. Indeed, after pullback along any $x: \text{Spec}(A) \rightarrow X$, the opens $h_x^{-1}(U_i)$ cover $\text{Spec}(A)$; quasi-compactness of the affine spectrum gives a finite principal refinement, and each member of that refinement lifts to the appropriate $G(U_i)$. By Step 1 in the affine case,

$$G(U_i) \simeq G(\Phi(\text{Spec}(R_i))) \simeq \text{Spec}(R_i).$$

Thus X has an affine open cover and is a scheme.

Finally consider the counit $\varepsilon_Y: \Phi(X) \rightarrow Y$. The affine-test calculation above identifies the inverse image of U_i with $\Phi(G(U_i))$, and on that open the counit is

$$\Phi(G(U_i)) \simeq \Phi(\text{Spec}(R_i)) \xrightarrow{\sim} U_i,$$

the affine counit. These open subsets cover both source and target and the local isomorphisms are restrictions of the same counit, so they glue to a global isomorphism ε_Y . This proves essential surjectivity. $\square$

6.4. Why Schemes Are Special

For an algebraic functor X , the category Mod_X of quasi-coherent modules (Definition 3.1.15) is by construction a limit of categories Mod_R over the category of elements, with transition functors given by base change f^* (Example 3.1.4).

For any ring R , Mod_R is a Grothendieck abelian category, and a limit of abelian categories with *exact* transition functors is again abelian. However, for a general ring map $\varphi: R \rightarrow S$, the base change $\varphi^* = (-) \otimes_R S$ is only right exact, not exact. Consequently, for a general algebraic functor X , the category Mod_X need not be abelian (kernels may not exist or behave well).

The situation improves dramatically for schemes. The key point: a scheme is glued from affines along open immersions, and an open immersion of affines $j: \text{Spec}(B) \hookrightarrow \text{Spec}(A)$ comes from a flat ring map $A \rightarrow B$ (e.g., a localization $B = A_f$). Hence the gluing functor $j^*: \text{Mod}_A \rightarrow \text{Mod}_B$ is exact, not merely right exact.

Using Zariski descent for quasi-coherent modules (Proposition 3.3.3), we can describe Mod_X locally; since the gluing functors are exact, we obtain favorable abelian-categorical properties.

Lemma 6.4.1 (Reducedness along affine open immersions). *Let $\text{Spec}(S) \hookrightarrow \text{Spec}(R)$ be an open immersion. If $I \subset R$ is radical, then $IS \subset S$ is radical.*

Proof. Base change gives an open immersion

$$\text{Spec}(S/IS) \hookrightarrow \text{Spec}(R/I).$$

The ring R/I is reduced. For every prime $\mathfrak{q} \subset S/IS$, the open immersion identifies $(S/IS)_{\mathfrak{q}}$ with a localization of R/I , hence with a reduced ring. If $\bar{s} \in S/IS$ is nilpotent, its image is therefore zero in every $(S/IS)_{\mathfrak{q}}$. An element of a module that vanishes at every prime localization is zero, so $\bar{s} = 0$. Thus S/IS is reduced and IS is radical. $\square$

Theorem 6.4.2 (Abelian Property of Mod_X). *Let X be a scheme, and let $(u_i: \text{Spec}(R_i) \hookrightarrow X)_{i \in I}$ be an affine open cover. Then:*

- (i) Mod_X is an abelian category.
- (ii) For any open immersion $j: U \hookrightarrow X$, the restriction functor $j^*: \text{Mod}_X \rightarrow \text{Mod}_U$ is exact.

- (iii) The family $(u_i^*)_{i \in I}$ is jointly exact and conservative. Consequently, a morphism f in Mod_X is a monomorphism (resp. epimorphism, isomorphism) iff each restriction $f|_{\text{Spec}(R_i)}$ is.
- (iv) A quasi-coherent module $M \in \text{Mod}_X$ is a vector bundle (resp. a line bundle) if and only if each $M(u_i)$ is a finitely generated projective R_i -module (resp. an invertible R_i -module).
- (v) There is an isomorphism of posets

$$\text{Id}_X \simeq \{\text{subobjects of } \mathcal{O}_X \text{ in } \text{Mod}_X\}.$$

- (vi) There is an isomorphism of posets

$$\text{Rad}_X \simeq \{K \in \text{Id}_X \mid \text{for all } i \in I, K(u_i) \subset R_i \text{ is radical}\}.$$

Proof. **(i) Local presentation as a limit of abelian categories with exact transitions.** Write $U_i = \text{Spec}(R_i)$. By Zariski descent (Proposition 3.3.3),

$$\text{Mod}_X \xrightarrow{\sim} \lim \left(\prod_i \text{Mod}_{U_i} \rightrightarrows \prod_{i,j} \text{Mod}_{U_i \cap U_j} \rightrightarrows \prod_{i,j,k} \text{Mod}_{U_i \cap U_j \cap U_k} \right).$$

Each pairwise intersection $U_i \cap U_j$ is open in the affine U_i , so admits an affine open cover by principal opens; iterating, we may refine the limit to one indexed by a diagram of affines in which every transition map comes from an open immersion $\text{Spec}(A_f) \hookrightarrow \text{Spec}(A)$.

Each such transition functor is base change along a localization $A \rightarrow A_f$, which is flat. Hence $\text{Mod}_A \rightarrow \text{Mod}_{A_f}$, $M \mapsto M \otimes_A A_f$, is exact. A limit of abelian categories along exact functors is abelian, with kernels, cokernels, and exact sequences computed termwise.

(ii) Exactness of j^* . Let $j: U \hookrightarrow X$ be an open immersion. Each intersection $U \cap U_i$ is open in U_i and so admits an affine cover by principal opens $\text{Spec}((R_i)_{f_{i,k}})_k$. Together these give an affine open cover of U , and the descent presentation of j^* reduces to base change along the family of flat localizations $R_i \rightarrow (R_i)_{f_{i,k}}$. Each such base change is exact, hence so is j^* .

(iii) Joint exactness and conservativity. The restriction $\prod_i u_i^*: \text{Mod}_X \rightarrow \prod_i \text{Mod}_{U_i}$ is the projection from the descent limit (Proposition 3.3.3) onto its first factor; in particular, a morphism in Mod_X is an isomorphism iff each component restriction is, so $\prod_i u_i^*$ is conservative. Each u_i^* is exact by (ii), so the product is exact. An exact and conservative functor preserves and detects monomorphisms, epimorphisms, and exact sequences.

(iv) Vector bundles. By definition, M is a vector bundle precisely when every affine pullback is a finite projective module, and it is a line bundle precisely when every affine pullback is invertible. It is enough to test this on the chosen affine cover because finite projectivity and invertibility are preserved by base change and descend along principal Zariski covers. Notice that an affine chart need not trivialize a projective module: finite projective is the correct conclusion on a fixed affine cover, while freeness holds only after a further Zariski refinement.

(v) and (vi) follow from the fact that, when X is a scheme, the forgetful functors $\text{Rad}_R \hookrightarrow \text{Id}_R \hookrightarrow (\text{Mod}_R)_{/R}$ are functorial in R along open immersions:

Claim 6.4.3. *The forgetful functors $\text{Rad}_R \hookrightarrow \text{Id}_R \hookrightarrow (\text{Mod}_R)_{/R}$ are functorial with respect to open immersions $\text{Spec}(S) \hookrightarrow \text{Spec}(R)$.*

Proof of Claim 6.4.3. Consider the diagram

$$\begin{array}{ccccc} \text{Rad}_R & \hookrightarrow & \text{Id}_R & \hookrightarrow & (\text{Mod}_R)_{/R} \\ \downarrow & (i) & \downarrow & (ii) & \downarrow \\ \text{Rad}_S & \hookrightarrow & \text{Id}_S & \hookrightarrow & (\text{Mod}_S)_{/S} \end{array}$$

We must verify that both (i) and (ii) commute.

For (i): given $I \in \text{Rad}_R$, the extension IS is radical by Lemma 6.4.1.

For (ii): this follows from open immersions being flat together with flat base change. $\square$

$\square$

Finally, we explain how to identify quasi-coherent modules over a scheme X with certain sheaves on the underlying topological space $|X|$.

Definition 6.4.4 (Quasi-coherent Sheaf). Let $(X, \mathcal{O}_X)$ be a ringed space and M an $\mathcal{O}_X$ -module in $\text{Shv}(X)$. We say that M is a *quasi-coherent sheaf* on $(X, \mathcal{O}_X)$ if it admits a presentation locally on X , i.e., if for every $x \in X$ there is an open neighborhood U of x and an exact sequence

$$\mathcal{O}_U^{(J)} \rightarrow \mathcal{O}_U^{(I)} \rightarrow M|_U \rightarrow 0$$

for some sets I and J .

When X is a scheme, the category of quasi-coherent modules over X is equivalent to the category of quasi-coherent sheaves on $(|X|, \mathcal{O}_X)$.

Proposition 6.4.5 (Quasi-coherent Modules as Sheaves). *Let X be a scheme. Then the functor*

$$\text{Mod}_X \rightarrow \text{Mod}_{\mathcal{O}_X}(\text{Shv}(|X|)), \quad M \mapsto (U \mapsto M(U))$$

— where $M(U) = \Gamma(U, M|_U)$ is the section module of Notation 3.1.21 — is fully faithful and identifies Mod_X with the category of quasi-coherent sheaves on $(|X|, \mathcal{O}_X)$.

Proof. Fix an affine open cover $(u_i: \text{Spec}(R_i) \hookrightarrow X)_{i \in I}$, and set $U_i = \text{Spec}(R_i)$. Over U_i the functor sends M to the sheaf $D(f) \mapsto M(u_i)_f$ on $|U_i|$, that is, to the classical sheaf $\widetilde{M(u_i)}$ associated to the R_i -module $M(u_i)$; by [The25, Tag 01IB] this is an equivalence $\text{Mod}_{R_i} \xrightarrow{\sim} \text{QCoh}(U_i)$ onto the quasi-coherent sheaves of Definition 6.4.4. Both sides of the asserted equivalence satisfy descent along the cover: the source by Proposition 3.3.3, and the target because $\mathcal{O}_X$ -modules glue along open covers and quasi-coherence is a local condition. As the functor restricts to the affine equivalence over each U_i and over each overlap $U_i \cap U_j$, it is an equivalence on descent data, hence on the limits, giving $\text{Mod}_X \xrightarrow{\sim} \text{QCoh}(|X|, \mathcal{O}_X)$. $\square$

This equivalence deserves to be read from two angles: *how* the two notions of quasi-coherence differ in spirit, and *what* each one computes on global sections. Throughout, write $\widetilde{M}$ for the sheaf $U \mapsto M(U)$ on $|X|$ attached to $M \in \text{Mod}_X$ by Proposition 6.4.5.

Remark 6.4.6 (Intrinsic versus Spatial Quasi-coherence). The two descriptions are built on opposite primitives — *points* on the left, *opens* on the right.

	Quasi-coherent module $M \in \text{Mod}_X$	Quasi-coherent sheaf $\tilde{M} \in \text{QCoh}(X , \mathcal{O}_X)$
Ambient category	the descent limit $\text{Mod}_X = \lim_{(R,x) \in \text{El}(X)^{\text{op}}} \text{Mod}_R$	a full subcategory of all $\mathcal{O}_X$ -modules $\text{Mod}_{\mathcal{O}_X}(\text{Shv}(X))$
Primitive datum	a module $M(x) \in \text{Mod}_R$ at each point $x: \text{Spec } R \rightarrow X$	a section module $\tilde{M}(U) \in \text{Mod}_{\mathcal{O}_X(U)}$ over each open U
Indexed over	$\text{El}(X)^{\text{op}}$, the probe category with arrows reversed for extension of scalars	the locale $\text{Open}(X)$ of opens
Glued by	base change $M(x) \otimes_R S \xrightarrow{\sim} M(y)$ and Zariski descent (Proposition 3.3.3)	restriction maps and the sheaf condition
Quasi-coherence is	<i>automatic</i> : descent is the definition	a <i>condition</i> : local presentability (Definition 6.4.4)
Available for	any algebraic functor X	any ringed space; agrees with the left column once X is a scheme

The left-hand description is affine-probe based: it records how M pulls back to every $x: \text{Spec}(R) \rightarrow X$. The opposite in $\text{El}(X)^{\text{op}}$ is essential, because a map of probes induces extension of scalars in the direction used by the limit diagram. Quasi-coherence — compatibility with this base change — is wired into the definition of Mod_X , so there is no non-quasi-coherent object to begin with. The right-hand description is spatial and contravariant on opens: it records the sections of $\tilde{M}$ over the opens of $|X|$, and quasi-coherence must be *imposed*, as the local-presentation condition of Definition 6.4.4 ([The25, Tag 01BD]) that carves $\text{QCoh}(|X|, \mathcal{O}_X)$ out of the much larger category of all $\mathcal{O}_X$ -modules. Proposition 6.4.5 says the two routes converge for schemes, and the bridge in both directions is evaluation on affine opens: an open probe $u: \text{Spec}(R) \hookrightarrow X$ and the section module over $U = \text{Spec}(R)$ carry the same R -module $M(u) = \tilde{M}(U)$. For a general algebraic functor only the left column survives — one reason the functorial definition is taken as primary.

We now compare what the two sides compute on global sections; reassuringly, they agree.

Proposition 6.4.7 (Global Sections in Both Descriptions). *Let X be a scheme and $M \in \text{Mod}_X$, with associated quasi-coherent sheaf $\tilde{M}$ on $(|X|, \mathcal{O}_X)$. Then the two global-section functors agree: there is a natural isomorphism of $\mathcal{O}(X)$ -modules*

$$\Gamma(X, M) \xrightarrow{\sim} \Gamma(|X|, \tilde{M}).$$

For any affine open cover $(u_i: \text{Spec}(R_i) \hookrightarrow X)_{i \in I}$, writing $U_i = \text{Spec}(R_i)$ and $U_{ij} = U_i \cap U_j$, this common value is computed by the equalizer

$$\Gamma(X, M) = \text{eq}\left(\prod_{i \in I} M(u_i) \rightrightarrows \prod_{i,j \in I} M(U_{ij})\right),$$

whose two arrows are the restrictions to double overlaps (on why the overlap term is the section module $M(U_{ij})$ and not a value $M(u_{ij})$ at a point, see Remark 6.4.8). In particular, for $X = \text{Spec}(R)$ affine the equalizer collapses and

$$\Gamma(\text{Spec}(R), M) = M(\text{id}_{\text{Spec}(R)}),$$

the R -module underlying M under $\text{Mod}_{\text{Spec}(R)} \simeq \text{Mod}_R$: global sections recover the module, so $M \mapsto \tilde{M}$ inverts Γ on affines.

Proof. By construction $\tilde{M}(U) = M(U)$ for every open $U \subseteq |X|$. Taking $U = |X|$ itself gives

$$\Gamma(|X|, \tilde{M}) = \tilde{M}(|X|) = M(X) = \Gamma(X, M),$$

the last equality being the definition of the global sections of a quasi-coherent module; this is the asserted isomorphism, manifestly natural in M and $\mathcal{O}(X)$ -linear.

It remains to compute this value from an affine cover. By definition $\Gamma(X, M) = \text{Hom}_{\text{Mod}_X}(\mathcal{O}_X, M)$. The cover (u_i) is a Zariski-local epimorphism, so Proposition 3.3.3 presents Mod_X as the descent limit

$$\text{Mod}_X \xrightarrow{\sim} \lim \left(\prod_i \text{Mod}_{U_i} \rightrightarrows \prod_{i,j} \text{Mod}_{U_{ij}} \rightrightarrows \cdots \right),$$

under which $\mathcal{O}_X$ and M correspond to the families $(\mathcal{O}_{U_i})_i$ and $(u_i^* M)_i$ with their canonical gluings. A morphism between two objects of a descent limit is exactly a family of morphisms over the U_i agreeing on the double overlaps U_{ij} , so

$$\text{Hom}_{\text{Mod}_X}(\mathcal{O}_X, M) = \text{eq} \left(\prod_i \text{Hom}_{\text{Mod}_{U_i}}(\mathcal{O}_{U_i}, u_i^* M) \rightrightarrows \prod_{i,j} \text{Hom}_{\text{Mod}_{U_{ij}}}(\mathcal{O}_{U_{ij}}, u_{ij}^* M) \right).$$

Each Hom out of the unit is a module of sections: $\text{Hom}_{\text{Mod}_{U_i}}(\mathcal{O}_{U_i}, u_i^* M) = \Gamma(U_i, u_i^* M) = M(u_i)$ since U_i is affine, and likewise the overlap terms are $\Gamma(U_{ij}, u_{ij}^* M) = M(u_{ij})$. This yields the displayed equalizer, which is visibly the sheaf condition for $\tilde{M}$ on the cover (U_i) of $|X|$, consistent with the identification above. Finally, for $X = \text{Spec}(R)$ the one-element cover by X itself collapses the equalizer to its single factor $\text{Hom}_{\text{Mod}_R}(R, M(\text{id}_{\text{Spec}(R)})) = M(\text{id}_{\text{Spec}(R)})$, recovering $\Gamma(\text{Spec}(R), \tilde{M}) = M$ (the affine case of Theorem A.3.1; cf. [The25, Tag 01I7]). $\square$

Remark 6.4.8 (Charts, Overlaps, and the Notation $M(U_{ij})$). The two factors of the equalizer in Proposition 6.4.7 are recorded asymmetrically on purpose. Both are sections over an open, $M(U) = \Gamma(U, M|_U)$ (Notation 3.1.21); on an affine chart this is the value at the point, $M(U_i) = M(u_i) \in \text{Mod}_{R_i}$ in the sense of Definition 3.1.15. Recording the overlap likewise as a single value $M(u_{ij})$ at an affine point $u_{ij}: \text{Spec}(R_{ij}) \hookrightarrow X$ is legitimate exactly when $U_{ij} = U_i \cap U_j$ is affine — which holds for *every* pair when X is separated (Remark 6.2.3). In general U_{ij} is only an open subscheme of the affine scheme U_i (hence is separated), but need not be quasi-compact or affine, and then carries no such single affine point. Refining U_{ij} by an affine cover $(W_k = \text{Spec}(R_{ijk}) \hookrightarrow U_{ij})_k$ recovers it as an equalizer,

$$M(U_{ij}) = \Gamma(U_{ij}, M|_{U_{ij}}) = \text{eq} \left(\prod_k M(W_k) \rightrightarrows \prod_{k,l} M(W_k \cap W_l) \right),$$

whose overlaps $W_k \cap W_l$ are themselves affine because U_{ij} is separated. Thus $M(U_{ij})$ is the refined value the point notation would suggest, written compactly as sections over the open U_{ij} ; a bare $M(u_{ij})$ presupposes U_{ij} affine. (The same reading applies to Proposition 6.4.5: the comparison functor $M \mapsto \tilde{M}$ sends M to $U \mapsto M(U)$ on *all* opens, the overlaps included.)

Remark 6.4.9 (Exactness, and Why Affine Is Special). As $\Gamma(X, -) = \text{Hom}_{\text{Mod}_X}(\mathcal{O}_X, -)$ is representable, it preserves all limits and is in particular *left exact*. On an affine scheme it is better still: Proposition 6.4.7 identifies $\Gamma(\text{Spec}(R), -)$ with the inverse of the equivalence $\text{Mod}_R \xrightarrow{\sim} \text{Mod}_{\text{Spec}(R)}$, so it is *exact*, and a short exact sequence of quasi-coherent modules stays exact on global sections.

Globally this fails. An epimorphism $M \twoheadrightarrow N$ in Mod_X — one surjective on every chart, $M(u_i) \twoheadrightarrow N(u_i)$ (Theorem 6.4.2(iii)) — need not be surjective on Γ , since a global section of N lifts only when the chartwise lifts can be made to agree on overlaps. The structure sheaf on $\mathbb{P}_k^1$ already shows the equalizer biting: with the cover of Example 6.1.12 by $\text{Spec}(k[t])$ and $\text{Spec}(k[s])$ glued along $s = t^{-1}$,

$$\Gamma(\mathbb{P}_k^1, \mathcal{O}) = \text{eq}(k[t] \times k[s] \rightrightarrows k[t, t^{-1}]) = k,$$

because a pair $(f(t), g(s))$ agrees on the overlap only when $f(t) = g(t^{-1})$, forcing both to a common constant. Thus $\mathbb{P}_k^1$ carries only constant global functions, even though each chart contributes the whole ring $k[t]$: the equalizer is strictly smaller than the product. More generally, for $n \geq 1$ and every $d \in \mathbb{Z}$ one has $\Gamma(\mathbb{P}_R^n, \mathcal{O}(d)) = R[T_0, \dots, T_n]_d$; the same formula holds for $n = 0$ when $d \geq 0$, whereas $\mathbb{P}_R^0 \simeq \text{Spec}(R)$ and $\Gamma(\mathbb{P}_R^0, \mathcal{O}(d)) = R$ for every d (see [The25, Tag 01XT]). The gap between the chartwise data

and the global sections is what the higher cohomology $H^{>0}(X, M)$ measures: affine charts are acyclic (Corollary A.3.2), so the equalizer above is the degree-zero part of the Čech complex of the cover ([The25, Tag 01X8]). This is the entry point to the cohomology of quasi-coherent sheaves, developed from the sheafification viewpoint in Chapter A.

Looking Ahead

The theory of schemes is now set up: Sch as a full subcategory of $\text{Shv}_{\text{Zar}}(\text{Aff})$ closed under immersions and many categorical constructions; gluing via locally trivial equivalence relations; separatedness, quasi-compactness, quasi-separatedness, and finite presentation/finite type as the natural finiteness conditions; the classical locally-ringed-space picture recovered as an equivalent viewpoint; and quasi-coherent modules forming an abelian category with good local properties.

With the foundations complete, Part II of these notes turns to the *geometry* of schemes — qualitative properties of schemes and their morphisms. The next chapter introduces the basic toolkit: local properties (reduced, noetherian, regular, normal), global properties (irreducible, integral, qcqs), and the dimension theory that underlies most geometric reasoning on schemes. It is followed by smooth, unramified, and étale morphisms; divisors and blowups; and finally properness, valuative criteria, projectivity, and finiteness.

Part II.

Algebraic Geometry II (Geometry of Schemes)

Chapter 7.

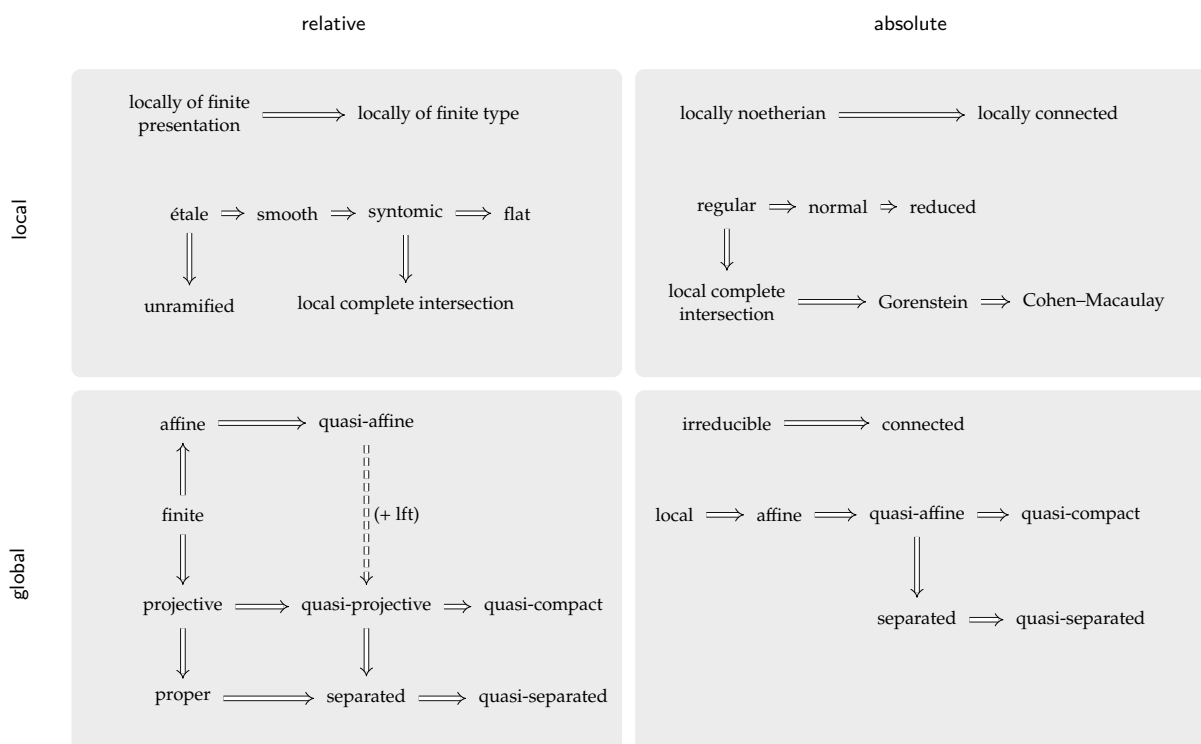
Properties of Schemes

7.1. Local Properties

To navigate the category of schemes one classifies its objects and morphisms by *properties*, organized along two orthogonal axes. A property is *local* when it can be checked in arbitrarily small Zariski neighborhoods of every point, and *global* otherwise; it is *absolute* when it pertains to a scheme on its own, and *relative* when it pertains to a morphism.

Local properties of schemes are entirely controlled by commutative algebra: they are determined by their restrictions to affine opens. Typical examples are “locally of finite presentation” (Definition 6.2.19), “reduced” (Definition 7.2.1), “locally noetherian” (Definition 7.3.4), regularity, and normality. Affineness, connectedness, and quasi-compactness are global. On the relative side, projectivity is a basic example, depending intrinsically on the base.

The following chart is a roadmap for *Part II*, not just for the present chapter. It lists the properties used throughout this part and organizes them by the two orthogonal axes “local / global” and “relative / absolute”. Every arrow is one-way. This chapter develops the absolute local properties through normality; the subsequent chapters define the relative properties and justify the remaining arrows.



In the lower-right panel, “local” means “the spectrum of a local ring”; it is an absolute global condition, not the adjective in “local property.”

Below the diagram, we record three convenient abbreviations that mix local and global:

of finite presentation	=	lfp + qcqs
of finite type	=	lft + qc
noetherian	=	locally noetherian + qc
integral	=	reduced + irreducible

Remark 7.1.1 (From Relative to Absolute). Every relative property has an underlying absolute version, since every scheme is canonically a $\mathrm{Spec}(\mathbb{Z})$ -scheme. The absolute versions of “affine”, “quasi-affine”, “quasi-compact”, “quasi-separated”, and “separated” are obtained in this way and are well-behaved. For other relative properties such as smoothness or properness the resulting absolute property is too restrictive to be useful (e.g. a nonempty $\mathbb{C}$ -scheme can never be smooth or proper over $\mathrm{Spec}(\mathbb{Z})$).

The bridge from commutative algebra to scheme theory is a single bootstrapping principle: any reasonable property of rings extends uniquely to schemes via Zariski covers. We now make “reasonable” precise.

Definition 7.1.2 (Local Property of Rings). A property P of (commutative) rings is *local* if:

- (i) for every R with property P and every $f \in R$, the localization R_f has property P ;
- (ii) whenever $f_1, \dots, f_n \in R$ generate the unit ideal and each R_{f_i} has property P , the ring R has property P .

Definition 7.1.3 (Local Property of Schemes). A property P of schemes is *local* if:

- (i) for every scheme X with property P and every open subscheme $U \subset X$, the scheme U has property P ;
- (ii) whenever $(U_i \subset X)_{i \in I}$ is an open cover and each U_i has property P , the scheme X has property P .

Remark 7.1.4 (Local Properties as Zariski Sheaves). There is a synthetic way to view Definition 7.1.3. Condition (i) says that, for every scheme X , the property assembles into a functor

$$\mathrm{Open}(X)^{\mathrm{op}} \xrightarrow{P} \{\perp \rightarrow \top\} \subset \mathrm{Set},$$

where $\perp = \emptyset$ encodes *false* and $\top = *$ encodes *true*. Condition (ii) then says that, for every X , this functor is a Zariski sheaf on $\mathrm{Open}(X)$.

The following bootstrapping result is the engine that produces local properties of schemes from properties of rings.

Proposition 7.1.5 (Extending Absolute Local Properties). Every local property P of rings extends uniquely to a local property $\widehat{P}$ of schemes such that $\widehat{P}(\mathrm{Spec}(R)) \iff P(R)$.

Proof. Define $\widehat{P}$ on schemes by declaring that, for every scheme X , $\widehat{P}(X)$ holds if and only if there exists an affine open cover $(\mathrm{Spec}(R_i) \hookrightarrow X)_i$ with $P(R_i)$ for every i . We must verify:

- (i) $\widehat{P}(\mathrm{Spec}(R)) \iff P(R)$;
- (ii) $\widehat{P}$ is a local property of schemes.

For (i), the implication $P(R) \Rightarrow \widehat{P}(\text{Spec}(R))$ is immediate from the singleton cover. For the converse, suppose $\widehat{P}(\text{Spec}(R))$ is witnessed by an affine open cover $(\text{Spec}(R_i) \hookrightarrow \text{Spec}(R))_i$ with $P(R_i)$ for each i . The principal opens $\{D(f)\}_{f \in R}$ form a basis of $\text{Open}(\text{Spec}(R))$, so the subfamily

$$\{D(f) \mid \exists i, D(f) \subset \text{Spec}(R_i)\}$$

is again an open cover of $\text{Spec}(R)$. Since $\text{Spec}(R)$ is quasi-compact, finitely many such basic opens $D(f_1), \dots, D(f_n)$ already cover $\text{Spec}(R)$, equivalently $(f_1, \dots, f_n) = R$. By Definition 7.1.2(ii) it suffices to show $P(R_{f_j})$ for every j . Fix j and choose i with $D(f_j) \subset \text{Spec}(R_i)$; then $R_{f_j} \cong (R_i)_{f_j}$, and Definition 7.1.2(i) applied to $P(R_i)$ yields $P((R_i)_{f_j}) = P(R_{f_j})$. Hence $P(R)$.

For (ii), combine Remark 7.1.4, Proposition 5.6.2, and the fact that $\text{Open}^{\text{aff}}(X) \subset \text{Open}(X)$ is a dense subcategory. $\square$

We now turn to relative versions. For morphisms $f : Y \rightarrow X$ there are two distinct “axes” along which a property can be local: along the source Y and along the base X .

Definition 7.1.6 (Locality for Morphisms of Schemes). Let P be a property of morphisms of schemes.

- (i) P is *local on the source* if for every morphism $f : Y \rightarrow X$:
 - if f has property P and $V \subset Y$ is open, then $f|_V : V \rightarrow X$ has property P ;
 - if $(V_i \subset Y)_{i \in I}$ is an open cover and each $f|_{V_i}$ has property P , then so does f .
- (ii) P is *local on the base* if for every morphism $f : Y \rightarrow X$:
 - if f has property P and $U \subset X$ is open, then the pullback $f_U : Y \times_X U \rightarrow U$ has property P ;
 - if $(U_i \subset X)_{i \in I}$ is an open cover and each f_{U_i} has property P , then so does f .
- (iii) P is *local* if it is local both on the source and on the base.
- (iv) P is *stable under base change* if for every cartesian square

$$\begin{array}{ccc} Y' & \longrightarrow & Y \\ f' \downarrow & \lrcorner & \downarrow f \\ X' & \longrightarrow & X \end{array}$$

the property $P(f)$ implies $P(f')$.

- (v) P is *stable under composition* if it contains all isomorphisms and if $g \circ f$ has property P whenever both f and g do.

Definition 7.1.6 has an evident analogue for morphisms of rings (just apply Spec). The relative version of the extension proposition is then:

Proposition 7.1.7 (Extending Relative Local Properties). Let P be a property of morphisms of (commutative) rings.

- (i) If P is local on the base, then P extends uniquely to a property $\widehat{P}$ of affine morphisms of schemes that is local on the base.
- (ii) If P is local, then P extends uniquely to a local property $\widehat{P}$ of morphisms of schemes.

In both cases, stability under base change and stability under composition are inherited by $\widehat{P}$.

Proof. For (i), recall that a morphism $f: Y \rightarrow X$ of schemes is *affine* if for every map $\text{Spec}(R) \rightarrow X$ the pullback $Y \times_X \text{Spec}(R)$ is an affine scheme (Proposition 3.2.15). Define

$$\widehat{P}(f) : \iff \text{there is an affine open cover } (\text{Spec}(R_i) \hookrightarrow X)_i \text{ with } P(R_i \rightarrow \mathcal{O}(Y \times_X \text{Spec}(R_i))) \text{ for all } i.$$

This reduces to Proposition 7.1.5 applied to X , with the following adaptation. Affineness of f ensures that each pullback $Y \times_X \text{Spec}(R_i)$ is again affine, say $\text{Spec}(S_i)$ for an R_i -algebra S_i ; refining the cover to a basic open $D(g) \subset \text{Spec}(R_i)$ replaces the pullback by $\text{Spec}((S_i)_g)$, and locality of P on the base relates $P(R_i \rightarrow S_i)$ to $P((R_i)_g \rightarrow (S_i)_g)$. The cover-refinement step in Proposition 7.1.5 then transports to the present setting and yields well-definedness and locality on the base.

For (ii), apply the construction of (i) on the source as well: $\widehat{P}(f)$ holds iff there exist affine open covers $(\text{Spec}(R_i) \hookrightarrow X)_i$ and $(\text{Spec}(S_{ij}) \hookrightarrow Y \times_X \text{Spec}(R_i))_j$ with $P(R_i \rightarrow S_{ij})$ for all i, j . Independence of the source cover and locality on the source follow by repeating the argument of (i) on Y , using locality of P on the source in place of locality on the base.

Stability under base change: given $\widehat{P}(f)$ and a cartesian square as in Definition 7.1.6(iv), pull the witnessing cover $(\text{Spec}(R_i) \hookrightarrow X)_i$ back to a cover $(\text{Spec}(R_i) \times_X X' \hookrightarrow X')_i$, refine to an affine cover, and apply stability of P under base change on each piece.

Stability under composition: given $\widehat{P}(f)$ and $\widehat{P}(g)$ for $f: Y \rightarrow X$ and $g: X \rightarrow Z$, refine the cover witnessing $\widehat{P}(g)$ to an affine open cover $(\text{Spec}(T_k) \hookrightarrow Z)_k$, pull back the cover witnessing $\widehat{P}(f)$ to each $\text{Spec}(T_k)$, and apply stability of P under composition on each affine piece. $\square$

Remark 7.1.8 (Zariski-Local = Local on the Base). A property of R -algebras is *Zariski-local* in the sense of Definition 1.3.8 if and only if it is local on the base in the sense of Definition 7.1.6. Both notions express the same descent statement: locality of the property along Zariski covers of the target.

Remark 7.1.9 (Source-Locality Often Suffices). Most relative properties of practical interest are stable under base change and composition and hold for open immersions. For such P , locality on the source implies locality on the base: given a base cover $(U_i \subset X)_i$, pull it back to $(Y \times_X U_i \subset Y)_i$, an open cover of Y . If P holds for each base change $f_{U_i}: Y \times_X U_i \rightarrow U_i$, then it also holds for each source restriction $f|_{Y \times_X U_i}: Y \times_X U_i \rightarrow X$ — the latter being f_{U_i} followed by the open immersion $U_i \hookrightarrow X$ — whence f has P by source-locality. The converse $P(f) \Rightarrow P(f_{U_i})$ is base change.

Example 7.1.10 (Local Finiteness Conditions). “Locally of finite presentation” and “locally of finite type” (Definition 1.4.1) are local properties of morphisms of rings, so by Proposition 7.1.7 they extend uniquely to local properties of morphisms of schemes. Unwinding the construction: $f: Y \rightarrow X$ is locally of finite presentation (resp. finite type) iff every $y \in |Y|$ admits affine open neighborhoods $V \ni y$ in Y and $U \ni f(y)$ in X with $f(V) \subset U$ such that $\mathcal{O}(V)$ is a finitely presented (resp. finitely generated) $\mathcal{O}(U)$ -algebra.

Example 7.1.11 (Flatness). Flatness is stable under base change and composition. It is moreover a local property: in fact, $\varphi: R \rightarrow S$ is flat if and only if, for every prime ideal $\mathfrak{q} \subset S$, the induced map of local rings $R_{\varphi^{-1}(\mathfrak{q})} \rightarrow S_{\mathfrak{q}}$ is flat (this follows immediately from the fact that, for any ring R , the functor $\text{Mod}_R \rightarrow \prod_{\mathfrak{p} \in \text{Spec}(R)} \text{Mod}_{R_{\mathfrak{p}}}$ is exact and conservative by Example 6.3.2). Hence, by Proposition 7.1.7, flatness extends uniquely to a local property of morphisms of schemes, which has the following characterization: $f: Y \rightarrow X$ is flat if and only if for every $y \in |Y|$, the map of local rings $f^*: \mathcal{O}_{X, f(y)} \rightarrow \mathcal{O}_{Y, y}$ is flat.

Example 7.1.12 (Properties Local on the Base). The following properties of morphisms of schemes are local on the base, but generally *not* on the source:

- (i) affine;
- (ii) integral;

- (iii) finite;
- (iv) closed immersion;
- (v) open immersion;
- (vi) immersion;
- (vii) quasi-affine;
- (viii) quasi-compact;
- (ix) quasi-separated;
- (x) separated;
- (xi) of finite presentation;
- (xii) of finite type.

All twelve are stable under base change and composition. Properties that fail to be local on the base are typically global in flavour: “projective” and “quasi-projective” are the standard examples. Properness, the local-on-base sibling of projectivity, repairs this defect.

7.2. Reduced Schemes

A ring R is *reduced* when its nilradical vanishes: $\text{Nil}(R) = \sqrt{0} = 0$. The *reduction* $R^{\text{red}} := R/\text{Nil}(R)$ is the universal reduced quotient. We extend both notions to schemes.

Pictorially, $\text{Spec}(k)$ is a single point; $\text{Spec}(k[\varepsilon]/\varepsilon^2)$ is that same point together with a *first-order infinitesimal thickening*, invisible to classical set-theoretic geometry but carrying nonzero nilpotents. Reducedness is precisely the absence of such thickenings.



Definition 7.2.1 (Reduced Scheme). A scheme X is *reduced* if there exists an affine open cover $(U_i \hookrightarrow X)_{i \in I}$ such that each $\mathcal{O}(U_i)$ is reduced.

Lemma 7.2.2 (Reduced is Local for Rings). The property “reduced” of rings is local in the sense of Definition 7.1.2.

Proof. For (i), localization commutes with the nilradical: $\text{Nil}(R_f) = \text{Nil}(R) \cdot R_f$, so $\text{Nil}(R) = 0$ forces $\text{Nil}(R_f) = 0$.

For (ii), suppose $(f_1, \dots, f_n) = R$ and each R_{f_i} is reduced. Given a nilpotent $a \in R$, its image in each R_{f_i} is also nilpotent, hence zero by reducedness. Now Zariski descent for elements applies: when (f_i) generates the unit ideal the canonical map $R \hookrightarrow \prod_i R_{f_i}$ is injective, so $a = 0$. $\square$

Proposition 7.2.3 (Reduced is Local for Schemes). The property “reduced” is a local property of schemes. Moreover, $\text{Spec}(R)$ is reduced if and only if R is reduced.

Proof. Apply Proposition 7.1.5 to Lemma 7.2.2. $\square$

Remark 7.2.4. (i) A scheme X is reduced if and only if, for every point $x \in |X|$, the local ring $\mathcal{O}_{X,x}$ is reduced.

- (ii) If X is a reduced scheme, then the ring $\mathcal{O}(X)$ is reduced. Indeed, if $(U_i \subset X)_{i \in I}$ is an open cover of X by reduced affine schemes, then $\mathcal{O}(X)$ is a subring of the reduced ring $\prod_{i \in I} \mathcal{O}(U_i)$.

Example 7.2.5 (Reduced Proj). Let A be an $\mathbb{N}$ -graded ring and $I \subset A_+$ a homogeneous subset with $A_+ \subset \sqrt{(I)}$. By Theorem 2.3.6 the affines $\text{Spec}(A_{(f)})$ with $f \in I$ cover $\text{Proj}(A)$, so reducedness of $\text{Proj}(A)$ amounts to reducedness of every $A_{(f)}$ with $f \in I$.

A clean sufficient condition is that A itself be reduced: then each localization A_f is reduced, and so is its degree-zero part $A_{(f)}$. In particular, if R is reduced then $\mathbb{P}_R^n$ is reduced for every $n \geq 0$ (with the convention $\mathbb{P}^{-1} = \emptyset$, the statement also holds trivially for $n = -1$).

For the next results we use Theorem 6.4.2, in particular the forgetful functor $\text{Rad}_X \hookrightarrow \text{Id}_X$, which expresses that “radical” is a well-defined property of quasi-coherent ideals on X : if $K \in \text{Id}_X$ and $(x_i: \text{Spec}(R_i) \hookrightarrow X)_{i \in I}$ is an affine open cover of X , then K is radical if and only if each $K(x_i) \subset R_i$ is radical.

Proposition 7.2.6 (Classification of Reduced Closed Subschemes). *Let X be a scheme. A closed subscheme $Z \hookrightarrow X$ is reduced if and only if the quasi-coherent ideal $I \in \text{Id}_X$ with $V(I) = Z$ is radical.*

Proof. Choose an affine open cover $(u_i: \text{Spec}(R_i) \hookrightarrow X)_{i \in I}$. Then $(U_i \cap Z \hookrightarrow Z)_{i \in I}$ is an open cover of Z , and $U_i \cap Z \simeq \text{Spec}(R_i/I(u_i))$. Since reducedness is a local property of schemes (Proposition 7.2.3), Z is reduced if and only if each $R_i/I(u_i)$ is reduced, i.e. each $I(u_i) \subset R_i$ is radical. By the local characterization of radical ideals above, this is equivalent to $I \in \text{Rad}_X$. $\square$

Corollary 7.2.7 (Closed-Reduced Bijection). *For any scheme X the composition*

$$\text{Closed}^{\text{red}}(X) \hookrightarrow \text{Closed}(X) \xrightarrow{|\cdot|} \text{Closed}(|X|)$$

is a bijection. In particular, the second map is surjective: every closed subset of $|X|$ is the underlying set of a unique reduced closed subscheme.

Proof. Three identifications combine. First, Proposition 7.2.6 gives $\text{Closed}^{\text{red}}(X) \simeq \text{Rad}_X$, since reducedness of a closed subscheme corresponds to radicality of its defining ideal. Second, the classification of closed subfunctors (Proposition 3.2.1), applied locally on an affine cover, gives $\text{Closed}(X) \simeq \text{Id}_X$. Finally, the realization $|\cdot|$ identifies Rad_X with $\text{Closed}(|X|)$: the locale of radical ideals of a ring is spatial (Theorem 4.1.1), and realization commutes with Zariski gluing. $\square$

Remark 7.2.8 (Subscheme Diagram). Combining Corollary 7.2.7 with the classification of open and closed subfunctors (Proposition 3.2.1) and the complementarity $U \leftrightarrow X \setminus U$ gives the following commutative diagram of order isomorphisms, where “ c ” denotes complementation. Note that Rad_X appears twice: on the left it parametrizes closed subschemes (via V), on the right it parametrizes open subschemes (via D , the complement of V).

$$\begin{array}{ccccc}
 & & \text{Closed}(|X|) & \xrightarrow{c} & \text{Open}(|X|) \\
 & \nearrow \simeq & \uparrow |\cdot| & & \uparrow \simeq \\
 \text{Closed}^{\text{red}}(X)^{\text{op}} & \hookrightarrow & \text{Closed}(X)^{\text{op}} & \xrightarrow{c} & \text{Open}(X) \\
 \uparrow \simeq & & \uparrow \simeq & & \uparrow \simeq \\
 \text{Rad}_X & \hookrightarrow & \text{Id}_X & \xrightarrow{\sqrt{}} & \text{Rad}_X
 \end{array}$$

Example 7.2.9 (Reduced Closed Subschemes of Proj). Let A be an $\mathbb{N}$ -graded ring. By Corollary 7.2.7 and Example 7.2.5, reduced closed subschemes of $\text{Proj}(A)$ are in bijection with closed subsets of $|\text{Proj}(A)|$, hence with saturated radical homogeneous ideals $I \subset A$ via the projective Nullstellensatz (Corollary 2.3.28):

$$\{\text{saturated radical homogeneous ideals of } A\} \xrightarrow{\sim} \text{Closed}^{\text{red}}(\text{Proj}(A)), \quad I \mapsto V(I).$$

Definition 7.2.10 (Reduction of a Scheme). Let X be a scheme. The *reduction* of X is the closed subscheme

$$X^{\text{red}} := V(\sqrt{0}) \subset X$$

where $\sqrt{0} \in \text{Rad}_X$ is the radical ideal corresponding to the entire underlying space $|X|$.

Proposition 7.2.11 (Reduction is a Right Adjoint). Let $\text{Sch}^{\text{red}} \subset \text{Sch}$ denote the full subcategory of reduced schemes. Then the inclusion $\text{Sch}^{\text{red}} \hookrightarrow \text{Sch}$ has a right adjoint given by $X \mapsto X^{\text{red}}$.

Proof. Concretely, we must show that for every reduced scheme Y and every morphism $f: Y \rightarrow X$, the postcomposition map

$$\text{Hom}_{\text{Sch}}(Y, X^{\text{red}}) \longrightarrow \text{Hom}_{\text{Sch}}(Y, X)$$

induced by the closed immersion $X^{\text{red}} \hookrightarrow X$ is a bijection: f factors uniquely through $X^{\text{red}} = V(\sqrt{0})$.

Such a factorization is equivalent to vanishing of the pullback ideal $f^*(\sqrt{0}) \in \text{Id}_Y$. Compute this pullback locally: on an affine open $\text{Spec}(S) \hookrightarrow Y$ mapping into $\text{Spec}(R) \hookrightarrow X$, it is the ideal $\sqrt{0_R} \cdot S$, which is contained in $\sqrt{0_S} \subset S$ (being generated by images of nilpotents). Since Y is reduced, every S in any affine cover is reduced, so $\sqrt{0_S} = 0$. Locality of Id_Y then promotes this to $f^*(\sqrt{0}) = 0$ globally, producing the desired unique factorization. $\square$

Remark 7.2.12 (Same Underlying Space, Smaller Structure Sheaf). The closed immersion $X^{\text{red}} \hookrightarrow X$ is an isomorphism on underlying topological spaces, since the open complement is $D(\sqrt{0}) = D(0) = \emptyset$. As a locally ringed space, X^{red} has the form $(|X|, \mathcal{O}_X^{\text{red}})$ where $\mathcal{O}_X^{\text{red}}(U) = \Gamma(U^{\text{red}}, \mathcal{O}_{U^{\text{red}}})$. The sheaf $\mathcal{O}_X^{\text{red}}$ is the sheafification of the presheaf $U \mapsto \Gamma(U, \mathcal{O}_X)^{\text{red}}$: on affine opens both presheaves agree, and “reduce-then-sheafify” equals “sheafify-then-take-the-corresponding-quotient”.

7.3. Noetherian Schemes

A ring R is *noetherian* when every ascending chain of ideals stabilizes; passing through V , this is equivalent to the descending chain condition (DCC) on closed subschemes of $\text{Spec}(R)$. We extend both formulations to schemes.

Proposition 7.3.1 (Characterizations of Noetherian Rings). For a ring R , the following are equivalent:

- (i) R is noetherian.
- (ii) Every submodule of a finitely generated R -module is finitely generated.
- (iii) The category Mod_R^{ft} of finitely generated R -modules is abelian and the inclusion $\text{Mod}_R^{\text{ft}} \hookrightarrow \text{Mod}_R$ is exact.
- (iv) Every R -module of finite type is of finite presentation: $\text{Mod}_R^{\text{ft}} = \text{Mod}_R^{\text{fp}}$.
- (v) Every R -algebra of finite type is of finite presentation: $\text{CAlg}_R^{\text{ft}} = \text{CAlg}_R^{\text{fp}}$.

Proof. We establish the chain $(i) \Leftrightarrow (ii) \Leftrightarrow (iii), (ii) \Leftrightarrow (iv) \Rightarrow (v) \Rightarrow (i)$.

For $(i) \Rightarrow (ii)$, take M finitely generated and $N \subset M$. Choose a surjection $R^n \twoheadrightarrow M$ and pull N back to a submodule $N' \subset R^n$; it suffices to show N' is finitely generated. Induct on n : the short exact sequence $0 \rightarrow R^{n-1} \rightarrow R^n \rightarrow R \rightarrow 0$ gives a short exact sequence $0 \rightarrow N' \cap R^{n-1} \rightarrow N' \rightarrow J \rightarrow 0$ with $J \subset R$ an ideal. ACC on ideals makes J finitely generated, the inductive hypothesis makes $N' \cap R^{n-1}$ finitely generated, and hence so is N' . The converse $(ii) \Rightarrow (i)$ is the case $M = R$.

The equivalence $(ii) \Leftrightarrow (iii)$ unfolds as follows: that $\text{Mod}_R^{\text{ft}} \hookrightarrow \text{Mod}_R$ is exact is the statement that finitely generated submodules and quotients of finitely generated modules are again finitely generated; the quotient direction is automatic, and the submodule direction is exactly (ii).

For $(ii) \Leftrightarrow (iv)$: if M is of finite type, choose a surjection $R^n \twoheadrightarrow M$. Its kernel is finitely generated by (ii), so M is finitely presented. The reverse runs the same argument backwards.

For $(iv) \Rightarrow (v)$, let A be a finite-type R -algebra, presented as $R[x_1, \dots, x_n] \twoheadrightarrow A$. The kernel is an ideal in $R[x_1, \dots, x_n]$, which is noetherian by Hilbert's basis theorem (Remark 7.3.3); hence the kernel is finitely generated and A is finitely presented.

For $(v) \Rightarrow (i)$, let $I \subset R$ be an arbitrary ideal. The quotient R/I is a finite-type R -algebra (presented by $R[\] \twoheadrightarrow R/I$), so by (v) it is finitely presented over R . A finite presentation of R/I as an R -algebra exhibits I as the ideal generated by finitely many elements. Hence ACC on ideals. $\square$

Remark 7.3.2 (Noetherian Rings in Practice). Most rings of geometric origin are noetherian — for instance, algebras of finite type over a field. An interesting non-noetherian example is the valuation ring $\mathcal{O}_{\mathbb{C}_p}$ of the p -adic complex numbers $\mathbb{C}_p$, whose prime spectrum consists of two points:



The closed point $\mathfrak{m}$ is the maximal ideal of elements of positive valuation; the generic point (0) is the zero ideal.

Remark 7.3.3 (Hilbert's Basis Theorem). Hilbert's basis theorem says that $R[x_1, \dots, x_n]$ is noetherian whenever R is. It is the input used to prove condition (v) of Proposition 7.3.1: if a finite-type R -algebra is presented as a quotient of such a polynomial ring, then the presentation ideal is finitely generated, so the algebra is finitely presented over R .

Definition 7.3.4 (Locally Noetherian, Noetherian Scheme). A scheme X is *locally noetherian* if there is an affine open cover $(\text{Spec}(R_i) \hookrightarrow X)_{i \in I}$ with each R_i noetherian. A scheme is *noetherian* if it is locally noetherian and quasi-compact.

Lemma 7.3.5 (Noetherian is Local for Rings). *The property “noetherian” of rings is local in the sense of Definition 7.1.2.*

Proof. For (i), localizations of noetherian rings are noetherian: every ideal of R_f is the extension of an ideal of R , and extension preserves inclusion, so any ascending chain in R_f pulls back to one in R that stabilizes by ACC.

For (ii), suppose $(f_1, \dots, f_n) = R$ and each R_{f_i} is noetherian. Given an ascending chain $J_0 \subset J_1 \subset \dots$ in R , each localized chain $J_0 R_{f_i} \subset J_1 R_{f_i} \subset \dots$ stabilizes at some index N_i . Let $N = \max_i N_i$. For $k \geq N$ the quotient J_k/J_N vanishes after every localization $R \rightarrow R_{f_i}$, and since (f_i) generates the unit ideal, Zariski descent for modules (Example 3.1.14) forces $J_k/J_N = 0$, i.e. $J_k = J_N$. $\square$

Proposition 7.3.6 (Locally Noetherian is Local for Schemes). *“Locally noetherian” is a local property of schemes. Moreover, $\text{Spec}(R)$ is locally noetherian if and only if R is noetherian.*

Proof. Apply Proposition 7.1.5 to Lemma 7.3.5. $\square$

Example 7.3.7 (Noetherian Proj). Let A be an $\mathbb{N}$ -graded ring and $I \subset A_+$ a homogeneous subset with $A_+ \subset \sqrt{(I)}$. Then:

- $\text{Proj}(A)$ is *locally noetherian* iff each $A_{(f)}$ with $f \in I$ is noetherian;
- $\text{Proj}(A)$ is moreover *quasi-compact* iff some *finite* such I exists.

If A itself is noetherian, both conditions are automatic, but one must not argue that an arbitrary degree-zero subring of a noetherian graded ring is noetherian. Instead, $A_0 = A/A_+$ is noetherian, and finite homogeneous generators of the ideal A_+ generate A as an A_0 -algebra (induct on degree). Fix a nonzero homogeneous f of degree d . The Veronese ring

$$A^{(d)} = \bigoplus_{m \geq 0} A_{md}$$

is a finitely generated graded A_0 -algebra: if A is generated by homogeneous elements $a_1, \dots, a_r$, then $A^{(d)}$ is generated by the a_i^d together with the finitely many monomials $a_1^{e_1} \cdots a_r^{e_r}$ for which every $e_i < d$ and the weighted degree is divisible by d . Moreover, f has degree 1 in $A^{(d)}$, and

$$A_{(f)} = (A_f^{(d)})_0.$$

If $b_1, \dots, b_r$ are homogeneous A_0 -algebra generators of $A^{(d)}$, then the finitely many fractions $b_i/f^{\deg(b_i)}$ generate this degree-zero ring over A_0 . Hence $A_{(f)}$ is a finite-type algebra over the noetherian ring A_0 , and is noetherian. Finally, A_+ is finitely generated as an ideal, so finitely many standard opens cover $\text{Proj}(A)$. Thus $\text{Proj}(A)$ is noetherian. In particular, $\mathbb{P}_R^n$ is noetherian whenever R is, for every $n \geq 0$.

Remark 7.3.8 (Coherent Rings). A ring R is *coherent* if the inclusion $\text{Mod}_R^{\text{fp}} \hookrightarrow \text{Mod}_R$ is closed under kernels — equivalently, Mod_R^{fp} is an abelian subcategory of Mod_R , and the inclusion is exact. Noetherian rings are coherent by Proposition 7.3.1(iii), but the converse fails: every valuation ring is coherent, and so is the polynomial ring $R[x_i \mid i \in I]$ for any noetherian R and any (possibly infinite) set I . “Coherent” is local, so by Proposition 7.1.5 it extends to a local property of schemes.

Proposition 7.3.9 (Noetherian iff DCC on Closed Subschemes). A scheme X is noetherian if and only if every descending chain $Z_0 \supset Z_1 \supset Z_2 \supset \cdots$ of closed subschemes in X stabilizes.

Proof. Suppose first that X is noetherian, and pick a finite affine open cover $(\text{Spec}(R_i))_{i=1}^n$ with each R_i noetherian. Given a descending chain $(Z_k)_k$ of closed subschemes, each restriction $Z_k \cap \text{Spec}(R_i)$ corresponds to an ascending chain of ideals in R_i , which stabilizes at some index N_i . Setting $N = \max_i N_i$, every restriction $Z_k \cap \text{Spec}(R_i)$ for $k \geq N$ already equals $Z_N \cap \text{Spec}(R_i)$, and Zariski descent for closed subschemes promotes this to $Z_k = Z_N$ globally.

Conversely, assume DCC for closed subschemes of X . We must show X is quasi-compact and that every affine open $\text{Spec}(R) \subset X$ has R noetherian.

For quasi-compactness, let $(U_\alpha)_{\alpha \in A}$ be any open cover, with closed complements $Z_\alpha = X \setminus U_\alpha$. Suppose for contradiction that no finite subfamily of (U_α) covers X , i.e. for every finite $F \subset A$ the intersection $Z_F := \bigcap_{\alpha \in F} Z_\alpha$ is nonempty. Build a strictly descending sequence inductively. Pick any $\alpha_0 \in A$ and set $F_0 = \{\alpha_0\}$, so $Z_{F_0} \neq \emptyset$. Having constructed F_k with $Z_{F_k} \neq \emptyset$, pick a point $p \in Z_{F_k}$. Since (U_α) covers X , there exists α_{k+1} with $p \in U_{\alpha_{k+1}}$, equivalently $p \notin Z_{\alpha_{k+1}}$. Then $F_{k+1} := F_k \cup \{\alpha_{k+1}\}$ satisfies $Z_{F_{k+1}} \subsetneq Z_{F_k}$, since the point $p \in Z_{F_k}$ is excluded from $Z_{F_{k+1}}$. The resulting strictly descending chain $Z_{F_0} \supsetneq Z_{F_1} \supsetneq \cdots$ contradicts DCC. Hence some finite subfamily covers X .

The same DCC applied to reduced closed subschemes shows that $|X|$ has DCC on closed subsets. Consequently every open subspace of $|X|$ is quasi-compact: the usual finite-subcover argument just given

applies inside any subspace. In particular, for an affine open immersion $j: U = \operatorname{Spec}(R) \hookrightarrow X$, the map j is quasi-compact (and it is automatically quasi-separated).

Now let $J_0 \subset J_1 \subset \cdots$ be an ascending chain of ideals in R . For every k define a quasi-coherent ideal on X by

$$\overline{\mathcal{J}}_k := \ker\left(\mathcal{O}_X \longrightarrow j_*(\mathcal{O}_U/\overline{J}_k)\right).$$

Because j is quasi-compact and quasi-separated, its pushforward preserves quasi-coherent modules [The25, Tag 01LC]; kernels of maps of quasi-coherent modules are quasi-coherent. Moreover $j^*\overline{\mathcal{J}}_k = \overline{J}_k$, and the construction is order-preserving in J_k . Thus the closed subschemes $V(\overline{\mathcal{J}}_k)$ form a descending chain on X . It stabilizes by hypothesis, and restriction to U then shows that the original chain $(J_k)_k$ stabilizes. Hence R is noetherian. $\square$

Definition 7.3.10 (Noetherian Space). A topological space T is *noetherian* if every descending sequence of closed subsets $Z_0 \supset Z_1 \supset Z_2 \supset \cdots$ in T stabilizes.

Remark 7.3.11 (Noetherian Implies Quasi-Compact). A noetherian space T is quasi-compact: any open cover gives by complementation a descending chain of closed sets under “remove one more open”; DCC forces the chain to stabilize, which happens exactly when a finite subcover has been reached. Moreover, any subspace of T is noetherian (the DCC restricts), hence also quasi-compact.

Remark 7.3.12 (Underlying Space of Noetherian Scheme). By Proposition 7.3.9 and Corollary 7.2.7, if X is a noetherian scheme then the underlying space $|X|$ is a noetherian space. The converse fails: for k a field, the ring $R = k[x_1, x_2, \dots]/(x_1^2, x_2^2, \dots)$ is non-noetherian (the chain $(x_1) \subsetneq (x_1, x_2) \subsetneq \cdots$ never stabilizes), yet $|\operatorname{Spec}(R)|$ is a single point since $R^{\text{red}} = k$.

Remark 7.3.13 (Noetherian Induction). To prove that every closed subset Z of a noetherian space T has a property P , it suffices to show that Z has P whenever every proper closed $W \subsetneq Z$ does. Indeed, if some closed subset failed P , the family of closed counterexamples would, by DCC, contain a minimal element Z_0 ; every proper closed $W \subsetneq Z_0$ would then have P , forcing Z_0 itself to have P by hypothesis — contradiction. The analogous statement holds for closed subschemes of a noetherian scheme (by Proposition 7.3.9). This method of proof is called *noetherian induction*.

Proposition 7.3.14 (Finite Type versus Finite Presentation). Let X be locally noetherian.

- (i) X is quasi-separated.
- (ii) If $f: Y \rightarrow X$ is locally of finite type, then Y is locally noetherian and f is locally of finite presentation.
- (iii) If X is noetherian and $f: Y \rightarrow X$ is of finite type, then Y is noetherian and f is of finite presentation.

Proof. For (i), by Proposition 6.2.10 quasi-separatedness asks that for every pair of affine opens $U, V \subset X$ the intersection $U \cap V$ be quasi-compact. Localizing on $U = \operatorname{Spec}(R)$ with R noetherian, $|U|$ is a noetherian space (Remark 7.3.12), so any open $|U \cap V| \subset |U|$ is again noetherian (Remark 7.3.11), in particular quasi-compact.

For (ii), both claims are local on X , so we may reduce to $X = \operatorname{Spec}(R)$ with R noetherian. Cover Y by affines $\operatorname{Spec}(A_i)$ with each A_i a finitely generated R -algebra; by Hilbert’s basis theorem each A_i is noetherian, so Y is locally noetherian, and by Proposition 7.3.1(v) each A_i is finitely presented, so f is locally of finite presentation.

For (iii), part (ii) gives Y locally noetherian and f locally of finite presentation. As f is quasi-compact (part of being of finite type) and X is quasi-compact, $Y = f^{-1}(X)$ is quasi-compact, so Y is noetherian; by (i) it is then quasi-separated, and hence so is f (the diagonal Δ_f is an immersion and every open of the noetherian Y is quasi-compact, so Δ_f is quasi-compact). Thus f is locally of finite presentation, quasi-compact and quasi-separated, i.e. of finite presentation. $\square$

7.4. Irreducible Components

A topological space T is *irreducible* if it is nonempty and any two nonempty opens meet (equivalently, T is not a union of two proper closed subsets). For sober spaces — e.g. underlying spaces of schemes — irreducibility is equivalent to the existence of a *generic point* η with $\overline{\{\eta\}} = T$. A scheme X is irreducible iff $|X|$ is (Definition 4.3.8).

Definition 7.4.1 (Irreducible Components). (i) The *irreducible components* of a topological space T are the maximal irreducible subsets of T .

(ii) The *irreducible components* of a scheme X are the irreducible components of $|X|$.

Remark 7.4.2 (Basic Facts on Irreducible Components). Let T be a topological space.

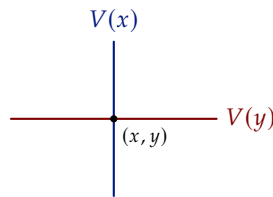
- (i) By Zorn's lemma, every irreducible subset of T is contained in a maximal one, i.e. in an irreducible component. Since singletons are irreducible, T is the union of its irreducible components.
- (ii) A subset $A \subset T$ is irreducible iff its closure $\overline{A}$ is. Consequently, every irreducible component of T is closed.

Example 7.4.3 (Irreducible Components of Affine Schemes). Let R be a ring. The bijection $\text{Rad}_R \xrightarrow{\sim} \text{Closed}(\text{Prim}(R)), I \mapsto V(I) = \{\mathfrak{p} \mid I \subset \mathfrak{p}\}$ (Proposition 4.2.4), restricts to an inclusion-reversing bijection between primes and irreducible closed subsets. Consequently, the irreducible components of $\text{Spec}(R)$ correspond to the *minimal* primes of R , and $\text{Spec}(R)$ is irreducible iff R has a unique minimal prime, iff $\text{Nil}(R)$ is prime.

Example 7.4.4 (Irreducible Components of Proj). Let A be an $\mathbb{N}$ -graded ring. By the projective Nullstellensatz (Corollary 2.3.28), the closed subsets of $|\text{Proj}(A)|$ correspond to saturated radical homogeneous ideals $I \subset A$. Under this bijection, irreducible closed subsets correspond to saturated homogeneous *prime* ideals, and hence irreducible components correspond to the minimal such. In particular, $\text{Proj}(A)$ is irreducible iff $\text{Nil}(A)^{\text{sat}}$ is prime.

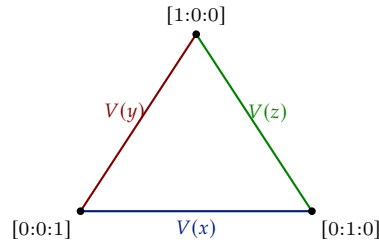
Example 7.4.5 (Two Concrete Examples). Let k be a field.

- (i) The affine k -scheme $\text{Spec}(k[x, y]/(xy))$ has exactly two irreducible components, corresponding to the minimal primes (x) and (y) in $k[x, y]/(xy)$. These are the two coordinate axes; they meet only at the origin (x, y) .



- (ii) The projective k -scheme $\text{Proj}(k[x, y, z]/(xyz))$ has three irreducible components, corresponding to the minimal saturated prime ideals (x) , (y) , (z) . These are three lines in $\mathbb{P}_k^2$ arranged in a triangle;

each pair meets at a single point away from the third component.



Proposition 7.4.6 (Noetherian Spaces Have Finitely Many Components). *A noetherian topological space has only finitely many irreducible components. In particular, a noetherian scheme has only finitely many irreducible components.*

Proof. We show by noetherian induction (Remark 7.3.13) on closed subsets $Z \subset T$ that every Z has only finitely many irreducible components; the case $Z = T$ then yields the proposition.

So fix Z and assume the conclusion for every proper closed $W \subsetneq Z$. If Z itself is irreducible there is nothing to prove. Otherwise, write $Z = Z_1 \cup Z_2$ with both pieces proper closed. Each Z_j has finitely many irreducible components by the inductive hypothesis. Any irreducible subset of Z lies entirely in Z_1 or in Z_2 (else its decomposition would contradict irreducibility), so the irreducible components of Z are precisely the maximal elements of the union of components of Z_1 and of Z_2 — a finite set. $\square$

Definition 7.4.7 (Integral Scheme, Function Field). A scheme X is *integral*¹ if it is irreducible and reduced. The *function field* (or *field of rational functions*) of an integral scheme X is the residue field $\kappa(\eta)$ at the generic point $\eta \in |X|$.

Example 7.4.8 (Integral Affine Schemes). A scheme $\text{Spec}(R)$ is integral iff R is an integral domain, in which case the function field is $\text{Frac}(R)$. Indeed, $\text{Spec}(R)$ is reduced iff R is reduced (Proposition 7.2.3), and irreducible iff $\text{Nil}(R)$ is prime (Example 7.4.3); together these say (0) is prime, i.e. R is an integral domain.

Proposition 7.4.9 (Properties of Integral Schemes). *Let X be an integral scheme with generic point η .*

- (i) *For every nonempty open subscheme $U \subset X$, the restriction map $\mathcal{O}_X(U) \rightarrow \kappa(\eta)$ is injective.*
- (ii) *For every nonempty affine open $U \subset X$, the map $\mathcal{O}_X(U) \rightarrow \kappa(\eta)$ exhibits $\kappa(\eta)$ as the fraction field of $\mathcal{O}_X(U)$.*
- (iii) *For every point $x \in |X|$, the map $\mathcal{O}_{X,x} \rightarrow \kappa(\eta)$ exhibits $\kappa(\eta)$ as the fraction field of the local ring $\mathcal{O}_{X,x}$.*

Proof. We start with the affine case (ii). Let $U = \text{Spec}(R)$ be nonempty. Irreducibility of X forces $\eta \in U$, and U inherits both irreducibility (any nonempty open of an irreducible space is irreducible) and reducedness (locality of reducedness, Proposition 7.2.3). Hence U is integral, so by Example 7.4.8 R is a domain with $\kappa(\eta) = \text{Frac}(R)$.

For (i), choose a nonempty affine open $V = \text{Spec}(R) \subset U$. We first prove that restriction $\mathcal{O}_X(U) \rightarrow \mathcal{O}_X(V)$ is injective. Let $s \in \mathcal{O}_X(U)$ restrict to zero on V , and cover U by affine opens $V_\alpha = \text{Spec}(R_\alpha)$. Irreducibility makes every intersection $V \cap V_\alpha$ nonempty. Choose a nonempty principal open $D(f_\alpha) \subset V \cap V_\alpha$. By (ii), R_α is a domain, and $f_\alpha \neq 0$; hence $R_\alpha \rightarrow (R_\alpha)_{f_\alpha}$ is injective. The restriction of $s|_{V_\alpha}$ to $D(f_\alpha)$ is zero, so $s|_{V_\alpha} = 0$. The sheaf condition now gives $s = 0$. Composing the resulting injection $\mathcal{O}_X(U) \hookrightarrow R$ with $R \hookrightarrow \text{Frac}(R) = \kappa(\eta)$ proves (i).

¹Here “integral” refers to “integral domain”, unrelated to the notion of integral element used in integral closure.

For (iii), fix $x \in |X|$ and an affine open $\text{Spec}(R) \ni x$ corresponding to a prime $\mathfrak{p}_x \subset R$. The stalk $\mathcal{O}_{X,x} = R_{\mathfrak{p}_x}$ is a localization of the domain R ; its fraction field equals $\text{Frac}(R) = \kappa(\eta)$, since localizations of a domain share the same fraction field. $\square$

Example 7.4.10 (Integral Proj). Let A be an integral $\mathbb{N}$ -graded ring with $A_+ \neq 0$. Then $\text{Proj}(A)$ is integral: reducedness comes from Example 7.2.5, and irreducibility from Example 7.4.4 (the hypothesis $A_+ \neq 0$ forces $(0)^{\text{sat}} = (0)$ in the integral ring A , so $(0)^{\text{sat}}$ is prime).

Let $Q^h(A)$ denote the *homogeneous fraction ring*, obtained by inverting every nonzero homogeneous element of A ; it is naturally $\mathbb{Z}$ -graded. By Proposition 7.4.9(ii), the function field of $\text{Proj}(A)$ equals $\text{Frac}(A_{(f)})$ for any nonzero homogeneous $f \in A_+$, and there is a canonical identification

$$\text{Frac}(A_{(f)}) \simeq Q^h(A)_0.$$

Here the subscript means the degree-zero part of the *graded* ring $Q^h(A)$; the ordinary field $\text{Frac}(A)$ need not carry a direct-sum grading. As a special case, for R a domain with fraction field K and any $n \geq 0$, $\mathbb{P}_R^n$ is integral with function field $K(x_1, \dots, x_n)$.

Remark 7.4.11 (Sheaf of Rational Functions). For any scheme X , one defines the *sheaf of rational functions* $\mathcal{K}_X$ on $|X|$ as the sheafification of the presheaf

$$U \mapsto \mathcal{O}_X(U)[S_U^{-1}],$$

where $S_U \subset \mathcal{O}_X(U)$ is the multiplicative subset of all sections f such that $f|_V$ is a non-zero-divisor in $\mathcal{O}_X(V)$ for every open $V \subset U$. Sheafification being left exact, the canonical map $\mathcal{O}_X \hookrightarrow \mathcal{K}_X$ is a monomorphism. If X is integral with generic point η , then $\mathcal{K}_X$ is the constant sheaf with value $\kappa(\eta)$ [The25, Tag 01X5]. An open immersion $j: U \hookrightarrow X$ is called *schematically dense* when $\mathcal{O}_X \rightarrow j_*\mathcal{O}_U$ is injective. For locally noetherian X , this is equivalent to U containing every associated point of X [The25, Tag 083P], and one can identify

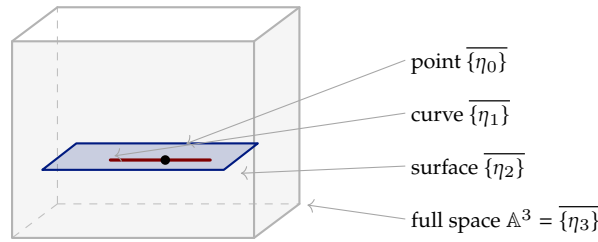
$$\mathcal{K}_X(X) = \text{colim}_{U \subset X} \mathcal{O}_X(U),$$

where the colimit ranges over schematically dense open subschemes $U \subset X$. Such an open is topologically dense, but the converse can fail. For example, with $A = k[x, y]/(x^2, xy)$ and $X = \text{Spec}(A)$, the open $D(y)$ is topologically dense (it contains the unique minimal prime (x)), while the nonzero section x restricts to zero on $D(y)$; hence $D(y) \hookrightarrow X$ is not schematically dense.

7.5. Dimension Theory

Geometrically, the dimension of a space is the length of the longest nested chain of “subspaces of decreasing size” it admits: a 3-dimensional space contains a surface, contains a curve, contains a point, in four strict layers. For schemes, where there is no a priori notion of subspace beyond closed subschemes, Krull translated this idea into pure topology: count the longest chain of irreducible closed subsets. The choice of irreducible closed sets (rather than arbitrary closed subsets, or open ones) is forced by sobriety: in a sober space such chains are exactly chains of points under specialization, so each layer of the chain has a unique generic point and dimension counts “how deep specialization can run”. The picture below

in $\mathbb{A}^3$ illustrates the prototypical chain of length 3.



$$\{\text{point}\} \subsetneq \text{curve} \subsetneq \text{surface} \subsetneq \mathbb{A}^3$$

codimensions 3, 2, 1, 0 (a chain of length 3 = $\dim \mathbb{A}^3$)

Regularity and normality are local conditions on schemes that refine “reduced”; both are most naturally explained through Krull dimension. We collect the dimension-theoretic background here. The deeper theorems — Krull’s principal ideal theorem and the dimension formula — are quoted from commutative algebra [Mat89, ch. 5].

Definition 7.5.1 (Chains and Length in a Poset). Let $(P, \leq)$ be a poset. A *chain of length n* in P is a sequence $x_0 \leq x_1 \leq \dots \leq x_n$ of strictly increasing elements. The *length* of P is

$$\ell(P) := \sup\{n \in \mathbb{N} \mid \text{there exists a chain of length } n \text{ in } P\} \in \mathbb{N} \cup \{\infty\}.$$

Definition 7.5.2 (Krull Dimension, Codimension). Let T be a topological space, ordered by inclusion on irreducible closed subsets.

(i) The *Krull dimension* of T is $\dim(T) := \ell(\{\text{irreducible closed subsets of } T\})$.

(ii) For $K \subset T$ irreducible closed, the *codimension* of K in T is

$$\text{codim}(K, T) := \ell(\{\text{irreducible closed subsets of } T \text{ containing } K\}) \in \mathbb{N} \cup \{\infty\}$$

(the poset is nonempty since it contains K).

For a scheme X , set $\dim(X) := \dim(|X|)$ and $\text{codim}(K, X) := \text{codim}(K, |X|)$ for $K \subset |X|$ irreducible closed; for a ring R , $\dim(R) := \dim(\text{Spec}(R))$, which equals the length of the longest chain of primes in R .

Remark 7.5.3 (Notational Convention for the Rest of the Chapter). Krull dimension and codimension are intrinsically topological invariants: rigorously they are computed on $|X|$, not on X itself. To keep notation light we will write $\dim(X)$ and $\text{codim}(K, X)$ in place of $\dim(|X|)$ and $\text{codim}(K, |X|)$ throughout this and the remaining sections of the chapter, and similarly speak of an “irreducible closed subset $K \subset X$ ” meaning a closed subscheme whose underlying space is irreducible (equivalently, an irreducible closed subset of $|X|$). The context makes the intended reading unambiguous.

Remark 7.5.4 (Specialization Order). Given $x, y \in T$, write $x \rightsquigarrow y$ and say that x *specializes to* y if $y \in \overline{\{x\}}$; this defines a preorder on the underlying set of T , which becomes a partial order precisely when T is T_0 (in particular when T is sober). For any sober space the closure map is an isomorphism of posets

$$(T, \rightsquigarrow) \xrightarrow{\sim} (\{\text{irreducible closed subsets of } T\}, \supset), \quad x \mapsto \overline{\{x\}}.$$

Definition 7.5.2 can therefore be rephrased in terms of the poset of points: for example,

$$\dim(T) = \ell(T, \rightsquigarrow) = \sup_{x \in T} \text{codim}(x, T).$$

Remark 7.5.5 (Dimension is Local). (i) If $(T_i \subset T)_{i \in I}$ is an open covering, then $\dim(T) = \sup_i \dim(T_i)$. The same formula holds for a *finite* closed covering. It also holds for an arbitrary closed covering when T is sober (in particular, when T is the underlying space of a scheme).

Indeed, the inequality $\dim(T_i) \leq \dim(T)$ follows by taking closures in T . Conversely, an open T_i meeting the smallest member of a chain $K_0 \subsetneq \cdots \subsetneq K_n$ cuts out a chain of the same length: each nonempty open trace $K_j \cap T_i$ is dense in K_j . For a finite closed cover, irreducibility forces the largest member K_n into one member of the cover. For an arbitrary closed cover of a sober space, choose the generic point of K_n ; any closed member containing that point contains all of K_n . Without finiteness or sobriety the closed-cover assertion is false.

(ii) If $T' \subset T$ is any subspace, then $\dim(T') \leq \dim(T)$.

(iii) For a scheme X and $x \in X$, one has $\text{codim}(x, X) = \dim(\mathcal{O}_{X,x})$ (exercise).

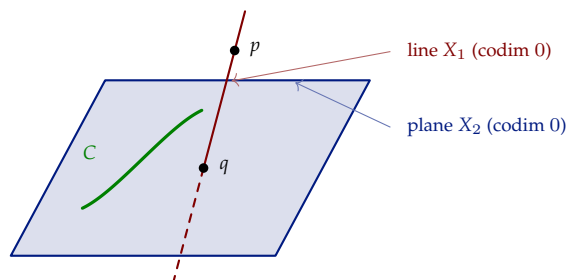
Remark 7.5.6 (Naive Desiderata Fail). Two innocent-looking desiderata about dimension are *false* in general:

(i) $\dim(U) = \dim(X)$ for every nonempty open $U \subset X$;

(ii) $\dim(V(f)) = \dim(X) - 1$ for every $f \in \mathcal{O}(X)$.

Both already fail for $X = \text{Spec}(k[x]_{(x)})$, the spectrum of a local DVR with $\dim X = 1$ (the unique nontrivial chain of primes is $(0) \subsetneq (x)$). Removing the closed point gives the field $U = \text{Spec } k(x)$ with $\dim U = 0$, breaking (i); choosing f to be a unit in $\mathcal{O}(X)$ produces $V(f) = \emptyset$, breaking (ii). Notice that the open immersion $U \hookrightarrow X$ is flat, so flatness alone does not restore a global equality of dimensions. The dimension formula below is instead a *pointwise codimension* statement for morphisms of locally noetherian schemes; the finite-type integral case supplies a separate global equality.

The same subtlety persists when irreducible components meet. The figure below shows $X = X_1 \cup X_2$ with X_1 a line meeting X_2 a plane transversally at a single point: the codimension of a closed subset of X depends genuinely on which irreducible component of X contains it.



Codimension in $X = X_1 \cup X_2$ depends on which component the subset sits inside:

- $C \subset X_2$: a curve in the plane, so $\text{codim}(C, X) = 1$;
- $p \in X_1$: a point on the line away from the plane, so $\text{codim}(p, X) = 1$;
- $q \in X_1 \cap X_2$: $\text{codim } 1$ in X_1 but $\text{codim } 2$ in X_2 , so $\text{codim}(q, X) = 2$.

We turn from these pathologies to the simplest positive computation: the dimension of a PID. This will be the workhorse base case for everything that follows.

Example 7.5.7 (Dimension of Principal Ideal Domains). For a principal ideal domain R ,

$$\dim(R) = \begin{cases} 0 & \text{if } R \text{ is a field,} \\ 1 & \text{otherwise.} \end{cases}$$

Indeed, in a PID every nonzero prime is maximal (generated by an irreducible element), so the longest chain of primes is $(0) \leq (\pi)$. Consequently, if X is a scheme all of whose local rings are PIDs, then $\dim(X) \leq 1$.

We quote the fundamental trio of noetherian dimension theorems.

Theorem 7.5.8 (Krull's Principal Ideal Theorem). *Let R be a noetherian ring, $X = \text{Spec}(R)$, and $f \in R$. Then every irreducible component $K = V(\mathfrak{p}) \subset V(f)$ satisfies $\text{codim}(K, X) \leq 1$, with*

$$\text{codim}(K, X) = \begin{cases} 0 & \text{if } \mathfrak{p} \text{ is a minimal prime of } R \text{ (equivalently, } K \text{ is also a component of } X), \\ 1 & \text{otherwise.} \end{cases}$$

In particular, if f is not a zero-divisor in R^{red} (equivalently, f lies in no minimal prime of R), then every irreducible component of $V(f)$ has codimension 1. See [The25, Tag 0BBZ].

Warning 7.5.9 (Per-Component Reading). The dichotomy in Theorem 7.5.8 is *per component*, not global. When f is a zero-divisor in R^{red} , some components of $V(f)$ have codimension 0 (those corresponding to minimal primes of R that happen to contain f) while others may still have codimension 1. For instance, $R = k[x, y, z]/(xy)$ and $f = xz$: f is a zero-divisor in $R = R^{\text{red}}$, $V(f)$ has component $V(x)$ of codimension 0 and component $V(y, z)$ of codimension 1.

Remark 7.5.10 (Local Form: A Proof Sketch). Localizing at the generic point of an irreducible component of $V(f)$ reduces Theorem 7.5.8 to the following local statement, whose proof we sketch since it conveys the geometric content of the theorem.

Claim. Let $(R, \mathfrak{m})$ be a noetherian local ring, $f \in \mathfrak{m}$ a non-zero-divisor with $\sqrt{(f)} = \mathfrak{m}$. Then $\dim R \leq 1$.

The geometric idea is that $\sqrt{(f)} = \mathfrak{m}$ makes $R/(f)$ artinian (a 0-dimensional “thickened point”), forcing chains of primes through $\mathfrak{m}$ to have length at most one. The proof formalizes this through symbolic powers and two applications of Nakayama’s lemma.

Let $\mathfrak{p} \subsetneq \mathfrak{m}$ be a prime; we will show $\mathfrak{p}$ is minimal in R , so the chain $\mathfrak{p} \subsetneq \mathfrak{m}$ has length one.

For each $n \geq 0$ define the *symbolic power* $\mathfrak{p}^{(n)} := \mathfrak{p}^n R_{\mathfrak{p}} \cap R$. Note that $f \notin \mathfrak{p}$, since $f \in \mathfrak{p}$ would give $\sqrt{(f)} \subseteq \mathfrak{p} \subsetneq \mathfrak{m}$, contradicting $\sqrt{(f)} = \mathfrak{m}$. By the definition of symbolic powers via the localization $R_{\mathfrak{p}}$, this means $f \cdot r \in \mathfrak{p}^{(n)} \Rightarrow r \in \mathfrak{p}^{(n)}$ for every $r \in R$.

Step 1: the symbolic powers stabilize. Since $R/(f)$ is artinian, the descending chain of ideals $(\mathfrak{p}^{(n)} + (f))/(f)$ in $R/(f)$ stabilizes at some index n_0 . Translating back to R : for $n \geq n_0$,

$$\mathfrak{p}^{(n_0)} \subseteq \mathfrak{p}^{(n_0+1)} + (f),$$

i.e. every $a \in \mathfrak{p}^{(n_0)}$ can be written $a = b + fc$ with $b \in \mathfrak{p}^{(n_0+1)}$ and $c \in R$. Then $fc = a - b \in \mathfrak{p}^{(n_0)}$, and since $f \notin \mathfrak{p}$ we obtain $c \in \mathfrak{p}^{(n_0)}$. Hence $\mathfrak{p}^{(n_0)} = \mathfrak{p}^{(n_0+1)} + f \cdot \mathfrak{p}^{(n_0)}$.

This expresses the finitely generated R -module $M := \mathfrak{p}^{(n_0)}/\mathfrak{p}^{(n_0+1)}$ as $M = (f) \cdot M$, with $f \in \mathfrak{m}$. Nakayama’s lemma applied to M over the local ring R yields $M = 0$, i.e. $\mathfrak{p}^{(n_0)} = \mathfrak{p}^{(n_0+1)}$. By induction, $\mathfrak{p}^{(n)} = \mathfrak{p}^{(n_0)}$ for every $n \geq n_0$.

Step 2: $\mathfrak{p}R_{\mathfrak{p}}$ is nilpotent. Localizing at $\mathfrak{p}$ converts the equality $\mathfrak{p}^{(n_0)} = \mathfrak{p}^{(n_0+1)}$ (whose definition was already in $R_{\mathfrak{p}}$) into $(\mathfrak{p}R_{\mathfrak{p}})^{n_0} = (\mathfrak{p}R_{\mathfrak{p}})^{n_0+1}$. The finitely generated $R_{\mathfrak{p}}$ -module $(\mathfrak{p}R_{\mathfrak{p}})^{n_0}$ is now fixed under multiplication by the maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$, so a second application of Nakayama (this time over the local ring $R_{\mathfrak{p}}$) yields $(\mathfrak{p}R_{\mathfrak{p}})^{n_0} = 0$.

Hence $\mathfrak{p}R_{\mathfrak{p}}$ is nilpotent, $\dim R_{\mathfrak{p}} = 0$, and $\mathfrak{p}$ is a minimal prime of R . Since this holds for every $\mathfrak{p} \subsetneq \mathfrak{m}$, the longest chain of primes ending at $\mathfrak{m}$ has length ≤ 1 , i.e. $\dim R \leq 1$.

The full theorem follows by localizing at the prime $\mathfrak{p}$ corresponding to K , so that $\text{codim}(K, X) = \dim(R_{\mathfrak{p}})$. The image $\bar{f} \in R_{\mathfrak{p}}$ satisfies $\sqrt{(\bar{f})} = \mathfrak{p}R_{\mathfrak{p}}$ since $\mathfrak{p}$ is minimal over (f) . Whether the claim’s hypothesis “ $\bar{f}$ is a non-zero-divisor” holds depends entirely on $\mathfrak{p}$:

- If $\mathfrak{p}$ is itself a minimal prime of R , then $\bar{f} \in \mathfrak{p}R_{\mathfrak{p}}$ is nilpotent (as $\mathfrak{p}R_{\mathfrak{p}}$ is the unique prime of $R_{\mathfrak{p}}$, hence its nilradical), so $\dim R_{\mathfrak{p}} = 0$ and $\text{codim}(K, X) = 0$.
- If $\mathfrak{p}$ is not a minimal prime of R , choose $\mathfrak{q} \subsetneq \mathfrak{p}$ a minimal prime contained in $\mathfrak{p}$. Every minimal prime of $R_{\mathfrak{p}}$ has the form $\mathfrak{q}'R_{\mathfrak{p}}$ for a minimal prime $\mathfrak{q}' \subseteq \mathfrak{p}$ of R , and $f \notin \mathfrak{q}'$ (otherwise f would lie in a prime strictly smaller than $\mathfrak{p}$, contradicting minimality of $\mathfrak{p}$ over (f)); hence $\bar{f}$ avoids all minimal primes of $R_{\mathfrak{p}}$ and is a non-zero-divisor in $R_{\mathfrak{p}}^{\text{red}}$. Replacing $R_{\mathfrak{p}}$ by $R_{\mathfrak{p}}/\sqrt{0}$ does not change the dimension, and the claim now applies, giving $\dim R_{\mathfrak{p}} \leq 1$. Combined with $\mathfrak{q} \subsetneq \mathfrak{p}$, we get $\text{codim}(K, X) = 1$.

Thus the codimension is determined component-by-component by whether the generic point of the component is a minimal prime of R , as stated.

Theorem 7.5.11 (Krull's Height Theorem). *Let R be noetherian, $f_1, \dots, f_n \in R$, and $Y = V(f_1, \dots, f_n) \subset X = \text{Spec}(R)$. For every irreducible closed subset $K \subset Y$,*

$$\text{codim}(K, X) \leq \text{codim}(K, Y) + n.$$

The case where K is an irreducible component of Y (so $\text{codim}(K, Y) = 0$) recovers the more common form $\text{codim}(K, X) \leq n$.

Theorem 7.5.12 (Converse to Height). *Let R be noetherian and $K \subset \text{Spec}(R)$ an irreducible closed subset of codimension n . Then there exist $f_1, \dots, f_n \in R$ such that K is an irreducible component of $V(f_1, \dots, f_n)$.*

Corollary 7.5.13 (Codimension via Generators). *Let R be noetherian, $X = \text{Spec}(R)$, and $\mathfrak{p} \subset R$ a prime ideal. Then*

$$\text{codim}(V(\mathfrak{p}), X) \leq (\text{minimum number of generators of } \mathfrak{p} \text{ as a radical ideal}) \leq \infty.$$

Proof. Apply Theorem 7.5.11 with $f_1, \dots, f_n$ a set of radical-ideal generators of $\mathfrak{p}$. □

Theorem 7.5.14 (Dimension Formula). *Let $f: Y \rightarrow X$ be a morphism of locally noetherian schemes, $y \in Y$, and $x = f(y) \in X$. Write $Y_x := \text{Spec}(\kappa(x)) \times_X Y$ for the fiber, containing y canonically. Then*

$$\text{codim}(y, Y) \leq \text{codim}(x, X) + \text{codim}(y, Y_x),$$

with equality whenever f is flat at y (i.e. the induced ring map $\mathcal{O}_{X,x} \rightarrow \mathcal{O}_{Y,y}$ is flat); the relevant input is that flat local homomorphisms of noetherian local rings satisfy the going-down property, which is what forces the codimensions to add.

Corollary 7.5.15 (Dimension Under Polynomial Extension). *If X is a locally noetherian scheme, then $\dim(X \times \mathbb{A}^n) = \dim(X) + n$.*

Proof. By induction it suffices to treat $n = 1$, and by Remark 7.5.5 we may work on an affine open $X = \text{Spec}(R)$. Let $\mathfrak{q} \subset R[t]$ lie over $\mathfrak{p} \subset R$. Flatness and Theorem 7.5.14 give

$$\text{ht}(\mathfrak{q}) = \text{ht}(\mathfrak{p}) + \text{ht}(\mathfrak{q}/\mathfrak{p}R[t]).$$

The last height is at most 1, because the fiber ring is $\kappa(\mathfrak{p})[t]$. This proves $\dim(R[t]) \leq \dim(R) + 1$. Conversely, for every prime $\mathfrak{p} \subset R$, the prime $\mathfrak{q} = \mathfrak{p}R[t] + (t)$ has fiber prime $(t) \subset \kappa(\mathfrak{p})[t]$ of height 1. The same equality gives $\text{ht}(\mathfrak{q}) = \text{ht}(\mathfrak{p}) + 1$; taking suprema over $\mathfrak{p}$ proves the reverse inequality. □

Example 7.5.16 (Dimension of Affine and Projective Bundles). *Let X be a locally noetherian scheme and V a vector bundle of rank $n \geq 1$ over X . Then*

$$\dim(\mathbb{A}(V)) = \dim(X) + n, \quad \dim(\mathbb{P}(V)) = \dim(X) + n - 1.$$

Both claims are local on X (Remark 7.5.5), so we may assume V is free, whence $\mathbb{A}(V) \simeq X \times \mathbb{A}^n$ and $\mathbb{P}(V) \simeq X \times \mathbb{P}^{n-1}$. The $\mathbb{A}$ -case is Corollary 7.5.15; the $\mathbb{P}$ -case reduces to the affine case via the standard cover of $\mathbb{P}^{n-1}$ by copies of $\mathbb{A}^{n-1}$.

Theorem 7.5.17 (Dimension of Integral k -Schemes of Finite Type). *Let k be a field and X an integral k -scheme of finite type, with generic point η . Then*

- (i) $\dim(X) = \text{tr.deg}(\kappa(\eta)/k)$.
- (ii) *For every irreducible closed $K \subset X$, $\dim(X) = \dim(K) + \text{codim}(K, X)$.*

Example 7.5.18 (Failure of $\dim + \text{codim}$ in General). The equality $\dim(K) + \text{codim}(K, X) = \dim(X)$ fails outside the finite-type-over-a-field setting. Take $X = \text{Spec}(\mathbb{Z}_{(p)}[x])$, which has $\dim(X) = 2$ via the chain $(0) \leq (p) \leq (p, x)$. The closed subscheme $K = V(px - 1) = \text{Spec}(\mathbb{Z}_{(p)}[x]/(px - 1)) = \text{Spec}(\mathbb{Z}_{(p)}[1/p]) = \text{Spec}(\mathbb{Q})$ has $\dim(K) = 0$. Yet Krull's principal ideal theorem gives $\text{codim}(K, X) = 1$, since $px - 1$ is a non-zero-divisor. Hence $\dim(K) + \text{codim}(K, X) = 1 < 2 = \dim(X)$.

Proposition 7.5.19 (Dimension of a Noetherian Local Ring). *Let $(R, \mathfrak{m})$ be a noetherian local ring. Then*

$$\dim(R) = \min\{n \in \mathbb{N} \mid \text{there exist } f_1, \dots, f_n \in \mathfrak{m} \text{ with } \sqrt{(f_1, \dots, f_n)} = \mathfrak{m}\}.$$

Proof. For the upper bound, any tuple $(f_1, \dots, f_n)$ with $\sqrt{(f_1, \dots, f_n)} = \mathfrak{m}$ makes $V(\mathfrak{m})$ an irreducible component of $V(f_1, \dots, f_n)$, so Theorem 7.5.11 gives $\dim(R) = \text{codim}(V(\mathfrak{m}), \text{Spec}(R)) \leq n$. The minimum on the right is therefore at least $\dim(R)$.

For the matching lower bound, apply Theorem 7.5.12 to $K = V(\mathfrak{m})$, which has codimension $d = \dim(R)$, to obtain $f_1, \dots, f_d \in R$ with $V(\mathfrak{m})$ an irreducible component of $V(f_1, \dots, f_d)$. Since $V(\mathfrak{m})$ is the unique closed point of $\text{Spec}(R)$, we must have $V(f_1, \dots, f_d) = V(\mathfrak{m})$, i.e. $\sqrt{(f_1, \dots, f_d)} = \mathfrak{m}$. Hence the minimum is at most d . $\square$

7.6. Tangent Spaces and Regularity

7.6.1. Tangent and Cotangent Spaces at a Point

There is an evocative way to package the Zariski tangent space, using the *ring of dual numbers* $k[\varepsilon] := k[t]/(t^2)$. An R -valued point $a + b\varepsilon \in \mathbb{A}^n(R[\varepsilon])$ has $a \in R^n$ recording a “base point” and $b \in R^n$ recording a “tangent direction” at a : the relation $\varepsilon^2 = 0$ is exactly what is needed to make this “infinitesimal displacement” interpretation algebraically rigorous. This upgrades to a functorial definition of tangent space for any algebraic k -functor.

Definition 7.6.1 (Tangent Space (Functor of Points)). Let X be an algebraic k -functor and $x \in X(k)$ a k -point. For each k -algebra R , the projection $\pi: R[\varepsilon] \twoheadrightarrow R$, $\varepsilon \mapsto 0$, induces $X(\pi): X(R[\varepsilon]) \rightarrow X(R)$; write $x_R \in X(R)$ for the image of x under $X(k \rightarrow R)$. The *tangent space to X at x* is the algebraic k -functor

$$T_x X: \text{CAlg}_k \rightarrow \text{Set}, \quad T_x X(R) := X(\pi)^{-1}(x_R),$$

defined by the pullback

$$\begin{array}{ccc} T_x X(R) & \longrightarrow & X(R[\varepsilon]) \\ \downarrow & \lrcorner & \downarrow X(\pi) \\ \{x_R\} & \hookrightarrow & X(R). \end{array}$$

In words, $T_x X(R)$ collects the “infinitesimal deformations” of x_R : those $R[\varepsilon]$ -points of X that degenerate to x_R when ε is set to zero.

For closed subfunctors of affine space the tangent space has a completely explicit description via derivatives, thanks to the following algebraic identity.

Lemma 7.6.2 (Dual-Number Evaluation). *Let $f \in k[x_1, \dots, x_n]$, $a \in k^n$, R a k -algebra, and $b \in R^n$. Then in $R[\varepsilon]$,*

$$f(a + b\varepsilon) = f(a) + \left(\sum_{i=1}^n \frac{\partial f}{\partial x_i}(a) \cdot b_i \right) \varepsilon,$$

where $\frac{\partial f}{\partial x_i}$ is the formal partial derivative.

Proof. Both sides are k -linear in f (the left via the evaluation homomorphism $\text{ev}_{a+b\varepsilon}$, the right via linearity of $\partial/\partial x_i$), so it suffices to check on monomials $f = \prod_i x_i^{m_i}$. Substituting $x_i = a_i + b_i\varepsilon$ and using $\varepsilon^2 = 0$, the binomial theorem gives $(a_i + b_i\varepsilon)^{m_i} = a_i^{m_i} + m_i a_i^{m_i-1} b_i \varepsilon$. Multiplying the factors and discarding all terms of order ≥ 2 in ε yields the claimed identity, the ε -coefficient being exactly $\sum_j \frac{\partial f}{\partial x_j}(a) \cdot b_j$. $\square$

Remark 7.6.3 (Derivatives Are Algebraic). The proof of Lemma 7.6.2 is purely algebraic: partial derivatives appear *not* because we are doing analytic Taylor expansion, but because the algebraic relation $\varepsilon^2 = 0$ automatically kills every higher-order term, leaving the first-order information — which is precisely what formal partial derivatives record. Formal partial derivatives are an algebraic consequence of $\varepsilon^2 = 0$, not an analytic input.

Corollary 7.6.4 (Tangent Space via Jacobian). *Let $f_1, \dots, f_m \in k[x_1, \dots, x_n]$ and $a \in V(f_1, \dots, f_m)(k)$. Then, as k -functors,*

$$T_a V(f_1, \dots, f_m) = \ker \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(a) & \cdots & \frac{\partial f_1}{\partial x_n}(a) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(a) & \cdots & \frac{\partial f_m}{\partial x_n}(a) \end{pmatrix},$$

the kernel of the Jacobian matrix interpreted as a k -linear map $\mathbb{A}^n \rightarrow \mathbb{A}^m$. In particular, for a single equation f , $T_a V(f) = \ker(\nabla f(a): \mathbb{A}^n \rightarrow \mathbb{A}^1)$.

Proof. For any k -algebra R , $T_a V(f_1, \dots, f_m)(R)$ consists of those $b \in R^n$ such that $f_j(a + b\varepsilon) = 0$ in $R[\varepsilon]$ for every j . By Lemma 7.6.2 this becomes $f_j(a) + (\nabla f_j(a) \cdot b)\varepsilon = 0$; the constant term vanishes since $a \in V(f_j)(k)$, so the condition reduces to $\nabla f_j(a) \cdot b = 0$, i.e. $b \in \ker(\text{Jac}(a))(R)$. The identification holds for every R , hence as k -functors. $\square$

The Jacobian description is concrete but unsatisfactory in two respects. First, it depends on a presentation of X as a closed subfunctor of $\mathbb{A}^n$, and a different embedding produces a different Jacobian and a different (yet canonically isomorphic) kernel. Second, the dual-numbers definition itself is set up only for an algebraic k -functor and a k -point, leaving aside the general setting of an arbitrary scheme (or locally ringed space) and an arbitrary point thereof, whose residue field need not equal k . Both restrictions can be lifted by a single intrinsic construction — the tangent space as a space of derivations of the local ring at x — which we now develop. The resulting framework is broad enough to apply to all schemes, to non-rational points, and to geometric categories built from rings of functions other than polynomial: formal schemes, complex-analytic spaces, C^∞ -manifolds.

Fix a locally ringed space $(X, \mathcal{O}_X)$ over k and a point $x \in X$. The local data at x consists of the stalk $\mathcal{O}_{X,x}$, which is a local k -algebra; its unique maximal ideal $\mathfrak{m}_x$, consisting of those germs vanishing at x ; and the residue field $\kappa(x) := \mathcal{O}_{X,x}/\mathfrak{m}_x$, an extension of k . The reader more comfortable with the affine algebraic case should keep the following dictionary in mind: when $X = \text{Spec}(A)$ and $x: A \rightarrow k$ is a k -point with kernel $\mathfrak{m} = \ker(x)$, one has $\mathcal{O}_{X,x} = A_{\mathfrak{m}}$ and $\kappa(x) = k$.

A tangent vector at x should be an infinitesimal deformation of the point, in the same sense as in Definition 7.6.1, but now phrased in stalk-theoretic terms. The point itself — as a piece of algebraic data on the

local ring — is the residue map $\mathcal{O}_{X,x} \rightarrow \kappa(x)$. To deform it infinitesimally is to lift it across the projection $\kappa(x)[\varepsilon] \rightarrow \kappa(x)$, $\varepsilon \mapsto 0$: that is, to give a k -algebra homomorphism

$$\tilde{x}: \mathcal{O}_{X,x} \longrightarrow \kappa(x)[\varepsilon]$$

whose composition with $\varepsilon \mapsto 0$ recovers the residue map. We accordingly define the tangent space at x in this stalk-theoretic setting to be the set of all such lifts. (When X is an affine k -scheme and x is a k -point, this set is in canonical bijection with $T_x X(k)$ from Definition 7.6.1: a k -algebra map $A \rightarrow k[\varepsilon]$ lifting x is the same data as a k -algebra map $A_{\mathfrak{m}} \rightarrow k[\varepsilon]$ lifting the residue map, since $k[\varepsilon]$ is local and $A \rightarrow A_{\mathfrak{m}}$ inverts everything outside $\mathfrak{m}$.)

We now extract from such a lift its essential algebraic content. Every element of $\kappa(x)[\varepsilon]$ admits a unique expression $a + b\varepsilon$ with $a, b \in \kappa(x)$; writing $\bar{f} \in \kappa(x)$ for the image of $f \in \mathcal{O}_{X,x}$ under the residue map, the lifting condition forces the constant part of $\tilde{x}(f)$ to be $\bar{f}$, leaving the ε -coefficient as the only freedom. We define

$$D(f) := \frac{\tilde{x}(f) - \bar{f}}{\varepsilon} \in \kappa(x), \quad \text{so that} \quad \tilde{x}(f) = \bar{f} + D(f)\varepsilon.$$

The map $D: \mathcal{O}_{X,x} \rightarrow \kappa(x)$ is k -linear, since both $\tilde{x}$ and the residue map are k -linear and the operation of extracting the ε -coefficient is $\kappa(x)$ -linear and a fortiori k -linear. Conversely, any k -linear map $D: \mathcal{O}_{X,x} \rightarrow \kappa(x)$ produces a k -linear lift $\tilde{x}$ via the same formula; the question is for which D this lift is moreover *multiplicative*.

The multiplicativity condition $\tilde{x}(fg) = \tilde{x}(f)\tilde{x}(g)$, expanded using $\varepsilon^2 = 0$, becomes

$$\overline{fg} + D(fg)\varepsilon = (\bar{f} + D(f)\varepsilon)(\bar{g} + D(g)\varepsilon) = \bar{f}\bar{g} + (\bar{f}D(g) + \bar{g}D(f))\varepsilon.$$

The constant terms agree automatically (the residue map is itself multiplicative), so the surviving condition is the equality of ε -coefficients — the *Leibniz rule*

$$D(fg) = \bar{f} \cdot D(g) + \bar{g} \cdot D(f).$$

This identifies tangent vectors at x with k -linear maps $D: \mathcal{O}_{X,x} \rightarrow \kappa(x)$ satisfying the Leibniz rule. Such a map is called a *k -derivation of $\mathcal{O}_{X,x}$ valued in $\kappa(x)$* , where $\kappa(x)$ is regarded as an $\mathcal{O}_{X,x}$ -module via the residue map; the space of all such derivations is denoted $\text{Der}_k(\mathcal{O}_{X,x}, \kappa(x))$.

The Leibniz rule has two pleasant immediate consequences for a derivation D . Setting $f = g = 1$ gives $D(1) = \bar{1} \cdot D(1) + \bar{1} \cdot D(1) = 2D(1)$, so $D(1) = 0$, and by k -linearity D vanishes on the image of k in $\mathcal{O}_{X,x}$. More substantively, for any $f, g \in \mathfrak{m}_x$ one has $\bar{f} = \bar{g} = 0$, and Leibniz gives $D(fg) = \bar{f} \cdot D(g) + \bar{g} \cdot D(f) = 0$; thus D vanishes on $\mathfrak{m}_x^2$. Since D vanishes on $\mathfrak{m}_x^2$, its restriction to $\mathfrak{m}_x$ factors through a $\kappa(x)$ -linear functional $\bar{D}: \mathfrak{m}_x/\mathfrak{m}_x^2 \rightarrow \kappa(x)$. When x is a k -rational point, so that $\kappa(x) = k$ and the residue map splits as $\mathcal{O}_{X,x} = k \oplus \mathfrak{m}_x$, the derivation D vanishes on k and is therefore determined by $\bar{D}$; conversely each such $\bar{D}$ extends to a unique derivation via $D(c + m) = \bar{D}(m \bmod \mathfrak{m}_x^2)$ for $c \in k$ and $m \in \mathfrak{m}_x$ (Leibniz being immediate). We thus obtain, at a k -rational point, a chain of canonical isomorphisms

$$T_x X \simeq \text{Der}_k(\mathcal{O}_{X,x}, \kappa(x)) \simeq \text{Hom}_{\kappa(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2, \kappa(x)) = (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee.$$

The first isomorphism — a tangent vector reread intrinsically as a derivation of the local ring — holds at *every* point. The second, identifying such derivations with functionals on the (in good cases finite-dimensional) $\kappa(x)$ -vector space $\mathfrak{m}_x/\mathfrak{m}_x^2$, requires care at non-rational points: the restriction map $\text{Der}_k(\mathcal{O}_{X,x}, \kappa(x)) \rightarrow (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee$ has kernel $\text{Der}_k(\kappa(x), \kappa(x))$ — the derivations of the residue field itself — and is an isomorphism when $\Omega_{\kappa(x)/k} = 0$, e.g. when $\kappa(x)/k$ is separable algebraic, in particular when $\kappa(x) = k$. (At the generic point of $\mathbb{A}_k^1$, by contrast, $\mathfrak{m}_x = 0$ while $T_x X = \text{Der}_k(k(t), k(t)) \neq 0$.)

The vector space that has emerged in the course of this analysis — attached directly to the local ring at x , without reference to dual numbers, embeddings, or presentations — is itself a fundamental geometric invariant, dual to the tangent space and equally directly defined. It deserves a name.

Definition 7.6.5 (Zariski Cotangent Space). Let $(X, \mathcal{O}_X)$ be a locally ringed space and $x \in X$. The *Zariski cotangent space* at x is the $\kappa(x)$ -vector space

$$T_x^*X := \mathfrak{m}_x / \mathfrak{m}_x^2 \simeq \mathfrak{m}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x),$$

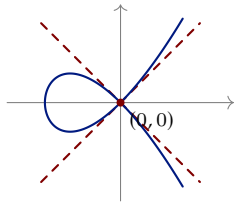
together with the canonical *differential* $d: \mathfrak{m}_x \rightarrow T_x^*X$, $f \mapsto df$. Elements of T_x^*X are called *cotangent vectors* or *differentials* at x .

The two definitions of (co)tangent space at x in our hands — Definition 7.6.1 for $T_x X$ and Definition 7.6.5 for T_x^*X — are formulated in two different ambient frameworks and extend in different directions beyond their common ground of an affine k -scheme at a k -point. The dual-numbers definition lives along the functor-of-points axis: it makes sense for any algebraic k -functor and any k -point, including settings such as formal schemes, ind-schemes, and derived schemes, where stalks are unavailable or insufficient. The local-ring definition lives along the locally ringed space axis: it makes sense for any $(X, \mathcal{O}_X)$ with local stalks and any point $x \in X$, including geometric categories far from algebraic geometry such as C^∞ -manifolds or complex-analytic spaces, and including non-rational points of schemes (whose residue field $\kappa(x)$ properly extends k). On the common ground where both apply, the calculation above identifies the two, exhibiting T_x^*X as the $\kappa(x)$ -linear dual of $T_x X$.

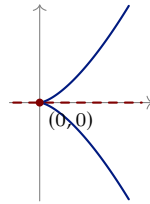
Example 7.6.6 (Plane Curve Singularities: Node, Cusp, Tacnode). Assume $\text{char}(k) \neq 2, 3$. Consider the plane curves

$$N = V(y^2 - x^3 - x^2), \quad C = V(y^2 - x^3), \quad T = V(y^2 - x^4)$$

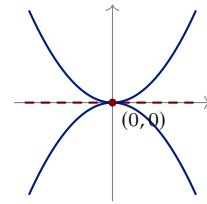
in $\mathbb{A}_k^2$. Using Corollary 7.6.4, the tangent space at (a, b) is the kernel of $\nabla f(a, b)$ where f is the defining polynomial.



Node $N : y^2 = x^2(x+1)$



Cusp $C : y^2 = x^3$



Tacnode $T : y^2 = x^4$

For the node N , $f(x, y) = y^2 - x^3 - x^2$ gives $\nabla f = (-3x^2 - 2x, 2y)$. The gradient vanishes iff $x(3x+2) = 0$ and $y = 0$, i.e. at $(0, 0)$ or at $(-2/3, 0)$. Only the first lies on N , since substituting $(-2/3, 0)$ into the equation gives $0 - (-2/3)^3 - (-2/3)^2 = 8/27 - 4/9 = -4/27 \neq 0$. Hence $(0, 0)$ is the unique singular point of N , with $T_{(0,0)}N = k^2$ (two-dimensional), and all other points are smooth with $\dim T_{(a,b)}N = 1$.

For the cusp C , $f(x, y) = y^2 - x^3$ gives $\nabla f = (-3x^2, 2y)$, which vanishes only at the origin. At $(0, 0) \in C$, $T_{(0,0)}C = k^2$; elsewhere the curve is smooth.

For the tacnode T , $f(x, y) = y^2 - x^4$ gives $\nabla f = (-4x^3, 2y)$, vanishing only at the origin. Again $(0, 0)$ is the sole singularity with $T_{(0,0)}T = k^2$.

All three curves thus have a single singular point at the origin with two-dimensional tangent space; the Zariski tangent space alone cannot distinguish them. The actual geometric types differ: the node consists of two smooth branches meeting transversally, the cusp is one branch “returning on itself”, and the tacnode is two branches tangent to the x -axis. Finer invariants (multiplicity, Milnor number, intersection numbers) are needed to tell them apart.

Example 7.6.7 (Tangent Vectors via Curves). On a C^∞ -manifold one usually represents tangent vectors by smooth curves:

$$\{\gamma: (-1, 1) \rightarrow X \text{ smooth}, \gamma(0) = x\} \rightarrow T_x X, \quad \gamma \mapsto ([f] \mapsto (f \circ \gamma)'(0)).$$

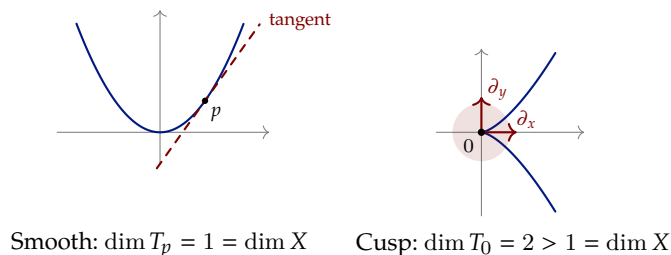
For general schemes there may be no such curves, but Definition 7.6.1 provides an algebraic replacement that agrees with this construction in the smooth case via the local-ring identification $T_x X \simeq (\mathfrak{m}_x / \mathfrak{m}_x^2)^\vee$ established above.

We close the section's introduction with a comment on why we bother naming the cotangent space at all, given that $T_x X$ has the more transparent pointwise definition — a tangent vector is, after all, simply an infinitesimal displacement, whereas a cotangent vector is the more abstract notion of a function vanishing at x taken modulo its second-order error term.

The reason is structural: at a single point the two definitions are mutually dual, but as soon as one passes to global, sheaf-theoretic, relative geometry, the symmetry breaks. The construction $\mathfrak{m}_x / \mathfrak{m}_x^2$ globalizes and relativizes in a way the dual-number tangent functor does not. For a closed subfunctor cut out by an ideal sheaf $\mathcal{I}$, the analogous quotient $\mathcal{I} / \mathcal{I}^2$ is a quasi-coherent module on the subscheme; for a morphism $X \rightarrow Y$, the analogous quotient on the diagonal yields a quasi-coherent sheaf $\Omega_{X/Y}$ on X , the *relative cotangent sheaf*. The dual-number tangent functor admits no comparable direct upgrade to a quasi-coherent “tangent sheaf” outside the smooth case — the natural candidate $\mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/Y}, \mathcal{O}_X)$ behaves badly when $\Omega_{X/Y}$ fails to be locally free. This asymmetry is invisible at a single point but decisive at the sheaf level, and it is the technical reason why the development to come, and the smoothness theory of the next chapter, are organized around cotangent and conormal objects, with their tangent and normal counterparts treated as duals.

7.6.2. Regular Local Rings

Geometrically, regularity is the algebraic counterpart of *smoothness*: the tangent space has the “correct” dimension matching $\dim X$. The figure shows a smooth point of an affine plane curve with its tangent line, contrasted with the singular cusp $y^2 = x^3$ at the origin where “too many” tangent directions exist.



The fundamental bound on Zariski cotangent dimension is the following.

Proposition 7.6.8 (Nakayama Bound). *Let $(R, \mathfrak{m})$ be a noetherian local ring with residue field κ . Then*

$$\dim(R) \leq \dim_{\kappa}(\mathfrak{m}/\mathfrak{m}^2).$$

Proof. By Nakayama's lemma, $\dim_{\kappa}(\mathfrak{m}/\mathfrak{m}^2)$ equals the minimal number of generators of $\mathfrak{m}$. The bound is then Corollary 7.5.13 applied to $\mathfrak{p} = \mathfrak{m}$. $\square$

Definition 7.6.9 (Regular Local Ring). A noetherian local ring $(R, \mathfrak{m})$ is *regular* if equality holds in Proposition 7.6.8: $\dim(R) = \dim_{\kappa}(\mathfrak{m}/\mathfrak{m}^2)$. Equivalently, $\mathfrak{m}$ can be generated by $\dim(R)$ elements.

Example 7.6.10 ($\mathbb{A}_k^n$ at the Origin). For a field k , the local ring $R = k[x_1, \dots, x_n]_{(x_1, \dots, x_n)}$ has $\dim(R) = n$ and $\mathfrak{m} = (x_1, \dots, x_n)$ generated by exactly n elements. Hence R is regular — the affine space $\mathbb{A}_k^n$ is smooth at every closed point, as expected.

The following standard structural results on regular local rings are quoted from commutative algebra; the proofs are nontrivial, but the statements are what one uses in practice.

Theorem 7.6.11 (Regular Local Rings: Standard Facts). *Let $(R, \mathfrak{m})$ be a regular local ring of dimension d with residue field $\kappa = R/\mathfrak{m}$. Then:*

(i) *the associated graded*

$$\mathrm{gr}_{\mathfrak{m}}(R) = \bigoplus_{i \geq 0} \mathfrak{m}^i / \mathfrak{m}^{i+1} \simeq \mathrm{Sym}_{\kappa}(\mathfrak{m}/\mathfrak{m}^2) \simeq \kappa[t_1, \dots, t_d]$$

is a polynomial ring;

(ii) *(Auslander–Buchsbaum 1959) R is a unique factorization domain;²*

(iii) *for every prime $\mathfrak{p} \subset R$, the localization $R_{\mathfrak{p}}$ is regular local.*

Morally:

(i) *is the “infinitesimal polynomial ring” picture at a smooth point (for proof, see [The25, Tag 00NO]);*

(ii) *is algebraic smoothness manifesting as unique factorization (for proof, see [The25, Tag 0AG0]);*

(iii) *is exactly what licenses the global definition of a regular scheme (for proof, see [The25, Tag 0AFS]).*

Theorem 7.6.12 (Serre’s Homological Characterization of Regularity). *For a noetherian ring R and a finite integer $n \in \mathbb{N}$, the following are equivalent:*

(i) *R is regular of Krull dimension $\leq n$;*

(ii) *every R -module has a finite projective resolution of length $\leq n$;*

(iii) *for every ideal $I \subset R$, the R -module R/I admits a finite projective resolution of length $\leq n$.*

The equivalence (i) $\Leftrightarrow$ (ii) is [The25, Tag 00OE], and (ii) $\Leftrightarrow$ (iii) is [The25, Tag 065T]. Consequently, a noetherian ring has finite global dimension if and only if it is regular and has finite Krull dimension; in that case the two dimensions agree. The finite-dimensional qualification is essential: a regular noetherian ring of unbounded Krull dimension need not have finite global dimension. Condition (ii), applied locally, also gives Theorem 7.6.11(iii).

7.6.3. Regular Rings and Regular Schemes

Definition 7.6.13 (Regular Ring, Regular Scheme). A ring R is *regular* if it is noetherian and every localization $R_{\mathfrak{p}}$ is regular local. A scheme X is *regular* if it is locally noetherian and every stalk $\mathcal{O}_{X,x}$ is regular local.

Remark 7.6.14 (Regularity is Local). “Regular” is a local property of rings: localization preservation (Theorem 7.6.11(iii)) gives axiom (i) of Definition 7.1.2, and gluing from a Zariski cover reduces to a stalk-by-stalk check. By Proposition 7.1.5, regularity extends uniquely to a local property of schemes, equivalent to the stalk formulation above.

²This 1959 theorem of Auslander–Buchsbaum should not be confused with the *Auslander–Buchsbaum formula* ($\mathrm{depth}(M) + \mathrm{pd}(M) = \mathrm{depth}(R)$ for finitely generated M of finite projective dimension), which is the result more commonly meant by “Auslander–Buchsbaum theorem” in modern usage.

7.7. Cohen–Macaulay Rings and Depth

Regularity is too restrictive in practice; Cohen–Macaulay-ness relaxes it by replacing “ $\mathfrak{m}$ generated by $\dim(R)$ elements” with “ $\mathfrak{m}$ contains a regular sequence of length $\dim(R)$ ”. The geometric content is best seen by contrast with dimension. The Krull dimension counts how many irreducible closed subsets of $\text{Spec}(R)$ can be nested through the closed point; the depth instead counts the longest sequence of cuts in $\mathfrak{m}$ that genuinely lower the dimension at each step — each cut a non-zerodivisor on what remains, refusing to be absorbed into existing components. When these counts match, this is the Cohen–Macaulay condition. Embedded associated primes are one mechanism that can force a depth defect, and the table below isolates that mechanism in a concrete case. They are not the only mechanism: absence of embedded primes is much weaker than Cohen–Macaulayness, and even domains can fail to be Cohen–Macaulay.

ideal J	$V(J)$	k -basis of $k[x, y]/J$	geometric content
(x)	y -axis	$1, y, y^2, \dots$	 values along the axis
(x^2, y)	origin	$1, x$	 value $+ \partial_x$ at origin
$(x^2, xy) = (x) \cap (x^2, y)$	y -axis	$1, x, y, y^2, \dots$	 axis values $+ \partial_x$ at origin

The third row decomposes as the union of the first two: $V(x^2, xy) = V(x) \cup V(x^2, y)$, mirroring the primary decomposition $(x^2, xy) = (x) \cap (x^2, y)$. The minimal prime (x) gives the clean y -axis; the embedded prime $(x, y) \supsetneq (x)$ is witnessed by the fat point $V(x^2, y)$, geometrically a thickening of the origin in the x -direction. The third scheme is *not* Cohen–Macaulay: every element of $\mathfrak{m} = (x, y)$ is annihilated by something supported at the embedded prime, so $\text{depth} = 0 < 1 = \dim$.

We now formalize the two notions made informal above — “cuts that lower dimension” and “regular sequence” — and define depth and Cohen–Macaulay-ness precisely.

Definition 7.7.1 (System of Parameters). A *system of parameters* in a noetherian local ring $(R, \mathfrak{m})$ of dimension d is a sequence $f_1, \dots, f_d \in \mathfrak{m}$ with $\sqrt{(f_1, \dots, f_d)} = \mathfrak{m}$. By Proposition 7.5.19, d is the minimum length of such a sequence, and systems of parameters always exist.

Definition 7.7.2 (Regular Sequence). A sequence $f_1, \dots, f_n \in \mathfrak{m}$ is a *regular sequence* if each f_i is a non-zerodivisor in $R/(f_1, \dots, f_{i-1})$ (the case $i = 1$ asking that f_1 be a non-zerodivisor in R).

Corollary 7.7.3 (Non-Zerodivisors Drop Dimension). Let $(R, \mathfrak{m})$ be a noetherian local ring. If $f \in \mathfrak{m}$ is a non-zerodivisor in R^{red} (equivalently, f lies in no minimal prime of R), then $\dim(R/f) = \dim(R) - 1$.

Proof. This is the local statement [The25, Tag 00KW]. For the upper bound: any chain of primes $\mathfrak{p}_0 \subsetneq \dots \subsetneq \mathfrak{p}_k = \mathfrak{m}/(f)$ in $R/(f)$ lifts to a chain $\mathfrak{q}_0 \subsetneq \dots \subsetneq \mathfrak{q}_k = \mathfrak{m}$ in R , all containing f ; since f lies in no minimal prime, $\mathfrak{q}_0$ properly contains some minimal prime, giving a length- $(k+1)$ chain in R , so $k+1 \leq \dim R$. For the matching lower bound, lift any system of parameters of $R/(f)$ to elements of R ; together with f this collection generates $\mathfrak{m}$ up to radical, so by Proposition 7.5.19, $\dim(R) \leq \dim(R/f) + 1$. $\square$

Corollary 7.7.4 (Regular Sequences Drop Dimension Additively). For a regular sequence $f_1, \dots, f_n \in \mathfrak{m}$, $\dim(R/(f_1, \dots, f_n)) = \dim(R) - n$.

Proof. Iterate Corollary 7.7.3. $\square$

Definition 7.7.5 (Depth, Cohen–Macaulay). Let $(R, \mathfrak{m})$ be a noetherian local ring.

- (i) The *depth* of R is the maximum length of a regular sequence in $\mathfrak{m}$:

$$\text{depth}(R) := \max\{n \mid \exists \text{ regular sequence } f_1, \dots, f_n \in \mathfrak{m}\}.$$

- (ii) R is *Cohen–Macaulay* (CM) if $\text{depth}(R) = \dim(R)$.

By Corollary 7.7.4, $\text{depth}(R) \leq \dim(R)$ always.

Proposition 7.7.6 (Regular Parameters Are a Regular Sequence). Let R be a regular local ring and let $f_1, \dots, f_d$ be a minimal set of generators for the maximal ideal $\mathfrak{m}$. Then $f_1, \dots, f_d$ is a regular sequence, and each $R/(f_1, \dots, f_c)$ is a regular local ring of dimension $d - c$. In particular, R is Cohen–Macaulay.

Proof. See [The25, Tag 00NQ]. □

Remark 7.7.7 (Regular $\Rightarrow$ Cohen–Macaulay). By Proposition 7.7.6, a regular local ring has a regular sequence of length $\dim(R)$ in its maximal ideal, so $\text{depth}(R) \geq \dim(R)$. Combined with the reverse inequality (Definition 7.7.5), equality holds: regular implies Cohen–Macaulay.

Example 7.7.8 (The Cross: Cohen–Macaulay but Not Regular). Consider the local ring $R = (k[x, y]/(xy))_{(x, y)}$, obtained by localizing the coordinate ring of the cross $V(xy) \subset \mathbb{A}_k^2$ at the origin. Geometrically R captures the infinitesimal neighborhood of the singular point where the two coordinate axes meet; one has $\dim(R) = 1$.

This ring is not regular: the maximal ideal $\mathfrak{m} = (x, y)$ requires two generators, since x and y live in distinct minimal primes (the two axes) and no single element of $\mathfrak{m}$ generates $\mathfrak{m}$. Equivalently, $\mathfrak{m}/\mathfrak{m}^2$ has dimension $2 > 1 = \dim(R)$.

Yet R is Cohen–Macaulay, because the line $V(x - y)$ meets the cross transversally: it crosses both axes only at the origin. Algebraically, the element $x - y$ is a non-zerodivisor in R . To see this, note that $R = (k[x, y]/(xy))_{(x, y)}$ is a localization of $k[x, y]/(xy)$, and the minimal primes of the latter are (x) and (y) . Since $x - y$ lies in neither (x) nor (y) (as $-y \notin (x)$ and $x \notin (y)$), it is a non-zerodivisor in $k[x, y]/(xy)$, hence also in any localization. Thus $(x - y)$ is a regular sequence of length 1 in $\mathfrak{m}$, and $\text{depth}(R) \geq 1 = \dim(R)$, forcing equality. The cross is singular but has no “hidden embedded components”.

Example 7.7.9 (The Cuspidal Cubic). Let k be a field with $\text{char}(k) \neq 2, 3$ and $X = \text{Spec}(k[x, y]/(y^2 - x^3))$. We compute T_p^*X at two points.

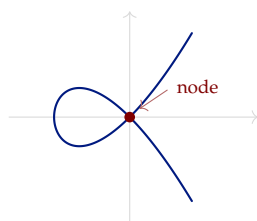
At the singular point $p = (0, 0)$, $\mathfrak{m} = (x, y)$ and the relation $y^2 - x^3$ lies in $\mathfrak{m}^2$, so $\mathfrak{m}/\mathfrak{m}^2 \simeq k\{x, y\}$ is two-dimensional. Hence $\dim T_{(0,0)}^*X = 2 > 1 = \dim X$, so X is not regular at the origin.

At the smooth point $p = (1, 1)$, substitute $z = x - 1$, $w = y - 1$. The relation $y^2 = x^3$ becomes $(1 + w)^2 = (1 + z)^3$, i.e. $w^2 + 2w = z^3 + 3z^2 + 3z$. Modulo $(z, w)^2$ this reduces to $2w = 3z$, and since $2, 3 \in k^\times$ this eliminates w in favor of z . So $T_{(1,1)}^*X$ has basis $\{z\}$ and $\dim T_{(1,1)}^*X = 1 = \dim X$, confirming that X is regular at $(1, 1)$.

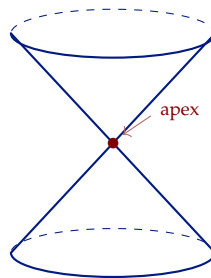
7.8. Normal Schemes

Normality is a weaker cousin of regularity, defined by integral-closedness of all local rings. Under the locally noetherian hypotheses used below, Serre’s criterion says that normality has two independent parts: regularity in codimension 1 and the depth condition (S_2) . Consequently a normal scheme can have singularities only in codimension at least 2, but placing a singular locus in codimension at least 2 does not by itself prove normality. Two complementary intuitions make this concrete.

The first is *Hartogs extension*: on a normal noetherian scheme X , every rational function regular on the complement of a codimension- ≥ 2 closed subset already extends regularly across that subset. This is the algebraic shadow of the classical Hartogs phenomenon for holomorphic functions on $\mathbb{C}^n$ ($n \geq 2$): codimension-2 “holes” cannot truly remove regularity. The second is *Serre’s criterion* Theorem 7.8.3: normality is exactly $(R_1) + (S_2)$. The (R_1) condition makes every codimension-1 local ring regular, while (S_2) supplies the depth needed for reconstruction from codimension-1 data. Nodes and cusps on curves violate (R_1) . The quadric cone below is normal for a different, two-step reason: it satisfies (R_1) away from its codimension-2 apex, and, being a hypersurface, it is Cohen–Macaulay and hence satisfies (S_2) .



Nodal curve $V(y^2 - x^3 - x^2)$
singular locus = origin (codim 1)
not normal



Quadric cone $V(xy - z^2) \subset \mathbb{A}^3$
singular locus = apex (codim 2)
normal

The nodal curve fails normality because its singular point is codimension 1 in a 1-dimensional scheme. For the quadric cone, codimension 2 of the apex proves only (R_1) . Its local rings are hypersurface quotients of regular local rings by a nonzerodivisor, so they are Cohen–Macaulay ([The25, Tags 00NQ and 0C6G]) and therefore satisfy (S_2) . Serre’s criterion then proves normality.

In dimension 1, the codim- ≥ 2 allowance is empty, so normal coincides with regular — this is Proposition 7.8.7 below.

The algebraic definition of normality goes through integral closure of rings: a domain A is normal when no “hidden algebraic functions” live just outside A in its fraction field. Concretely, integral elements over A are roots of monic polynomials with coefficients in A , and the condition asks that no such root in $\text{Frac}(A) \setminus A$ exist. The connection to codimension-1 regularity in Serre’s criterion runs through valuation theory. A codimension-1 failure of regularity obstructs normality, but higher-codimension singularities can do so as well when (S_2) fails; codimension alone is not a normality test.

Definition 7.8.1 (Integrally Closed Extension, Integrally Closed Domain). (i) A ring extension $A \subset B$ is *integrally closed* if every $b \in B$ satisfying a monic polynomial $f \in A[x]$ with $f(b) = 0$ already lies in A .
(ii) A domain A is *integrally closed* (or *normal*) if A is integrally closed in its fraction field $\text{Frac}(A)$.

Remark 7.8.2 (UFDs and Regular Local Rings Are Normal). Every UFD is integrally closed in its fraction field. By Theorem 7.6.11(ii), every regular local ring is a UFD, hence normal.

Theorem 7.8.3 (Serre’s Criterion for Normality). *Let A be a noetherian domain. The following are equivalent:*

- (i) A is integrally closed;
- (ii) for every prime $\mathfrak{p} \subset A$ of codimension 1, the localization $A_{\mathfrak{p}}$ is a DVR, and $A = \bigcap_{\text{codim}(\mathfrak{p})=1} A_{\mathfrak{p}}$ inside $\text{Frac}(A)$.

Proof. The $(R_1) + (S_2)$ characterization of normality is [The25, Tag 031S]; in particular, integrally closedness gives that every codimension-1 localization is regular of dimension 1, hence a DVR. The intersection equality $A = \bigcap_{\text{codim}(\mathfrak{p})=1} A_{\mathfrak{p}}$ for noetherian normal domains is [The25, Tag 031T]. Conversely, if every codimension-1 localization is a DVR and A equals their intersection, then A is the intersection of integrally closed subrings of $\text{Frac}(A)$ and hence integrally closed. $\square$

Definition 7.8.4 (Normal Scheme). A scheme X is *normal* if for every $x \in X$ the local ring $\mathcal{O}_{X,x}$ is an integrally closed domain.

Definition 7.8.5 (Valuation Ring, DVR). A *valuation ring* is a domain R in which divisibility is a total order. A *discrete valuation ring* (DVR) is a noetherian valuation ring that is not a field; equivalently, a local PID that is not a field.

Theorem 7.8.6 (Krull's Intersection Theorem). Let $(R, \mathfrak{m})$ be a noetherian local ring. Then $\bigcap_{n \geq 0} \mathfrak{m}^n = 0$. See [Mat89, Theorem 8.10].

Proposition 7.8.7 (Regular Local of Dimension 1 = DVR). Let $(R, \mathfrak{m})$ be a noetherian local ring of dimension 1. Then R is regular if and only if R is a DVR.

Proof. Suppose R is a DVR. Then $\mathfrak{m} = (\pi)$ is principal, so R is regular of dimension 1.

Conversely, suppose R is regular of dimension 1. By Theorem 7.6.11(ii), R is a UFD — in particular a domain — so $|\text{Spec}(R)| = \{(0), \mathfrak{m}\}$ and $\mathfrak{m} = (\pi)$ for a single uniformiser π . We must show every nonzero $a \in R$ has a unique expression $a = u\pi^n$ with $u \in R^\times$ and $n \geq 0$. By Theorem 7.8.6, $\bigcap_n \mathfrak{m}^n = 0$, so there is a maximal n with $a \in \mathfrak{m}^n$; write $a = u\pi^n$. If $u \in \mathfrak{m}$, then $a = u\pi^n \in \mathfrak{m}^{n+1}$, contradicting maximality of n ; hence $u \notin \mathfrak{m}$, and since R is local this forces $u \in R^\times$. Uniqueness of n and u follow by the same maximality argument. Thus R is a DVR. $\square$

Example 7.8.8 (Regular Local Rings of Dimension ≤ 1). (i) R is regular of dimension 0 iff R is a field.
(ii) R is regular of dimension 1 iff R is a DVR (Proposition 7.8.7).

When an integral scheme fails to be normal, integral closure produces a canonical normal replacement over it, the *normalization*. In dimension 1, this replacement is regular provided it is locally noetherian, by Proposition 7.8.7; no such regularity conclusion holds for normalization in higher dimension.

Construction 7.8.9. Let X be an integral scheme. The *normalization* of X is a morphism $X^\nu \rightarrow X$, which is uniquely determined by the following properties:

- (i) X^ν is a normal integral scheme;
- (ii) $X^\nu \rightarrow X$ is integral;
- (iii) $X^\nu \rightarrow X$ induces an isomorphism of function fields.

The next example carries this out by hand for the cusp, illustrating how a rational function “invented” on X can become regular on the cover.

Example 7.8.10 (Normalization of the Cuspidal Cubic). Let k be a field, and consider the cuspidal cubic $X = \operatorname{Spec}(A)$ with $A = k[x, y]/(y^2 - x^3)$, an integral 1-dimensional k -scheme with a unique singular point at the origin (Example 7.7.9). The defining polynomial $y^2 - x^3$ is irreducible, so A is a domain; its fraction field $K = \operatorname{Frac}(A)$ is generated by x and y subject to $y^2 = x^3$.

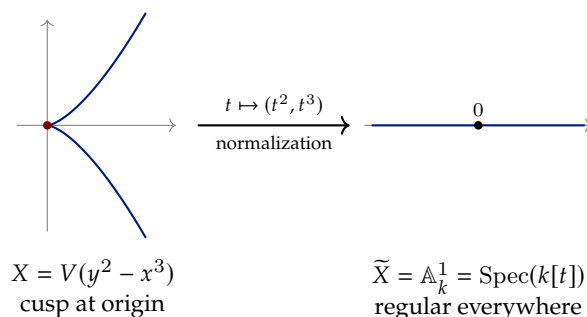
Set $t := y/x \in K$. The relation $y^2 = x^3$ rewrites as

$$t^2 = \frac{y^2}{x^2} = \frac{x^3}{x^2} = x,$$

so t satisfies the monic polynomial $T^2 - x = 0$ over A , hence is integral over A . Moreover $y = tx = t \cdot t^2 = t^3$. Thus inside K we have $A \subseteq A[t] = k[t]$ with $x = t^2, y = t^3$, and $k[t]$ is integral over A . Conversely, if $z \in K$ is integral over A , then the same monic equation also shows that z is integral over $k[t]$, because $A \subseteq k[t]$. Since the PID $k[t]$ is integrally closed in $K = \operatorname{Frac}(k[t])$, it follows that $z \in k[t]$. Therefore

$$\tilde{A} = k[t], \quad \tilde{X} = \mathbb{A}_k^1 \xrightarrow{t \mapsto (t^2, t^3)} X.$$

Geometrically, the normalization map separates the cusp: $\mathbb{A}_k^1$ is regular everywhere, while X has its cuspidal singularity at the origin; the map is bijective on points but fails to be an isomorphism precisely there, where $t = 0$ pulls back the cusp to the regular point $0 \in \mathbb{A}_k^1$. The cusp is, in this sense, “parametrized away” by the new variable $t = y/x$ — which is rational on X but regular on $\tilde{X}$.



Remark 7.8.11 (Regular versus Normal). For locally noetherian schemes of dimension at most 1, regular and normal coincide (Proposition 7.8.7). In higher dimensions normal is strictly weaker than regular, and the quadric cone $V(xy - z^2) \subset \mathbb{A}^3$ pictured at the start of this section is the prototypical witness: it is normal everywhere but not regular at the apex $(0, 0, 0)$, where the maximal ideal $\mathfrak{m} = (x, y, z)$ requires three generators while the local ring has dimension 2. This singularity is known as the A_1 singularity or ordinary double point. Normality rules out codimension-1 singularities and imposes (S_2) ; subject to that depth condition, it can tolerate isolated singularities such as this codimension-2 apex.

Looking Ahead

The preceding chapters built the foundational categorical machinery; this chapter added the first substantial geometric vocabulary: local versus global properties, reducedness, noetherianity, irreducibility, integrality, dimension (in the Krull sense), and the three central conditions on singularities — regular, Cohen–Macaulay, and normal — with the implications

$$\text{regular} \Rightarrow \text{Cohen–Macaulay} \quad \text{and} \quad \text{regular} \Rightarrow \text{normal}$$

under the usual noetherian hypotheses. For noetherian local domains of dimension at most 1, regularity and normality coincide, but Cohen–Macaulayness is still strictly weaker: the one-dimensional cross of Example 7.7.8 is already a counterexample to a three-way equivalence. Zariski tangent spaces and

Nakayama's lemma provide the concrete tools for detecting regularity at a point, and Serre's criteria (homological for regularity, $R_1 + S_2$ for normality) reduce these conditions to data that are computable in practice.

The remaining chapters of Part II move beyond properties of individual schemes to the geometry of *morphisms*. Smooth, unramified, and étale morphisms are followed by effective, Cartier, and Weil divisors, blowups, and properness. Topics such as Chern classes and intersection theory lie beyond the present volume. The local-to-global technology built here — extension of properties along Zariski covers, the interplay between Mod_X and subfunctors, and dimension-theoretic control — will be used repeatedly in these chapters.

A categorical reformulation deepens the dual-numbers picture and points beyond the static setting. Given an A -module M , the *trivial square-zero extension* $A \oplus M$ is the ring with multiplication $(a, m)(a', m') = (aa', am' + a'm)$ and canonical projection to A ; a k -derivation $D: A \rightarrow M$ is precisely a section of this projection in the slice $(\text{CAlg}_k)_{/A}$, giving $\text{Der}_k(A, M) \simeq \text{Hom}_{(\text{CAlg}_k)_{/A}}(A, A \oplus M)$. Taking $M = k$ recovers the tangent space at a k -point; taking $M = \Omega_{A/k}$ gives the universal derivation $d: A \rightarrow \Omega_{A/k}$. In the derived refinement of algebraic geometry (animated rings, ∞ -categories), the static Kähler module $\Omega_{A/k}$ is replaced by the *cotangent complex* $\mathbb{L}_{A/k}$, characterized by the analogous mapping-space equivalence; concretely, $\mathbb{L}_{-/k}$ is the left-derived (animated) extension of $\Omega_{-/k}$. The cotangent complex governs deformation theory and obstruction theory uniformly across smooth, singular, and derived settings, and will reappear when we discuss smoothness via lifting properties.

Chapter 8.

Smoothness

In this chapter, we introduce smooth, unramified, and étale morphisms.

Smooth morphisms in algebraic geometry are the analogue of *submersions* in differential geometry, namely those $f: X \rightarrow Y$ such that the differential $TX \rightarrow f^*TY$ is surjective, so that tangent vectors can be lifted. In particular, this has no relation to the use of “smooth” to mean “ C^∞ ” in analysis (in that sense, every map in algebraic geometry is “smooth”, since polynomial maps are). Étale morphisms are smooth morphisms of relative dimension zero; they are analogous to *local diffeomorphisms* in differential geometry (indeed, by the inverse function theorem, a submersion of relative dimension zero is a local diffeomorphism). A third class, the *unramified* morphisms, are the analogues of immersions: the differential is injective rather than surjective, so that an infinitesimal lift is unique when it exists.

8.1. Submersions and Infinitesimal Lifting

We begin with some recollections from differential geometry. If $f_1, \dots, f_m$ are C^∞ -functions on $\mathbb{R}^n$, the implicit function theorem gives a sufficient condition under which the vanishing locus

$$X = V(f_1, \dots, f_m) \subset \mathbb{R}^n$$

acquires the structure of a C^∞ -manifold: it suffices that, at every point $p \in X$, the Jacobian matrix

$$\left(\frac{\partial f_i}{\partial x_j}(p) \right) \in \text{Mat}_{m \times n}(\mathbb{R})$$

have rank m . In that case, the m rows of this matrix form a basis of the normal space to X at p , whose orthogonal complement in $\mathbb{R}^n$ is the tangent space to X at p . As in algebraic geometry, the vanishing locus $X = V(f_1, \dots, f_m) \subset \mathbb{R}^n$ is a priori a (possibly singular) geometric object called a “ C^∞ -scheme”, and the Jacobian criterion is then more precisely a sufficient condition for the map of C^∞ -schemes $X \rightarrow *$ to be a submersion.

A peculiar feature of the real numbers is that the standard scalar product on $\mathbb{R}^n$ remains non-degenerate on every subspace — this is not the case over other fields, not even over $\mathbb{C}$. Consequently, every subspace of $\mathbb{R}^n$ is canonically isomorphic to its dual space, which can cause some confusion. Generalizing the above discussion away from the real numbers reveals that the row space of the Jacobian matrix is more canonically the *conormal* space, i.e., the dual of the normal space, whose elements are the linear forms vanishing on tangent vectors. Concretely, the rows are the differentials df_i of the defining functions, and a tangent vector at p is by definition a vector on which all of these differentials vanish.

This conormal interpretation of the Jacobian rows is the local shadow of a structural pattern that pervades algebraic geometry. Algebraic geometry, as developed in this text, is the study of *functions on algebraic functors*: the intrinsic datum of an affine scheme Sol_Σ is its ring of functions $k[\Sigma] = O(\text{Sol}_\Sigma)$, the system Σ itself being merely a presentation (Remark 1.2.13). Studying X near a point x thus means studying the functions vanishing at x to first order, which assemble into the cotangent space $T_x^*X = \mathfrak{m}_x / \mathfrak{m}_x^2$ (Definition 7.6.5); the tangent space $T_xX = (T_x^*X)^\vee$ appears as its dual. We will accordingly define cotangent

and conormal objects first throughout this chapter, with their tangent and normal counterparts treated as duals.

It remains to pin down what the morphisms of this chapter — smooth, unramified, étale — actually are. The implicit function theorem is unavailable in the Zariski topology, so we cannot define a submersion by exhibiting a local product structure, as one does in differential geometry. We define it instead by what a submersion *does*: it lifts infinitesimal motion. The first task is to make “infinitesimal motion” algebraic.

Consider the tower of truncated polynomial rings

$$k = k[x]/(x) \leftarrow k[x]/(x^2) \leftarrow k[x]/(x^3) \leftarrow \cdots, \quad \varprojlim_n k[x]/(x^n) = k[[x]],$$

in which each map forgets the top-degree coefficient. Dually, the *fat points* $D_n := \operatorname{Spec} k[x]/(x^{n+1})$ form an increasing chain of infinitesimal thickenings of the origin,

$$\{0\} = D_0 \hookrightarrow D_1 \hookrightarrow D_2 \hookrightarrow \cdots, \quad \widehat{\mathbb{A}}_0^1 := \operatorname{colim}_n D_n = \operatorname{Spf} k[[x]],$$

the *formal disk*. The last equality is notation in the category of formal schemes: the ordinary scheme $\operatorname{Spec} k[[x]]$ is not this colimit and has, for example, a generic point. None of the D_n is the affine line $\mathbb{A}^1$: each is a single point carrying only the nilpotent fuzz $x^{n+1} = 0$, so the inclusions add infinitesimal structure without adding points.

A morphism from a fat point into a scheme is a *jet*. For an affine scheme $X = \operatorname{Spec} A$, a k -morphism $\gamma: D_n \rightarrow X$ is the same datum as a k -algebra homomorphism $\gamma^*: A \rightarrow k[x]/(x^{n+1})$; writing $\gamma^*(f) = \sum_{i=0}^n c_i(f) x^i$, the coefficient maps c_i are the Hasse–Schmidt higher derivatives along γ . When the relevant factorials are invertible they agree, after division by $i!$, with ordinary successive derivatives; this qualification is essential in positive characteristic. The case $n = 1$ is precisely the dual-number identity $f(a + bx) \equiv f(a) + f'(a)bx \pmod{x^2}$ of Lemma 7.6.2: the first-order fat point D_1 reads off the first derivative, and the tower reads off all of them. The formal disk thus carries the complete Taylor expansion of an arc, organized order by order, and a jet $D_n \rightarrow X$ remembers exactly the Hasse–Schmidt derivatives of order $\leq n$.

This recasts the notion of submersion as one that survives the absence of an implicit function theorem. We shall call a morphism $f: X \rightarrow Y$ *formally smooth* when every infinitesimal motion in Y whose restriction has already been lifted to X admits a lift to X . The condition is genuinely first order, imposed at every order: each rung $D_n \hookrightarrow D_{n+1}$ adds a single *square-zero* shell — its ideal $(x^{n+1})/(x^{n+2})$ has square contained in $(x^{2n+2}) \subseteq (x^{n+2})$ — so climbing the whole tower amounts to lifting, one order at a time, against an arbitrary *square-zero thickening* $T_0 \hookrightarrow T$: a closed immersion of affine schemes whose defining ideal I satisfies $I^2 = 0$. Lifting against these thickenings is the algebraic shadow of “ df is surjective,” promoted from tangent vectors to jets of every order. Geometric smoothness additionally imposes local finite presentation; without it, for example, a polynomial algebra on infinitely many variables is formally smooth but not smooth.

That last word — *arbitrary* — is essential, and is a genuine strengthening over the tower itself. A square-zero thickening is the inclusion $\operatorname{Spec} R \hookrightarrow \operatorname{Spec} T$ of the closed subscheme cut out by a square-zero ideal $I = \ker(T \twoheadrightarrow R)$, which is thereby an R -module; its datum thus involves a base ring R and a coefficient module I . (In the split case $T = R \oplus M$ this is exactly the pair (R, M) , but a general square-zero extension need not split.) The tower fixes both — it sits over the single field k , with one-dimensional ideals $(x^{n+1})/(x^{n+2}) \cong k$ — and each must be allowed to vary. The coefficient module must vary because the lifts will be controlled by the module of differentials $\Omega_{B/A}$ of Section 8.2, and a module is detected only by pairing it against *all* coefficient modules, never the single $M = k$ the tower supplies. The base ring must vary because smoothness has to hold over every base and at every point, rational or not: the purely inseparable extension $k = \mathbb{F}_p(t) \subset k[x]/(x^p - t)$ has no k -points at all, yet is not étale — a failure no jet over k can see, though a single square-zero thickening over its residue field does.

Three formal notions emerge, according as such a lift always *exists*, is *unique*, or both. The geometric classes are obtained by adding the indicated finiteness hypothesis. We record the dictionary now as a signpost; each entry is made precise, and the equivalences proved, over the course of the chapter.

morphism $f: X \rightarrow Y$	finiteness	infinitesimal lift	df	differential geometry
smooth	locally finitely presented	a lift exists	surjective	submersion
unramified	locally finite type	at most one lift	injective	immersion
étale	locally finitely presented	a unique lift exists	bijective	local diffeomorphism

Here “infinitesimal lift” means a lift against a square-zero thickening as above. The map $df: TX \rightarrow f^*TY$ is dual to the cotangent map $f^*\Omega_Y \rightarrow \Omega_X$; its relative kernel is dual to $\Omega_{X/Y}$ when the latter is finite locally free. One should read the finiteness and lifting columns together as the geometric definition; the lifting statement is given its precise torsor-theoretic form in Proposition 8.2.4.

8.2. Kähler Differentials

The conormal philosophy of the preceding pages instructs us to build, for a morphism of schemes, a quasi-coherent sheaf of differentials before any tangent object, recovering the tangent side only afterwards by dualizing. We begin with the affine prototype. To a homomorphism of commutative rings $R \rightarrow A$ we attach an A -module $\Omega_{A/R}$ together with a universal derivation $d: A \rightarrow \Omega_{A/R}$, which linearizes A over R to first order. This single module — and the two exact sequences relating it across compositions and quotients — generates the whole chapter: the cotangent and conormal sheaves, the Jacobian criterion, and the characterizations of smooth, unramified, and étale. The material is classical. We prove the elementary algebraic steps in full and explicitly mark deeper inputs with a proof roadmap and a precise external reference.

The starting point is the notion of derivation, already met in the previous chapter for the local ring at a point (Definition 7.6.5). Having just used square-zero thickenings to probe morphisms infinitesimally (Section 8.1), we adopt the same device as the definition: a derivation is a section of the trivial square-zero extension. Throughout, $R \rightarrow A$ is a homomorphism of commutative rings, and “module” means a module over the indicated commutative ring.

Derivations via square-zero extensions

Definition 8.2.1 (Square-zero extension; derivation). Let $R \rightarrow A$ be a ring homomorphism and M an A -module. The *trivial square-zero extension* of A by M is the commutative ring $A \oplus M$ with multiplication

$$(a, m)(a', m') = (aa', am' + a'm),$$

in which $M = 0 \oplus M$ is an ideal of square zero. The inclusion $a \mapsto (a, 0)$ makes $A \oplus M$ an A -algebra, hence an R -algebra, and the projection $\text{pr}: A \oplus M \rightarrow A$, $(a, m) \mapsto a$, is a homomorphism of R -algebras. An R -linear derivation of A valued in M is an R -algebra section of pr , that is, an R -algebra homomorphism $s: A \rightarrow A \oplus M$ with $\text{pr} \circ s = \text{id}_A$. The set of all such sections is denoted $\text{Der}_R(A, M)$.

Remark 8.2.2 (The square-zero and cotangent functors). The trivial square-zero extension is functorial in the pair (A, M) : a ring homomorphism $A \rightarrow A'$ together with a map $M \rightarrow M'$ linear over it induces $A \oplus M \rightarrow A' \oplus M'$. Assemble these pairs into a category Mod — an object is a ring together with a module over it, a morphism $(A, M) \rightarrow (A', M')$ a ring map $A \rightarrow A'$ together with an A -linear map $M \rightarrow M'$ (equivalently an A' -linear map $M \otimes_A A' \rightarrow M'$) — fibered over CAlg by the forgetful functor $(A, M) \mapsto A$, the cocartesian fibration classifying $A \mapsto \text{Mod}_A$, with fiber Mod_A over each ring A . The square-zero extension is then a functor into the arrow category $\text{Fun}([1], \text{CAlg})$,

$$\text{Mod} \longrightarrow \text{Fun}([1], \text{CAlg}), \quad (A, M) \longmapsto (A \oplus M \xrightarrow{\text{pr}} A),$$

landing in the split surjections. In this language Definition 8.2.1 is a single slice-categorical identity,

$$\mathrm{Der}_R(A, M) = \mathrm{Hom}_{(\mathrm{CAlg}_R)_/A}(A, A \oplus M),$$

a derivation being a section of the square-zero extension — a map in the slice over A from id_A to $A \oplus M$. By the universal property of Definition 8.2.6, the functor $M \mapsto \mathrm{Der}_R(A, M)$ on Mod_A is corepresented by $\Omega_{A/R}$, the value at id_A of the left adjoint to $M \mapsto A \oplus M$.

The cotangent module runs in the opposite direction, and in doing so packages its own functoriality. A commutative square of rings

$$\begin{array}{ccc} R & \longrightarrow & A \\ \downarrow & & \downarrow \\ R' & \longrightarrow & A' \end{array}$$

makes the composite $A \rightarrow A' \xrightarrow{d} \Omega_{A'/R'}$ an R -derivation of A — it kills R because d kills R' — so Definition 8.2.6 factors it through a canonical A -linear comparison map $\Omega_{A/R} \rightarrow \Omega_{A'/R'}$, equivalently an A' -linear map $\Omega_{A/R} \otimes_A A' \rightarrow \Omega_{A'/R'}$. Being compatible with vertical composition of squares, this exhibits the assignment $(R \rightarrow A) \mapsto (A, \Omega_{A/R})$ as a functor Ω from $\mathrm{Fun}([1], \mathrm{CAlg})$ to Mod lying over CAlg :

$$\begin{array}{ccc} & & \mathrm{Mod} \\ & \nearrow \Omega & \downarrow \text{forget} \\ \mathrm{Fun}([1], \mathrm{CAlg}) & \xrightarrow{\text{target}} & \mathrm{CAlg}. \end{array}$$

The triangle commutes because the underlying ring of $(A, \Omega_{A/R})$ is the target A of the arrow $R \rightarrow A$. The cases in which the comparison map is an *isomorphism* — base change along a cocartesian square, and localization of A — are the content of Proposition 8.2.14. With the test ring allowed to vary, this same functor of points globalizes $\Omega_{A/R}$ to the cotangent sheaf $\Omega_{X/Y}$ in the next section.

This square-zero construction is the same dual-numbers device that organized the tangent space in Definition 7.6.1, now run with coefficients in an arbitrary module: it converts the multiplicative Leibniz rule into the single statement that a map is a ring homomorphism, as the following dictionary records.

Lemma 8.2.3 (Leibniz form of a derivation). *A section s of pr has the form $s(a) = (a, D(a))$ for a unique map $D: A \rightarrow M$, and s is an R -algebra homomorphism if and only if D is additive, vanishes on the image of R , and satisfies the Leibniz rule*

$$D(aa') = a D(a') + a' D(a).$$

Identified with these maps D , the set $\mathrm{Der}_R(A, M)$ is an A -module under the pointwise operations $(D + D')(a) = D(a) + D'(a)$ and $(c \cdot D)(a) = c D(a)$, and is functorial in M : an A -linear map $M \rightarrow N$ sends $\mathrm{Der}_R(A, M) \rightarrow \mathrm{Der}_R(A, N)$ by post-composition. Maps D of this kind are the derivations of [The25, Tag 00RN].

Proof. Since $\mathrm{pr} \circ s = \mathrm{id}_A$, we have $s(a) = (a, D(a))$ for a unique $D(a) \in M$. Now s is additive iff D is; it preserves 1 and the image of R iff $D(1) = 0$ and $D(r) = 0$ for $r \in R$; and, comparing

$$s(aa') = (aa', D(aa')) \quad \text{with} \quad s(a)s(a') = (aa', a D(a') + a' D(a)),$$

s is multiplicative iff D obeys the Leibniz rule. Additivity and Leibniz already force $D(1) = 2D(1) = 0$, so vanishing on the image of R is equivalent to the stronger R -linearity $D(ra) = r D(a)$. The A -module structure and the functoriality in M are immediate from the operations on M . $\square$

The reason derivations belong to a chapter on smoothness is that they measure the ambiguity in solving an *infinitesimal lifting problem*. The next proposition is the algebraic heart of the entire theory; the reader should read it as the precise content of the slogan from the introduction that for a submersion “tangent vectors can be lifted”.

Proposition 8.2.4 (Derivations control infinitesimal lifts). *Let $R \rightarrow A$ be a ring homomorphism and $\pi: T' \twoheadrightarrow T$ a surjection of R -algebras whose kernel I satisfies $I^2 = 0$, so that I is naturally a T -module. Let $u: A \rightarrow T$ be an R -algebra homomorphism, and regard I as an A -module through u . A lift of u is an R -algebra homomorphism $\tilde{u}$ — the dashed arrow — making the square*

$$\begin{array}{ccc} R & \longrightarrow & T' \\ \downarrow & \nearrow \tilde{u} & \downarrow \pi \\ A & \xrightarrow{u} & T \end{array}$$

commute. Then the set L of lifts of u is either empty or a torsor under the additive group $\text{Der}_R(A, I)$: the formula $(\tilde{u}, D) \mapsto \tilde{u} + D$, with $(\tilde{u} + D)(a) := \tilde{u}(a) + D(a)$, defines a free and transitive action.

Proof. Because $I^2 = 0$, the ideal I annihilates itself, so its T' -module structure factors through π and makes I a module over $T = T'/I$ and, via u , over A ; for $a \in A$ and $\xi \in I$ we have $\tilde{u}(a)\xi = u(a)\xi = a \cdot \xi$ for any lift $\tilde{u}$, since $\tilde{u}(a)$ and $u(a)$ differ by an element of I and $I \cdot I = 0$.

Suppose $\tilde{u} \in L$ and $D \in \text{Der}_R(A, I)$. Each value $(\tilde{u} + D)(a) = \tilde{u}(a) + D(a)$ lies in T' , and $\pi(\tilde{u}(a) + D(a)) = u(a)$ since $D(a) \in I = \ker \pi$; thus $\tilde{u} + D$ lifts u and is additive and R -linear. For multiplicativity, expand

$$(\tilde{u} + D)(a)(\tilde{u} + D)(a') = \tilde{u}(a)\tilde{u}(a') + \tilde{u}(a)D(a') + \tilde{u}(a')D(a) + D(a)D(a').$$

The last term lies in $I^2 = 0$, and by the previous paragraph $\tilde{u}(a)D(a') = a D(a')$ and $\tilde{u}(a')D(a) = a' D(a)$. Hence the right-hand side equals $\tilde{u}(aa') + a D(a') + a' D(a) = \tilde{u}(aa') + D(aa')$ exactly when D satisfies Leibniz, which it does. So $\tilde{u} + D \in L$.

Conversely, if $\tilde{u}, \tilde{u}' \in L$, set $D := \tilde{u} - \tilde{u}'$. Both lifts reduce to u modulo I , so D takes values in I ; it is additive and kills R . Running the displayed computation backwards (with $\tilde{u} = \tilde{u}' + D$ and using $\tilde{u}'$ multiplicative) shows that $\tilde{u}$ is multiplicative iff D obeys Leibniz, so $D \in \text{Der}_R(A, I)$ by Lemma 8.2.3, and $\tilde{u} = \tilde{u}' + D$. The action is therefore transitive; it is free because $\tilde{u}' + D = \tilde{u}'$ forces $D = 0$. $\square$

Remark 8.2.5 (Toward formal smoothness). Proposition 8.2.4 isolates the two independent questions hidden in an infinitesimal lifting problem. *Existence* of a lift ($L \neq \emptyset$) is an obstruction-theoretic matter; *uniqueness* of a lift is governed entirely by the group $\text{Der}_R(A, I)$ acting on it. We will declare $R \rightarrow A$ *formally smooth* when a lift always exists, *formally unramified* when a lift is unique whenever it exists (equivalently $\text{Der}_R(A, I) = 0$ for all such I), and *formally étale* when a lift exists and is unique. Since $\text{Der}_R(A, I) = \text{Hom}_A(\Omega_{A/R}, I)$ for the module $\Omega_{A/R}$ defined next, the uniqueness side is controlled by that single A -module, the universal recipient of derivations.

The module of differentials

We single out the universal derivation — the one through which every other factors — and take its universal property as the definition of the differentials.

Definition 8.2.6 (Module of Kähler differentials). A *module of Kähler differentials* of A over R is an A -module $\Omega_{A/R}$ together with an R -derivation $d: A \rightarrow \Omega_{A/R}$ that is *universal*: for every A -module M , precomposition with d is a bijection

$$\text{Hom}_A(\Omega_{A/R}, M) \xrightarrow{\sim} \text{Der}_R(A, M), \quad \varphi \mapsto \varphi \circ d.$$

Its elements $da := d(a)$ are the *differentials* of A over R .

Being characterized by a universal property, $(\Omega_{A/R}, d)$ is unique up to a unique isomorphism if it exists, and we speak of *the* module of differentials. For existence we give the construction that globalizes, guided by one principle: *a differential is an infinitesimal difference*. To compare a at a point and at an infinitesimally near one we pass to *pairs* of points, in the functor-of-points sense: a pair of T -points $x, y: A \rightarrow T$ valued in a common test ring T is precisely a T -point of $\mathrm{Spec} A \times_{\mathrm{Spec} R} \mathrm{Spec} A = \mathrm{Spec}(A \otimes_R A)$, the tensor over R being what makes the resulting differentials *relative* to R . On such a pair the element

$$\delta a := 1 \otimes a - a \otimes 1$$

takes the value $y(a) - x(a) \in T$, the difference of a between the two points; the family $(\delta a)_{a \in A}$ vanishes exactly when $x = y$, that is, on the diagonal. The differences are thus the functions vanishing on the diagonal — the kernel I_μ of the multiplication map $\mu: A \otimes_R A \rightarrow A$ — and keeping only the *first-order* part of a difference means discarding what vanishes to second order along the diagonal, namely I_μ^2 . The quotient I_μ/I_μ^2 is therefore the module of first-order infinitesimal differences, the *conormal module of the diagonal*; its element da , the class of δa , is the universal differential. With the diagonal in place of a single point this is precisely the Zariski cotangent space $\mathfrak{m}_x/\mathfrak{m}_x^2$ of Definition 7.6.5 — functions vanishing on the diagonal, modulo those vanishing to second order — and, the diagonal being available for every morphism, it is this construction, rather than the generators-and-relations description of Remark 8.2.9, that globalizes in the next section to the sheaf $\Omega_{X/Y}$.

Proposition 8.2.7 (Existence: the diagonal construction). *Let $\mu: A \otimes_R A \rightarrow A$, $a \otimes a' \mapsto aa'$, be the multiplication map and $I_\mu = \ker \mu$ its kernel, the diagonal ideal. The quotient $(A \otimes_R A)/I_\mu \simeq A$ makes I_μ/I_μ^2 an A -module, and the map*

$$d: A \longrightarrow I_\mu/I_\mu^2, \quad da := (1 \otimes a - a \otimes 1) \bmod I_\mu^2,$$

exhibits it as a module of Kähler differentials of A over R ([The25, Tag 00RO]). In particular $\Omega_{A/R}$ exists.

Proof. d is a well-defined R -derivation. First, $1 \otimes a - a \otimes 1 \in I_\mu$ since $\mu(1 \otimes a - a \otimes 1) = 0$. The ring $A \otimes_R A$ carries two A -module structures, through its two factors; they coincide on I_μ/I_μ^2 , because for $z \in I_\mu$ the difference $(a \otimes 1 - 1 \otimes a)z$ of the two actions lies in I_μ^2 . So I_μ/I_μ^2 is canonically an A -module. The map d kills R (as $1 \otimes r - r \otimes 1 = 0$ over R) and is additive, and the Leibniz rule holds modulo I_μ^2 by the identity

$$d(aa') - a da' - a' da = (1 \otimes a - a \otimes 1)(1 \otimes a' - a' \otimes 1) \in I_\mu^2,$$

verified by expanding in $A \otimes_R A$ (with the A -action taken in the left factor). By Lemma 8.2.3, d is a derivation.

d is universal. Fix an A -module M ; the map $\varphi \mapsto \varphi \circ d$ is A -linear and natural, and we exhibit a two-sided inverse. The elements δa generate I_μ as a left A -module: for $\sum_i a_i \otimes a'_i \in I_\mu$, so $\sum_i a_i a'_i = 0$,

$$\sum_i (a_i \otimes 1) \delta a'_i = \sum_i (a_i \otimes a'_i - a_i a'_i \otimes 1) = \sum_i a_i \otimes a'_i,$$

so the classes da generate $\Omega_{A/R}$ and $\varphi \mapsto \varphi \circ d$ is injective. For surjectivity, given $D \in \mathrm{Der}_R(A, M)$ define $\psi: A \otimes_R A \rightarrow A \oplus M$ by $\psi(a \otimes a') = (aa', a D(a'))$. It is R -bilinear and multiplicative, since

$$\psi(a_1 \otimes a'_1) \psi(a_2 \otimes a'_2) = (a_1 a'_1 a_2 a'_2, a_1 a'_1 a_2 D(a'_2) + a_2 a'_2 a_1 D(a'_1))$$

agrees, by Leibniz, with $\psi((a_1 \otimes a'_1)(a_2 \otimes a'_2)) = (a_1 a_2 a'_1 a'_2, a_1 a_2 D(a'_1 a'_2))$. Thus ψ is an R -algebra map with $\mathrm{pr} \circ \psi = \mu$, so it carries I_μ into $\ker \mathrm{pr} = M$ and I_μ^2 into $M^2 = 0$, inducing $\bar{\psi}: \Omega_{A/R} \rightarrow M$. It is A -linear — $\psi(a \otimes 1) = (a, 0)$ acts on $(0, m)$ by $(0, am)$ — and

$$\bar{\psi}(da) = \psi(1 \otimes a - a \otimes 1) = (a, D(a)) - (a, 0) = (0, D(a)),$$

so $\bar{\psi} \circ d = D$. Hence $\varphi \mapsto \varphi \circ d$ is bijective. $\square$

Remark 8.2.8 (The diagonal over the affine line). The picture is concrete for $A = k[x]$: then $A \otimes_k A = k[x, y]$ (with $y = 1 \otimes x$), the map μ is the substitution $y \mapsto x$, and $I_\mu = (y - x)$ is the ideal of the diagonal $\{y = x\} \subset \mathbb{A}^2$. The generator $\delta x = y - x$ is literally the displacement between the two points, and reduction modulo $I_\mu^2 = (y - x)^2$ reads off the first-order term of a difference:

$$\delta a = a(y) - a(x) \equiv a'(x)(y - x) = a'(x) dx \pmod{(y - x)^2},$$

as in $y^n - x^n = (y - x)(y^{n-1} + \cdots + x^{n-1}) \equiv nx^{n-1}(y - x)$. Thus $\Omega_{k[x]/k} = k[x] dx$ with $da = a' dx$: the universal derivation is the derivative, recovered as the linear part of the difference.

Remark 8.2.9 (Presentation by generators and relations). Unwinding the universal property, $\Omega_{A/R}$ is equivalently the A -module generated by symbols $\{da : a \in A\}$ subject to the relations

$$d(a + a') = da + da', \quad dr = 0 \ (r \in R), \quad d(aa') = a da' + a' da,$$

with d the tautological map $a \mapsto da$. Indeed this module manifestly satisfies the same universal property. The two descriptions are therefore canonically isomorphic; we shall use whichever is convenient.

The generators-and-relations description makes the first computation immediate and reveals where the formal partial derivatives of Lemma 7.6.2 come from — not from analysis, but from the Leibniz relation.

Proposition 8.2.10 (Differentials of a polynomial algebra). *For any set S , the module $\Omega_{R[x_i : i \in S]/R}$ is free on the symbols $\{dx_i\}_{i \in S}$ (cf. [The25, Tag 00RX]), and the universal derivation is the total formal derivative*

$$df = \sum_{i \in S} \frac{\partial f}{\partial x_i} dx_i.$$

Proof. By Leibniz, additivity, and R -linearity, any derivation D on the polynomial ring $P = R[x_i]$ satisfies $D(f) = \sum_i \frac{\partial f}{\partial x_i} D(x_i)$, so D is determined by the family $(D(x_i))_i$; conversely any family $(m_i)_i$ in a P -module M defines a derivation by this formula (a monomial computation, as in Lemma 7.6.2). Thus $\text{Der}_R(P, M) \simeq \prod_i M = \text{Hom}_P(\bigoplus_i P dx_i, M)$ naturally in M , so the free module $\bigoplus_i P dx_i$ with $dx_i \mapsto dx_i$ represents the functor. By uniqueness it is $(\Omega_{P/R}, d)$, and the formula for df is the derivation property. $\square$

The two fundamental exact sequences

Differentials interact with composition of ring maps and with passage to a quotient through two right-exact sequences — the algebraic counterparts of the cotangent and conormal sequences of differential geometry. The first we prove by dualizing: applying $\text{Hom}_B(-, M)$ and rewriting the result, through the universal property of the cotangent module, as a sequence of derivation modules that is manifestly exact for every M . The second is then not a separate theorem but the first prolonged — along a surjection it extends the cotangent sequence one term to the left, by the conormal module — and we read it off directly.

Proposition 8.2.11 (First exact sequence: a composition $R \rightarrow A \rightarrow B$). *For ring homomorphisms $R \rightarrow A \rightarrow B$ there is an exact sequence of B -modules*

$$B \otimes_A \Omega_{A/R} \xrightarrow{\alpha} \Omega_{B/R} \xrightarrow{\beta} \Omega_{B/A} \rightarrow 0,$$

where $\beta(d_{B/R} b) = d_{B/A} b$ and $\alpha(b \otimes d_{A/R} a) = b d_{B/R}(a)$ (with a denoting also its image in B); see [The25, Tag 00RS].

Proof. Apply $\text{Hom}_B(-, M)$ to the candidate sequence and use the universal property (Definition 8.2.6) together with the extension–restriction adjunction $\text{Hom}_B(B \otimes_A \Omega_{A/R}, M) \simeq \text{Hom}_A(\Omega_{A/R}, M)$. The resulting sequence is

$$0 \rightarrow \text{Der}_A(B, M) \xrightarrow{\beta^*} \text{Der}_R(B, M) \xrightarrow{\alpha^*} \text{Der}_R(A, M),$$

where β^* is the inclusion of those R -derivations of B that are A -linear, and α^* is restriction along $A \rightarrow B$. Both exactness claims are immediate: β^* is injective, and a derivation of B kills the image of A — i.e. lies in $\ker \alpha^*$ — exactly when it is A -linear, i.e. in $\text{im } \beta^*$. As this holds for every M , the original sequence is right-exact, by the universal property of the cokernel. $\square$

The second sequence is this one prolonged, in the case of a surjection. Fix an ideal $J \subseteq A$ and set $B = A/J$. As an A -algebra B is generated by the image of A — by nothing new — so $d_{B/A} \bar{a} = 0$ for every a , whence $\Omega_{B/A} = 0$. The first sequence (Proposition 8.2.11) therefore collapses on the right to the surjectivity of its map $\gamma: B \otimes_A \Omega_{A/R} \rightarrow \Omega_{B/R}$ (the α of that proposition), and only $\ker \gamma$ remains to be found. By the generators-and-relations description (Remark 8.2.9), $\Omega_{B/R}$ is presented on the same generators $\{da\}_{a \in A}$ as $B \otimes_A \Omega_{A/R}$ — those of $\Omega_{B/R}$ indexed through $A \rightarrow B$, and in both cases subject to additivity, R -linearity, and Leibniz — with a single further family of relations: over B the symbol da depends only on the residue $\bar{a} = a \bmod J$, that is, $1 \otimes df = 0$ for every $f \in J$. So γ is precisely the quotient by the span of these elements:

$$\ker \gamma = \langle 1 \otimes d_{A/R} f : f \in J \rangle.$$

This span is the image of $\delta: f + J^2 \mapsto 1 \otimes d_{A/R} f$, a map well-defined on J/J^2 because Leibniz makes it kill J^2 : for $f, g \in J$,

$$\delta(fg + J^2) = 1 \otimes d(fg) = \bar{f}(1 \otimes dg) + \bar{g}(1 \otimes df) = 0 \quad (\bar{f} = \bar{g} = 0 \text{ in } B).$$

Hence $\ker \gamma = \text{im } \delta$, and the cotangent sequence, extended one term to the left by the conormal module J/J^2 , becomes the second of our two fundamental sequences.

Proposition 8.2.12 (Second exact sequence: a quotient $B = A/J$). *Let $R \rightarrow A$ be a ring homomorphism, $J \subseteq A$ an ideal, and $B = A/J$. There is an exact sequence of B -modules*

$$J/J^2 \xrightarrow{\delta} B \otimes_A \Omega_{A/R} \xrightarrow{\gamma} \Omega_{B/R} \rightarrow 0,$$

where $\delta(f + J^2) = 1 \otimes d_{A/R} f$ and $\gamma(b \otimes d_{A/R} a) = b d_{B/R} \bar{a}$. The module J/J^2 is the conormal module of J , and δ is the conormal map; see [The25, Tag 00RU].

Combining the two sequences with the polynomial computation gives the link back to the Jacobian matrix of the introduction: differentials of a finitely presented algebra are the cokernel of the (transposed) Jacobian.

Corollary 8.2.13 (Differentials of a finitely presented algebra). *Let $A = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$. Then*

$$\Omega_{A/R} \simeq \text{coker} \left(A^m \xrightarrow{\text{Jac}^T} A^n \right) = \left(\bigoplus_{i=1}^n A dx_i \right) / (df_1, \dots, df_m), \quad df_j = \sum_{i=1}^n \frac{\partial f_j}{\partial x_i} dx_i,$$

where $\text{Jac} = (\partial f_j / \partial x_i)_{j,i}$ is the Jacobian matrix.

Proof. Apply Proposition 8.2.12 to $P = R[x_1, \dots, x_n] \rightarrow A = P/J$ with $J = (f_1, \dots, f_m)$. By Proposition 8.2.10, $A \otimes_P \Omega_{P/R} = \bigoplus_i A dx_i$ is free, and the conormal map sends the generator $f_j + J^2$ of J/J^2 to $1 \otimes df_j = \sum_i \frac{\partial f_j}{\partial x_i} dx_i$. Right exactness of the conormal sequence identifies $\Omega_{A/R}$ with the cokernel of the map $A^m \rightarrow A^n$ whose columns are the df_j , that is, with $\text{coker}(\text{Jac}^T)$. $\square$

Base change, localization, and first examples

For the affine module $\Omega_{A/R}$ to glue into a quasi-coherent sheaf in the next section, it must be compatible with the two operations out of which schemes are assembled: base change of the ground ring and localization of A . Both compatibilities are formal consequences of the universal property.

Proposition 8.2.14 (Base change and localization). *Let $R \rightarrow A$ be a ring homomorphism.*

- (i) (Base change; [The25, Tag 00RV]) *For any $R \rightarrow R'$, writing $A' = A \otimes_R R'$, the natural map is an isomorphism*

$$\Omega_{A'/R'} \simeq \Omega_{A/R} \otimes_A A' = \Omega_{A/R} \otimes_R R'.$$

- (ii) (Localization; [The25, Tag 00RT]) *For any multiplicative subset $S \subseteq A$, the natural map is an isomorphism*

$$\Omega_{S^{-1}A/R} \simeq S^{-1}\Omega_{A/R} = \Omega_{A/R} \otimes_A S^{-1}A,$$

$$\text{and } d(a/s) = (s da - a ds)/s^2.$$

Proof. (i) For an A' -module M — equally an A -module by restriction — an R' -derivation $A' \rightarrow M$ restricts to an R -derivation $A \rightarrow M$, and is determined by that restriction because A' is generated over R' by the image of A and a derivation kills R' ; conversely any R -derivation of A extends uniquely to an R' -derivation of A' by the Leibniz rule. Hence $\text{Der}_{R'}(A', M) \simeq \text{Der}_R(A, M)$ naturally, i.e. $\text{Hom}_{A'}(\Omega_{A'/R'}, M) \simeq \text{Hom}_{A'}(\Omega_{A/R} \otimes_A A', M)$; the claim follows by the Yoneda lemma.

(ii) For an $S^{-1}A$ -module M , the quotient rule $D(a/s) = (s D(a) - a D(s))/s^2$ shows that every R -derivation of A valued in M extends uniquely to one of $S^{-1}A$; that is, $\text{Der}_R(S^{-1}A, M) \simeq \text{Der}_R(A, M) = \text{Hom}_{S^{-1}A}(S^{-1}\Omega_{A/R}, M)$, and Yoneda again gives the isomorphism. (See [Mat89] for the elementary verification that this extension is well-defined.) $\square$

We close the section with the two examples that will animate the rest of the chapter: differentials detect, on the one hand, the failure of separability for a field extension — the seed of étaleness — and, on the other hand, the singularities of a variety — the seed of the Jacobian criterion.

Example 8.2.15 (Separable extensions, and μ_n). Let k be a field and $A = k[x]/(f)$ for a nonzero f . By Corollary 8.2.13,

$$\Omega_{A/k} = A dx / (f'(x) dx) \simeq k[x]/(f, f'),$$

which vanishes if and only if $(f, f') = (1)$ in $k[x]$, i.e. if and only if f is separable (has no repeated root in $\bar{k}$). Thus $\Omega_{A/k} = 0$ exactly when A is a finite separable k -algebra — the prototype of an étale k -algebra.

For a concrete instance, take $f = x^n - 1$, so that $\text{Spec } A = \mu_n$ is the group scheme of n -th roots of unity. Here $f' = nx^{n-1}$, and since x is a unit in A (with inverse x^{n-1}) the ideal (f, f') contains the unit nx^{n-1} precisely when n is invertible in k . Hence

$$\Omega_{\mu_n/k} = 0 \iff \text{char}(k) \nmid n,$$

so μ_n is étale over k away from the characteristic, and ramifies when $\text{char}(k) = p \mid n$: for $n = p$ one has $f = (x - 1)^p$ and $f' = 0$, giving $\Omega_{\mu_p/k} = A dx$, free of rank 1. The purely inseparable extension $k = \mathbb{F}_p(t) \subset \mathbb{F}_p(t^{1/p}) = k[x]/(x^p - t)$ is detected the same way: there $f' = 0$, so $\Omega_{A/k} = A dx \neq 0$.

Example 8.2.16 (Differentials see singularities: the cuspidal cubic). Let $\text{char}(k) \neq 2, 3$ and $A = k[x, y]/(g)$ with $g = y^2 - x^3$, the cuspidal cubic of Example 7.6.6. By Corollary 8.2.13,

$$\Omega_{A/k} = (A dx \oplus A dy) / (dg), \quad dg = -3x^2 dx + 2y dy.$$

The curve is integral of dimension 1, and over its function field the single relation dg is nonzero, so $\Omega_{A/k}$ has generic rank 1. At the cusp $\mathfrak{m} = (x, y)$, however, the relation reduces to 0 modulo $\mathfrak{m}$, and the fiber

$$\Omega_{A/k} \otimes_A \kappa(\mathfrak{m}) = (k dx \oplus k dy)/(0) \simeq k^2$$

is two-dimensional — matching the two-dimensional Zariski cotangent space computed in Example 7.6.6. Thus $\Omega_{A/k}$ is locally free of rank 1 on the smooth locus but fails to be locally free at the cusp, where its fiber jumps. The rank of the fiber of $\Omega_{A/R}$ is precisely the obstruction the Jacobian criterion will measure, as developed below.

The conormal module and the cotangent complex

The conormal sequence (Proposition 8.2.12) is only right-exact: its conormal map δ may have a kernel, and passing to the cokernel $\Omega_{A/R}$ discards it. That discarded kernel is a second invariant, independent of $\Omega_{A/R}$, and the two together — not $\Omega_{A/R}$ alone — govern smoothness. We isolate it.

Fix a *presentation* of A over R : a surjection $\pi: P \twoheadrightarrow A$ from a polynomial R -algebra $P = R[x_i : i \in I]$, with kernel $J = \ker \pi$, so that $A = P/J$. Since $\Omega_{P/R} = \bigoplus_i P dx_i$ is free (Proposition 8.2.10), the conormal sequence of $P \twoheadrightarrow A$ reads

$$J/J^2 \xrightarrow{\delta_\pi} A \otimes_P \Omega_{P/R} = \bigoplus_i A dx_i \longrightarrow \Omega_{A/R} \rightarrow 0, \quad \delta_\pi(f + J^2) = \sum_i \frac{\partial f}{\partial x_i} dx_i.$$

Here the partial derivatives are computed in the polynomial ring P and their coefficients then reduced to A ; thus $\delta_\pi(f + J^2) = 0$ requires every $\partial f / \partial x_i$ to lie in J — not merely f itself, which lies in J by definition. Its cokernel is $\Omega_{A/R}$ by right exactness; its kernel we now name.

Definition 8.2.17 (Conormal module; truncated cotangent complex). Retain a presentation π as above. The two-term complex of A -modules

$$\left[J/J^2 \xrightarrow{\delta_\pi} \bigoplus_i A dx_i \right] \quad (\text{degrees } 1, 0)$$

is classically the *naïve cotangent complex*. Up to canonical homotopy equivalence it is independent of π , and it represents the 1-truncation $\tau_{\leq 1} \mathbb{L}_{A/R}$ of the *cotangent complex* $\mathbb{L}_{A/R} \in D(\text{Mod}_A)$ — the derived refinement of $\Omega_{A/R}$, a connective complex with $H_0 \mathbb{L}_{A/R} = \Omega_{A/R}$ ([The25, Tags 08P5, 00S0, 08RB]). We write $\tau_{\leq 1} \mathbb{L}_{A/R}$ for the object it represents. Its homology is

$$H_0(\tau_{\leq 1} \mathbb{L}_{A/R}) = \text{coker } \delta_\pi = \Omega_{A/R}, \quad H_1(\tau_{\leq 1} \mathbb{L}_{A/R}) = \ker \delta_\pi,$$

the degree-one part, the *conormal module* $N_{A/R} := \ker \delta_\pi = H_1 \mathbb{L}_{A/R}$, being independent of π as well. This usage extends — but does not literally coincide with — the conormal module J/J^2 of Proposition 8.2.12: for a *surjection* $A \rightarrow B$, choosing the zero-variable presentation $A \twoheadrightarrow B$ gives $\delta_\pi = 0$ and $N_{B/A} = J/J^2$. For a general presentation, $N_{A/R} = \ker \delta_\pi$ is a submodule (possibly all) of the presentation-dependent J/J^2 ; unlike that chosen ideal quotient, the kernel is intrinsic (in Example 8.2.20, J/J^2 is free of rank 1 over A while $N_{A/k}$ is not).

The degree-zero homology $\Omega_{A/R}$ is a *cokernel*, corepresenting the derivation functor (Definition 8.2.6), and is intrinsic for that reason alone — indeed it was built with no presentation (Proposition 8.2.7). The conormal module $N_{A/R}$ is by contrast a *kernel*: H_1 is no left adjoint, so $N_{A/R}$ corepresents nothing and carries no universal property of its own. Its independence of π is inherited instead from $\mathbb{L}_{A/R}$, of which it is the first homology and which is constructed with no presentation at all; for an elementary verification not using $\mathbb{L}$, see Remark 8.2.18.

Remark 8.2.18 (Independence by hand: adding variables). The presentation-independence can be checked elementarily, without invoking $\mathbb{L}_{A/R}$. Let S be any set and put $P' = P[y_s \mid s \in S]$. Extend π by prescribing $y_s \mapsto c_s \in A$, choose lifts $\hat{c}_s \in P$, and write $g_s = y_s - \hat{c}_s$. Then $P' = P[g_s \mid s \in S]$ and

$$J' = \ker(P' \twoheadrightarrow A) = (J, g_s \mid s \in S).$$

The classes of the polynomial variables g_s split off free summands in both degrees: $J'/J'^2 \simeq J/J^2 \oplus \bigoplus_{s \in S} A \cdot [g_s]$, while the degree-zero term gains the basis elements $dg_s = dy_s - d\hat{c}_s$, and $\delta_{\pi'}([g_s]) = dg_s$. Hence the complex for π' is that for π plus an acyclic direct sum,

$$[J'/J'^2 \rightarrow \bigoplus_i A dx_i \oplus \bigoplus_{s \in S} A dy_s] \simeq [J/J^2 \rightarrow \bigoplus_i A dx_i] \oplus \bigoplus_{s \in S} [A \xrightarrow{\sim} A],$$

so H_0 and H_1 are unchanged. Given two arbitrary polynomial presentations of A , their tensor product over R is a common presentation obtained from either one by adjoining a set of variables, so the preceding comparison applies without a finiteness hypothesis. The displayed splitting depends on the chosen lifts $\hat{c}_s$; different choices induce chain-homotopic comparisons, while the canonical homotopy class is the one supplied intrinsically by $\tau_{\leq 1} \mathbb{L}_{A/R}$. This is the elementary shadow of the comparison [The25, Tag 08RB].

Cotangent and conormal modules are not separate data: across a tower $R \rightarrow A \rightarrow B$ they are woven into a single long exact sequence, unifying and prolonging the two right-exact sequences of Propositions 8.2.11 and 8.2.12.

Proposition 8.2.19 (Fundamental exact sequence of cotangent and conormal modules). *Let $R \rightarrow A \rightarrow B$ be ring maps with $\mathrm{Tor}_1^A(\Omega_{A/R}, B) = 0$; this holds, in particular, if $A \rightarrow B$ is flat or if $\Omega_{A/R}$ is flat over A . Then there is an exact sequence of B -modules*

$$N_{A/R} \otimes_A B \rightarrow N_{B/R} \rightarrow N_{B/A} \xrightarrow{\partial} \Omega_{A/R} \otimes_A B \rightarrow \Omega_{B/R} \rightarrow \Omega_{B/A} \rightarrow 0,$$

prolonging the cotangent sequence of Proposition 8.2.11 to the left by the conormal modules. For a surjection $A \rightarrow B$ it reduces to the conormal sequence of Proposition 8.2.12: then $\Omega_{B/A} = 0$, the module $N_{B/A}$ is the conormal module of $\ker(A \rightarrow B)$, and ∂ is the conormal map.

The same sequence is the long exact homology sequence of the transitivity (Jacobi–Zariski) triangle $\mathbb{L}_{A/R} \otimes_A^{\mathbb{L}} B \rightarrow \mathbb{L}_{B/R} \rightarrow \mathbb{L}_{B/A}$ in $D(\mathrm{Mod}_B)$ ([The25, Tag 08QX]); that viewpoint also fixes the leftmost term, $H_1(\mathbb{L}_{A/R} \otimes_A^{\mathbb{L}} B)$, when the flatness hypothesis is dropped.

Although Proposition 8.2.19 links them across towers, $\Omega_{A/R}$ and $N_{A/R}$ are independent invariants of A over R , each blind to what the other sees, as the running examples show.

Example 8.2.20 (Fat point versus cusp: Ω and N are independent). *The fat point $A = k[x]/(x^2)$, presented by $P = k[x] \twoheadrightarrow A$ with $J = (x^2)$. Here $J/J^2 = (x^2)/(x^4)$ is free of rank 1 over A on the class of x^2 , and $\delta(x^2 + J^2) = 2x dx$. For $\mathrm{char}(k) \neq 2$ this gives $\Omega_{A/k} = A dx / (2x dx) \simeq k$ and*

$$N_{A/k} = \ker \delta = k \cdot (x^3 + J^2) \neq 0, \quad \text{because } \delta(x^3 + J^2) = 3x^2 dx = 0 \text{ in } A dx.$$

The nonzero N records the nilpotent. In *characteristic 2* the point sharpens: there $\delta = 0$, so $\Omega_{A/k} = A dx$ is free of rank 1 — locally free, hence invisible to the cotangent module alone — while $N_{A/k} = J/J^2 \simeq A \neq 0$ still reports that the fat point is not smooth.

The cusp $A = k[x, y]/(y^2 - x^3)$ (Example 8.2.16) shows the reverse. With $P = k[x, y]$ and $J = (g)$, $g = y^2 - x^3$, the module $J/J^2 \simeq A$ is free on the class of g and $\delta(g + J^2) = dg = -3x^2 dx + 2y dy$; as A is a domain and $2y \neq 0$, the annihilator of dg vanishes, so $N_{A/k} = 0$. Yet $\Omega_{A/k}$ is not locally free, its fiber jumping at the cusp (Example 8.2.16). Here $N = 0$ while Ω sees the singularity — the exact opposite of the fat point.

Remark 8.2.21 (What the pair $(\Omega_{A/R}, N_{A/R})$ will decide). These two examples are the shadow of the chapter's main classification, proved later: for a finitely presented $R \rightarrow A$, the morphism is *smooth* exactly when $\Omega_{A/R}$ is locally free and $N_{A/R} = 0$; *unramified* exactly when $\Omega_{A/R} = 0$; and *étale* exactly when both vanish, equivalently when $\tau_{\leq 1} \mathbb{L}_{A/R} \simeq 0$. The fat point fails smoothness through N , the cusp through Ω , and a finite separable field extension (Example 8.2.15) realizes the étale case.

We have used only the bottom two homologies. The full cotangent complex $\mathbb{L}_{A/R}$ carries higher homology beyond $\tau_{\leq 1}$, obtained by repeating the construction of $\Omega_{A/R}$ with the category of modules replaced by its derived category; it governs deformations and obstructions uniformly across smooth, singular, and derived settings. It is the object promised at the end of the previous chapter, and will return when we phrase smoothness through lifting.

With the module of differentials and its formal properties in hand — the conormal module among them — we now globalize: in the next section the affine assignment $A \mapsto \Omega_{A/R}$ is upgraded, via the diagonal, to a quasi-coherent sheaf $\Omega_{X/Y}$ on any morphism of schemes $X \rightarrow Y$, and the conormal module J/J^2 to the conormal sheaf of an immersion.

8.3. Cotangent Modules of Algebraic Functors

The affine theory has two features that survive globalization. First, $\Omega_{A/R}$ is characterized by what it does to every square-zero extension $A \oplus M$, rather than by a choice of generators and relations. Second, the scalar-extension isomorphism for differentials is precisely the descent datum needed to make these representing modules quasi-coherent. This section carries out that globalization. It also records, in a form that we shall use repeatedly, the Jacobian definitions of smooth, unramified, and étale morphisms announced in Section 8.1. The affine calculations of the preceding section supply the local models; the only additional input is their compatibility with localization and Zariski descent.

For an arbitrary algebraic functor the lifting universal property is the definition; for a morphism of schemes it agrees with the conormal module of the diagonal, so this construction is exactly the diagonal globalization promised at the end of the preceding section.

The relative cotangent module

Let $f: Y \rightarrow X$ be a map of algebraic functors, let $y \in Y(R)$, and let M be an R -module. The inclusion $R \rightarrow R \oplus M$, $r \mapsto (r, 0)$, gives a distinguished extension of $f(y)$ to an $(R \oplus M)$ -point of X . Write

$$\text{Lift}_f(y, M) = \left\{ \begin{array}{ccc} \text{Spec } R & \xrightarrow{y} & Y \\ \downarrow & \nearrow & \downarrow f \\ \text{Spec}(R \oplus M) & \longrightarrow & X \end{array} \right\}$$

for the set of dotted arrows making the diagram commute. Thus $\text{Lift}_f(y, M)$ measures infinitesimal motions of y that are vertical over X and have coefficients in M .

Definition 8.3.1 (Cotangent module of a map of algebraic functors). We say that $f: Y \rightarrow X$ admits a cotangent module if:

- (i) for every R and $y \in Y(R)$, the functor

$$\text{Mod}_R \longrightarrow \text{Set}, \quad M \longmapsto \text{Lift}_f(y, M)$$

is represented by an R -module, denoted $\Omega_f(y)$, so that

$$\text{Lift}_f(y, M) \simeq \text{Hom}_R(\Omega_f(y), M)$$

naturally in M ; and

(ii) for every ring map $R \rightarrow S$, the canonical comparison map

$$\Omega_f(y) \otimes_R S \longrightarrow \Omega_f(y_S)$$

is an isomorphism, where $y_S \in Y(S)$ is the scalar extension of y .

The resulting quasi-coherent module on Y is written

$$\Omega_f = \Omega_{Y/X} \in \text{Mod}_Y$$

and is called the *relative cotangent module*.

The qualification *admits* matters for a general algebraic functor: a set-valued deformation functor need not be linear, and even a linear one need not be represented by a module compatible with scalar extension. For schemes no such qualification is needed.

Proposition 8.3.2 (Existence and affine identification). *Every schematic morphism $f : Y \rightarrow X$ of algebraic functors admits a cotangent module. If*

$$\begin{array}{ccc} \text{Spec } B & \hookrightarrow & Y \\ \downarrow & & \downarrow f \\ \text{Spec } A & \longrightarrow & X \end{array}$$

is cartesian and the upper arrow identifies $\text{Spec } B$ with an affine open of $Y \times_X \text{Spec } A$, then

$$\Omega_f|_{\text{Spec } B} \simeq \widetilde{\Omega_{B/A}}.$$

In particular, for a morphism of schemes the modules $\Omega_{B/A}$ on affine charts glue uniquely to $\Omega_{Y/X}$.

Proof. For $Y = \text{Spec } B$ and $X = \text{Spec } A$, a dotted arrow in Definition 8.3.1 is, after reversing arrows, an A -algebra section of the appropriate split square-zero extension. By Definitions 8.2.1 and 8.2.6, these sections are naturally

$$\text{Der}_A(B, M) \simeq \text{Hom}_B(\Omega_{B/A}, M).$$

After a test point $\text{Spec } T \rightarrow \text{Spec } B$, the representing module becomes $\Omega_{B/A} \otimes_B T$, by Proposition 8.2.14. This proves the affine assertion and the required compatibility with scalar extension. For a schematic f , cover every pullback $Y \times_X \text{Spec } A$ by affine opens. Localization of differentials makes the affine modules agree on overlaps, so Zariski descent for quasi-coherent modules glues them. The universal property glues as well, and uniqueness follows from Yoneda. $\square$

Example 8.3.3 (Affine and projective space). Let $M \in \text{Mod}_X$ and use the quotient convention for the affine-space functor $\mathbb{A}(M)$ adopted earlier in the book. For the structure map $p : \mathbb{A}(M) \rightarrow X$ one has

$$\Omega_{\mathbb{A}(M)/X} \simeq p^*M.$$

If M is a vector bundle and $q : \mathbb{P}(M) \rightarrow X$, the universal quotient $q^*M \twoheadrightarrow \mathcal{O}_{\mathbb{P}(M)}(1)$ gives the Euler sequence

$$0 \longrightarrow \Omega_{\mathbb{P}(M)/X} \longrightarrow q^*M \otimes \mathcal{O}_{\mathbb{P}(M)}(-1) \longrightarrow \mathcal{O}_{\mathbb{P}(M)} \longrightarrow 0.$$

At an R -point represented by a quotient line $M(x) \twoheadrightarrow L$, the fiber of the first term is $\ker(M(x) \twoheadrightarrow L) \otimes_R L^{-1}$. This pointwise description both represents the lifting functor and makes its base-change compatibility transparent. For $M = \mathcal{O}_X^{\oplus 2}$, the Euler sequence gives $\Omega_{\mathbb{P}_X^1/X} \simeq \mathcal{O}_{\mathbb{P}_X^1}(-2)$.

Proposition 8.3.4 (Base change, restriction, and transitivity). *Let $f : Y \rightarrow X$ admit a cotangent module.*

(i) *In a cartesian square*

$$\begin{array}{ccc} Y' & \xrightarrow{h} & Y \\ f' \downarrow & & \downarrow f \\ X' & \longrightarrow & X, \end{array}$$

the map f' admits a cotangent module and $\Omega_{f'} \simeq h^ \Omega_f$.*

(ii) *If $j : U \hookrightarrow Y$ is open, then $\Omega_{f \circ j} \simeq j^* \Omega_f$.*

(iii) *For $Z \xrightarrow{g} Y \xrightarrow{f} X$, whenever the indicated cotangent modules exist, there is a natural right-exact sequence*

$$g^* \Omega_f \longrightarrow \Omega_{f \circ g} \longrightarrow \Omega_g \longrightarrow 0. \quad (\star)$$

Proof. A lift after base change is exactly a lift before base change together with the prescribed X' -structure. Consequently the two lift functors are naturally the same, and their representing modules are related by pullback; this proves (i). Part (ii) follows either from (i) and the fact that an open immersion is stable under base change, or directly from the unique lifting of a point through an open subfunctor.

For (iii), fix $z \in Z(R)$ and an R -module M . A vertical first-order motion of z over X first determines a vertical motion of $g(z)$ over X and then, after that motion is fixed, a motion of z over Y . Expressing this composition law by the three representing functors gives a cocartesian square of R -modules

$$\begin{array}{ccc} g^* \Omega_f(z) & \longrightarrow & \Omega_{f \circ g}(z) \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \Omega_g(z). \end{array}$$

In an abelian category a square of this form is cocartesian exactly when the associated sequence is right exact. These pointwise sequences commute with scalar extension and therefore glue. $\square$

Conormal modules and the tangent functor

For schemes we also globalize the degree-one invariant from Definition 8.2.17. It is useful to distinguish two related pieces of notation. The cotangent complex $\mathbb{L}_{Y/X} \in D(\mathcal{O}_Y)$ restricts on every compatible affine chart $\text{Spec } B \rightarrow \text{Spec } A$ to $\mathbb{L}_{B/A}$; its homology sheaves are quasi-coherent and $H_0(\mathbb{L}_{Y/X}) = \Omega_{Y/X}$.

Definition 8.3.5 (Intrinsic and geometric conormal modules). *For a morphism of schemes $f : Y \rightarrow X$, the affine modules $N_{B/A} = H_1(\mathbb{L}_{B/A})$ glue, using localization, to a quasi-coherent module*

$$N_f = N_{Y/X} = H_1(\mathbb{L}_{Y/X}) \in \text{Mod}_Y.$$

This is the *intrinsic conormal module* of f , compatible with the notation $N_{A/R}$ of Definition 8.2.17.

If $i : Z \hookrightarrow Y$ is a closed immersion with ideal $\mathcal{I}$, we also write

$$\mathcal{N}_{Z/Y}^* := \mathcal{I} / \mathcal{I}^2$$

for its *geometric conormal sheaf*. In this special case $N_i \simeq \mathcal{N}_{Z/Y}^*$. The superscript in $\mathcal{N}_{Z/Y}^*$ is a reminder that its dual, when locally free, is the normal bundle. For a general f , the symbol N_f always means $H_1(\mathbb{L}_f)$ and not an arbitrarily chosen ideal quotient.

The diagonal gives a useful bridge between the two meanings. The diagonal $\Delta_f: Y \rightarrow Y \times_X Y$ is an immersion, and its conormal module is canonically

$$N_{\Delta_f} \simeq \Omega_f. \quad (\dagger)$$

If f is separated, Δ_f is closed and the left-hand side is literally $\mathcal{I}_\Delta/\mathcal{I}_\Delta^2$. In general the same statement is read on open neighborhoods on which the diagonal is closed. The two identifications obtained from the projections $Y \times_X Y \rightrightarrows Y$ differ by a sign; the swap of the two factors acts by -1 on $\mathcal{I}_\Delta/\mathcal{I}_\Delta^2$. At an ordinary point $x \in |Y|$, the other special case is

$$N_{\text{Spec } \kappa(x)/Y} \simeq \mathfrak{m}_{Y,x}/\mathfrak{m}_{Y,x}^2 = T_x^*Y,$$

so the Zariski cotangent space is the conormal module of the inclusion of the point.

Theorem 8.3.6 (Global fundamental sequences). *Let $Z \xrightarrow{g} Y \xrightarrow{f} X$ be morphisms of schemes and set $h = f \circ g$. Here g^* on a cotangent complex means derived pullback; on an ordinary quasi-coherent module it means the usual pullback.*

(i) *There is a transitivity triangle*

$$g^*\mathbb{L}_f \longrightarrow \mathbb{L}_h \longrightarrow \mathbb{L}_g \longrightarrow g^*\mathbb{L}_f[1]. \quad (\text{T})$$

Its low-degree homology sequence ends in

$$H_1(g^*\mathbb{L}_f) \longrightarrow N_h \longrightarrow N_g \longrightarrow g^*\Omega_f \longrightarrow \Omega_h \longrightarrow \Omega_g \longrightarrow 0. \quad (\text{F})$$

If g is flat, or if Ω_f is flat, then $H_1(g^\mathbb{L}_f) \simeq g^*N_f$.*

(ii) *In a cartesian square as in Proposition 8.3.4, there is always a comparison $h^*N_f \rightarrow N_{f'}$. It is an isomorphism if the base change $X' \rightarrow X$ is flat; it is also an isomorphism when both f and Ω_f are flat.*

(iii) *If $i: Z \hookrightarrow Y$ is a closed immersion with ideal $\mathcal{I}$ and $Y \rightarrow X$ is any morphism, then*

$$\mathcal{I}/\mathcal{I}^2 \longrightarrow i^*\Omega_{Y/X} \longrightarrow \Omega_{Z/X} \longrightarrow 0 \quad (\text{C})$$

is exact. If both $Y \rightarrow X$ and $Z \rightarrow X$ are smooth, it extends to a short exact sequence

$$0 \longrightarrow \mathcal{I}/\mathcal{I}^2 \longrightarrow i^*\Omega_{Y/X} \longrightarrow \Omega_{Z/X} \longrightarrow 0. \quad (\text{C}_{\text{sm}})$$

Proof. On compatible affine charts, (T) is the transitivity triangle quoted after Proposition 8.2.19; its compatibility with localization lets the triangles glue. Taking homology gives (F), and the stated flatness assumptions remove the relevant Tor terms. The same affine calculation and flat base change give (ii).

For (iii), apply the degree-zero part of (F) to $Z \rightarrow Y \rightarrow X$ and identify $\Omega_{Z/Y} = 0$ and $N_{Z/Y} = \mathcal{I}/\mathcal{I}^2$. If the two structure morphisms are smooth, their intrinsic conormal modules vanish and their cotangent modules are locally free by the criterion below; the preceding part of (F) therefore supplies injectivity on the left. $\square$

Definition 8.3.7 (Absolute and relative tangent functors). Put $R[\epsilon] = R[t]/(t^2)$. For an algebraic functor X , define

$$T(X)(R) = X(R[\epsilon]).$$

The projection and zero section of the dual numbers give $T(X) \rightarrow X$ and $X \rightarrow T(X)$. For $f: Y \rightarrow X$, its *relative tangent functor* is

$$T(Y/X) = T(Y) \times_{T(X)} X, \quad (\text{Tan})$$

where $X \rightarrow T(X)$ is the zero section. A point of the fiber over $y \in Y(R)$ is a tangent vector at y that maps to the zero tangent vector at $f(y)$.

Proposition 8.3.8 (Cotangent–tangent duality). *If f admits a cotangent module, then there is a canonical isomorphism of algebraic functors over Y*

$$T(Y/X) \simeq \mathbb{A}(\Omega_f).$$

Consequently the relative tangent module at $y \in Y(R)$ is

$$T_y(Y/X) = \text{Hom}_R(\Omega_f(y), R).$$

More generally, vertical motions with coefficients in M form $\text{Hom}_R(\Omega_f(y), M)$.

Proof. An R -point of the fiber of $T(Y/X) \rightarrow Y$ at y is exactly a lift through the split extension $R[\epsilon] = R \oplus R$. The universal property of $\Omega_f(y)$ identifies it with $\text{Hom}_R(\Omega_f(y), R)$. By the functor-of-points definition of $\mathbb{A}(\Omega_f)$, these bijections are functorial in R and y and therefore assemble into the claimed isomorphism. Replacing the coefficient copy of R by M gives the final assertion. $\square$

Example 8.3.9 (The tangent equations are the linearized equations). Let

$$Z = \text{Spec } k[x_i \mid i \in I]/(h_j \mid j \in J)$$

and let R be a k -algebra. An R -point of $T(Z/k)$ is a pair $(a, u) \in R^I \times R^I$ such that $h_j(a) = 0$ and

$$\sum_{i \in I} \frac{\partial h_j}{\partial x_i}(a) u_i = 0 \quad (j \in J).$$

Indeed, modulo ϵ^2 one has

$$h_j(a + \epsilon u) = h_j(a) + \epsilon \sum_i \frac{\partial h_j}{\partial x_i}(a) u_i.$$

Thus the tangent fiber is the kernel of the Jacobian, while the cotangent fiber is the cokernel of its transpose.

The postponed Jacobian definition

We can now state the concrete definition promised at the beginning of the chapter. If H is a possibly infinite set of polynomials in $A[x_1, \dots, x_n]$, the assertion that its $H \times n$ Jacobian has rank n at a point means that some n rows have a nonzero $n \times n$ minor there.

Definition 8.3.10 (Smooth, unramified, and étale: Jacobian form). Let $f: Y \rightarrow X$ be a morphism of schemes.

- (i) It is *smooth* if every $y \in |Y|$ has affine neighborhoods $y \in V \subset Y$ and $U \subset X$, with $f(V) \subset U$, and an isomorphism over U

$$V \simeq V(h_1, \dots, h_m) \subset \mathbb{A}_U^n, \quad m \leq n,$$

such that the $m \times n$ matrix

$$J_H = \left(\frac{\partial h_i}{\partial x_j} \right)_{i,j}$$

has rank m at every point of V .

- (ii) It is *unramified* if the same local description is available with $V = V(H) \subset \mathbb{A}_U^n$ for a set H whose Jacobian has rank n at every point of V .
- (iii) It is *étale* if the description is available with exactly n equations and the square Jacobian determinant is a unit on V .

Thus smooth and étale morphisms are locally of finite presentation, whereas unramified morphisms are required only to be locally of finite type.

The rank condition is local. After shrinking V , a chosen maximal minor becomes a unit. It is therefore equivalent to use *standard smooth charts*: localized finite presentations for which one $m \times m$ Jacobian minor is invertible. This formulation also shows why extra equations are harmless in the unramified case: only n of their differentials need span the n coordinate differentials.

Theorem 8.3.11 (Jacobian and cotangent-complex criteria). *Let $f : Y \rightarrow X$ be a morphism of schemes.*

(i) *The following are equivalent:*

- f is smooth;
- f is locally of finite presentation, Ω_f is a vector bundle, and $N_f = 0$;
- f is locally of finite presentation and $\mathbb{L}_f$ is a finite locally free module in degree 0.

(ii) *The morphism f is unramified if and only if it is locally of finite type and $\Omega_f = 0$.*

(iii) *The following are equivalent:*

- f is étale;
- f is locally of finite presentation and $\Omega_f = N_f = 0$;
- f is locally of finite presentation and $\mathbb{L}_f \simeq 0$.

Proof. The assertions are local on source and target, so take a presentation

$$B = A[x_1, \dots, x_n]/J.$$

At a point of $\operatorname{Spec} B$, choose finitely many elements $h_1, \dots, h_m \in J$ whose classes generate the required part of J/J^2 . The truncated cotangent complex is

$$\left[J/J^2 \xrightarrow{dh} \bigoplus_{j=1}^n B dx_j \right], \quad dh_i = \sum_j \frac{\partial h_i}{\partial x_j} dx_j. \quad (\text{J})$$

If the Jacobian has full row rank, a maximal minor is invertible after localization. Elementary row and column operations then split dh as the inclusion of a free direct summand. Hence its kernel $N_{B/A}$ vanishes and its cokernel $\Omega_{B/A}$ is finite locally free. Full column rank makes the cokernel vanish; in the square case both kernel and cokernel vanish.

Conversely, suppose first that B is finitely presented, $N_{B/A} = 0$, and $\Omega_{B/A}$ is finite projective. The exact sequence in (J) is then locally split. Nakayama's lemma lets us choose finitely many relations whose differentials form a basis of the kernel after passing to a residue field; shrinking makes the corresponding maximal minor a unit. This is a standard smooth chart. If merely $\Omega_{B/A} = 0$, finitely many differentials of relations span all dx_j near the chosen point, which is exactly the unramified Jacobian condition. Combining the two conclusions yields the square étale case. These two-term criteria are recorded in [The25, Tags 0D0N and 0G7Z]. For a smooth map the full cotangent complex is $\Omega_{B/A}$ in degree zero, and for an étale map it vanishes, respectively by [The25, Tags 08R5 and 08R2]; the converses follow by taking the two lowest homology modules. $\square$

Corollary 8.3.12 (Locality and closure properties). *Smooth, unramified, and étale morphisms are local on source and target and are stable under base change and composition. Moreover,*

$$\text{étale} \iff \text{smooth and unramified}.$$

An immersion is unramified, and a smooth immersion is an open immersion. More generally, every monomorphism locally of finite type is unramified; for a monomorphism, smooth and étale are equivalent.

Proof. Locality follows from Theorem 8.3.11 and the restriction isomorphisms for Ω and N . Base change follows from standard Jacobian charts: an invertible minor remains invertible after scalar extension. For composition use the transitivity triangle Theorem 8.3.6: extensions of vector bundles split locally, while the vanishing statements pass through its exact homology sequence. The displayed equivalence is then immediate.

For a monomorphism, every infinitesimal lifting set has at most one element, so its cotangent module is zero. The finite-type criterion then makes it unramified. For an immersion, the same vanishing is visible locally because its ring map is a localization followed by a surjection. Finally let a smooth immersion be locally the closed immersion cut out by a finitely generated ideal I . Smoothness gives $I/I^2 = 0$. Nakayama's lemma produces $a \in 1 + I$ with $aI = 0$, and then $V(I) = D(a)$; hence the immersion is open. $\square$

Corollary 8.3.13 (The diagonal criterion for unramifiedness). *A morphism $f: Y \rightarrow X$ is unramified if and only if it is locally of finite type and its diagonal*

$$\Delta_f: Y \longrightarrow Y \times_X Y$$

is an open immersion.

Proof. The diagonal is an immersion and, because f is locally of finite type, it is locally of finite presentation. Moreover, $N_{\Delta_f} \simeq \Omega_f$. If f is unramified, then this module vanishes. Moreover $\Omega_{\Delta_f} = 0$ because the diagonal is a monomorphism. Thus the cotangent criterion says that Δ_f is a smooth immersion, hence open by Corollary 8.3.12. Conversely, if Δ_f is open, its conormal module vanishes, so $\Omega_f = 0$; local finite type and Theorem 8.3.11 imply that f is unramified. $\square$

Proposition 8.3.14 (Local forms of smooth and unramified maps). *Let $f: Y \rightarrow X$ be a morphism of schemes.*

- (i) *The map f is smooth at y of relative dimension d if and only if, after shrinking around y and $f(y)$, it factors as*

$$V \xrightarrow{e} \mathbb{A}_U^d \longrightarrow U$$

with e étale.

- (ii) *The map f is unramified at y if and only if, after shrinking, it factors as*

$$V \hookrightarrow V' \xrightarrow{e} U$$

with the first map a closed immersion and e étale.

- (iii) *If $Y \rightarrow X$ is smooth, $Z \hookrightarrow Y$ is closed, and $Z \rightarrow X$ is smooth, then near any $z \in Z$ there is an étale map $V \rightarrow \mathbb{A}_X^n$ for which*

$$Z \cap V = V \times_{\mathbb{A}_X^n} \mathbb{A}_X^{n-m},$$

where $n = \text{rk}_z \Omega_{Y/X}$ and $m = \text{rk}_z \mathcal{N}_{Z/Y}^$.*

Proof. For (i), take a standard smooth presentation with m equations in n variables and choose an invertible $m \times m$ minor. After reordering variables, the map defined by

$$(h_1, \dots, h_m, x_{m+1}, \dots, x_n): \mathbb{A}_U^n \longrightarrow \mathbb{A}_U^n$$

is étale along V . Restricting to $h_1 = \dots = h_m = 0$ gives an étale map $V \rightarrow \mathbb{A}_U^{n-m}$. Conversely, affine space is smooth and smoothness is stable under composition.

For (ii), select n relations whose differentials span the coordinate differentials. Keeping only these relations gives an étale ambient scheme V' , and imposing the remaining relations cuts out V as a closed subscheme. The converse follows from stability under composition. For (iii), the short exact conormal sequence (C_{sm}) shows that $\mathcal{N}_{Z/Y}^*$ is locally a direct summand generated by m equations with independent differentials. Extending these differentials to a local frame of $\Omega_{Y/X}$ gives étale coordinates and the stated coordinate linear subspace. $\square$

8.4. Smoothness over a Field

Over a field, the infinitesimal rank conditions can be compared with the intrinsic commutative-algebraic notion of regularity. There is one essential caveat: over an imperfect field, a regular scheme need not be smooth. The missing datum is separability of residue fields. This section makes that caveat precise and then gives the relative, fiberwise form of the result.

Definition 8.4.1 (Separability for finitely generated extensions). A finitely generated field extension K/k is *separable* if it has a separating transcendence basis: there are algebraically independent $t_1, \dots, t_r \in K$ such that

$$k \subset k(t_1, \dots, t_r) \subset K$$

and the second extension is finite separable. The integer $r = \text{trdeg}_k K$ is independent of the chosen basis.

Proposition 8.4.2 (Differentials of a separable field extension). If K/k is a finitely generated separable field extension and $t_1, \dots, t_r$ is a separating transcendence basis, then

$$\Omega_{K/k} \simeq \bigoplus_{i=1}^r K dt_i, \quad N_{K/k} = 0. \quad (\text{Sep})$$

In particular, a finite field extension K/k is étale if and only if it is separable.

Proof. Localization of the polynomial calculation gives

$$\Omega_{k(t_1, \dots, t_r)/k} \simeq \bigoplus_i k(t_1, \dots, t_r) dt_i, \quad N_{k(t_1, \dots, t_r)/k} = 0.$$

A finite separable extension is generated by one element after passing to a primitive element. If $K = F[a]$ with separable minimal polynomial p , then $p'(a) \neq 0$, hence is a unit in the field K . Corollary 8.2.13 and Theorem 8.3.11 show that $F \rightarrow K$ is étale and its cotangent complex vanishes. The transitivity triangle for $k \rightarrow F \rightarrow K$ now gives (Sep). Conversely, for a finite extension we use the standard field-theoretic criterion

$$\Omega_{K/k} = 0 \iff K/k \text{ is separable.}$$

It is proved by passing to the maximal separable subextension and observing that every nontrivial finite purely inseparable extension has a generator whose differential is nonzero. This supplies the converse without assuming in advance that every element occurs in a simple presentation over k . $\square$

Proposition 8.4.3 (The point sequence). Let X be locally of finite presentation over a field k , let $x \in |X|$, put $K = \kappa(x)$, and let $\mathfrak{m} = \mathfrak{m}_{X,x}$. The triangle

$$\text{Spec } K \xrightarrow{x} X \longrightarrow \text{Spec } k$$

gives an exact sequence of K -vector spaces

$$H_1(x^* \mathbb{L}_{X/k}) \longrightarrow N_{K/k} \longrightarrow \mathfrak{m}/\mathfrak{m}^2 \longrightarrow \Omega_{X/k}(x) \longrightarrow \Omega_{K/k} \longrightarrow 0. \quad (\text{P})$$

If K/k is separable and $r = \text{trdeg}_k K$, this becomes the short exact sequence

$$0 \longrightarrow \mathfrak{m}/\mathfrak{m}^2 \longrightarrow \Omega_{X/k}(x) \longrightarrow K^r \longrightarrow 0. \quad (\text{P}_{\text{sep}})$$

Proof. The relative cotangent module of $\text{Spec } K \rightarrow X$ is zero, because the map on local rings $\mathcal{O}_{X,x} \rightarrow K$ is surjective. Its intrinsic conormal module is $\mathfrak{m}/\mathfrak{m}^2$. Thus (P) is the low-degree part of the global fundamental sequence Theorem 8.3.6. Under separability, Proposition 8.4.2 gives $N_{K/k} = 0$ and $\Omega_{K/k} \simeq K^r$, proving the second assertion. $\square$

For later comparison, recall the dimension identity for a finite-type k -scheme:

$$\dim_x X = \dim \mathcal{O}_{X,x} + \text{trdeg}_k \kappa(x). \quad (\text{Dim})$$

Here $\dim_x X$ is the maximum of the dimensions of the irreducible components passing through x . Combining (P_{sep}) , (Dim) , and the regularity criterion

$$\mathcal{O}_{X,x} \text{ regular} \iff \dim_K(\mathfrak{m}/\mathfrak{m}^2) = \dim \mathcal{O}_{X,x}$$

immediately yields the following numerical test.

Lemma 8.4.4 (Regularity from the cotangent rank). *If $\kappa(x)/k$ is separable, then*

$$\mathcal{O}_{X,x} \text{ is regular} \iff \dim_{\kappa(x)} \Omega_{X/k}(x) = \dim_x X.$$

Theorem 8.4.5 (Smoothness and regularity over a field). *Let X be a k -scheme locally of finite presentation and $x \in |X|$. Consider:*

- (i) $X \rightarrow \text{Spec } k$ is smooth on an open neighborhood of x ;
- (ii) $\dim_{\kappa(x)} \Omega_{X/k}(x) = \dim_x X$;
- (iii) $\mathcal{O}_{X,x}$ is regular.

Then (i) and (ii) are equivalent, and either implies (iii). If $\kappa(x)/k$ is separable, all three conditions are equivalent.

Proof. If X is smooth near x , its étale local form over $\mathbb{A}_k^d$ shows simultaneously that $\Omega_{X/k}$ has rank d , that $\dim_x X = d$, and that $\mathcal{O}_{X,x}$ is regular. Thus (i) $\Rightarrow$ (ii),(iii).

Conversely, write a neighborhood of x as $X = V(I) \subset \mathbb{A}_k^n$. The fiber rank of $\Omega_{X/k}$ is n minus the rank of the Jacobian of I at x . Under (ii), choose

$$c = n - \dim_x X$$

elements $f_1, \dots, f_c \in I$ whose differentials are independent at x , and put $C = V(f_1, \dots, f_c)$. A corresponding Jacobian minor is invertible after shrinking, so C is smooth of dimension $n - c = \dim_x X$ near x . There is a surjection of local rings

$$\mathcal{O}_{C,x} \twoheadrightarrow \mathcal{O}_{X,x}.$$

The two local rings have the same Krull dimension by (Dim), since they have the same residue field and $\dim_x C = \dim_x X$. The source is a regular local domain. A surjection from a finite-dimensional domain to a ring of at least the same dimension has zero kernel: any nonzero kernel lies in a nonzero prime and strictly lowers dimension. Hence the displayed map is an isomorphism. Shrinking once more gives $X = C$ near x , so X is smooth there. This proves (ii) $\Rightarrow$ (i), and hence (iii) as well.

Finally, if $\kappa(x)/k$ is separable, Lemma 8.4.4 gives (iii) $\Rightarrow$ (ii). □

Corollary 8.4.6 (Perfect fields and geometric regularity). *Let X be locally of finite presentation over a field k .*

- (i) *If k is perfect, then X is smooth over k if and only if X is regular.*
- (ii) *For arbitrary k , the scheme X is smooth over k if and only if $X_{k'}$ is regular for every field extension k'/k . Equivalently, smooth finite-type k -schemes are precisely the geometrically regular ones.*
- (iii) *If k is perfect and X is reduced, then the smooth locus contains a dense open subscheme.*

Proof. Every finitely generated extension of a perfect field is separable, so (i) follows pointwise from Theorem 8.4.5. Smoothness is stable under base change, giving the forward implication of (ii). For the converse one may take an algebraic closure k'/k ; the residue fields of closed points of $X_{k'}$ are then separable over k' , and the theorem gives smoothness at every closed point after base change. Since a scheme locally of finite type over the algebraically closed field k' is Jacobson, a nonempty closed nonsmooth locus would contain a closed point. Thus the whole base change is smooth. Smoothness descends along the faithfully flat map $\text{Spec } k' \rightarrow \text{Spec } k$, giving the claim. For (iii), the local rings at the generic points of a reduced finite-type scheme are fields, hence regular. Apply (i) there and use openness of the smooth locus. $\square$

Warning 8.4.7 (Regular need not mean smooth over an imperfect field). Let $k = \mathbb{F}_p(s)$ and $K = k[t]/(t^p - s)$. Then $\text{Spec } K$ is regular because K is a field, but

$$\Omega_{K/k} = K dt \neq 0$$

since $d(t^p - s) = 0$ in characteristic p . Hence K/k is not unramified and therefore not étale or smooth. This is exactly the failure of the final implication in Theorem 8.4.5 without separability.

Remark 8.4.8 (Smooth points can have inseparable residue fields). The separability hypothesis is needed only for the implication from regularity to smoothness at a specified point. Smoothness itself does not force $\kappa(x)/k$ to be separable: every finitely generated field extension of k occurs as the residue field of a point of some affine space $\mathbb{A}_k^n$, and affine space is smooth over k .

Corollary 8.4.9 (Zero-dimensional smooth schemes). For a k -scheme X , the following are equivalent:

- (i) X is étale over k ;
- (ii) X is unramified over k ;
- (iii) there is a family of finite separable field extensions $(k_i/k)_{i \in I}$ such that

$$X \simeq \bigsqcup_{i \in I} \text{Spec } k_i.$$

Proof. An étale morphism is unramified. If X is unramified over k , it is locally of finite type and has zero cotangent module. Its local Jacobians have full column rank, so its local rings are fields and its underlying space is discrete. Each point therefore has an affine open neighborhood $\text{Spec } k_i$, where k_i/k is finite; vanishing of differentials makes it separable by Proposition 8.4.2. These neighborhoods are disjoint and exhaust X . Conversely, every finite separable field extension is étale and coproducts preserve the property. $\square$

Theorem 8.4.10 (Fiberwise criterion for smoothness). Let $f: Y \rightarrow X$ be flat and locally of finite presentation. The following are equivalent:

- (i) f is smooth;
- (ii) every geometric fiber $Y_{\bar{x}} = Y \times_X \text{Spec } \Omega$, with Ω algebraically closed, is regular;
- (iii) every geometric fiber is smooth over its residue field.

Proof. Smoothness is preserved by base change, and a scheme smooth over an algebraically closed field is regular by Theorem 8.4.5; hence (i) $\Rightarrow$ (ii). Over an algebraically closed, hence perfect, field, (ii) and (iii) are equivalent by Corollary 8.4.6.

For (iii) $\Rightarrow$ (i), fix $x \in X$ and an algebraic closure $\Omega/\kappa(x)$. Smoothness of Y_Ω descends along the faithfully flat field extension $\Omega/\kappa(x)$, so the ordinary fiber Y_x is smooth over $\kappa(x)$. The flat, locally finitely presented fiber criterion now applies directly [The25, Tag 01V8]. $\square$

8.5. Formal Smoothness and Infinitesimal Geometry

The Jacobian criterion is concrete, but its coordinate-free meaning is a lifting property. We now formulate that property for arbitrary algebraic functors. Along the way we distinguish three notions that are often conflated: a finite infinitesimal neighborhood, the functorial completion obtained by taking all such neighborhoods, and the inverse-limit rings used to describe an affine formal scheme. The two-term truncation $\tau_{\leq 1}\mathbb{L}_f$ of the cotangent complex controls the existence and uniqueness questions considered here. Its homology modules are N_f and Ω_f ; higher homology is important for derived deformation problems, but is not needed for the classical square-zero lifting criteria below.

Thickenings, completions, and jets

Definition 8.5.1 (Thickening and its order). A *thickening of rings* is a surjection

$$\tau: R' \twoheadrightarrow R$$

whose kernel I is a nilideal, meaning that each element of I is nilpotent. It has *order at most n* if $I^{n+1} = 0$. A thickening of order at most 1 is a *square-zero extension*.

A *thickening of algebraic functors* is a closed immersion $Z \hookrightarrow Z'$ defined by a quasi-coherent nilideal. It has order at most n when that ideal has $(n+1)$ st power zero.

A nilideal need not be nilpotent, so an arbitrary thickening need not have finite order. In contrast, every finitely generated nilideal is nilpotent. The split extension $R \oplus M \rightarrow R$ is the universal model with a chosen section. Two useful nonsplit examples are

$$\mathbb{Z}/p^{n+1} \twoheadrightarrow \mathbb{F}_p \quad \text{and} \quad W(k)/p^{n+1} \twoheadrightarrow k$$

for a perfect $\mathbb{F}_p$ -algebra k , where $W(k)$ denotes the ring of p -typical Witt vectors.

Definition 8.5.2 (Infinitesimal neighborhoods and completion). Let $Z \hookrightarrow X$ be closed with quasi-coherent ideal $\mathcal{I}$. Its *n th infinitesimal neighborhood* in X is

$$Z^{(n)} = V(\mathcal{I}^{n+1}).$$

Thus $Z \hookrightarrow Z^{(n)}$ has order at most n .

More generally, let $K \subset |X|$ be closed and let $\mathcal{I}$ be the radical quasi-coherent ideal sheaf defining K . The *completion of X along K* is the algebraic subfunctor $\widehat{X}_K \subset X$ defined by

$$\widehat{X}_K(R) = \{x \in X(R) \mid \mathcal{I}(x) \subseteq \sqrt{(0)} \subset R\}. \quad (\widehat{})$$

For a closed subfunctor Z , we also write $\widehat{X}_Z$ for $\widehat{X}_{|Z|}$.

Proposition 8.5.3 (Completion as all finite neighborhoods). *There is always a monomorphism*

$$\operatorname{colim}_n Z^{(n)} \longrightarrow \widehat{X}_Z.$$

It is an isomorphism when $Z \hookrightarrow X$ is of finite presentation.

Proof. An R -point factors through $Z^{(n)}$ precisely when the pulled-back ideal $\mathcal{I}(x)$ satisfies $\mathcal{I}(x)^{n+1} = 0$. Every local section of $\sqrt{\mathcal{I}}$ has a power in $\mathcal{I}$, so its image is nilpotent and the point belongs to the completion. Conversely, finite presentation makes $\mathcal{I}$ finitely generated. The completion condition makes the images of a finite generating set nilpotent; one common sufficiently high power of the generated ideal therefore vanishes, and the point factors through some $Z^{(n)}$. No finite-generation assertion for the radical $\sqrt{\mathcal{I}}$ is needed. $\square$

Warning 8.5.4 (The completion is generally not a scheme). The functor

$$R \longmapsto \{a \in R \mid a \text{ is nilpotent}\}$$

is $\widehat{\mathbb{A}}_0^1$, but it is not represented by an ordinary scheme. It is a formal scheme. If $X = \operatorname{Spec} A$ and $Z = V(I)$ with I finitely generated, the tower of ordinary schemes $\operatorname{Spec}(A/I^{n+1})$ is governed by the complete topological ring $\widehat{A} = \varprojlim_n A/I^{n+1}$, but $\operatorname{Spec} \widehat{A}$ should not be confused with the functorial completion $\widehat{X}_Z$. The former has ordinary points away from $V(I\widehat{A})$; the latter remembers only points on which I becomes nilpotent.

Definition 8.5.5 (Tangential data and jet functors). Let $x \in X(R)$ and let $\tau: R' \twoheadrightarrow R$ be a thickening. A τ -tangential datum at x is an R' -point $\tilde{x} \in X(R')$ whose restriction to R is x .

For $f: Y \rightarrow X$ and $n \geq 0$, the relative n -jet functor is

$$J_n(f)(R) = Y(R[t]/(t^{n+1})) \times_{X(R[t]/(t^{n+1}))} X(R). \quad (\text{Jet})$$

There is a natural tower

$$J_\infty(f) = \varprojlim_n J_n(f) \longrightarrow \cdots \longrightarrow J_2(f) \longrightarrow J_1(f) = T(Y/X) \longrightarrow J_0(f) = Y.$$

The inverse limit is the *relative arc functor*.

If

$$X = \operatorname{Spec} k[x_i \mid i \in I]/(h_j \mid j \in J),$$

then an R -point of $J_n(X/k)$ is a tuple $(a_0, \dots, a_n) \in (R^I)^{n+1}$ satisfying

$$h_j(a_0 + a_1 t + \cdots + a_n t^n) = 0 \quad \text{in } R[t]/(t^{n+1})$$

for every j . Equating coefficients gives $(n+1)|J|$ polynomial equations in $(n+1)|I|$ variables when I and J are finite. Thus finite-order jet functors of an affine finite-presentation scheme are again affine schemes of finite presentation.

Formal lifting properties

Definition 8.5.6 (Formally smooth, unramified, and étale). Let $f: Y \rightarrow X$ be a map of algebraic functors. For every square-zero extension $\tau: R' \twoheadrightarrow R$ and every $y \in Y(R)$, there is a map

$$\{\tau\text{-tangential data at } y \text{ in } Y\} \longrightarrow \{\tau\text{-tangential data at } f(y) \text{ in } X\}. \quad (\text{Lift}_\tau)$$

The map f is:

- (i) *formally smooth* if this map is always surjective;
- (ii) *formally unramified* if it is always injective;
- (iii) *formally étale* if it is always bijective.

Equivalently, these are respectively existence, uniqueness, and unique existence of a dotted arrow in every square

$$\begin{array}{ccc} \operatorname{Spec} R & \xrightarrow{y} & Y \\ \operatorname{Spec}(\tau) \downarrow & \nearrow \text{dotted} & \downarrow f \\ \operatorname{Spec} R' & \longrightarrow & X. \end{array}$$

Example 8.5.7 (Three elementary lifting calculations). (i) An open immersion is formally étale. Indeed, $\mathrm{Spec} R \hookrightarrow \mathrm{Spec} R'$ is a homeomorphism for a nilpotent thickening, so a map to X lands in an open subfunctor before the thickening if and only if it does afterwards.

(ii) If M is a projective quasi-coherent module on X , then $\mathbb{A}(M) \rightarrow X$ is formally smooth: maps out of a projective module lift through surjections. The same holds for $\mathbb{P}(M) \rightarrow X$; a quotient line lifts across a square-zero extension because projective rank-one modules lift and the surjective locus is open.

(iii) Let $p \in R'[t]$ and let $\bar{a} \in R$ be a simple root of its reduction. If $I = \ker(R' \rightarrow R)$ has square zero, then for $h \in I$ Taylor's formula truncates to

$$p(a + h) = p(a) + p'(a)h.$$

Since $p'(\bar{a})$ is a unit, there is a unique h killing $p(a + h)$. This is the square-zero form of Hensel's lemma and proves directly that separable algebraic field extensions are formally étale.

Example 8.5.8 (Nilinvariant functors and the infinitesimal stack). An algebraic functor F is *nilinvariant* if it sends every thickening of rings to a bijection. Every map between nilinvariant functors is formally étale. The inclusion of nilinvariant functors has a left adjoint

$$F \longmapsto F_{\mathrm{inf}}, \quad F_{\mathrm{inf}}(R) = F(R_{\mathrm{red}}).$$

The functor F_{inf} is called the *infinitesimal stack* of F , or its *de Rham stack* when F is defined over $\mathbb{Q}$. It deliberately forgets all nilpotent directions, so maps between such nilinvariant functors have unique infinitesimal lifts for the tautological reason that both source and target see only reduced rings.

Definition 8.5.9 (Abstract lifting terminology). Let $\mathcal{C}$ be a category and $\mathcal{T}$ a class of maps. A map $f : Y \rightarrow X$ is a $\mathcal{T}$ -epimorphism, $\mathcal{T}$ -monomorphism, or $\mathcal{T}$ -isomorphism if every lifting square with left arrow in $\mathcal{T}$ has, respectively, at least one, at most one, or exactly one filler.

Proposition 8.5.10 (Formal calculus of lifting properties). For any $\mathcal{T}$:

- (i) the three classes in Definition 8.5.9 are stable under base change and composition;
- (ii) $\mathcal{T}$ -monomorphisms and $\mathcal{T}$ -isomorphisms are stable under limits in the arrow category;
- (iii) replacing $\mathcal{T}$ by its closure under pushout and composition does not change any of the three classes;
- (iv) replacing $\mathcal{T}$ by its closure under colimits in the arrow category does not change the monomorphisms or isomorphisms.

Consequently the formal properties are stable under base change and composition, and their definition is unchanged if square-zero extensions are replaced by arbitrary finite-order thickenings.

Proof. Fillers pull back, and two successive fillers compose, proving (i). A map into a limit is the same as a compatible family of maps, proving (ii). A filler against a pushout is the same as a filler against the original arrow, and a filler against a composite is built one stage at a time, proving (iii). Finally, maps out of a colimit are compatible families of maps; uniqueness can therefore be tested before taking the colimit, proving (iv).

If $I^{n+1} = 0$, the chain

$$R' \twoheadrightarrow R'/I^n \twoheadrightarrow \cdots \twoheadrightarrow R'/I = R$$

factors the thickening into square-zero extensions, since $(I^j/I^{j+1})^2 = 0$ in R'/I^{j+1} . Apply (iii). □

Remark 8.5.11 (From finite-order to arbitrary thickenings). Every thickening $R' \rightarrow R$ is a filtered colimit of finite-order quotients $R' \rightarrow R_\alpha$. Hence a formally smooth or formally étale map that is locally of finite presentation has the expected lifting property for arbitrary thickenings of rings. For formal unramifiedness the analogous conclusion holds when the diagonal is locally of finite presentation. Without a finiteness condition, a map from a filtered colimit need not factor through a finite stage, so the conclusion is not formal.

Every finite-order thickening of algebraic functors is a colimit of finite-order thickenings of affine schemes. The uniqueness parts of the lifting calculus commute with such colimits. Hence formally unramified and formally étale maps also have their respective lifting properties against finite-order thickenings of algebraic functors. Existence alone does not enjoy this conclusion without an additional finiteness or descent hypothesis.

Definition 8.5.12 (Cohesive morphisms). A map $f: Y \rightarrow X$ of algebraic functors is *cohesive* if, for every pullback square of rings $A \times_C B$ with both $A \twoheadrightarrow C$ and $B \twoheadrightarrow C$ surjective, the natural square

$$\begin{array}{ccc} Y(A \times_C B) & \longrightarrow & Y(A) \times_{Y(C)} Y(B) \\ \downarrow & & \downarrow \\ X(A \times_C B) & \longrightarrow & X(A) \times_{X(C)} X(B) \end{array} \quad (\text{Coh})$$

is cartesian. It is *infinitesimally cohesive* if the same is required only when both surjections are square-zero extensions.

Proposition 8.5.13 (Schemes are cohesive). *Every schematic morphism of algebraic functors is cohesive.*

Proof. Put $D = A \times_C B$. Because $A \twoheadrightarrow C$ and $B \twoheadrightarrow C$ are surjective, the Ferrand–Milnor square is a pushout in schemes:

$$\text{Spec } D \simeq \text{Spec } A \amalg_{\text{Spec } C} \text{Spec } B.$$

This affine pushout is [The25, Tag 0ET0]. Fix an $x \in X(D)$ and base-change the schematic morphism to the scheme $Y_x = Y \times_X \text{Spec } D$. A compatible pair on the right-hand side of (Coh) is exactly a pair of maps $\text{Spec } A \rightarrow Y_x$ and $\text{Spec } B \rightarrow Y_x$ agreeing on $\text{Spec } C$. The displayed pushout supplies a unique map $\text{Spec } D \rightarrow Y_x$, which proves that (Coh) is cartesian. This is closed gluing, not an application of Zariski-open descent. $\square$

Remark 8.5.14 (Why cohesiveness supplies addition). Scalar multiplication on tangent vectors comes from

$$\mathbb{Z}[\epsilon] \longrightarrow \mathbb{Z}[\epsilon, t], \quad a + u\epsilon \longmapsto a + ut\epsilon.$$

If f is infinitesimally cohesive, the pullback $R[\epsilon] \times_R R[\epsilon]$ is sent to a pullback of point sets, and the ring map

$$R[\epsilon] \times_R R[\epsilon] \longrightarrow R[\epsilon], \quad (a + u\epsilon, a + v\epsilon) \longmapsto a + (u + v)\epsilon$$

defines addition. Thus the vertical tangent set at y is canonically an R -module, agreeing with $\text{Hom}_R(\Omega_f(y), R)$ whenever Ω_f exists. The same argument with $R \oplus M$ gives an R -module structure on all split coefficient-valued lifts.

The deformation ring and nonsplit extensions

For a split square-zero extension, the cotangent module describes all lifts directly. A nonsplit extension has no preferred zero lift, so the set of solutions is not naturally a module. When nonempty, it should instead be a torsor under the module of solutions to the split problem. The deformation ring interpolates between these two situations.

Definition 8.5.15 (Deformation ring). Let $I \subset R$ be an ideal and put $\bar{R} = R/I$. The *deformation ring of I in R* , also called the *extended Rees algebra*, is

$$\text{Def}_I(R) = \bigoplus_{n \in \mathbb{Z}} I^n t^{-n} \subset R[t, t^{-1}], \quad I^n = R \text{ for } n \leq 0. \quad (\text{Def})$$

It is an $R[t]$ -algebra and has natural maps

$$R[t] \hookrightarrow \text{Def}_I(R) \twoheadrightarrow \bar{R}[t].$$

The fiber at $t = 1$ is R , while the fiber at $t = 0$ is

$$\text{gr}_I(R) = \bigoplus_{n \geq 0} I^n / I^{n+1},$$

the coordinate algebra of the normal cone.

Construction 8.5.16 (Deforming a square-zero extension). Let $\tau: R' \twoheadrightarrow R$ be square-zero with kernel I . Since $I^2 = 0$,

$$\text{Def}_I(R') = R'[t] \oplus It^{-1} \twoheadrightarrow R[t]$$

is itself a square-zero extension of $R'[t]$ -algebras. For $a \in R$, choose any lift $a' \in R'$ and set

$$R'(a) = (R'[t] \oplus It^{-1}) / (t - a') \twoheadrightarrow R[t] / (t - a) \simeq R. \quad (\tau_a)$$

The R' -module $R'(a)$ is the quotient of $R' \oplus It^{-1}$ by $i = a'it^{-1}$ for $i \in I$. Because the R' -action on I factors through R , the result is independent of the chosen lift a' . The family $\tau_a: R'(a) \twoheadrightarrow R$ satisfies:

$$\tau_1 = \tau, \quad \tau_0 = (R \oplus I \twoheadrightarrow R).$$

It also has scalar and addition maps

$$\begin{aligned} R'(a) &\longrightarrow R'(ua), & r + it^{-1} &\longmapsto r + uit^{-1}, \\ R'(a) \times_R R'(b) &\longrightarrow R'(a+b), & (r + it^{-1}, r + jt^{-1}) &\longmapsto r + (i+j)t^{-1}. \end{aligned} \quad (\text{DA})$$

Proposition 8.5.17 (The solution set is a torsor). Let $f: Y \rightarrow X$ be infinitesimally cohesive and consider a lifting problem

$$\begin{array}{ccc} \text{Spec } R & \xrightarrow{y} & Y \\ \downarrow & \nearrow & \downarrow f \\ \text{Spec } R' & \xrightarrow{x'} & X \end{array}$$

for a square-zero extension with kernel I . If a lift exists, the set of all lifts is canonically a torsor under the R -module of lifts for the split extension $R \oplus I \twoheadrightarrow R$. If f admits a cotangent module, this acting module is

$$\text{Hom}_R(\Omega_f(y), I).$$

Proof. Apply the lifting problem to the family τ_a of Construction 8.5.16. Infinitesimal cohesiveness turns the addition and scalar maps in (DA) into addition and scalar operations on the disjoint union of the corresponding solution sets. The map remembering $a \in R$ is R -linear. Its kernel is the solution module for τ_0 , while its fiber over 1 is the solution set for the original extension $\tau_1 = \tau$. A nonempty fiber of a linear map is a torsor under its kernel. Finally, Definition 8.3.1 identifies that kernel with $\text{Hom}_R(\Omega_f(y), I)$. $\square$

Proposition 8.5.18 (Formal unramifiedness and differentials). If a map of algebraic functors $f: Y \rightarrow X$ is formally unramified, it admits a cotangent module and $\Omega_f = 0$. Conversely, $\Omega_f = 0$ implies formal unramifiedness whenever f is infinitesimally cohesive or locally separated; in particular, the converse holds for morphisms of schemes.

Proof. Fix the special split lifting problem used to define $\Omega_f(y)$ in Definition 8.3.1: its bottom datum is the scalar extension of $f(y)$. This problem has a distinguished zero lift, obtained by scalar-extending y . Formal unramifiedness says that this is the only lift. Hence this particular lift functor is the one-point functor, represented compatibly by the zero module. We do not assert that an arbitrary bottom datum over $R \oplus M$ has a zero lift.

Conversely, if f is infinitesimally cohesive, Proposition 8.5.17 says that any two lifts differ by an element of $\text{Hom}_R(\Omega_f(y), I) = 0$, so they coincide. If f is locally separated, its diagonal is an immersion. Work on a neighborhood where it is closed with ideal J . The equality $\Omega_f \simeq N_{\Delta_f} = J/J^2 = 0$ says $J = J^2$. Given two putative lifts, the image of J in the square-zero difference ideal I is therefore contained in the image of J^2 , hence in $I^2 = 0$. Their joint map factors through the diagonal, so the two lifts are equal. $\square$

Lemma 8.5.19 (The local solvability ideal). *Let X and Y be Zariski sheaves, let $f: Y \rightarrow X$ be infinitesimally cohesive, and let $R' \twoheadrightarrow R$ be square-zero. Assume that Ω_f exists and is locally projective. For a fixed lifting problem, the set of elements $h \in R'$ for which the localized problem over*

$$R'_h \twoheadrightarrow R_h$$

has a solution is a radical ideal of R' .

Proof. Let $S \subset R'$ be the set in the statement. We have $0 \in S$, and $h \in S$ implies $rh \in S$ for every $r \in R'$, because a solution restricts to every smaller principal open. Also $R'_h = R'_{h^n}$ canonically, so $h \in S$ if and only if $h^n \in S$.

It remains to prove closure under addition. Suppose $h, k \in S$ and localize the entire problem at $h+k$. The two elements h, k then generate the unit ideal, so $D(h)$ and $D(k)$ cover $\text{Spec } R'_{h+k}$. Choose lifts on these two opens. On $D(hk)$ their difference is, by Proposition 8.5.17, an element of

$$\text{Hom}(\Omega_f(y)_{hk}, I_{hk}), \quad I = \ker(R'_{h+k} \rightarrow R_{h+k}).$$

The principal-open Čech sequence

$$0 \longrightarrow I \longrightarrow I_h \oplus I_k \longrightarrow I_{hk} \longrightarrow 0$$

is exact. Projectivity of $\Omega_f(y)_{h+k}$ makes $\text{Hom}(\Omega_f(y)_{h+k}, -)$ exact, so the difference on $D(hk)$ is the difference of corrections defined on $D(h)$ and $D(k)$. Translate the two local lifts by these corrections. They then agree on the overlap, and the Zariski sheaf property glues them to a lift over R'_{h+k} . Thus $h+k \in S$. We have proved that S is an ideal and that it is radical. $\square$

Theorem 8.5.20 (Formal smoothness via the cotangent complex). *Let $f: Y \rightarrow X$ be a morphism of schemes.*

- (i) *f is formally smooth if and only if Ω_f is locally projective as a quasi-coherent module and $N_f = 0$.*
- (ii) *f is formally unramified if and only if $\Omega_f = 0$.*
- (iii) *f is formally étale if and only if $\Omega_f = N_f = 0$.*

Equivalently, in deformation-theoretic language, a lifting problem against a square-zero ideal I has an obstruction class

$$\text{ob}(y, x') \in \text{Ext}_R^1(\mathbb{L}y^*\mathbb{L}_f, I);$$

it has a solution exactly when this class vanishes, and then the solutions form a torsor under

$$\text{Ext}_R^0(\mathbb{L}y^*\mathbb{L}_f, I) = \text{Hom}_R(\Omega_f(y), I).$$

Proof roadmap and reference. The precise two-term criteria are [The25, Tags 0D0L, 02H9, and 0D0M]; the affine obstruction-and-torsor calculation is [The25, Tag 08S5].

The split square-zero case is the universal property of Ω_f . For a general extension, the construction $\tau_0 \rightsquigarrow \tau_1$ above makes every nonempty solution set a torsor under the split solution module. Resolving an affine algebra by a polynomial algebra, the failure to lift the relations defines the displayed Ext^1 class. This class depends only on $\tau_{\leq 1}\mathbb{L}_f$. If $N_f = 0$, that truncation is $\Omega_f[0]$, and local projectivity of Ω_f makes the relevant Ext^1 group vanish locally. When a solution exists, the difference between two solutions is the displayed Ext^0 , equivalently a homomorphism from $\Omega_f(y)$.

The subtle local-to-global step is Lemma 8.5.19: the elements on whose principal opens the lifting problem is solvable form a radical ideal. Local projectivity is Zariski-local, so local solutions patch to give formal smoothness. Uniqueness is the zero-cotangent calculation of Proposition 8.5.18. Combining existence and uniqueness gives the étale statement. $\square$

Corollary 8.5.21 (Locality of the formal properties). *Formally smooth, formally unramified, and formally étale are local properties of morphisms of schemes.*

Proof. The criteria of Theorem 8.5.20 reduce the claim to locality of vanishing and projectivity of quasi-coherent modules, together with localization of Ω_f and N_f . $\square$

Theorem 8.5.22 (Formal plus finiteness equals geometric). *For a morphism of schemes $f : Y \rightarrow X$:*

$$\begin{aligned} f \text{ is smooth} &\iff f \text{ is formally smooth and locally of finite presentation,} \\ f \text{ is unramified} &\iff f \text{ is formally unramified and locally of finite type,} \\ f \text{ is étale} &\iff f \text{ is formally étale and locally of finite presentation.} \end{aligned}$$

Proof. Compare the three parts of Theorem 8.5.20 with the three parts of Theorem 8.3.11. Under local finite presentation, the cotangent module is finitely presented; if it is also locally projective, it is therefore a vector bundle. The unramified statement needs only finite type because its criterion is the degree-zero equality $\Omega_f = 0$. $\square$

Remark 8.5.23 (What the finiteness hypothesis buys). Formal smoothness sees only infinitesimal lifting; it imposes no finite-generation condition. For example, a polynomial algebra in infinitely many variables is formally smooth but its spectrum is not smooth in the geometric sense of Definition 8.3.10, because it is not locally of finite presentation. The word *formal* must therefore never be dropped without checking the accompanying finiteness hypothesis.

8.6. Local Structure of Étale Morphisms

An étale morphism is infinitesimally a local isomorphism. It need not be a Zariski-local isomorphism: a nontrivial finite separable field extension is the smallest counterexample. The correct local normal form is a standard étale algebra, and the key finiteness input needed to reach it is Zariski's main theorem.

Locally quasi-finite morphisms

Recall that a finite morphism is affine and, on affine charts, corresponds to an algebra that is finite as a module over the base ring. The definition is revisited with the properness comparisons in Definition 10.5.1.

Definition 8.6.1 (Locally quasi-finite and quasi-finite). A morphism of schemes $f: Y \rightarrow X$ is *locally quasi-finite* if it is locally of finite type and every fiber

$$Y_x = Y \times_X \operatorname{Spec} \kappa(x)$$

has discrete underlying space. It is *quasi-finite* if, in addition, it is quasi-compact.

Proposition 8.6.2 (Equivalent fiberwise finiteness). *A morphism is quasi-finite if and only if it is of finite type and has finite fibers. Every unramified morphism is locally quasi-finite, and every finite morphism is quasi-finite.*

Proof. Let T be of finite type over a field k . Then T is noetherian, and the following are equivalent:

$$\begin{aligned} |T| \text{ is discrete,} \quad \dim T = 0, \quad |T| \text{ is finite,} \\ T \text{ is finite over } k. \end{aligned} \tag{QF}$$

Indeed, a quasi-compact discrete space is finite. A noetherian zero-dimensional ring is artinian, and an artinian finite-type k -algebra has a finite filtration whose factors are finite field extensions of k , hence is finite-dimensional over k . The reverse implications follow from integrality and the dimension theorem. Apply (QF) to every fiber and use quasi-compactness to pass from local finite type to finite type.

The geometric fibers of an unramified morphism are disjoint unions of spectra of finite separable fields by Corollary 8.4.9, so they are discrete. A finite morphism is of finite type and has finite fibers, proving the final claim. $\square$

Theorem 8.6.3 (Local Zariski main theorem). *Let $A \rightarrow B$ be of finite type, let $\mathfrak{p} \in \operatorname{Spec} A$, and let $\mathfrak{q} \in \operatorname{Spec} B$ lie above $\mathfrak{p}$. Suppose $\mathfrak{q}$ is isolated in the fiber $\operatorname{Spec}(B \otimes_A \kappa(\mathfrak{p}))$. Then there are a finite A -subalgebra $B' \subset B$ and an element $g \in B' \setminus \mathfrak{q}$ such that*

$$B'_g \xrightarrow{\sim} B_g. \tag{ZMT}_{\text{loc}}$$

Proof roadmap and reference. The isolated-fiber hypothesis says precisely that $A \rightarrow B$ is quasi-finite at $\mathfrak{q}$ [The25, Tag 00PK]. Let $C \subset B$ be the integral closure of A in B . The local form of Zariski's main theorem supplies $g \in C \setminus \mathfrak{q}$ with $C_g \simeq B_g$ [The25, Tag 00Q9]. Since $B_g = C_g$ is a finite-type A -algebra, choose finitely many elements $c_i \in C$ whose fractions, together with g^{-1} , generate C_g over A . Set $B' = A[g, c_1, \dots, c_n] \subset C$. Every generator is integral over A , so B' is finite over A , and the choice of the c_i gives $B'_g = C_g = B_g$. This is $(\text{ZMT}_{\text{loc}})$. $\square$

Remark 8.6.4 (Global Zariski main theorem). Under the qcqs hypotheses used later, the global finite-factorization form says that a separated quasi-finite morphism factors as

$$Y \xrightarrow{j} \bar{Y} \xrightarrow{p} X,$$

where j is a quasi-compact open immersion and p is finite. This is proved in Theorem 10.5.5; an integral-envelope variant constructs $\bar{Y}$ from the integral closure of $\mathcal{O}_X$ in $f_*\mathcal{O}_Y$. The difficult point is showing that j is globally open and descending the data to a finite algebra; the local theorem only proves the needed statement near one point. We record the link for orientation and do not use it in the proof below.

Standard étale algebras

Definition 8.6.5 (Standard étale algebra). An A -algebra B is *standard étale* if

$$B \simeq (A[t]/(p))_g, \quad (\text{SE})$$

where $p \in A[t]$ is monic and the image of $p'(t)$ is a unit in B . Here $g \in A[t]$ is any polynomial used for the localization.

Proposition 8.6.6 (Standard étale algebras are étale and flat). *Every standard étale A -algebra is finitely presented, étale, and flat over A .*

Proof. Introduce a variable u to write the localization as

$$B \simeq A[t, u]/(p, ug - 1).$$

The Jacobian matrix of the two relations with respect to (t, u) is

$$\begin{pmatrix} p' & 0 \\ ug' & g \end{pmatrix},$$

whose determinant is $p'g$. Both factors are units in B , so Definition 8.3.10 proves that B/A is étale. It is visibly finitely presented. Since p is monic, $A[t]/(p)$ is finite free over A on $1, t, \dots, t^{\deg p-1}$. Localization is flat, and a composite of flat maps is flat; hence B is flat over A . $\square$

Theorem 8.6.7 (Local structure of étale morphisms). *Let $f: Y \rightarrow X$ be étale and let $y \in |Y|$. There are affine open neighborhoods $y \in V \subset Y$ and $f(y) \in U \subset X$, with $f(V) \subset U$, for which $\mathcal{O}(V)$ is a standard étale $\mathcal{O}(U)$ -algebra.*

Proof roadmap and reference. This is [The25, Tag 02GT]; its affine algebra input is [The25, Tag 00UE]. We indicate where Theorem 8.6.3 enters. An étale map is unramified, hence locally quasi-finite, so the point y is isolated in its fiber. Local Zariski main theorem replaces a neighborhood of y by an open subalgebra of a finite algebra over $\mathcal{O}(U)$. One then isolates the factor of the special fiber containing y . Its residue-field extension is finite separable, so it has a primitive element with separable minimal polynomial. In the proof of Tag 00UE, one first lifts such a generator to the finite algebra and uses Nakayama's lemma to obtain a monogenic algebra near the chosen point. One then constructs a monic relation whose reduction has the chosen separable factor and localizes where its derivative is a unit. The resulting presentation is precisely $(A[t]/(p))_g$ with p monic.

The primitive-element step alone controls only the residue field. Finiteness from Zariski main theorem is what lets Nakayama promote that generator from the fiber to an entire neighborhood. $\square$

Corollary 8.6.8 (Smooth morphisms are flat). *Every smooth morphism of schemes is flat.*

Proof. By Proposition 8.3.14, locally on source and target a smooth morphism factors as

$$V \longrightarrow \mathbb{A}_U^d \longrightarrow U$$

with the first arrow étale. By Theorem 8.6.7 and Proposition 8.6.6, the first arrow is flat. The second is flat because a polynomial algebra is free over its coefficient ring. Their composite is flat, and flatness is local. $\square$

Combining this corollary with Theorem 8.4.10 completes the promised global classification.

Theorem 8.6.9 (The three classification tables). *Let $f: Y \rightarrow X$ be a morphism of schemes.*

- (i) If f is locally of finite presentation, the following are equivalent:
- (i) f is smooth in the Jacobian sense;
 - (ii) f is formally smooth;
 - (iii) Ω_f is a vector bundle and $N_f = 0$;
 - (iv) f is flat and Ω_f is a vector bundle whose rank at y is the dimension of the fiber through y ;
 - (v) f is flat and every geometric fiber is regular.
- (ii) If f is locally of finite presentation, the following are equivalent:
- (i) f is étale;
 - (ii) f is formally étale;
 - (iii) $\Omega_f = N_f = 0$;
 - (iv) f is flat and $\Omega_f = 0$;
 - (v) every geometric fiber is a discrete reduced scheme and f is flat.
- (iii) If f is locally of finite type, the following are equivalent:
- (i) f is unramified;
 - (ii) f is formally unramified;
 - (iii) $\Omega_f = 0$;
 - (iv) every geometric fiber is a disjoint union of reduced points.

Proof. The equivalences between the first three entries are Theorems 8.3.11, 8.5.20 and 8.5.22. Smoothness implies flatness by Corollary 8.6.8; the rank of its cotangent bundle is read on a standard smooth chart. Conversely, if f is flat and Ω_f has the stated fiber rank, then after geometric base change Theorem 8.4.5 makes every fiber regular; Theorem 8.4.10 then makes f smooth. The étale table is the relative-dimension-zero specialization. The unramified fiber description follows from Corollary 8.4.9 after geometric base change and from the Jacobian criterion in the reverse direction. $\square$

The étale topology

Definition 8.6.10 (Étale covers and the small étale site). An *étale covering family* of an affine scheme $\text{Spec } A$ is a finite family of étale maps

$$\{\text{Spec } A_i \longrightarrow \text{Spec } A\}_{i \in I}$$

whose images jointly cover $\text{Spec } A$. These families define the *étale topology* on CAlg^{op} . Equivalently, on schemes one allows jointly surjective families of étale morphisms; over a quasi-compact affine base a finite subfamily suffices.

For a scheme X , the *small étale site* $(\text{Et}/X, \text{et})$ has étale X -schemes as objects and étale covering families as covers. A sheaf on this site is an *étale sheaf* on X .

Proposition 8.6.11 (Position among standard topologies). The inclusions of covering classes are

$$\text{Zar} \subset \text{et} \subset \text{fppf} \subset \text{fpqc}.$$

The topology generated by jointly surjective smooth morphisms is the same as the étale topology.

Proof. Open immersions are étale, so every Zariski cover is étale. Étale maps are flat and locally of finite presentation, so every étale cover is fppf; the last inclusion is definitional.

A smooth cover is locally an étale map to an affine space by Proposition 8.3.14. After an étale base change, the chosen point of the affine-space fiber extends to a local section; these local sections give an étale refinement of the smooth cover. Thus smooth covers generate no additional covering sieves. Precisely, every smooth covering admits an étale refinement [The25, Tag 055V]. $\square$

Remark 8.6.12 (Why the étale topology is genuinely finer). The cover $\mathrm{Spec} K \rightarrow \mathrm{Spec} k$ attached to a finite separable field extension generally has no Zariski-local section, so the étale topology detects arithmetic that the underlying topological space of $\mathrm{Spec} k$ cannot see. It supports a cohomology theory with finite coefficients that, over $\mathbb{C}$, compares with singular cohomology, while still making sense over finite and imperfect fields. It also provides the natural local setting in which free finite group actions have geometric quotients.

8.7. Hyperplanes, Incidence, and Bertini's Theorem

Bertini's theorem turns a statement about a generic choice into a geometric statement about an open subset of a parameter space. The parameter space for hyperplanes is the dual projective space, and the single family containing all hyperplane sections is the incidence correspondence. We retain the quotient-line convention for $\mathbb{P}(V)$ used throughout this book. The incidence construction below packages the hyperplanes and their first-order singularity conditions into a single finite-type family.

Hyperplanes and their parameter space

Definition 8.7.1 (Hyperplane and hyperplane section). Let k be a ring and V a k -module. A *hyperplane* in $\mathbb{P}(V)$ is a closed subscheme $\mathbb{P}(W)$ arising from a locally split exact sequence

$$0 \longrightarrow L \longrightarrow V \longrightarrow W \longrightarrow 0$$

whose kernel L is a line. Equivalently, the kernel is a universally injective line subbundle of V . When V is finite locally free of rank r , this is equivalent to requiring W to be finite locally free of rank $r - 1$. If $X \subset \mathbb{P}(V)$, the corresponding *hyperplane section* is

$$X \cap \mathbb{P}(W) = X \times_{\mathbb{P}(V)} \mathbb{P}(W).$$

Assume now that V is a vector space, that is, a finite locally free k -module of constant rank. Dualizing the kernel line of $V \twoheadrightarrow W$ in the locally split sequence identifies hyperplanes after any scalar extension with quotient lines of $V^\vee$. Hence

$$\mathbb{P}(V^\vee)(R) \simeq \{\text{hyperplanes in } \mathbb{P}(V)_R\}. \quad (\text{Dual})$$

This explains the name *dual projective space*. For an arbitrary, possibly nonprojective V , the hyperplane functor still exists but need not be a scheme.

Construction 8.7.2 (The incidence hyperplane). Let π_1, π_2 be the projections from $\mathbb{P}(V) \times \mathbb{P}(V^\vee)$ and set

$$\mathcal{O}(1, 1) = \pi_1^* \mathcal{O}_{\mathbb{P}(V)}(1) \otimes \pi_2^* \mathcal{O}_{\mathbb{P}(V^\vee)}(1).$$

There is a canonical global section s of this line bundle. At an R -point given by quotient lines

$$q_1: V_R \twoheadrightarrow L_1, \quad q_2: V_R^\vee \twoheadrightarrow L_2,$$

its value is the image of 1 under

$$R \xrightarrow{\text{coev}} V_R \otimes_R V_R^\vee \xrightarrow{q_1 \otimes q_2} L_1 \otimes_R L_2. \quad (\text{Inc})$$

The zero scheme

$$\mathcal{H} = V(s) \subset \mathbb{P}(V) \times \mathbb{P}(V^\vee)$$

is the *incidence hyperplane* or *universal hyperplane*.

Proposition 8.7.3 (Universal property of incidence). *For every k -algebra R ,*

$$\mathcal{H}(R) = \{(x, H) \mid x \in \mathbb{P}(V)(R), H \subset \mathbb{P}(V)_R \text{ a hyperplane}, x \in H\}.$$

If $h: \text{Spec } R \rightarrow \mathbb{P}(V^\vee)$ classifies H , then H is the pullback of $\mathcal{H}$ along

$$\text{id}_{\mathbb{P}(V)} \times h: \mathbb{P}(V)_R \longrightarrow \mathbb{P}(V) \times \mathbb{P}(V^\vee),$$

and h is unique.

Proof. A hyperplane classified by q_2 consists of those quotient lines q_1 for which the canonical pairing between their corresponding line and covector line vanishes. Formula (Inc) says exactly that $s(q_1, q_2) = 0$. This proves the pointwise equality, naturally in R . Yoneda then gives both the pullback assertion and uniqueness. $\square$

Flatness of the universal section

Proposition 8.7.4 (Fiberwise non-zero-divisors and flat quotients). *Let $A \rightarrow B$ be a flat map of noetherian rings and $b \in B$. Suppose that for every prime ideal $\mathfrak{p} \subset A$, the image of b is a nonzerodivisor in $B \otimes_A \kappa(\mathfrak{p})$. Then b is a nonzerodivisor in B and B/bB is flat over A .*

Proof. Localize at a prime ideal $\mathfrak{n} \subset B$ and put $\mathfrak{p} = \mathfrak{n} \cap A$. The local homomorphism

$$A_{\mathfrak{p}} \longrightarrow B_{\mathfrak{n}}$$

has noetherian target, and $B_{\mathfrak{n}}$ is flat over $A_{\mathfrak{p}}$. Multiplication by b is injective after reduction modulo $\mathfrak{p}A_{\mathfrak{p}}$, by the hypothesis and localization. The local flatness lemma therefore says that multiplication by b on $B_{\mathfrak{n}}$ is injective and that its cokernel is flat over $A_{\mathfrak{p}}$ [The25, Tag 00ME]. Since this holds at every prime ideal of B , multiplication by b on B is injective and B/bB is flat over A . The noetherian hypothesis enters exactly through the cited local lemma. $\square$

The bad-hyperplane locus

Let k be a field, let V have rank r , and let $X \subset \mathbb{P}(V)$ be a locally closed k -subscheme smooth over k . Restrict the incidence hyperplane to obtain

$$\begin{array}{ccc} & \mathcal{Y} = (X \times \mathbb{P}(V^\vee)) \cap \mathcal{H} & \\ p \swarrow & & \searrow q \\ X & & \mathbb{P}(V^\vee). \end{array} \quad (\text{Fam})$$

The fiber of q at a point h classifying a hyperplane H is $X_{\kappa(h)} \cap H$. Let

$$\Sigma = |\mathcal{Y}| \setminus \text{Sm}(q)$$

be the closed complement of the smooth locus of q , equipped below with its reduced induced closed subscheme structure. Points of $q(\Sigma)$ are precisely the hyperplanes whose section is singular somewhere, together with the hyperplanes for which flatness fails by containing a component.

Fix a field extension κ/k and a κ -point x of X . Let $\tilde{x}$ be the induced κ -rational, hence closed, point of X_κ . Put

$$d_x = \dim_{\tilde{x}} X_\kappa = \dim \mathcal{O}_{X_\kappa, \tilde{x}}.$$

Evaluation to first order defines a κ -linear map

$$\begin{aligned} \phi_x: V_\kappa &\simeq \Gamma(\mathbb{P}(V_\kappa), \mathcal{O}(1)) \longrightarrow \mathcal{O}_{X_\kappa}(1)_{\tilde{x}} \\ &\longrightarrow \mathcal{O}_{X_\kappa}(1)_{\tilde{x}} / \mathfrak{m}_{\tilde{x}}^2 \mathcal{O}_{X_\kappa}(1)_{\tilde{x}}. \end{aligned} \quad (\phi_x)$$

The first isomorphism uses the quotient convention for projective space. Locally trivializing $\mathcal{O}(1)$ identifies the final target with $\mathcal{O}_{X_\kappa, \tilde{x}} / \mathfrak{m}_{\tilde{x}}^2$, a vector space of dimension $d_x + 1$ because X_κ is regular.

Lemma 8.7.5 (Hyperplanes singular at a fixed point). *The map ϕ_x is surjective. As a reduced closed subscheme of $\mathbb{P}(V_\kappa^\vee)$, the locus of hyperplanes H for which $\tilde{x} \in X_\kappa \cap H$ and the section is not regular of dimension $d_x - 1$ at $\tilde{x}$ is*

$$\mathbb{P}(\ker(\phi_x)^\vee).$$

It has dimension $r - d_x - 2$, with dimension -1 interpreted as the empty locus.

Proof. Because X is locally closed in $\mathbb{P}(V)$, linear forms separate the value and the first-order directions at $\tilde{x}$; equivalently, $V_\kappa \rightarrow \mathcal{O}_{X_\kappa, \tilde{x}}(1) / \mathfrak{m}_{\tilde{x}}^2 \mathcal{O}_{X_\kappa, \tilde{x}}(1)$ is surjective.

Let $e \in V_\kappa$ define a hyperplane through $\tilde{x}$. After trivializing $\mathcal{O}(1)$ near $\tilde{x}$, its local section is an element $e \in \mathfrak{m}_{\tilde{x}}$, and

$$\mathcal{O}_{X_\kappa \cap V(e), \tilde{x}} \simeq \mathcal{O}_{X_\kappa, \tilde{x}} / (e).$$

In a regular local ring of dimension d_x , this quotient is regular of dimension $d_x - 1$ exactly when $e \notin \mathfrak{m}_{\tilde{x}}^2$. That condition is $\phi_x(e) \neq 0$. Hence the bad hyperplanes are precisely the quotient lines annihilating the image, namely $\mathbb{P}(\ker(\phi_x)^\vee)$. Since ϕ_x has target rank $d_x + 1$, its kernel has rank $r - d_x - 1$, and the displayed projective space has dimension $r - d_x - 2$. $\square$

Theorem 8.7.6 (Bertini's theorem). *Let k be a field, V a finite-dimensional k -vector space, and $X \subset \mathbb{P}(V)$ a locally closed subscheme smooth over k . Assume $\dim_k(V) \geq 1$. There is a nonempty open subscheme*

$$U \subset \mathbb{P}(V^\vee)$$

such that, for every $h \in |U|$ classifying a hyperplane $H \subset \mathbb{P}(V)_{\kappa(h)}$,

$$X_{\kappa(h)} \cap H$$

is smooth over $\kappa(h)$ and contains no irreducible component of $X_{\kappa(h)}$.

Proof roadmap and reference. If $X = \emptyset$, take $U = \mathbb{P}(V^\vee)$, so assume that X is nonempty. The standard incidence proof is given in [The25, Tag 0FD6]. We give the dimension argument; the dimension-of-fibers input is [The25, Tag 0B2L].

For the family (Fam), the failure of a hyperplane to meet a component properly is exactly the failure of the incidence equation to be a fiberwise nonzerodivisor. On the complementary locus, Proposition 8.7.4 makes q flat, and Theorem 8.4.10 identifies its nonsmooth locus with the locus of singular fiber sections.

By Lemma 8.7.5, the fiber of $p: \Sigma \rightarrow X$ over x has dimension at most $r - d_x - 2$. The dimension-of-fibers estimate therefore gives

$$\begin{aligned} \dim \overline{q(\Sigma)} &\leq \dim \Sigma \\ &\leq \sup_{x \in |X|} (d_x + \dim(p^{-1}(x) \cap \Sigma)) \\ &\leq \sup_x (d_x + r - d_x - 2) = r - 2. \end{aligned}$$

But $\dim \mathbb{P}(V^\vee) = r - 1$. Hence $\overline{q(\Sigma)}$ is a proper closed subset, and its open complement U is nonempty. By construction, every hyperplane parameterized by U meets every component properly and has a smooth section. The heavy inputs are the dimension-of-image theorem, openness of the smooth locus, and the fiberwise flatness criterion; the incidence and tangent-space computations themselves are contained above. $\square$

Remark 8.7.7 (Geometric genericity versus a rational hyperplane). The theorem produces a nonempty open subscheme of the dual projective space. It need not contain a k -rational point when k is finite. Thus it does not, over an arbitrary field, assert the existence of a hyperplane defined over k .

Lemma 8.7.8 (Rational points over an infinite field). *If k is infinite, then $\mathbb{A}^n(k)$ is dense in $|\mathbb{A}_k^n|$ for every $n \geq 0$. Consequently every nonempty open of $\mathbb{P}_k^n$ contains a k -rational point.*

Proof. A polynomial in one variable that vanishes at every element of an infinite field is zero. Inducting on n , view a polynomial in $k[x_1, \dots, x_n]$ as a polynomial in x_n with coefficients in the first $n - 1$ variables. If it vanishes on all of k^n , the one-variable assertion and the induction hypothesis kill every coefficient. Thus no proper closed subset contains $\mathbb{A}^n(k)$. Every nonempty open in projective space meets some standard affine chart, proving the final statement. $\square$

Corollary 8.7.9 (Bertini over an infinite field). *Under the hypotheses of Theorem 8.7.6, if k is infinite there exists a hyperplane $H \subset \mathbb{P}(V)$ defined over k such that $X \cap H$ is smooth over k and contains no irreducible component of X .*

Proof. Apply Lemma 8.7.8 to the nonempty open $U \subset \mathbb{P}(V^\vee)$ supplied by Bertini, and choose $h \in U(k)$. $\square$

Summary. The chapter's chain of ideas is now complete:

$$\begin{aligned} \text{square-zero lifts} + \text{finiteness} &\xLeftrightarrow{\Omega_f, N_f} \text{Jacobian rank} \iff \text{smooth} / \text{unramified} / \text{étale} \\ &\implies \text{étale local forms and flatness.} \end{aligned}$$

Over a field, separability is the bridge from regularity to smoothness. In projective geometry, the universal incidence hyperplane packages all sections into one finite-type family, which is flat away from the component-containing locus, and a dimension estimate removes the singular members generically.

Chapter 9.

Divisors and Blowups

9.1. Effective divisors

Divisors turn a codimension-one equation into an object that can be transported and added. Blowups solve the complementary problem: instead of asking an ideal to remain principal, they modify the ambient scheme until its inverse image is an effective Cartier divisor. The two constructions are therefore best developed together.

We use relative affine and projective spaces, the quotient-line convention, and relative Proj from Definitions 3.2.11 and 3.2.13 and Remark 3.2.12, and the conormal and cotangent-complex conventions from Definition 8.3.5 and Theorem 8.3.6.

Why replace a point by directions?

Start with the rational slope y/x . It defines the morphism

$$\sigma: \mathbb{A}_k^2 - \{0\} \longrightarrow \mathbb{P}_k^1, \quad (x, y) \longmapsto [x : y],$$

but no value at the origin is compatible with every approach to that point: the restriction of σ to the line $y = mx$ has limiting value $[1 : m]$. The missing datum is therefore not one extra point but the whole projective line of one-dimensional directions.

This description is already the quotient-line functor of the blowup. For a k -algebra R , an R -point of $B = \text{Bl}_0(\mathbb{A}_k^2)$ is a factorization

$$R^2 \xrightarrow[\text{quotient line}]{q} L \xrightarrow{s} R, \quad s \circ q = (a, b), \quad (*)$$

where $(a, b) \in R^2$ is the underlying point of $\mathbb{A}^2$ and L is invertible of rank one. Over $\mathbb{A}^2 - \{0\}$, the map s is locally an isomorphism, so the quotient q is forced and B is unchanged there. Over the origin, $s = 0$ and q is arbitrary, giving the exceptional fiber $E \simeq \mathbb{P}_k^1$. Equivalently, the two fiber-product identities are

$$B \times_{\mathbb{A}_k^2} \{0\} = E \simeq \mathbb{P}_k^1, \quad B \times_{\mathbb{A}_k^2} (\mathbb{A}_k^2 - \{0\}) \simeq \mathbb{A}_k^2 - \{0\}.$$

They are the two cartesian squares in Proposition 9.2.3 for this center. Remembering q extends σ to $B \rightarrow \mathbb{P}_k^1$, and remembering s identifies

$$\text{Bl}_0(\mathbb{A}_k^2) \simeq \mathbb{A}_{\mathbb{P}_k^1}(\mathcal{O}(1)) \simeq \mathbb{P}_{\mathbb{A}_k^2}((x, y)). \quad (**)$$

The first model is the total affine space of the tautological quotient line; the second is the projectivization of the ideal (x, y) , which is of linear type. Its two standard charts have coordinates $k[x, y/x]$ and $k[x/y, y]$.

The same modification can separate branches. For this curve example assume that k is a field, and let

$$C = V(y^2 - x^3 - x^2) \subset \mathbb{A}_k^2.$$

On the x -chart put $t = y/x$. Removing the exceptional factor x^2 from the total-transform equation leaves

$$t^2 = x + 1, \quad x = t^2 - 1, \quad y = t(t^2 - 1).$$

Thus the strict transform is $\mathbb{A}_k^1$, and its map to C is the normalization

$$t \mapsto (t^2 - 1, t(t^2 - 1)).$$

If $\text{char}(k) \neq 2$, the two points $t = 1$ and $t = -1$ over the node record its two distinct tangent directions $y = x$ and $y = -x$. In characteristic two the same formula remains valid, but the two tangent directions and the two points coalesce scheme-theoretically; one must not read a two-branch picture literally in that case.

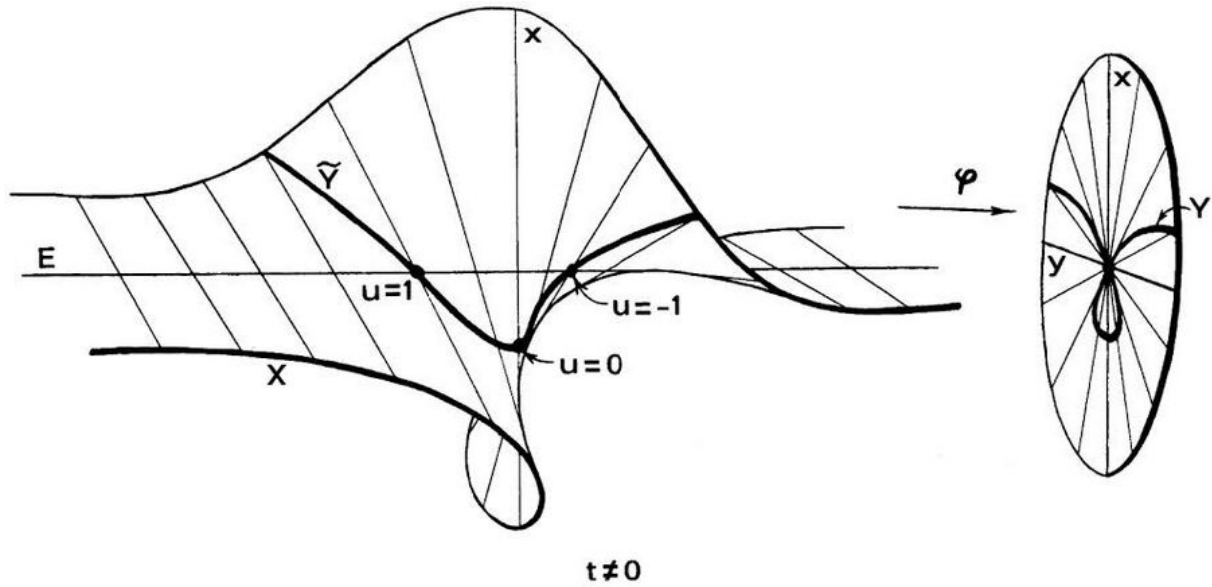


Figure 9.1.: A classical local schematic of a blowup. The morphism $\varphi: \tilde{X} \rightarrow X$ is an isomorphism away from the center y , while the exceptional divisor $E = \varphi^{-1}(y)$ records approach directions; $\tilde{Y}$ denotes the strict transform of Y . The drawing is heuristic, not a metric or global model.

Abstractly, the universal property proved below says that $b: \text{Bl}_Z(X) \rightarrow X$ is final among maps for which the inverse image of Z is an effective Cartier divisor. Its exceptional divisor is $\mathbb{P}_Z(C_Z X)$, the projectivized normal cone. These two facts explain both examples: a formerly ambiguous limiting direction becomes a point of the exceptional divisor, and the strict transform keeps distinct directions apart.

Blowups are consequently basic tools in elimination of indeterminacies and resolution of singularities. A graph closure is the first modification attached to a rational map, and resolution algorithms in characteristic zero use sequences of blowups along carefully chosen regular centers. Turning those ideas into general extension or resolution theorems requires additional finiteness, excellence, characteristic, and principalization hypotheses. No such general theorem is proved from the material here; the two precise external inputs are recorded next. This chapter instead establishes the Cartier, Rees, normal-cone, and regular-center interfaces on which those deeper arguments rely.

The term “proper” in the following preview is defined in Definition 10.1.1; no result from the properness chapter is used in the proofs below.

Theorem 9.1.1 (Guiding applications: a cited theorem and an accepted input).

- (i) *Elimination of indeterminacies (cited).* Let S be a scheme, let X and P be S -schemes with X qcqs and P proper over S , and let $U \subset X$ be a quasi-compact open subscheme. Every S -morphism $f: U \rightarrow P$ extends to an

S-morphism

$$\widehat{f}: \mathrm{Bl}_Z(X) \longrightarrow P$$

for some finitely presented closed subscheme $Z \subset X$ disjoint from U .

- (ii) *Resolution by one blowup (accepted).* Let X be of finite type over a complete noetherian local $\mathbb{Q}$ -algebra, and let $U \subset X$ be a regular open subscheme. There is a closed subscheme $Z \subset X$, disjoint from U , such that $\mathrm{Bl}_Z(X)$ is regular.

Sources and proof roadmaps. Neither statement is used to prove the elementary results in this chapter.

For (1), let Γ be the scheme-theoretic closure of the graph of f in $X \times_S P$. Since $P \rightarrow S$ is proper, the projection $\Gamma \rightarrow X$ is proper; since $P \rightarrow S$ is separated, its restriction over U is the graph and hence isomorphic to U . The theorem that a proper modification over a qcqs scheme is dominated by a blowup away from a prescribed quasi-compact open then gives a blowup $\mathrm{Bl}_Z(X) \rightarrow X$ through Γ , with Z finitely presented and disjoint from U [The25, Tag 081T]. Composing with $\Gamma \rightarrow P$ gives the asserted extension.

Statement (2) is accepted here as a heavy input; no independent reference for this precise single-blowup formulation is supplied. Its proof roadmap is to apply characteristic-zero resolution, which lowers a singularity invariant by a sequence of blowups with permissible centers outside U , and then use a principalization/composition theorem to express the resulting modification as the blowup of one ideal supported outside U . These roadmaps indicate where the stated finiteness, properness, completeness, and characteristic-zero hypotheses enter; they do not assert that every hypothesis is logically minimal. $\square$

Definition 9.1.2 (Effective virtual and effective Cartier divisors). Let X be an algebraic functor. An *effective virtual divisor* on X is a line bundle $L \in \mathrm{Line}_X$ together with a morphism

$$s: L \longrightarrow \mathcal{O}_X.$$

We write

$$\mathrm{vDiv}^+(X) = ((\mathrm{Line}_X)_{/\mathcal{O}_X})^{\mathrm{op}}.$$

If X is a scheme, the pair (L, s) is an *effective Cartier divisor*, or simply an *effective divisor*, when s is a monomorphism. Their full subcategory is denoted $\mathrm{Div}^+(X)$.

For an effective divisor, the image of s is an invertible quasi-coherent ideal $\mathcal{I} \subset \mathcal{O}_X$, and conversely every such ideal gives an effective divisor. Thus $\mathrm{Div}^+(X)$ is a poset: its order is chosen so that $D \leq D'$ precisely when the associated closed subschemes satisfy $V(D) \subseteq V(D')$.

Proposition 9.1.3 (Local equation criterion). Let $Z \hookrightarrow X$ be a closed subscheme with quasi-coherent ideal $\mathcal{I}$. The following are equivalent.

- (i) $\mathcal{I}$ is a line bundle, so Z is an effective Cartier divisor.
- (ii) There is an affine open cover $U_i = \mathrm{Spec}(A_i)$ and elements $f_i \in A_i$ such that

$$Z \cap U_i = \mathrm{Spec}(A_i/(f_i))$$

and every f_i is a non-zero-divisor in A_i .

Proof. Suppose first that $\mathcal{I}$ is invertible. After restricting to a suitable affine neighborhood $U = \mathrm{Spec}(A)$, choose an isomorphism $\mathcal{I}|_U \simeq \widetilde{A}$. The inclusion into $\mathcal{O}_U$ is multiplication by an element $f \in A$, and its injectivity says exactly that f is a non-zero-divisor. Its cokernel is $A/(f)$, hence $Z \cap U = V(f)$.

Conversely, on a chart as in (2), multiplication by f_i identifies A_i with the ideal (f_i) . Thus $\mathcal{I}|_{U_i} \simeq \mathcal{O}_{U_i}$, and local freeness of rank one glues. Compare the scheme-theoretic criteria in [The25, Tags 01WR and 01WS]. $\square$

Warning 9.1.4 (Affine does not mean principal). On $X = \operatorname{Spec}(A)$, a line bundle corresponds to an invertible, finitely generated projective A -module of rank one. It need not be isomorphic to A . Consequently an effective divisor on an affine scheme need not be globally $V(f)$, even though it has this form on a Zariski cover. No argument below will replace an affine line bundle by a free module without first passing to such a cover.

Proposition 9.1.5 (Addition, pullback, and descent). *Let X be an algebraic functor.*

(i) *Effective virtual divisors form a symmetric monoidal category under*

$$(L, s) + (L', s') = (L \otimes L', s \otimes s'), \quad 0 = (\mathcal{O}_X, 1).$$

If X is a scheme, this restricts to the ordered commutative monoid $\operatorname{Div}^+(X)$.

(ii) *Every map $f: Y \rightarrow X$ pulls effective virtual divisors back. If X, Y are schemes and D has ideal $\mathcal{I}$, then f^*D is effective exactly when*

$$\operatorname{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_X/\mathcal{I}) = 0, \quad (*)$$

where the assertion is checked on affine charts. In particular flat pullback preserves effective divisors.

(iii) *Effective virtual divisors satisfy Zariski descent. On a scheme, effective Cartier divisors form a Zariski sheaf of posets.*

Proof. Tensoring a monomorphism with an invertible module remains a monomorphism, which proves the first assertion. For the second, pull back $0 \rightarrow \mathcal{I} \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X/\mathcal{I} \rightarrow 0$. The beginning of the Tor sequence identifies the obstruction to the injectivity of $f^*\mathcal{I} \rightarrow \mathcal{O}_Y$ with the group in (*). Flatness kills it. Finally, line bundles and their maps glue, while being a monomorphism can be checked on a Zariski cover. $\square$

Example 9.1.6. Every $f \in \mathcal{O}_X(X)$ gives the virtual divisor $(\mathcal{O}_X, f)$; it is an effective Cartier divisor precisely when f is a non-zero-divisor locally on X . If A is a UFD, every invertible ideal is principal and

$$\operatorname{Div}^+(\operatorname{Spec} A) \simeq (A - \{0\})/A^\times$$

with addition given by multiplication and order given by divisibility. If A is a Dedekind domain, the effective divisors are instead the finite formal sums of nonzero prime ideals. These examples already show why a line bundle, rather than one global equation, belongs in the definition.

There is also a projective version. If an $\mathbb{N}$ -graded ring A is generated in degrees at most one and $f \in A_d$ is homogeneous, multiplication by f gives an effective virtual divisor $(\mathcal{O}(-d), f)$ on $\operatorname{Proj}(A)$. It is Cartier exactly when the fractions f/h^d are non-zero-divisors on the standard charts $D_+(h)$, for $h \in A_1$; in particular this holds if f is a non-zero-divisor in A .

Remark 9.1.7 (Optional stack-theoretic viewpoint). For a ring R , let $R//R$ be the divisibility category: its objects are elements of R , and a morphism $r \rightarrow s$ is an element t with $tr = s$. Sending r to (R, r) identifies effective virtual divisors locally with this category. Since line bundles are only locally trivial, the global statement is that vDiv^+ is the Zariski sheafification of $\mathcal{O}//\mathcal{O}$; on maximal subgroupoids this is the quotient stack $\mathcal{O}/\mathcal{O}^\times$, not merely an orbit set.

The next construction removes precisely the torsion that prevents a virtual divisor from being Cartier.

Definition 9.1.8 (Cartier locus). Let (L, s) be an effective virtual divisor on a scheme X . Locally, after trivializing L and writing $s = f$, form the localization $\mathcal{O}_X[s^{-1}]$, whose restriction to an affine chart is A_f . The *Cartier locus* of s is the closed subscheme

$$X_s^{\operatorname{Car}} = V(\ker(\mathcal{O}_X \rightarrow \mathcal{O}_X[s^{-1}])).$$

Proposition 9.1.9 (Universal property of the Cartier locus). *The scheme X_s^{Car} is the scheme-theoretic closure of the open locus $D(s)$ where s is an epimorphism. Its pulled-back divisor is an effective Cartier divisor, and it is final among X -schemes on which the pullback of (L, s) is an effective Cartier divisor.*

*For a map $f : Y \rightarrow X$, there is always a closed immersion from the Cartier locus of f^*s into $Y \times_X X_s^{\text{Car}}$. It is an isomorphism whenever pullback by f preserves the relevant localization kernel. Explicitly, on an affine chart $A \rightarrow B$ where s is represented by $a \in A$, this is equivalent to*

$$\ker(A \rightarrow A_a)B = \ker(B \rightarrow B_a).$$

Flat base change satisfies this equality.

Proof. The claim is affine-local. Write $X = \text{Spec}(A)$, $L \simeq \widetilde{A}$, and $s = f \in A$. Put $K = \ker(A \rightarrow A_f) = \{a \mid f^n a = 0 \text{ for some } n\}$. Multiplication by f on A/K is injective, so the restricted divisor is Cartier. If an A -algebra B has multiplication by f injective, then every f -power-torsion element of A maps to zero in B . Thus $A \rightarrow B$ factors uniquely through A/K . This gives finality. The same quotient is the scheme-theoretic closure of $D(f)$, because its ideal is the kernel of $A \rightarrow A_f$. The base-change assertion follows by comparing kernels before and after tensor product. $\square$

9.2. Blowups and deformation to the normal cone

Blowups as Cartier loci

We begin with the functor-of-points construction. It records the one additional datum that a blowup is meant to create: a quotient line through which a chosen equation factors.

Definition 9.2.1 (Virtual blowup and its exceptional divisor). Let X be an algebraic functor and $z : M \rightarrow \mathcal{O}_X$ a map in Mod_X . The *virtual blowup* $\text{vBl}_z(X)$ is the subfunctor of $\mathbb{P}_X(M)$ whose R -points over $x \in X(R)$ are diagrams

$$\begin{array}{ccc} M(x) & \xrightarrow{z(x)} & R \\ & \searrow q & \nearrow s \\ & L & \end{array}$$

with L an invertible rank-one R -module. If $b : \text{vBl}_z(X) \rightarrow X$ is the structure map, the universal factorization

$$b^*M \twoheadrightarrow \mathcal{O}(1) \xrightarrow{s} \mathcal{O}_{\text{vBl}_z(X)}$$

defines its *exceptional effective virtual divisor* $(\mathcal{O}(1), s)$.

For a closed subscheme $Z \hookrightarrow X$ with quasi-coherent ideal $\mathcal{I}$, write $\text{vBl}_Z(X) = \text{vBl}_{\mathcal{I} \hookrightarrow \mathcal{O}_X}(X)$.

The factorization condition cuts out a closed subfunctor of $\mathbb{P}_X(M)$. Hence the virtual blowup is schematic over X , and it is projective in the conventions of this book when M is of finite type. No finite-generation hypothesis is needed merely to define it.

Definition 9.2.2 (Blowup). When X is a scheme, the *blowup* $\text{Bl}_Z(X)$ is the Cartier locus of the exceptional virtual divisor on $\text{vBl}_Z(X)$. Its restricted divisor is the *exceptional divisor*. For a closed subscheme $Z \subset X$, we use the notation $\text{Bl}_Z(X)$.

Proposition 9.2.3 (The open part, the center, and base change). *Let $z : M \rightarrow \mathcal{O}_X$.*

(i) *There are cartesian squares*

$$\begin{array}{ccccc} \mathbb{P}_{V(z)}(M|_{V(z)}) & \longrightarrow & \mathrm{vBl}_z(X) & \longleftarrow & D(z) \\ \downarrow & & \downarrow b & & \downarrow \\ V(z) & \longrightarrow & X & \longleftarrow & D(z). \end{array}$$

If X is a scheme, $\mathrm{Bl}_z(X)$ is the closure of $D(z)$ inside $\mathrm{vBl}_z(X)$, and its exceptional divisor is the boundary over $V(z)$.

- (ii) If $f: Y \rightarrow X$ and $f^*M \rightarrow N \rightarrow \mathcal{O}_Y$ is compatible with z , there is a natural map $\mathrm{vBl}_N(Y) \rightarrow \mathrm{vBl}_z(X)$. If the quotient $q: f^*M \rightarrow N$ is an isomorphism, virtual blowup commutes with this base change. For actual blowups the same conclusion holds when f is flat.
- (iii) For a closed $Z \subset X$, virtual blowup commutes with a base change $Y \rightarrow X$ provided Y and Z are Tor-independent over X , in the sense of Definition 9.3.1 below. Actual blowup commutes with every flat base change.
- (iv) Blowing up an effective Cartier divisor does nothing: $\mathrm{Bl}_Z(X) \simeq X$. Blowing up the whole scheme gives the empty scheme.
- (v) Actual blowup depends only on the scheme-theoretic vanishing locus: for any $z: M \rightarrow \mathcal{O}_X$, the natural closed immersion

$$\mathrm{Bl}_{V(z)}(X) \longrightarrow \mathrm{Bl}_z(X)$$

is an isomorphism. The analogous assertion for virtual blowups is false.

Proof. Over $V(z)$, the map z is zero, so every quotient line is allowed; over $D(z)$, an epimorphism to $\mathcal{O}$ has a unique factorization through a quotient line. This proves the two cartesian statements. The closure description is the universal property of the Cartier locus.

The functorial statements follow by pulling quotient lines back. For a closed subscheme with ideal $\mathcal{I}$, the only possible defect is the failure of $f^*\mathcal{I} \rightarrow \mathcal{O}_Y$ to remain injective; its kernel is controlled by $\mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z)$. Flatness also commutes with the subsequent Cartier-locus construction. Finally, if $\mathcal{I}$ is already an invertible ideal, the identity of X has the required quotient line and is universal. The zero ideal has exceptional section zero, whose Cartier locus is empty.

For the last assertion, the image ideal of z gives a closed immersion of the virtual blowup at $V(z)$ into the virtual blowup at z . Both actual blowups are the scheme-theoretic closure of the same open $D(z)$, so this immersion restricts to an isomorphism on their Cartier loci. $\square$

Example 9.2.4 (A zero-divisor center). Let $f \in \mathcal{O}_X(X)$. The virtual blowup at $V(f)$ is the closed subscheme of X cut out by $\mathrm{Ann}(f)$, the ideal of f -torsion, while the actual blowup is cut out by $\bigcup_{n \geq 1} \mathrm{Ann}(f^n)$, the ideal of f -power torsion. Thus the distinction between virtual and actual blowup is already visible in rank one. For example, in $X = \mathrm{Spec} k[x, y]/(xy)$, blowing up $V(x)$ gives the component $V(y)$.

Definition 9.2.5 (Total and strict transforms). Let $Z \subset X$ be closed and $Y \rightarrow X$ a scheme over X . Its *total transform* is

$$Y^{\mathrm{tot}} = Y \times_X \mathrm{Bl}_Z(X),$$

whereas its *strict transform* is

$$Y^{\mathrm{str}} = \mathrm{Bl}_{Z \times_X Y}(Y).$$

There is a natural closed immersion $Y^{\mathrm{str}} \hookrightarrow Y^{\mathrm{tot}}$. It is an isomorphism if $Y \rightarrow X$ is flat; in general the strict transform is the scheme-theoretic closure of $(X - Z) \times_X Y$ in the total transform. These descriptions agree with [The25, Tags 080D, 080E, and 080F].

Theorem 9.2.6 (Universal property of the blowup). Let $Z \hookrightarrow X$ be a closed subscheme. For $b: \mathrm{Bl}_Z(X) \rightarrow X$:

- (i) $b^{-1}(Z)$ is the exceptional effective Cartier divisor;
- (ii) if $f : Y \rightarrow X$ has $f^{-1}(Z) \hookrightarrow Y$ an effective Cartier divisor, then there is a unique X -map

$$Y \longrightarrow \mathrm{Bl}_Z(X).$$

Thus the blowup is final among X -schemes on which the inverse image of Z is an effective Cartier divisor.

Proof. Let I be the ideal of Z . The ideal of the inverse image of Z in Y is the image of $f^*I \rightarrow \mathcal{O}_Y$. By hypothesis this image is a line bundle, so it is a quotient line of f^*I , and the inclusion into $\mathcal{O}_Y$ supplies a point of the virtual blowup. Because that last map is a monomorphism, the point factors through its Cartier locus, which is the actual blowup. Both the quotient and its inclusion are forced, so the factorization is unique. Applying this to $Y = \mathrm{Bl}_Z(X)$ identifies the exceptional divisor with $b^{-1}(Z)$. Compare [The25, Tags 0806 and 0807]. $\square$

Theorem 9.2.7 (Smooth center). *Let S be a scheme, let $X \rightarrow S$ be smooth, and let $Z \hookrightarrow X$ be a closed subscheme that is smooth over S . Then $\mathrm{Bl}_Z(X) \rightarrow S$ is smooth.*

Proof roadmap. This is a local-structure theorem rather than a formal consequence of the universal property. Smooth-locally on X , the pair (X, Z) is the pair consisting of affine space and a coordinate linear subspace, by Proposition 8.3.14. Blowup commutes with the resulting étale, hence flat, base change by Corollary 8.6.8. The coordinate model is covered by affine spaces, as the chart computation below shows, and is smooth. Descent of smoothness finishes the argument. The regularity of an immersion between smooth schemes and flat base change for blowups are recorded in [The25, Tags 067U and 0805]. $\square$

Example 9.2.8 (The origin and the cusp). Assume in this example that k is a field. (The linear-center formulas below make sense more generally, but the final use of *normalization* needs this hypothesis.) For $I = (x_1, \dots, x_n) \subset k[x_1, \dots, x_n]$, the blowup of the origin in $\mathbb{A}_k^n$ is the total affine space

$$\mathbb{A}_{\mathbb{P}_k^{n-1}}(\mathcal{O}(1)).$$

Its exceptional divisor is the zero section $\mathbb{P}_k^{n-1}$. On the chart corresponding to x_i , its coordinate ring is

$$k[x_i, x_1/x_i, \dots, \widehat{x_i/x_i}, \dots, x_n/x_i].$$

With the same quotient-line convention, blowing up the origin in projective space gives

$$\mathrm{Bl}_0(\mathbb{P}_k^n) \simeq \mathbb{P}_{\mathbb{P}_k^{n-1}}(\mathcal{O}(1) \oplus \mathcal{O}).$$

More generally, for quasi-coherent modules M, N on a scheme X , the linear-center calculations are

$$\begin{aligned} \mathrm{Bl}_{\mathbb{A}_X(N)}(\mathbb{A}_X(M \oplus N)) &\simeq \mathbb{A}_{\mathbb{P}_X(M)}(\mathcal{O}(1) \oplus N), \\ \mathrm{Bl}_{\mathbb{P}_X(N)}(\mathbb{P}_X(M \oplus N)) &\simeq \mathbb{P}_{\mathbb{P}_X(M)}(\mathcal{O}(1) \oplus N). \end{aligned}$$

For the cusp $C = V(y^2 - x^3) \subset \mathbb{A}_k^2$, the strict transform on the x -chart is $V(\tilde{y}^2 - x)$, where $\tilde{y} = y/x$. It is a copy of $\mathbb{A}_k^1$, so blowing up the singular point normalizes this cusp. All the formulas follow by writing the quotient-line coordinates on the standard charts.

Rees algebras and deformation to the normal cone

We now pass from the moduli description to graded algebras. The latter is the efficient way to calculate.

Definition 9.2.9 (Rees, extended Rees, and associated graded algebras). Let $\mathcal{I} \subset \mathcal{O}_X$ be a quasi-coherent ideal, and put $Z = V(\mathcal{I})$. Define

$$\begin{aligned} \text{Rees}_{\mathcal{I}}(\mathcal{O}_X) &= \bigoplus_{n \geq 0} \mathcal{I}^n, \\ \text{Def}_{\mathcal{I}}(\mathcal{O}_X) &= \bigoplus_{n \in \mathbb{Z}} \mathcal{I}^n, \quad \mathcal{I}^n = \mathcal{O}_X \quad (n < 0), \\ \text{gr}_{\mathcal{I}}(\mathcal{O}_X) &= \bigoplus_{n \geq 0} \mathcal{I}^n / \mathcal{I}^{n+1}. \end{aligned}$$

The first is $\mathbb{N}$ -graded, the second is $\mathbb{Z}$ -graded, and the last is an $\mathbb{N}$ -graded quasi-coherent algebra over Z . Multiplication is induced by $\mathcal{I}^m \mathcal{I}^n \subseteq \mathcal{I}^{m+n}$. On an affine open $U = \text{Spec}(R)$, the middle algebra is the sheafification of

$$\text{Def}_I(R) = \bigoplus_{n \in \mathbb{Z}} I^n t^{-n}$$

from Definition 8.5.15. Thus the element $t = 1 \in I^{-1}$ has degree -1 , and the inclusion

$$\mathcal{O}_X[t] = \text{Def}_{\mathcal{I}}(\mathcal{O}_X)_{\leq 0} \hookrightarrow \text{Def}_{\mathcal{I}}(\mathcal{O}_X)$$

defines a $\mathbb{G}_m$ -equivariant structure morphism $\text{Def}_Z(X) \rightarrow \mathbb{A}_X^1$.

Definition 9.2.10 (Deformation space and normal cone). The *deformation space* and *normal cone* are

$$\text{Def}_Z(X) = \underline{\text{Spec}}_X \text{Def}_{\mathcal{I}}(\mathcal{O}_X), \quad C_Z X = \underline{\text{Spec}}_Z \text{gr}_{\mathcal{I}}(\mathcal{O}_X).$$

The grading gives a $\mathbb{G}_m$ -action on the deformation space and an $\mathbb{A}^1$ -action on the normal cone. The degree-one quotient $\mathcal{I} / \mathcal{I}^2$ is the conormal module in the sense of Definition 8.3.5. Since the associated graded algebra is generated in degree one, there is a canonical closed immersion

$$C_Z X \hookrightarrow \mathbb{A}_Z(\mathcal{I} / \mathcal{I}^2).$$

Here $\mathbb{A}^1$ acts through multiplication, not through the additive group $\mathbb{G}_a$. More generally, a *cone* over an algebraic functor T is an affine T -functor with such a multiplicative $\mathbb{A}^1$ -action whose graded algebra is generated in degree one. If $C = \underline{\text{Spec}}_T(A)$, its projectivization is $\mathbb{P}_T(C) = \underline{\text{Proj}}_T(A)$. Every affine space $\mathbb{A}_T(M)$ is a cone, and a multiplicative $\mathbb{A}^1$ -equivariant affine functor is a cone exactly when it admits an equivariant closed immersion into some $\mathbb{A}_T(M)$.

Example 9.2.11 (The tangent cone of a node). Let

$$R = k[x, y] / (y^2 - x^3 - x^2), \quad I = (x, y).$$

Then $I^n = (x^n, x^{n-1}y)$, and the initial relation is $y^2 - x^2$. Hence

$$\text{gr}_I(R) \simeq k[x, y] / (y^2 - x^2),$$

with x, y in degree one. The normal cone at the node is therefore the two tangent lines $V(y^2 - x^2) \subset \mathbb{A}_k^2$; in characteristic two these coincide and retain their doubled scheme structure.

Theorem 9.2.12 (Deformation to the normal cone). The deformation space has general fiber X and special fiber $C_Z X$. More precisely, there are $\mathbb{G}_m$ -equivariant cartesian squares

$$\begin{array}{ccccc} C_Z X & \longrightarrow & \text{Def}_Z(X) & \longleftarrow & X \times \mathbb{G}_m \\ \downarrow & & \downarrow & & \downarrow \\ \{0\} & \longrightarrow & \mathbb{A}^1 & \longleftarrow & \mathbb{G}_m. \end{array}$$

Proof. This can be checked on $X = \operatorname{Spec}(A)$, $Z = V(I)$. Inverting the parameter $t = 1 \in I^{-1}$ of degree -1 identifies the extended Rees algebra with $A[t, t^{-1}]$, giving $X \times_{\mathbb{G}_m}$. Quotienting by that parameter has degree- n piece I^n/I^{n+1} , giving $\operatorname{gr}_I(A)$. Both identifications respect the grading and hence the group action. $\square$

Theorem 9.2.13 (Rees description of blowups). *Let $Z \hookrightarrow X$ have quasi-coherent ideal I . Then*

$$\mathbf{vBl}_Z(X) = \mathbb{P}_X(I), \quad \operatorname{Bl}_Z(X) = \underline{\operatorname{Proj}}_X \operatorname{Rees}_I(\mathcal{O}_X). \quad (+)$$

If $X = \operatorname{Spec}(A)$ and I is generated by $a_1, \dots, a_r$, the blowup is covered by

$$\operatorname{Spec} A[I/a_i], \quad A[I/a_i] = A[b/a_i \mid b \in I]$$

for $1 \leq i \leq r$. If I is of finite type, the blowup is projective over X ; without that hypothesis, formula (+) still defines the blowup but the morphism need not be quasi-compact.

Equivalently, a quotient line $I(x) \twoheadrightarrow L$ in the virtual blowup belongs to the actual blowup precisely when, for every $n \geq 2$, the quotient $I(x)^{\otimes n} \twoheadrightarrow L^{\otimes n}$ factors through the image $I^n(x)$.

Proof roadmap. For any base ring $R \rightarrow S$, the multiplication map $I \otimes_R S \rightarrow S$ factors through every quotient line $q: I \otimes_R S \twoheadrightarrow L$. Indeed, locally trivialize $L \simeq S$. Since q is surjective, choose finitely many $a_j \in I$ and $s_j \in S$ such that $1 = \sum_j s_j q(a_j)$, and put $u = \sum_j a_j s_j \in S$. For $b \in I$, S -linearity and the identity $a_j \otimes b = b \otimes a_j$ give

$$uq(b) = \sum_j s_j a_j q(b) = b \sum_j s_j q(a_j) = b.$$

Thus multiplication factors through q by the map $L \simeq S \rightarrow S$ sending 1 to u . The factor is unique because q is an epimorphism, so these local factors glue. This identifies the virtual blowup with $\mathbb{P}(I)$. A quotient line $I \otimes S \twoheadrightarrow L$ lies in the Cartier locus exactly when its tensor powers respect the images $I^{\otimes n} \rightarrow I^n$. By the functorial characterization of relative Proj , these compatibility conditions are precisely the homogeneous relations defining $\operatorname{Proj}(\bigoplus I^n)$. Finally, the standard opens $D_+(a_i)$ have degree-zero coordinate rings $A[I/a_i]$.

This proves the comparison; the same affine charts, flat base change, universal property, and exceptional divisor are recorded respectively in [The25, Tags 0804, 0805, 0806, and 0807]. $\square$

Corollary 9.2.14 (Exceptional divisor and normal cone). *The exceptional divisor of $\operatorname{Bl}_Z(X)$ is*

$$\mathbb{P}_Z(\mathcal{C}_Z X) = \underline{\operatorname{Proj}}_Z \operatorname{gr}_I(\mathcal{O}_X).$$

In particular, both squares in

$$\begin{array}{ccccc} \mathbb{P}_Z(\mathcal{C}_Z X) & \hookrightarrow & \operatorname{Bl}_Z(X) & \longleftarrow & X - Z \\ \downarrow & & \downarrow & & \downarrow \\ Z & \hookrightarrow & X & \longleftarrow & X - Z \end{array}$$

are cartesian.

Proof. Pulling the Rees algebra back to Z replaces each I^n by I^n/I^{n+1} , hence gives the associated graded algebra. Relative Proj commutes with this base change. The right square follows because the ideal becomes the unit ideal off Z . $\square$

Theorem 9.2.15 (Blowup and deformation space). *Here $\text{Def}_Z(Z) \hookrightarrow \text{Def}_Z(X)$ is the closed deformation subspace induced degreewise by the quotient*

$$\text{Def}_I(\mathcal{O}_X) \twoheadrightarrow \text{Def}_0(\mathcal{O}_Z).$$

The $\mathbb{G}_m$ -action on its complement is free, and the quotient below is taken in Zariski sheaves. There are canonical descriptions

$$\text{Bl}_Z(X) \simeq (\text{Def}_Z(X) - \text{Def}_Z(Z))/\mathbb{G}_m$$

as a Zariski-sheaf quotient over X , and

$$\text{Def}_Z(X) \simeq \text{Bl}_{Z \times \{0\}}(X \times \mathbb{A}^1) - \text{Bl}_{Z \times \{0\}}(X \times \{0\}).$$

Proof. Work on $X = \text{Spec}(A)$, with center I . The extended Rees algebra is

$$B = A[t, I/t] \subset A[t, t^{-1}],$$

where t has degree -1 and a/t has degree 1 for $a \in I$. The complement of $\text{Def}_Z(Z)$ is covered by the invariant opens $D(a/t)$. On such an open the degree-zero subring is

$$(B[(a/t)^{-1}])_0 = A[I/a].$$

The action is free there because the invertible homogeneous element a/t trivializes the grading locally. Taking these degree-zero rings and gluing therefore gives the affine cover of $\text{Bl}_Z(X)$ from Theorem 9.2.13, which proves the first formula as a Zariski-sheaf quotient.

For the second, form the ordinary Rees algebra of the ideal $J = I + (t) \subset A[t]$. The standard chart selected by the generator t has coordinate ring

$$A[t][J/t] = A[t, I/t] = B.$$

On every other standard chart, the strict transform of the zero fiber is cut out by the degree-zero ratio t/a ; deleting that strict transform is the open where t/a is invertible, and this open identifies with the corresponding localization of the (t) -chart. Consequently the complements of the strict transform glue to $\text{Spec}(B)$. The construction is compatible on affine overlaps, so it globalizes to the asserted isomorphism. $\square$

9.3. Tor-independence, regular immersions, and lci maps

Tor independence and Koszul regularity

Flat base change is often stronger than necessary. The correct weakening for intersections and blowups is the vanishing of higher Tor.

Definition 9.3.1 (Tor-independence). Let R be a ring. Two R -modules M, N are *Tor-independent* if

$$\text{Tor}_i^R(M, N) = 0 \quad (i > 0).$$

For X -schemes Y, Z , and quasi-coherent modules $\mathcal{M}$ on Y , $\mathcal{N}$ on Z , the pair $(\mathcal{M}, \mathcal{N})$ is Tor-independent over X if this condition holds on every compatible triple of affine charts. The schemes Y, Z are Tor-independent when their structure sheaves are. A cartesian square of schemes is *Tor-independent* when its two arms are Tor-independent over the lower-right corner.

Equivalently, the ordinary fiber product of a Tor-independent pair already computes their derived fiber product; there are no hidden higher intersection terms.

Proposition 9.3.2 (Tor calculus). *Let R be a ring and M an R -module.*

- (i) *If $N' \subset N$ and M is Tor-independent from N/N' , then M is Tor-independent from N if and only if it is Tor-independent from N' .*
- (ii) *Let $R \rightarrow S$ be a ring map and N an S -module. If M and S are Tor-independent over R , then M, N are Tor-independent over R if and only if $M \otimes_R S, N$ are Tor-independent over S .*
- (iii) *Tor-independent squares satisfy the pasting law: if the left square in a horizontal composite is Tor-independent, then the right square is Tor-independent exactly when the outer rectangle is.*

The same assertions hold affine-locally for schemes.

Proof. Apply the long exact Tor sequence to $0 \rightarrow N' \rightarrow N \rightarrow N/N' \rightarrow 0$ for (1). Under the hypothesis in (2), a flat resolution of M remains a resolution after tensoring with S , and the change-of-rings spectral sequence collapses to

$$\mathrm{Tor}_i^R(M, N) \simeq \mathrm{Tor}_i^S(M \otimes_R S, N).$$

Part (3) is (2) applied successively to the two base changes. Localization preserves all the displayed Tor groups, so the argument globalizes. $\square$

Definition 9.3.3 (Regular map). Let $E \in \mathrm{Mod}_X$, and let $s: E \rightarrow \mathcal{O}_X$. View s as a section of $\mathbb{A}_X(E) = \mathrm{Spec}_X \mathrm{Sym}(E)$. The map s is *regular* when the square

$$\begin{array}{ccc} V(s) & \longrightarrow & X \\ \downarrow & & \downarrow s \\ X & \xrightarrow{0} & \mathbb{A}_X(E) \end{array}$$

is Tor-independent.

Theorem 9.3.4 (Koszul test). *Assume E is flat over X . The augmentation of $\mathrm{Sym}(E)$ has a canonical flat Koszul resolution*

$$\cdots \longrightarrow \bigwedge^2 E \otimes \mathrm{Sym}(E) \longrightarrow E \otimes \mathrm{Sym}(E) \longrightarrow \mathrm{Sym}(E) \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

After extension of scalars along $\mathrm{Sym}(E) \xrightarrow{s} \mathcal{O}_X$, this becomes the Koszul complex $K(E, s)$. The map s is regular if and only if $H_i K(E, s) = 0$ for all $i > 0$.

If $E = \mathcal{O}_X^{\oplus n}$ and $s = (f_1, \dots, f_n)$, an ordinary regular sequence

$$f_i \text{ a non-zero-divisor on } \mathcal{O}_X / (f_1, \dots, f_{i-1})$$

is regular in this sense. Conversely, over a noetherian local ring $(R, \mathfrak{m})$, a Koszul-regular tuple contained in $\mathfrak{m}$ is an ordinary regular sequence.

Proof roadmap. The differential is the graded derivation determined by multiplication $E \otimes \mathrm{Sym}(E) \rightarrow \mathrm{Sym}(E)$; this formulation does not require E to be dualizable. For finite free E , tensoring the one-variable complexes $0 \rightarrow R[t] \xrightarrow{t} R[t] \rightarrow R \rightarrow 0$ proves exactness. For an arbitrary flat module, Lazard's theorem writes it as a filtered colimit of finite free modules; symmetric and exterior powers commute with that colimit, and filtered colimits of modules are exact. This yields the stated flat resolution. Its base change along s computes exactly the Tor groups in the defining square, which proves the criterion.

For an ordinary regular sequence, tensor the one-variable complexes in order; injectivity at every stage makes the tensor product exact in positive degrees. The noetherian converse uses Nakayama's lemma and induction on the length. That converse is genuinely noetherian and is not asserted for an arbitrary ring. The ordinary implication and the noetherian equivalence are recorded in [The25, Tags 062F and 063L]. $\square$

Proposition 9.3.5 (Rank one recovers Cartier divisors). *Let $s : L \rightarrow \mathcal{O}_X$ be an effective virtual divisor. Then s is a regular map if and only if it is an effective Cartier divisor.*

Proof. Zariski-locally $L \simeq \mathcal{O}_X$, and the Koszul complex is

$$0 \longrightarrow \mathcal{O}_X \xrightarrow{s} \mathcal{O}_X \longrightarrow \mathcal{O}_X/(s) \longrightarrow 0.$$

Its only possible positive homology is $\ker(s)$. Thus Koszul exactness is exactly injectivity of s . $\square$

Regular closed immersions

Definition 9.3.6 (Regular closed immersion). A closed immersion $i : Z \hookrightarrow X$ is *regular* if X has an open cover U on which $Z \cap U = V(s)$ for a regular map $s : E \rightarrow \mathcal{O}_U$ with E a finite-rank vector bundle.

Warning 9.3.7 (Regular immersion versus regular scheme). The adjective “regular” describes the equations of the immersion, not the singularity type of its source. Even inside a regular scheme, a regular closed immersion can have a singular source. In the locally noetherian case, the definition is equivalent to asking locally that the ideal be generated by an ordinary regular sequence. Outside the noetherian setting, that familiar criterion is sufficient but need not be necessary; the Tor/Koszul formulation is the stable one. Compare [The25, Tags 063J and 063L]. In the terminology of the Stacks Project, our non-noetherian notion is a *Koszul-regular immersion*; this qualifier matters outside the noetherian setting.

Theorem 9.3.8 (Geometry of regular immersions). *Let $i : Z \hookrightarrow X$ be a regular closed immersion with ideal $\mathcal{I}$.*

(i) *The conormal module $\mathcal{I} / \mathcal{I}^2$ is a vector bundle, and multiplication induces an isomorphism*

$$\mathrm{Sym}_{\mathcal{O}_Z}(\mathcal{I} / \mathcal{I}^2) \xrightarrow{\sim} \mathrm{gr}_{\mathcal{I}}(\mathcal{O}_X).$$

Consequently

$$C_Z X \simeq \mathbb{A}_Z(\mathcal{I} / \mathcal{I}^2).$$

The rank of this conormal bundle is locally constant on $|Z|$; it is called the codimension of the regular immersion.

(ii) *Composites of regular closed immersions are regular.*

(iii) *In a Tor-independent cartesian square, base change of a regular closed immersion is regular. In particular this holds after flat base change.*

(iv) *Effective Cartier divisors are precisely the regular closed immersions of codimension one.*

Proof roadmap. Locally write the ideal as the image of a regular map from a finite free module. Koszul exactness first identifies $\mathcal{I} / \mathcal{I}^2$ with the corresponding free module after restriction to Z . Filtering the Koszul complex by powers of $\mathcal{I}$ then proves that the symmetric algebra maps isomorphically to the associated graded algebra. This is the substantial step. Composition and Tor-independent base change follow from the pasting law for the defining Tor-independent squares. The rank-one assertion is Proposition 9.3.5. Compare the normal-cone, flat-base-change, and composition statements in [The25, Tags 0636, 067P, and 067Q]. $\square$

Theorem 9.3.9 (Blowing up a regular immersion). *Let $E \in \mathrm{Mod}_X$ be flat and $s : E \rightarrow \mathcal{O}_X$ regular. Then*

$$\mathrm{Bl}_{V(s)}(X) = \mathrm{vBl}_s(X).$$

In particular, if $Z \hookrightarrow X$ is a regular closed immersion with ideal $\mathcal{I}$, write $E_{Z/X} = b^{-1}(Z)$ for the exceptional divisor. Then

$$\mathrm{Bl}_Z(X) = \mathbb{P}_X(\mathcal{I}), \quad E_{Z/X} = \mathbb{P}_Z(\mathcal{I} / \mathcal{I}^2).$$

Proof roadmap. Koszul exactness is preserved after passage to every quotient-line chart. On such a chart, the exceptional section is therefore a non-zero-divisor, so its Cartier locus is the whole virtual blowup. The second statement is obtained by choosing local regular presentations and applying the first; the formula for the exceptional divisor follows from Corollary 9.2.14 and Theorem 9.3.8. The corresponding regular-sequence chart calculation is [The25, Tag 0BIQ]. $\square$

Proposition 9.3.10 (Regular immersions detected by smoothness). *Let S be a scheme and let $i: Z \hookrightarrow X$ be a closed immersion of S -schemes. Assume that $X \rightarrow S$ is locally of finite presentation and that $Z \rightarrow S$ is smooth. Then i is a regular closed immersion if and only if some open neighborhood of Z in X is smooth over S .*

Proof roadmap. The assertion is local along Z . If the neighborhood is smooth, the local structure theorem for a closed immersion between smooth S -schemes identifies the ideal with a regular sequence. Conversely, the nontrivial input is the local criterion for flatness along a regular immersion: regularity of $Z \hookrightarrow X$, together with smoothness of Z/S , makes X/S flat near Z . The conormal sequence and the cotangent criterion then show that X/S is smooth along Z , and openness of the smooth locus supplies the required neighborhood. The finite-presentation hypothesis on $X \rightarrow S$ is used in this package. The two nonformal inputs are the flat-neighborhood criterion and regularity of an immersion between smooth schemes; see [The25, Tags 063V and 067U]. $\square$

lci and syntomic morphisms

Definition 9.3.11 (Local complete intersections and syntomic maps). Let $f: Y \rightarrow X$ be a morphism of schemes.

- (i) It is a *local complete intersection morphism*, abbreviated *lci*, if every $y \in Y$ has affine neighborhoods $y \in V \subset Y$ and $U \subset X$, with $f(V) \subset U$, for which $f|_V$ factors as

$$V \hookrightarrow \mathbb{A}_U^n \rightarrow U,$$

the first arrow being a regular closed immersion and $n < \infty$.

- (ii) It is *syntomic* if it is both flat and lci.

In particular, an lci morphism is locally of finite presentation. No separatedness or quasi-compactness assumption is part of either definition.

Theorem 9.3.12 (Permanence and cotangent criterion).

- (i) The lci property is local on source and target, stable under composition, and stable under Tor-independent base change.
- (ii) Syntomic morphisms are local on source and target, stable under composition, and stable under arbitrary base change.
- (iii) A closed immersion is lci if and only if it is regular.
- (iv) With the homological convention for the cotangent complex used in the smoothness chapter, an lci morphism f is locally of finite presentation and $\mathbb{L}_f$ is perfect of Tor-amplitude contained in $[0, 1]$. Conversely, this condition characterizes lci morphisms when f is a perfect morphism between locally noetherian schemes; here “perfect” means pseudo-coherent and of finite Tor-dimension.

Proof roadmap. Locality is built into the factorization. For base change, regular immersions use Theorem 9.3.8; the projection from affine space is flat, so an arbitrary base change of a syntomic map is automatically Tor-independent. Composition is proved by embedding the two successive factorizations into one affine space and applying composition of regular immersions. A closed immersion has no affine-space part after its graph factorization, giving (3).

For (4), the regular-immersion term of the transitivity triangle is a finite vector bundle placed in homological degree 1, while the smooth affine projection contributes a vector bundle in degree 0. This proves the forward implication. Under the stated perfectness and noetherian hypotheses, the converse reduces the amplitude condition to regularity of the ideal in a local polynomial presentation; that is the heavy input. Here the transitivity convention is the one fixed in Theorem 8.3.6. See [The25, Tags 069L, 08SL, and 0FK2]. $\square$

Definition 9.3.13 (Global complete intersection). A ring map $R \rightarrow B$ is a *global complete intersection* if

$$B \simeq R[x_1, \dots, x_n]/(f_1, \dots, f_m)$$

and every fiber has dimension either $n - m$ or $-\infty$ (the latter meaning that the fiber is empty).

Theorem 9.3.14 (Fiber-dimension test for syntomicity). A morphism $f : Y \rightarrow X$ is syntomic if and only if, for every $y \in Y$, there are affine neighborhoods $V \ni y$ and $U \ni f(y)$, with $f(V) \subset U$, on which the induced ring map is a global complete intersection in this sense.

Proof roadmap. For a syntomic presentation, Koszul regularity and flatness force the stated fiber dimension. The reverse implication over an arbitrary base ring is not an immediate application of noetherian miracle flatness. The standard proof first uses absolute noetherian approximation to descend the presentation and the finite dimension data to a finite-type $\mathbb{Z}$ -algebra. At that level one proves successively that $f_1, \dots, f_m$ is a regular sequence and that every intermediate quotient is flat over the base; these properties then return under the filtered colimit. The approximation, regular-sequence, and syntomic conclusions are [The25, Tags 00SU, 00SV, and 00SW]. $\square$

Corollary 9.3.15 (Smooth maps are syntomic). Every smooth morphism is syntomic. More generally, in a triangle

$$\begin{array}{ccc} Z & \xrightarrow{g} & Y \\ & \searrow h & \swarrow f \\ & X & \end{array}$$

if f is smooth and h is lci, then g is lci. In particular, every morphism between smooth X -schemes is lci.

Proof. A smooth morphism is flat by Corollary 8.6.8 and is syntomic by the standard local form Proposition 8.3.14. For the triangle, the graph

$$\Gamma_g : Z \longrightarrow Z \times_X Y$$

is a section of the smooth projection $Z \times_X Y \rightarrow Z$, hence a regular immersion; after restricting to a neighborhood of each point it is a regular closed immersion. The other projection $Z \times_X Y \rightarrow Y$ is the base change of the lci map h . Their composite is g , so permanence under base change and composition proves that g is lci without invoking the restricted converse in part (4) of Theorem 9.3.12. Compare [The25, Tags 01UC, 01VD, and 067R]. $\square$

Remark 9.3.16 (Maps between regular schemes). In the locally noetherian setting, one further consequence is that a locally finite type morphism between regular schemes is lci. On affine neighborhoods it factors as a closed immersion into a finite-dimensional affine space over the target. The ambient and source are locally noetherian regular, and the local complete-intersection criterion for a quotient of regular local rings makes the immersion regular. The latter commutative-algebra input is [The25, Tag 00NR].

9.4. Cartier divisors and rational functions

Effective divisors only record zeros. To subtract them and hence record poles as well, we use the sheaf of rational functions. Recall Remark 7.4.11: the inclusion $\mathcal{O}_X \hookrightarrow \mathcal{K}_X$ is the maximal localization of $\mathcal{O}_X$ that remains a monomorphism of sheaves. Set

$$K(X) = \Gamma(X, \mathcal{K}_X).$$

Warning 9.4.1 (Two features of $\mathcal{K}_X$). Even for affine $X = \operatorname{Spec}(A)$, the ring $K(X)$ can strictly contain the total quotient ring of A ; equality holds when X is noetherian. Also, $\mathcal{K}_X$ need not be quasi-coherent, even on a noetherian scheme. Thus Cartier divisors live naturally among sheaves of $\mathcal{O}_X$ -modules, even though their line bundles remain quasi-coherent; no quasi-coherence argument may be applied to $\mathcal{K}_X$ without an additional hypothesis. See the definition and the noetherian affine calculation in [The25, Tags 01X2 and 02OV].

Definition 9.4.2 (Virtual and Cartier divisors). Let X be a scheme. A *virtual divisor* is a pair (L, s) , with $L \in \operatorname{Line}_X$ and an $\mathcal{O}_X$ -linear map

$$s: L \longrightarrow \mathcal{K}_X.$$

It is a *Cartier divisor*, or simply a *divisor*, if s is a monomorphism. All maps here are taken in sheaves of $\mathcal{O}_X$ -modules; explicitly,

$$\operatorname{vDiv}(X) = ((\operatorname{Line}_X)_{/\mathcal{K}_X})^{\operatorname{op}},$$

and $\operatorname{Div}(X)$ is its full subcategory on monomorphisms.

Tensor product followed by multiplication in $\mathcal{K}_X$ defines addition:

$$(L, s) + (M, t) = (L \otimes M, st), \quad 0 = (\mathcal{O}_X, 1).$$

An effective virtual divisor maps into this category through $\mathcal{O}_X \hookrightarrow \mathcal{K}_X$.

The four related notions can be read from the following table; the row is the target of the line-bundle map and the column records whether it is required to be injective:

target	arbitrary map	monomorphism
$\mathcal{O}_X$	$\operatorname{vDiv}^+(X)$	$\operatorname{Div}^+(X)$
$\mathcal{K}_X$	$\operatorname{vDiv}(X)$	$\operatorname{Div}(X)$.

Proposition 9.4.3 (Three tests for a Cartier divisor). For a virtual divisor (L, s) , the following are equivalent.

- (i) $s: L \rightarrow \mathcal{K}_X$ is a monomorphism.
- (ii) (L, s) is invertible for the tensor product on $\operatorname{vDiv}(X)$.
- (iii) The induced map

$$\mathcal{K}_X \otimes_{\mathcal{O}_X} L \xrightarrow{\sim} \mathcal{K}_X$$

is an isomorphism.

When these conditions hold, the inverse divisor is represented by

$$L^\vee \longrightarrow \mathcal{K}_X \otimes L^\vee \xrightarrow{(s^\vee)^{-1}} \mathcal{K}_X.$$

Consequently $\operatorname{Div}(X)$ is an ordered abelian group, $\operatorname{Div}^+(X)$ is its positive cone, and

$$\operatorname{Div}^+(X) = \operatorname{Div}(X) \cap \operatorname{vDiv}^+(X).$$

Proof. The assertion is local on X , so trivialize L . The map is then multiplication by a local section a/t of the total quotient sheaf, with t a non-zero-divisor. It is injective on $\mathcal{O}_X$ precisely when a is also a non-zero-divisor. In that case a/t is a unit after passing to $\mathcal{K}_X$, which proves (1) $\Rightarrow$ (3). Condition (3) constructs the displayed inverse using the dual line, so it implies (2). Tensoring an inverse identity back to $L \rightarrow \mathcal{K}_X$ shows that the latter cannot have a kernel, proving (2) $\Rightarrow$ (1). The remaining statements follow from the same inverse and from the fact that the categories of genuine divisors are posets. $\square$

Proposition 9.4.4 (Pullback and group completion). *Let $f : Y \rightarrow X$ be a map of schemes.*

(i) *A pullback $f^* : \text{vDiv}(X) \rightarrow \text{vDiv}(Y)$ is defined whenever the localization map*

$$\mathcal{O}_Y \longrightarrow f^*\mathcal{K}_X$$

is a monomorphism. Under this hypothesis $f^\mathcal{K}_X$ maps to the maximal localization $\mathcal{K}_Y$, and Cartier divisors pull back to Cartier divisors. Flat f always satisfies the hypothesis.*

(ii) *On the open subsets of X , the sheaf of abelian groups Div is the sheafwise group completion of the sheaf of commutative monoids Div^+ .*

Proof. The first assertion follows from maximality of $\mathcal{K}_Y$ and Proposition 9.4.3; flat pullback preserves the monomorphism $\mathcal{O}_X \hookrightarrow \mathcal{K}_X$. For the second, trivialize the line bundle and clear a denominator locally. Every Cartier divisor then becomes a difference of two effective ones. Any monoid map out of effective divisors therefore extends locally and uniquely to Cartier divisors, and uniqueness makes the extensions glue. $\square$

Definition 9.4.5 (Principal virtual divisor). For $f \in K(X)$, its *divisor* is the virtual divisor

$$\text{div}(f) = (\mathcal{O}_X \xrightarrow{1 \mapsto f} \mathcal{K}_X).$$

A virtual divisor is *principal* if it is isomorphic to $\text{div}(f)$ for some rational function f . The particular virtual divisor $\text{div}(f)$ is Cartier exactly when $f \in K(X)^\times$.

Proposition 9.4.6 (Zeros, poles, and local equations). *Let X be a scheme and $f, g \in K(X)$.*

(i) *There are canonical identities*

$$\text{div}(1) = 0, \quad \text{div}(fg) = \text{div}(f) + \text{div}(g).$$

(ii) *$\text{div}(f)$ is an effective virtual divisor if and only if $f \in \mathcal{O}_X(X)$.*

(iii) *$\text{div}(f)$ is Cartier if and only if $f \in K(X)^\times$.*

(iv) *Local equations identify the group of Cartier divisors with*

$$\text{Div}(X) \simeq \Gamma(X, \mathcal{K}_X^\times / \mathcal{O}_X^\times). \quad (\dagger)$$

Proof. Multiplication in $\mathcal{K}_X$ gives (1). An effective virtual presentation of $\text{div}(f)$ is precisely a factorization of $\mathcal{O}_X \rightarrow \mathcal{K}_X$ through $\mathcal{O}_X$; since $\mathcal{O}_X$ is a subsheaf, this is equivalent to a global regular section, proving (2). Part (3) is Proposition 9.4.3 applied to the trivial line.

For (4), trivialize the line of a Cartier divisor on an open cover. The inclusion into $\mathcal{K}_X$ is then multiplication by a unit of $\mathcal{K}_X$. Changing a trivialization multiplies that unit by a unit of $\mathcal{O}_X$, so the local functions define a section of the quotient sheaf. Conversely such local representatives glue their trivial lines by the transition units. The two constructions are inverse. $\square$

Example 9.4.7. On $\mathbb{P}_k^1$, with affine coordinate $t = x/y$,

$$\operatorname{div}(t^m) = m[0] - m[\infty], \quad m \in \mathbb{Z}.$$

This equality is Cartier-theoretic: on the $y \neq 0$ chart the local equation is t^m , while on the $x \neq 0$ chart it is $(t^{-1})^{-m}$. Their ratio is a regular unit on the overlap.

For comparison with the effective examples, if R is a UFD with fraction field K , then

$$\operatorname{Div}(\operatorname{Spec} R) \simeq K^\times / R^\times$$

as an ordered abelian group, with order induced by divisibility. If R is Dedekind, this is the free abelian group on the nonzero prime ideals; the positive cone recovers effective divisors.

Definition 9.4.8 (Fractional ideal). A *fractional ideal* on X is a quasi-coherent $\mathcal{O}_X$ -submodule $\mathcal{I} \subset \mathcal{K}_X$, where the inclusion is taken in the category of sheaves. Let $\mathcal{I}^n$ be the image of $\mathcal{I}^{\otimes n} \rightarrow \mathcal{K}_X$, and define

$$\operatorname{Bl}_{\mathcal{I}}(X) = \operatorname{Proj}_X \bigoplus_{n \geq 0} \mathcal{I}^n.$$

Proposition 9.4.9 (Universal property for a fractional ideal). *The fractional-ideal blowup is final among maps $f: Y \rightarrow X$ for which*

- (i) $\mathcal{O}_Y \rightarrow f^*\mathcal{K}_X$ is a monomorphism, so rational functions pull back; and
- (ii) the image of $f^*\mathcal{I}$ in $\mathcal{K}_Y$ is an invertible fractional ideal, equivalently a Cartier divisor.

Proof roadmap. The image line gives a quotient of $f^*\mathcal{I}$, and multiplication in $\mathcal{K}_Y$ makes all tensor powers factor through $f^*\mathcal{I}^n$. The universal property of relative Proj gives the unique map to the displayed blowup. Conversely the tautological quotient on that Proj is a Cartier fractional ideal. This is exactly the Rees argument of Theorem 9.2.13, now performed inside $\mathcal{K}_X$. $\square$

9.5. Cycles and Weil divisors

Cartier divisors remember local equations. Weil divisors retain only the codimension-one components and their integer multiplicities. The latter description is simpler, but it behaves well only under hypotheses that rule out pathologies in codimension one.

Definition 9.5.1 (Locally finite cycle). Let T be a topological space and A a commutative monoid. An A -valued *cycle* on T is a formal sum

$$\alpha = \sum_V a_V [V], \quad a_V \in A,$$

over irreducible closed subsets $V \subset T$, subject to local finiteness: every point has a neighborhood meeting only finitely many V for which $a_V \neq 0$. Denote the monoid of cycles by $Z_*(T, A)$, and its submonoid supported in codimension n by $Z^n(T, A)$, where $n \in \mathbb{N} \cup \{\infty\}$.

If T is sober, an irreducible closed subset may be replaced by its generic point. If T is quasi-compact, local finiteness forces the sum to be finite. Restriction sends $[V]$ to $[V \cap U]$, or to zero when the intersection is empty; restriction and gluing preserve local finiteness, so cycles form a sheaf on T . Codimension is the one fixed in Definition 7.5.2; each irreducible support has a unique codimension, which gives the grading below. If A is an abelian group or an ordered monoid, the same structure is inherited coefficientwise. When T is quasi-compact, one also has the finite direct-sum decomposition

$$Z_*(T, A) = \bigoplus_{n \in \mathbb{N} \cup \{\infty\}} Z^n(T, A).$$

Definition 9.5.2 (Weil divisor). A *Weil divisor* on a scheme X is an element of $Z^1(|X|, \mathbb{Z})$. It is *effective* if all coefficients are nonnegative, that is, if it lies in $Z^1(|X|, \mathbb{N})$.

No noetherian assumption is needed for the definition, but it enters as soon as multiplicities are assigned.

Definition 9.5.3 (Fundamental cycle). Let X be locally noetherian. Its codimension-zero points are locally finite, and their local rings are artinian. The *fundamental cycle* is

$$[X] = \sum_{\text{codim}(x, X)=0} \text{length}_{\mathcal{O}_{X,x}}(\mathcal{O}_{X,x})[x].$$

For a closed subscheme $Z \subset X$, its fundamental cycle $[Z]$ is defined by the same formula on Z and then viewed as a cycle on X .

The lengths record nilpotent thickness. Thus a reduced irreducible closed subscheme contributes coefficient one at its generic point, while a doubled component contributes coefficient two when its generic local ring has length two.

Construction 9.5.4 (Multiplicity of a Cartier divisor). Let X be locally noetherian and let x have codimension one. The local ring $R = \mathcal{O}_{X,x}$ has dimension at most one. If $f \in R$ is a non-zero-divisor, then $R/(f)$ is artinian and

$$f \mapsto \text{length}_R(R/(f))$$

is additive under multiplication. Indeed, for non-zero-divisors f, g there is an exact sequence

$$0 \longrightarrow R/(f) \xrightarrow{g} R/(fg) \longrightarrow R/(g) \longrightarrow 0.$$

It vanishes on units and hence extends to the group completion $\text{Div}(\text{Spec } R)$, using Proposition 9.4.4. Pullback along the flat localization $\text{Spec } R \rightarrow X$ defines

$$\text{mult}_x: \text{Div}(X) \longrightarrow \mathbb{Z}.$$

Proposition 9.5.5 (The cycle of a Cartier divisor). Let X be locally noetherian. For every $D \in \text{Div}(X)$, only locally finitely many codimension-one points have $\text{mult}_x(D) \neq 0$. Therefore

$$\text{cyc}(D) = \sum_{\text{codim}(x, X)=1} \text{mult}_x(D)[x] \quad (\star)$$

defines an order-preserving group homomorphism

$$\text{cyc}: \text{Div}(X) \longrightarrow Z^1(|X|, \mathbb{Z}).$$

If D is effective, then $\text{cyc}(D) = [V(D)]$.

Proof. The assertion is local, so write an effective divisor as a single non-zero-divisor f on an affine noetherian chart. The codimension-one points with positive multiplicity are generic points of irreducible components of $V(f)$; a noetherian space has only finitely many of them. An arbitrary Cartier divisor is locally a difference of two effective ones, which proves local finiteness in general. Additivity follows from the exact length sequence in Construction 9.5.4. For an effective divisor, the coefficient at x is by definition the length of $\mathcal{O}_{X,x}/(f)$, exactly the generic multiplicity in the fundamental cycle of its vanishing scheme. $\square$

Proposition 9.5.6 (Flat pullback of codimension-graded cycles). Let $f: Y \rightarrow X$ be flat and locally of finite type, with X, Y locally noetherian. For a cycle $\alpha = \sum_V a_V [V]$, regard V as a reduced closed subscheme and set

$$f^* \alpha = \sum_V a_V [f^{-1}(V)].$$

This is locally finite and preserves codimension, so it restricts to maps

$$f^*: Z^n(|X|, \mathbb{Z}) \longrightarrow Z^n(|Y|, \mathbb{Z}).$$

It satisfies

$$f^*[Z] = [f^{-1}(Z)]$$

for closed subschemes $Z \subset X$. Moreover the square

$$\begin{array}{ccc} \mathrm{Div}(X) & \xrightarrow{\mathrm{cyc}} & Z^1(|X|, \mathbb{Z}) \\ f^* \downarrow & & \downarrow f^* \\ \mathrm{Div}(Y) & \xrightarrow{\mathrm{cyc}} & Z^1(|Y|, \mathbb{Z}) \end{array}$$

commutes.

Proof. Let p be the generic point of V , and let q be the generic point of an irreducible component of $f^{-1}(V)$. Work on compatible affine neighborhoods, writing $p \leftrightarrow \mathfrak{p} \subset A$ and $q \leftrightarrow \mathfrak{q} \subset B$. Then $\mathfrak{q}$ is minimal over $\mathfrak{p}B$, and going down gives $\mathfrak{q} \cap A = \mathfrak{p}$. Hence $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}}$ is a flat local map whose closed fiber is zero-dimensional. The noetherian dimension inequality together with going down now gives, in the notation of Theorem 7.5.14,

$$\dim B_{\mathfrak{q}} = \dim A_{\mathfrak{p}} + \dim(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}) = \dim A_{\mathfrak{p}}.$$

This proves codimension preservation. For local finiteness, choose an affine neighborhood W of a point of Y . Its image is quasi-compact and hence lies in a quasi-compact open of X , which meets only finitely many supports occurring in α ; each inverse image has only finitely many irreducible components on the noetherian open W .

For the multiplicities, the flat local map $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}}$ has artinian closed fiber. If M has finite length over $A_{\mathfrak{p}}$, then

$$\mathrm{length}_{B_{\mathfrak{q}}}(M \otimes_{A_{\mathfrak{p}}} B_{\mathfrak{q}}) = \mathrm{length}_{A_{\mathfrak{p}}}(M) \mathrm{length}_{B_{\mathfrak{q}}}(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}).$$

Apply this first to the generic local modules of a closed subscheme Z to obtain $f^*[Z] = [f^{-1}(Z)]$, and then to the quotient by a local equation of an effective Cartier divisor. This proves commutativity for effective divisors; group completion proves it for every Cartier divisor. The same definition and local multiplicity calculation appear in [The25, Tags 02RB, 02RE, and 02SM]. $\square$

Theorem 9.5.7 (Cartier–Weil comparison). *Let X be a locally noetherian normal scheme. Then*

$$\mathrm{cyc}: \mathrm{Div}(X) \longrightarrow Z^1(|X|, \mathbb{Z})$$

is injective and reflects the order. It is surjective if and only if every local ring of X is a UFD. In particular it is an isomorphism of ordered abelian groups when X is regular.

Proof roadmap. Normality makes every codimension-one local ring a DVR. Hence a Cartier divisor is measured there by a valuation, and these valuations are exactly the coefficients in $(\star)$. The intersection description of a noetherian normal domain inside its fraction field shows that nonnegative valuation at every height-one prime forces a rational local equation to be regular. Applied to D and $-D$, this gives order reflection and injectivity.

Surjectivity says, locally, that each height-one prime cycle is cut out by one rational equation. This is equivalent to height-one primes being principal in every local ring, hence to unique factorization. A regular local ring is a UFD. These are the heavy commutative-algebra inputs; see [The25, Tags 0AGA, 0BE8, and 0BE9]. The cited Stacks statements are written for integral normal schemes; the result above follows componentwise because the irreducible components of a locally noetherian normal scheme are open and closed locally. $\square$

9.6. The Picard group

Definition 9.6.1 (Picard group). For an algebraic functor X , its *Picard group* is

$$\mathrm{Pic}(X) = \{\text{isomorphism classes of line bundles on } X\},$$

with addition $[L] + [M] = [L \otimes M]$ and inverse $-[L] = [L^\vee]$.

On an affine scheme, this is the group of isomorphism classes of invertible finitely generated projective modules. Once again, these modules need not be free.

Proposition 9.6.2 (Cohomological interpretation). For every scheme X , there is a canonical isomorphism

$$\mathrm{Pic}(X) \simeq H^1(|X|, \mathcal{O}_X^\times).$$

Proof. A line bundle is Zariski-locally isomorphic to $\mathcal{O}_X$, and the automorphism sheaf of $\mathcal{O}_X$ is $\mathcal{O}_X^\times$. Choosing local bases therefore gives a $\mathcal{O}_X^\times$ -valued Čech cocycle. Changing the bases changes it by a coboundary, and gluing trivial lines from a cocycle gives the inverse construction. Equivalently, line bundles are $\mathcal{O}_X^\times$ -torsors. This torsor identification is [The25, Tag 09NU]. $\square$

Theorem 9.6.3 (Unit–Picard exact sequence). For every scheme X , there is an exact sequence

$$0 \longrightarrow \mathcal{O}_X(X)^\times \longrightarrow K(X)^\times \xrightarrow{\mathrm{div}} \mathrm{Div}(X) \xrightarrow{\rho} \mathrm{Pic}(X), \quad (\heartsuit)$$

where $\rho(L, s) = [L^\vee]$. If X is reduced and its irreducible components are locally finite, then ρ is also surjective.

Proof. The first arrow is injective because $\mathcal{O}_X \hookrightarrow \mathcal{K}_X$. A rational unit has zero divisor exactly when it and its inverse are regular, namely when it belongs to $\mathcal{O}_X(X)^\times$.

If $D = (L, s)$ is principal, then $L \simeq \mathcal{O}_X$, so $\rho(D) = 0$. Conversely, if $[L^\vee] = 0$, choose a trivialization of L ; under it s becomes a rational unit f , and $D = \mathrm{div}(f)$. This proves exactness at $\mathrm{Div}(X)$.

Under the final hypotheses, a line bundle has a rational trivialization: choose bases at the generic points of the irreducible components and extend them to dense opens; local finiteness lets these choices be made simultaneously near every point. Reducedness makes the resulting map into $\mathcal{K}_X$ injective. Applied to $L^\vee$, it constructs a divisor whose image under ρ is $[L]$, proving surjectivity. $\square$

Remark 9.6.4 (The sheaf sequence behind $(\heartsuit)$). The unit–Picard sequence begins the cohomology sequence of

$$1 \longrightarrow \mathcal{O}_X^\times \longrightarrow \mathcal{K}_X^\times \longrightarrow \mathcal{K}_X^\times / \mathcal{O}_X^\times \longrightarrow 1.$$

The identifications in degree zero are Propositions 9.4.6 and 9.6.2. If X is reduced and its irreducible components form a locally finite family, $\mathcal{K}_X^\times$ is flasque, so its first cohomology vanishes and explains the last surjectivity. The description of $\mathcal{K}_X$ as the product of the pushforwards of the generic-point fields, which gives this flasqueness, is [The25, Tag 02OW].

Example 9.6.5 (Dedekind domains). If A is a domain, $\mathrm{Pic}(A)$ is the group of invertible fractional ideals modulo principal fractional ideals. For a Dedekind domain every nonzero ideal is invertible, so this is its ideal class group. It vanishes exactly when A is a PID, equivalently a UFD.

The next two computations are structural theorems. They are quoted with proof roadmaps because their key inputs go beyond the divisor calculus developed here.

Theorem 9.6.6 (Picard group of a projective bundle). *Let X be a scheme, let V be a vector bundle whose rank is at least two at every point, and let $p: \mathbb{P}_X(V) \rightarrow X$ be the quotient-line projective bundle of Definition 3.2.11. Then*

$$\mathrm{Pic}(X) \oplus \mathrm{Map}_{\mathrm{lc}}(|X|, \mathbb{Z}) \xrightarrow{\sim} \mathrm{Pic}(\mathbb{P}_X(V)), \quad ([L], n) \mapsto [p^*L \otimes \mathcal{O}(n)].$$

Here $\mathrm{Map}_{\mathrm{lc}}$ means locally constant functions, so the twist is taken separately on each clopen level set of n . If X is connected, this becomes

$$\mathrm{Pic}(\mathbb{P}_X(V)) \simeq \mathrm{Pic}(X) \oplus \mathbb{Z}.$$

Proof roadmap. Restriction to every geometric projective-space fiber assigns an integer degree to a line bundle; this degree is locally constant on X . Twisting by the negative of that degree reduces to a bundle of fiberwise degree zero. For such a bundle N , cohomology and base change on projective space show that p_*N is a line bundle and that the adjunction map

$$p^*p_*N \rightarrow N$$

is an isomorphism: it is so on every geometric fiber, where a degree-zero line bundle is trivial. The same calculation gives uniqueness because $p_*\mathcal{O} = \mathcal{O}_X$. These constructions commute with restriction to clopen pieces, producing the locally constant function in the statement. The heavy inputs are cohomology and base change together with the projective-bundle cohomology calculation, the latter recorded in [The25, Tag 01XX]. The rank-one case is excluded because then $\mathbb{P}_X(V) \simeq X$ and there is no independent twisting degree. $\square$

Theorem 9.6.7 (Picard invariance for the affine space of a flat module). *Let X be a normal scheme and let $M \in \mathrm{Mod}_X$ be quasi-coherent and flat. For the associated affine space $p: \mathbb{A}_X(M) \rightarrow X$, pullback is an isomorphism*

$$p^*: \mathrm{Pic}(X) \xrightarrow{\sim} \mathrm{Pic}(\mathbb{A}_X(M)).$$

No finite-generation assumption on M is required.

Proof roadmap. The heavy affine input is the Traverso–Swan theorem: for a normal ring A , polynomial extension induces $\mathrm{Pic}(A) \simeq \mathrm{Pic}(A[t_1, \dots, t_n])$. The noetherian normal-domain, one-variable case and the contrasting cuspidal example are recorded in [The25, Tag 02AN]. Normality implies seminormality, which is the general ring-theoretic hypothesis behind the Traverso–Swan input.

Affine-locally on X , Lazard’s theorem writes the flat module M as a filtered colimit of finite free modules. Symmetric algebra commutes with this colimit, and an invertible module, its inverse, and the two inverse isomorphisms descend to one finite stage. At that stage the polynomial result shows that the line bundle comes uniquely from the base. Units of a symmetric algebra over a normal, hence reduced, ring are pulled back from the base; this identifies the descent cocycles on overlaps. Zariski descent therefore globalizes the affine calculation. The hypothesis cannot simply be omitted: polynomial extension can create new line bundles on a non-seminormal scheme, as the cuspidal example in the cited tag shows. $\square$

Remark 9.6.8 (Dependency map for later chapters). The stable interfaces supplied by this chapter are:

- effective and Cartier divisors and fractional ideals Definitions 9.1.2, 9.4.2 and 9.4.8;
- the universal modification Theorem 9.2.6;
- the computable Rees model Theorem 9.2.13;
- the exceptional divisor as the projectivized normal cone Corollary 9.2.14;
- the smooth-center statement Theorem 9.2.7; and
- the regular-center formulas Theorems 9.3.8 and 9.3.9.

These constructions build on the earlier relative- Proj formalism and provide the stated interfaces for the later properness chapter and other applications.

Chapter 10.

Properness

Properness packages the existence and uniqueness of geometric limits. Classical algebraic geometers called a proper variety *complete*; the word reflects the same limit intuition as completeness in analysis, although it is not a metric notion. The topological model separates the two roles cleanly: quasi-compactness supplies limiting points, while the Hausdorff condition makes a limit unique.

Topological model (motivation). The topological discussion in this paragraph and the next is motivational; no later proof will use it as an unproved criterion for schemes. For a continuous map $p: E \rightarrow B$, call p *universally closed* if every base change

$$E \times_B B' \longrightarrow B'$$

is a closed map. The tube lemma gives the precise intrinsic characterization

$$p \text{ is universally closed} \iff p \text{ is closed and every fiber } E_b \text{ is quasi-compact.}$$

Indeed, quasi-compactness of a fiber reduces an open cover of that fiber to finitely many tubes, and closedness shrinks the base so that these tubes cover all nearby fibers. Conversely, base change to b , followed by arbitrary products, reduces to the standard fact that a space K is quasi-compact exactly when every projection $K \times Z \rightarrow Z$ is closed. In particular, $E \rightarrow *$ is universally closed exactly when E is quasi-compact.

The map p is *separated* in the topological sense when its diagonal $E \rightarrow E \times_B E$ is closed. Thus $E \rightarrow *$ is separated exactly when E is Hausdorff. A topological map is called *proper* when it is both universally closed and separated, so a proper object of Top is precisely a quasi-compact Hausdorff space.

Nets and the limit heuristic. A net is a generalized sequence e_i indexed by a directed set; a subnet is understood in the usual Moore–Smith sense. We write $e_i \rightarrow e$ when the net is eventually in every neighborhood of e . A point e is a *cluster point* when every neighborhood of e is visited arbitrarily late; equivalently, some subnet converges to e . Thus a space is quasi-compact if and only if every net has a convergent subnet, while it is Hausdorff if and only if no net has two distinct limits. There is also a useful relative consequence: if p is universally closed and $p(e_i) \rightarrow b$, then some subnet of e_i converges to a point of E_b . To see this, suppose that E_b contains no cluster point. Quasi-compactness gives a finite union U of neighborhoods of points of E_b that the net eventually avoids. Since $p(E \setminus U)$ is closed and misses b , its complement is a neighborhood W of b with $p^{-1}(W) \subset U$, contradicting $p(e_i) \rightarrow b$. A closed relative diagonal similarly says that one net cannot converge to two distinct points in the same fiber.

This resembles a lifting property, but it is only a heuristic: passing to a subnet changes the source of the proposed lift, and different subnets may have different limits even in a compact Hausdorff space. Two elementary test spaces show the distinction. Let $\mathbb{S} = \{\eta, s\}$ be the Sierpiński space with opens $\emptyset, \{\eta\}, \mathbb{S}$. A map $\mathbb{S} \rightarrow B$ records one specialization $b_\eta \rightsquigarrow b_s$, so a lifting square against $\{\eta\} \hookrightarrow \mathbb{S}$ asks a chosen point over b_η to specialize over b_s . This single test does not detect quasi-compactness: for a map to a point, the constant extension always fills the square. At the other extreme, $[0, 1) \hookrightarrow [0, 1]$ models completion along one specified parameter, but its right lifting property is stronger than compactness. For example, the map

$$[0, 1) \longrightarrow [-1, 1], \quad t \longmapsto \sin(1/(1-t)),$$

has no extension at 1, although $[-1, 1]$ is compact Hausdorff.

The algebraic replacement. For schemes, the spectrum of a valuation ring is the rigid test object that replaces an arbitrarily reindexed approach to a limit. Its points are totally ordered by specialization, and the inclusion

$$\mathrm{Spec}(K) \hookrightarrow \mathrm{Spec}(V), \quad K = \mathrm{Frac}(V),$$

packages a generic point together with all intermediate specializations. Let $\mathcal{T}_{\mathrm{val}}$ be the class of these inclusions. Then existence, at-most-one, and unique existence of fillers are called formal primness, formal separatedness, and formal properness below. The valuative criteria show that the corresponding geometric notions are obtained after adding the requisite finite-type or quasi-separatedness hypotheses.

This is parallel to the infinitesimal lifting theory of Definition 8.5.6. If $\mathcal{T}_{\mathrm{inf}}$ denotes the square-zero thickenings, the abstract terminology of Definition 8.5.9 gives the following three-line dictionary. The table intentionally suppresses the finiteness hypotheses.

Abstract filler condition	Infinitesimal side $\mathcal{T}_{\mathrm{inf}}$	Valuative side $\mathcal{T}_{\mathrm{val}}$
$\mathcal{T}$ -epimorphism (at least one)	smooth	prim
$\mathcal{T}$ -monomorphism (at most one)	unramified	separated
$\mathcal{T}$ -isomorphism (exactly one)	étale	proper

Precisely, the missing hypotheses on the infinitesimal side are listed in Theorem 8.5.22; those on the valuative side are listed in Corollary 10.3.5. Thus the table is a memory aid for the common filler pattern, not an assertion that finiteness can be discarded.

All schemes below may equivalently be regarded as their functors of points. Thus every cartesian square and every lifting problem is a diagram of algebraic functors, while the words *closed*, *quasi-compact*, and so on refer to the corresponding scheme morphisms. We use the conventions on universal topological properties from Definition 4.3.10, on finite type from Definition 6.2.19, and on relative projectivity from Definition 3.2.13. We also use effective and Cartier divisors, fractional ideals, and the blowup universal property from Definitions 9.1.2, 9.2.2, 9.4.2 and 9.4.8 and Theorem 9.2.6. The chapter proceeds from the topological definition to valuation-ring tests. It first isolates the existence part of properness, then develops Prüfer domains and valuation rings, proves the classical valuative criteria, and finally compares proper morphisms with projective, integral, and finite morphisms.

We use the standard convention that a finite morphism is affine and is given on affine charts by an algebra finite as a module over the base ring; Definition 10.5.1 revisits this together with integral and quasi-finite morphisms.

10.1. Prim and proper morphisms

Definition 10.1.1 (Universally closed, prim, separated, and proper). Let $f: Y \rightarrow X$ be a morphism of schemes.

- (i) The morphism f is *universally closed* if, for every $X' \rightarrow X$, the base change $f_{X'}: Y \times_X X' \rightarrow X'$ is a closed map on underlying spaces.
- (ii) It is *prim* if it is locally of finite type and universally closed.
- (iii) It is *separated* if its diagonal

$$\Delta_f: Y \longrightarrow Y \times_X Y$$

is prim.

- (iv) It is *proper* if it is both prim and separated.

The word *prim* is useful because it isolates the existence half of properness. The locally-finite-type clause can be checked on small charts, whereas universal closedness is a genuinely global condition on the source.

Proposition 10.1.2 (Agreement with the usual definitions). *For a morphism $f: Y \rightarrow X$ of schemes, the separatedness in Definition 10.1.1 is equivalent to the condition that Δ_f be a closed immersion, hence agrees with Definition 6.2.1. Moreover,*

$$f \text{ is proper} \iff f \text{ is separated, of finite type, and universally closed.}$$

In particular, every prim morphism is of finite type.

Proof. The diagonal of a scheme morphism is an immersion by Proposition 6.2.2, and every immersion is locally of finite type. An immersion is universally closed if and only if it is a closed immersion: one implication follows from stability of closed immersions under base change, and the other follows by working on an open set through which the immersion factors as a closed immersion and using that its image is closed in the ambient scheme. This proves the first assertion.

The only apparent difference in the second assertion is quasi-compactness. A universally closed morphism is quasi-compact by Theorem 10.1.3 below. Therefore a prim morphism, being locally of finite type and quasi-compact, is of finite type under the convention of Definition 6.2.19. The stated characterization of properness now follows directly. $\square$

Theorem 10.1.3 (Universal closedness forces quasi-compactness). *Every universally closed morphism of schemes is quasi-compact.*

Reference and proof roadmap. This nontrivial theorem is [The25, Tag 04XU]. After restricting to an affine target $\text{Spec}(A)$, choose a point s such that Y_U is not quasi-compact for any neighborhood U of s , choose an affine cover $Y = \bigcup_{i \in I} Y_i$, and make the base change

$$\text{Spec}(A[y_i \mid i \in I]) \longrightarrow \text{Spec}(A).$$

Inside the base change of Y , take the complement of $\bigcup_i (Y_i \times D(y_i))$. It is closed, and its image misses the point t over s at which every y_i equals 1. If that image were closed, a single polynomial g would vanish on it while remaining nonzero at t . Only finitely many variables occur in g , and one coefficient is invertible on a neighborhood U of s . Since Y_U is not quasi-compact, the corresponding finite family of the Y_i does not cover it. A point outside that family, together with a suitable point of the polynomial base change, then lies in the closed subset where g is nonzero. This contradiction proves quasi-compactness. The infinitely many variables package the failure of every finite subcover into one base change. $\square$

Proposition 10.1.4 (Permanence properties). *The properties prim, separated, and proper are*

- (i) *local on the base for the Zariski topology;*
- (ii) *stable under composition; and*
- (iii) *stable under arbitrary base change.*

Proof. Universal closedness is local on the target: after any base change, a subset of the new target is closed precisely when its intersections with an open cover are closed. It is stable under base change by definition, and it is stable under composition because a composite of closed maps is closed after every base change. Being locally of finite type has the same three permanence properties. This proves the assertions for prim morphisms.

For separatedness, use the diagonal. The diagonal of a base change is the base change of the original diagonal. For a composite, the diagonal factors through a base change of one diagonal followed by the

other diagonal; equivalently one may apply Proposition 6.2.5. Locality on the base is checked after restricting the same diagonal to an open cover. Combining the prim and separated cases proves the claims for proper morphisms. $\square$

Remark 10.1.5 (Descent and failure of source locality). These three properties are in fact fpqc-local on the base; for universal closedness, local finite type, separatedness, and properness, respectively, see [The25, Tag 02KS, Tag 02KX, Tag 02KU, and Tag 02L1]. None is local on the source. For primness and properness, the map $\coprod_{n \geq 0} \text{Spec}(k) \rightarrow \text{Spec}(k)$ is covered on its source by proper components but is not quasi-compact, hence cannot be universally closed. For separatedness, the affine line with two origins is covered by two copies of $\mathbb{A}_k^1$, each mapping isomorphically to $\mathbb{A}_k^1$, while the glued map is not separated.

Proposition 10.1.6 (First examples). (i) Every closed immersion is proper.

(ii) A prim immersion is a closed immersion.

(iii) Every integral morphism is universally closed and separated. Consequently every finite morphism is proper.

Proof. A closed immersion stays a closed immersion after base change, is locally of finite type, and has a closed-immersion diagonal. This proves (1).

For (2), factor the immersion $i: Y \rightarrow X$ as a closed immersion $Y \hookrightarrow U$ followed by an open immersion $U \hookrightarrow X$. Since i is universally closed, its image is closed in X . The two opens U and $X \setminus i(Y)$ cover X ; over the first i is closed by construction and over the second it is empty. Hence i is a closed immersion.

An integral morphism is affine. The going-up theorem shows that an integral map of affine schemes is closed, and integrality is preserved by base change, so it is universally closed. Every affine morphism is separated by Example 6.2.4. Finally, a finite morphism is integral and locally of finite type, so it is proper. These implications are also recorded in [The25, Tag 01WM and Tag 01WN]. $\square$

Remark 10.1.7 (A stronger monomorphism theorem). The conclusion of Proposition 10.1.6(2) remains true with *immersion* replaced by *monomorphism*: a prim monomorphism is a closed immersion. The passage from an arbitrary monomorphism to an immersion is the difficult part, so we treat this as a cited theorem rather than silently folding it into the elementary argument above; see [The25, Tag 04XV].

Theorem 10.1.8 (Nagata compactification). Let X be quasi-compact and quasi-separated, and let $f: Y \rightarrow X$ be separated and of finite type. There is a factorization

$$Y \xrightarrow{j} \bar{Y} \xrightarrow{p} X$$

in which j is an open immersion and p is proper.

Reference and proof roadmap. This is the Nagata compactification theorem [The25, Tag 0F41]. Its proof is substantially deeper than the basic theory in this chapter. One first replaces the morphism by one of finite presentation and descends it, together with separatedness, to a noetherian approximation of the qcqs base. Over a noetherian base, choose a finite affine cover of Y whose total intersection is dense, compactify each affine chart in projective space, and glue the compactifications successively. The gluing theorem, which uses admissible blowups and the valuative criterion to reconcile the overlaps, is the heavy input. Base change of the resulting compactification returns the asserted factorization over X ; taking the scheme-theoretic image then handles the initial finite-type enlargement. $\square$

10.2. Prüfer pairs and valuation rings

The points of the spectrum of a valuation ring form a chain under specialization, with a unique generic point and a unique closed point; intermediate points may also occur. Prüfer pairs abstract the condition that modifications supported away from a dense locus create no additional directions of approach.

Prüfer pairs

Definition 10.2.1 (Prüfer pairs, domains, and schemes). Let $j: X' \rightarrow X$ be a monomorphism of schemes. The pair (X, X') is a *Prüfer pair* if all of the following hold.

- (i) The morphism j is quasi-compact.
- (ii) The subfunctor X' is the intersection, in the poset of subfunctors of X , of all open subfunctors containing it.
- (iii) The restriction map $\mathcal{O}_X \rightarrow j_*\mathcal{O}_{X'}$ is injective.
- (iv) If $U \subset X$ is open and $Z \hookrightarrow U$ is a finitely presented closed subscheme with $Z \times_X X' = \emptyset$, then Z is an effective Cartier divisor on U .

The final condition uses effective Cartier divisors in the sense of Definition 9.1.2 and Proposition 9.1.3.

An integral domain R is a *Prüfer domain* if

$$(\mathrm{Spec}(R), \mathrm{Spec}(\mathrm{Frac}(R)))$$

is a Prüfer pair. A scheme is a *Prüfer scheme* if it has an affine open cover by spectra of Prüfer domains.

Theorem 10.2.2 (Affine criterion for a Prüfer pair). Let $R \rightarrow R'$ be a localization. Then $(\mathrm{Spec}(R), \mathrm{Spec}(R'))$ is a Prüfer pair if and only if $R \rightarrow R'$ is injective and every finitely generated ideal $I \subset R$ satisfying $IR' = R'$ is an invertible R -module of rank one. In particular, a domain R is Prüfer if and only if every nonzero finitely generated ideal of R is invertible.

Proof. Write $R' = S^{-1}R$. The morphism $j: \mathrm{Spec}(R') \rightarrow \mathrm{Spec}(R)$ is affine and hence quasi-compact. Moreover, each $D(s)$, $s \in S$, contains $\mathrm{Spec}(R')$, and a T -point of $\mathrm{Spec}(R)$ factors through every $D(s)$ if and only if every image of s is a unit on T . By the universal property of localization, such a point factors uniquely through $\mathrm{Spec}(R')$. Thus the intersection condition in Definition 10.2.1 is automatic. Finally, $\mathcal{O}_X \rightarrow j_*\mathcal{O}_{X'}$ is injective if and only if $R \rightarrow R'$ is injective.

Suppose first that $(\mathrm{Spec}(R), \mathrm{Spec}(R'))$ is a Prüfer pair. If I is finitely generated and $IR' = R'$, then $V(I)$ is a finitely presented closed subscheme disjoint from $\mathrm{Spec}(R')$. It is therefore an effective Cartier divisor, so its ideal I is an invertible R -module of rank one.

Conversely, assume injectivity and the stated condition on ideals. Let $Z \hookrightarrow U \subset \mathrm{Spec}(R)$ be a finitely presented closed subscheme disjoint from $\mathrm{Spec}(R')$. It suffices to check the Cartier condition on a principal affine $D(f) \subset U$. Write the ideal of $Z \cap D(f)$ as IR_f for a finitely generated ideal $I \subset R$. Disjointness says

$$IR'_f = R'_f.$$

Clearing the finitely many denominators in an expression for 1 gives some $s \in S$ and $N \geq 0$ with $sf^N \in I$. Set $L = I + (s)$. Since s is a unit in R' , we have $LR' = R'$, so L is invertible by hypothesis. After localizing at f , the relation $sf^N \in I$ gives $s \in IR_f$, and hence

$$LR_f = IR_f.$$

Thus the ideal of $Z \cap D(f)$ is invertible. An invertible ideal is locally generated by a nonzerodivisor, so $Z \cap D(f)$ is an effective Cartier divisor. The principal opens $D(f) \subset U$ cover U , proving the required condition.

For a domain, localize at the set of all nonzero elements. Injectivity is automatic, and precisely the nonzero finitely generated ideals become the unit ideal after localization. This gives the final assertion. $\square$

Proposition 10.2.3 (Prüfer domains are normal). *Every Prüfer domain is integrally closed. Consequently every Prüfer scheme is normal.*

Proof. Let R be Prüfer, let $K = \text{Frac}(R)$, and suppose $x = r/s \in K$ is integral over R , with $s \neq 0$. The ideal (r, s) is invertible by Theorem 10.2.2. On a principal open cover of $\text{Spec}(R)$, choose a generator d and write

$$r = ad, \quad s = bd, \quad (a, b) = 1.$$

It suffices to show a/b belongs to the corresponding localized ring. If

$$(a/b)^n + c_{n-1}(a/b)^{n-1} + \cdots + c_0 = 0,$$

then multiplication by b^{n-1} gives a^n/b in that ring. Thus b divides a^n . Since $(a, b) = 1$, the image of a is a unit modulo b ; but its n -th power is zero modulo b . Hence the quotient by b is the zero ring, so b is a unit and a/b belongs to the localized ring. Thus x is a regular function on every member of the cover. The sheaf property glues these local sections to show $x \in R$. Normality is local on a scheme, which proves the final assertion. $\square$

Valuation rings

Definition 10.2.4 (Valuation and its valuation ring). Let K be a field and let Γ be a totally ordered abelian group, written additively. Extend its order and addition by $\gamma < \infty$ and $\gamma + \infty = \infty$. A *valuation* is a map

$$v: K \longrightarrow \Gamma \cup \{\infty\}$$

such that

$$\begin{aligned} v(1) &= 0, & v(xy) &= v(x) + v(y), \\ v(0) &= \infty, & v(x + y) &\geq \min\{v(x), v(y)\}. \end{aligned}$$

Its valuation ring, maximal ideal, unit group, and residue field are

$$\begin{aligned} \mathcal{O}_v &= \{x \in K : v(x) \geq 0\}, & \mathfrak{m}_v &= \{x \in K : v(x) > 0\}, \\ \mathcal{O}_v^\times &= \{x \in K : v(x) = 0\}, & \kappa(v) &= \mathcal{O}_v / \mathfrak{m}_v. \end{aligned}$$

After replacing Γ by the image $v(K^\times)$, we call Γ the *value group*.

Proposition 10.2.5 (Equivalent descriptions of valuation rings). *Let R be a domain with fraction field K . The following are equivalent.*

- (i) R is a local Prüfer domain.
- (ii) Divisibility in R is total: for nonzero $a, b \in R$, either $a \mid b$ or $b \mid a$.
- (iii) The ideals of R are totally ordered by inclusion.
- (iv) Cartier divisors on $\text{Spec}(R)$ are totally ordered by the relation $D \leq E$ when $E - D$ is effective.
- (v) For each $x \in K^\times$, either $x \in R$ or $x^{-1} \in R$.
- (vi) There is a valuation v of K with $R = \mathcal{O}_v$.

A domain satisfying these conditions is called a valuation ring. This agrees with Definition 7.8.5.

Proof. We prove $(1) \Rightarrow (2) \Leftrightarrow (3) \Rightarrow (1)$, then $(2) \Leftrightarrow (5) \Leftrightarrow (6)$, and finally $(4) \Leftrightarrow (5)$.

Assume (1). For $a, b \neq 0$, the ideal (a, b) is invertible and therefore free of rank one over the local ring R . Write $(a, b) = (d)$, $a = du$, and $b = dv$. Since $(u, v) = R$, one of u, v is a unit, and hence one of a, b divides the other. This proves $(1) \Rightarrow (2)$.

Suppose (2), and let I, J be ideals with $I \not\subseteq J$. Choose $a \in I \setminus J$. For every $b \in J$, the alternative $b \mid a$ would put a in J ; hence $a \mid b$, so $b \in I$. Thus $J \subset I$, proving (3). Conversely, (3) makes the principal ideals comparable and gives (2). It also shows that R has a unique maximal ideal and that every nonzero finitely generated ideal is the largest among the principal ideals generated by a finite generating set. Such an ideal is principal and invertible. By Theorem 10.2.2, (3) therefore implies (1).

For $x = a/b \in K^\times$, total divisibility says either $b \mid a$, so $x \in R$, or $a \mid b$, so $x^{-1} \in R$. This proves $(2) \Rightarrow (5)$. Conversely, apply (5) to a/b for nonzero $a, b \in R$: if $a/b \in R$, then $b \mid a$, while if $b/a \in R$, then $a \mid b$. Thus $(5) \Rightarrow (2)$.

Assume (5). Put $\Gamma = K^\times/R^\times$, written additively so that $[xy] = [x] + [y]$, and order it by

$$[x] \geq [y] \iff x/y \in R.$$

Condition (5) makes this a well-defined total order compatible with the group law. Define $v(0) = \infty$ and $v(x) = [x]$ for $x \neq 0$. If $v(x) \geq v(y)$, then $x/y \in R$, whence $(x+y)/y \in R$ and $v(x+y) \geq v(y) = \min\{v(x), v(y)\}$. Thus v is a valuation and $R = \mathcal{O}_v$, proving (6). Conversely, totality of the order on a value group says that $v(x) \geq 0$ or $v(x^{-1}) = -v(x) \geq 0$; hence (6) implies (5).

Finally, over the local scheme $\text{Spec}(R)$, every invertible fractional ideal is principal. Comparing two Cartier divisors therefore amounts to asking whether the quotient of two elements of $K^\times$ or its inverse lies in R . This identifies (4) with (5). $\square$

Remark 10.2.6 (The intrinsic value group). For a valuation ring $R \subset K$, the construction in the preceding proof gives the canonical valuation

$$K^\times \twoheadrightarrow K^\times/R^\times.$$

Every valuation with valuation ring R factors through this quotient; after replacing its target by its value group, the induced map is an order isomorphism. The positive cone consists of classes represented by nonzero elements of R . It is often helpful to think of this ordered group as the group of principal Cartier divisors of $\text{Spec}(R)$.

Example 10.2.7 (Basic valuation rings). (i) A field is a valuation ring with trivial value group.

(ii) A discrete valuation ring has value group $\mathbb{Z}$. If π is a uniformizer, every nonzero element is $u\pi^n$ with u a unit and $n \geq 0$.

(iii) Let k be a field and Γ a totally ordered abelian group. The Hahn field $k((t^\Gamma))$ consists of formal sums $\sum_\gamma a_\gamma t^\gamma$ with well-ordered support. Taking the least exponent defines a valuation with value group Γ ; the elements of nonnegative value form its valuation ring.

Proposition 10.2.8 (Noetherian valuation rings). A noetherian valuation ring is either a field or a discrete valuation ring.

Proof. Every finitely generated ideal in a valuation ring is generated by one of any finite set of generators, because principal ideals are totally ordered. If the ring is noetherian, all ideals are therefore principal. A local principal ideal domain is a field when its maximal ideal is zero, and otherwise is a discrete valuation ring. $\square$

Lemma 10.2.9 (Local criterion for Prüfer domains). *An integral domain R is a Prüfer domain if and only if $R_{\mathfrak{p}}$ is a valuation ring for every prime ideal $\mathfrak{p} \subset R$.*

Proof. If R is Prüfer, every nonzero finitely generated ideal becomes invertible and hence principal after localization. Applied to (a, b) , this says that the principal ideals of each local ring $R_{\mathfrak{p}}$ are totally ordered, so Proposition 10.2.5 makes it a valuation ring. Conversely, if every $R_{\mathfrak{p}}$ is a valuation ring, every nonzero finitely generated ideal of R is locally principal of rank one. Such an ideal is an invertible module, which is the affine criterion of Theorem 10.2.2. $\square$

Corollary 10.2.10 (The noetherian Prüfer case). *The locally noetherian Prüfer schemes are precisely the Dedekind schemes: the locally noetherian normal schemes whose local rings are fields or discrete valuation rings.*

Proof. Localizations of a Prüfer domain at prime ideals are local Prüfer domains and hence valuation rings by Proposition 10.2.5. In the noetherian case Proposition 10.2.8 makes each such local ring a field or a DVR. Conversely, fields and DVRs are valuation rings. Refining to integral affine neighborhoods, Lemma 10.2.9 gives a Prüfer affine cover. $\square$

Dominating local rings

Definition 10.2.11 (Domination). If $A \subset B \subset K$ are local subrings of a field, then B dominates A if $\mathfrak{m}_A = A \cap \mathfrak{m}_B$. Equivalently, the inclusion $A \rightarrow B$ is a local homomorphism.

Theorem 10.2.12 (Valuation rings as maximal local subrings). *Let K be a field and $R \subset K$ a local subring with $\text{Frac}(R) = K$. Then R is a valuation ring if and only if it is maximal, under domination, among local subrings of K . More generally, every local subring $A \subset K$ is dominated by some valuation ring $V \subset K$ with fraction field K .*

Reference and proof roadmap. These statements are recorded in [The25, Tag 00I9, Tag 00IA, Tag 00IB, and Tag 052K]. For existence, order the local subrings of K which dominate A ; the union of a chain is again a dominating local subring, so Zorn's lemma produces a maximal one. Maximality is then converted into the test $x \in V$ or $x^{-1} \in V$: adjoining a missing element and localizing at a prime over the old maximal ideal would otherwise give a strictly larger dominating local ring. Conversely, the same x -or- x^{-1} test prevents any proper dominating enlargement. This is the heavy valuation-existence input used below. $\square$

10.3. Valutive criteria

The classical valutive criteria

Definition 10.3.1 (Formal valutive properties). Let $\mathcal{T}$ be the collection of inclusions

$$\text{Spec}(K) \longrightarrow \text{Spec}(V),$$

where V ranges over valuation rings and $K = \text{Frac}(V)$. Given a morphism $f : Y \rightarrow X$ of algebraic functors, consider every solid commutative square

$$\begin{array}{ccc} \text{Spec}(K) & \xrightarrow{u} & Y \\ \downarrow & \nearrow \tilde{u} & \downarrow f \\ \text{Spec}(V) & \xrightarrow{v} & X. \end{array}$$

We call f

- (i) *formally prim* if a dashed lift always exists;

- (ii) *formally separated* if there is at most one dashed lift; and
- (iii) *formally proper* if there is exactly one dashed lift.

Thus formally proper means formally prim and formally separated. These adjectives refer only to valuation-ring lifting problems; they should not be confused with infinitesimal formal smoothness or formal schemes.

Theorem 10.3.2 (Specializations are witnessed by valuations). *Let X be a scheme, let $y \rightsquigarrow x$ be a specialization in $|X|$, and fix a field extension $\kappa(y) \hookrightarrow K$. There are a valuation ring $V \subset K$, with fraction field K , and a morphism*

$$g: \operatorname{Spec}(V) \longrightarrow X$$

such that the restriction of g to the generic point is the prescribed map $\operatorname{Spec}(K) \rightarrow \operatorname{Spec}(\kappa(y)) \rightarrow X$, while the closed point of $\operatorname{Spec}(V)$ maps to x .

Reference and proof roadmap. This is [The25, Tag 01J8]. The specialization gives a map from the local domain $\mathcal{O}_{X,x}/\mathfrak{p}_y$ into K , where $\mathfrak{p}_y \subset \mathcal{O}_{X,x}$ is the prime corresponding to the generization y . Apply Theorem 10.2.12 to choose a valuation ring of K dominating this local subring. The resulting local homomorphism $\mathcal{O}_{X,x} \rightarrow V$ defines $\operatorname{Spec}(V) \rightarrow X$. Domination sends the closed point to x , while the prescribed inclusion into K fixes the generic-point map. The Zorn-lemma step hidden in the choice of V is why this theorem is cited as a heavy valuation result. $\square$

Lemma 10.3.3 (A quasi-compact specialization-closed image is closed). *Let $g: Z \rightarrow T$ be quasi-compact. If $g(|Z|)$ is stable under specialization in $|T|$, then $g(|Z|)$ is closed.*

Proof. The claim is local on T , so take $T = \operatorname{Spec}(A)$. Quasi-compactness gives a finite affine cover $Z = \bigcup_{i=1}^n \operatorname{Spec}(B_i)$. The image of Z equals the image of

$$\operatorname{Spec}\left(\prod_{i=1}^n B_i\right) \longrightarrow \operatorname{Spec}(A).$$

For a map of affine schemes, an image which is stable under specialization is closed; this is the affine prime-ideal lemma used in [The25, Tag 05JL]. Applying it to the displayed map proves the result. $\square$

Theorem 10.3.4 (Valuative criterion for universal closedness). *For a morphism $f: Y \rightarrow X$ of schemes, the following are equivalent:*

- (i) *f is universally closed;*
- (ii) *f is formally prim and quasi-compact.*

Proof. Assume first that f is universally closed. It is quasi-compact by Theorem 10.1.3. Given a valuation square, base-change f to $\operatorname{Spec}(V)$, and let $\xi \in Y_V$ be the image of the specified K -point. The closure $\overline{\{\xi\}} \subset Y_V$ has closed image in $\operatorname{Spec}(V)$, and that image contains the generic point; it therefore contains the closed point. Choose $x \in \overline{\{\xi\}}$ above the closed point and put

$$B = \operatorname{im}(\mathcal{O}_{Y_V,x} \longrightarrow \kappa(\xi) \longrightarrow K).$$

Then $B \subset K$ is a local subring containing and dominating V . The maximality of the *given* valuation ring V among local subrings of K , Theorem 10.2.12, forces $B = V$. The resulting local map $\mathcal{O}_{Y_V,x} \rightarrow V$ defines a morphism $\operatorname{Spec}(V) \rightarrow Y_V$ whose restriction to the generic point is the original K -point. Hence f is formally prim.

Conversely, assume that f is quasi-compact and formally prim. These properties survive arbitrary base change, so fix $T \rightarrow X$, a closed subset $Z \subset Y_T$, and a point $z \in Z$ whose image $t \in T$ specializes to t' .

Apply Theorem 10.3.2 to $t \leadsto t'$, using a field extension which contains $\kappa(z)$. Formal primness produces a map from the resulting valuation spectrum to Y_T whose generic point maps to z . Since Z is closed and the generic point is dense in the spectrum of a domain, the whole valuation spectrum maps into Z . Its closed point supplies a point of Z over t' . Thus the image of $Z \rightarrow T$ is stable under specialization. The map $Z \rightarrow T$ is quasi-compact, so Lemma 10.3.3 makes its image closed. Therefore every base change of f is closed.

This criterion is also [The25, Tag 01KF]; the proof above spells out where its two valuation inputs enter. $\square$

Corollary 10.3.5 (Valuative criteria for primness, separatedness, and properness). *Let $f: Y \rightarrow X$ be a morphism of schemes.*

- (i) f is prim if and only if it is formally prim and of finite type.
- (ii) f is separated if and only if it is formally separated and quasi-separated.
- (iii) f is proper if and only if it is formally proper, of finite type, and quasi-separated.

Proof. For (1), a prim morphism is of finite type by Proposition 10.1.2 and formally prim by Theorem 10.3.4. Conversely, finite type implies quasi-compact, so the same theorem turns formal primness into universal closedness; local finite type is already part of finite type.

If f is separated, two extensions $\text{Spec}(V) \rightrightarrows Y$ agree on the generic point. Their equalizer is the inverse image of the closed diagonal, hence is closed in $\text{Spec}(V)$; as it contains the dense generic point, it is all of $\text{Spec}(V)$. Thus f is formally separated, and its diagonal is quasi-compact, so f is quasi-separated.

Conversely, suppose f is quasi-separated and formally separated. Then Δ_f is quasi-compact. A valuation lifting problem for Δ_f is a pair of maps $\text{Spec}(V) \rightrightarrows Y$ which agree generically; formal separatedness says that the pair agrees everywhere. Hence Δ_f is formally prim. By Theorem 10.3.4, Δ_f is universally closed. It is also an immersion and hence locally of finite type, so it is a prim immersion; therefore Proposition 10.1.6(2) shows that it is a closed immersion. This proves (2), matching [The25, Tag 01L0].

Finally, formal properness is the conjunction of the two formal conditions. Combining (1) and (2) proves (3). The explicit finite-type and quasi-separated hypotheses are indispensable in the reverse direction; compare [The25, Tag 0BX5]. $\square$

Example 10.3.6 (The affine line with two origins). Let D be obtained by gluing two copies of $\mathbb{A}_k^1$ along the punctured line, and let $D \rightarrow \mathbb{A}_k^1$ be induced by the identity on both charts. This morphism is of finite presentation and formally prim, but it is not formally separated. Indeed, if V is a valuation ring and a nonzero $t \in \mathfrak{m}_V$ is used as the coordinate, the generic point lands in the common punctured open, while the extension over the closed point may choose either origin. Existence always holds: if the generic coordinate is nonzero, choose either affine chart, and if it is zero the chosen generic origin determines the chart. Hence $D \rightarrow \mathbb{A}_k^1$ is prim but not proper. This is the valuative form of the non-separated example in Example 6.2.4.

Extension across Prüfer pairs

Theorem 10.3.7 (Accepted theorem: extension across Prüfer schemes). *Let C be a Prüfer scheme and let $j: C' \rightarrow C$ be a scheme monomorphism through which every codimension-zero point of C factors. Suppose that j identifies C' , as a subfunctor of C , with*

$$C' = \bigcap_{\alpha} U_{\alpha}$$

for a cofiltered family of open subschemes $U_{\alpha} \subset C$ whose inclusions $U_{\alpha} \hookrightarrow C$ are quasi-compact, and suppose that each complement $|C| \setminus |U_{\alpha}|$ is locally finite: every point of C has a neighborhood meeting only finitely many

points of that complement. Then every prim morphism $f : Y \rightarrow X$ has the right lifting property with respect to j . Explicitly, every solid commutative square

$$\begin{array}{ccc} C' & \longrightarrow & Y \\ j \downarrow & \nearrow & \downarrow f \\ C & \longrightarrow & X \end{array}$$

admits a dashed lift. Here quasi-compactness is a condition on each displayed open immersion $U_\alpha \hookrightarrow C$; it does not assert that the monomorphism j is itself an open immersion.

Status and proof roadmap. We take this global extension theorem as a heavy input. The following paragraph records its proof architecture, not a complete proof.

First, one combines the cofiltered presentation of C' with a genuine approximation lemma to spread the prescribed map from C' to a controlled open neighborhood U_α whose inclusion in C is quasi-compact. This step is subtler than the usual finite-presentation limit lemma: a prim morphism is only assumed of finite type, and the open transition maps in the displayed intersection need not form an affine inverse system. Second, after localizing C at a point outside C' , the local ring is a valuation ring by Lemma 10.2.9. Formal primness, supplied by Corollary 10.3.5, gives an extension from its fraction field. A Prüfer–Cartier compatibility lemma is then needed to make this choice agree with the map already prescribed on all intermediate points belonging to C' ; generic-point extension alone does not give that agreement for a non-separated target. Finally, local finiteness reduces the missing locus to finitely many points near any chosen point of C . After a common refinement of the U_α , the compatible local extensions glue, and cofilteredness makes the result independent of that refinement.

For comparison, Temkin–Tyomkin’s characterization of Prüfer pairs and their semilocal Prüfer valuative criterion isolate closely related local inputs [TT17, Theorem 4.1.11 and Proposition 5.2.9]. Those results do not themselves state the global extension theorem above with a prescribed map on C' and a locally finite complement; this is why the theorem is explicitly marked as accepted here. $\square$

Example 10.3.8 (Completing Dedekind schemes). Let C be a Dedekind scheme. Two basic instances of Theorem 10.3.7 are:

- (i) If $x \in C$ has codimension one, then

$$C \setminus \{x\} \hookrightarrow C.$$

For example, one obtains

$$\mathrm{Spec}(\mathbb{Z}[1/p]) \hookrightarrow \mathrm{Spec}(\mathbb{Z}) \quad \text{and} \quad \mathbb{A}_k^1 \hookrightarrow \mathbb{P}_k^1,$$

where the latter inclusion removes the point at infinity.

- (ii) If $C^{(0)}$ is the set of codimension-zero points, then the generic-point inclusion is

$$\coprod_{\eta \in C^{(0)}} \mathrm{Spec}(\kappa(\eta)) \longrightarrow C.$$

This specializes to $\mathrm{Spec}(\mathbb{Q}) \longrightarrow \mathrm{Spec}(\mathbb{Z})$ and $\mathrm{Spec}(k(t)) \longrightarrow \mathbb{P}_k^1$.

For either inclusion, a square against a prim morphism admits an extension. The next remark supplies the at-most-one statement for a separated morphism and hence unique extension for a proper morphism.

Remark 10.3.9 (Existence, uniqueness, and the general Prüfer-pair form). For a monomorphism $C' \rightarrow C$ covered by Theorem 10.3.7, primness gives existence. Applying that theorem to the diagonal gives at most one lift for a separated morphism: two candidate lifts define a map to $Y \times_X Y$, and a lift through the prim diagonal forces them to coincide. A proper morphism therefore gives a unique lift.

We also accept the stronger Prüfer-pair form: for an arbitrary Prüfer pair $j: X' \rightarrow X$ in the sense of Definition 10.2.1, separated morphisms have at most one lift and proper morphisms have exactly one lift in every square with left edge j . This last assertion is an additional heavy input; it is not a formal consequence of the preceding theorem because a general Prüfer pair need not come equipped with the displayed locally finite open presentation.

10.4. Properness and projectivity

Theorem 10.4.1 (Projective morphisms are proper). *Every projective morphism of schemes is proper.*

Proof. Properness is local on the base, so take an affine open of the target. For a projective morphism in the sense of Definition 3.2.13, choose finitely many generators of the degree-one part of its graded algebra. They give a surjection from a polynomial algebra and hence a closed immersion

$$Y \hookrightarrow \mathbb{P}_X^n.$$

By Proposition 10.1.4 and Proposition 10.1.6, it remains to prove $\mathbb{P}_X^n \rightarrow X$ proper.

Projective space is of finite type and separated. To verify the existence part of the valuative criterion, take a valuation ring V with fraction field K . A K -point of $\mathbb{P}_V^n$ is represented by a nonzero tuple $(a_0, \dots, a_n) \in K^{n+1}$, up to common scaling. The principal fractional ideals $a_i V$ are totally ordered. Choose a_j which generates their sum. After scaling by a_j^{-1} , every coordinate belongs to V and the j -th coordinate is 1. This normalized tuple defines a V -point extending the original K -point. Therefore $\mathbb{P}_X^n \rightarrow X$ is formally prim; it is quasi-compact, so Theorem 10.3.4 makes it universally closed. Separatedness and finite type now give properness.

The same conclusion is recorded in [The25, Tag 01WC]. □

Remark 10.4.2 (Why the finiteness in projectivity matters). Finite generation in the definition of projectivity supplies local finite type and quasi-compactness, and it is also used directly in the valuative extension argument. That argument fails for infinite projective space. For example, let I be infinite, choose an unbounded map $\phi: I \rightarrow \mathbb{N}$, and put $V = \mathbb{Z}_{(p)}$, $K = \mathbb{Q}$. The quotient

$$K^{(I)} \twoheadrightarrow K, \quad e_i \mapsto p^{-\phi(i)},$$

defines a point of the infinite projective space. No common scaling makes all coordinates lie in V with one a unit, because their valuations are unbounded below. Thus the point does not extend over V ; infinite projective space is not formally prim.

Remark 10.4.3 (Two blowup consequences). For a finitely presented closed subscheme $Z \subset X$, the standard blowup $\text{Bl}_Z(X) \rightarrow X$ is projective and therefore proper. Its image is the closure of $X \setminus Z$; consequently it is surjective exactly when $X \setminus Z$ is dense in X . The construction, Rees description, geometry, and universal property are given in Definition 9.2.2, Theorems 9.2.6 and 9.2.13, and Proposition 9.2.3.

Corollary 10.4.4 (Proper quasi-projective morphisms). *A quasi-projective morphism is projective if and only if it is proper.*

Proof. The forward implication is Theorem 10.4.1. For the converse, write Y as a quasi-compact open subscheme of a projective X -scheme P , as in Definition 3.2.13. Because $P \rightarrow X$ is separated and $Y \rightarrow X$ is proper, the map $Y \rightarrow P$ is proper: factor it as its closed graph

$$Y \hookrightarrow Y \times_X P$$

followed by the projection, which is a base change of $Y \rightarrow X$. The open immersion $Y \hookrightarrow P$ is therefore prim and hence closed by Proposition 10.1.6(2). A closed subscheme of a projective scheme is projective, so $Y \rightarrow X$ is projective. □

Remark 10.4.5 (Elimination and the missing converse). The closedness of $\mathbb{P}_R^n \rightarrow \operatorname{Spec}(R)$ is the geometric form of elimination: the condition that a finite homogeneous system acquire a nontrivial solution after extending a residue field defines a closed subset of the base. The converse is false: proper need not mean projective, and proper nonprojective schemes exist even over fields. Chow's lemma explains the precise sense in which proper schemes nevertheless admit projective models.

Theorem 10.4.6 (Chow's lemma). *Let S be quasi-compact and quasi-separated, let $X \rightarrow S$ be separated and of finite type. There exist an integer $n \geq 0$, a proper surjective morphism*

$$\pi: X' \longrightarrow X,$$

and an S -immersion $X' \hookrightarrow \mathbb{P}_S^n$. In particular, $X' \rightarrow S$ is quasi-projective. If X is proper over S , then $X' \rightarrow S$ is projective.

Reference and proof roadmap. The existence statement is [The25, Tag 0202]. The proof first enlarges X to a separated scheme of finite presentation over S , descends this model to a noetherian approximation of the qcqs base, and applies the noetherian Chow lemma [The25, Tag 0200]. In the noetherian case, one chooses finitely many affine charts with a common dense open, compactifies the charts in projective spaces, and uses the scheme-theoretic closure of the diagonal of that dense open to build X' . Checking properness over each affine chart of X completes the construction. These approximation and closure arguments are the heavy content and are cited rather than reproduced here.

The scheme X' is quasi-compact, and a quasi-compact immersion into $\mathbb{P}_S^n$ is quasi-projective. If $X \rightarrow S$ is proper, then the composite $X' \rightarrow X \rightarrow S$ is proper by Proposition 10.1.4. It is also quasi-projective, so Corollary 10.4.4 makes it projective. $\square$

10.5. Properness and finiteness

Definition 10.5.1 (Integral and finite morphisms). Let $f: Y \rightarrow X$ be a morphism of schemes.

- (i) It is *integral* if it is affine and, for every affine open $\operatorname{Spec}(A) \subset X$, writing $f^{-1}(\operatorname{Spec}(A)) = \operatorname{Spec}(B)$, every element of B is integral over A .
- (ii) It is *finite* if it is affine and every such B is a finite A -module. Equivalently, it is integral and locally of finite type.

The locally quasi-finite and quasi-finite terminology remains that of Definition 8.6.1 and Proposition 8.6.2: the former has discrete fibers, while the latter is quasi-compact and has finite underlying fiber sets. In either case the fibers may retain nonreduced scheme structure.

Lemma 10.5.2 (Dominance for affine schemes). *For a ring map $\varphi: A \rightarrow B$, the morphism $\operatorname{Spec}(B) \rightarrow \operatorname{Spec}(A)$ is dominant if and only if $\ker(\varphi)$ is a nil ideal.*

Proof. The closure of the image is $V(\ker(\varphi))$. Indeed, the ideal of that closure is the intersection of the inverse images of all prime ideals of B , which is $\varphi^{-1}(\sqrt{(0)_B}) = \sqrt{\ker(\varphi)}$. Thus the image is dense precisely when $V(\ker(\varphi)) = \operatorname{Spec}(A)$, equivalently when every element of the kernel is nilpotent. $\square$

Theorem 10.5.3 (Valuative criterion for integrality). *A morphism $f: Y \rightarrow X$ is integral if and only if it is affine and formally prim.*

Cited algebraic input and proof roadmap. If f is integral, the going-up theorem and stability of integrality under base change make it universally closed. Hence it is formally prim by Theorem 10.3.4.

Conversely, an affine morphism is quasi-compact, so formal primness and Theorem 10.3.4 make it universally closed. The algebraic theorem

$$\text{integral} \iff \text{affine and universally closed}$$

is [The25, Tag 01WM], and proves the converse. Concretely, for $A \rightarrow B$ and $b \in B$, one applies closedness after the polynomial base change $A[T] \rightarrow B[T]$ to the locus $bT = 1$. Its closed image avoids the zero section. Translating this disjointness into an equation in $A[T]$, and then reversing the coefficients, produces a monic polynomial satisfied by b . This elimination step is the substantial affine input, so it is not presented here as an uncited one-line argument. $\square$

Theorem 10.5.4 (Affine characterizations of finiteness). *For a morphism $f: Y \rightarrow X$, the following are equivalent:*

- (i) f is finite;
- (ii) f is affine and proper;
- (iii) f is affine and projective.

Proof. If f is finite, it is affine and proper by Proposition 10.1.6. Conversely, suppose f is affine and proper. Properness implies formal primness, so Theorem 10.5.3 makes f integral. It is also of finite type. On affine opens this says that the corresponding algebra is generated by finitely many integral elements. Adjoining one integral element at a time gives a finite module at each step, so the algebra is finite. Thus f is finite. This proves (1) $\Leftrightarrow$ (2) and recovers [The25, Tag 01WN].

It remains to connect projectivity. If $Y = \text{Spec}_X(\mathcal{B}) \rightarrow X$ is finite, multiplication gives a surjection of quasi-coherent algebras

$$\text{Sym}_{\mathcal{O}_X}(\mathcal{B}) \twoheadrightarrow \mathcal{B}$$

and therefore a closed immersion

$$Y \hookrightarrow \mathbb{A}_X(\mathcal{B}).$$

Because $\mathcal{B}$ is a finite $\mathcal{O}_X$ -module, this affine space is the standard quasi-projective open chart in $\mathbb{P}_X(\mathcal{B} \oplus \mathcal{O}_X)$. If $\bar{Y}$ is the scheme-theoretic closure of Y in this projective bundle, then closedness inside the affine chart gives

$$Y = \bar{Y} \cap \mathbb{A}_X(\mathcal{B}).$$

Thus Y is a standard quasi-projective open of the projective X -scheme $\bar{Y}$. It is already proper, so Corollary 10.4.4 makes it projective in the convention of Definition 3.2.13. Finally, an affine projective morphism is affine proper by Theorem 10.4.1. This proves (3) equivalent to the first two conditions. $\square$

Theorem 10.5.5 (Zariski's Main Theorem, factorization form). *Let S be quasi-compact and quasi-separated and let $f: Y \rightarrow S$ be quasi-finite and separated. There is a factorization*

$$Y \xhookrightarrow{j} \bar{Y} \xrightarrow{\pi} S$$

where j is a quasi-compact open immersion and π is finite. For an arbitrary base, this factorization is available after restricting to an affine open of the base.

Reference and proof roadmap. This is the form of Zariski's Main Theorem in [The25, Tag 05K0], globalizing the pointwise form Theorem 8.6.3. The proof first places a separated quasi-finite morphism inside an integral affine envelope, then writes the integral algebra as a filtered union of finite subalgebras. Because Y is quasi-compact and of finite type, its open immersion and the relations defining it descend to one finite stage. Factoring the resulting quasi-compact immersion into an open immersion followed by a closed immersion gives $\bar{Y}$. The approximation and descent steps are the heavy content; they are not reproved here. $\square$

Corollary 10.5.6 (Finiteness from properness and quasi-finiteness). *For a morphism $f : Y \rightarrow X$, the three conditions of Theorem 10.5.4 are further equivalent to:*

- (iv) f is quasi-finite and proper;
- (v) f is locally quasi-finite, universally closed, and separated.

Proof. A finite morphism is quasi-finite, universally closed, and separated, so (1) implies both new conditions.

Assume (4). The assertion that f is finite is local on X , so restrict to an affine open. Apply Theorem 10.5.5 to factor $f = \pi \circ j$, with j open and π finite. Since f is proper and $\bar{Y} \rightarrow X$ is separated, the graph argument from Corollary 10.4.4 shows that j is proper. A proper open immersion is closed by Proposition 10.1.6(2). Hence Y is open and closed in $\bar{Y}$, and the restriction of the finite morphism π to this open-and-closed subscheme is finite. Thus (4) implies (1).

Finally assume (5). Universal closedness gives quasi-compactness by Theorem 10.1.3. Therefore locally quasi-finite becomes quasi-finite, and locally finite type becomes finite type. Together with separatedness, universal closedness now says that f is proper. Thus (5) implies (4), while the reverse implication is immediate.

The Zariski-main-theorem package behind these equivalences is also [The25, Tag 02LS]. $\square$

Appendices

Appendix A.

Sheaf Cohomology as Sheafification

Status: This is an auxiliary derived supplement, not part of the original lecture course. It grew out of discussions on understanding sheaf cohomology from a derived/animated perspective. The treatment follows Lurie’s Spectral Algebraic Geometry [Lur18] and Mathew’s work on Galois groups in stable homotopy theory [Mat16]; whenever comparison with ordinary sheaf cohomology is used, the site and completeness hypotheses are stated explicitly.

A.1. Conventions

Throughout this appendix we work in the derived setting and *drop the R-prefix* on derived functors. We use two Zariski sites, and will always say which one is meant: X_{Zar} is the small Zariski site of a fixed scheme X , while $(\text{Aff}/\text{Spec } R)_{\text{Zar}} \simeq (\text{CAlg}_R)_{\text{Zar}}^{\text{op}}$ is the big affine Zariski site over R . Concretely:

- All limits, colimits, and tensor products are derived. The symbol $\otimes$ stands for derived tensor product. If R , M , and N are connective, then

$$\pi_0(M \otimes_R N) \simeq \pi_0 M \otimes_{\pi_0 R} \pi_0 N.$$

For nonconnective inputs there is no such blanket formula; the classical tensor product is recovered either from discrete inputs or under the displayed connectivity hypotheses.

- $\Gamma(X, \mathcal{F})$ denotes derived global sections; the classical Γ is $H^0(X, \mathcal{F}) := \pi_0 \Gamma(X, \mathcal{F})$.
- $\text{Hom}_R(M, N)$ denotes derived Hom , so $\text{Ext}_R^i(M, N) = \pi_{-i} \text{Hom}_R(M, N)$.
- For a stable ∞ -category $\mathcal{C}$, the symbol Mod_R refers to the ∞ -category of (left) module spectra over a connective $\mathbb{E}_\infty$ -ring R (equivalently, $D(R)$ when R is discrete).
- Presheaves take values in a stable presentable category $\mathcal{D}$ with small limits — typically $D(\mathbb{Z})$, Sp , or Mod_R for a base ring. When hyperdescent matters, we either pass explicitly to hypercomplete $\mathcal{D}$ -valued sheaves or restrict to a bounded-below, left-complete t -structured setting where the required comparison is valid; these are two hypotheses, not a blanket identification.

For discrete coefficients, passing to the heart recovers the classical 1-categorical constructions; higher homotopy groups retain the higher cohomology rather than disappearing.

A.2. The Thesis: Cohomology = Sheafification + Global Sections

Fix a scheme X and let $L_X : \text{PShv}(X_{\text{Zar}}, \mathcal{D}) \rightarrow \text{Shv}(X_{\text{Zar}}, \mathcal{D})$ denote derived (and, under the preceding convention, hypercomplete) sheafification. Since X is terminal in its small Zariski site, a presheaf $\mathcal{F}$ determines the derived global-sections object

$$\Gamma(X, L_X \mathcal{F}) = (L_X \mathcal{F})(X).$$

For $\mathcal{D} = D(\mathbb{Z})$ and a bounded-below complex of ordinary sheaves, this is the usual $R\Gamma$. A presheaf on a big site must first be sheafified and then restricted to X_{Zar} ; identifying that restriction with a familiar

small-site sheaf is a separate big–small comparison. For quasi-coherent modules this comparison is the one used below. Thus the displayed formula is not a claim that evaluation of an arbitrary functor on Sch^{op} is automatically derived global sections.

Write $\mathcal{G} = L_X \mathcal{F}$. Three consequences of the sheaf condition illustrate the formalism:

- (i) *Mayer–Vietoris*. For an open cover $X = U \cup V$, the sheaf condition is exactly the fiber square

$$\begin{array}{ccc} \Gamma(X, \mathcal{G}) & \longrightarrow & \Gamma(U, \mathcal{G}) \\ \downarrow & \lrcorner & \downarrow \\ \Gamma(V, \mathcal{G}) & \longrightarrow & \Gamma(U \cap V, \mathcal{G}). \end{array}$$

- (ii) *Čech cohomology*. For an open cover $\mathfrak{U} = (U_i \rightarrow X)_{i \in I}$, write

$$U_n := \coprod_{(i_0, \dots, i_n) \in I^{n+1}} U_{i_0} \cap \dots \cap U_{i_n}.$$

Here $\Gamma(U_n, \mathcal{G})$ means the product of the section objects on the displayed multiple intersections (equivalently, sections after extending $\mathcal{G}$ to disjoint unions of objects of the site). The sheaf condition along $\mathfrak{U}$ unfolds into the equivalence

$$\Gamma(X, \mathcal{G}) \xrightarrow{\sim} \lim_{[n] \in \Delta} \Gamma(U_n, \mathcal{G}).$$

We call the right-hand side the *derived Čech complex* of $\mathfrak{U}$.

- (iii) *Pushforward*. For a morphism $f: Y \rightarrow X$, the pushforward $f_*: \text{Shv}_{\text{Zar}}(Y, \mathcal{D}) \rightarrow \text{Shv}_{\text{Zar}}(X, \mathcal{D})$ is the right adjoint to f^* , characterized by $\Gamma(U, f_* \mathcal{G}) \simeq \Gamma(f^{-1}(U), \mathcal{G})$ naturally in the open $U \subseteq X$ (the global-sections identity $\Gamma(X, f_* \mathcal{G}) \simeq \Gamma(Y, \mathcal{G})$ being the case $U = X$). The higher direct images are the homotopy sheaves $R^i f_* \mathcal{G} = \pi_{-i} f_* \mathcal{G}$ on X in the appropriate t -structure; they should not be confused with global cohomology.

This packages the formal origin of the standard cohomological tools in the sheaf condition and the universal property of sheafification. Injective resolutions and spectral sequences remain useful computational models; the point is that they compute an intrinsically defined derived object.

Remark A.2.1 (Where classical Čech can fail). The classical *naïve* Čech complex of a cover is obtained from the cosimplicial abelian group $[n] \mapsto H^0(U_n, \mathcal{G})$, whereas derived Čech uses the full objects $\Gamma(U_n, \mathcal{G})$. Nonzero higher cohomology on an intersection can therefore contribute information that the naïve complex does not see. We will use in Section A.5 the important sufficient criterion that every $\Gamma(U_n, \mathcal{G})$ be concentrated in degree zero; accidental agreement can occur without this criterion.

A.3. Quasi-coherent Sheaves are Sheaves

For a scheme X , recall from Definition 3.1.15 that the category of quasi-coherent modules is the limit

$$\text{Mod}_X = \lim_{(R, x) \in \text{El}(X)^{\text{op}}} \text{Mod}_R$$

with derived base change as transition functors. For an R -module M , the associated presheaf on CAlg_R is

$$\mathcal{F}_M: \text{CAlg}_R \rightarrow \mathcal{D}, \quad A \mapsto M \otimes_R A.$$

The main theorem of this appendix is:

Theorem A.3.1 (Quasi-coherent Modules are Zariski Sheaves). *For any ring R and any derived R -module M , the presheaf $\mathcal{F}_M$ is a sheaf on the big affine Zariski site $(\text{Aff}/\text{Spec } R)_{\text{Zar}}$. Equivalently, for every R -algebra A and every finite principal cover $(A \rightarrow A_{f_i})_i$, the canonical map*

$$M \otimes_R A \longrightarrow \lim_{[n] \in \Delta} \left(M \otimes_R \prod_{i_0, \dots, i_n} A_{f_{i_0} \cdots f_{i_n}} \right)$$

is an equivalence. In particular, sheafification does not change $\mathcal{F}_M$, and at the initial object $R \in \text{CAlg}_R$

$$\Gamma(\text{Spec}(R), \mathcal{F}_M) \simeq \mathcal{F}_M(R) = M.$$

Since M is concentrated in degree zero (for $M \in \text{Mod}_R^\vee$), the higher cohomology vanishes:

Corollary A.3.2 (Affine Cohomology Vanishing). *For an affine scheme $X = \text{Spec}(R)$ and a discrete quasi-coherent module M ,*

$$H^i(X, \mathcal{F}_M) = 0 \quad \text{for all } i > 0.$$

The strategy for proving Theorem A.3.1 is to establish the stronger statement that $\mathcal{F}_M$ satisfies *faithfully flat* descent. This is the content of the next section, via Mathew's framework of descendable algebras.

A.4. Derived Faithfully Flat Descent under a Size Hypothesis

The classical 1-categorical theorem Theorem 5.2.30 has no presentation or cardinality hypothesis. The argument below proves the stronger assertion that a faithfully flat connective algebra is *descendable*, hence that its Amitsur totalization is controlled by a pro-constant tower. This particular derived proof uses an $\aleph_n$ -presentability hypothesis. For the finite Zariski covers used in Theorem A.3.1, the product of localizations satisfies that hypothesis.

Throughout this section, $\mathcal{C}$ is a stable presentable symmetric monoidal category. In our application $\mathcal{C} = \text{Mod}_R$.

A.4.1. Descendable Algebras

Definition A.4.1 (Thick tensor ideal). For $A \in \text{CAlg}(\mathcal{C})$, the *thick tensor ideal generated by A* , denoted $\langle A \rangle \subset \mathcal{C}$, is the smallest full subcategory containing A and closed under finite (co)limits, retracts, and tensor products $- \otimes X$ for arbitrary $X \in \mathcal{C}$.

Definition A.4.2 (Descendable algebra, [Mat16, Def. 3.18]). $A \in \text{CAlg}(\mathcal{C})$ is *descendable* if $\mathbb{1}_{\mathcal{C}} \in \langle A \rangle$.

The cobar construction associated to A is the cosimplicial algebra

$$A^{\otimes \bullet}: \Delta \rightarrow \text{CAlg}(\mathcal{C}), \quad [n] \mapsto A^{\otimes(n+1)}.$$

Definition A.4.3 (Pro-constant cosimplicial object). A cosimplicial object $M^\bullet: \Delta \rightarrow \mathcal{C}$ determines the inverse tower

$$\cdots \longrightarrow \text{Tot}_{\leq 2}(M^\bullet) \longrightarrow \text{Tot}_{\leq 1}(M^\bullet) \longrightarrow \text{Tot}_{\leq 0}(M^\bullet), \quad \text{Tot}_{\leq n}(M^\bullet) := \lim_{[m] \in \Delta_{\leq n}} M^m.$$

It is *pro-constant with value M* if this cofiltered diagram is isomorphic in $\text{Pro}(\mathcal{C})$ to the constant pro-object on M . This is a statement in the pro-category, not a claim that the transition maps eventually become equivalences.

Example A.4.4 (Split cosimplicial objects). Let $M^{-1} \rightarrow M^\bullet$ be a split augmented cosimplicial object, meaning that it admits an extra codegeneracy (equivalently, a contracting homotopy compatible with the cosimplicial operators). Then its partial-totalization tower is pro-equivalent to the constant tower on M^{-1} . Thus it is pro-constant, although the individual objects $\mathrm{Tot}_{\leq n}(M^\bullet)$ need not stabilize at any finite stage.

Example A.4.5 (Tensoring with pro-constants). In any stable symmetric monoidal category $- \otimes X$ is exact, hence preserves *finite* limits (which coincide with finite colimits in the stable setting). It therefore carries the partial-totalization tower of a pro-constant cosimplicial object to the corresponding pro-constant tower after tensoring with X . When the original tower is pro-constant with value M , its limit is M , the tensor tower has limit $M \otimes X$, and hence

$$\left(\lim_{\Delta} M^\bullet \right) \otimes X \simeq \lim_{\Delta} (M^\bullet \otimes X).$$

In particular this applies to $C = \mathrm{Mod}_R$.

The two key theorems of Mathew connect descendability with pro-constancy:

Theorem A.4.6 (Mathew, [Mat16, Prop. 3.20]). $A \in \mathrm{CAlg}(C)$ is descendable iff the cobar $A^{\otimes \bullet}$ is pro-constant with limit $\mathbb{1}_C$.

Proof sketch. ($\Rightarrow$) Let $\mathcal{X} \subset C$ be the full subcategory of X such that $X \otimes A^{\otimes \bullet}$ is pro-constant with limit X . Example A.4.5 shows $\mathcal{X}$ is closed under finite (co)limits, retracts, and tensor products. By Example A.4.4, $A \in \mathcal{X}$ — the cosimplicial object $A \otimes A^{\otimes \bullet}$ is split via the obvious extra degeneracy. Hence $\langle A \rangle \subset \mathcal{X}$, in particular $\mathbb{1} \in \mathcal{X}$.

($\Leftarrow$) If $A^{\otimes \bullet}$ is pro-constant with limit $\mathbb{1}$, some $\mathrm{Tot}_{\leq n}(A^{\otimes \bullet})$ admits $\mathbb{1}$ as a retract. But $\mathrm{Tot}_{\leq n}(A^{\otimes \bullet})$ is a finite limit of $A, A^{\otimes 2}, \dots$, hence in $\langle A \rangle$. $\square$

Theorem A.4.7 (Mathew, [Mat16, Prop. 3.22]). If $A \in \mathrm{CAlg}(C)$ is descendable, then the canonical functor

$$C \xrightarrow{\sim} \lim_{\Delta} \mathrm{Mod}_{A^{\otimes(\bullet+1)}}(C)$$

is an equivalence.

Proof sketch. The adjunction $- \otimes A : C \rightleftarrows \mathrm{Mod}_A(C) : F$ satisfies the Barr–Beck–Lurie criterion. Conservativity of $- \otimes A$: if $X \otimes A \simeq 0$, the subcategory $\mathcal{Y} = \{Y \mid X \otimes Y \simeq 0\}$ contains A and is closed under finite (co)limits, retracts, and tensor products, so $\mathbb{1} \in \langle A \rangle \subset \mathcal{Y}$, giving $X \simeq 0$. Limit-exchange for split cosimplicial objects is similar, using Example A.4.5 and Theorem A.4.6. $\square$

A.4.2. Faithfully Flat Maps Are Descendable

We specialize $C = \mathrm{Mod}_R$, with $A = S$ for a connective ring map $R \rightarrow S$.

Definition A.4.8 (Flat / Faithfully flat). A connective R -module M is *flat* if $-\otimes_R M$ is t -exact, equivalently if $\pi_0 M$ is a flat $\pi_0 R$ -module and the canonical map $\pi_n R \otimes_{\pi_0 R} \pi_0 M \rightarrow \pi_n M$ is an isomorphism for every n . A connective ring map $R \rightarrow S$ is *faithfully flat* if S is a flat R -module and $\pi_0 R \rightarrow \pi_0 S$ is faithfully flat in the classical sense (equivalently, $\mathrm{Spec} \pi_0 S \rightarrow \mathrm{Spec} \pi_0 R$ is surjective).

The technical engine is a flat-Ext vanishing lemma due to Lurie:

Lemma A.4.9 (Flat Ext vanishing, [Lur18, Lem. D.3.3.6]). *Let R be a connective ring and M a flat R -module such that $\pi_0 M$ is $\aleph_n$ -presentable as a $\pi_0 R$ -module. Then for any connective $N \in \text{Mod}_R$ and any $k > n$,*

$$\text{Ext}_R^k(M, N) = 0.$$

Theorem A.4.10 (Derived faithfully flat descent under a size hypothesis). *Let $R \rightarrow S$ be a faithfully flat connective ring map. Assume that the module $\pi_0 S$ is $\aleph_n$ -presentable over $\pi_0 R$ for some finite n . Then S is descendable as an R -algebra, and consequently*

$$\text{Mod}_R \xrightarrow{\sim} \lim_{\Delta} \text{Mod}_{S^{\otimes(\bullet+1)}}.$$

A finite presentation of $\pi_0 S$ as a $\pi_0 R$ -module permits $n = 0$. A finitely presented algebra need not be finite as a module (for example $R[x]$); it nevertheless has a countable module presentation, so $n = 1$ suffices in that case. These are examples of the hypothesis, not interchangeable meanings of “finitely presented.”

Proof. By Theorem A.4.7, descendability of S implies the equivalence; we show S is descendable.

Set $K := \text{fib}(R \rightarrow S)$ and $C := \text{cofib}(R \rightarrow S)$, related by $K \simeq C[-1]$. The structure map $\rho: K \rightarrow R$ assembles, for each $m \geq 1$, into the m -fold tensor power

$$\rho^{(m)}: K^{\otimes m} \xrightarrow{\rho \otimes \dots \otimes \rho} R.$$

Claim: $\rho^{(m)}$ is null-homotopic for m sufficiently large.

By the shift $K \simeq C[-1]$,

$$\rho^{(m)} \in [K^{\otimes m}, R]_{\text{Mod}_R} = [C^{\otimes m}[-m], R]_{\text{Mod}_R} = \text{Ext}_R^m(C^{\otimes m}, R).$$

Since $R \rightarrow S$ is faithfully flat, $C = \text{cofib}(R \rightarrow S)$ is flat: base change along $R \rightarrow S$ carries the map $R \rightarrow S$ to the unit $S \rightarrow S \otimes_R S$, which is split by the multiplication, so $C \otimes_R S \simeq \text{cofib}(S \rightarrow S \otimes_R S)$ is a retract of the flat S -module $S \otimes_R S$; flatness of C then descends along the faithfully flat $R \rightarrow S$. (Mere flatness of S would not suffice: for $R = \mathbb{Z} \rightarrow S = \mathbb{Q}$ the cofiber is $\mathbb{Q}/\mathbb{Z}$, which is not flat.) Hence $C^{\otimes m}$ is flat as well. The cardinality assumption makes $\pi_0(C^{\otimes m})$ $\aleph_n$ -presentable, so by Lemma A.4.9, $\text{Ext}_R^m(C^{\otimes m}, R) = 0$ for $m > n$.

Concluding descendability: The factorization $\rho^{(m+1)}: K^{\otimes(m+1)} \xrightarrow{\rho \otimes \text{id}} K^{\otimes m} \xrightarrow{\rho^{(m)}} R$ produces a cofiber sequence

$$K^{\otimes m} \otimes_R S \rightarrow \text{cofib}(\rho^{(m+1)}) \rightarrow \text{cofib}(\rho^{(m)}).$$

Inductively, $\text{cofib}(\rho^{(m)}) \in \langle S \rangle$ for every m , since $K^{\otimes m} \otimes_R S \in \langle S \rangle$ and $\langle S \rangle$ is closed under cofibers. When $\rho^{(m)}$ is null-homotopic,

$$\text{cofib}(\rho^{(m)}) \simeq R \oplus K^{\otimes m}[1],$$

exhibiting R as a retract of an object in $\langle S \rangle$. Thus $R \in \langle S \rangle$, i.e., S is descendable. $\square$

A.4.3. Proof of the Main Theorem

Proof of Theorem A.3.1. We must check covers over *every* object of the big affine site. Fix an R -algebra A and a Zariski cover $(A \rightarrow A_{f_i})_{i \in I}$, where the f_i generate the unit ideal. A finite subfamily still generates the unit ideal, so we may assume I finite. Put

$$N := M \otimes_R A, \quad S := \prod_i A_{f_i}.$$

The map $A \rightarrow S$ is faithfully flat: each localization is flat, and $\text{Spec}(S) = \coprod_i \text{Spec}(A_{f_i}) \rightarrow \text{Spec}(A)$ is surjective. As an A -module, each A_f has a countable presentation (use generators e_m representing f^{-m}

and relations $f e_{m+1} = e_m$, so the finite product S is $\aleph_1$ -presentable. Thus the cardinality hypothesis of Theorem A.4.10 holds with $n = 1$.

Theorem A.4.10 gives an equivalence

$$\mathrm{Mod}_A \xrightarrow{\sim} \lim_{\Delta} \mathrm{Mod}_{S^{\otimes_A(\bullet+1)}}.$$

Tracing N through this equivalence, the unit of the adjunction sends it to the cosimplicial $S^{\otimes_A(\bullet+1)}$ -module $[n] \mapsto N \otimes_A S^{\otimes_A(n+1)}$, and the equivalence reads

$$N \xrightarrow{\sim} \lim_{[n] \in \Delta} (N \otimes_A S^{\otimes_A(n+1)}).$$

At level n ,

$$S^{\otimes_A(n+1)} \simeq \prod_{i_0, \dots, i_n} A_{f_{i_0} \dots f_{i_n}},$$

so this is exactly the derived Čech descent condition for $\mathcal{F}_M$ at the object A and the chosen cover. Since A and its cover were arbitrary, $\mathcal{F}_M$ is a sheaf on the whole big affine Zariski site. $\square$

A.5. Naïve Čech as a Special Case

Theorem A.3.1 immediately clarifies the relationship between derived and classical Čech cohomology.

Derived Čech is sheaf cohomology, always. Let $\mathcal{F}$ be a derived sheaf on X_{Zar} (hypercomplete when required by our convention) and $\mathfrak{U} = (U_i \rightarrow X)_{i \in I}$ any cover. The sheaf condition along $\mathfrak{U}$ is the equivalence

$$\Gamma(X, \mathcal{F}) \xrightarrow{\sim} \lim_{[n] \in \Delta} \Gamma(U_n, \mathcal{F}).$$

No truncation, no spectral sequence, no acyclicity hypothesis. The right-hand side is the *derived Čech complex*.

Naïve Čech recovers derived Čech when intersections are acyclic. The classical naïve Čech complex comes from the cosimplicial discrete abelian group $[n] \mapsto H^0(U_n, \mathcal{F})$. A sufficient condition for it to agree with derived Čech is that every $\Gamma(U_n, \mathcal{F})$ be concentrated in degree zero. By Corollary A.3.2, this holds when $\mathcal{F}$ is quasi-coherent and U_n is a coproduct of affine schemes. By the affine-intersection property of separated schemes (Remark 6.2.3), the condition is automatic for affine covers of separated schemes:

Corollary A.5.1 (Naïve Čech for Separated Schemes). *Let X be a separated scheme, $\mathfrak{U}$ an affine open cover, and $\mathcal{F}$ a quasi-coherent sheaf on X . Then naïve and derived Čech cohomology along $\mathfrak{U}$ coincide, and both equal sheaf cohomology:*

$$\check{H}^i(\mathfrak{U}, \mathcal{F}) \xrightarrow{\sim} H^i(X, \mathcal{F}) \quad \text{for all } i \geq 0.$$

Proof. For X separated, the diagonal $X \rightarrow X \times X$ is a closed immersion, so any finite intersection of affine opens is affine; hence each U_n is a coproduct of affine schemes. By Corollary A.3.2 the sections over each affine piece are concentrated in degree zero, and $\Gamma(U_n, \mathcal{F})$ — their product over the indexing set — is therefore concentrated in degree zero, where it agrees with $H^0(U_n, \mathcal{F})$. Hence the cosimplicial object $[n] \mapsto \Gamma(U_n, \mathcal{F})$ is discrete, and its limit is the Moore complex of the cosimplicial abelian group $[n] \mapsto H^0(U_n, \mathcal{F})$ — i.e., the naïve Čech complex. $\square$

Cartan–Leray, reinterpreted. Assume for simplicity that $\mathcal{F}$ is bounded below, so the totalization filtration has the usual first-quadrant convergence. The Bousfield–Kan spectral sequence for the cosimplicial derived sections has

$$E_1^{p,q} = H^q(U_p, \mathcal{F}), \quad d_1 = \sum_{i=0}^{p+1} (-1)^i d^i,$$

and hence

$$E_2^{p,q} = H^p(E_1^{\bullet,q}, d_1) \implies H^{p+q}(X, \mathcal{F}).$$

Thus the q -th row is formed from the groups $H^q(U_p, \mathcal{F})$ on the actual multiple intersections; it is not obtained merely by writing ordinary Čech cohomology of a sheaf $\mathcal{H}^q(\mathcal{F})$ without a comparison argument. If all higher $H^q(U_p, \mathcal{F})$ vanish, only the row $q = 0$ remains and the spectral sequence recovers naïve Čech cohomology. For unbounded coefficients one must state an appropriate completeness/convergence hypothesis rather than invoke this first-quadrant argument verbatim.

A.6. Connection with the Locally Ringed Space Picture

The category Mod_X defined as a limit

$$\text{Mod}_X = \lim_{(R,x) \in \text{El}(X)^{\text{op}}} \text{Mod}_R$$

agrees, when X is a scheme, with quasi-coherent $\mathcal{O}_X$ -modules on the locally ringed space $(|X|, \mathcal{O}_X)$; this is Proposition 6.4.5. The comparison of global sections is the separate statement Proposition 6.4.7. In particular, the fact that Mod_X is a limit of categories should not by itself be mistaken for a proof that an arbitrary limit of module values computes $\text{R}\Gamma$.

After passing to the derived enhancement and invoking that comparison, the formula

$$\Gamma(X, M) \simeq \lim_{(R,x) \in \text{El}(X)^{\text{op}}} M(x)$$

from Definition 3.1.20 is interpreted as a homotopy limit. For a diagram of discrete modules, its π_0 is the ordinary limit $\lim^0$, hence the classical $H^0(X, M)$, while the lower homotopy groups record the higher derived limits. On an affine scheme they vanish by Corollary A.3.2; on a general scheme they give the higher cohomology. In the bounded-below classical setting fixed in Section A.1, this object agrees with $\text{R}\Gamma$ computed by injective resolutions. Thus the resolution and sheafification viewpoints compute the same object once the site comparison and derived interpretation have been supplied.

Appendix B.

Schemes in Topos Theory

The main text constructs schemes as functors of points assembled from affine schemes. This appendix explains the complementary, point-free formulation: replace a topological space by its topos of sheaves, replace a sheaf of local rings by a local ring object in that topos, and then isolate the abstract “affine objects and open maps” that make the construction work. The resulting language accommodates derived and spectral geometries and leads naturally to fractured topoi, six-functor geometries, and analytic stacks.

Our baseline for terminology and scope is Chapter 7 of the Higher Topos Theory notes [Hoy26]. The basic definitions used below are stated explicitly. Results whose titles begin “Accepted” are deliberate inputs: we state the exact version used and give a proof roadmap, but do not present the roadmap as a proof. The final Gestalten section preserves the explicitly informal status of the source’s outlook.

The chapter follows one continuous arc. Ringed and locally ringed topoi recover ordinary schemes; a geometry packages affine tests, admissible maps, and covers; a fractured topos remembers the same structure intrinsically; a six-functor formalism supplies a new geometry; and analytic stacks provide the principal example. The closing outlook records how iterated categories of kernels are expected to assemble these levels into Gestalten.

Throughout this appendix:

- Categories refer to $(\infty, 1)$ -categories; classical categories are called 1-categories. We generally denote categories by $\mathcal{C}$.
- 2-Categories refer to $(\infty, 2)$ -categories, generally denoted by $\mathbb{C}$.
- Topoi refer to $(\infty, 1)$ -topoi. In a topos, coverings are effective epimorphisms.
- $\mathbf{An}$ denotes the ∞ -category of anima (spaces).

Recall that in Theorem 6.3.4 there exists an adjunction:

$$\mathrm{LocRingSpace} \begin{array}{c} \xrightarrow{\Gamma} \\ \underset{\mathrm{Spec}}{\xleftarrow{\perp}} \end{array} \mathrm{CAlg}^{\mathrm{op}}$$

where the left side is the category of locally ringed spaces, Γ is the global sections functor, and Spec is the prime spectrum functor, i.e., $\mathrm{Spec}(R) = (\mathrm{Prim}(R), \mathcal{O}_{\mathrm{Spec}(R)})$. The spectrum functor Spec is fully faithful, with essential image consisting of affine locally ringed spaces.

The definition of schemes follows: let Aff denote the full subcategory of affine locally ringed spaces, then the category of schemes Sch can be identified with those objects in $\mathrm{LocRingSpace}$ that can be covered by objects from Aff along open immersions.

To replicate this construction in topos theory, we need to accomplish three tasks:

- (i) Rewrite “locally ringed spaces” in the language of topos theory;
- (ii) Construct the spectrum functor Spec in the topos-theoretic framework;
- (iii) Define schemes as “objects covered by affine objects along admissible morphisms.”

B.1. Topos-Theoretic Preliminaries

The constructions in this appendix use a small amount of topos-theoretic vocabulary. We recall the definitions and key results we need without proof, following [Hoy26, Ch. 7] and the general conventions of Lurie [Lur09; Lur18].

B.1.1. Topoi

A colimit $X = \operatorname{colim}_i X_i$ in a presentable category $\mathcal{T}$ with pullbacks is *van Kampen* if the slice functor $\mathcal{T}^{\operatorname{op}} \rightarrow \operatorname{Cat}_\infty$, $X \mapsto \mathcal{T}_{/X}$, sends it to a limit. Equivalently, the colimit is both *universal* (preserved by every base change) and *effective* (every cartesian natural transformation $Y_\bullet \rightarrow X_\bullet$ is recovered from its colimit by pulling back along $X_i \rightarrow X$).

Definition B.1.1. A *topos* is a presentable category in which every small colimit is van Kampen.

B.1.2. Geometric and algebraic morphisms; points

A *geometric morphism* $\varphi: \mathcal{S} \rightarrow \mathcal{T}$ is a functor $\varphi_*: \mathcal{S} \rightarrow \mathcal{T}$ admitting a left exact left adjoint φ^* , called the *algebraic morphism* from $\mathcal{T}$ to $\mathcal{S}$. The resulting categories of topoi satisfy $\operatorname{Topos} \simeq \operatorname{Logos}^{\operatorname{op}}$, and both upgrade naturally to 2-categories $\operatorname{Topos}, \operatorname{Logos}$ (with algebraic morphisms as 1-morphisms in Logos , and $\operatorname{Hom}_{\operatorname{Logos}}(\mathcal{S}, \mathcal{T}) \subset \operatorname{Fun}(\mathcal{S}, \mathcal{T})$ on the nose). A *point* of $\mathcal{T}$ is a geometric morphism $\operatorname{An} \rightarrow \mathcal{T}$; equivalently, a left exact, colimit-preserving “stalk” functor $\mathcal{T} \rightarrow \operatorname{An}$. We write $\operatorname{Pt}(\mathcal{T}) = \operatorname{Hom}_{\operatorname{Topos}}(\operatorname{An}, \mathcal{T})$.

B.1.3. Étale geometric morphisms

Definition B.1.2. A geometric morphism $\varphi: \mathcal{T} \rightarrow \mathcal{S}$ is *étale* if there is an equivalence $\mathcal{T} \simeq \mathcal{S}_{/X}$ for some $X \in \mathcal{S}$ under which φ_* becomes the forgetful functor $\mathcal{S}_{/X} \rightarrow \mathcal{S}$.

We will use the following standard facts, all proved in [Hoy26, Prop. 4.33–4.34]: étale morphisms are closed under composition and base change, satisfy left cancellation, and yield a 2-categorical equivalence $\mathcal{S} \xrightarrow{\sim} \operatorname{Topos}_{/\mathcal{S}}^{\operatorname{ét}}$ via $X \mapsto \mathcal{S}_{/X}$. The prototypical examples are slice topoi: for any topological space X and open $U \subseteq X$, the inclusion induces an étale morphism $\operatorname{Shv}(U) \rightarrow \operatorname{Shv}(X)$; for any Zariski scheme $(\mathcal{T}, \mathcal{O})$ and $U \in \mathcal{T}$, the map $(\mathcal{T}_{/U}, \mathcal{O}_{|U}) \rightarrow (\mathcal{T}, \mathcal{O})$ is étale.

B.1.4. Congruences

Let $\mathcal{T}$ be a topos and $\Sigma \subseteq \operatorname{Ar}(\mathcal{T})$.

- Σ is *saturated* if it contains the isomorphisms and is closed under composition and colimits in $\operatorname{Ar}(\mathcal{T})$.
- Σ is *strongly saturated* if it is saturated and satisfies the 2-out-of-3 property. It is *of small generation* if it is the smallest strongly saturated class containing a set of morphisms. Under this additional hypothesis, [Hoy26, Prop. A.15] constructs an accessible localization $L: \mathcal{T} \rightarrow \mathcal{T}'$ whose inverted morphisms are exactly Σ . Without small generation, neither accessibility nor even the existence of the stated presentable localization follows from strong saturation alone.
- Σ is a *modality* if it is the left class of a factorization system that is stable under base change.

- Σ is a *congruence* if it is strongly saturated and stable under base change. A congruence of *small generation* is equivalently ([Hoy26, Cor. 4.16]) the kernel of a left exact accessible localization, or the class of φ^* -isomorphisms for an algebraic morphism $\varphi^*: \mathcal{T} \rightarrow \mathcal{S}$. It is these small-generated congruences that parametrize accessible quotients of $\mathcal{T}$ inside Topos; the unqualified class-theoretic notion does not automatically do so.

Inside the lattice of saturated classes one has $\text{Cong}(\mathcal{T}) = \text{SSat}(\mathcal{T}) \cap \text{Mdl}(\mathcal{T})$, with both $\text{SSat}, \text{Mdl} \hookrightarrow \text{Sat}(\mathcal{T})$.

Definition B.1.3. For a modality Σ , set $D(\Sigma) := \{u \in \Sigma \mid \Delta_u \in \Sigma\}$, and iterate: $D^{n+1} = D \circ D^n$, $D^\infty = \bigcap_n D^n$. The operation $\Sigma \mapsto D^\infty(\Sigma)$, called *décalage*, is right adjoint to the inclusion $\text{Cong}(\mathcal{T}) \hookrightarrow \text{Mdl}(\mathcal{T})$ ([Hoy26, Prop. 5.30]).

Theorem B.1.4 (Accepted theorem: Anel–Biedermann–Finster–Joyal). *For a small subset $\Sigma \subseteq \text{Ar}(\mathcal{T})$, the congruence generated by Σ equals the modality generated by the closure of Σ under iterated diagonals.*

Proof roadmap. Let M be the modality generated by the iterated diagonals of the maps in Σ . The décalage theorem identifies $D^\infty(M)$ with the largest congruence contained in M . Every iterated diagonal of a map in Σ lies in M , so $\Sigma \subseteq D^\infty(M)$. Minimality of the congruence generated by Σ gives $\Sigma^c \subseteq D^\infty(M) \subseteq M$; conversely, Σ^c is a modality containing every iterated diagonal of Σ , whence $M \subseteq \Sigma^c$. We accept the décalage input and its size bookkeeping from [Hoy26, §5.3]; the resulting statement is Theorem 5.36 there. $\square$

B.1.5. Lax slice 2-categories

For a 2-category $\mathbb{C}$ and $X \in \mathbb{C}$, the *lax slice* $\mathbb{C}_{//X}$ relaxes the strict slice by allowing comparison triangles to commute up to a *not-necessarily-invertible* 2-cell.

Definition B.1.5. The (left-)lax slice $\mathbb{C}_{//X}$ has

- objects: 1-morphisms $t: Y \rightarrow X$ in $\mathbb{C}$;
- 1-morphisms $(Y, t) \rightarrow (Z, u)$: pairs (f, ε) with $f: Y \rightarrow Z$ and a 2-cell $\varepsilon: u \circ f \rightarrow t$;
- 2-morphisms $(f, \varepsilon) \rightarrow (f', \varepsilon')$: pairs (ϕ, α) with $\phi: f \rightarrow f'$ a 2-cell and α an identification $\varepsilon \simeq \varepsilon' \circ (u \cdot \phi)$.

The strict slice $\mathbb{C}_{/X}$ embeds as the wide subcategory where ε is invertible ([GR17, §A.2.5]).

The crucial feature for our purposes is that right adjoints in $\mathbb{C}$ lift automatically into the lax slice: if $f \dashv g$ with counit $\varepsilon: fg \rightarrow \text{id}_X$, then $(f, \text{id}_f) \dashv (g, \varepsilon)$ in $\mathbb{C}_{//X}$ ([BH21, Lemma 8.10]). The counit ε is typically non-invertible, so (g, ε) does *not* live in the strict slice $\mathbb{C}_{/X}$ — only in the lax one. This is the reason lax slices appear in our treatment of structured topoi.

B.2. Ringed Topoi and Schemes

A scheme, classically, is a locally ringed space. To separate the structure sheaf from any particular presentation by points, one replaces “space” by “topos” and begins with a *ringed topos*. Classical schemes arise from static local ring objects on spatial topoi. Derived, spectral, and other geometries retain the same pattern but replace the coefficient objects and admissible maps; they are not being identified with static ringed topoi.

Definition B.2.1 (Ringed Topos). A *ringed topos* is a pair $(\mathcal{T}, \mathcal{O})$, where $\mathcal{T}$ is a topos and $\mathcal{O}$ is a static ring sheaf on $\mathcal{T}$, i.e., a small-limit-preserving functor

$$\mathcal{O}: \mathcal{T}^{\text{op}} \rightarrow \mathbf{CAlg}^{\nabla}.$$

A morphism of ringed topoi $(f, f^{\sharp}): (\mathcal{T}, \mathcal{O}_{\mathcal{T}}) \rightarrow (\mathcal{S}, \mathcal{O}_{\mathcal{S}})$ consists of a geometric morphism $f: \mathcal{T} \rightarrow \mathcal{S}$ and a map of ring sheaves $f^{\sharp}: f^* \mathcal{O}_{\mathcal{S}} \rightarrow \mathcal{O}_{\mathcal{T}}$. We write $\mathbf{RingTopos}$ for the category of ringed topoi and all such morphisms.

Remark B.2.2. We use the intrinsic notion of C -valued sheaves on a topos: for a topos $\mathcal{T}$ and a category C with small limits, a C -valued sheaf is a small-limit-preserving functor $\mathcal{T}^{\text{op}} \rightarrow C$. When $\mathcal{T} \simeq \mathbf{Shv}_{\tau}(\mathcal{D})$ for a small site $\mathcal{D}$, this matches the classical notion of a C -valued τ -sheaf on $\mathcal{D}$.

Definition B.2.3 (Locally Ringed Topos). A ringed topos $(\mathcal{T}, \mathcal{O})$ is *locally ringed* if $\mathcal{O}$ is a local ring object. Concretely, for every finitely presented ring S and every finite family $f_1, \dots, f_n \in S$ generating the unit ideal, the canonical map of objects of $\mathcal{T}$

$$\coprod_{i=1}^n \underline{\text{Hom}}_{\mathbf{CAlg}}(S[f_i^{-1}], \mathcal{O}) \longrightarrow \underline{\text{Hom}}_{\mathbf{CAlg}}(S, \mathcal{O})$$

is an effective epimorphism. Equivalently, $\mathcal{O}$ is a structure for the classical Zariski geometry of Definition B.3.3. If $\mathcal{T}$ has enough points, this is equivalent to every stalk of $\mathcal{O}$ being a local ring.

Definition B.2.4 (Local Morphism). Let $(f, f^{\sharp}): (\mathcal{T}, \mathcal{O}_{\mathcal{T}}) \rightarrow (\mathcal{S}, \mathcal{O}_{\mathcal{S}})$ be a morphism of ringed topoi, where $f: \mathcal{T} \rightarrow \mathcal{S}$ is a geometric morphism and $f^{\sharp}: f^* \mathcal{O}_{\mathcal{S}} \rightarrow \mathcal{O}_{\mathcal{T}}$ is a morphism of ring sheaves on $\mathcal{T}$. We say $(f, f^{\sharp})$ is *local* if $f^{\sharp}$ detects units: for each object U in $\mathcal{T}$ and the induced map $\varphi: (f^* \mathcal{O}_{\mathcal{S}})(U) \rightarrow \mathcal{O}_{\mathcal{T}}(U)$, the following square is a pullback:

$$\begin{array}{ccc} (f^* \mathcal{O}_{\mathcal{S}})(U)^{\times} & \xrightarrow{\varphi} & \mathcal{O}_{\mathcal{T}}(U)^{\times} \\ \downarrow & & \downarrow \\ (f^* \mathcal{O}_{\mathcal{S}})(U) & \xrightarrow{\varphi} & \mathcal{O}_{\mathcal{T}}(U) \end{array}$$

i.e., $\varphi^{-1}(\mathcal{O}_{\mathcal{T}}(U)^{\times}) = (f^* \mathcal{O}_{\mathcal{S}})(U)^{\times}$. We write $\mathbf{LocRingTopos}$ for the category whose objects are *locally* ringed topoi and whose morphisms are local morphisms. Thus $\mathbf{LocRingTopos}$ is not obtained merely by changing the morphisms of $\mathbf{RingTopos}$: its object class is restricted as well.

Remark B.2.5 (Classical analogy). For locally ringed spaces, the corresponding condition is the familiar one: $(f, f^{\sharp})$ is local iff for every point x , the induced stalk map

$$\mathcal{O}_{S, f(x)} \rightarrow \mathcal{O}_{\mathcal{T}, x}$$

is a local ring homomorphism. Definition B.2.4 is the topos-theoretic counterpart, expressed via the unit-detecting square.

Construction B.2.6 (Zariski Spectrum). For a static ring R , define its *Zariski spectrum* $\text{Spec}(R)_{\text{Zar}}$ as the following ringed topos.

Consider the locale $\text{Open}(\text{Prim}(R))$ of open sets of the prime spectrum $\text{Prim}(R)$, which is isomorphic to the locale Rad_R of radical ideals of R ordered by inclusion. Equipping it with the canonical topology gives the topos:

$$\mathcal{T}_R := \mathbf{Shv}(\text{Prim}(R)).$$

The structure sheaf $\mathcal{O}_R \in \mathbf{CAlg}^{\nabla}(\mathcal{T}_R)$ sends the basic open $D(f) = \text{Prim}(f)$ to the localization $R[f^{-1}]$; the assignment already satisfies Zariski descent, so it is a sheaf without further sheafification. Set $\text{Spec}(R)_{\text{Zar}} := (\mathcal{T}_R, \mathcal{O}_R)$. The resulting functor

$$\text{Spec}(-)_{\text{Zar}}: \mathbf{CAlg}^{\nabla, \text{op}} \hookrightarrow \mathbf{LocRingTopos}$$

is right adjoint to global sections $\Gamma: \text{LocRingTopos} \rightarrow \text{CAlg}^{\heartsuit, \text{op}}, (\mathcal{T}, \mathcal{O}) \mapsto \mathcal{O}(*_{\mathcal{T}})$, and fully faithful by Zariski descent.

Remark B.2.7. Similarly, one can replace the Zariski topology with the étale topology to define the *étale spectrum* $\text{Spec}(R)_{\text{ét}}$. In this case, $\mathcal{T}_R = \text{Shv}_{\text{ét}}(\text{Spec}(R))$, and the structure sheaf $\mathcal{O}_R$ is given by étale descent.

Definition B.2.8 (Zariski Scheme). A *Zariski scheme* is a ringed topos $(\mathcal{T}, \mathcal{O})$ such that there exists a covering family $\{U_i \rightarrow *_{\mathcal{T}}\}_{i \in I}$ in $\mathcal{T}$ such that for each $i \in I$, there is an equivalence of locally ringed topoi:

$$(\mathcal{T}_{|U_i}, \mathcal{O}|_{U_i}) \simeq \text{Spec}(R_i)_{\text{Zar}}$$

for some static ring R_i . Morphisms between Zariski schemes are local morphisms, and we denote their category by Sch_{Zar} .

Remark B.2.9. Similarly, one can define *étale schemes* as ringed topoi locally equivalent to $\text{Spec}(R)_{\text{ét}}$, denoting their category by $\text{Sch}_{\text{ét}}$. Étale schemes are also called Deligne–Mumford ∞ -stacks.

Example B.2.10. • Classical schemes are precisely 0-localic Zariski schemes, i.e., the case where $\mathcal{T}$ is a 0-localic topos (a topos equivalent to the category of sheaves on some locale).

- Classical Deligne–Mumford stacks are precisely étale schemes with 1-localic topoi.
- If $(\mathcal{T}, \mathcal{O})$ is a Zariski scheme and $X \in \mathcal{T}$, then $(\mathcal{T}_{|X}, \mathcal{O}|_X)$ is still a Zariski scheme.

Remark B.2.11 (Relationship with the Functor-of-Points Definition). The topos-theoretic Definition B.2.8 agrees with the functor-of-points definition used in the main text. Concretely, given a classical scheme X , its small Zariski topos $\text{Shv}(\text{Prim}(X))$ together with the structure sheaf is a Zariski scheme in the sense above, and every 0-localic Zariski scheme arises in this way (Example B.2.10). The topos formulation has the additional virtue of accommodating higher-localic objects such as Deligne–Mumford stacks.

B.3. Geometries and Structured Topoi

Classical algebraic geometry begins with three ingredients: affine test objects, basic open immersions, and covers by basic opens. A geometry records exactly these ingredients without presupposing rings.

Definition B.3.1 (Admissibility Structure). Let $\mathcal{C}$ be a category with finite limits. An *admissibility structure* is a wide subcategory $\mathcal{C}^{\text{ad}} \subseteq \mathcal{C}$ such that admissible morphisms are stable under base change, left cancellable, and closed under retracts. Left cancellation means that, for composable maps $X \xrightarrow{f} Y \xrightarrow{g} Z$, admissibility of g and $g \circ f$ implies admissibility of f .

Definition B.3.2 (Geometric Site and Geometry). Let $(\mathcal{C}, \mathcal{C}^{\text{ad}})$ carry an admissibility structure. A Grothendieck topology τ is *compatible* if every τ -covering sieve contains a covering sieve generated by admissible maps. The triple

$$\mathbf{G} = (\mathcal{C}, \mathcal{C}^{\text{ad}}, \tau)$$

is then a *geometric site*; its admissibly generated covers are called *admissible covers*. If $\mathcal{C}$ is also small and idempotent complete, we call $\mathbf{G}$ a *geometry*.

The classical Zariski geometry has $\mathcal{C} = (\text{CAlg}^{\text{fp}})^{\text{op}}$, with admissible morphisms opposite to principal localizations $R \rightarrow R[f^{-1}]$ and covers generated by finite families for which the chosen elements generate the unit ideal. Derived, spectral, and étale geometries replace the affine category and the admissible maps while retaining the same axioms.

Definition B.3.3 (Structured Topos). Let $G = (C, C^{\text{ad}}, \tau)$ be a geometry and $\mathcal{T}$ a topos. A G -structure on $\mathcal{T}$ is a left exact functor $\mathcal{O}: C \rightarrow \mathcal{T}$ such that every admissible cover $(U_i \rightarrow X)_{i \in I}$ is sent to an effective epimorphism

$$\coprod_{i \in I} \mathcal{O}(U_i) \longrightarrow \mathcal{O}(X).$$

We write $\text{Str}_G(\mathcal{T})$ for the full subcategory of $\text{Fun}^{\text{lex}}(C, \mathcal{T})$ spanned by these structures.

For the classical Zariski geometry and $\mathcal{T} = \text{An}$, a structure is exactly a local ring. Indeed, a ring A determines $S \mapsto \text{Hom}_{\text{CAlg}}(S, A)$, and the cover condition says that whenever the f_i generate (1), every map $S \rightarrow A$ sends at least one f_i to a unit. This is equivalent to the nonunits of A forming its unique maximal ideal. The same argument with π_0 characterizes local connective $\mathbb{E}_\infty$ -rings in spectral Zariski geometry.

Definition B.3.4 (Local Map of Structures). For $\mathcal{O}, \mathcal{O}' \in \text{Str}_G(\mathcal{T})$, a natural transformation $\alpha: \mathcal{O} \rightarrow \mathcal{O}'$ is *local* if, for every admissible map $U \rightarrow X$ of C , the square

$$\begin{array}{ccc} \mathcal{O}(U) & \xrightarrow{\alpha_U} & \mathcal{O}'(U) \\ \downarrow & \lrcorner & \downarrow \\ \mathcal{O}(X) & \xrightarrow{\alpha_X} & \mathcal{O}'(X) \end{array}$$

is a pullback. For the classical Zariski geometry this says precisely that a map of local rings detects units, recovering Definition B.2.4.

Construction B.3.5 (The two sheaf topoi of a geometry). Let (C, C^{ad}, τ) be a geometric site. The subcategory C^{ad} inherits the topology whose covering sieves are those generating τ -covering sieves in C . Restriction has a left adjoint obtained from left Kan extension followed by sheafification:

$$j_! : \text{Shv}_\tau(C^{\text{ad}}) \longrightarrow \text{Shv}_\tau(C), \quad j_! \dashv j^*.$$

The functor $j_!$ is faithful but generally not full. In particular, its essential image must not be called a full subtopos.

B.3.1. Classical scheme theory as a geometry

Recast in topos-theoretic terms, classical scheme theory exhibits the following interconnected structures. We use them here as standard accepted inputs, in the precise scope recorded in [Hoy26, §7.1].

- *Spec is fully faithful.* The functor $\text{Spec}(-)_{\text{Zar}} : \text{CAlg}^{\heartsuit, \text{op}} \hookrightarrow \text{LocRingTopos}$ exhibits affine schemes as a full subcategory of locally ringed topoi (Construction B.2.6).
- *Schemes embed into the Zariski topos.* The functor

$$\text{Sch}_{\text{Zar}} \rightarrow \text{Shv}_{\text{Zar}}(\text{CAlg}^{\text{op}}), \quad (\mathcal{T}, \mathcal{O}) \mapsto \text{Hom}(\text{Spec}(-)_{\text{Zar}}, (\mathcal{T}, \mathcal{O}))$$

is fully faithful.

- *Classifying topos for local rings.* The category $\mathcal{E} := \text{Shv}_{\text{Zar}}(\text{CAlg}^{\text{fp}, \text{op}})$ is the classifying topos for local rings: the category of algebraic morphisms $\mathcal{E} \rightarrow \mathcal{X}$ is equivalent to the category of local ring sheaves on $\mathcal{X}$.
- *Accepted result: étale slices.* A morphism of Zariski schemes is *étale* if the underlying geometric morphism is étale (Definition B.1.2); writing $\text{Sch}_{\text{Zar}}^{\text{ét}} := \text{Sch}_{\text{Zar}} \times_{\text{Topos}} \text{Topos}^{\text{ét}}$, this category is, after fixing universes, a topos, and

$$\text{Sch}_{\text{Zar}}^{\text{ét}} \simeq \text{Shv}_{\text{Zar}}(\text{CAlg}^{\text{ad}, \text{op}}),$$

where $\text{CAlg}^{\text{ad}} \subset \text{CAlg}$ is the wide subcategory of Zariski localizations $R \rightarrow R[s^{-1}]$. This is Proposition 7.7 of [Hoy26], including its size caveat. (This “étale” is the topos-theoretic notion; on affines it coincides with Zariski open immersion, not with the algebraic étale topology.)

- *Factorization system.* On any topos $\mathcal{T}$, every morphism $\alpha: A \rightarrow B$ of static ring sheaves factors uniquely as $A \rightarrow A[S^{-1}] \rightarrow B$, with the first arrow a localization and the second a local morphism.

The preceding facts motivate four pieces of additional structure on the classifying topos $\mathcal{E}$:

- (i) *Corporeal topos.* Left Kan extension and sheafification induce a faithful but non-full functor

$$j_! : \mathcal{E}^{\text{corp}} := \text{Shv}_{\text{Zar}}(\text{CAlg}^{\text{fp,ad,op}}) \longrightarrow \mathcal{E},$$

whose source is itself a topos. Objects in $\mathcal{E}^{\text{corp}}$ are called *corporeal objects*; the terminology refers to the source category, not to a full subcategory of $\mathcal{E}$.

- (ii) *Admissibility structure.* The wide subcategory $\mathcal{E}^{\text{ad}} \subset \mathcal{E}$ consists of representable étale morphisms.
- (iii) *Factorization system.* The functor $\text{Fun}_{\text{Topos}}(-, \mathcal{E}) : \text{Topos}^{\text{op}} \rightarrow \text{Cat}$ lifts to $\text{Topos}^{\text{op}} \rightarrow \text{Cat}^{\text{fact}}$.
- (iv) *Local morphisms.* The lax slice 2-category $\text{Topos}_{//\mathcal{E}}$ (see Definition B.1.5) has a wide subcategory $(\text{Topos}_{//\mathcal{E}})^{\text{loc}}$ consisting of local morphisms.

The accepted results in the next section make precise the sense in which these structures determine one another and recover Sch_{Zar} . Lurie’s theory of *fractured topoi* axiomatizes this phenomenon.

B.4. Fractured Topoi

The four structures just listed — the corporeal topos, the admissibility structure, the functorial factorization system, and the local morphisms in the lax slice — share a common pattern. The word “corporeal” does not mean that the distinguished topos is fully embedded in the ambient one.

Definition B.4.1 (Fractured Topos). A *fractured topos* is a topos $\mathcal{E}$, a category $\mathcal{E}^{\text{corp}}$, and a faithful functor $j_! : \mathcal{E}^{\text{corp}} \rightarrow \mathcal{E}$ satisfying:

- (i) $\mathcal{E}^{\text{corp}}$ has pullbacks and $j_!$ preserves them;
- (ii) $j_!$ has a conservative, colimit-preserving right adjoint $j^* : \mathcal{E} \rightarrow \mathcal{E}^{\text{corp}}$;
- (iii) for every map $U \rightarrow V$ in $\mathcal{E}^{\text{corp}}$, the square

$$\begin{array}{ccc} j_! j^* j_! U & \longrightarrow & j_! j^* j_! V \\ \downarrow & \lrcorner & \downarrow \\ j_! U & \longrightarrow & j_! V \end{array}$$

is a pullback;

- (4*) every retract in $\mathcal{E}$ of an object $j_! U$ is equivalent to $j_! U'$ for some $U' \in \mathcal{E}^{\text{corp}}$.

We call objects of $\mathcal{E}^{\text{corp}}$ *corporeal*. We write $\mathcal{E}^{\text{corp,full}} \subseteq \mathcal{E}$ for the *full* subcategory spanned by objects equivalent to some $j_! U$. This notation keeps the non-full functor $j_!$ distinct from the full subcategory on its objects.

Theorem B.4.2 (Accepted theorem: basic properties of fractured topoi). For a fractured topos $(\mathcal{E}, \mathcal{E}^{\text{corp}}, j_!)$:

- (i) $\mathcal{E}^{\text{corp}}$ is a topos and j^* is left exact as well as colimit-preserving;
- (ii) for every $X \in \mathcal{E}^{\text{corp}}$, the induced functor

$$j_! : (\mathcal{E}^{\text{corp}})_{/X} \longrightarrow \mathcal{E}_{/j_! X}$$

is fully faithful;

(iii) the objects $j_!X$ generate $\mathcal{E}$ under colimits.

Proof roadmap. As a right adjoint, j^* preserves finite limits; by axiom it also preserves colimits. The pullback axiom yields, after passing to slices, the mapping-space comparison in (ii). Conservativity and preservation of colimits then show that the objects $j_!X$ detect equivalences, which gives generation. The full higher-categorical argument is the result recorded after Definition 7.11 of [Hoy26]. $\square$

Theorem B.4.3 (Accepted theorem: a geometric site produces a fracture). *For a geometric site (C, C^{ad}, τ) , the adjunction of Construction B.3.5*

$$j_! : \text{Shv}_\tau(C^{\text{ad}}) \rightleftarrows \text{Shv}_\tau(C) : j^*$$

satisfies Definition B.4.1, including axiom (4*).

Proof roadmap. Restriction preserves limits and colimits; compatibility of the topology with admissible maps identifies its left adjoint with sheafified left Kan extension. Base-change stability gives the pullback axiom, and retract closure supplies (4*). This is Theorem 7.13 of [Hoy26]. $\square$

Example B.4.4. Classical, derived, and spectral Zariski geometries; classical and derived étale geometries; and geometries based on C^∞ -rings all give fractured topoi by Theorem B.4.3. In the classical Zariski case the functor is exactly the faithful non-full $j_!$ displayed above.

B.4.1. Equivalence of Fractured and Admissibility Structures

The two notions — fractured structure and (geometric) admissibility structure — are equivalent.

Definition B.4.5. Let $(\mathcal{E}, \mathcal{E}^{\text{corp}}, j_!)$ be a fractured topos. A morphism $f : Y \rightarrow X$ in $\mathcal{E}$ is *admissible* if, for every $Z \in \mathcal{E}^{\text{corp}}$ and every map $j_!Z \rightarrow X$, the pullback projection

$$Y \times_X j_!Z \longrightarrow j_!Z$$

is equivalent in the slice to $j_!h : j_!W \rightarrow j_!Z$ for some morphism $h : W \rightarrow Z$ in $\mathcal{E}^{\text{corp}}$. Thus it is the pullback map, not merely its source object, that must come from the corporeal category. These morphisms form a wide subcategory $\mathcal{E}^{\text{ad}} \subseteq \mathcal{E}$.

Definition B.4.6. An admissibility structure $\mathcal{E}^{\text{ad}}$ on a topos $\mathcal{E}$ is called *local* if:

- $\mathcal{E}^{\text{ad}}$ is a local class of morphisms, that is, it satisfies the following properties:
 - (i) It is closed under base change;
 - (ii) It is closed under composition;
 - (iii) Given a pullback square in $\mathcal{E}$ of the form

$$\begin{array}{ccc} Y' & \xrightarrow{g'} & Y \\ \downarrow f' & \lrcorner & \downarrow f \\ X' & \xrightarrow{g} & X \end{array}$$

in which $f' \in \mathcal{E}^{\text{ad}}$ and g is an effective epimorphism, we also have $f \in \mathcal{E}^{\text{ad}}$.

- For each $X \in \mathcal{E}$, the slice $\mathcal{E}_{/X}^{\text{ad}} := (\mathcal{E}^{\text{ad}})_{/X}$ is a presentable category, and the fully faithful inclusion $\mathcal{E}_{/X}^{\text{ad}} \hookrightarrow \mathcal{E}_{/X}$ preserves colimits (in particular, it has a right adjoint $\rho_X : \mathcal{E}_{/X} \rightarrow \mathcal{E}_{/X}^{\text{ad}}$).

An object $X \in \mathcal{E}$ is called *corporeal* if ρ_X preserves colimits. A local admissibility structure is called *geometric* if $\mathcal{E}$ can be generated by corporeal objects along colimits.

Theorem B.4.7 (Accepted theorem: fractures and geometric admissibility). *Let $\mathcal{E}$ be a topos. Then the assignments:*

$$(\mathcal{E}, \mathcal{E}^{\text{corp}}) \mapsto (\mathcal{E}, \mathcal{E}^{\text{ad}}) \quad (\text{Definition B.4.5})$$

and:

$$(\mathcal{E}, \mathcal{E}^{\text{ad}}) \mapsto (\mathcal{E}, \mathcal{E}^{\text{corp}}) \quad (\text{Definition B.4.6})$$

determine an isomorphism of posets:

$$\{\text{fractured structures on } \mathcal{E}\} \simeq \{\text{geometric admissibility structures on } \mathcal{E}\}.$$

Proof roadmap. Starting with a fracture, one checks the locality and slice-presentability conditions after pullback to corporeal objects. Conversely, a geometric local admissibility structure reconstructs the corporeal objects as those X for which ρ_X preserves colimits; the two constructions are inverse. This is Theorem 7.17 of [Hoy26]; the reconstruction uses the complete list of axioms in Definition B.4.6, not merely base-change stability. $\square$

B.4.2. Local Morphisms and Schemes

A third equivalent way to package the data of a fractured topos uses *local morphisms* in the lax slice 2-category.

Definition B.4.8. Let $(\mathcal{E}, \mathcal{E}^{\text{corp}}, j_!)$ be a fractured topos. Consider two geometric morphisms $f, g: \mathcal{T} \rightarrow \mathcal{E}$ and a natural transformation $\alpha: f^* \rightarrow g^*$. We say α is *local* if for each morphism $h: U \rightarrow V$ in $\mathcal{E}^{\text{corp}}$, the following square is a pullback:

$$\begin{array}{ccc} f^* j_! U & \xrightarrow{\alpha_{j_! U}} & g^* j_! U \\ \downarrow f^* j_! h & & \downarrow g^* j_! h \\ f^* j_! V & \xrightarrow{\alpha_{j_! V}} & g^* j_! V \end{array}$$

Theorem B.4.9 (Accepted theorem: local factorization system). *For every topos $\mathcal{T}$, the local transformations of Definition B.4.8 form the right class of a factorization system on $\text{Fun}_{\text{Topos}}(\mathcal{T}, \mathcal{E})$. As $\mathcal{T}$ varies, they determine a wide subcategory*

$$(\text{Topos}_{//\mathcal{E}})^{\text{loc}} \subseteq \text{Topos}_{//\mathcal{E}}$$

of the lax slice.

Proof roadmap. Use the maps $j_! h$ with h in $\mathcal{E}^{\text{corp}}$ as tests for an orthogonal factorization system on natural transformations. The fracture axioms identify the right-orthogonality condition with the pullback squares in Definition B.4.8; corporeal generation makes these tests detect the right class on all of $\mathcal{E}$. Compatibility with change of the test topos then assembles the right classes into the displayed wide subcategory. We accept the higher-categorical orthogonal-factorization construction recorded in [Hoy26, §7.3.2]. $\square$

Remark B.4.10. There are two different classifying topoi here. For

$$\mathcal{E}_{\text{ring}} = \text{PSh}(\text{CAlg}^{\text{fp,op}}),$$

the lax slice $\text{Topos}_{//\mathcal{E}_{\text{ring}}}$ classifies ringed topoi. Passing to

$$\mathcal{E}_{\text{loc}} = \text{Shv}_{\text{Zar}}(\text{CAlg}^{\text{fp,op}})$$

is a left exact localization that imposes the local-ring axiom; accordingly, $\mathbf{Topos}_{//\mathcal{E}_{\text{loc}}}$ is already the full subcategory on locally ringed topoi. The superscript “loc” on $(\mathbf{Topos}_{//\mathcal{E}_{\text{loc}}})^{\text{loc}}$ then restricts the *morphisms* to local morphisms between those locally ringed topoi. The “detects units” condition of Definition B.2.4 is the unpacking of Definition B.4.8 in this case.

Local morphisms allow us to recover the notion of scheme intrinsically.

Theorem B.4.11 (Accepted theorem: recovering schemes from a fracture). *Let $(\mathcal{E}, \mathcal{E}^{\text{corp}}, j_!)$ be a fractured topos and set*

$$\mathcal{L}_{\mathcal{E}} := (\mathbf{Topos}_{//\mathcal{E}})^{\text{loc}}.$$

For $X \in \mathcal{E}$, pullback and global sections define the presheaf

$$\Gamma^X: \mathcal{L}_{\mathcal{E}}^{\text{op}} \longrightarrow \mathbf{An}, \quad (a: \mathcal{T} \rightarrow \mathcal{E}) \longmapsto \Gamma_{\mathcal{T}}(a^*X).$$

The opposite is forced by our convention that arrows in $\mathbf{Topos}$ are geometric morphisms: a local morphism of structured topoi pulls sections back.

(i) *If $X \in \mathcal{E}^{\text{corp,full}}$, then Γ^X is represented by the object*

$$\text{Re}(X) := (\mathcal{E}_{/X}^{\text{ad}}, \mathcal{E}_{/X}^{\text{ad}} \hookrightarrow \mathcal{E}_{/X} \longrightarrow \mathcal{E}) \in \mathcal{L}_{\mathcal{E}};$$

that is, $\Gamma^X(-) \simeq \text{Map}_{\mathcal{L}_{\mathcal{E}}}(-, \text{Re}(X))$.

(ii) *The assignment $X \mapsto \text{Re}(X)$ is a fully faithful functor*

$$\text{Re}: \mathcal{E}^{\text{corp,full}} \longrightarrow \mathcal{L}_{\mathcal{E}}.$$

(iii) *A map $u: Y \rightarrow X$ in $\mathcal{E}$ is admissible if and only if the following condition holds. For every map $a: X' \rightarrow X$, set $Y' = Y \times_X X'$. Whenever $\Gamma^{X'}$ is represented by some $X' \in \mathcal{L}_{\mathcal{E}}$, the presheaf $\Gamma^{Y'}$ is represented by some $\mathcal{Y}'$, and the natural transformation $\Gamma^{Y'} \rightarrow \Gamma^{X'}$ induced by $Y' \rightarrow X'$ corresponds under Yoneda to an étale geometric morphism $\mathcal{Y}' \rightarrow X'$.*

Consequently the wide subcategory $\mathcal{L}_{\mathcal{E}}$ inside the lax slice determines both the corporeal objects and the admissible morphisms, and hence the fractured structure.

Proof roadmap. For a corporeal X , the admissible slice $\mathcal{E}_{/X}^{\text{ad}}$ has the universal property of a local structured topos equipped with a section of X , giving (i). Applying Yoneda to this universal property gives (ii). Representable étale morphisms are stable under arbitrary base change, so the same universal property detects admissibility after every $X' \rightarrow X$, which gives (iii). This is the variance-explicit form of Theorem 7.19 in [Hoy26]. Its displayed functor suppresses the opposite; the target $\text{Pt}(\text{PSh}(C))^{\text{op}}$ in Construction 7.20 confirms the contravariance used here. $\square$

Thus a geometric site produces its own spectrum functor and notion of scheme.

Construction B.4.12 (Accepted construction: the G-spectrum). Let $G = (C, C^{\text{ad}}, \tau)$ be a geometric site, and let

$$(\mathcal{E}, \mathcal{E}^{\text{corp}}) = (\text{Shv}_{\tau}(C), \text{Shv}_{\tau}(C^{\text{ad}}))$$

be the corresponding fractured topos. Evaluation of a local structure on the objects of C gives a functor

$$\Gamma_G: (\mathbf{Topos}_{//\mathcal{E}})^{\text{loc}} \longrightarrow \text{Pt}(\text{PSh}(C))^{\text{op}} \simeq \text{Pro}(C).$$

It has a right adjoint

$$\text{Spec}_G: \text{Pro}(C) \longrightarrow (\mathbf{Topos}_{//\mathcal{E}})^{\text{loc}};$$

when τ is subcanonical, Spec_G is fully faithful ([Hoy26, Construction 7.20]).

Construction roadmap. The values of a local G-structure define a left exact functor on C , hence a point of its presheaf topos. The pro-Yoneda equivalence identifies the opposite category of such points with $\text{Pro}(C)$. The adjoint functor theorem then produces Spec_G ; subcanonicity identifies its unit on pro-representables and gives full faithfulness. We accept the coherence and size bookkeeping of Construction 7.20 in [Hoy26]. $\square$

Definition B.4.13 (G-schemes). The category Sch_G is the full subcategory of $(\text{Topos}_{//\mathcal{E}})^{\text{loc}}$ on objects $(\mathcal{T}, \mathcal{O})$ admitting an effective-epimorphic family $\coprod_i U_i \rightarrow *_\mathcal{T}$ such that every restriction $(\mathcal{T}|_{U_i}, \mathcal{O}|_{U_i})$ lies in the image of Spec_G . Equivalently, in the functor-of-points description of [Hoy26, Definition 7.21], this is the essential image of

$$\text{Shv}_\tau(\text{Pro}(C)^{\text{ad}}) \longrightarrow \text{Shv}_\tau(\text{Pro}(C)).$$

B.5. Geometries from Six-Functor Formalisms

Six-functor formalisms provide a systematic source of geometries. The essential mechanism is that universal $!$ -descent singles out the covers; the associated sheaf topos is then fractured along the admissible objects. We follow [Hoy26, §7.4]; [Sch25] gives the broader six-functor framework.

Definition B.5.1 (Presentable six-functor formalism). Let (C, C^{ad}) be an admissibility structure. A *presentable six-functor formalism* is a lax symmetric monoidal functor

$$D: \text{Span}(C, C^{\text{ad}}) \longrightarrow \text{Pr}^{\text{L}}.$$

Here a span has arbitrary left leg and admissible right leg. The two orientations of a map encode the familiar $(-)^*$ and $(-)_!$ operations; their adjoints supply the other operations when defined.

Definition B.5.2 (D-topology). The *D-topology* τ_D on C is the Grothendieck topology generated by those families $(X_i \rightarrow X)_i$ in C^{ad} for which $D_!$ satisfies descent after every base change of the family. Such a family is said to satisfy *universal $!$ -descent*. Saying “generated by” is important: the displayed families are generators for the topology, not a second definition of every covering sieve.

Theorem B.5.3 (Accepted theorem: the D-geometry). *The triple $(C, C^{\text{ad}}, \tau_D)$ is a geometric site. Consequently*

$$(\text{Shv}_{\tau_D}(C), \text{Shv}_{\tau_D}(C^{\text{ad}}))$$

is a fractured topos, and it induces a local admissibility structure on $\text{Shv}_{\tau_D}(C)$.

Proof roadmap. Base-change stability of universal $!$ -descent gives the topology axioms. The span identities give base-change stability and left cancellation for admissible maps. One then applies the geometric-site construction of Theorem B.4.3 to obtain the fracture. This is Proposition 7.25 and Corollary 7.26 of [Hoy26]. $\square$

Theorem B.5.4 (Accepted theorem: Mann–Scholze extension). *There is a unique extension of D to the local admissibility structure*

$$D: \text{Span}(\text{Shv}_{\tau_D}(C), \text{Shv}_{\tau_D}(C^{\text{ad}})) \longrightarrow \text{Pr}^{\text{L}}$$

for which both

$$D^*: \text{Shv}_{\tau_D}(C)^{\text{op}} \longrightarrow \text{Cat} \quad \text{and} \quad D_!: (\text{Shv}_{\tau_D}(C^{\text{ad}}))^{\text{op}} \longrightarrow \text{Cat}$$

preserve limits.

Proof roadmap. First extend the two coefficient systems from representables by right Kan extension. Universal $!$ -descent makes the resulting values insensitive to the chosen presentations by covers; base change then reconstructs the span functor and its monoidal structure. Uniqueness follows because representable objects generate the sheaf topos under colimits. The full coherence argument is the heavy input recorded as Proposition 7.28 of [Hoy26]. $\square$

B.5.1. The Σ_D -localization

Ordinary τ_D -sheafification need not invert every morphism that the coefficient system $D_!$ cannot distinguish. The following second localization removes precisely the congruence that is invisible after all base changes and all iterated diagonals.

Construction B.5.5 (The D -congruence). On the corporeal topos consider

$$D_! : \mathrm{Shv}_{\tau_D}(C^{\mathrm{ad}})^{\mathrm{op}} \longrightarrow \mathrm{Cat}.$$

Let Σ_D be the largest congruence contained in the inverse image of the equivalences under this functor. Equivalently,

$$\Sigma_D = \left\{ f \mid \begin{array}{c} \text{every base change of every iterated diagonal } \Delta^n f \\ \text{is a } D_!\text{-isomorphism} \end{array} \right\}.$$

This is Construction 7.29 of [Hoy26]. In the conventions of Section B.1, the accessible localization used below requires Σ_D to be of small generation. We make that size hypothesis explicit; equivalently, one may take the existence of the stated accessible localization as part of the accepted construction.

Remark B.5.6. The definition of the D -topology implies that a $D_!$ -isomorphism is an effective epimorphism. Hence $\Sigma_D \subseteq \mathrm{Conn}_\infty$, the class of ∞ -connected morphisms.

Definition B.5.7 (Fractured D -localization). Assume that Σ_D is of small generation, and let i_Σ denote the fully faithful right adjoint of its accessible left exact localization. Define $\mathrm{Shv}_{\tau_D}^+(C)$ by the pullback

$$\begin{array}{ccc} \mathrm{Shv}_{\tau_D}^+(C) & \xrightarrow{\quad} & \mathrm{Shv}_{\tau_D}(C) \\ \downarrow & \lrcorner & \downarrow j^* \\ \mathrm{Shv}_{\tau_D}(C^{\mathrm{ad}})[\Sigma_D^{-1}] & \xrightarrow{i_\Sigma} & \mathrm{Shv}_{\tau_D}(C^{\mathrm{ad}}). \end{array}$$

This line is only a definition of the pullback category. The topos and fracture assertions are the content of the next accepted theorem.

Theorem B.5.8 (Accepted theorem: fractured localization). *The upper horizontal functor in Definition B.5.7 is the fully faithful right adjoint of an accessible left exact localization. Moreover,*

$$(\mathrm{Shv}_{\tau_D}^+(C), \mathrm{Shv}_{\tau_D}(C^{\mathrm{ad}})[\Sigma_D^{-1}])$$

is a fractured topos, with the admissible maps inherited from the original D -fracture.

Proof roadmap. The small-generation hypothesis produces a presentable reflective localization of the corporeal topos. One pulls that localization back along j^* and verifies that the resulting reflector is left exact. The unit, counit, and base-change identities for the original fracture then descend to the pullback and verify the four fractured-topos axioms. This is the concrete construction asserted after Construction 7.29 and in Remark 7.30 of [Hoy26]; the verification is not a formal consequence of the pullback diagram alone. $\square$

Theorem B.5.9 (Accepted theorem: descent of the formalism). *The extended six-functor formalism descends further to*

$$D: \text{Span}(\text{Shv}_{\tau_D}^+(C), \text{Shv}_{\tau_D}(C^{\text{ad}})[\Sigma_D^{-1}]) \longrightarrow \text{Pr}^L.$$

Proof roadmap. If every base change of an admissible map is a $D_!$ -isomorphism, self-duality of $D(Y)$ as a $D(X)$ -module exchanges $f_!$ with f^* and shows that f is also a D^* -isomorphism. The same duality turns universal $!$ -descent into universal $*$ -descent. Thus all operations and exchange maps factor through the localization. The complete coherence statement is Lemma 7.31 and Remark 7.32 of [Hoy26]. $\square$

Remark B.5.10. The preceding proof roadmap explains why Σ_D must be defined using all base changes: without that universal condition, the exchange maps needed for a span formalism would not necessarily survive localization.

B.6. Analytic Stacks

As the principal example, we now apply the construction to analytic rings in the sense of Clausen–Scholze. This section follows [Hoy26, §7.5] and deliberately records the scope of the condensed site, since changing its covers changes the theory.

Definition B.6.1 (Condensed anima). A *condensed anima* is a hypercomplete sheaf on the category CHaus of compact Hausdorff spaces, where a *finite* family $(X_i \rightarrow X)_{i \in I}$ covers precisely when $\coprod_{i \in I} X_i \rightarrow X$ is surjective. Equivalently, one may use ProFin with the induced topology. Arbitrary jointly surjective families are not the covers of this site.

For size control one often uses the light version

$$\text{CondAn} := \text{Shv}(\{\text{countably profinite sets}\})^{\text{hyp}}.$$

From now on, *analytic ring* means a light analytic ring, and “ring” means animated commutative ring.

Definition B.6.2 (Analytic ring). An *analytic ring* is a pair $A = (A^\bullet, \text{Mod}_A)$ consisting of a condensed ring $A^\bullet$ and a reflective inclusion

$$\iota: \text{Mod}_A \hookrightarrow \text{Mod}_{A^\bullet}, \quad L \dashv \iota,$$

such that

- (i) Mod_A is closed under colimits;
- (ii) it is closed under $\underline{\text{Hom}}(M, -)$ for every $M \in \text{Mod}_{A^\bullet}$;
- (iii) ιL preserves connective objects for the canonical t -structure; and
- (iv) $A^\bullet \in \text{Mod}_A$.

A morphism $f: A \rightarrow B$ is a homomorphism $A^\bullet \rightarrow B^\bullet$ whose pushforward on underlying modules restricts to $f_*: \text{Mod}_B \rightarrow \text{Mod}_A$. Its left adjoint is denoted f^* ; explicitly, include an A -module among the underlying $A^\bullet$ -modules, base-change to $B^\bullet$, and then apply the reflector to Mod_B . These objects and morphisms form AnRing .

Remark B.6.3 (Light versus non-light). In the non-light theory one additionally requires the Frobenius map $A \rightarrow A/p$ to be a morphism of analytic rings for every prime p . This is automatic in the light setting used here; it is not known to be automatic in the unrestricted non-light setting. This is the scope distinction of [Hoy26, Remark 7.35].

Definition B.6.4 (Proper, open, and !-able maps). Let $f: A \rightarrow B$ be a morphism of analytic rings.

- It is *proper* if f_* induces an equivalence

$$\mathrm{Mod}_B \xrightarrow{\sim} \mathrm{Mod}_{B^\flat} \times_{\mathrm{Mod}_{A^\flat}} \mathrm{Mod}_A.$$

- It is an *open immersion* if f^* admits a further left adjoint $f_!$ satisfying, for $M \in \mathrm{Mod}_A$ and $N \in \mathrm{Mod}_B$, the projection formula

$$f_!(f^*M \otimes_B N) \simeq M \otimes_A f_!N.$$

- It is *!-able* if the identity on $B^\flat$ defines an open immersion of analytic rings

$$(B^\flat, \mathrm{Mod}_{B^\flat} \times_{\mathrm{Mod}_{A^\flat}} \mathrm{Mod}_A) \longrightarrow (B^\flat, \mathrm{Mod}_B).$$

Write $\mathrm{AnRing}^!$ for the wide subcategory of !-able maps.

Lemma B.6.5 (Accepted lemma: compactification criterion). *A morphism $f: A \rightarrow B$ of analytic rings is !-able if and only if its opposite geometric arrow (the arrow represented in $\mathrm{AnRing}^{\mathrm{op}}$) factors as an open immersion followed by a proper morphism. Equivalently, in the analytic-ring direction there is a factorization*

$$A \longrightarrow C \longrightarrow B$$

whose first arrow is proper and whose second arrow is open.

Proof roadmap. For a !-able map, take $C = (B^\flat, \mathrm{Mod}_{B^\flat} \times_{\mathrm{Mod}_{A^\flat}} \mathrm{Mod}_A)$. The ring-direction map $A \rightarrow C$ is proper by construction and $C \rightarrow B$ is open by the definition of !-ability; reversing arrows gives the order in the first sentence of the lemma. Conversely, the projection formula and the pullback description of complete modules are stable under composing such a factorization. This is Lemma 7.38 of [Hoy26]. $\square$

Theorem B.6.6 (Accepted theorem: analytic six-functor formalism). *There is a presentable six-functor formalism*

$$\mathrm{D}: \mathrm{Span}(\mathrm{AnRing}^{\mathrm{op}}, \mathrm{AnRing}^{!,\mathrm{op}}) \longrightarrow \mathrm{Pr}^{\mathrm{L}}, \quad A \longmapsto \mathrm{Mod}_A.$$

Proof roadmap. Base change of complete modules gives f^* , restriction gives f_* , and the definition of a !-able map supplies $f_!$; the remaining adjoints and exchange morphisms are then organized by the universal property of span 2-categories. Verifying coherent base change and the projection formula is the heavy step. We accept Proposition 7.39 of [Hoy26], which points to [CLL25] for this construction. $\square$

For the formalism of Theorem B.6.6, we take the existence of the accessible Σ_{D} -localization as part of the accepted analytic stack construction; in the terminology of Construction B.5.5, this means that the required small-generation hypothesis has been fixed.

Definition B.6.7 (Analytic stacks). Define

$$\mathrm{AnStk} := \mathrm{Shv}_{\tau_{\mathrm{D}}}^+(\mathrm{AnRing}^{\mathrm{op}}).$$

Thus analytic stacks form the topos obtained first from universal !-descent and then from the Σ_{D} -localization of Definition B.5.7.

Remark B.6.8. This is the corrected definition recorded in Remark 7.41 of [Hoy26]. By Theorems B.5.4 and B.5.9, the analytic six-functor formalism extends to this topos.

B.7. Gestalten: an Informal Outlook

The final section of the baseline notes is explicitly schematic. We retain its definitions and organizing ideas because they explain why fractured localization appears, but none of the claims in this section is used as a proved theorem elsewhere in the book.

Definition B.7.1 (Stefanich rings and Gestalten). Let ${}^n\text{Pr}$ denote the proposed presentably symmetric monoidal higher category at categorical height n , with ${}^0\text{Pr} = \text{An}$ and ${}^1\text{Pr} = \text{Pr}$. A *Stefanich ring* is a categorical spectrum

$$\mathcal{A} = (\mathcal{A}_0, \mathcal{A}_1, \dots), \quad \mathcal{A}_n \in \text{CAlg}({}^n\text{Pr}),$$

together with equivalences

$$\mathcal{A}_n \simeq \text{End}_{\mathcal{A}_{n+1}}(\mathbb{1}) \quad (n \geq 0).$$

Write StRing for the resulting category and set

$$\text{Gest} := \text{StRing}^{\text{op}}.$$

Warning B.7.2 (Status of the Gestalten topos). Section 7.6 of [Hoy26] describes Gest as a topos only “up to various issues.” Those coherence, size, and presentability issues are not resolved in this appendix. We therefore do not invoke topos axioms for Gest below; statements about effective epimorphisms record the intended formalism rather than a theorem established here.

B.7.1. Affine levels and the relative sphere

The outlook proposes a fully faithful functor

$$\text{CAlg}({}^n\text{Pr})^{\text{op}} \longrightarrow \text{Gest}$$

by iterated delooping: starting with $\mathcal{A}_n \in \text{CAlg}({}^n\text{Pr})$, set

$$\mathcal{A}_{n+1} := \text{Mod}_{\mathcal{A}_n}({}^n\text{Pr})$$

and repeat. A Gestalt is called *n-affine* when it belongs to the essential image, equivalently when $\mathcal{A}_{k+1} \simeq \text{Mod}_{\mathcal{A}_k}({}^k\text{Pr})$ for all $k \geq n$.

Absolute 0-affine objects come from commutative monoids in anima . Stable presentable categories also attach Gestalten to ring spectra. To make these 0-affine one works relative to the sphere:

$$\text{Gest}_{\mathbb{S}} := \text{Gest}/_{\text{Spec}(\mathbb{S})}.$$

In this relative sense, the Gestalt associated with every $R \in \text{CAlg}(\mathbb{S})$ is 0-affine. This is the content and intended scope of Constructions 7.46–7.48 and Remark 7.49 of [Hoy26].

B.7.2. Delooping a six-functor formalism

Let $D: \text{Span}(C, C^{\text{ad}}) \rightarrow \text{Pr}^{\text{L}}$ be presentable. For $X \in C$, restriction to the admissible slice gives D_X , and the proposed next categorical level is the 2-category of kernels

$$K_D(X) := \text{Mod}_{D_X}(\text{Span}(C_{/X}^{\text{ad}}), \text{Pr}) \in {}^2\text{Pr}.$$

The outlook asserts that $X \mapsto K_D(X)$ is functorial and ambidextrous on $C^{\text{ad,op}}$, so that it assembles into a new six-functor formalism

$$K_D: \text{Span}(C, C^{\text{ad}}) \longrightarrow {}^2\text{Pr}.$$

It is designed so that

$$\mathrm{End}_{K_D(X)}(\mathbb{1}) \simeq D(X).$$

Iterating the kernel construction is therefore expected to produce

$$[X]_D = (\mathrm{End}_{D(X)}(\mathbb{1}), D(X), K_D(X), K_D^2(X), \dots) \in \mathrm{Gest}.$$

Definition B.7.3 (Künneth condition). The formalism D satisfies *Künneth* if

$$D^*: \mathcal{C}^{\mathrm{ad}} \longrightarrow \mathrm{CAlg}(\mathrm{Pr})^{\mathrm{op}}$$

preserves pullbacks.

Within the informal scope of this section, the analytic formalism of Theorem B.6.6 is asserted to satisfy Künneth; this is Example 7.51 of [Hoy26].

Under this condition the outlook identifies $K_D^n(X)$ with the iterated module category over $D(X)$, so that $[X]_D$ is 1-affine. It further predicts that $[-]_D: \mathcal{C}^{\mathrm{ad}} \rightarrow \mathrm{Gest}$ preserves pullbacks and that a family $(X_i \rightarrow X)_i$ is a D -cover exactly when

$$\coprod_i [X_i]_D \longrightarrow [X]_D$$

is an effective epimorphism. If these claims and the topos status of Gest are supplied, the universal property of sheafification gives a left exact colimit-preserving functor

$$F: \mathrm{Shv}_{\tau_D}(\mathcal{C}^{\mathrm{ad}}) \longrightarrow \mathrm{Gest}.$$

The notes compare the kernel $F^{-1}(\mathrm{Iso})$ with Σ_D and indicate a further map out of the Σ_D -localization. This comparison remains part of the schematic outlook: the displayed containment in the notes, by itself, is not enough to prove the required factorization through a localization. Equality is reported as an expectation in general and as a claim for analytic rings. We use neither assertion in Definition B.6.7.

Conclusion

The appendix has now completed the promised passage from ordinary schemes to topos-theoretic geometry. Locally ringed topoi isolate the classical model; geometric sites and fractured topoi abstract its affine and open pieces; local structured morphisms recover schemes; and six-functor formalisms build new fractures, culminating in analytic stacks. The Gestalten outlook points toward an iteration of this story, but its informal assertions have been kept strictly separate from the accepted results used in the preceding sections.

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